European Network of
Transmission System Operators
for Electricity

entsoe

Detail description of the CGMES profiles

Version: 2.4.15

Common Grid Model Exchange Standard (CGMES)

7 August 2014

Remark 30 May 2016:
This document shall not be used as a reference regarding the type of stereotypes assigned to classes and attribites which belong to teh CGMES.
The HTML documentation is the right reference regarding such information.

Contents

Contents 2
1 EquipmentBoundaryProfile 13
1.1 General 13
1.2 (Entsoe) EquipmentBoundaryVersion(Abstract) 13
1.3 Core 14
1.3.1 General 14
1.3.2 ConductingEquipment(Abstract) 14
1.3.3 ConnectivityNode 15
1.3.4 ConnectivityNodeContainer(Abstract) 17
1.3.5 Equipment(Abstract) 17
1.3.6 EquipmentContainer(Abstract) 18
1.3.7 PowerSystemResource(Abstract) 18
1.3.8 Terminal 19
1.3.9 BaseVoltage 19
1.3.10 GeographicalRegion 19
1.3.11 IdentifiedObject(Abstract) 19
1.3.12 (Entsoe) SubGeographicalRegion 20
1.4 Wires 21
1.4.1 General 21
1.4.2 Connector(Abstract) 21
1.4.3 EnergySchedulingType 21
1.4.4 EnergySource(Abstract) 22
1.4.5 Junction 22
1.4.6 Line 23
1.5 Global Used Datatypes 24
1.5.1 Datatypes 24
1.5.2 Enumerations 24
1.5.3 Primitives 25
2 TopologyBoundaryProfile 25
2.1 General 25
2.2 (Entsoe) TopologyBoundaryVersion(Abstract) 26
2.3 Core 27
2.3.1 General 27
2.3.2 BaseVoltage(Abstract) 27
2.3.3 ConnectivityNode 28
2.3.4 ConnectivityNodeContainer(Abstract) 28
2.3.5 IdentifiedObject(Abstract) 28
2.4 Topology 29
2.4.1 General 29
2.4.2 (Entsoe) TopologicalNode 29
2.5 Global Used Datatypes 31
2.5.1 Primitives 31
3 EquipmentProfile 32

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

3.1 General 32
3.2 (Entsoe) EquipmentVersion(Abstract) 32
3.3 DC 34
3.3.1 General 34
3.3.2 ACDCConverter(Abstract) 38
3.3.3 ACDCConverterDCTerminal 40
3.3.4 CsConverter 41
3.3.5 DCBaseTerminal(Abstract) 42
3.3.6 DCBreaker 43
3.3.7 DCBusbar 44
3.3.8 DCChopper 44
3.3.9 DCConductingEquipment(Abstract) 45
3.3.10 DCConverterUnit 46
3.3.11 DCDisconnector 46
3.3.12 DCEquipmentContainer 47
3.3.13 DCGround 47
3.3.14 DCLine 48
3.3.15 DCLineSegment 49
3.3.16 DCNode 50
3.3.17 DCSeriesDevice 50
3.3.18 DCShunt 51
3.3.19 DCSwitch 52
3.3.20 DCTerminal 52
3.3.21 PerLengthDCLineParameter 53
3.3.22 VsCapabilityCurve 53
3.3.23 VsConverter 54
3.3.24 Enumerations 55
3.4 Topology 56
3.4.1 General 56
3.4.2 BusNameMarker 56
3.5 Meas 56
3.5.1 General 56
3.5.2 Accumulator 60
3.5.3 AccumulatorLimit 60
3.5.4 (Operation) AccumulatorLimitSet 61
3.5.5 (Operation) AccumulatorReset 62
3.5.6 AccumulatorValue 62
3.5.7 Analog 63
3.5.8 (Operation) AnalogControl(Abstract) 64
3.5.9 AnalogLimit 65
3.5.10 (Operation) AnalogLimitSet 66
3.5.11 (Operation) AnalogValue 66
3.5.12 Command 67
3.5.13 (Operation) Control(Abstract) 68
3.5.14 (Operation) Discrete 69

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

3.5.15 DiscreteValue 70
3.5.16 (Operation) Limit(Abstract) 71
3.5.17 (Operation) LimitSet(Abstract) 71
3.5.18 (Operation) Measurement(Abstract) 72
3.5.19 (Operation) MeasurementValue(Abstract) 74
3.5.20 (Operation) MeasurementValueQuality 74
3.5.21 (Operation) MeasurementValueSource 75
3.5.22 (Operation) Quality61850(Abstract) 76
3.5.23 (Operation) RaiseLowerCommand 78
3.5.24 SetPoint 78
3.5.25 StringMeasurement 79
3.5.26 StringMeasurementValue 80
3.5.27 ValueAliasSet 81
3.5.28 ValueToAlias 81
3.5.29 Enumerations 82
3.6 Production 82
3.6.1 General 82
3.6.2 (Operation) EnergySchedulingType 84
3.6.3 (Operation) EnergySource 84
3.6.4 FossilFuel 86
3.6.5 GeneratingUnit 86
3.6.6 GrossToNetActivePowerCurve 89
3.6.7 (Operation) HydroGeneratingUnit 89
3.6.8 HydroPowerPlant 91
3.6.9 HydroPump 91
3.6.10 NuclearGeneratingUnit 92
3.6.11 SolarGeneratingUnit 93
3.6.12 ThermalGeneratingUnit 94
3.6.13 WindGeneratingUnit 95
3.6.14 Enumerations 96
3.7 Core 97
3.7.1 General 97
3.7.2 ACDCTerminal(Abstract) 100
3.7.3 BaseVoltage 100
3.7.4 BasicIntervalSchedule(Abstract) 101
3.7.5 Bay 101
3.7.6 ConductingEquipment(Abstract) 102
3.7.7 ConnectivityNode 103
3.7.8 ConnectivityNodeContainer(Abstract) 104
3.7.9 Curve(Abstract) 104
3.7.10 CurveData 105
3.7.11 Equipment(Abstract) 105
3.7.12 EquipmentContainer(Abstract) 106
3.7.13 GeographicalRegion 107
3.7.14 IdentifiedObject(Abstract) 107

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

3.7.15 PowerSystemResource(Abstract) 108
3.7.16 RegularIntervalSchedule(Abstract) 108
3.7.17 RegularTimePoint 109
3.7.18 (Operation) ReportingGroup 110
3.7.19 (Operation) SubGeographicalRegion 110
3.7.20 Substation 111
3.7.21 Terminal 111
3.7.22 VoltageLevel 112
3.7.23 Enumerations 113
3.8 OperationalLimits 114
3.8.1 General 114
3.8.2 ActivePowerLimit 116
3.8.3 ApparentPowerLimit 117
3.8.4 CurrentLimit 118
3.8.5 OperationalLimit(Abstract) 118
3.8.6 OperationalLimitSet 119
3.8.7 OperationalLimitType 120
3.8.8 VoltageLimit 121
3.8.9 Enumerations 122
3.9 Wires 123
3.9.1 General 123
3.9.2 ACLineSegment 127
3.9.3 AsynchronousMachine 129
3.9.4 Breaker 131
3.9.5 BusbarSection 132
3.9.6 Conductor(Abstract) 133
3.9.7 Connector(Abstract) 133
3.9.8 Disconnector 134
3.9.9 (ShortCircuit) EarthFaultCompensator(Abstract) 135
3.9.10 EnergyConsumer 136
3.9.11 ExternalNetworkInjection 137
3.9.12 Ground 140
3.9.13 GroundDisconnector 141
3.9.14 GroundingImpedance 141
3.9.15 Junction 142
3.9.16 Line 143
3.9.17 LinearShuntCompensator 143
3.9.18 LoadBreakSwitch 145
3.9.19 MutualCoupling 145
3.9.20 (ShortCircuit) NonlinearShuntCompensator 147
3.9.21 NonlinearShuntCompensatorPoint 148
3.9.22 PetersenCoil 149
3.9.23 PhaseTapChanger(Abstract) 150
3.9.24 PhaseTapChangerAsymmetrical 151
3.9.25 PhaseTapChangerLinear 152

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

3.9.26 PhaseTapChangerNonLinear(Abstract) 153
3.9.27 PhaseTapChangerSymmetrical 154
3.9.28 PhaseTapChangerTable 155
3.9.29 PhaseTapChangerTablePoint 156
3.9.30 PhaseTapChangerTabular 156
3.9.31 PowerTransformer 157
3.9.32 PowerTransformerEnd 159
3.9.33 ProtectedSwitch(Abstract) 162
3.9.34 RatioTapChanger 162
3.9.35 RatioTapChangerTable 163
3.9.36 RatioTapChangerTablePoint 164
3.9.37 ReactiveCapabilityCurve 164
3.9.38 RegulatingCondEq(Abstract) 165
3.9.39 RegulatingControl 166
3.9.40 RegulationSchedule 167
3.9.41 RotatingMachine(Abstract) 167
3.9.42 SeriesCompensator 169
3.9.43 ShuntCompensator(Abstract) 170
3.9.44 StaticVarCompensator 171
3.9.45 Switch 173
3.9.46 SwitchSchedule 174
3.9.47 SynchronousMachine 175
3.9.48 TapChanger(Abstract) 178
3.9.49 TapChangerControl 179
3.9.50 TapChangerTablePoint 180
3.9.51 TapSchedule 181
3.9.52 TransformerEnd(Abstract) 182
3.9.53 Enumerations 183
3.10 LoadModel 185
3.10.1 General 185
3.10.2 ConformLoad 188
3.10.3 ConformLoadGroup 189
3.10.4 ConformLoadSchedule 189
3.10.5 DayType 190
3.10.6 (Operation) EnergyArea(Abstract) 191
3.10.7 (Operation) LoadArea 191
3.10.8 LoadGroup(Abstract) 191
3.10.9 LoadResponseCharacteristic 192
3.10.10 NonConformLoad 194
3.10.11 NonConformLoadGroup 195
3.10.12 NonConformLoadSchedule 196
3.10.13 Season 196
3.10.14 (Operation) SeasonDayTypeSchedule(Abstract) 197
3.10.15 (Operation) StationSupply 198
3.10.16 SubLoadArea 199

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

3.11 Equivalents 199
3.11.1 General 199
3.11.2 (Operation) EquivalentBranch 200
3.11.3 EquivalentEquipment(Abstract) 204
3.11.4 EquivalentInjection 205
3.11.5 EquivalentNetwork 207
3.11.6 EquivalentShunt 207
3.12 ControlArea 208
3.12.1 General 208
3.12.2 ControlArea 209
3.12.3 (Operation) ControlAreaGeneratingUnit 210
3.12.4 TieFlow 211
3.12.5 Enumerations 212
3.13 Global Used Datatypes 212
3.13.1 Datatypes 212
3.13.2 Enumerations 220
3.13.3 Primitives 228
4 TopologyProfile 228
4.1 General 228
4.2 (Entsoe) TopologyVersion(Abstract) 230
4.3 Core 231
4.3.1 General 231
4.3.2 ACDCTerminal(Abstract) 232
4.3.3 BaseVoltage(Abstract) 232
4.3.4 ConnectivityNode 233
4.3.5 ConnectivityNodeContainer(Abstract) 233
4.3.6 ReportingGroup(Abstract) 233
4.3.7 IdentifiedObject(Abstract) 233
4.3.8 (Entsoe) Terminal 234
4.4 DC 235
4.4.1 General 235
4.4.2 (Description) ACDCConverterDCTerminal 235
4.4.3 DCBaseTerminal(Abstract) 236
4.4.4 DCEquipmentContainer(Abstract) 236
4.4.5 DCNode 236
4.4.6 (Description) DCTerminal 237
4.4.7 DCTopologicalNode 237
4.5 Topology 238
4.5.1 General 238
4.5.2 TopologicalNode 238
4.6 Global Used Datatypes 239
4.6.1 Primitives 239
5 SteadyStateHypothesisProfile 239
5.1 General 239
5.2 (Entsoe) SteadyStateHypothesisVersion(Abstract) 240

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

5.3 Core 241
5.3.1 General 241
5.3.2 ACDCTerminal(Abstract) 242
5.3.3 ConductingEquipment(Abstract) 243
5.3.4 Equipment(Abstract) 243
5.3.5 IdentifiedObject(Abstract) 243
5.3.6 PowerSystemResource(Abstract) 244
5.3.7 Terminal 244
5.4 ControlArea 245
5.4.1 General 245
5.4.2 ControlArea 245
5.5 DC 245
5.5.1 General 245
5.5.2 ACDCConverter(Abstract) 246
5.5.3 ACDCConverterDCTerminal 247
5.5.4 CsConverter 247
5.5.5 DCBaseTerminal(Abstract) 248
5.5.6 DCTerminal 248
5.5.7 VsConverter 249
5.5.8 Enumerations 249
5.6 Equivalents 250
5.6.1 General 250
5.6.2 EquivalentEquipment(Abstract) 250
5.6.3 EquivalentInjection 251
5.7 Generation 251
5.7.1 General 251
5.7.2 (Description) GeneratingUnit 251
5.7.3 (Description) HydroGeneratingUnit 252
5.7.4 (Description) NuclearGeneratingUnit 252
5.7.5 (Description) SolarGeneratingUnit 252
5.7.6 (Description) ThermalGeneratingUnit 253
5.7.7 (Description) WindGeneratingUnit 253
5.8 Wires 253
5.8.1 General 253
5.8.2 (Description) AsynchronousMachine 253
5.8.3 (Description) Breaker 254
5.8.4 (Description) Disconnector 254
5.8.5 (Description) EnergyConsumer 254
5.8.6 (Description) EnergySource 255
5.8.7 (Description) ExternalNetworkInjection 255
5.8.8 (Description) GroundDisconnector 256
5.8.9 (Description) LoadBreakSwitch 256
5.8.10 PhaseTapChanger(Abstract) 257
5.8.11 PhaseTapChangerAsymmetrical 257
5.8.12 PhaseTapChangerLinear 257

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

5.8.13 PhaseTapChangerNonLinear(Abstract) 258
5.8.14 PhaseTapChangerSymmetrical 258
5.8.15 PhaseTapChangerTabular 258
5.8.16 ProtectedSwitch(Abstract) 259
5.8.17 RatioTapChanger 259
5.8.18 RegulatingCondEq(Abstract) 259
5.8.19 RotatingMachine(Abstract) 260
5.8.20 StaticVarCompensator 260
5.8.21 (Description) Switch 261
5.8.22 (Description) SynchronousMachine 261
5.8.23 (Description) TapChangerControl 262
5.8.24 (Description) LinearShuntCompensator 262
5.8.25 (Description) NonlinearShuntCompensator 262
5.8.26 (Description) RegulatingControl 263
5.8.27 ShuntCompensator(Abstract) 264
5.8.28 TapChanger(Abstract) 264
5.8.29 Enumerations 265
5.9 LoadModel 265
5.9.1 General 265
5.9.2 ConformLoad 265
5.9.3 NonConformLoad 266
5.9.4 StationSupply 266
5.10 Global Used Datatypes 266
5.10.1 Datatypes 266
5.10.2 Enumerations 269
5.10.3 Primitives 271
6 StateVariablesProfile 271
6.1 General 271
6.2 (Entsoe) StateVariablesVersion(Abstract) 272
6.3 Core 273
6.3.1 General 273
6.3.2 ACDCTerminal(Abstract) 274
6.3.3 ConductingEquipment(Abstract) 274
6.3.4 IdentifiedObject(Abstract) 274
6.3.5 Terminal(Abstract) 275
6.4 DC 275
6.4.1 General 275
6.4.2 ACDCConverter(Abstract) 276
6.4.3 CsConverter 277
6.4.4 (Description) DCTopologicalIsland 277
6.4.5 VsConverter 278
6.5 StateVariables 278
6.5.1 General 278
6.5.2 (Description) SvStatus 278
6.5.3 (Operation) SvInjection 279

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

6.5.4 SvPowerFlow 279
6.5.5 SvShuntCompensatorSections 280
6.5.6 SvTapStep 281
6.5.7 SvVoltage 281
6.6 Topology 282
6.6.1 General 282
6.6.2 DCTopologicalNode(Abstract) 283
6.6.3 TopologicalNode(Abstract) 283
6.6.4 TopologicalIsland 283
6.7 Wires 284
6.7.1 General 284
6.7.2 ShuntCompensator(Abstract) 284
6.7.3 TapChanger(Abstract) 284
6.8 Global Used Datatypes 284
6.8.1 Datatypes 284
6.8.2 Enumerations 286
6.8.3 Primitives 287
7 DiagramLayoutProfile 287
7.1 General 287
7.2 (Entsoe) DiagramLayoutVersion(Abstract) 288
7.3 DiagramLayout 289
7.3.1 General 289
7.3.2 (Entsoe) DiagramStyle 289
7.3.3 (Entsoe) Diagram 289
7.3.4 DiagramObject 290
7.3.5 DiagramObjectGluePoint 292
7.3.6 DiagramObjectPoint 292
7.3.7 DiagramObjectStyle 293
7.3.8 TextDiagramObject 293
7.3.9 VisibilityLayer 294
7.4 Core 295
7.4.1 General 295
7.4.2 IdentifiedObject(Abstract) 295
7.5 Global Used Datatypes 295
7.5.1 Datatypes 295
7.5.2 Enumerations 296
7.5.3 Primitives 297
8 GeographicalLocationProfile 298
8.1 General 298
8.2 (Entsoe) GeographicalLocationVersion(Abstract) 298
8.3 Common 299
8.3.1 General 299
8.3.2 (Entsoe) CoordinateSystem 299
8.3.3 Location 300
8.3.4 PositionPoint 301

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

8.4 Core 301
8.4.1 General 301
8.4.2 IdentifiedObject(Abstract) 302
8.4.3 PowerSystemResource(Abstract) 303
8.5 Global Used Datatypes 303
8.5.1 Primitives 303
9 DynamicsProfile 303
9.1 General 303
9.2 (Entsoe) DynamicsVersion(Abstract) 303
9.3 Core 304
9.3.1 General 304
9.3.2 ACDCTerminal(Abstract) 305
9.3.3 ConductingEquipment(Abstract) 306
9.3.4 Equipment(Abstract) 306
9.3.5 IdentifiedObject(Abstract) 306
9.3.6 PowerSystemResource(Abstract) 307
9.3.7 Terminal(Abstract) 307
9.4 Wires 308
9.4.1 General 308
9.4.2 AsynchronousMachine(Abstract) 308
9.4.3 (Description) EnergyConsumer(Abstract) 309
9.4.4 EnergySource(Abstract) 309
9.4.5 RegulatingCondEq(Abstract) 310
9.4.6 RotatingMachine(Abstract) 310
9.4.7 SynchronousMachine(Abstract) 310
9.5 StandardInterconnections 310
9.5.1 General 310
9.5.2 RemoteInputSignal 324
9.5.3 Enumerations 325
9.6 StandardModels 326
9.6.1 General 326
9.6.2 DynamicsFunctionBlock(Abstract) 326
9.6.3 RotatingMachineDynamics(Abstract) 326
9.6.4 SynchronousMachineDynamics 327
9.6.5 AsynchronousMachineDynamics 355
9.6.6 TurbineGovernorDynamics 363
9.6.7 TurbineLoadControllerDynamics 481
9.6.8 MechanicalLoadDynamics 484
9.6.9 ExcitationSystemDynamics 489
9.6.10 OverexcitationLimiterDynamics 610
9.6.11 UnderexcitationLimiterDynamics 619
9.6.12 PowerSystemStabilizerDynamics 629
9.6.13 DiscontinuousExcitationControlDynamics 662
9.6.14 PFVArControllerType1Dynamics 668
9.6.15 VoltageAdjusterDynamics 672

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

9.6.16 PFVArControllerType2Dynamics 674
9.6.17 VoltageCompensatorDynamics 679
9.6.18 WindDynamics 684
9.6.19 LoadDynamics 719
9.7 UserDefinedModels 732
9.7.1 General 732
9.7.2 WindPlantUserDefined 733
9.7.3 WindType1or2UserDefined 734
9.7.4 WindType3or4UserDefined 735
9.7.5 SynchronousMachineUserDefined 736
9.7.6 AsynchronousMachineUserDefined 737
9.7.7 TurbineGovernorUserDefined 738
9.7.8 TurbineLoadControllerUserDefined 739
9.7.9 MechanicalLoadUserDefined 740
9.7.10 ExcitationSystemUserDefined 741
9.7.11 OverexcitationLimiterUserDefined 742
9.7.12 UnderexcitationLimiterUserDefined 743
9.7.13 PowerSystemStabilizerUserDefined 744
9.7.14 DiscontinuousExcitationControlUserDefined 745
9.7.15 PFVArControllerType1UserDefined 746
9.7.16 VoltageAdjusterUserDefined 746
9.7.17 PFVArControllerType2UserDefined 747
9.7.18 VoltageCompensatorUserDefined 748
9.7.19 LoadUserDefined 749
9.7.20 ProprietaryParameterDynamics 750
9.8 Global Used Datatypes 753
9.8.1 Datatypes 753
9.8.2 Enumerations 756
9.8.3 Primitives 761

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

1 EquipmentBoundaryProfile

1.1 General

This profile has been built on the basis of the IEC 61970-452 document and adjusted to fit the purpose of the ENTSO-E boundary profile.

Figure 1-1 shows the diagram.

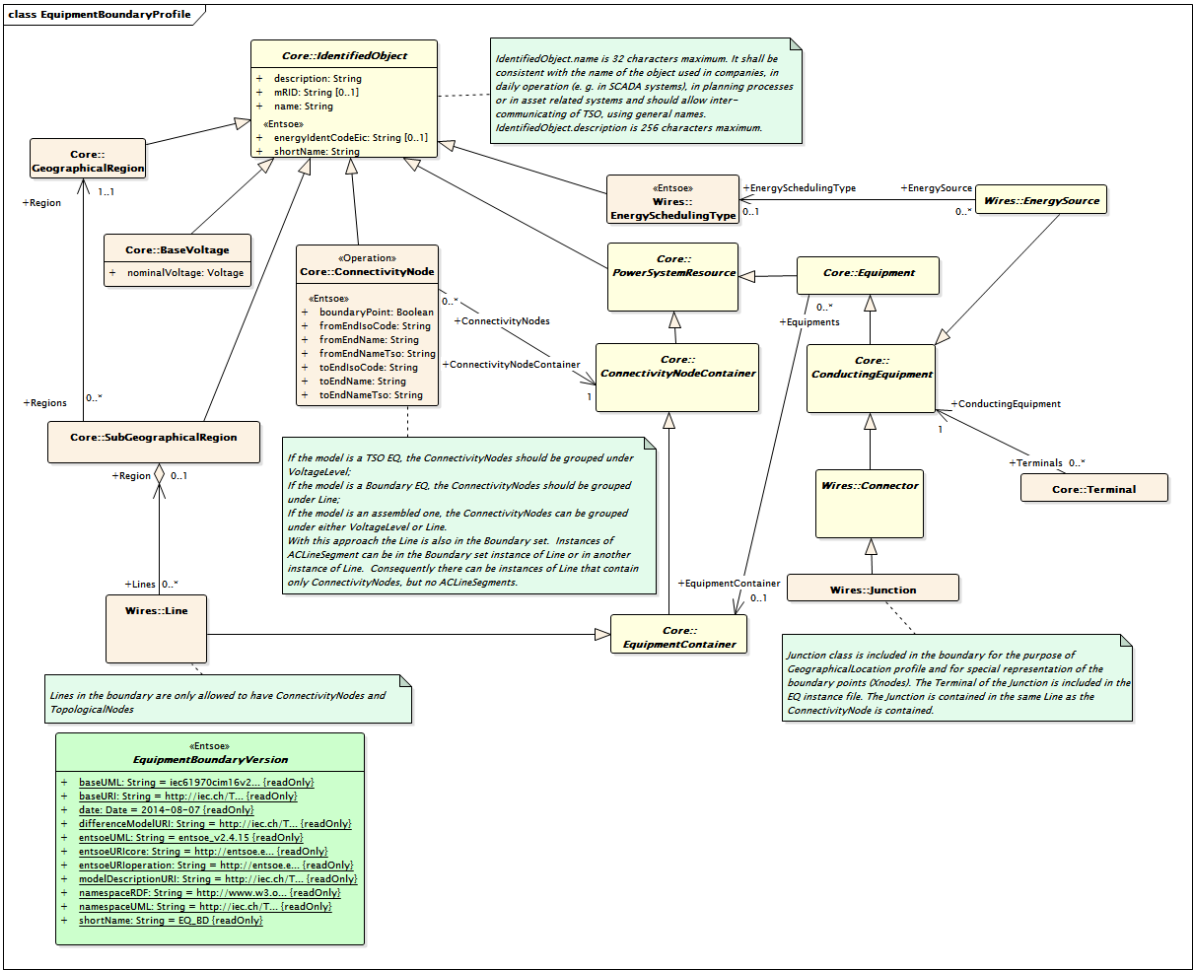


Figure 1-1 - diagram EquipmentBoundaryProfile

1.2 (Entsoe) EquipmentBoundaryVersion(Abstract)

Profile version details.

Table 1-1 shows all attributes of EquipmentBoundaryVersion.

Table 1-1 - Attributes of EquipmentBoundaryProfile::EquipmentBoundaryVersion

Attribute name	mult.	Attribute type	Description
baseUML (Entsoe)	1..1	String	ConstantValue= iec61970cim16v28_iec61968cim12v08_iec62325cim03v01a Base UML provided by CIM model manager.
baseURI (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/2013/61970-452/EquipmentBoundary/3

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			Profile URI used in the Model Exchange header and defined in IEC standards. It uniquely identifies the Profile and its version. It is given for information only and to identify the closest IEC profile to which this CGMES profile is based on.
date (Entsoe)	1..1	Date	ConstantValue= 2014-08-07 Profile creation date Form is YYYY-MM-DD for example for January 5, 2009 it is 2009-01-05.
differenceModelURI (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/61970-552/DifferenceModel/1# Difference model URI defined by IEC 61970-552.
entsoeUML (Entsoe)	1..1	String	ConstantValue= entsoe_v2.4.15 UML provided by ENTSO-E.
entsoeURIcore (Entsoe)	1..1	String	ConstantValue= http://entsoe.eu/CIM/EquipmentBoundary/3/1 Profile URI defined by ENTSO-E and used in the Model Exchange header. It uniquely identifies the Profile and its version. The last two elements in the URI (http://entsoe.eu/CIM/EquipmentBoundary/yy/zzz) indicate major and minor versions where: - yy - indicates a major version; - zzz - indicates a minor version.
entsoeURIoperation (Entsoe)	1..1	String	ConstantValue= http://entsoe.eu/CIM/EquipmentBoundaryOperation/3/1 Profile URI defined by ENTSO-E and used in the Model Exchange header. It uniquely identifies the Profile and its version. The last two elements in the URI (http://entsoe.eu/CIM/EquipmentBoundaryOperation/yy/zzz) indicate major and minor versions where: - yy - indicates a major version; - zzz - indicates a minor version.
modelDescriptionURI (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/61970-552/ModelDescription/1# Model Description URI defined by IEC 61970-552.
namespaceRDF (Entsoe)	1..1	String	ConstantValue= http://www.w3.org/1999/02/22-rdf-syntax-ns# RDF namespace.
namespaceUML (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/2013/CIM-schema-cim16# CIM UML namespace.
shortName (Entsoe)	1..1	String	ConstantValue= EQ_BD The short name of the profile used in profile documentation.

1.3 Core

1.3.1 General

1.3.2 ConductingEquipment(Abstract)

The parts of the AC power system that are designed to carry current or that are conductively connected through terminals.

Inheritance path: ->Equipment->PowerSystemResource->IdentifiedObject

Table 1-2 shows all attributes of ConductingEquipment.

Table 1-2 - Attributes of Core::ConductingEquipment

Attribute name	mult.	Attribute type	Description

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

description	1..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	1..1	String	see IdentifiedObject

Table 1-3 shows all association ends of Equipment with other classes.

Table 1-3 - Association ends of Core::Equipment with other classes

mult.	mult.	Class type name	Description
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

1.3.3 ConnectivityNode

Connectivity nodes are points where terminals of AC conducting equipment are connected together with zero impedance.

If the model is a TSO EQ, the ConnectivityNodes should be grouped under VoltageLevel; If the model is a Boundary EQ, the ConnectivityNodes should be grouped under Line; If the model is an assembled one, the ConnectivityNodes can be grouped under either VoltageLevel or Line. With this approach the Line is also in the Boundary set. Instances of ACLineSegment can be in the Boundary set instance of Line or in another instance of Line. Consequently there can be instances of Line that contain only ConnectivityNodes, but no ACLineSegments.

Inheritance path: ->IdentifiedObject

Table 1-4 shows all attributes of ConnectivityNode.

Table 1-4 - Attributes of Core::ConnectivityNode

Attribute name	mult.	Attribute type	Description
boundaryPoint (Entsoe)	1..1	Boolean	Identifies if a node is a BoundaryPoint. If boundaryPoint=true the ConnectivityNode or the TopologicalNode represents a BoundaryPoint.
fromEndIsoCode (Entsoe)	1..1	String	The attribute is used for an exchange of the ISO code of the region to which the “From” side of the Boundary point belongs to or it is connected to. The ISO code is two characters country code as defined by ISO 3166 (http://www.iso.org/iso/country_codes). The length of the string is 2 characters maximum. The attribute is a required for the Boundary Model Authority Set where this attribute is used only for the TopologicalNode in the Boundary Topology profile and ConnectivityNode in the Boundary Equipment profile.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

fromEndName (Entsoe)	1..1	String	The attribute is used for an exchange of a human readable name with length of the string 32 characters maximum. The attribute covers two cases: if the Boundary point is **placed on a tie-line** the attribute is used for exchange of the geographical name of the **substation** to which the "From" side of the tie-line is connected to. if the Boundary point is **placed in a substation** the attribute is used for exchange of the name of the **element** (e.g. PowerTransformer, ACLineSegment, Switch, etc) to which the "From" side of the Boundary point is connected to. The attribute is required for the Boundary Model Authority Set where it is used only for the TopologicalNode in the Boundary Topology profile and ConnectivityNode in the Boundary Equipment profile.
fromEndNameTso (Entsoe)	1..1	String	The attribute is used for an exchange of the name of the TSO to which the "From" side of the Boundary point belongs to or it is connected to. The length of the string is 32 characters maximum. The attribute is required for the Boundary Model Authority Set where it is used only for the TopologicalNode in the Boundary Topology profile and ConnectivityNode in the Boundary Equipment profile.
toEndIsoCode (Entsoe)	1..1	String	The attribute is used for an exchange of the ISO code of the region to which the "To" side of the Boundary point belongs to or it is connected to. The ISO code is two characters country code as defined by ISO 3166 (http://www.iso.org/iso/country_codes). The length of the string is 2 characters maximum. The attribute is a required for the Boundary Model Authority Set where this attribute is used only for the TopologicalNode in the Boundary Topology profile and ConnectivityNode in the Boundary Equipment profile.
toEndName (Entsoe)	1..1	String	The attribute is used for an exchange of a human readable name with length of the string 32 characters maximum. The attribute covers two cases: if the Boundary point is **placed on a tie-line** the attribute is used for exchange of the geographical name of the **substation** to which the "To" side of the tie-line is connected to. if the Boundary point is **placed in a substation** the attribute is used for exchange of the name of the **element** (e.g. PowerTransformer, ACLineSegment, Switch, etc) to which the "To" side of the Boundary point is connected to. The attribute is required for the

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			Boundary Model Authority Set where it is used only for the TopologicalNode in the Boundary Topology profile and ConnectivityNode in the Boundary Equipment profile.
toEndNameTso (Entsoe)	1..1	String	The attribute is used for an exchange of the name of the TSO to which the “To” side of the Boundary point belongs to or it is connected to. The length of the string is 32 characters maximum. The attribute is required for the Boundary Model Authority Set where it is used only for the TopologicalNode in the Boundary Topology profile and ConnectivityNode in the Boundary Equipment profile.
description	1..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	1..1	String	see IdentifiedObject

Table 1-5 shows all association ends of ConnectivityNode with other classes.

Table 1-5 - Association ends of Core::ConnectivityNode with other classes

mult.	mult.	Class type name	Description
ConnectivityNodeContainer (Operation)	[1..1]	ConnectivityNodeContainer	Container of this connectivity node. Association Based On : Core:: ConnectivityNode.ConnectivityNodes[0..*] ----- Core:: ConnectivityNodeContainer.ConnectivityNodeContainer[1]

1.3.4 ConnectivityNodeContainer(Abstract)

A base class for all objects that may contain connectivity nodes or topological nodes.

Inheritance path: ->PowerSystemResource->IdentifiedObject

Table 1-6 shows all attributes of ConnectivityNodeContainer.

Table 1-6 - Attributes of Core::ConnectivityNodeContainer

Attribute name	mult.	Attribute type	Description
description	1..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	1..1	String	see IdentifiedObject

1.3.5 Equipment(Abstract)

The parts of a power system that are physical devices, electronic or mechanical.

Inheritance path: ->PowerSystemResource->IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 1-7 shows all attributes of Equipment.

Table 1-7 - Attributes of Core::Equipment

Attribute name	mult.	Attribute type	Description
description	1..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	1..1	String	see IdentifiedObject

Table 1-8 shows all association ends of Equipment with other classes.

Table 1-8 - Association ends of Core::Equipment with other classes

mult.	mult.	Class type name	Description
EquipmentContainer	[0..1]	EquipmentContainer	Container of this equipment. Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] --- Core:: Equipment.Equipments[0..*]

1.3.6 EquipmentContainer(Abstract)

A modeling construct to provide a root class for containing equipment.

Inheritance path: ->ConnectivityNodeContainer->PowerSystemResource->IdentifiedObject

Table 1-9 shows all attributes of EquipmentContainer.

Table 1-9 - Attributes of Core::EquipmentContainer

Attribute name	mult.	Attribute type	Description
description	1..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	1..1	String	see IdentifiedObject

1.3.7 PowerSystemResource(Abstract)

A power system resource can be an item of equipment such as a switch, an equipment container containing many individual items of equipment such as a substation, or an organisational entity such as sub-control area. Power system resources can have measurements associated.

Inheritance path: ->IdentifiedObject

Table 1-10 shows all attributes of PowerSystemResource.

Table 1-10 - Attributes of Core::PowerSystemResource

Attribute name	mult.	Attribute type	Description
description	1..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	1..1	String	see IdentifiedObject

1.3.8 Terminal

An AC electrical connection point to a piece of conducting equipment. Terminals are connected at physical connection points called connectivity nodes.

1.3.9 BaseVoltage

Defines a system base voltage which is referenced.

Inheritance path: ->IdentifiedObject

Table 1-11 shows all attributes of BaseVoltage.

Table 1-11 - Attributes of Core::BaseVoltage

Attribute name	mult.	Attribute type	Description
nominalVoltage	1..1	Voltage	The power system resource's base voltage.
description	1..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	1..1	String	see IdentifiedObject

1.3.10 GeographicalRegion

A geographical region of a power system network model.

Inheritance path: ->IdentifiedObject

Table 1-12 shows all attributes of GeographicalRegion.

Table 1-12 - Attributes of Core::GeographicalRegion

Attribute name	mult.	Attribute type	Description
description	1..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	1..1	String	see IdentifiedObject

1.3.11 IdentifiedObject(Abstract)

This is a root class to provide common identification for all classes needing identification and naming attributes.

IdentifiedObject.name is 32 characters maximum. It shall be consistent with the name of the object used in companies, in daily operation (e. g. in SCADA systems), in planning processes or in asset

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

related systems and should allow inter-communicating of TSO, using general names. IdentifiedObject.description is 256 characters maximum.

Table 1-13 shows all attributes of IdentifiedObject.

Table 1-13 - Attributes of Core::IdentifiedObject

Attribute name	mult.	Attribute type	Description
description	1..1	String	The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy.
energyIdentCodeEic (Entsoe)	0..1	String	The attribute is used for an exchange of the EIC code (Energy identification Code). The length of the string is 16 characters as defined by the EIC code. References: Local issuing offices for EIC: https://www.entsoe.eu/publications/edi-library/links-to-eic-websites/ EIC description: https://www.entsoe.eu/index.php?id=73&libCat=eic .
mRID	0..1	String	Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552 Edition 1, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name	1..1	String	The name is any free human readable and possibly non unique text naming the object.
shortName (Entsoe)	1..1	String	The attribute is used for an exchange of a human readable short name with length of the string 12 characters maximum.

1.3.12 (Entsoe) SubGeographicalRegion

A subset of a geographical region of a power system network model.

Inheritance path: ->IdentifiedObject

Table 1-14 shows all attributes of SubGeographicalRegion.

Table 1-14 - Attributes of Core::SubGeographicalRegion

Attribute name	mult.	Attribute type	Description
description	1..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	1..1	String	see IdentifiedObject

Table 1-15 shows all association ends of SubGeographicalRegion with other classes.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 1-15 - Association ends of Core::SubGeographicalRegion with other classes

mult.	mult.	Class type name	Description
Region	[1..1]	GeographicalRegion	The geographical region to which this sub-geographical region is within. Association Based On : Core:: GeographicalRegion.Region[0..1] ----- Core:: SubGeographicalRegion.Regions[0..*]

1.4 Wires

1.4.1 General

1.4.2 Connector(Abstract)

A conductor, or group of conductors, with negligible impedance, that serve to connect other conducting equipment within a single substation and are modelled with a single logical terminal.

Inheritance path: ->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 1-16 shows all attributes of Connector.

Table 1-16 - Attributes of Wires::Connector

Attribute name	mult.	Attribute type	Description
description	1..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	1..1	String	see IdentifiedObject

Table 1-17 shows all association ends of Equipment with other classes.

Table 1-17 - Association ends of Core::Equipment with other classes

mult.	mult.	Class type name	Description
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

1.4.3 EnergySchedulingType

Used to define the type of generation for scheduling purposes.

Inheritance path: ->IdentifiedObject

Table 1-18 shows all attributes of EnergySchedulingType.

Table 1-18 - Attributes of Wires::EnergySchedulingType

Attribute name	mult.	Attribute type	Description
description	1..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	1..1	String	see IdentifiedObject

1.4.4 EnergySource(Abstract)

A generic equivalent for an energy supplier on a transmission or distribution voltage level.

Inheritance path: ->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 1-19 shows all attributes of EnergySource.

Table 1-19 - Attributes of Wires::EnergySource

Attribute name	mult.	Attribute type	Description
description	1..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	1..1	String	see IdentifiedObject

Table 1-20 shows all association ends of EnergySource with other classes.

Table 1-20 - Association ends of Wires::EnergySource with other classes

mult.	mult.	Class type name	Description
EnergySchedulingType	[0..1]	EnergySchedulingType	Energy Scheduling Type of an Energy Source Association Based On : Wires:: EnergySource.EnergySource[0..*] ----- Extension:: EnergySchedulingType.EnergySchedulingType[0..1]
EquipmentContainer	0..1	EquipmentContainer	 Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

1.4.5 Junction

A point where one or more conducting equipments are connected with zero resistance.

Junction class is included in the boundary for the purpose of GeographicalLocation profile and for special representation of the boundary points (Xnodes). The Terminal of the Junction is included in the EQ instance file. The Junction is contained in the same Line as the ConnectivityNode is contained.

Inheritance path: ->Connector->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 1-21 shows all attributes of Junction.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 1-21 - Attributes of Wires::Junction

Attribute name	mult.	Attribute type	Description
description	1..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	1..1	String	see IdentifiedObject

Table 1-22 shows all association ends of Equipment with other classes.

Table 1-22 - Association ends of Core::Equipment with other classes

mult.	mult.	Class type name	Description
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

1.4.6 Line

Contains equipment beyond a substation belonging to a power transmission line.

Lines in the boundary are only allowed to have ConnectivityNodes and TopologicalNodes

Inheritance path: ->EquipmentContainer->ConnectivityNodeContainer->PowerSystemResource->IdentifiedObject

Table 1-23 shows all attributes of Line.

Table 1-23 - Attributes of Wires::Line

Attribute name	mult.	Attribute type	Description
description	1..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	1..1	String	see IdentifiedObject

Table 1-24 shows all association ends of Line with other classes.

Table 1-24 - Association ends of Wires::Line with other classes

mult.	mult.	Class type name	Description
Region	[0..1]	SubGeographicalRegion	The lines within the sub-geographical region. Association Based On : Core:: SubGeographicalRegion.Region[0..1] ----- Wires:: Line.Lines[0..*]

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

1.5 Global Used Datatypes

1.5.1 Datatypes

1.5.1.1 Voltage

Electrical voltage, can be both AC and DC.

Value type is IEEE 754 simple precision floating point

Table 1-25 shows all attributes of Voltage.

Table 1-25 - Attributes of DomainProfile::Voltage.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= V
multiplier	0..1	UnitMultiplier	ConstantValue= k

1.5.2 Enumerations

1.5.2.1 UnitSymbol

The units defined for usage in the CIM.

Literal	Description	Code
VA	Apparent power in volt ampere.	
W	Active power in watt.	
VAr	Reactive power in volt ampere reactive.	
VAh	Apparent energy in volt ampere hours.	
Wh	Real energy in what hours.	
VArh	Reactive energy in volt ampere reactive hours.	
V	Voltage in volt.	
ohm	Resistance in ohm.	
A	Current in ampere.	
F	Capacitance in farad.	
H	Inductance in henry.	
degC	Relative temperature in degrees Celsius. In the SI unit system the symbol is ºC. Electric charge is measured in coulomb that has the unit symbol C. To distinguish degree Celsius form coulomb the symbol used in the UML is degC. Reason for not using ºC is the special character º is difficult to manage in software.	
s	Time in seconds.	
min	Time in minutes.	
h	Time in hours.	
deg	Plane angle in degrees.	
rad	Plane angle in radians.	
J	Energy in joule.	
N	Force in newton.	

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

S	Conductance in siemens.	
none	Dimension less quantity, e.g. count, per unit, etc.	
Hz	Frequency in hertz.	
g	Mass in gram.	
Pa	Pressure in pascal (n/m2).	
m	Length in meter.	
m2	Area in square meters.	
m3	Volume in cubic meters.	

1.5.2.2 UnitMultiplier

The unit multipliers defined for the CIM.

Literal	Description	Code
p	Pico 10**-12.	
n	Nano 10**-9.	
micro	Micro 10**-6.	
m	Milli 10**-3.	
c	Centi 10**-2.	
d	Deci 10**-1.	
k	Kilo 10**3.	
M	Mega 10**6.	
G	Giga 10**9.	
T	Tera 10**12.	
none	No multiplier or equivalently multiply by 1.	

1.5.3 Primitives

1.5.3.1 String

A string consisting of a sequence of characters. The character encoding is UTF-8. The string length is unspecified and unlimited.

1.5.3.2 Date

Date as "yyyy-mm-dd", which conforms with ISO 8601. UTC time zone is specified as "yyyy-mm-ddZ". A local timezone relative UTC is specified as "yyyy-mm-dd(+/-)hh:mm".

1.5.3.3 Boolean

A type with the value space "true" and "false".

1.5.3.4 Float

A floating point number. The range is unspecified and not limited.

2 TopologyBoundaryProfile

2.1 General

This profile has been built on the basis of the IEC 61970-456 document and adjusted to fit the purpose of the ENTSO-E boundary profile.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Figure 2-1 shows the diagram.

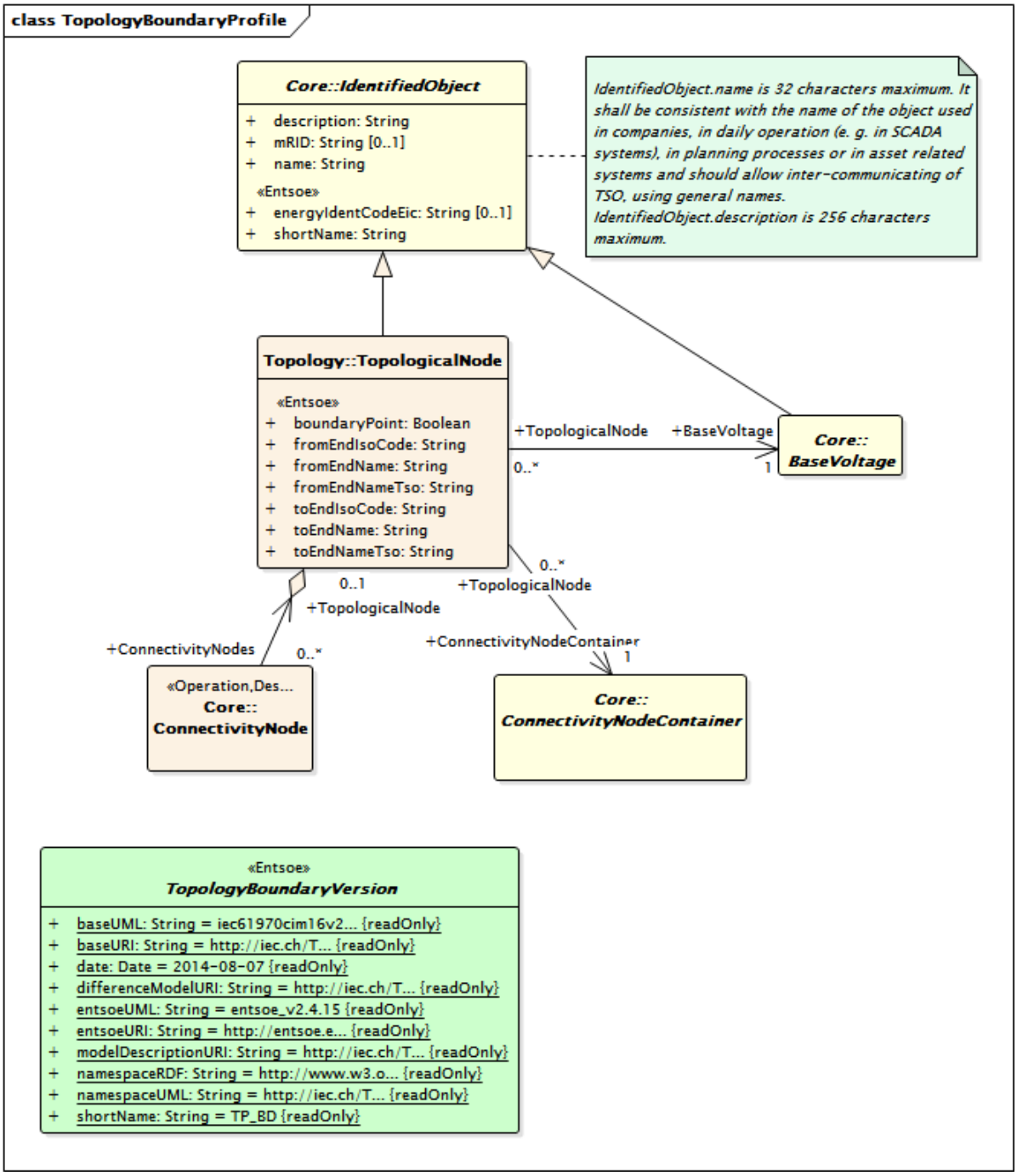


Figure 2-1 - diagram TopologyBoundaryProfile

2.2 (Entsoe) TopologyBoundaryVersion(Abstract)

Version details.

Table 2-1 shows all attributes of TopologyBoundaryVersion.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 2-1 - Attributes of TopologyBoundaryProfile::TopologyBoundaryVersion

Attribute name	mult.	Attribute type	Description
baseUML (Entsoe)	1..1	String	ConstantValue= iec61970cim16v28_iec61968cim12v08_iec62325cim03v01a Base UML provided by CIM model manager.
baseURI (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/2013/61970-456/TopologyBoundary/3 Profile URI used in the Model Exchange header and defined in IEC standards. It uniquely identifies the Profile and its version. It is given for information only and to identify the closest IEC profile to which this CGMES profile is based on.
date (Entsoe)	1..1	Date	ConstantValue= 2014-08-07 Profile creation date Form is YYYY-MM-DD for example for January 5, 2009 it is 2009-01-05.
differenceModelURI (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/61970-552/DifferenceModel/1# Difference model URI defined by IEC 61970-552.
entsoeUML (Entsoe)	1..1	String	ConstantValue= entsoe_v2.4.15 UML provided by ENTSO-E.
entsoeURI (Entsoe)	1..1	String	ConstantValue= http://entsoe.eu/CIM/TopologyBoundary/3/1 Profile URI defined by ENTSO-E and used in the Model Exchange header. It uniquely identifies the Profile and its version. The last two elements in the URI (http://entsoe.eu/CIM/TopologyBoundary/yy/zzz) indicate major and minor versions where: - yy - indicates a major version; - zzz - indicates a minor version.
modelDescriptionURI (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/61970-552/ModelDescription/1# Model Description URI defined by IEC 61970-552.
namespaceRDF (Entsoe)	1..1	String	ConstantValue= http://www.w3.org/1999/02/22-rdf-syntax-ns# RDF namespace.
namespaceUML (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/2013/CIM-schema-cim16# CIM UML namespace.
shortName (Entsoe)	1..1	String	ConstantValue= TP_BD The short name of the profile used in profile documentation.

2.3 Core

2.3.1 General

2.3.2 BaseVoltage(Abstract)

Defines a system base voltage which is referenced.

Inheritance path: ->IdentifiedObject

Table 2-2 shows all attributes of BaseVoltage.

Table 2-2 - Attributes of Core::BaseVoltage

Attribute name	mult.	Attribute type	Description

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

description	1..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	1..1	String	see IdentifiedObject

2.3.3 ConnectivityNode

Connectivity nodes are points where terminals of AC conducting equipment are connected together with zero impedance.

OCL constraint:R.4.10.11. Description length restriction
OCL constraint:R.4.10.11. Energy Ident Code length restriction (optional)
OCL constraint:R.4.10.11. From End ISO Code length restriction
OCL constraint:R.4.10.11. From End Name length restriction
OCL constraint:R.4.10.11. From End Name TSO length restriction
OCL constraint:R.4.10.11. Name length restriction
OCL constraint:R.4.10.11. ShortName length restriction
OCL constraint:R.4.10.11. To End ISO Code length restriction
OCL constraint:R.4.10.11. To End Name length restriction
OCL constraint:R.4.10.11. To End Name TSO length restriction

2.3.4 ConnectivityNodeContainer(Abstract)

A base class for all objects that may contain connectivity nodes or topological nodes.

2.3.5 IdentifiedObject(Abstract)

This is a root class to provide common identification for all classes needing identification and naming attributes.

IdentifiedObject.name is 32 characters maximum. It shall be consistent with the name of the object used in companies, in daily operation (e. g. in SCADA systems), in planning processes or in asset related systems and should allow inter-communicating of TSO, using general names. IdentifiedObject.description is 256 characters maximum.

Table 2-3 shows all attributes of IdentifiedObject.

Table 2-3 - Attributes of Core::IdentifiedObject

Attribute name	mult.	Attribute type	Description
description	1..1	String	The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy.
energyIdentCodeEic (Entsoe)	0..1	String	The attribute is used for an exchange of the EIC code (Energy identification Code). The length of the string is 16 characters as defined by the EIC code. References: Local issuing offices for EIC: https://www.entsoe.eu/publications/edi-library/links-to-eic-websites/ EIC description: https://www.entsoe.eu/index.php?id=73&libCat=eic .
mRID	0..1	String	Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in RFC

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552 Edition 1, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name	1..1	String	The name is any free human readable and possibly non unique text naming the object.
shortName (Entsoe)	1..1	String	The attribute is used for an exchange of a human readable short name with length of the string 12 characters maximum.

2.4 Topology

2.4.1 General

2.4.2 (Entsoe) TopologicalNode

For a detailed substation model a topological node is a set of connectivity nodes that, in the current network state, are connected together through any type of closed switches, including jumpers. Topological nodes change as the current network state changes (i.e., switches, breakers, etc. change state).
For a planning model, switch statuses are not used to form topological nodes. Instead they are manually created or deleted in a model builder tool. Topological nodes maintained this way are also called "busses".

OCL constraint:R.4.10.11. Description length restriction
OCL constraint:R.4.10.11. Energy Ident Code length restriction (optional)
OCL constraint:R.4.10.11. From End ISO Code length restriction
OCL constraint:R.4.10.11. From End Name length restriction
OCL constraint:R.4.10.11. From End Name TSO length restriction
OCL constraint:R.4.10.11. Name length restriction
OCL constraint:R.4.10.11. ShortName length restriction
OCL constraint:R.4.10.11. To End ISO Code length restriction
OCL constraint:R.4.10.11. To End Name length restriction
OCL constraint:R.4.10.11. To End Name TSO length restriction

Inheritance path: ->IdentifiedObject

Table 2-4 shows all attributes of TopologicalNode.

Table 2-4 - Attributes of Topology::TopologicalNode

Attribute name	mult.	Attribute type	Description
boundaryPoint (Entsoe)	1..1	Boolean	Identifies if a node is a BoundaryPoint. If boundaryPoint=true the ConnectivityNode or the TopologicalNode represents a BoundaryPoint.
fromEndIsoCode (Entsoe)	1..1	String	The attribute is used for an exchange of the ISO code of the region to which the “From” side of the Boundary point belongs to or it is connected to. The ISO code is two characters country code as defined by ISO 3166 (http://www.iso.org/iso/country_codes). The length of the string is 2 characters maximum. The attribute is a required for the Boundary Model Authority Set where this attribute is used only for the TopologicalNode in the Boundary Topology profile and ConnectivityNode

			in the Boundary Equipment profile.
fromEndName (Entsoe)	1..1	String	The attribute is used for an exchange of a human readable name with length of the string 32 characters maximum. The attribute covers two cases: if the Boundary point is **placed on a tie-line** the attribute is used for exchange of the geographical name of the **substation** to which the “From” side of the tie-line is connected to. if the Boundary point is **placed in a substation** the attribute is used for exchange of the name of the **element** (e.g. PowerTransformer, ACLineSegment, Switch, etc) to which the “From” side of the Boundary point is connected to. The attribute is required for the Boundary Model Authority Set where it is used only for the TopologicalNode in the Boundary Topology profile and ConnectivityNode in the Boundary Equipment profile.
fromEndNameTso (Entsoe)	1..1	String	The attribute is used for an exchange of the name of the TSO to which the “From” side of the Boundary point belongs to or it is connected to. The length of the string is 32 characters maximum. The attribute is required for the Boundary Model Authority Set where it is used only for the TopologicalNode in the Boundary Topology profile and ConnectivityNode in the Boundary Equipment profile.
toEndIsoCode (Entsoe)	1..1	String	The attribute is used for an exchange of the ISO code of the region to which the “To” side of the Boundary point belongs to or it is connected to. The ISO code is two characters country code as defined by ISO 3166 (http://www.iso.org/iso/country_codes). The length of the string is 2 characters maximum. The attribute is a required for the Boundary Model Authority Set where this attribute is used only for the TopologicalNode in the Boundary Topology profile and ConnectivityNode in the Boundary Equipment profile.
toEndName (Entsoe)	1..1	String	The attribute is used for an exchange of a human readable name with length of the string 32 characters maximum. The attribute covers two cases: if the Boundary point is **placed on a tie-line** the attribute is used for exchange of the geographical name of the **substation** to which the “To” side of the tie-line is connected to. if the Boundary point is **placed in a substation** the attribute is used for exchange of the name of the **element** (e.g. PowerTransformer, ACLineSegment, Switch, etc) to which the “To” side of the Boundary point is

			connected to. The attribute is required for the Boundary Model Authority Set where it is used only for the TopologicalNode in the Boundary Topology profile and ConnectivityNode in the Boundary Equipment profile.
toEndNameTso (Entsoe)	1..1	String	The attribute is used for an exchange of the name of the TSO to which the “To” side of the Boundary point belongs to or it is connected to. The length of the string is 32 characters maximum. The attribute is required for the Boundary Model Authority Set where it is used only for the TopologicalNode in the Boundary Topology profile and ConnectivityNode in the Boundary Equipment profile.
description	1..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	1..1	String	see IdentifiedObject

Table 2-5 shows all association ends of TopologicalNode with other classes.

Table 2-5 - Association ends of Topology::TopologicalNode with other classes

mult.	mult.	Class type name	Description
ConnectivityNodeContainer	[1..1]	ConnectivityNodeContainer	The connectivity node container to which the toplogical node belongs. Association Based On : Topology:: TopologicalNode.TopologicalNode[0..*] ----- Core:: ConnectivityNodeContainer.ConnectivityNodeContainer[0..1]
BaseVoltage	[1..1]	BaseVoltage	The base voltage of the topologocial node. Association Based On : Topology:: TopologicalNode.TopologicalNode[0..*] ----- Core:: BaseVoltage.BaseVoltage[0..1]

2.5 Global Used Datatypes

2.5.1 Primitives

2.5.1.1 String

A string consisting of a sequence of characters. The character encoding is UTF-8. The string length is unspecified and unlimited.

2.5.1.2 Date

Date as "yyyy-mm-dd", which conforms with ISO 8601. UTC time zone is specified as "yyyy-mm-ddZ". A local timezone relative UTC is specified as "yyyy-mm-dd(+/-)hh:mm".

2.5.1.3 Boolean

A type with the value space "true" and "false".

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

3 EquipmentProfile

3.1 General

This profile has been built on the basis of the IEC 61970-452 document and adjusted to fit the purpose of the ENTSO-E CGMES.

Figure 3-1 shows the diagram.

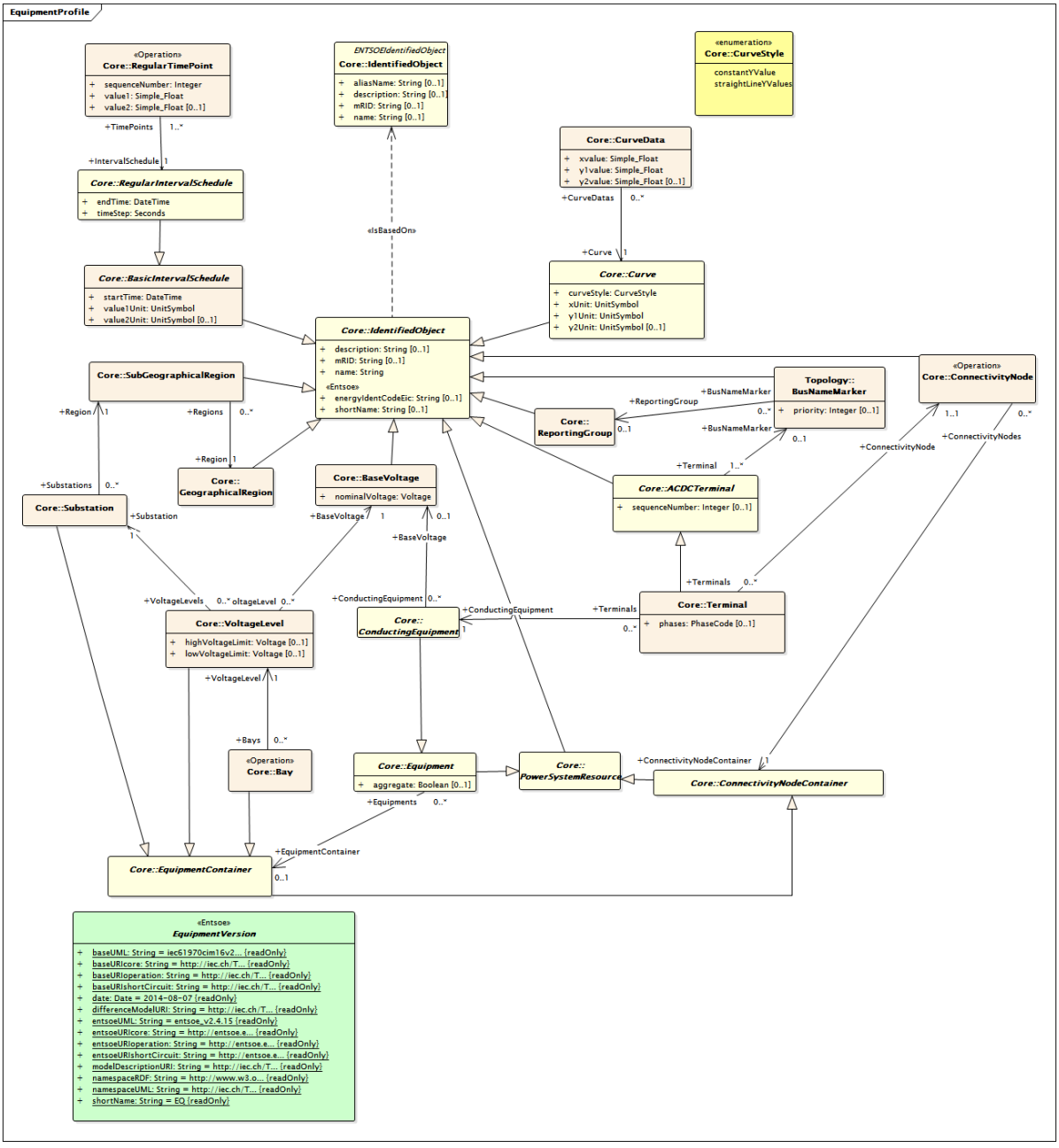


Figure 3-1 - diagram EquipmentProfile

3.2 (Entsoe) EquipmentVersion(Abstract)

Version details.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-1 shows all attributes of EquipmentVersion.

Table 3-1 - Attributes of EquipmentProfile::EquipmentVersion

Attribute name	mult.	Attribute type	Description
baseUML (Entsoe)	1..1	String	ConstantValue= iec61970cim16v28_iec61968cim12v08_iec62325cim03v01a Base UML provided by CIM model manager.
baseURIcore (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/2013/61970-452/EquipmentCore/4 Profile URI used in the Model Exchange header and defined in IEC standards. It uniquely identifies the Profile and its version. It is given for information only and to identify the closest IEC profile to which this CGMES profile is based on.
baseURIoperation (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/2013/61970-452/EquipmentOperation/4 Profile URI used in the Model Exchange header and defined in IEC standards. It uniquely identifies the Profile and its version. It is given for information only and to identify the closest IEC profile to which this CGMES profile is based on.
baseURIshortCircuit (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/2013/61970-452/EquipmentShortCircuit/4 Profile URI used in the Model Exchange header and defined in IEC standards. It uniquely identifies the Profile and its version. It is given for information only and to identify the closest IEC profile to which this CGMES profile is based on.
date (Entsoe)	1..1	Date	ConstantValue= 2014-08-07 Profile creation date Form is YYYY-MM-DD for example for January 5, 2009 it is 2009-01-05.
differenceModelURI (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/61970-552/DifferenceModel/1# Difference model URI defined by IEC 61970-552.
entsoeUML (Entsoe)	1..1	String	ConstantValue= entsoe_v2.4.15 UML provided by ENTSO-E.
entsoeURIcore (Entsoe)	1..1	String	ConstantValue= http://entsoe.eu/CIM/EquipmentCore/3/1 Profile URI defined by ENTSO-E and used in the Model Exchange header. It uniquely identifies the Profile and its version. The last two elements in the URI (http://entsoe.eu/CIM/EquipmentCore/yy/zzz) indicate major and minor versions where: - yy - indicates a major version; - zzz - indicates a minor version.
entsoeURIoperation (Entsoe)	1..1	String	ConstantValue= http://entsoe.eu/CIM/EquipmentOperation/3/1 Profile URI defined by ENTSO-E and used in the Model Exchange header. It uniquely identifies the Profile and its version. The last two elements in the URI (http://entsoe.eu/CIM/EquipmentOperation/yy/zzz) indicate major and minor versions where: - yy - indicates a major version; - zzz - indicates a minor version.
entsoeURIshortCircuit (Entsoe)	1..1	String	ConstantValue= http://entsoe.eu/CIM/EquipmentShortCircuit/3/1 Profile URI defined by ENTSO-E and used in the Model Exchange header. It uniquely identifies the Profile and its version. The last two elements in the URI (http://entsoe.eu/CIM/EquipmentShortCircuit/yy/zzz)

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			indicate major and minor versions where: - yy - indicates a major version; - zzz - indicates a minor version.
modelDescriptionURI (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/61970-552/ModelDescription/1# Model Description URI defined by IEC 61970-552.
namespaceRDF (Entsoe)	1..1	String	ConstantValue= http://www.w3.org/1999/02/22-rdf-syntax-ns# RDF namespace.
namespaceUML (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/2013/CIM-schema-cim16# CIM UML namespace.
shortName (Entsoe)	1..1	String	ConstantValue= EQ The short name of the profile used in profile documentation.

3.3 DC

3.3.1 General

Figure 3-2 shows the diagram.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

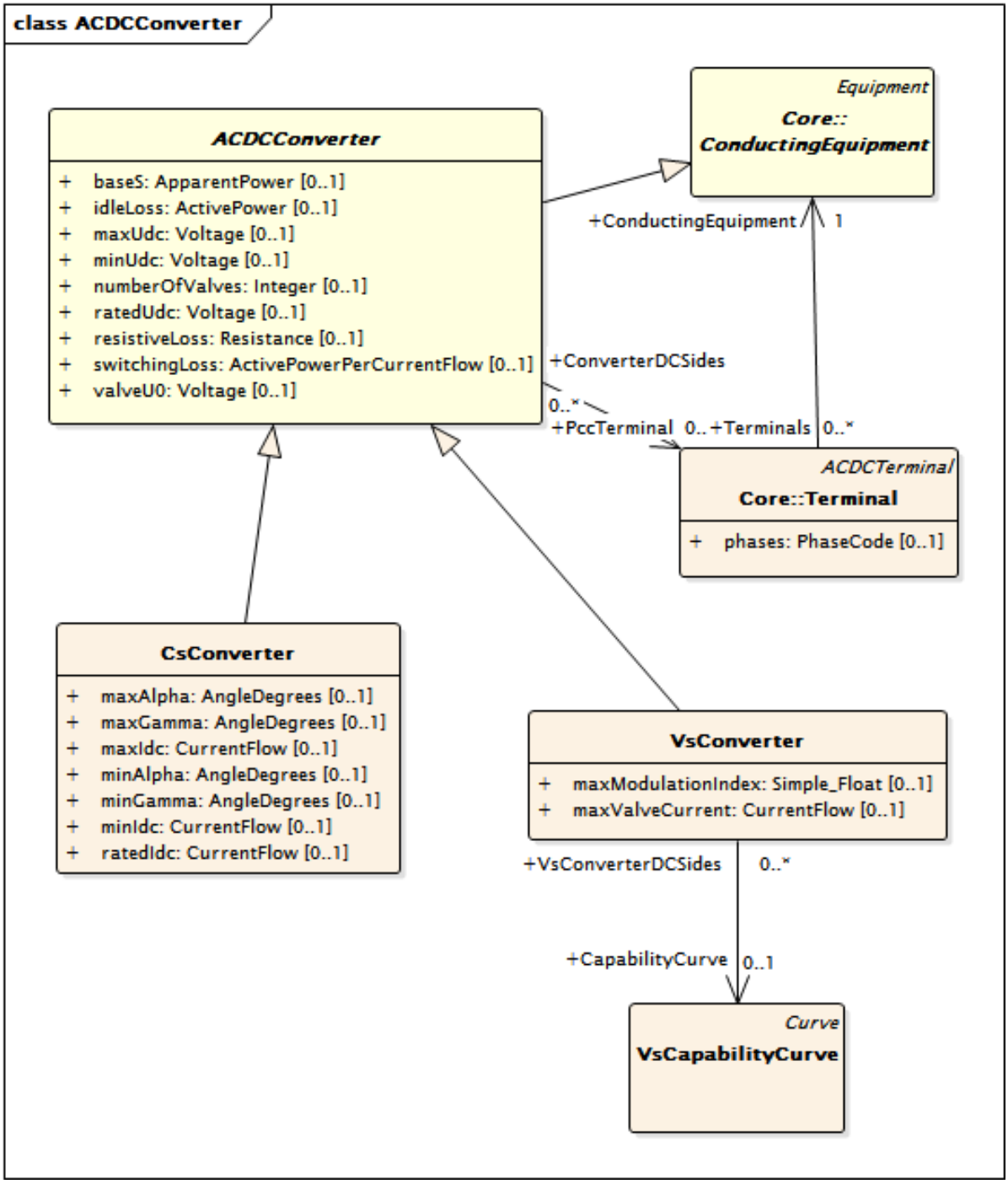


Figure 3-2 - diagram ACDCConverter

This diagram shows classes used to model DC equipment.

Figure 3-3 shows the diagram.

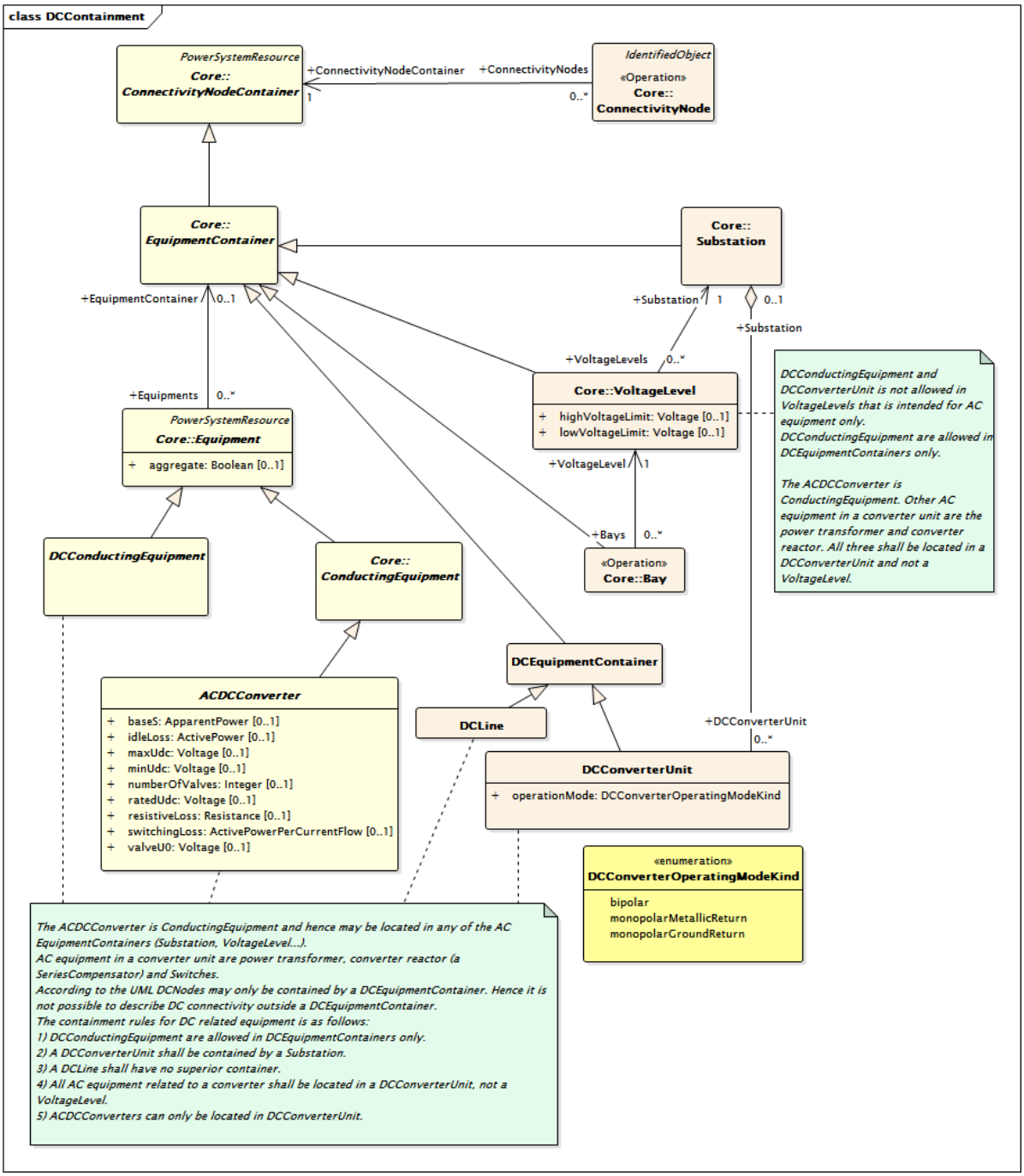


Figure 3-3 - diagram DCContainment

Figure 3-4 shows the diagram.

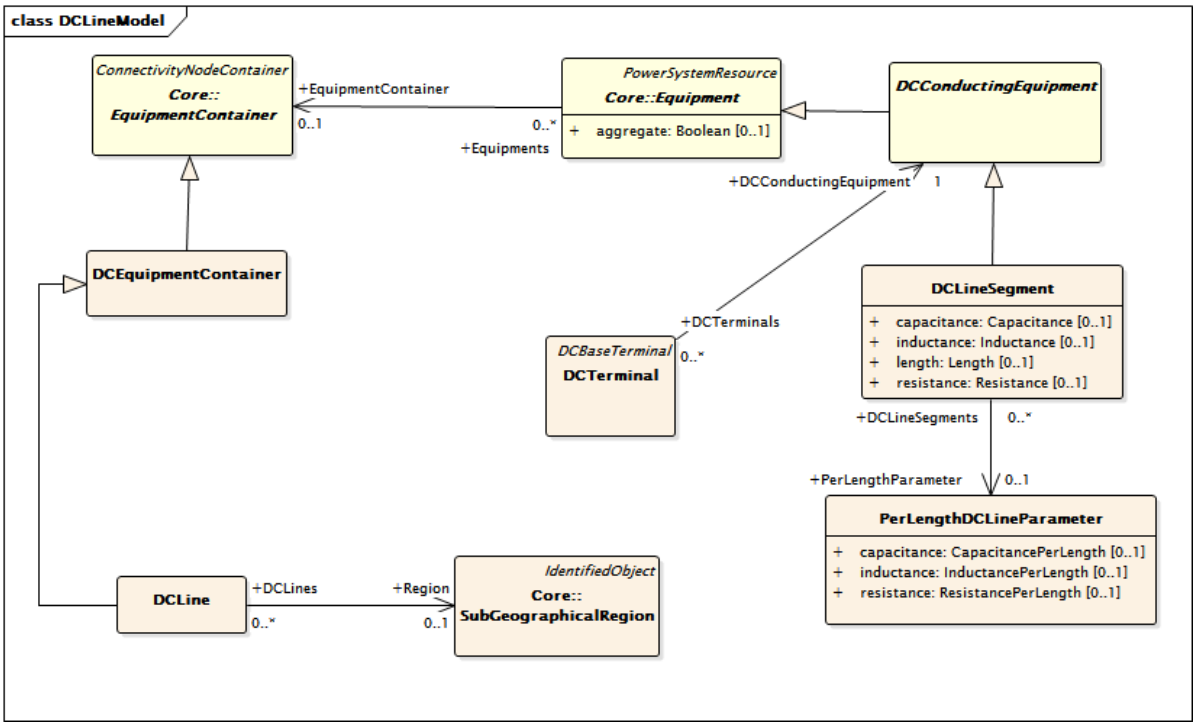


Figure 3-4 - diagram DCLineModel

Figure 3-5 shows the diagram.

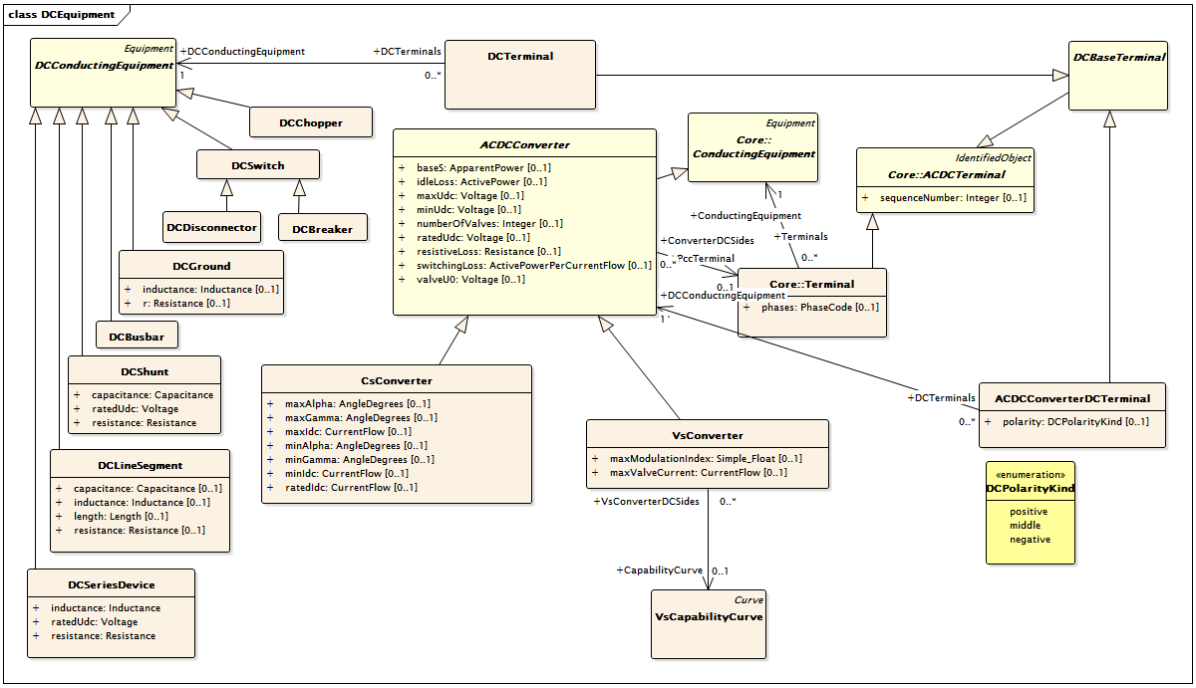


Figure 3-5 - diagram DCEquipment

This diagram shows classes used to model DC equipment.

Figure 3-6 shows the diagram.

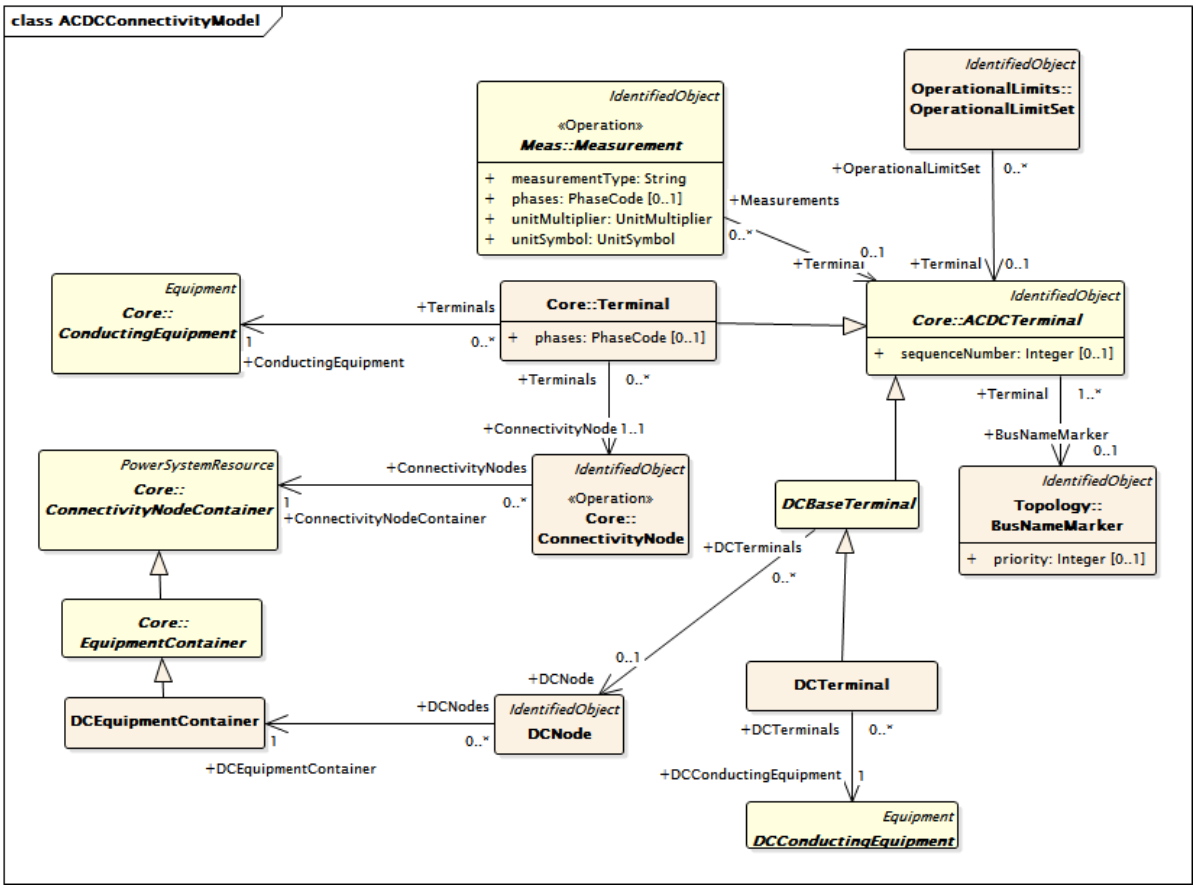


Figure 3-6 - diagram ACDCConnectivityModel

This diagram shows connectivity and topology model for both AC and DC networks.

3.3.2 ACDCConverter(Abstract)

A unit with valves for three phases, together with unit control equipment, essential protective and switching devices, DC storage capacitors, phase reactors and auxiliaries, if any, used for conversion.

OCL constraint: An ACDCConverter must be in a DCConveterUnit

The ACDCConverter is ConductingEquipment and hence may be located in any of the AC EquipmentContainers (Substation, VoltageLevel...). AC equipment in a converter unit are power transformer, converter reactor (a SeriesCompensator) and Switches. According to the UML DCNodes may only be contained by a DCEquipmentContainer. Hence it is not possible to describe DC connectivity outside a DCEquipmentContainer. The containment rules for DC related equipment is as follows: 1) DCConductingEquipment are allowed in DCEquipmentContainers only. 2) A DCConverterUnit shall be contained by a Substation. 3) A DCLine shall have no superior container. 4) All AC equipment related to a converter shall be located in a DCConverterUnit, not a VoltageLevel. 5) ACDCConverters can only be located in DCConverterUnit.

Inheritance path: ->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-2 shows all attributes of ACDCConverter.

Table 3-2 - Attributes of DC::ACDCConverter

Attribute name	mult.	Attribute type	Description
baseS	0..1	ApparentPower	Base apparent power of the converter pole.
idleLoss	0..1	ActivePower	Active power loss in pole at no power transfer. Converter configuration data used in power flow.
maxUdc	0..1	Voltage	The maximum voltage on the DC side at which the converter should operate. Converter configuration data used in power flow.
minUdc	0..1	Voltage	Min allowed converter DC voltage. Converter configuration data used in power flow.
numberOfValves	0..1	Integer	Number of valves in the converter. Used in loss calculations.
ratedUdc	0..1	Voltage	Rated converter DC voltage, also called UdN. Converter configuration data used in power flow.
resistiveLoss	0..1	Resistance	Converter configuration data used in power flow. Refer to poleLossP.
switchingLoss	0..1	ActivePowerPerCurrentFlow	Switching losses, relative to the base apparent power 'baseS'. Refer to poleLossP.
valveU0	0..1	Voltage	Valve threshold voltage. Forward voltage drop when the valve is conducting. Used in loss calculations, i.e. the switchLoss depend on numberOfValves * valveU0.
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-3 shows all association ends of ACDCConverter with other classes.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-3 - Association ends of DC::ACDCConverter with other classes

mult.	mult.	Class type name	Description
PccTerminal	[0..1]	Terminal	All converters' DC sides linked to this point of common coupling terminal. Association Based On : DC:: ACDCConverter.ConverterDCSides[0..*] ----- Core:: Terminal.PccTerminal[0..1]
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.3.3 ACDCConverterDCTerminal

A DC electrical connection point at the AC/DC converter. The AC/DC converter is electrically connected also to the AC side. The AC connection is inherited from the AC conducting equipment in the same way as any other AC equipment. The AC/DC converter DC terminal is separate from generic DC terminal to restrict the connection with the AC side to AC/DC converter and so that no other DC conducting equipment can be connected to the AC side.

Inheritance path: ->DCBaseTerminal->ACDCTerminal->IdentifiedObject

Table 3-4 shows all attributes of ACDCConverterDCTerminal.

Table 3-4 - Attributes of DC::ACDCConverterDCTerminal

Attribute name	mult.	Attribute type	Description
polarity	0..1	DCPolarityKind	Represents the normal network polarity condition.
sequenceNumber	0..1	Integer	see ACDCTerminal
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-5 shows all association ends of ACDCConverterDCTerminal with other classes.

Table 3-5 - Association ends of DC::ACDCConverterDCTerminal with other classes

mult.	mult.	Class type name	Description
DCConductingEquipment	[1..1]	ACDCConverter	Association Based On : DC:: ACDCConverterDCTerminal.DCTerminals[0..*]

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			----- DC:: ACDCConverter.DCConductingEquipment[1]
DCNode	0..1	DCNode	Inherited Association Based On : DC:: DCNode.DCNode[0..1] ----- DC:: DCBaseTerminal.DCTerminals[0..*]
BusNameMarker	0..1	BusNameMarker	Inherited Association Based On : Topology:: BusNameMarker.BusNameMarker[0..1] ----- Core:: ACDCTerminal.Terminal[1..*]

3.3.4 CsConverter

DC side of the current source converter (CSC).

Inheritance path: ->ACDCConverter->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-6 shows all attributes of CsConverter.

Table 3-6 - Attributes of DC::CsConverter

Attribute name	mult.	Attribute type	Description
maxAlpha	0..1	AngleDegrees	Maximum firing angle. CSC configuration data used in power flow.
maxGamma	0..1	AngleDegrees	Maximum extinction angle. CSC configuration data used in power flow.
maxIdc	0..1	CurrentFlow	The maximum direct current (Id) on the DC side at which the converter should operate. Converter configuration data use in power flow.
minAlpha	0..1	AngleDegrees	Minimum firing angle. CSC configuration data used in power flow.
minGamma	0..1	AngleDegrees	Minimum extinction angle. CSC configuration data used in power flow.
minIdc	0..1	CurrentFlow	The minimum direct current (Id) on the DC side at which the converter should operate. CSC configuration data used in power flow.
ratedIdc	0..1	CurrentFlow	Rated converter DC current, also called IdN. Converter configuration data used in power

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			flow.
baseS	0..1	ApparentPower	see ACDCConverter
idleLoss	0..1	ActivePower	see ACDCConverter
maxUdc	0..1	Voltage	see ACDCConverter
minUdc	0..1	Voltage	see ACDCConverter
numberOfValves	0..1	Integer	see ACDCConverter
ratedUdc	0..1	Voltage	see ACDCConverter
resistiveLoss	0..1	Resistance	see ACDCConverter
switchingLoss	0..1	ActivePowerPerCurrentFlow	see ACDCConverter
valveU0	0..1	Voltage	see ACDCConverter
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-7 shows all association ends of ACDCConverter with other classes.

Table 3-7 - Association ends of DC::ACDCConverter with other classes

mult.	mult.	Class type name	Description
PccTerminal	0..1	Terminal	Inherited Association Based On : DC:: ACDCConverter.ConverterDCSides[0..*] ----- Core:: Terminal.PccTerminal[0..1]
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.3.5 DCBaseTerminal(Abstract)

An electrical connection point at a piece of DC conducting equipment. DC terminals are connected at one physical DC node that may have multiple DC terminals connected. A DC node is similar to an AC connectivity node. The model enforces that DC connections are distinct from AC connections.

Inheritance path: ->ACDCTerminal->IdentifiedObject

Table 3-8 shows all attributes of DCBaseTerminal.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-8 - Attributes of DC::DCBaseTerminal

Attribute name	mult.	Attribute type	Description
sequenceNumber	0..1	Integer	see ACDCTerminal
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-9 shows all association ends of DCBaseTerminal with other classes.

Table 3-9 - Association ends of DC::DCBaseTerminal with other classes

mult.	mult.	Class type name	Description
DCNode	[0..1]	DCNode	Association Based On : DC:: DCNode.DCNode[0..1] ----- DC:: DCBaseTerminal.DCTerminals[0..*]
BusNameMarker	0..1	BusNameMarker	Inherited Association Based On : Topology:: BusNameMarker.BusNameMarker[0..1] ----- Core:: ACDCTerminal.Terminal[1..*]

3.3.6 DCBreaker

A breaker within a DC system.

Inheritance path: ->DCSwitch->DCConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-10 shows all attributes of DCBreaker.

Table 3-10 - Attributes of DC::DCBreaker

Attribute name	mult.	Attribute type	Description
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-11 shows all association ends of Equipment with other classes.

Table 3-11 - Association ends of Core::Equipment with other classes

mult.	mult.	Class type name	Description

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.3.7 DCBusbar

A busbar within a DC system.

Inheritance path: ->DCConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-12 shows all attributes of DCBusbar.

Table 3-12 - Attributes of DC::DCBusbar

Attribute name	mult.	Attribute type	Description
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-13 shows all association ends of Equipment with other classes.

Table 3-13 - Association ends of Core::Equipment with other classes

mult.	mult.	Class type name	Description
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.3.8 DCChopper

Low resistance equipment used in the internal DC circuit to balance voltages. It has typically positive and negative pole terminals and a ground.

Inheritance path: ->DCConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-14 shows all attributes of DCChopper.

Table 3-14 - Attributes of DC::DCChopper

Attribute name	mult.	Attribute type	Description
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-15 shows all association ends of Equipment with other classes.

Table 3-15 - Association ends of Core::Equipment with other classes

mult.	mult.	Class type name	Description
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.3.9 DCConductingEquipment(Abstract)

The parts of the DC power system that are designed to carry current or that are conductively connected through DC terminals.

The ACDCConverter is ConductingEquipment and hence may be located in any of the AC EquipmentContainers (Substation, VoltageLevel...). AC equipment in a converter unit are power transformer, converter reactor (a SeriesCompensator) and Switches. According to the UML DCNodes may only be contained by a DCEquipmentContainer. Hence it is not possible to describe DC connectivity outside a DCEquipmentContainer. The containment rules for DC related equipment is as follows: 1) DCConductingEquipment are allowed in DCEquipmentContainers only. 2) A DCConverterUnit shall be contained by a Substation. 3) A DCLine shall have no superior container. 4) All AC equipment related to a converter shall be located in a DCConverterUnit, not a VoltageLevel. 5) ACDCConverters can only be located in DCConverterUnit.

Inheritance path: ->Equipment->PowerSystemResource->IdentifiedObject

Table 3-16 shows all attributes of DCConductingEquipment.

Table 3-16 - Attributes of DC::DCConductingEquipment

Attribute name	mult.	Attribute type	Description
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-17 shows all association ends of Equipment with other classes.

Table 3-17 - Association ends of Core::Equipment with other classes

mult.	mult.	Class type name	Description
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1]

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			----- Core:: Equipment.Equipments[0..*]

3.3.10 DCConverterUnit

Indivisible operative unit comprising all equipment between the point of common coupling on the AC side and the point of common coupling – DC side, essentially one or more converters, together with one or more converter transformers, converter control equipment, essential protective and switching devices and auxiliaries, if any, used for conversion.

The ACDCConverter is ConductingEquipment and hence may be located in any of the AC EquipmentContainers (Substation, VoltageLevel...). AC equipment in a converter unit are power transformer, converter reactor (a SeriesCompensator) and Switches. According to the UML DCNodes may only be contained by a DCEquipmentContainer. Hence it is not possible to describe DC connectivity outside a DCEquipmentContainer. The containment rules for DC related equipment is as follows: 1) DCConductingEquipment are allowed in DCEquipmentContainers only. 2) A DCConverterUnit shall be contained by a Substation. 3) A DCLine shall have no superior container. 4) All AC equipment related to a converter shall be located in a DCConverterUnit, not a VoltageLevel. 5) ACDCConverters can only be located in DCConverterUnit.

Inheritance path: ->DCEquipmentContainer->EquipmentContainer->ConnectivityNodeContainer->PowerSystemResource->IdentifiedObject

Table 3-18 shows all attributes of DCConverterUnit.

Table 3-18 - Attributes of DC::DCConverterUnit

Attribute name	mult.	Attribute type	Description
operationMode	1..1	DCConverterOperatingModeKind	
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-19 shows all association ends of DCConverterUnit with other classes.

Table 3-19 - Association ends of DC::DCConverterUnit with other classes

mult.	mult.	Class type name	Description
Substation	[0..1]	Substation	Association Based On : DC:: DCConverterUnit.DCConverterUnit[0..*] ----- Core:: Substation.Substation[0..1]

3.3.11 DCDisconnector

A disconnector within a DC system.

Inheritance path: ->DCSwitch->DCConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-20 shows all attributes of DCDisconnector.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-20 - Attributes of DC::DCDisconnector

Attribute name	mult.	Attribute type	Description
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-21 shows all association ends of Equipment with other classes.

Table 3-21 - Association ends of Core::Equipment with other classes

mult.	mult.	Class type name	Description
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.3.12 DCEquipmentContainer

A modeling construct to provide a root class for containment of DC as well as AC equipment. The class differ from the EquipmentContaner for AC in that it may also contain DCNodes. Hence it can contain both AC and DC equipment.

Inheritance path: ->EquipmentContainer->ConnectivityNodeContainer->PowerSystemResource->IdentifiedObject

Table 3-22 shows all attributes of DCEquipmentContainer.

Table 3-22 - Attributes of DC::DCEquipmentContainer

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

3.3.13 DCGround

A ground within a DC system.

OCL constraint:A DCGround must be in a DCConverterUnit

Inheritance path: ->DCConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-23 shows all attributes of DCGround.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-23 - Attributes of DC::DCGround

Attribute name	mult.	Attribute type	Description
inductance	0..1	Inductance	Inductance to ground.
r	0..1	Resistance	Resistance to ground.
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-24 shows all association ends of Equipment with other classes.

Table 3-24 - Association ends of Core::Equipment with other classes

mult.	mult.	Class type name	Description
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.3.14 DCLine

Overhead lines and/or cables connecting two or more HVDC substations.

The ACDCConverter is ConductingEquipment and hence may be located in any of the AC EquipmentContainers (Substation, VoltageLevel...). AC equipment in a converter unit are power transformer, converter reactor (a SeriesCompensator) and Switches. According to the UML DCNodes may only be contained by a DCEquipmentContainer. Hence it is not possible to describe DC connectivity outside a DCEquipmentContainer. The containment rules for DC related equipment is as follows: 1) DCConductingEquipment are allowed in DCEquipmentContainers only. 2) A DCConverterUnit shall be contained by a Substation. 3) A DCLine shall have no superior container. 4) All AC equipment related to a converter shall be located in a DCConverterUnit, not a VoltageLevel. 5) ACDCConverters can only be located in DCConverterUnit.

Inheritance path: ->DCEquipmentContainer->EquipmentContainer->ConnectivityNodeContainer->PowerSystemResource->IdentifiedObject

Table 3-25 shows all attributes of DCLine.

Table 3-25 - Attributes of DC::DCLine

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-26 shows all association ends of DCLine with other classes.

Table 3-26 - Association ends of DC::DCLine with other classes

mult.	mult.	Class type name	Description
Region	[0..1]	SubGeographicalRegion	Association Based On : DC:: DCLine.DCLines[0..*] ----- Core:: SubGeographicalRegion.Region[0..1]

3.3.15 DCLineSegment

A wire or combination of wires not insulated from one another, with consistent electrical characteristics, used to carry direct current between points in the DC region of the power system.

OCL constraint:A DCLineSegment must be contained in a DCLine

Inheritance path: ->DCConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-27 shows all attributes of DCLineSegment.

Table 3-27 - Attributes of DC::DCLineSegment

Attribute name	mult.	Attribute type	Description
capacitance	0..1	Capacitance	Capacitance of the DC line segment. Significant for cables only.
inductance	0..1	Inductance	Inductance of the DC line segment. Neglectable compared with DCSeriesDevice used for smoothing.
resistance	0..1	Resistance	Resistance of the DC line segment.
length	0..1	Length	Segment length for calculating line section capabilities.
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-28 shows all association ends of DCLineSegment with other classes.

Table 3-28 - Association ends of DC::DCLineSegment with other classes

mult.	mult.	Class type name	Description
PerLengthParameter	[0..1]	PerLengthDCLineParameter	Set of per-length parameters for this line segment. Association Based On : DC:: PerLengthDCLineParameter.PerLengthParameter[0..1]

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			----- DC:: DCLineSegment.DCLineSegments[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.3.16 DCNode

DC nodes are points where terminals of DC conducting equipment are connected together with zero impedance.

Inheritance path: ->IdentifiedObject

Table 3-29 shows all attributes of DCNode.

Table 3-29 - Attributes of DC::DCNode

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-30 shows all association ends of DCNode with other classes.

Table 3-30 - Association ends of DC::DCNode with other classes

mult.	mult.	Class type name	Description
DCEquipmentContainer	[1..1]	DCEquipmentContainer	Association Based On : DC:: DCNode.DCNodes[0..*] ----- DC:: DCEquipmentContainer.DCEquipmentContainer[1]

3.3.17 DCSeriesDevice

A series device within the DC system, typically a reactor used for filtering or smoothing. Needed for transient and short circuit studies.

Inheritance path: ->DCConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-31 shows all attributes of DCSeriesDevice.

Table 3-31 - Attributes of DC::DCSeriesDevice

Attribute name	mult.	Attribute type	Description
inductance	1..1	Inductance	Inductance of the device.
resistance	1..1	Resistance	Resistance of the DC device.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

ratedUdc	1..1	Voltage	Rated DC device voltage. Converter configuration data used in power flow.
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-32 shows all association ends of Equipment with other classes.

Table 3-32 - Association ends of Core::Equipment with other classes

mult.	mult.	Class type name	Description
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.3.18 DCShunt

A shunt device within the DC system, typically used for filtering. Needed for transient and short circuit studies.

Inheritance path: ->DCConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-33 shows all attributes of DCShunt.

Table 3-33 - Attributes of DC::DCShunt

Attribute name	mult.	Attribute type	Description
capacitance	1..1	Capacitance	Capacitance of the DC shunt.
resistance	1..1	Resistance	Resistance of the DC device.
ratedUdc	1..1	Voltage	Rated DC device voltage. Converter configuration data used in power flow.
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-34 shows all association ends of Equipment with other classes.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-34 - Association ends of Core::Equipment with other classes

mult.	mult.	Class type name	Description
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.3.19 DCSwitch

A switch within the DC system.

Inheritance path: ->DCConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-35 shows all attributes of DCSwitch.

Table 3-35 - Attributes of DC::DCSwitch

Attribute name	mult.	Attribute type	Description
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-36 shows all association ends of Equipment with other classes.

Table 3-36 - Association ends of Core::Equipment with other classes

mult.	mult.	Class type name	Description
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.3.20 DCTerminal

An electrical connection point to generic DC conducting equipment.

Inheritance path: ->DCBaseTerminal->ACDCTerminal->IdentifiedObject

Table 3-37 shows all attributes of DCTerminal.

Table 3-37 - Attributes of DC::DCTerminal

Attribute name	mult.	Attribute type	Description
sequenceNumber	0..1	Integer	see ACDCTerminal
description	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-38 shows all association ends of DCTerminal with other classes.

Table 3-38 - Association ends of DC::DCTerminal with other classes

mult.	mult.	Class type name	Description
DCConductingEquipment	[1..1]	DCConductingEquipment	Association Based On : DC:: DCConductingEquipment.DCConductingEquipment[1] ----- DC:: DCTerminal.DCTerminals[0..*]
DCNode	0..1	DCNode	Inherited Association Based On : DC:: DCNode.DCNode[0..1] ----- DC:: DCBaseTerminal.DCTerminals[0..*]
BusNameMarker	0..1	BusNameMarker	Inherited Association Based On : Topology:: BusNameMarker.BusNameMarker[0..1] ----- Core:: ACDCTerminal.Terminal[1..*]

3.3.21 PerLengthDCLineParameter

Table 3-39 shows all attributes of PerLengthDCLineParameter.

Table 3-39 - Attributes of DC::PerLengthDCLineParameter

Attribute name	mult.	Attribute type	Description
capacitance	0..1	CapacitancePerLength	Capacitance per unit of length of the DC line segment; significant for cables only.
inductance	0..1	InductancePerLength	Inductance per unit of length of the DC line segment.
resistance	0..1	ResistancePerLength	Resistance per length of the DC line segment.

3.3.22 VsCapabilityCurve

The P-Q capability curve for a voltage source converter, with P on x-axis and Qmin and Qmax on y1-axis and y2-axis.

Inheritance path: ->Curve->IdentifiedObject

Table 3-40 shows all attributes of VsCapabilityCurve.

Table 3-40 - Attributes of DC::VsCapabilityCurve

Attribute name	mult.	Attribute type	Description

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

curveStyle	1..1	CurveStyle	see Curve
xUnit	1..1	UnitSymbol	see Curve
y1Unit	1..1	UnitSymbol	see Curve
y2Unit	0..1	UnitSymbol	see Curve
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

3.3.23 VsConverter

DC side of the voltage source converter (VSC).

Inheritance path: ->ACDCConverter->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-41 shows all attributes of VsConverter.

Table 3-41 - Attributes of DC::VsConverter

Attribute name	mult.	Attribute type	Description
maxModulationIndex	0..1	Simple_Float	The max quotient between the AC converter voltage (Uc) and DC voltage (Ud). A factor typically less than 1. VSC configuration data used in power flow.
maxValveCurrent	0..1	CurrentFlow	The maximum current through a valve. This current limit is the basis for calculating the capability diagram. VSC configuration data.
baseS	0..1	ApparentPower	see ACDCConverter
idleLoss	0..1	ActivePower	see ACDCConverter
maxUdc	0..1	Voltage	see ACDCConverter
minUdc	0..1	Voltage	see ACDCConverter
numberOfValves	0..1	Integer	see ACDCConverter
ratedUdc	0..1	Voltage	see ACDCConverter
resistiveLoss	0..1	Resistance	see ACDCConverter
switchingLoss	0..1	ActivePowerPerCurrentFlow	see ACDCConverter
valveU0	0..1	Voltage	see ACDCConverter
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-42 shows all association ends of VsConverter with other classes.

Table 3-42 - Association ends of DC::VsConverter with other classes

mult.	mult.	Class type name	Description
CapabilityCurve	[0..1]	VsCapabilityCurve	All converters with this capability curve. Association Based On : DC:: VsConverter.VsConverterDCSides[0..*] ----- DC:: VsCapabilityCurve.CapabilityCurve[0..1]
PccTerminal	0..1	Terminal	Inherited Association Based On : DC:: ACDCConverter.ConverterDCSides[0..*] ----- Core:: Terminal.PccTerminal[0..1]
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.3.24 Enumerations

3.3.24.1 DCConverterOperatingModeKind

The operating mode of an HVDC bipole.

Literal	Description	Code
bipolar	Bipolar operation.	
monopolarMetallicReturn	Monopolar operation with metallic return	
monopolarGroundReturn	Monopolar operation with ground return	

3.3.24.2 DCPolarityKind

Polarity for DC circuits.

Literal	Description	Code
positive	Positive pole.	
middle	Middle pole, potentially grounded.	
negative	Negative pole.	

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

3.4 Topology

3.4.1 General

3.4.2 BusNameMarker

Used to apply user standard names to topology buses. Typically used for "bus/branch" case generation. Associated with one or more terminals that are normally connected with the bus name. The associated terminals are normally connected by non-retained switches. For a ring bus station configuration, all busbar terminals in the ring are typically associated. For a breaker and a half scheme, both busbars would normally be associated. For a ring bus, all busbars would normally be associated. For a "straight" busbar configuration, normally only the main terminal at the busbar would be associated.

Inheritance path: ->IdentifiedObject

Table 3-43 shows all attributes of BusNameMarker.

Table 3-43 - Attributes of Topology::BusNameMarker

Attribute name	mult.	Attribute type	Description
priority	0..1	Integer	Priority of bus name marker for use as topology bus name. Use 0 for don t care. Use 1 for highest priority. Use 2 as priority is less than 1 and so on.
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-44 shows all association ends of BusNameMarker with other classes.

Table 3-44 - Association ends of Topology::BusNameMarker with other classes

mult.	mult.	Class type name	Description
ReportingGroup	[0..1]	ReportingGroup	The bus name markers that belong to this reporting group. Association Based On : Topology:: BusNameMarker.BusNameMarker[0..*] ----- Core:: ReportingGroup.ReportingGroup[0..1]

3.5 Meas

3.5.1 General

Figure 3-7 shows the diagram.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

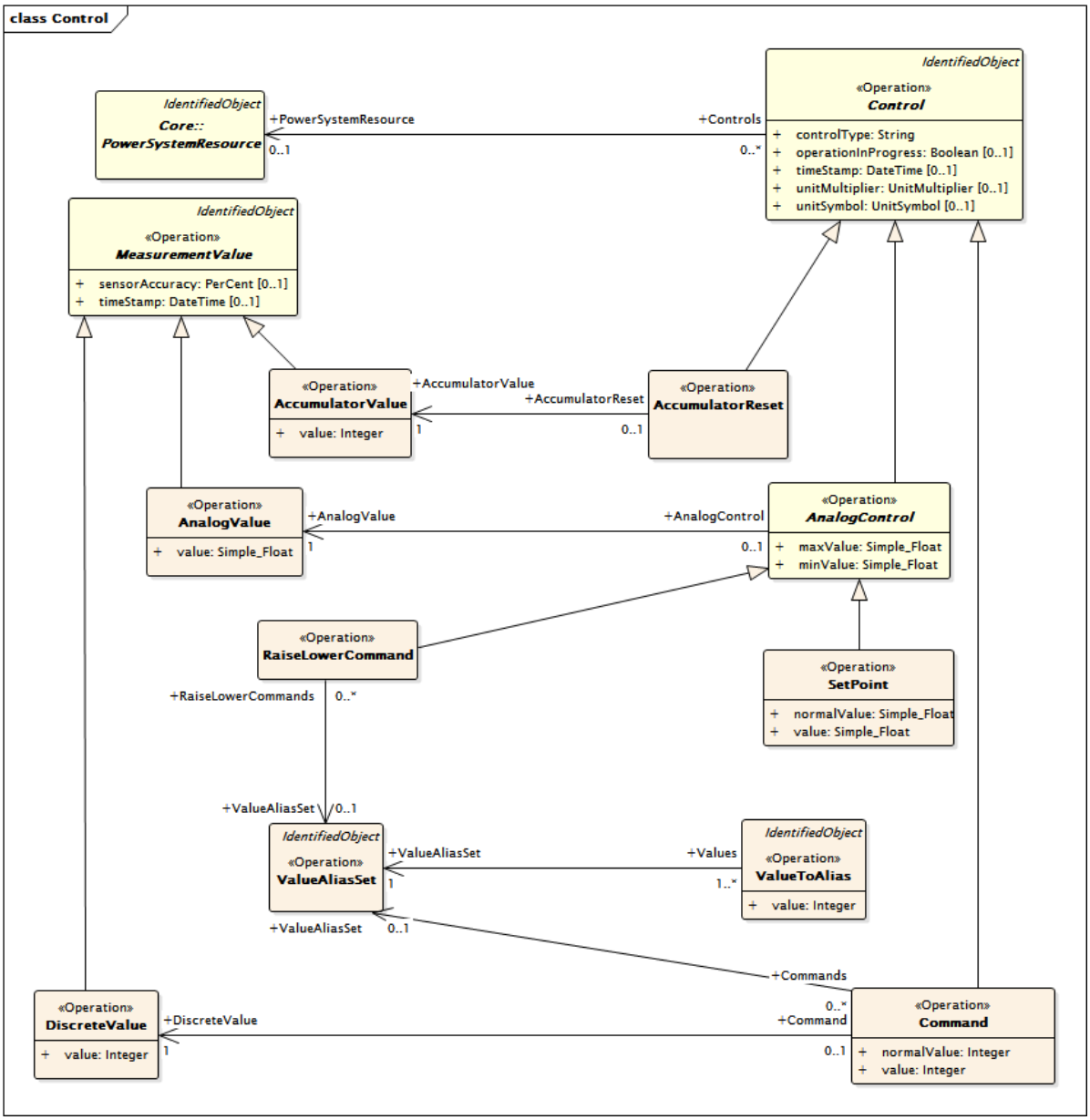


Figure 3-7 - diagram Control

Figure 3-8 shows the diagram.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

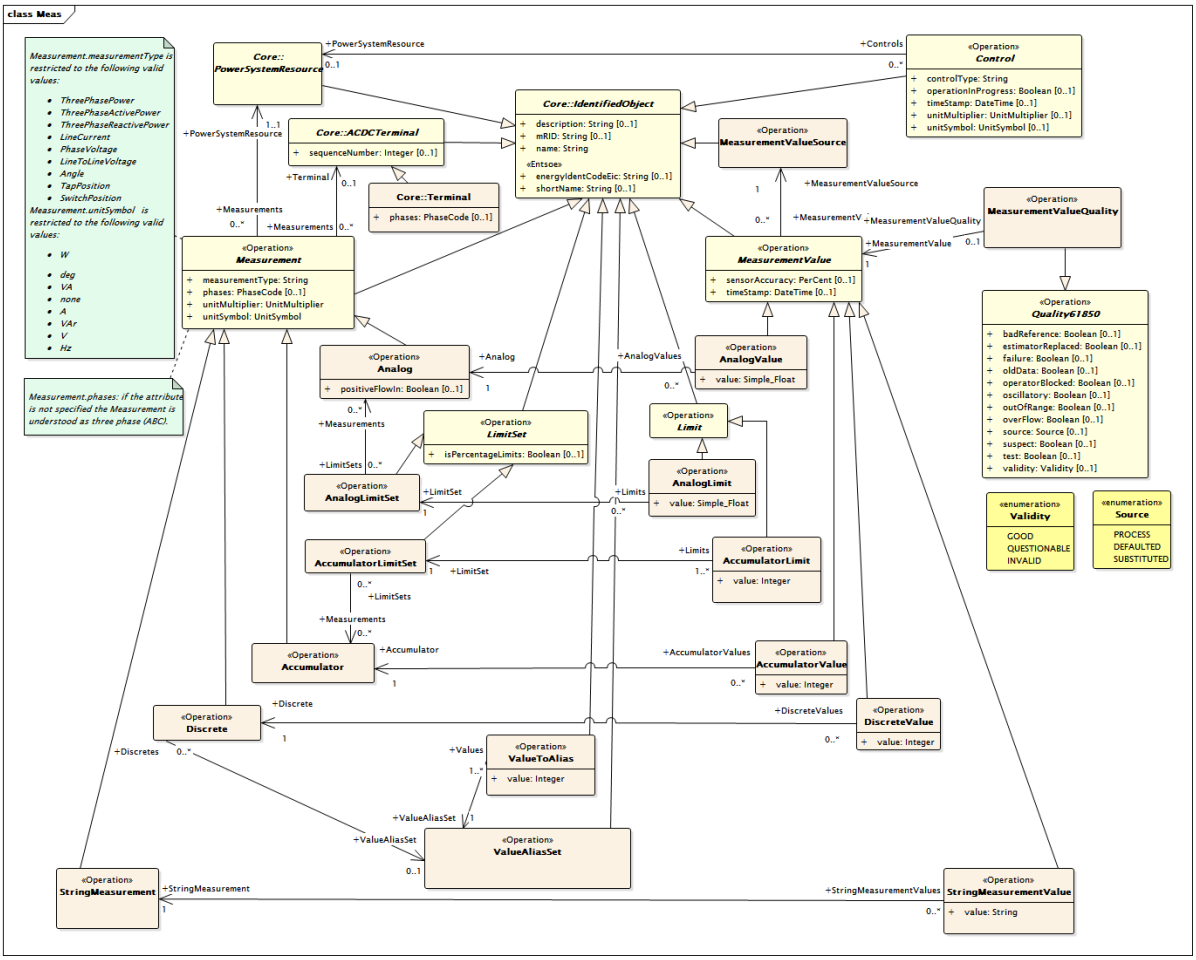


Figure 3-8 - diagram Meas

Figure 3-9 shows the diagram.

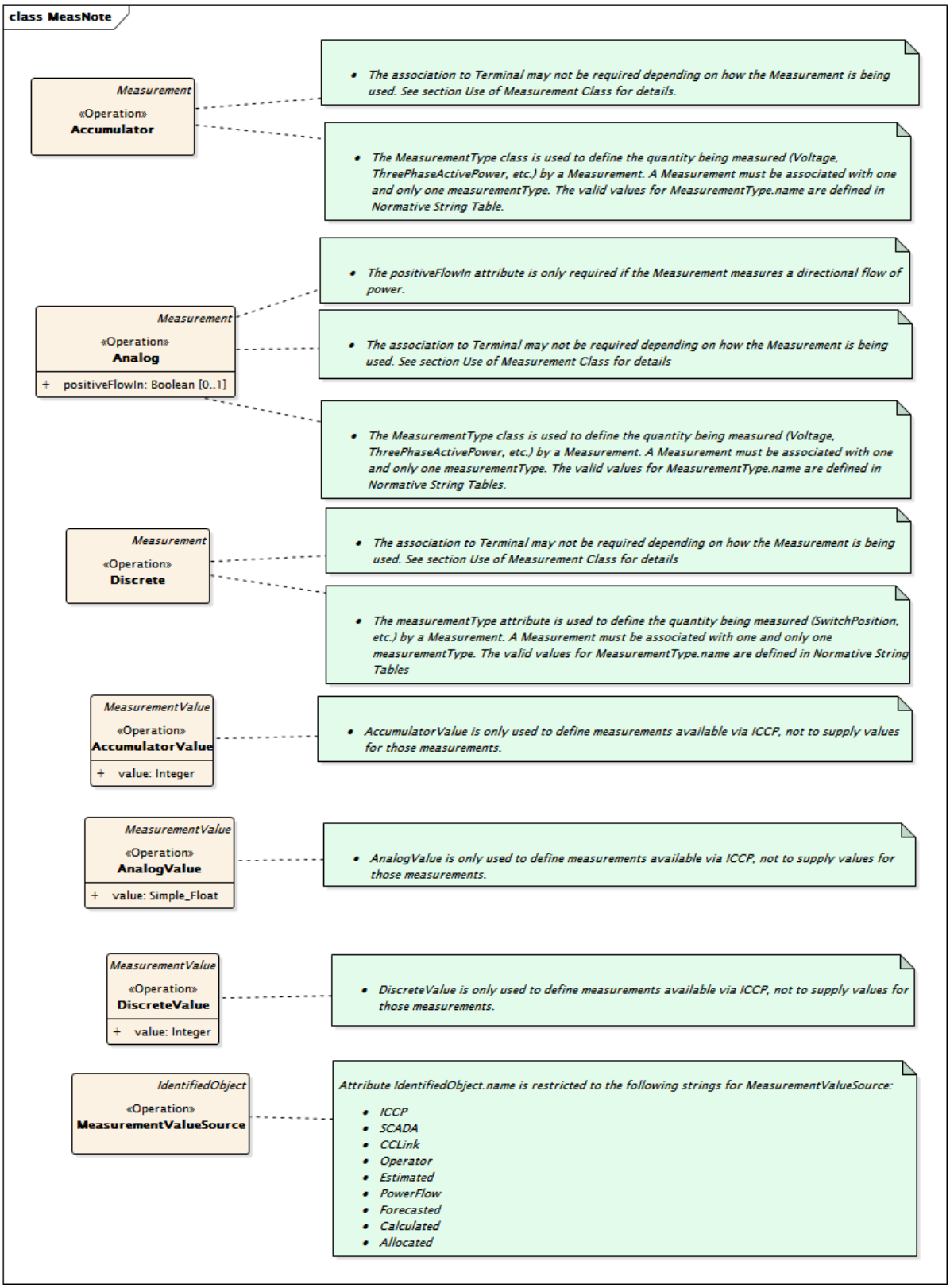


Figure 3-9 - diagram MeasNote

3.5.2 Accumulator

Accumulator represents an accumulated (counted) Measurement, e.g. an energy value.

- The association to Terminal may not be required depending on how the Measurement is being used. See section Use of Measurement Class for details.
- The MeasurementType class is used to define the quantity being measured (Voltage, ThreePhaseActivePower, etc.) by a Measurement. A Measurement must be associated with one and only one measurementType. The valid values for MeasurementType.name are defined in Normative String Table.

Inheritance path: ->Measurement->IdentifiedObject

Table 3-45 shows all attributes of Accumulator.

Table 3-45 - Attributes of Meas::Accumulator

Attribute name	mult.	Attribute type	Description
measurementType	1..1	String	see Measurement
phases	0..1	PhaseCode	see Measurement
unitSymbol	1..1	UnitSymbol	see Measurement
unitMultiplier	1..1	UnitMultiplier	see Measurement
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-46 shows all association ends of Measurement with other classes.

Table 3-46 - Association ends of Meas::Measurement with other classes

mult.	mult.	Class type name	Description
Terminal (Operation)	0..1	ACDCTerminal	Inherited Association Based On : Meas:: Measurement.Measurements[0..*] ----- Core:: ACDCTerminal.Terminal[0..1]
PowerSystemResource (Operation)	1..1	PowerSystemResource	Inherited Association Based On : Core:: PowerSystemResource.PowerSystemResource[0..1] ----- Meas:: Measurement.Measurements[0..*]

3.5.3 AccumulatorLimit

Limit values for Accumulator measurements.

Inheritance path: ->Limit->IdentifiedObject

Table 3-47 shows all attributes of AccumulatorLimit.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-47 - Attributes of Meas::AccumulatorLimit

Attribute name	mult.	Attribute type	Description
value (Operation)	1..1	Integer	The value to supervise against. The value is positive.
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-48 shows all association ends of AccumulatorLimit with other classes.

Table 3-48 - Association ends of Meas::AccumulatorLimit with other classes

mult.	mult.	Class type name	Description
LimitSet (Operation)	[1..1]	AccumulatorLimitSet	The limit values used for supervision of Measurements. Association Based On : Meas:: AccumulatorLimitSet.LimitSet[1] ----- Meas:: AccumulatorLimit.Limits[1..*]

3.5.4 (Operation) AccumulatorLimitSet

An AccumulatorLimitSet specifies a set of Limits that are associated with an Accumulator measurement.

Inheritance path: ->LimitSet->IdentifiedObject

Table 3-49 shows all attributes of AccumulatorLimitSet.

Table 3-49 - Attributes of Meas::AccumulatorLimitSet

Attribute name	mult.	Attribute type	Description
isPercentageLimits	0..1	Boolean	see LimitSet
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-50 shows all association ends of AccumulatorLimitSet with other classes.

Table 3-50 - Association ends of Meas::AccumulatorLimitSet with other classes

mult.	mult.	Class type name	Description
Measurements (Operation)	[0..*]	Accumulator	A measurement may have zero or more limit ranges defined for it. Association Based On : Meas::

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			Accumulator.Measurements[0..*] ----- Meas:: AccumulatorLimitSet.LimitSets[0..*]

3.5.5 (Operation) AccumulatorReset

This command reset the counter value to zero.

Inheritance path: ->Control->IdentifiedObject

Table 3-51 shows all attributes of AccumulatorReset.

Table 3-51 - Attributes of Meas::AccumulatorReset

Attribute name	mult.	Attribute type	Description
controlType	1..1	String	see Control
operationInProgress	0..1	Boolean	see Control
timeStamp	0..1	DateTime	see Control
unitMultiplier	0..1	UnitMultiplier	see Control
unitSymbol	0..1	UnitSymbol	see Control
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-52 shows all association ends of AccumulatorReset with other classes.

Table 3-52 - Association ends of Meas::AccumulatorReset with other classes

mult.	mult.	Class type name	Description
AccumulatorValue (Operation)	[1..1]	AccumulatorValue	The accumulator value that is reset by the command. Association Based On : Meas:: AccumulatorReset.AccumulatorReset[0..1] ----- Meas:: AccumulatorValue.AccumulatorValue[1]
PowerSystemResource (Operation)	0..1	PowerSystemResource	Inherited Association Based On : Core:: PowerSystemResource.PowerSystemResource[0..1] ----- Meas:: Control.Controls[0..*]

3.5.6 AccumulatorValue

AccumulatorValue represents an accumulated (counted) MeasurementValue.

- AccumulatorValue is only used to define measurements available via ICCP, not to supply values for those measurements.

Inheritance path: ->MeasurementValue->IdentifiedObject

Table 3-53 shows all attributes of AccumulatorValue.

Table 3-53 - Attributes of Meas::AccumulatorValue

Attribute name	mult.	Attribute type	Description
value (Operation)	1..1	Integer	The value to supervise. The value is positive.
timeStamp	0..1	DateTime	see MeasurementValue
sensorAccuracy	0..1	PerCent	see MeasurementValue
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-54 shows all association ends of AccumulatorValue with other classes.

Table 3-54 - Association ends of Meas::AccumulatorValue with other classes

mult.	mult.	Class type name	Description
Accumulator (Operation)	[1..1]	Accumulator	The values connected to this measurement. Association Based On : Meas:: Accumulator.Accumulator[1] ----- Meas:: AccumulatorValue.AccumulatorValues[0..*]
MeasurementValueSource (Operation)	1..1	MeasurementValueSource	Inherited Association Based On : Meas:: MeasurementValueSource.MeasurementValueSource[1] ----- Meas:: MeasurementValue.MeasurementValues[0..*]

3.5.7 Analog

Analog represents an analog Measurement.

- The positiveFlowIn attribute is only required if the Measurement measures a directional flow of power.
- The association to Terminal may not be required depending on how the Measurement is being used. See section Use of Measurement Class for details
- The MeasurementType class is used to define the quantity being measured (Voltage, ThreePhaseActivePower, etc.) by a Measurement. A Measurement must be associated with one and only one measurementType. The valid values for MeasurementType.name are defined in Normative String Tables.

Inheritance path: ->Measurement->IdentifiedObject

Table 3-55 shows all attributes of Analog.

Table 3-55 - Attributes of Meas::Analog

Attribute name	mult.	Attribute type	Description

positiveFlowIn (Operation)	0..1	Boolean	If true then this measurement is an active power, reactive power or current with the convention that a positive value measured at the Terminal means power is flowing into the related PowerSystemResource.
measurementType	1..1	String	see Measurement
phases	0..1	PhaseCode	see Measurement
unitSymbol	1..1	UnitSymbol	see Measurement
unitMultiplier	1..1	UnitMultiplier	see Measurement
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-56 shows all association ends of Measurement with other classes.

Table 3-56 - Association ends of Meas::Measurement with other classes

mult.	mult.	Class type name	Description
Terminal (Operation)	0..1	ACDCTerminal	Inherited Association Based On : Meas:: Measurement.Measurements[0..*] ----- Core:: ACDCTerminal.Terminal[0..1]
PowerSystemResource (Operation)	1..1	PowerSystemResource	Inherited Association Based On : Core:: PowerSystemResource.PowerSystemResource[0..1] ----- Meas:: Measurement.Measurements[0..*]

3.5.8 (Operation) AnalogControl(Abstract)

An analog control used for supervisory control.

Inheritance path: ->Control->IdentifiedObject

Table 3-57 shows all attributes of AnalogControl.

Table 3-57 - Attributes of Meas::AnalogControl

Attribute name	mult.	Attribute type	Description
maxValue (Operation)	1..1	Simple_Float	Normal value range maximum for any of the Control.value. Used for scaling, e.g. in bar graphs.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

minValue (Operation)	1..1	Simple_Float	Normal value range minimum for any of the Control.value. Used for scaling, e.g. in bar graphs.
controlType	1..1	String	see Control
operationInProgress	0..1	Boolean	see Control
timeStamp	0..1	DateTime	see Control
unitMultiplier	0..1	UnitMultiplier	see Control
unitSymbol	0..1	UnitSymbol	see Control
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-58 shows all association ends of AnalogControl with other classes.

Table 3-58 - Association ends of Meas::AnalogControl with other classes

mult.	mult.	Class type name	Description
AnalogValue (Operation)	[1..1]	AnalogValue	The Control variable associated with the MeasurementValue. Association Based On : Meas:: AnalogValue.AnalogValue[1] ----- Meas:: AnalogControl.AnalogControl[0..1]
PowerSystemResource (Operation)	0..1	PowerSystemResource	Inherited Association Based On : Core:: PowerSystemResource.PowerSystemResource[0..1] ----- Meas:: Control.Controls[0..*]

3.5.9 AnalogLimit

Limit values for Analog measurements.

Inheritance path: ->Limit->IdentifiedObject

Table 3-59 shows all attributes of AnalogLimit.

Table 3-59 - Attributes of Meas::AnalogLimit

Attribute name	mult.	Attribute type	Description
value (Operation)	1..1	Simple_Float	The value to supervise against.
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-60 shows all association ends of AnalogLimit with other classes.

Table 3-60 - Association ends of Meas::AnalogLimit with other classes

mult.	mult.	Class type name	Description
LimitSet (Operation)	[1..1]	AnalogLimitSet	The limit values used for supervision of Measurements. Association Based On : Meas:: AnalogLimitSet.LimitSet[1] ----- Meas:: AnalogLimit.Limits[0..*]

3.5.10 (Operation) AnalogLimitSet

An AnalogLimitSet specifies a set of Limits that are associated with an Analog measurement.

Inheritance path: ->LimitSet->IdentifiedObject

Table 3-61 shows all attributes of AnalogLimitSet.

Table 3-61 - Attributes of Meas::AnalogLimitSet

Attribute name	mult.	Attribute type	Description
isPercentageLimits	0..1	Boolean	see LimitSet
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-62 shows all association ends of AnalogLimitSet with other classes.

Table 3-62 - Association ends of Meas::AnalogLimitSet with other classes

mult.	mult.	Class type name	Description
Measurements (Operation)	[0..*]	Analog	A measurement may have zero or more limit ranges defined for it. Association Based On : Meas:: Analog.Measurements[0..*] ----- Meas:: AnalogLimitSet.LimitSets[0..*]

3.5.11 (Operation) AnalogValue

AnalogValue represents an analog MeasurementValue.

- AnalogValue is only used to define measurements available via ICCP, not to supply values for those measurements.

Inheritance path: ->MeasurementValue->IdentifiedObject

Table 3-63 shows all attributes of AnalogValue.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-63 - Attributes of Meas::AnalogValue

Attribute name	mult.	Attribute type	Description
value (Operation)	1..1	Simple_Float	The value to supervise.
timeStamp	0..1	DateTime	see MeasurementValue
sensorAccuracy	0..1	PerCent	see MeasurementValue
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-64 shows all association ends of AnalogValue with other classes.

Table 3-64 - Association ends of Meas::AnalogValue with other classes

mult.	mult.	Class type name	Description
Analog (Operation)	[1..1]	Analog	The values connected to this measurement. Association Based On : Meas:: Analog.Analog[1] ----- Meas:: AnalogValue.AnalogValues[0..*]
MeasurementValueSource (Operation)	1..1	MeasurementValueSource	Inherited Association Based On : Meas:: MeasurementValueSource.MeasurementValueSource[1] ----- Meas:: MeasurementValue.MeasurementValues[0..*]

3.5.12 Command

A Command is a discrete control used for supervisory control.

Inheritance path: ->Control->IdentifiedObject

Table 3-65 shows all attributes of Command.

Table 3-65 - Attributes of Meas::Command

Attribute name	mult.	Attribute type	Description
normalValue (Operation)	1..1	Integer	Normal value for Control.value e.g. used for percentage scaling.
value (Operation)	1..1	Integer	The value representing the actuator output.
controlType	1..1	String	see Control
operationInProgress	0..1	Boolean	see Control
timeStamp	0..1	DateTime	see Control
unitMultiplier	0..1	UnitMultiplier	see Control
unitSymbol	0..1	UnitSymbol	see Control

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-66 shows all association ends of Command with other classes.

Table 3-66 - Association ends of Meas::Command with other classes

mult.	mult.	Class type name	Description
DiscreteValue (Operation)	[1..1]	DiscreteValue	The Control variable associated with the MeasurementValue. Association Based On : Meas:: DiscreteValue.DiscreteValue[1] ----- Meas:: Command.Command[0..1]
ValueAliasSet (Operation)	[0..1]	ValueAliasSet	The ValueAliasSet used for translation of a Control value to a name. Association Based On : Meas:: Command.Commands[0..*] ----- Meas:: ValueAliasSet.ValueAliasSet[0..1]
PowerSystemResource (Operation)	0..1	PowerSystemResource	Inherited Association Based On : Core:: PowerSystemResource.PowerSystemResource[0..1] ----- Meas:: Control.Controls[0..*]

3.5.13 (Operation) Control(Abstract)

Control is used for supervisory/device control. It represents control outputs that are used to change the state in a process, e.g. close or open breaker, a set point value or a raise lower command.

Inheritance path: ->IdentifiedObject

Table 3-67 shows all attributes of Control.

Table 3-67 - Attributes of Meas::Control

Attribute name	mult.	Attribute type	Description
controlType (Operation)	1..1	String	Specifies the type of Control, e.g. BreakerOn/Off, GeneratorVoltageSetPoint, TieLineFlow etc. The ControlType.name shall be unique among all specified types and describe the type.
operationInProgress (Operation)	0..1	Boolean	Indicates that a client is currently sending control commands that has not completed.
timeStamp (Operation)	0..1	DateTime	The last time a control

			output was sent.
unitMultiplier (Operation)	0..1	UnitMultiplier	The unit multiplier of the controlled quantity.
unitSymbol (Operation)	0..1	UnitSymbol	The unit of measure of the controlled quantity.
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-68 shows all association ends of Control with other classes.

Table 3-68 - Association ends of Meas::Control with other classes

mult.	mult.	Class type name	Description
PowerSystemResource (Operation)	[0..1]	PowerSystemResource	The controller outputs used to actually govern a regulating device, e.g. the magnetization of a synchronous machine or capacitor bank breaker actuator. Association Based On : Core:: PowerSystemResource.PowerSystemResource[0..1] ----- Meas:: Control.Controls[0..*]

3.5.14 (Operation) Discrete

Discrete represents a discrete Measurement, i.e. a Measurement representing discrete values, e.g. a Breaker position.

- The measurementType attribute is used to define the quantity being measured (SwitchPosition, etc.) by a Measurement. A Measurement must be associated with one and only one measurementType. The valid values for MeasurementType.name are defined in Normative String Tables

- The association to Terminal may not be required depending on how the Measurement is being used. See section Use of Measurement Class for details

Inheritance path: ->Measurement->IdentifiedObject

Table 3-69 shows all attributes of Discrete.

Table 3-69 - Attributes of Meas::Discrete

Attribute name	mult.	Attribute type	Description
measurementType	1..1	String	see Measurement
phases	0..1	PhaseCode	see Measurement
unitSymbol	1..1	UnitSymbol	see Measurement
unitMultiplier	1..1	UnitMultiplier	see Measurement
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-70 shows all association ends of Discrete with other classes.

Table 3-70 - Association ends of Meas::Discrete with other classes

mult.	mult.	Class type name	Description
ValueAliasSet (Operation)	[0..1]	ValueAliasSet	The ValueAliasSet used for translation of a MeasurementValue.value to a name. Association Based On : Meas:: ValueAliasSet.ValueAliasSet[0..1] ----- Meas:: Discrete.Discretes[0..*]
Terminal (Operation)	0..1	ACDCTerminal	Inherited Association Based On : Meas:: Measurement.Measurements[0..*] ----- Core:: ACDCTerminal.Terminal[0..1]
PowerSystemResource (Operation)	1..1	PowerSystemResource	Inherited Association Based On : Core:: PowerSystemResource.PowerSystemResource[0..1] ----- Meas:: Measurement.Measurements[0..*]

3.5.15 DiscreteValue

DiscreteValue represents a discrete MeasurementValue.

- DiscreteValue is only used to define measurements available via ICCP, not to supply values for those measurements.

Inheritance path: ->MeasurementValue->IdentifiedObject

Table 3-71 shows all attributes of DiscreteValue.

Table 3-71 - Attributes of Meas::DiscreteValue

Attribute name	mult.	Attribute type	Description
value (Operation)	1..1	Integer	The value to supervise.
timeStamp	0..1	DateTime	see MeasurementValue
sensorAccuracy	0..1	PerCent	see MeasurementValue
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-72 shows all association ends of DiscreteValue with other classes.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-72 - Association ends of Meas::DiscreteValue with other classes

mult.	mult.	Class type name	Description
Discrete (Operation)	[1..1]	Discrete	The values connected to this measurement. Association Based On : Meas:: Discrete.Discrete[1] ----- Meas:: DiscreteValue.DiscreteValues[0..*]
MeasurementValueSource (Operation)	1..1	MeasurementValueSource	Inherited Association Based On : Meas:: MeasurementValueSource.MeasurementValueSource[1] ----- Meas:: MeasurementValue.MeasurementValues[0..*]

3.5.16 (Operation) Limit(Abstract)

Specifies one limit value for a Measurement. A Measurement typically has several limits that are kept together by the LimitSet class. The actual meaning and use of a Limit instance (i.e., if it is an alarm or warning limit or if it is a high or low limit) is not captured in the Limit class. However the name of a Limit instance may indicate both meaning and use.

Inheritance path: ->IdentifiedObject

Table 3-73 shows all attributes of Limit.

Table 3-73 - Attributes of Meas::Limit

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

3.5.17 (Operation) LimitSet(Abstract)

Specifies a set of Limits that are associated with a Measurement. A Measurement may have several LimitSets corresponding to seasonal or other changing conditions. The condition is captured in the name and description attributes. The same LimitSet may be used for several Measurements. In particular percentage limits are used this way.

Inheritance path: ->IdentifiedObject

Table 3-74 shows all attributes of LimitSet.

Table 3-74 - Attributes of Meas::LimitSet

Attribute name	mult.	Attribute type	Description
isPercentageLimits (Operation)	0..1	Boolean	Tells if the limit values are in percentage of normalValue or the specified Unit for Measurements and Controls.
description	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

3.5.18 (Operation) Measurement(Abstract)

A Measurement represents any measured, calculated or non-measured non-calculated quantity. Any piece of equipment may contain Measurements, e.g. a substation may have temperature measurements and door open indications, a transformer may have oil temperature and tank pressure measurements, a bay may contain a number of power flow measurements and a Breaker may contain a switch status measurement.
The PSR - Measurement association is intended to capture this use of Measurement and is included in the naming hierarchy based on EquipmentContainer. The naming hierarchy typically has Measurements as leafs, e.g. Substation-VoltageLevel-Bay-Switch-Measurement. Some Measurements represent quantities related to a particular sensor location in the network, e.g. a voltage transformer (PT) at a busbar or a current transformer (CT) at the bar between a breaker and an isolator. The sensing position is not captured in the PSR - Measurement association. Instead it is captured by the Measurement - Terminal association that is used to define the sensing location in the network topology. The location is defined by the connection of the Terminal to ConductingEquipment.
If both a Terminal and PSR are associated, and the PSR is of type ConductingEquipment, the associated Terminal should belong to that ConductingEquipment instance. When the sensor location is needed both Measurement-PSR and Measurement-Terminal are used. The Measurement-Terminal association is never used alone.

OCL constraint: If a Terminal exists its ConductingEquipment must be the same as PSR
OCL constraint:measurementType is restricted to the values: ThreePhasePower, ThreePhaseActivePower, ThreePhaseReactivePower, LineCurrent, PhaseVoltage, LineToLineVoltage, Angle, TapPosition, SwitchPosition
OCL constraint:unitSymbol is limited to W, deg, VA, none, A, VAr, V, Hz

Measurement.measurementType is restricted to the following valid values:

- ThreePhasePower
- ThreePhaseActivePower
- ThreePhaseReactivePower
- LineCurrent
- PhaseVoltage
- LineToLineVoltage
- Angle
- TapPosition
- SwitchPosition

Measurement.unitSymbol is restricted to the following valid values:

- W

- deg
- VA
- none
- A
- VAr
- V
- Hz

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Measurement.phases: if the attribute is not specified the Measurement is understood as three phase (ABC).

Inheritance path: ->IdentifiedObject

Table 3-75 shows all attributes of Measurement.

Table 3-75 - Attributes of Meas::Measurement

Attribute name	mult.	Attribute type	Description
measurementType (Operation)	1..1	String	Specifies the type of measurement. For example, this specifies if the measurement represents an indoor temperature, outdoor temperature, bus voltage, line flow, etc.
phases (Operation)	0..1	PhaseCode	Indicates to which phases the measurement applies and avoids the need to use 'measurementType' to also encode phase information (which would explode the types). The phase information in Measurement, along with 'measurementType' and 'phases' uniquely defines a Measurement for a device, based on normal network phase. Their meaning will not change when the computed energizing phasing is changed due to jumpers or other reasons. If the attribute is missing three phases (ABC) shall be assumed.
unitSymbol (Operation)	1..1	UnitSymbol	The unit of measure of the measured quantity.
unitMultiplier (Operation)	1..1	UnitMultiplier	The unit multiplier of the measured quantity.
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-76 shows all association ends of Measurement with other classes.

Table 3-76 - Association ends of Meas::Measurement with other classes

mult.	mult.	Class type name	Description

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Terminal (Operation)	[0..1]	ACDCTerminal	One or more measurements may be associated with a terminal in the network. Association Based On : Meas:: Measurement.Measurements[0..*] ----- Core:: ACDCTerminal.Terminal[0..1]
PowerSystemResource (Operation)	[1..1]	PowerSystemResource	The measurements associated with this power system resource. Association Based On : Core:: PowerSystemResource.PowerSystemResource[0..1] ----- Meas:: Measurement.Measurements[0..*]

3.5.19 (Operation) MeasurementValue(Abstract)

The current state for a measurement. A state value is an instance of a measurement from a specific source. Measurements can be associated with many state values, each representing a different source for the measurement.

Inheritance path: ->IdentifiedObject

Table 3-77 shows all attributes of MeasurementValue.

Table 3-77 - Attributes of Meas::MeasurementValue

Attribute name	mult.	Attribute type	Description
timeStamp (Operation)	0..1	DateTime	The time when the value was last updated
sensorAccuracy (Operation)	0..1	PerCent	The limit, expressed as a percentage of the sensor maximum, that errors will not exceed when the sensor is used under reference conditions.
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-78 shows all association ends of MeasurementValue with other classes.

Table 3-78 - Association ends of Meas::MeasurementValue with other classes

mult.	mult.	Class type name	Description
MeasurementValueSource (Operation)	[1..1]	MeasurementValueSource	The MeasurementValues updated by the source. Association Based On : Meas:: MeasurementValueSource.MeasurementValueSource[1] ----- Meas:: MeasurementValue.MeasurementValues[0..*]

3.5.20 (Operation) MeasurementValueQuality

Measurement quality flags. Bits 0-10 are defined for substation automation in draft IEC 61850 part 7-3. Bits 11-15 are reserved for future expansion by that document. Bits 16-31 are reserved for EMS applications.

Inheritance path: ->Quality61850

Table 3-79 shows all attributes of MeasurementValueQuality.

Table 3-79 - Attributes of Meas::MeasurementValueQuality

Attribute name	mult.	Attribute type	Description
badReference	0..1	Boolean	see Quality61850
estimatorReplaced	0..1	Boolean	see Quality61850
failure	0..1	Boolean	see Quality61850
oldData	0..1	Boolean	see Quality61850
operatorBlocked	0..1	Boolean	see Quality61850
oscillatory	0..1	Boolean	see Quality61850
outOfRange	0..1	Boolean	see Quality61850
overFlow	0..1	Boolean	see Quality61850
source	0..1	Source	see Quality61850
suspect	0..1	Boolean	see Quality61850
test	0..1	Boolean	see Quality61850
validity	0..1	Validity	see Quality61850

Table 3-80 shows all association ends of MeasurementValueQuality with other classes.

Table 3-80 - Association ends of Meas::MeasurementValueQuality with other classes

mult.	mult.	Class type name	Description
MeasurementValue (Operation)	[1..1]	MeasurementValue	A MeasurementValue has a MeasurementValueQuality associated with it. Association Based On : Meas:: MeasurementValueQuality.MeasurementValueQuality[0..1] ----- Meas:: MeasurementValue.MeasurementValue[1]

3.5.21 (Operation) MeasurementValueSource

MeasurementValueSource describes the alternative sources updating a MeasurementValue. User conventions for how to use the MeasurementValueSource attributes are described in the introduction to IEC 61970-301.

Attribute IdentifiedObject.name is restricted to the following strings for MeasurementValueSource:

- ICCP
- SCADA
- CCLink
- Operator
- Estimated
- PowerFlow
- Forecasted
- Calculated
- Allocated

Inheritance path: ->IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-81 shows all attributes of MeasurementValueSource.

Table 3-81 - Attributes of Meas::MeasurementValueSource

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

3.5.22 (Operation) Quality61850(Abstract)

Quality flags in this class are as defined in IEC 61850, except for estimatorReplaced, which has been included in this class for convenience.

Table 3-82 shows all attributes of Quality61850.

Table 3-82 - Attributes of Meas::Quality61850

Attribute name	mult.	Attribute type	Description
badReference (Operation)	0..1	Boolean	Measurement value may be incorrect due to a reference being out of calibration.
estimatorReplaced (Operation)	0..1	Boolean	Value has been replaced by State Estimator. estimatorReplaced is not an IEC61850 quality bit but has been put in this class for convenience.
failure (Operation)	0..1	Boolean	This identifier indicates that a supervision function has detected an internal or external failure, e.g. communication failure.
oldData (Operation)	0..1	Boolean	Measurement value is old and possibly invalid, as it has not been successfully updated during a specified time interval.
operatorBlocked (Operation)	0..1	Boolean	Measurement value is blocked and hence unavailable for transmission.
oscillatory (Operation)	0..1	Boolean	To prevent some overload of the communication it is sensible to detect and suppress oscillating (fast changing) binary inputs. If a signal changes in a defined time (tosc) twice in the same direction (from 0

			to 1 or from 1 to 0) then oscillation is detected and the detail quality identifier "oscillatory" is set. If it is detected a configured numbers of transient changes could be passed by. In this time the validity status "questionable" is set. If after this defined numbers of changes the signal is still in the oscillating state the value shall be set either to the opposite state of the previous stable value or to a defined default value. In this case the validity status "questionable" is reset and "invalid" is set as long as the signal is oscillating. If it is configured such that no transient changes should be passed by then the validity status "invalid" is set immediately in addition to the detail quality identifier "oscillatory" (used for status information only).
outOfRange (Operation)	0..1	Boolean	Measurement value is beyond a predefined range of value.
overFlow (Operation)	0..1	Boolean	Measurement value is beyond the capability of being represented properly. For example, a counter value overflows from maximum count back to a value of zero.
source (Operation)	0..1	Source	Source gives information related to the origin of a value. The value may be acquired from the process, defaulted or substituted.
suspect (Operation)	0..1	Boolean	A correlation function has detected that the value is not consitent with other values. Typically set by a network State Estimator.
test (Operation)	0..1	Boolean	Measurement value is transmitted for test purposes.
validity (Operation)	0..1	Validity	Validity of the measurement value.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

3.5.23 (Operation) RaiseLowerCommand

An analog control that increase or decrease a set point value with pulses.

Inheritance path: ->AnalogControl->Control->IdentifiedObject

Table 3-83 shows all attributes of RaiseLowerCommand.

Table 3-83 - Attributes of Meas::RaiseLowerCommand

Attribute name	mult.	Attribute type	Description
maxValue	1..1	Simple_Float	see AnalogControl
minValue	1..1	Simple_Float	see AnalogControl
controlType	1..1	String	see Control
operationInProgress	0..1	Boolean	see Control
timeStamp	0..1	DateTime	see Control
unitMultiplier	0..1	UnitMultiplier	see Control
unitSymbol	0..1	UnitSymbol	see Control
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-84 shows all association ends of RaiseLowerCommand with other classes.

Table 3-84 - Association ends of Meas::RaiseLowerCommand with other classes

mult.	mult.	Class type name	Description
ValueAliasSet (Operation)	[0..1]	ValueAliasSet	The ValueAliasSet used for translation of a Control value to a name. Association Based On : Meas:: RaiseLowerCommand.RaiseLowerCommands[0..*] - --- Meas:: ValueAliasSet.ValueAliasSet[0..1]
AnalogValue (Operation)	1..1	AnalogValue	Inherited Association Based On : Meas:: AnalogValue.AnalogValue[1] ----- Meas:: AnalogControl.AnalogControl[0..1]
PowerSystemResource (Operation)	0..1	PowerSystemResource	Inherited Association Based On : Core:: PowerSystemResource.PowerSystemResource[0..1] ----- Meas:: Control.Controls[0..*]

3.5.24 SetPoint

An analog control that issue a set point value.

Inheritance path: ->AnalogControl->Control->IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-85 shows all attributes of SetPoint.

Table 3-85 - Attributes of Meas::SetPoint

Attribute name	mult.	Attribute type	Description
normalValue (Operation)	1..1	Simple_Float	Normal value for Control.value e.g. used for percentage scaling.
value (Operation)	1..1	Simple_Float	The value representing the actuator output.
maxValue	1..1	Simple_Float	see AnalogControl
minValue	1..1	Simple_Float	see AnalogControl
controlType	1..1	String	see Control
operationInProgress	0..1	Boolean	see Control
timeStamp	0..1	DateTime	see Control
unitMultiplier	0..1	UnitMultiplier	see Control
unitSymbol	0..1	UnitSymbol	see Control
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-86 shows all association ends of AnalogControl with other classes.

Table 3-86 - Association ends of Meas::AnalogControl with other classes

mult.	mult.	Class type name	Description
AnalogValue (Operation)	1..1	AnalogValue	Inherited Association Based On : Meas:: AnalogValue.AnalogValue[1] ----- Meas:: AnalogControl.AnalogControl[0..1]
PowerSystemResource (Operation)	0..1	PowerSystemResource	Inherited Association Based On : Core:: PowerSystemResource.PowerSystemResource[0..1] ----- Meas:: Control.Controls[0..*]

3.5.25 StringMeasurement

StringMeasurement represents a measurement with values of type string.

Inheritance path: ->Measurement->IdentifiedObject

Table 3-87 shows all attributes of StringMeasurement.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-87 - Attributes of Meas::StringMeasurement

Attribute name	mult.	Attribute type	Description
measurementType	1..1	String	see Measurement
phases	0..1	PhaseCode	see Measurement
unitSymbol	1..1	UnitSymbol	see Measurement
unitMultiplier	1..1	UnitMultiplier	see Measurement
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-88 shows all association ends of Measurement with other classes.

Table 3-88 - Association ends of Meas::Measurement with other classes

mult.	mult.	Class type name	Description
Terminal (Operation)	0..1	ACDCTerminal	Inherited Association Based On : Meas:: Measurement.Measurements[0..*] ----- Core:: ACDCTerminal.Terminal[0..1]
PowerSystemResource (Operation)	1..1	PowerSystemResource	Inherited Association Based On : Core:: PowerSystemResource.PowerSystemResource[0..1] ----- Meas:: Measurement.Measurements[0..*]

3.5.26 StringMeasurementValue

StringMeasurementValue represents a measurement value of type string.

Inheritance path: ->MeasurementValue->IdentifiedObject

Table 3-89 shows all attributes of StringMeasurementValue.

Table 3-89 - Attributes of Meas::StringMeasurementValue

Attribute name	mult.	Attribute type	Description
value (Operation)	1..1	String	The value to supervise.
timeStamp	0..1	DateTime	see MeasurementValue
sensorAccuracy	0..1	PerCent	see MeasurementValue
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-90 shows all association ends of StringMeasurementValue with other classes.

Table 3-90 - Association ends of Meas::StringMeasurementValue with other classes

mult.	mult.	Class type name	Description
StringMeasurement (Operation)	[1..1]	StringMeasurement	Measurement to which this value is connected. Association Based On : Meas:: StringMeasurementValue.StringMeasurementValues[0..*] ----- Meas:: StringMeasurement.StringMeasurement[1]
MeasurementValueSource (Operation)	1..1	MeasurementValueSource	Inherited Association Based On : Meas:: MeasurementValueSource.MeasurementValueSource[1] ----- Meas:: MeasurementValue.MeasurementValues[0..*]

3.5.27 ValueAliasSet

Describes the translation of a set of values into a name and is intendend to facilitate cusom translations. Each ValueAliasSet has a name, description etc. A specific Measurement may represent a discrete state like Open, Closed, Intermediate etc. This requires a translation from the MeasurementValue.value number to a string, e.g. 0->"Invalid", 1->"Open", 2->"Closed", 3->"Intermediate". Each ValueToAlias member in ValueAliasSet.Value describe a mapping for one particular value to a name.

Inheritance path: ->IdentifiedObject

Table 3-91 shows all attributes of ValueAliasSet.

Table 3-91 - Attributes of Meas::ValueAliasSet

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

3.5.28 ValueToAlias

Describes the translation of one particular value into a name, e.g. 1 as "Open".

Inheritance path: ->IdentifiedObject

Table 3-92 shows all attributes of ValueToAlias.

Table 3-92 - Attributes of Meas::ValueToAlias

Attribute name	mult.	Attribute type	Description
value (Operation)	1..1	Integer	The value that is mapped.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-93 shows all association ends of ValueToAlias with other classes.

Table 3-93 - Association ends of Meas::ValueToAlias with other classes

mult.	mult.	Class type name	Description
ValueAliasSet (Operation)	[1..1]	ValueAliasSet	The ValueToAlias mappings included in the set. Association Based On : Meas:: ValueAliasSet.ValueAliasSet[1] ----- Meas:: ValueToAlias.Values[1..*]

3.5.29 Enumerations

3.5.29.1 Source

Source gives information related to the origin of a value.

Literal	Description	Code
PROCESS	The value is provided by input from the process I/O or being calculated from some function.	
DEFAULTED	The value contains a default value.	
SUBSTITUTED	The value is provided by input of an operator or by an automatic source.	

3.5.29.2 Validity

Validity for MeasurementValue.

Literal	Description	Code
GOOD	The value is marked good if no abnormal condition of the acquisition function or the information source is detected.	
QUESTIONABLE	The value is marked questionable if a supervision function detects an abnormal behaviour, however the value could still be valid. The client is responsible for determining whether or not values marked "questionable" should be used.	
INVALID	The value is marked invalid when a supervision function recognises abnormal conditions of the acquisition function or the information source (missing or non-operating updating devices). The value is not defined under this condition. The mark invalid is used to indicate to the client that the value may be incorrect and shall not be used.	

3.6 Production

3.6.1 General

The production package is responsible for classes which describe various kinds of generators. These classes also provide production costing information which is used to economically allocate demand among committed units and calculate reserve quantities.

Figure 3-10 shows the diagram.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

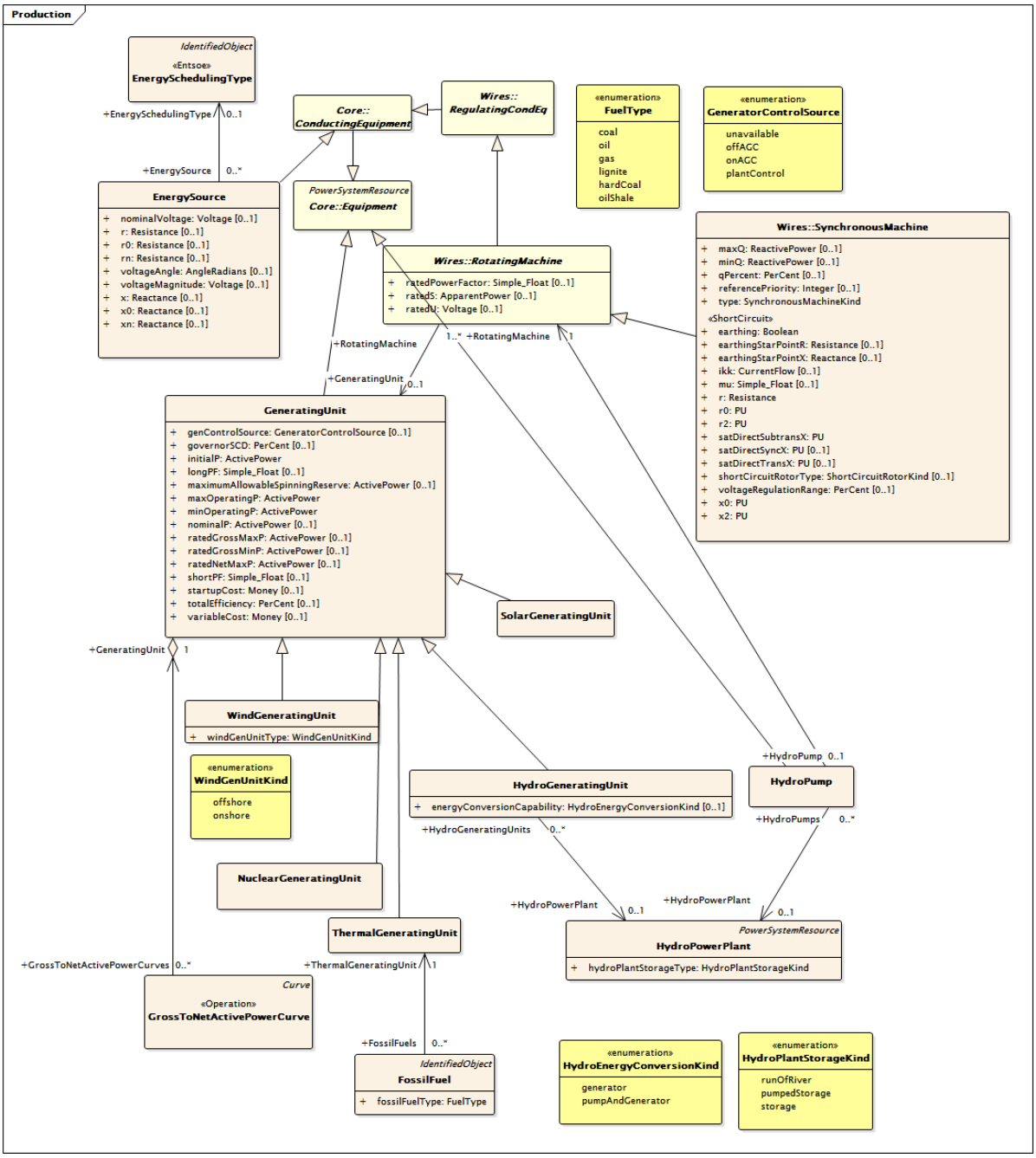


Figure 3-10 - diagram Production

Figure 3-11 shows the diagram.

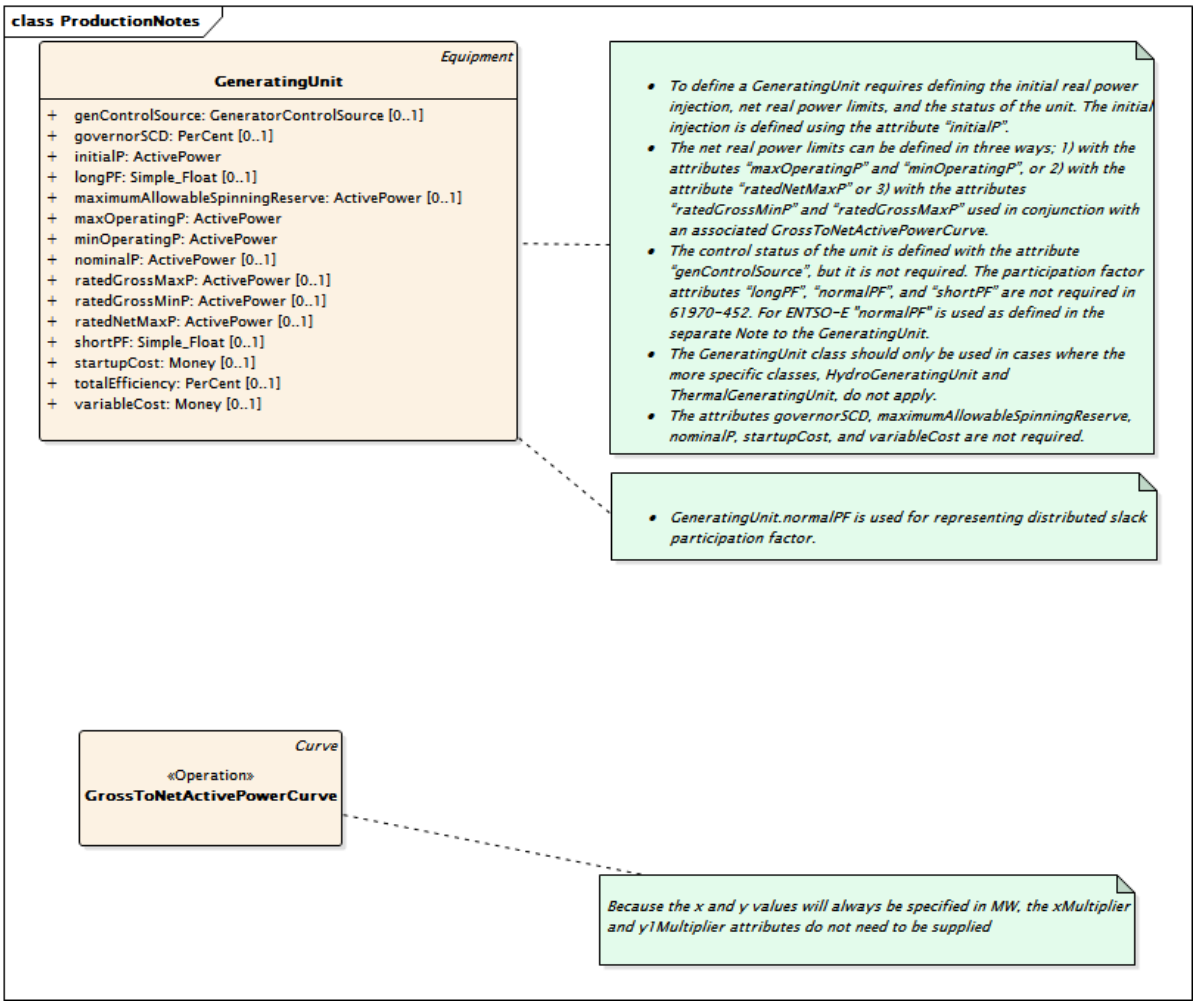


Figure 3-11 - diagram ProductionNotes

3.6.2 (Operation) EnergySchedulingType

Used to define the type of generation for scheduling purposes.

Inheritance path: ->IdentifiedObject

Table 3-94 shows all attributes of EnergySchedulingType.

Table 3-94 - Attributes of Production::EnergySchedulingType

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

3.6.3 (Operation) EnergySource

A generic equivalent for an energy supplier on a transmission or distribution voltage level.

Inheritance path: ->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-95 shows all attributes of EnergySource.

Table 3-95 - Attributes of Production::EnergySource

Attribute name	mult.	Attribute type	Description
nominalVoltage	0..1	Voltage	Phase-to-phase nominal voltage.
r	0..1	Resistance	Positive sequence Thevenin resistance.
r0	0..1	Resistance	Zero sequence Thevenin resistance.
rn	0..1	Resistance	Negative sequence Thevenin resistance.
voltageAngle	0..1	AngleRadians	Phase angle of a-phase open circuit.
voltageMagnitude	0..1	Voltage	Phase-to-phase open circuit voltage magnitude.
x	0..1	Reactance	Positive sequence Thevenin reactance.
x0	0..1	Reactance	Zero sequence Thevenin reactance.
xn	0..1	Reactance	Negative sequence Thevenin reactance.
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-96 shows all association ends of EnergySource with other classes.

Table 3-96 - Association ends of Production::EnergySource with other classes

mult.	mult.	Class type name	Description
EnergySchedulingType	[0..1]	EnergySchedulingType	Energy Source of a particular Energy Scheduling Type Association Based On : Wires:: EnergySource.EnergySource[0..*] ----- Extension:: EnergySchedulingType.EnergySchedulingType[0..1]
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	

			Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.6.4 FossilFuel

The fossil fuel consumed by the non-nuclear thermal generating unit. For example, coal, oil, gas, etc. This a the specific fuels that the generating unit can consume.

Inheritance path: ->IdentifiedObject

Table 3-97 shows all attributes of FossilFuel.

Table 3-97 - Attributes of Production::FossilFuel

Attribute name	mult.	Attribute type	Description
fossilFuelType	1..1	FuelType	The type of fossil fuel, such as coal, oil, or gas.
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-98 shows all association ends of FossilFuel with other classes.

Table 3-98 - Association ends of Production::FossilFuel with other classes

mult.	mult.	Class type name	Description
ThermalGeneratingUnit	[1..1]	ThermalGeneratingUnit	A thermal generating unit may have one or more fossil fuels. Association Based On : Production:: FossilFuel.FossilFuels[0..*] ----- Production:: ThermalGeneratingUnit.ThermalGeneratingUnit[1]

3.6.5 GeneratingUnit

A single or set of synchronous machines for converting mechanical power into alternating-current power. For example, individual machines within a set may be defined for scheduling purposes while a single control signal is derived for the set. In this case there would be a GeneratingUnit for each member of the set and an additional GeneratingUnit corresponding to the set.

- To define a GeneratingUnit requires defining the initial real power injection, net real power limits, and the status of the unit. The initial injection is defined using the attribute “initialP”.
- The net real power limits can be defined in three ways; 1) with the attributes “maxOperatingP” and “minOperatingP”, or 2) with the attribute “ratedNetMaxP” or 3) with the attributes “ratedGrossMinP” and “ratedGrossMaxP” used in conjunction with an associated GrossToNetActivePowerCurve.
- The control status of the unit is defined with the attribute “genControlSource”, but it is not required. The participation factor attributes “longPF”, “normalPF”, and “shortPF” are not required in 61970-452. For ENTSO-E "normalPF" is used as defined in the separate Note to the GeneratingUnit.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

- The GeneratingUnit class should only be used in cases where the more specific classes, HydroGeneratingUnit and ThermalGeneratingUnit, do not apply.
- The attributes governorSCD, maximumAllowableSpinningReserve, nominalP, startupCost, and variableCost are not required.

- GeneratingUnit.normalPF is used for representing distributed slack participation factor.

Inheritance path: ->Equipment->PowerSystemResource->IdentifiedObject

Table 3-99 shows all attributes of GeneratingUnit.

Table 3-99 - Attributes of Production::GeneratingUnit

Attribute name	mult.	Attribute type	Description
genControlSource	0..1	GeneratorControlSource	The source of controls for a generating unit.
governorSCD	0..1	PerCent	Governor Speed Changer Droop. This is the change in generator power output divided by the change in frequency normalized by the nominal power of the generator and the nominal frequency and expressed in percent and negated. A positive value of speed change droop provides additional generator output upon a drop in frequency.
initialP	1..1	ActivePower	Default initial active power which is used to store a powerflow result for the initial active power for this unit in this network configuration.
longPF	0..1	Simple_Float	Generating unit long term economic participation factor.
maximumAllowableSpinningReserve	0..1	ActivePower	Maximum allowable spinning reserve. Spinning reserve will never be considered greater than this value regardless of the current operating point.
maxOperatingP	1..1	ActivePower	This is the maximum operating active power limit the dispatcher can enter for this unit.
minOperatingP	1..1	ActivePower	This is the minimum operating active power limit the dispatcher can enter for this unit.
nominalP	0..1	ActivePower	The nominal power of the generating unit. Used to give precise meaning to percentage

			based attributes such as the governor speed change droop (governorSCD attribute). The attribute shall be a positive value equal or less than RotatingMachine.ratedS.
ratedGrossMaxP	0..1	ActivePower	The unit's gross rated maximum capacity (book value).
ratedGrossMinP	0..1	ActivePower	The gross rated minimum generation level which the unit can safely operate at while delivering power to the transmission grid.
ratedNetMaxP	0..1	ActivePower	The net rated maximum capacity determined by subtracting the auxiliary power used to operate the internal plant machinery from the rated gross maximum capacity.
shortPF	0..1	Simple_Float	Generating unit short term economic participation factor.
startupCost	0..1	Money	The initial startup cost incurred for each start of the GeneratingUnit.
variableCost	0..1	Money	The variable cost component of production per unit of ActivePower.
totalEfficiency	0..1	PerCent	The efficiency of the unit in converting the fuel into electrical energy.
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-100 shows all association ends of Equipment with other classes.

Table 3-100 - Association ends of Core::Equipment with other classes

mult.	mult.	Class type name	Description
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

3.6.6 GrossToNetActivePowerCurve

Relationship between the generating unit's gross active power output on the X-axis (measured at the terminals of the machine(s)) and the generating unit's net active power output on the Y-axis (based on utility-defined measurements at the power station). Station service loads, when modeled, should be treated as non-conforming bus loads. There may be more than one curve, depending on the auxiliary equipment that is in service.

Because the x and y values will always be specified in MW, the xMultiplier and y1Multiplier attributes do not need to be supplied

Inheritance path: ->Curve->IdentifiedObject

Table 3-101 shows all attributes of GrossToNetActivePowerCurve.

Table 3-101 - Attributes of Production::GrossToNetActivePowerCurve

Attribute name	mult.	Attribute type	Description
curveStyle	1..1	CurveStyle	see Curve
xUnit	1..1	UnitSymbol	see Curve
y1Unit	1..1	UnitSymbol	see Curve
y2Unit	0..1	UnitSymbol	see Curve
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-102 shows all association ends of GrossToNetActivePowerCurve with other classes.

Table 3-102 - Association ends of Production::GrossToNetActivePowerCurve with other classes

mult.	mult.	Class type name	Description
GeneratingUnit (Operation)	[1..1]	GeneratingUnit	A generating unit may have a gross active power to net active power curve, describing the losses and auxiliary power requirements of the unit. Association Based On : Production:: GrossToNetActivePowerCurve.GrossToNetActivePowerCurves[0..*] ----- Production:: GeneratingUnit.GeneratingUnit[1]

3.6.7 (Operation) HydroGeneratingUnit

A generating unit whose prime mover is a hydraulic turbine (e.g., Francis, Pelton, Kaplan).

Inheritance path: ->GeneratingUnit->Equipment->PowerSystemResource->IdentifiedObject

Table 3-103 shows all attributes of HydroGeneratingUnit.

Table 3-103 - Attributes of Production::HydroGeneratingUnit

Attribute name	mult.	Attribute type	Description

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

energyConversionCapability	0..1	HydroEnergyConversionKind	Energy conversion capability for generating.
genControlSource	0..1	GeneratorControlSource	see GeneratingUnit
governorSCD	0..1	PerCent	see GeneratingUnit
initialP	1..1	ActivePower	see GeneratingUnit
longPF	0..1	Simple_Float	see GeneratingUnit
maximumAllowableSpinningReserve	0..1	ActivePower	see GeneratingUnit
maxOperatingP	1..1	ActivePower	see GeneratingUnit
minOperatingP	1..1	ActivePower	see GeneratingUnit
nominalP	0..1	ActivePower	see GeneratingUnit
ratedGrossMaxP	0..1	ActivePower	see GeneratingUnit
ratedGrossMinP	0..1	ActivePower	see GeneratingUnit
ratedNetMaxP	0..1	ActivePower	see GeneratingUnit
shortPF	0..1	Simple_Float	see GeneratingUnit
startupCost	0..1	Money	see GeneratingUnit
variableCost	0..1	Money	see GeneratingUnit
totalEfficiency	0..1	PerCent	see GeneratingUnit
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-104 shows all association ends of HydroGeneratingUnit with other classes.

Table 3-104 - Association ends of Production::HydroGeneratingUnit with other classes

mult.	mult.	Class type name	Description
HydroPowerPlant	[0..1]	HydroPowerPlant	The hydro generating unit belongs to a hydro power plant. Association Based On : Production:: HydroPowerPlant.HydroPowerPlant[0..1] ----- Production:: HydroGeneratingUnit.HydroGeneratingUnits[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

3.6.8 HydroPowerPlant

A hydro power station which can generate or pump. When generating, the generator turbines receive water from an upper reservoir. When pumping, the pumps receive their water from a lower reservoir.

Inheritance path: ->PowerSystemResource->IdentifiedObject

Table 3-105 shows all attributes of HydroPowerPlant.

Table 3-105 - Attributes of Production::HydroPowerPlant

Attribute name	mult.	Attribute type	Description
hydroPlantStorageType	1..1	HydroPlantStorageKind	The type of hydro power plant water storage.
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

3.6.9 HydroPump

A synchronous motor-driven pump, typically associated with a pumped storage plant.

Inheritance path: ->Equipment->PowerSystemResource->IdentifiedObject

Table 3-106 shows all attributes of HydroPump.

Table 3-106 - Attributes of Production::HydroPump

Attribute name	mult.	Attribute type	Description
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-107 shows all association ends of HydroPump with other classes.

Table 3-107 - Association ends of Production::HydroPump with other classes

mult.	mult.	Class type name	Description
RotatingMachine	[1..1]	RotatingMachine	The synchronous machine drives the turbine which moves the water from a low elevation to a higher elevation. The direction of machine rotation for pumping may or may not be the same as for generating. Association Based On : Wires:: RotatingMachine.RotatingMachine[1] ----- Production:: HydroPump.HydroPump[0..1]

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

HydroPowerPlant	[0..1]	HydroPowerPlant	The hydro pump may be a member of a pumped storage plant or a pump for distributing water. Association Based On : Production:: HydroPowerPlant.HydroPowerPlant[0..1] ----- Production:: HydroPump.HydroPumps[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.6.10 NuclearGeneratingUnit

A nuclear generating unit.

Inheritance path: ->GeneratingUnit->Equipment->PowerSystemResource->IdentifiedObject

Table 3-108 shows all attributes of NuclearGeneratingUnit.

Table 3-108 - Attributes of Production::NuclearGeneratingUnit

Attribute name	mult.	Attribute type	Description
genControlSource	0..1	GeneratorControlSource	see GeneratingUnit
governorSCD	0..1	PerCent	see GeneratingUnit
initialP	1..1	ActivePower	see GeneratingUnit
longPF	0..1	Simple_Float	see GeneratingUnit
maximumAllowableSpinningReserve	0..1	ActivePower	see GeneratingUnit
maxOperatingP	1..1	ActivePower	see GeneratingUnit
minOperatingP	1..1	ActivePower	see GeneratingUnit
nominalP	0..1	ActivePower	see GeneratingUnit
ratedGrossMaxP	0..1	ActivePower	see GeneratingUnit
ratedGrossMinP	0..1	ActivePower	see GeneratingUnit
ratedNetMaxP	0..1	ActivePower	see GeneratingUnit
shortPF	0..1	Simple_Float	see GeneratingUnit
startupCost	0..1	Money	see GeneratingUnit
variableCost	0..1	Money	see GeneratingUnit
totalEfficiency	0..1	PerCent	see GeneratingUnit
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-109 shows all association ends of Equipment with other classes.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-109 - Association ends of Core::Equipment with other classes

mult.	mult.	Class type name	Description
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.6.11 SolarGeneratingUnit

A solar thermal generating unit.

Inheritance path: ->GeneratingUnit->Equipment->PowerSystemResource->IdentifiedObject

Table 3-110 shows all attributes of SolarGeneratingUnit.

Table 3-110 - Attributes of Production::SolarGeneratingUnit

Attribute name	mult.	Attribute type	Description
genControlSource	0..1	GeneratorControlSource	see GeneratingUnit
governorSCD	0..1	PerCent	see GeneratingUnit
initialP	1..1	ActivePower	see GeneratingUnit
longPF	0..1	Simple_Float	see GeneratingUnit
maximumAllowableSpinningReserve	0..1	ActivePower	see GeneratingUnit
maxOperatingP	1..1	ActivePower	see GeneratingUnit
minOperatingP	1..1	ActivePower	see GeneratingUnit
nominalP	0..1	ActivePower	see GeneratingUnit
ratedGrossMaxP	0..1	ActivePower	see GeneratingUnit
ratedGrossMinP	0..1	ActivePower	see GeneratingUnit
ratedNetMaxP	0..1	ActivePower	see GeneratingUnit
shortPF	0..1	Simple_Float	see GeneratingUnit
startupCost	0..1	Money	see GeneratingUnit
variableCost	0..1	Money	see GeneratingUnit
totalEfficiency	0..1	PerCent	see GeneratingUnit
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-111 shows all association ends of Equipment with other classes.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-111 - Association ends of Core::Equipment with other classes

mult.	mult.	Class type name	Description
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.6.12 ThermalGeneratingUnit

A generating unit whose prime mover could be a steam turbine, combustion turbine, or diesel engine.

Inheritance path: ->GeneratingUnit->Equipment->PowerSystemResource->IdentifiedObject

Table 3-112 shows all attributes of ThermalGeneratingUnit.

Table 3-112 - Attributes of Production::ThermalGeneratingUnit

Attribute name	mult.	Attribute type	Description
genControlSource	0..1	GeneratorControlSource	see GeneratingUnit
governorSCD	0..1	PerCent	see GeneratingUnit
initialP	1..1	ActivePower	see GeneratingUnit
longPF	0..1	Simple_Float	see GeneratingUnit
maximumAllowableSpinningReserve	0..1	ActivePower	see GeneratingUnit
maxOperatingP	1..1	ActivePower	see GeneratingUnit
minOperatingP	1..1	ActivePower	see GeneratingUnit
nominalP	0..1	ActivePower	see GeneratingUnit
ratedGrossMaxP	0..1	ActivePower	see GeneratingUnit
ratedGrossMinP	0..1	ActivePower	see GeneratingUnit
ratedNetMaxP	0..1	ActivePower	see GeneratingUnit
shortPF	0..1	Simple_Float	see GeneratingUnit
startupCost	0..1	Money	see GeneratingUnit
variableCost	0..1	Money	see GeneratingUnit
totalEfficiency	0..1	PerCent	see GeneratingUnit
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-113 shows all association ends of Equipment with other classes.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-113 - Association ends of Core::Equipment with other classes

mult.	mult.	Class type name	Description
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.6.13 WindGeneratingUnit

A wind driven generating unit. May be used to represent a single turbine or an aggregation.

Inheritance path: ->GeneratingUnit->Equipment->PowerSystemResource->IdentifiedObject

Table 3-114 shows all attributes of WindGeneratingUnit.

Table 3-114 - Attributes of Production::WindGeneratingUnit

Attribute name	mult.	Attribute type	Description
windGenUnitType	1..1	WindGenUnitKind	The kind of wind generating unit
genControlSource	0..1	GeneratorControlSource	see GeneratingUnit
governorSCD	0..1	PerCent	see GeneratingUnit
initialP	1..1	ActivePower	see GeneratingUnit
longPF	0..1	Simple_Float	see GeneratingUnit
maximumAllowableSpinningReserve	0..1	ActivePower	see GeneratingUnit
maxOperatingP	1..1	ActivePower	see GeneratingUnit
minOperatingP	1..1	ActivePower	see GeneratingUnit
nominalP	0..1	ActivePower	see GeneratingUnit
ratedGrossMaxP	0..1	ActivePower	see GeneratingUnit
ratedGrossMinP	0..1	ActivePower	see GeneratingUnit
ratedNetMaxP	0..1	ActivePower	see GeneratingUnit
shortPF	0..1	Simple_Float	see GeneratingUnit
startupCost	0..1	Money	see GeneratingUnit
variableCost	0..1	Money	see GeneratingUnit
totalEfficiency	0..1	PerCent	see GeneratingUnit
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-115 shows all association ends of Equipment with other classes.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-115 - Association ends of Core::Equipment with other classes

mult.	mult.	Class type name	Description
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.6.14 Enumerations

3.6.14.1 FuelType

Type of fuel.

Literal	Description	Code
coal	Generic coal, not including lignite type.	
oil	Oil.	
gas	Natural gas.	
lignite	The fuel is lignite coal. Note that this is a special type of coal, so the other enum of coal is reserved for hard coal types or if the exact type of coal is not known.	
hardCoal	Hard coal	
oilShale	Oil Shale	

3.6.14.2 GeneratorControlSource

The source of controls for a generating unit.

Literal	Description	Code
unavailable	Not available.	
offAGC	Off of automatic generation control (AGC).	
onAGC	On automatic generation control (AGC).	
plantControl	Plant is controlling.	

3.6.14.3 HydroEnergyConversionKind

Specifies the capability of the hydro generating unit to convert energy as a generator or pump.

Literal	Description	Code
generator	Able to generate power, but not able to pump water for energy storage.	
pumpAndGenerator	Able to both generate power and pump water for energy storage.	

3.6.14.4 HydroPlantStorageKind

The type of hydro power plant.

Literal	Description	Code
runOfRiver	Run of river.	
pumpedStorage	Pumped storage.	

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

storage	Storage.	

3.6.14.5 WindGenUnitKind

Kind of wind generating unit.

Literal	Description	Code
offshore	The wind generating unit is located offshore.	
onshore	The wind generating unit is located onshore.	

3.7 Core

3.7.1 General

Contains the core PowerSystemResource and ConductingEquipment entities shared by all applications plus common collections of those entities. Not all applications require all the Core entities. This package does not depend on any other package except the Domain package, but most of the other packages have associations and generalizations that depend on it.

Figure 3-12 shows the diagram.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

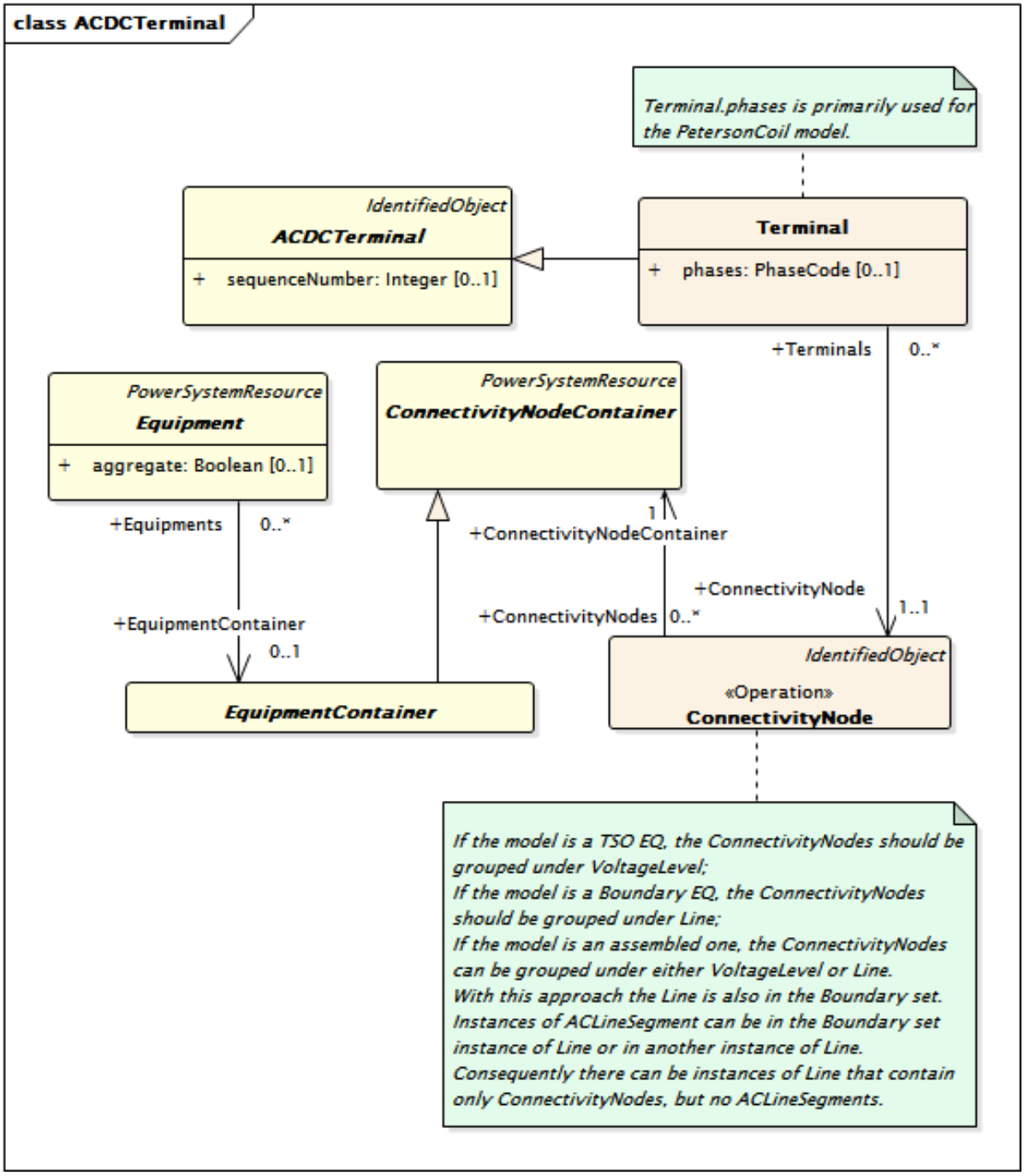


Figure 3-12 - diagram ACDCTerminal

Figure 3-13 shows the diagram.

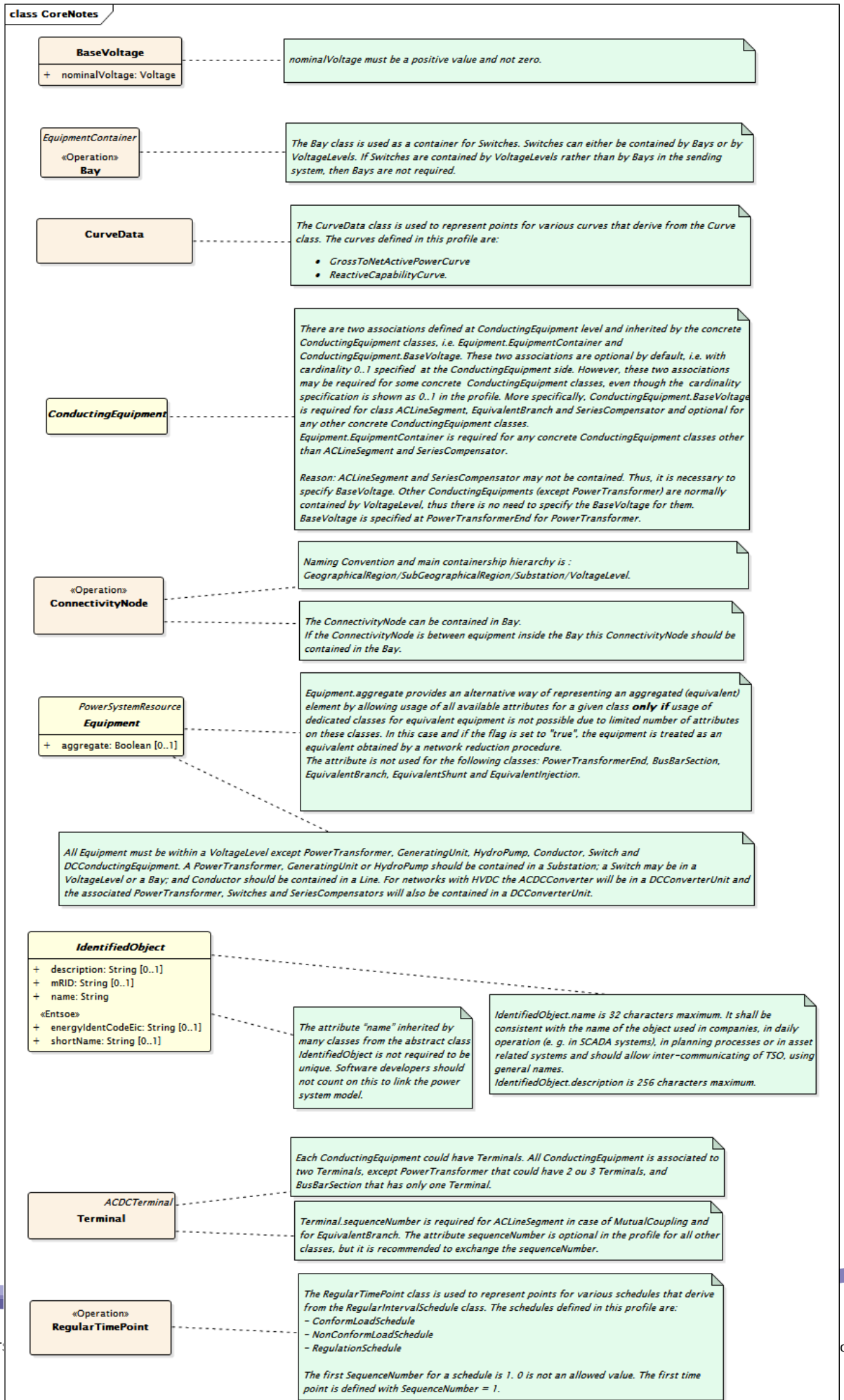

class CoreNotes
BaseVoltage
+ nominalVoltage: Voltage
nominalVoltage must be a positive value and not zero.
EquipmentContainer
«Operation»
Bay
The Bay class is used as a container for Switches. Switches can either be contained by Bays or by VoltageLevels. If Switches are contained by VoltageLevels rather than by Bays in the sending system, then Bays are not required.
CurveData
The CurveData class is used to represent points for various curves that derive from the Curve class. The curves defined in this profile are:
GrossToNetActivePowerCurve
ReactiveCapabilityCurve.
ConductingEquipment
There are two associations defined at ConductingEquipment level and inherited by the concrete ConductingEquipment classes, i.e. Equipment.EquipmentContainer and ConductingEquipment.BaseVoltage. These two associations are optional by default, i.e. with cardinality 0..1 specified at the ConductingEquipment side. However, these two associations may be required for some concrete ConductingEquipment classes, even though the cardinality specification is shown as 0..1 in the profile. More specifically, ConductingEquipment.BaseVoltage is required for class ACLineSegment, EquivalentBranch and SeriesCompensator and optional for any other concrete ConductingEquipment classes.
Equipment.EquipmentContainer is required for any concrete ConductingEquipment classes other than ACLineSegment and SeriesCompensator.
Reason: ACLineSegment and SeriesCompensator may not be contained. Thus, it is necessary to specify BaseVoltage. Other ConductingEquipments (except PowerTransformer) are normally contained by VoltageLevel, thus there is no need to specify the BaseVoltage for them.
BaseVoltage is specified at PowerTransformerEnd for PowerTransformer.
«Operation»
ConnectivityNode
Naming Convention and main containership hierarchy is :
GeographicalRegion/SubGeographicalRegion/Substation/VoltageLevel.
The ConnectivityNode can be contained in Bay.
If the ConnectivityNode is between equipment inside the Bay this ConnectivityNode should be contained in the Bay.
PowerSystemResource
Equipment
+ aggregate: Boolean [0..1]
Equipment.aggregate provides an alternative way of representing an aggregated (equivalent) element by allowing usage of all available attributes for a given class only if usage of dedicated classes for equivalent equipment is not possible due to limited number of attributes on these classes. In this case and if the flag is set to "true", the equipment is treated as an equivalent obtained by a network reduction procedure.
The attribute is not used for the following classes: PowerTransformerEnd, BusBarSection, EquivalentBranch, EquivalentShunt and EquivalentInjection.
All Equipment must be within a VoltageLevel except PowerTransformer, GeneratingUnit, HydroPump, Conductor, Switch and DCConductingEquipment. A PowerTransformer, GeneratingUnit or HydroPump should be contained in a Substation; a Switch may be in a VoltageLevel or a Bay; and Conductor should be contained in a Line. For networks with HVDC the ACDCConverter will be in a DCConverterUnit and the associated PowerTransformer, Switches and SeriesCompensators will also be contained in a DCConverterUnit.
IdentifiedObject
+ description: String [0..1]
+ mRID: String [0..1]
+ name: String
«Entsoe»
+ energyIdentCodeEic: String [0..1]
+ shortName: String [0..1]
The attribute "name" inherited by many classes from the abstract class IdentifiedObject is not required to be unique. Software developers should not count on this to link the power system model.
IdentifiedObject.name is 32 characters maximum. It shall be consistent with the name of the object used in companies, in daily operation (e. g. in SCADA systems), in planning processes or in asset related systems and should allow inter-communicating of TSO, using general names.
IdentifiedObject.description is 256 characters maximum.
ACDCTerminal
Terminal
Each ConductingEquipment could have Terminals. All ConductingEquipment is associated to two Terminals, except PowerTransformer that could have 2 ou 3 Terminals, and BusBarSection that has only one Terminal.
Terminal.sequenceNumber is required for ACLineSegment in case of MutualCoupling and for EquivalentBranch. The attribute sequenceNumber is optional in the profile for all other classes, but it is recommended to exchange the sequenceNumber.
«Operation»
RegularTimePoint
The RegularTimePoint class is used to represent points for various schedules that derive from the RegularIntervalSchedule class. The schedules defined in this profile are:
- ConformLoadSchedule
- NonConformLoadSchedule
- RegulationSchedule
The first SequenceNumber for a schedule is 1. 0 is not an allowed value. The first time point is defined with SequenceNumber = 1.

Figure 3-13 - diagram CoreNotes

3.7.2 ACDCTerminal(Abstract)

An electrical connection point (AC or DC) to a piece of conducting equipment. Terminals are connected at physical connection points called connectivity nodes.

Inheritance path: ->IdentifiedObject

Table 3-116 shows all attributes of ACDCTerminal.

Table 3-116 - Attributes of Core::ACDCTerminal

Attribute name	mult.	Attribute type	Description
sequenceNumber	0..1	Integer	The orientation of the terminal connections for a multiple terminal conducting equipment. The sequence numbering starts with 1 and additional terminals should follow in increasing order. The first terminal is the "starting point" for a two terminal branch.
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-117 shows all association ends of ACDCTerminal with other classes.

Table 3-117 - Association ends of Core::ACDCTerminal with other classes

mult.	mult.	Class type name	Description
BusNameMarker	[0..1]	BusNameMarker	The bus name marker used to name the bus (topological node). Association Based On : Topology:: BusNameMarker.BusNameMarker[0..1] ----- Core:: ACDCTerminal.Terminal[1..*]

3.7.3 BaseVoltage

Defines a system base voltage which is referenced.

OCL constraint: NominalVoltage must be positive

nominalVoltage must be a positive value and not zero.

Inheritance path: ->IdentifiedObject

Table 3-118 shows all attributes of BaseVoltage.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-118 - Attributes of Core::BaseVoltage

Attribute name	mult.	Attribute type	Description
nominalVoltage	1..1	Voltage	The power system resource's base voltage.
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

3.7.4 BasicIntervalSchedule(Abstract)

Schedule of values at points in time.

Inheritance path: ->IdentifiedObject

Table 3-119 shows all attributes of BasicIntervalSchedule.

Table 3-119 - Attributes of Core::BasicIntervalSchedule

Attribute name	mult.	Attribute type	Description
startTime	1..1	DateTime	The time for the first time point.
value1Unit	1..1	UnitSymbol	Value1 units of measure.
value2Unit	0..1	UnitSymbol	Value2 units of measure.
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

3.7.5 Bay

A collection of power system resources (within a given substation) including conducting equipment, protection relays, measurements, and telemetry. A bay typically represents a physical grouping related to modularization of equipment.

The Bay class is used as a container for Switches. Switches can either be contained by Bays or by VoltageLevels. If Switches are contained by VoltageLevels rather than by Bays in the sending system, then Bays are not required.

Inheritance path: ->EquipmentContainer->ConnectivityNodeContainer->PowerSystemResource->IdentifiedObject

Table 3-120 shows all attributes of Bay.

Table 3-120 - Attributes of Core::Bay

Attribute name	mult.	Attribute type	Description

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-121 shows all association ends of Bay with other classes.

Table 3-121 - Association ends of Core::Bay with other classes

mult.	mult.	Class type name	Description
VoltageLevel (Operation)	[1..1]	VoltageLevel	The voltage level containing this bay. Association Based On : Core:: VoltageLevel.VoltageLevel[0..1] ----- Core:: Bay.Bays[0..*]

3.7.6 ConductingEquipment(Abstract)

The parts of the AC power system that are designed to carry current or that are conductively connected through terminals.

OCL constraint: The ConductingEquipment must either have a BaseVoltage association or be within a VoltageLevel. If both are set the VoltageLevel's BaseVoltage must equal that of the ConductingEquipment

There are two associations defined at ConductingEquipment level and inherited by the concrete ConductingEquipment classes, i.e. Equipment.EquipmentContainer and ConductingEquipment.BaseVoltage. These two associations are optional by default, i.e. with cardinality 0..1 specified at the ConductingEquipment side. However, these two associations may be required for some concrete ConductingEquipment classes, even though the cardinality specification is shown as 0..1 in the profile. More specifically, ConductingEquipment.BaseVoltage is required for class ACLineSegment, EquivalentBranch and SeriesCompensator and optional for any other concrete ConductingEquipment classes. Equipment.EquipmentContainer is required for any concrete ConductingEquipment classes other than ACLineSegment and SeriesCompensator. Reason: ACLineSegment and SeriesCompensator may not be contained. Thus, it is necessary to specify BaseVoltage. Other ConductingEquipments (except PowerTransformer) are normally contained by VoltageLevel, thus there is no need to specify the BaseVoltage for them. BaseVoltage is specified at PowerTransformerEnd for PowerTransformer.

Inheritance path: ->Equipment->PowerSystemResource->IdentifiedObject

Table 3-122 shows all attributes of ConductingEquipment.

Table 3-122 - Attributes of Core::ConductingEquipment

Attribute name	mult.	Attribute type	Description
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-123 shows all association ends of ConductingEquipment with other classes.

Table 3-123 - Association ends of Core::ConductingEquipment with other classes

mult.	mult.	Class type name	Description
BaseVoltage	[0..1]	BaseVoltage	All conducting equipment with this base voltage. Use only when there is no voltage level container used and only one base voltage applies. For example, not used for transformers. Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.7.7 ConnectivityNode

Connectivity nodes are points where terminals of AC conducting equipment are connected together with zero impedance.

If the model is a TSO EQ, the ConnectivityNodes should be grouped under VoltageLevel; If the model is a Boundary EQ, the ConnectivityNodes should be grouped under Line; If the model is an assembled one, the ConnectivityNodes can be grouped under either VoltageLevel or Line. With this approach the Line is also in the Boundary set. Instances of ACLineSegment can be in the Boundary set instance of Line or in another instance of Line. Consequently there can be instances of Line that contain only ConnectivityNodes, but no ACLineSegments.

Naming Convention and main containership hierarchy is : GeographicalRegion/SubGeographicalRegion/Substation/VoltageLevel.
The ConnectivityNode can be contained in Bay. If the ConnectivityNode is between equipment inside the Bay this ConnectivityNode should be contained in the Bay.

Inheritance path: ->IdentifiedObject

Table 3-124 shows all attributes of ConnectivityNode.

Table 3-124 - Attributes of Core::ConnectivityNode

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-125 shows all association ends of ConnectivityNode with other classes.

Table 3-125 - Association ends of Core::ConnectivityNode with other classes

mult.	mult.	Class type name	Description

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

ConnectivityNodeContainer (Operation)	[1..1]	ConnectivityNodeContainer	Container of this connectivity node. Association Based On : Core:: ConnectivityNode.ConnectivityNodes[0..*] ----- Core:: ConnectivityNodeContainer.ConnectivityNodeContainer[1]

3.7.8 ConnectivityNodeContainer(Abstract)

A base class for all objects that may contain connectivity nodes or topological nodes.

Inheritance path: ->PowerSystemResource->IdentifiedObject

Table 3-126 shows all attributes of ConnectivityNodeContainer.

Table 3-126 - Attributes of Core::ConnectivityNodeContainer

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

3.7.9 Curve(Abstract)

A multi-purpose curve or functional relationship between an independent variable (X-axis) and dependent (Y-axis) variables.

Inheritance path: ->IdentifiedObject

Table 3-127 shows all attributes of Curve.

Table 3-127 - Attributes of Core::Curve

Attribute name	mult.	Attribute type	Description
curveStyle	1..1	CurveStyle	The style or shape of the curve.
xUnit	1..1	UnitSymbol	The X-axis units of measure.
y1Unit	1..1	UnitSymbol	The Y1-axis units of measure.
y2Unit	0..1	UnitSymbol	The Y2-axis units of measure.
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

3.7.10 CurveData

Multi-purpose data points for defining a curve. The use of this generic class is discouraged if a more specific class can be used to specify the x and y axis values along with their specific data types.

The CurveData class is used to represent points for various curves that derive from the Curve class. The curves defined in this profile are:

- GrossToNetActivePowerCurve
- ReactiveCapabilityCurve.

Table 3-128 shows all attributes of CurveData.

Table 3-128 - Attributes of Core::CurveData

Attribute name	mult.	Attribute type	Description
xvalue	1..1	Simple_Float	The data value of the X-axis variable, depending on the X-axis units.
y1value	1..1	Simple_Float	The data value of the first Y-axis variable, depending on the Y-axis units.
y2value	0..1	Simple_Float	The data value of the second Y-axis variable (if present), depending on the Y-axis units.

Table 3-129 shows all association ends of CurveData with other classes.

Table 3-129 - Association ends of Core::CurveData with other classes

mult.	mult.	Class type name	Description
Curve	[1..1]	Curve	The point data values that define this curve. Association Based On : Core:: CurveData.CurveDatas[0..*] ----- Core:: Curve.Curve[1]

3.7.11 Equipment(Abstract)

The parts of a power system that are physical devices, electronic or mechanical.

OCL constraint:OCL on containment. Please refer to the Note linked to the Equipment class.

Equipment.aggregate provides an alternative way of representing an aggregated (equivalent) element by allowing usage of all available attributes for a given class **only if** usage of dedicated classes for equivalent equipment is not possible due to limited number of attributes on these classes. In this case and if the flag is set to "true", the equipment is treated as an equivalent obtained by a network reduction procedure. The attribute is not used for the following classes: PowerTransformerEnd, BusBarSection, EquivalentBranch, EquivalentShunt and EquivalentInjection.

All Equipment must be within a VoltageLevel except PowerTransformer, GeneratingUnit, HydroPump, Conductor, Switch and DCConductingEquipment. A PowerTransformer, GeneratingUnit or HydroPump should be contained in a Substation; a Switch may be in a VoltageLevel or a Bay; and Conductor should be contained in a Line. For networks with HVDC the

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

ACDCConverter will be in a DCConverterUnit and the associated PowerTransformer, Switches and SeriesCompensators will also be contained in a DCConverterUnit.

Inheritance path: ->PowerSystemResource->IdentifiedObject

Table 3-130 shows all attributes of Equipment.

Table 3-130 - Attributes of Core::Equipment

Attribute name	mult.	Attribute type	Description
aggregate	0..1	Boolean	The single instance of equipment represents multiple pieces of equipment that have been modeled together as an aggregate. Examples would be power transformers or synchronous machines operating in parallel modeled as a single aggregate power transformer or aggregate synchronous machine. This is not to be used to indicate equipment that is part of a group of interdependent equipment produced by a network production program.
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-131 shows all association ends of Equipment with other classes.

Table 3-131 - Association ends of Core::Equipment with other classes

mult.	mult.	Class type name	Description
EquipmentContainer	[0..1]	EquipmentContainer	Container of this equipment. Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ---- Core:: Equipment.Equipments[0..*]

3.7.12 EquipmentContainer(Abstract)

A modeling construct to provide a root class for containing equipment.

Inheritance path: ->ConnectivityNodeContainer->PowerSystemResource->IdentifiedObject

Table 3-132 shows all attributes of EquipmentContainer.

Table 3-132 - Attributes of Core::EquipmentContainer

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

3.7.13 GeographicalRegion

A geographical region of a power system network model.

Inheritance path: ->IdentifiedObject

Table 3-133 shows all attributes of GeographicalRegion.

Table 3-133 - Attributes of Core::GeographicalRegion

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

3.7.14 IdentifiedObject(Abstract)

This is a root class to provide common identification for all classes needing identification and naming attributes.

OCL constraint:R.4.10.11. Description length restriction (optional)
OCL constraint:R.4.10.11. Energy Ident Code length restriction (optional)
OCL constraint:R.4.10.11. Name length restriction
OCL constraint:R.4.10.11. ShortName length restriction (optional)

IdentifiedObject.name is 32 characters maximum. It shall be consistent with the name of the object used in companies, in daily operation (e. g. in SCADA systems), in planning processes or in asset related systems and should allow inter-communicating of TSO, using general names. IdentifiedObject.description is 256 characters maximum.

The attribute “name” inherited by many classes from the abstract class IdentifiedObject is not required to be unique. Software developers should not count on this to link the power system model.

Table 3-134 shows all attributes of IdentifiedObject.

Table 3-134 - Attributes of Core::IdentifiedObject

Attribute name	mult.	Attribute type	Description
description	0..1	String	The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

energyIdentCodeEic (Entsoe)	0..1	String	The attribute is used for an exchange of the EIC code (Energy identification Code). The length of the string is 16 characters as defined by the EIC code. References: Local issuing offices for EIC: https://www.entsoe.eu/publications/edi-library/links-to-eic-websites/ EIC description: https://www.entsoe.eu/index.php?id=73&libCat=eic .
mRID	0..1	String	Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552 Edition 1, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name	1..1	String	The name is any free human readable and possibly non unique text naming the object.
shortName (Entsoe)	0..1	String	The attribute is used for an exchange of a human readable short name with length of the string 12 characters maximum.

3.7.15 PowerSystemResource(Abstract)

A power system resource can be an item of equipment such as a switch, an equipment container containing many individual items of equipment such as a substation, or an organisational entity such as sub-control area. Power system resources can have measurements associated.

Inheritance path: ->IdentifiedObject

Table 3-135 shows all attributes of PowerSystemResource.

Table 3-135 - Attributes of Core::PowerSystemResource

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

3.7.16 RegularIntervalSchedule(Abstract)

The schedule has time points where the time between them is constant.

Inheritance path: ->BasicIntervalSchedule->IdentifiedObject

Table 3-136 shows all attributes of RegularIntervalSchedule.

Table 3-136 - Attributes of Core::RegularIntervalSchedule

Attribute name	mult.	Attribute type	Description
timeStep	1..1	Seconds	The time between each pair of subsequent regular time points in

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			sequence order.
endTime	1..1	DateTime	The time for the last time point.
startTime	1..1	DateTime	see BasicIntervalSchedule
value1Unit	1..1	UnitSymbol	see BasicIntervalSchedule
value2Unit	0..1	UnitSymbol	see BasicIntervalSchedule
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

3.7.17 RegularTimePoint

Time point for a schedule where the time between the consecutive points is constant.

The RegularTimePoint class is used to represent points for various schedules that derive from the RegularIntervalSchedule class. The schedules defined in this profile are: - ConformLoadSchedule - NonConformLoadSchedule - RegulationSchedule The first SequenceNumber for a schedule is 1. 0 is not an allowed value. The first time point is defined with SequenceNumber = 1.

Table 3-137 shows all attributes of RegularTimePoint.

Table 3-137 - Attributes of Core::RegularTimePoint

Attribute name	mult.	Attribute type	Description
sequenceNumber (Operation)	1..1	Integer	The position of the regular time point in the sequence. Note that time points don't have to be sequential, i.e. time points may be omitted. The actual time for a RegularTimePoint is computed by multiplying the associated regular interval schedule's time step with the regular time point sequence number and adding the associated schedules start time.
value1 (Operation)	1..1	Simple_Float	The first value at the time. The meaning of the value is defined by the derived type of the associated schedule.
value2 (Operation)	0..1	Simple_Float	The second value at the time. The meaning of the value is defined by the derived type of the associated schedule.

Table 3-138 shows all association ends of RegularTimePoint with other classes.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-138 - Association ends of Core::RegularTimePoint with other classes

mult.	mult.	Class type name	Description
IntervalSchedule (Operation)	[1..1]	RegularIntervalSchedule	Regular interval schedule containing this time point. Association Based On : Core:: RegularIntervalSchedule.IntervalSchedule[1] ----- Core:: RegularTimePoint.TimePoints[1..*]

3.7.18 (Operation) ReportingGroup

A reporting group is used for various ad-hoc groupings used for reporting.

Inheritance path: ->IdentifiedObject

Table 3-139 shows all attributes of ReportingGroup.

Table 3-139 - Attributes of Core::ReportingGroup

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

3.7.19 (Operation) SubGeographicalRegion

A subset of a geographical region of a power system network model.

Inheritance path: ->IdentifiedObject

Table 3-140 shows all attributes of SubGeographicalRegion.

Table 3-140 - Attributes of Core::SubGeographicalRegion

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-141 shows all association ends of SubGeographicalRegion with other classes.

Table 3-141 - Association ends of Core::SubGeographicalRegion with other classes

mult.	mult.	Class type name	Description
Region	[1..1]	GeographicalRegion	The geographical region to which this sub-geographical region is within. Association Based On : Core:: GeographicalRegion.Region[0..1] -----

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			Core:: SubGeographicalRegion.Regions[0..*]

3.7.20 Substation

A collection of equipment for purposes other than generation or utilization, through which electric energy in bulk is passed for the purposes of switching or modifying its characteristics.

Inheritance path: ->EquipmentContainer->ConnectivityNodeContainer->PowerSystemResource->IdentifiedObject

Table 3-142 shows all attributes of Substation.

Table 3-142 - Attributes of Core::Substation

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-143 shows all association ends of Substation with other classes.

Table 3-143 - Association ends of Core::Substation with other classes

mult.	mult.	Class type name	Description
Region	[1..1]	SubGeographicalRegion	The SubGeographicalRegion containing the substation. Association Based On : Core:: SubGeographicalRegion.Region[0..1] ----- Core:: Substation.Substations[0..*]

3.7.21 Terminal

An AC electrical connection point to a piece of conducting equipment. Terminals are connected at physical connection points called connectivity nodes.

OCL constraint: Sequence Number is required for EquivalentBranch and ACLineSegments with MutualCoupling

Terminal.phases is primarily used for the PetersonCoil model.

Terminal.sequenceNumber is required for ACLineSegment in case of MutualCoupling and for EquivalentBranch. The attribute sequenceNumber is optional in the profile for all other classes, but it is recommended to exchange the sequenceNumber.
Each ConductingEquipment could have Terminals. All ConductingEquipment is associated to two Terminals, except PowerTransformer that could have 2 ou 3 Terminals, and BusBarSection that has only one Terminal.

Inheritance path: ->ACDCTerminal->IdentifiedObject

Table 3-144 shows all attributes of Terminal.

Table 3-144 - Attributes of Core::Terminal

Attribute name	mult.	Attribute type	Description
phases	0..1	PhaseCode	Represents the normal

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			network phasing condition. If the attribute is missing three phases (ABC or ABCN) shall be assumed.
sequenceNumber	0..1	Integer	see ACDCTerminal
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-145 shows all association ends of Terminal with other classes.

Table 3-145 - Association ends of Core::Terminal with other classes

mult.	mult.	Class type name	Description
ConnectivityNode	[1..1]	ConnectivityNode	Terminals interconnected with zero impedance at a this connectivity node. Association Based On : Core:: Terminal.Terminals[0..*] ----- Core:: ConnectivityNode.ConnectivityNode[0..1]
ConductingEquipment	[1..1]	ConductingEquipment	The conducting equipment of the terminal. Conducting equipment have terminals that may be connected to other conducting equipment terminals via connectivity nodes or topological nodes. Association Based On : Core:: Terminal.Terminals[0..*] ----- Core:: ConductingEquipment.ConductingEquipment[1]
BusNameMarker	0..1	BusNameMarker	Inherited Association Based On : Topology:: BusNameMarker.BusNameMarker[0..1] ----- Core:: ACDCTerminal.Terminal[1..*]

3.7.22 VoltageLevel

A collection of equipment at one common system voltage forming a switchgear. The equipment typically consist of breakers, busbars, instrumentation, control, regulation and protection devices as well as assemblies of all these.

DCConductingEquipment and DCConverterUnit is not allowed in VoltageLevels that is intended for AC equipment only. DCConductingEquipment are allowed in DCEquipmentContainers only. The ACDCConverter is ConductingEquipment. Other AC equipment in a converter unit are the power transformer and converter reactor. All three shall be located in a DCConverterUnit and not a VoltageLevel.

Inheritance path: ->EquipmentContainer->ConnectivityNodeContainer->PowerSystemResource->IdentifiedObject

Table 3-146 shows all attributes of VoltageLevel.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-146 - Attributes of Core::VoltageLevel

Attribute name	mult.	Attribute type	Description
highVoltageLimit	0..1	Voltage	The bus bar's high voltage limit
lowVoltageLimit	0..1	Voltage	The bus bar's low voltage limit
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-147 shows all association ends of VoltageLevel with other classes.

Table 3-147 - Association ends of Core::VoltageLevel with other classes

mult.	mult.	Class type name	Description
Substation	[1..1]	Substation	The substation of the voltage level. Association Based On : Core:: Substation.Substation[1] ----- Core:: VoltageLevel.VoltageLevels[0..*]
BaseVoltage	[1..1]	BaseVoltage	The base voltage used for all equipment within the voltage level. Association Based On : Core:: VoltageLevel.VoltageLevel[0..*] ----- Core:: BaseVoltage.BaseVoltage[1]

3.7.23 Enumerations

3.7.23.1 CurveStyle

Style or shape of curve.

Literal	Description	Code
constantYValue	The Y-axis values are assumed constant until the next curve point and prior to the first curve point.	
straightLineYValues	The Y-axis values are assumed to be a straight line between values. Also known as linear interpolation.	

3.7.23.2 PhaseCode

Enumeration of phase identifiers. Allows designation of phases for both transmission and distribution equipment, circuits and loads. Residential and small commercial loads are often served from single-phase, or split-phase, secondary circuits. For example of s12N, phases 1 and 2 refer to hot wires that are 180 degrees out of phase, while N refers to the neutral wire. Through single-phase transformer connections, these secondary circuits may be served from one or two of the primary phases A, B, and C. For three-phase loads, use the A, B, C phase codes instead of s12N.

Literal	Description	Code
ABCN	Phases A, B, C, and N.	

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

ABC	Phases A, B, and C.	
ABN	Phases A, B, and neutral.	
ACN	Phases A, C and neutral.	
BCN	Phases B, C, and neutral.	
AB	Phases A and B.	
AC	Phases A and C.	
BC	Phases B and C.	
AN	Phases A and neutral.	
BN	Phases B and neutral.	
CN	Phases C and neutral.	
A	Phase A.	
B	Phase B.	
C	Phase C.	
N	Neutral phase.	
s1N	Secondary phase 1 and neutral.	
s2N	Secondary phase 2 and neutral.	
s12N	Secondary phases 1, 2, and neutral.	
s1	Secondary phase 1.	
s2	Secondary phase 2.	
s12	Secondary phase 1 and 2.	

3.8 OperationalLimits

3.8.1 General

The OperationalLimits package models a specification of limits associated with equipment and other operational entities.

Figure 3-14 shows the diagram.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

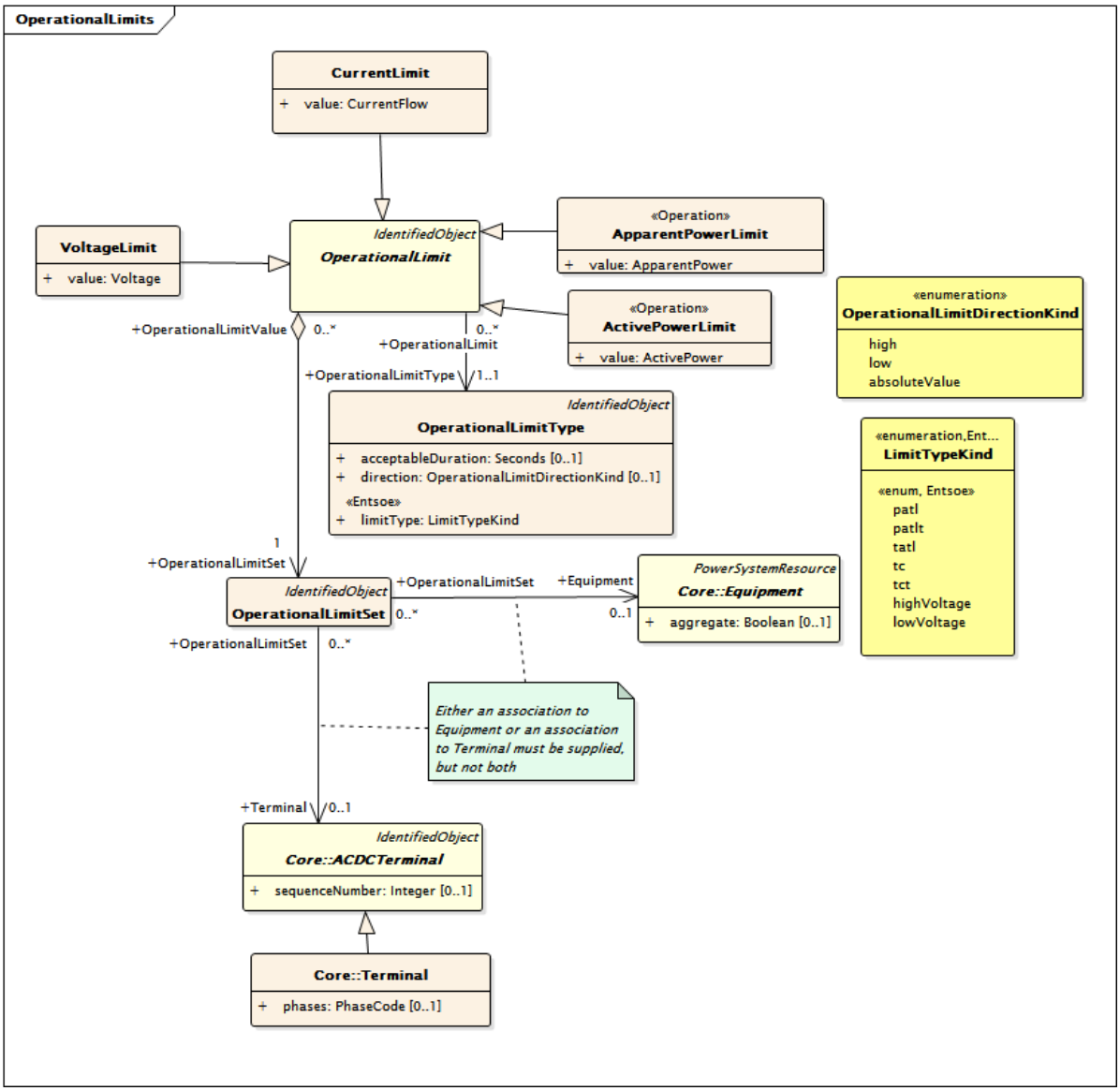


Figure 3-14 - diagram OperationalLimits

Figure 3-15 shows the diagram.

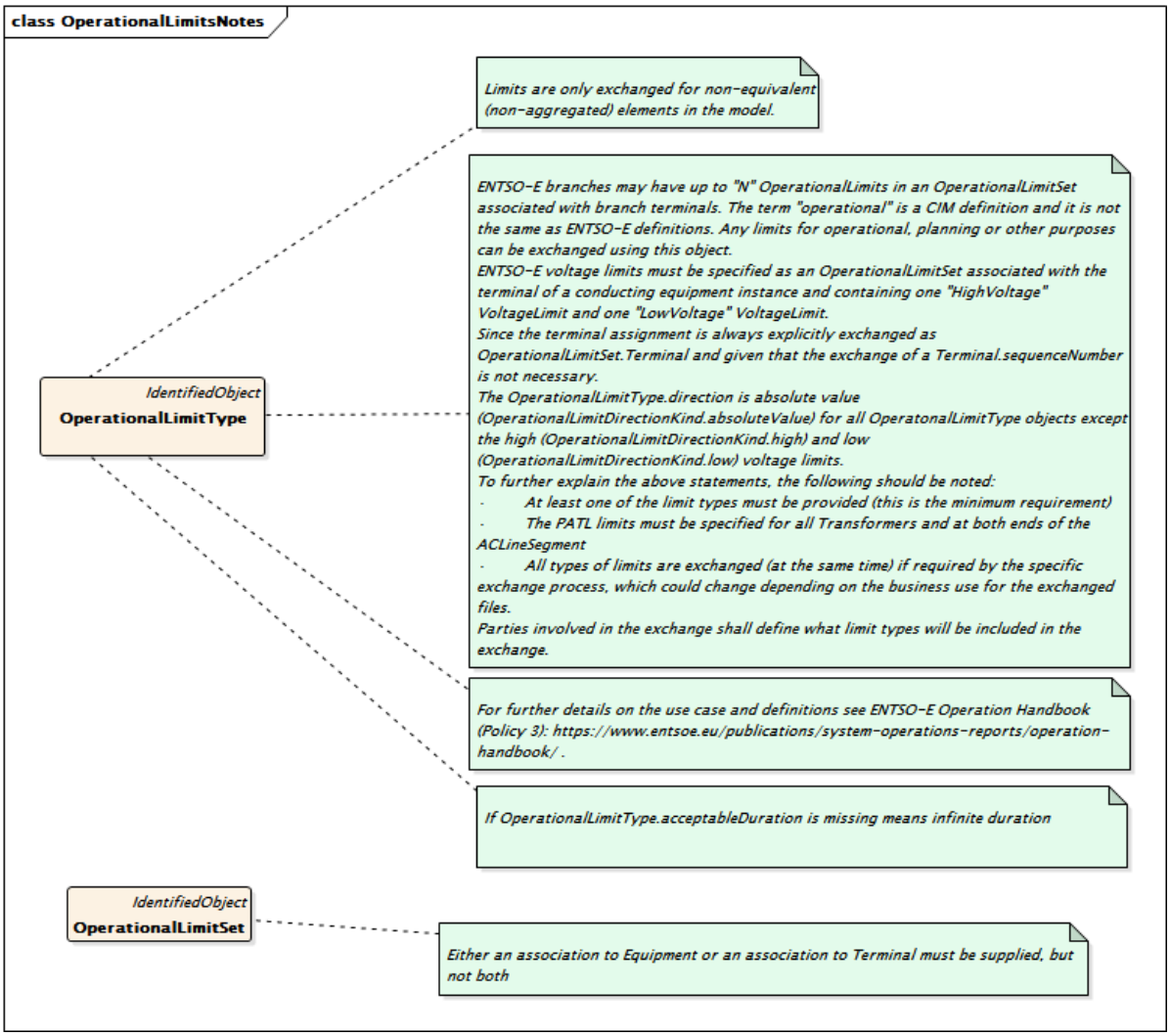


Figure 3-15 - diagram OperationalLimitsNotes

3.8.2 ActivePowerLimit

Limit on active power flow.

Inheritance path: ->OperationalLimit->IdentifiedObject

Table 3-148 shows all attributes of ActivePowerLimit.

Table 3-148 - Attributes of OperationalLimits::ActivePowerLimit

Attribute name	mult.	Attribute type	Description
value (Operation)	1..1	ActivePower	Value of active power limit.
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject

shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-149 shows all association ends of OperationalLimit with other classes.

Table 3-149 - Association ends of OperationalLimits::OperationalLimit with other classes

mult.	mult.	Class type name	Description
OperationalLimitSet	1..1	OperationalLimitSet	Inherited Association Based On : OperationalLimits:: OperationalLimit.OperationalLimitValue[0..*] ----- OperationalLimits:: OperationalLimitSet.OperationalLimitSet[1]
OperationalLimitType	1..1	OperationalLimitType	Inherited Association Based On : OperationalLimits:: OperationalLimit.OperationalLimit[0..*] ----- OperationalLimits:: OperationalLimitType.OperationalLimitType[0..1]

3.8.3 ApparentPowerLimit

Apparent power limit.

Inheritance path: ->OperationalLimit->IdentifiedObject

Table 3-150 shows all attributes of ApparentPowerLimit.

Table 3-150 - Attributes of OperationalLimits::ApparentPowerLimit

Attribute name	mult.	Attribute type	Description
value (Operation)	1..1	ApparentPower	The apparent power limit.
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-151 shows all association ends of OperationalLimit with other classes.

Table 3-151 - Association ends of OperationalLimits::OperationalLimit with other classes

mult.	mult.	Class type name	Description
OperationalLimitSet	1..1	OperationalLimitSet	Inherited Association Based On : OperationalLimits:: OperationalLimit.OperationalLimitValue[0..*] ----- OperationalLimits:: OperationalLimitSet.OperationalLimitSet[1]
OperationalLimitType	1..1	OperationalLimitType	Inherited Association Based On : OperationalLimits:: OperationalLimit.OperationalLimit[0..*] -----

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			OperationalLimits:: OperationalLimitType.OperationalLimitType[0..1]

3.8.4 CurrentLimit

Operational limit on current.

Inheritance path: ->OperationalLimit->IdentifiedObject

Table 3-152 shows all attributes of CurrentLimit.

Table 3-152 - Attributes of OperationalLimits::CurrentLimit

Attribute name	mult.	Attribute type	Description
value	1..1	CurrentFlow	Limit on current flow.
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-153 shows all association ends of OperationalLimit with other classes.

Table 3-153 - Association ends of OperationalLimits::OperationalLimit with other classes

mult.	mult.	Class type name	Description
OperationalLimitSet	1..1	OperationalLimitSet	Inherited Association Based On : OperationalLimits:: OperationalLimit.OperationalLimitValue[0..*] ----- OperationalLimits:: OperationalLimitSet.OperationalLimitSet[1]
OperationalLimitType	1..1	OperationalLimitType	Inherited Association Based On : OperationalLimits:: OperationalLimit.OperationalLimit[0..*] ----- OperationalLimits:: OperationalLimitType.OperationalLimitType[0..1]

3.8.5 OperationalLimit(Abstract)

A value associated with a specific kind of limit.
The sub class value attribute shall be positive.
The sub class value attribute is inversely proportional to OperationalLimitType.acceptableDuration (acceptableDuration for short). A pair of value_x and acceptableDuration_x are related to each other as follows:
if value_1 > value_2 > value_3 >... then acceptableDuration_1 < acceptableDuration_2 < acceptableDuration_3 < ...
A value_x with direction="high" shall be greater than a value_y with direction="low".

Inheritance path: ->IdentifiedObject

Table 3-154 shows all attributes of OperationalLimit.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-154 - Attributes of OperationalLimits::OperationalLimit

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-155 shows all association ends of OperationalLimit with other classes.

Table 3-155 - Association ends of OperationalLimits::OperationalLimit with other classes

mult.	mult.	Class type name	Description
OperationalLimitSet	[1..1]	OperationalLimitSet	Values of equipment limits. Association Based On : OperationalLimits:: OperationalLimit.OperationalLimitValue[0..*] ---- - OperationalLimits:: OperationalLimitSet.OperationalLimitSet[1]
OperationalLimitType	[1..1]	OperationalLimitType	The limit type associated with this limit. Association Based On : OperationalLimits:: OperationalLimit.OperationalLimit[0..*] ----- OperationalLimits:: OperationalLimitType.OperationalLimitType[0..1]

3.8.6 OperationalLimitSet

A set of limits associated with equipment. Sets of limits might apply to a specific temperature, or season for example. A set of limits may contain different severities of limit levels that would apply to the same equipment. The set may contain limits of different types such as apparent power and current limits or high and low voltage limits that are logically applied together as a set.

Either an association to Equipment or an association to Terminal must be supplied, but not both

Inheritance path: ->IdentifiedObject

Table 3-156 shows all attributes of OperationalLimitSet.

Table 3-156 - Attributes of OperationalLimits::OperationalLimitSet

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-157 shows all association ends of OperationalLimitSet with other classes.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-157 - Association ends of OperationalLimits::OperationalLimitSet with other classes

mult.	mult.	Class type name	Description
Equipment	[0..1]	Equipment	The equipment to which the limit set applies. Association Based On : OperationalLimits:: OperationalLimitSet.OperationalLimitSet[0..*] ----- Core:: Equipment.Equipment[0..1]
Terminal	[0..1]	ACDCTerminal	Association Based On : OperationalLimits:: OperationalLimitSet.OperationalLimitSet[0..*] ----- Core:: ACDCTerminal.Terminal[0..1]

3.8.7 OperationalLimitType

The operational meaning of a category of limits.

If OperationalLimitType.acceptableDuration is missing means infinite duration

Limits are only exchanged for non-equivalent (non-aggregated) elements in the model.
For further details on the use case and definitions see ENTSO-E Operation Handbook (Policy 3): https://www.entsoe.eu/publications/system-operations-reports/operation-handbook/ .
ENTSO-E branches may have up to "N" OperationalLimits in an OperationalLimitSet associated with branch terminals. The term "operational" is a CIM definition and it is not the same as ENTSO-E definitions. Any limits for operational, planning or other purposes can be exchanged using this object. ENTSO-E voltage limits must be specified as an OperationalLimitSet associated with the terminal of a conducting equipment instance and containing one "HighVoltage" VoltageLimit and one "LowVoltage" VoltageLimit. Since the terminal assignment is always explicitly exchanged as OperationalLimitSet.Terminal and given that the exchange of a Terminal.sequenceNumber is not necessary. The OperationalLimitType.direction is absolute value (OperationalLimitDirectionKind.absoluteValue) for all OperatonalLimitType objects except the high (OperationalLimitDirectionKind.high) and low (OperationalLimitDirectionKind.low) voltage limits. To further explain the above statements, the following should be noted: · At least one of the limit types must be provided (this is the minimum requirement) · The PATL limits must be specified for all Transformers and at both ends of the ACLineSegment · All types of limits are exchanged (at the same time) if required by the specific exchange process, which could change depending on the business use for the exchanged files. Parties involved in the exchange shall define what limit types will be included in the exchange.

Inheritance path: ->IdentifiedObject

Table 3-158 shows all attributes of OperationalLimitType.

Table 3-158 - Attributes of OperationalLimits::OperationalLimitType

Attribute name	mult.	Attribute type	Description
acceptableDuration	0..1	Seconds	The nominal acceptable duration of the limit. Limits are commonly expressed in terms of the a time limit for which the limit is normally acceptable. The actual acceptable duration of a specific limit may depend on other local factors such as temperature or wind speed.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

limitType (Entsoe)	1..1	LimitTypeKind	Types of limits defined in the ENTSO-E Operational Handbook Policy 3.
direction	0..1	OperationalLimitDirectionKind	The direction of the limit.
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

3.8.8 VoltageLimit

Operational limit applied to voltage.

Inheritance path: ->OperationalLimit->IdentifiedObject

Table 3-159 shows all attributes of VoltageLimit.

Table 3-159 - Attributes of OperationalLimits::VoltageLimit

Attribute name	mult.	Attribute type	Description
value	1..1	Voltage	Limit on voltage. High or low limit nature of the limit depends upon the properties of the operational limit type.
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-160 shows all association ends of OperationalLimit with other classes.

Table 3-160 - Association ends of OperationalLimits::OperationalLimit with other classes

mult.	mult.	Class type name	Description
OperationalLimitSet	1..1	OperationalLimitSet	Inherited Association Based On : OperationalLimits:: OperationalLimit.OperationalLimitValue[0..*] --- OperationalLimits:: OperationalLimitSet.OperationalLimitSet[1]
OperationalLimitType	1..1	OperationalLimitType	Inherited Association Based On : OperationalLimits:: OperationalLimit.OperationalLimit[0..*] --- OperationalLimits:: OperationalLimitType.OperationalLimitType[0..1]

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

3.8.9 Enumerations

3.8.9.1 LimitTypeKind

The enumeration defines the kinds of the limit types.

Literal	Description	Code
patl	The Permanent Admissible Transmission Loading (PATL) is the loading in Amps, MVA or MW that can be accepted by a network branch for an unlimited duration without any risk for the material. The duration attribute is not used and shall be excluded for the PATL limit type. Hence only one limit value exists for the PATL type.	
patlt	Permanent Admissible Transmission Loading Threshold (PATLT) is a value in engineering units defined for PATL and calculated using percentage less than 100 of the PATL type intended to alert operators of an arising condition. The percentage should be given in the name of the OperationalLimitSet. The aceptableDuration is another way to express the severity of the limit.	
tatl	Temporarily Admissible Transmission Loading (TATL) which is the loading in Amps, MVA or MW that can be accepted by a branch for a certain limited duration. The TATL can be defined in different ways: as a fixed percentage of the PATL for a given time (for example, 115% of the PATL that can be accepted during 15 minutes), pairs of TATL type and Duration calculated for each line taking into account its particular configuration and conditions of functioning (for example, it can define a TATL acceptable during 20 minutes and another one acceptable during 10 minutes). Such a definition of TATL can depend on the initial operating conditions of the network element (sag situation of a line). The duration attribute can be used define several TATL limit types. Hence multiple TATL limit values may exist having different durations.	
tc	Tripping Current (TC) is the ultimate intensity without any delay. It is defined as the threshold the line will trip without any possible remedial actions. The tripping of the network element is ordered by protections against short circuits or by overload protections, but in any case, the activation delay of these protections is not compatible with the reaction delay of an operator (less than one minute). The duration is always zero and the duration attribute may be left out. Hence only one limit value exists for the TC type.	
tct	Tripping Current Threshold (TCT) is a value in engineering units defined for TC and calculated using percentage less than 100 of the TC type intended to alert operators of an arising condition. The percentage should be given in the name of the OperationalLimitSet. The aceptableDuration is another way to express the severity of the limit.	
highVoltage	Referring to the rating of the equipments, a voltage too high can lead to accelerated ageing or the destruction of the equipment. This limit type may or may not have duration.	
lowVoltage	A too low voltage can disturb the normal operation of some protections and transformer equipped with on-load tap changers, electronic power devices or can affect the behaviour of the auxiliaries of generation units. This limit type may or may not have duration.	

3.8.9.2 OperationalLimitDirectionKind

The direction attribute describes the side of a limit that is a violation.

Literal	Description	Code
high	High means that a monitored value above the limit value is a	

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

	violation. If applied to a terminal flow, the positive direction is into the terminal.	
low	Low means a monitored value below the limit is a violation. If applied to a terminal flow, the positive direction is into the terminal.	
absoluteValue	An absoluteValue limit means that a monitored absolute value above the limit value is a violation.	

3.9 Wires

3.9.1 General

An extension to the Core and Topology package that models information on the electrical characteristics of Transmission and Distribution networks. This package is used by network applications such as State Estimation, Load Flow and Optimal Power Flow.

Figure 3-16 shows the diagram.

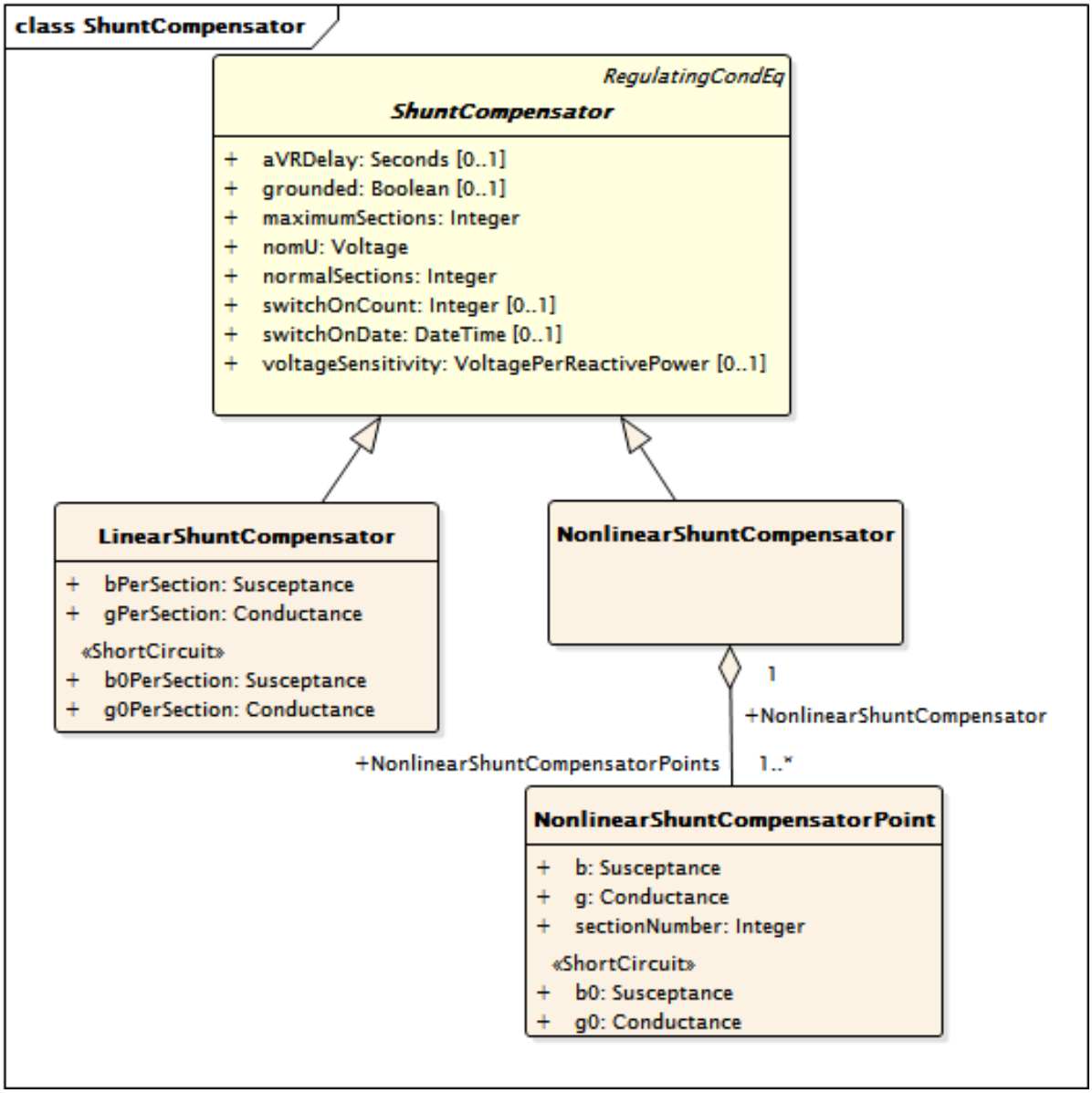


Figure 3-16 - diagram ShuntCompensator

Figure 3-17 shows the diagram.

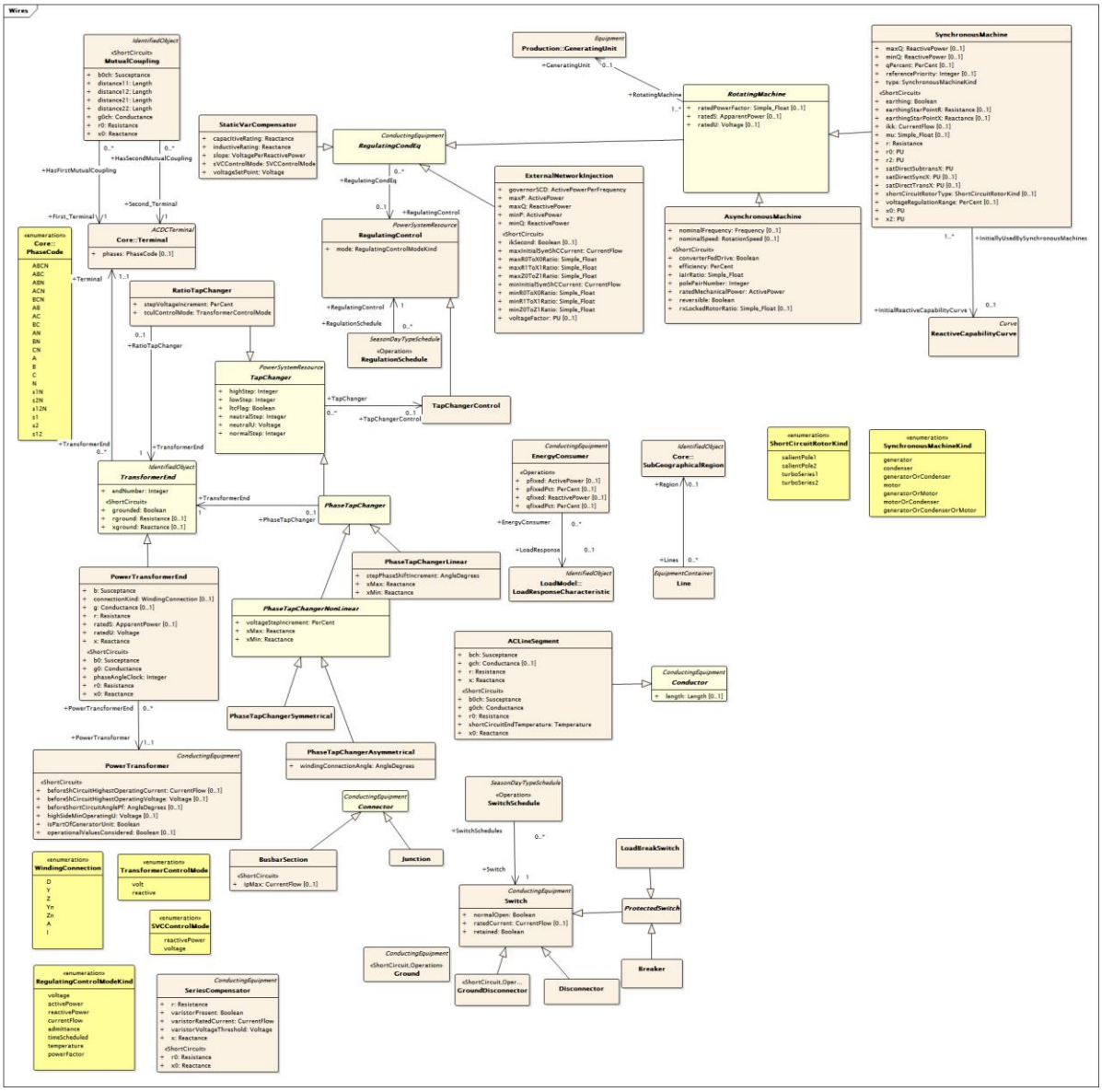


Figure 3-17 - diagram Wires

Figure 3-18 shows the diagram.

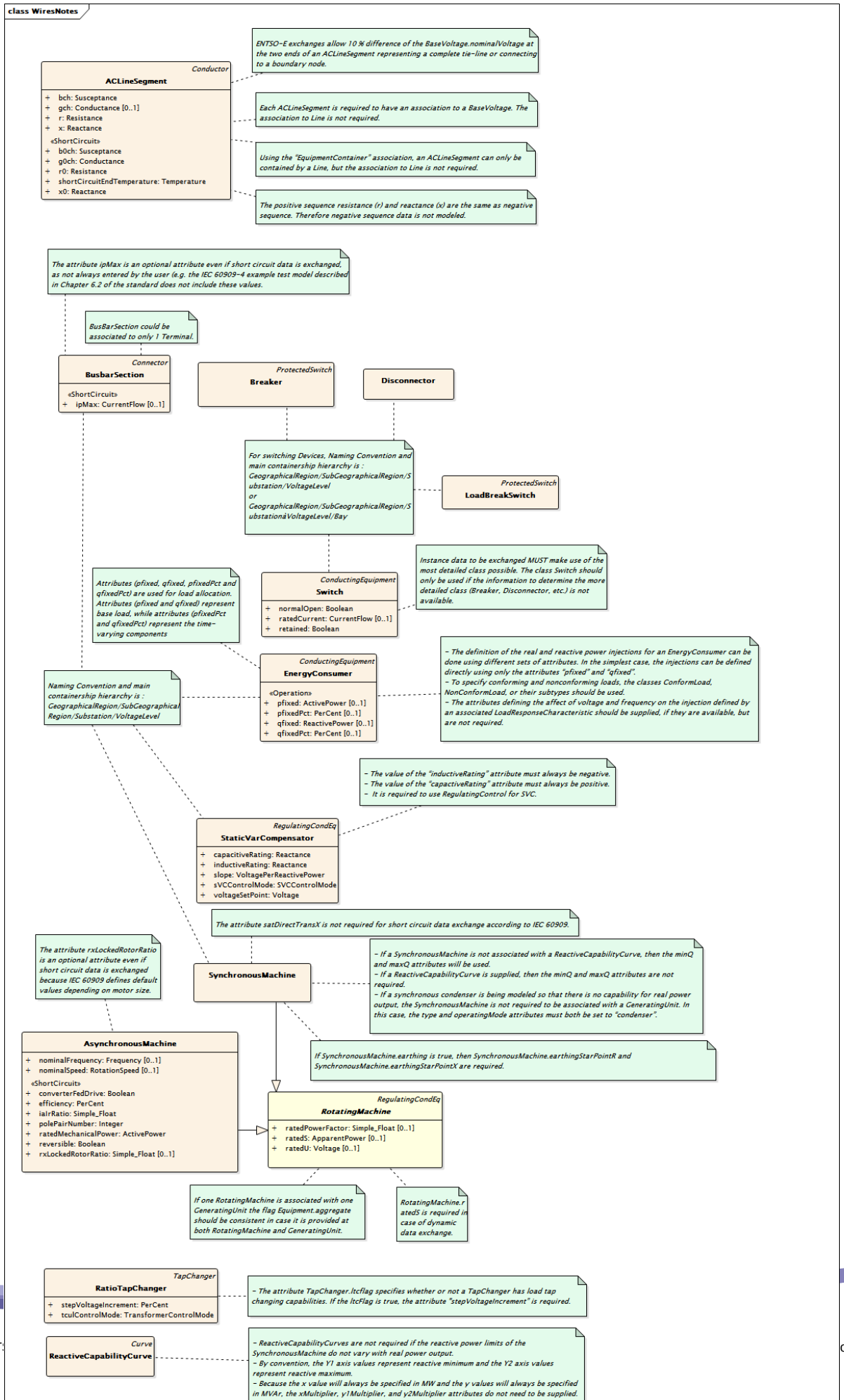

class WiresNotes
ENTSO-E exchanges allow 10 % difference of the BaseVoltage.nominalVoltage at the two ends of an ACLineSegment representing a complete tie-line or connecting to a boundary node.
Conductor
ACLineSegment
+ bch: Susceptance
+ gch: Conductance [0..1]
+ r: Resistance
+ x: Reactance
«ShortCircuit»
+ b0ch: Susceptance
+ g0ch: Conductance
+ r0: Resistance
+ shortCircuitEndTemperature: Temperature
+ x0: Reactance
Each ACLineSegment is required to have an association to a BaseVoltage. The association to Line is not required.
Using the "EquipmentContainer" association, an ACLineSegment can only be contained by a Line, but the association to Line is not required.
The positive sequence resistance (r) and reactance (x) are the same as negative sequence. Therefore negative sequence data is not modeled.
The attribute ipMax is an optional attribute even if short circuit data is exchanged, as not always entered by the user (e.g. the IEC 60909-4 example test model described in Chapter 6.2 of the standard does not include these values.
BusBarSection could be associated to only 1 Terminal.
Connector
BusbarSection
«ShortCircuit»
+ ipMax: CurrentFlow [0..1]
ProtectedSwitch
Breaker
Disconnector
For switching Devices, Naming Convention and main containership hierarchy is : GeographicalRegion/SubGeographicalRegion/Substation/VoltageLevel or GeographicalRegion/SubGeographicalRegion/SubstationàVoltageLevel/Bay
ProtectedSwitch
LoadBreakSwitch
Attributes (pfixed, qfixed, pfixedPct and qfixedPct) are used for load allocation. Attributes (pfixed and qfixed) represent base load, while attributes (pfixedPct and qfixedPct) represent the time-varying components
ConductingEquipment
Switch
+ normalOpen: Boolean
+ ratedCurrent: CurrentFlow [0..1]
+ retained: Boolean
Instance data to be exchanged MUST make use of the most detailed class possible. The class Switch should only be used if the information to determine the more detailed class (Breaker, Disconnector, etc.) is not available.
Naming Convention and main containership hierarchy is : GeographicalRegion/SubGeographicalRegion/Substation/VoltageLevel
ConductingEquipment
EnergyConsumer
«Operation»
+ pfixed: ActivePower [0..1]
+ pfixedPct: PerCent [0..1]
+ qfixed: ReactivePower [0..1]
+ qfixedPct: PerCent [0..1]
- The definition of the real and reactive power injections for an EnergyConsumer can be done using different sets of attributes. In the simplest case, the injections can be defined directly using only the attributes "pfixed" and "qfixed".
- To specify conforming and nonconforming loads, the classes ConformLoad, NonConformLoad, or their subtypes should be used.
- The attributes defining the affect of voltage and frequency on the injection defined by an associated LoadResponseCharacteristic should be supplied, if they are available, but are not required.
- The value of the "inductiveRating" attribute must always be negative.
- The value of the "capacitiveRating" attribute must always be positive.
- It is required to use RegulatingControl for SVC.
RegulatingCondEq
StaticVarCompensator
+ capacitiveRating: Reactance
+ inductiveRating: Reactance
+ slope: VoltagePerReactivePower
+ sVCControlMode: SVCControlMode
+ voltageSetPoint: Voltage
The attribute satDirectTransX is not required for short circuit data exchange according to IEC 60909.
The attribute rxLockedRotorRatio is an optional attribute even if short circuit data is exchanged because IEC 60909 defines default values depending on motor size.
SynchronousMachine
- If a SynchronousMachine is not associated with a ReactiveCapabilityCurve, then the minQ and maxQ attributes will be used.
- If a ReactiveCapabilityCurve is supplied, then the minQ and maxQ attributes are not required.
- If a synchronous condenser is being modeled so that there is no capability for real power output, the SynchronousMachine is not required to be associated with a GeneratingUnit. In this case, the type and operatingMode attributes must both be set to "condenser".
AsynchronousMachine
+ nominalFrequency: Frequency [0..1]
+ nominalSpeed: RotationSpeed [0..1]
«ShortCircuit»
+ converterFedDrive: Boolean
+ efficiency: PerCent
+ iaIrRatio: Simple_Float
+ polePairNumber: Integer
+ ratedMechanicalPower: ActivePower
+ reversible: Boolean
+ rxLockedRotorRatio: Simple_Float [0..1]
If SynchronousMachine.earthing is true, then SynchronousMachine.earthingStarPointR and SynchronousMachine.earthingStarPointX are required.
RegulatingCondEq
RotatingMachine
+ ratedPowerFactor: Simple_Float [0..1]
+ ratedS: ApparentPower [0..1]
+ ratedU: Voltage [0..1]
If one RotatingMachine is associated with one GeneratingUnit the flag Equipment.aggregate should be consistent in case it is provided at both RotatingMachine and GeneratingUnit.
RotatingMachine.ratedS is required in case of dynamic data exchange.
TapChanger
RatioTapChanger
+ stepVoltageIncrement: PerCent
+ tculControlMode: TransformerControlMode
- The attribute TapChanger.ltcflag specifies whether or not a TapChanger has load tap changing capabilities. If the ltcFlag is true, the attribute "stepVoltageIncrement" is required.
Curve
ReactiveCapabilityCurve
- ReactiveCapabilityCurves are not required if the reactive power limits of the SynchronousMachine do not vary with real power output.
- By convention, the Y1 axis values represent reactive minimum and the Y2 axis values represent reactive maximum.
- Because the x value will always be specified in MW and the y values will always be specified in MVAr, the xMultiplier, y1Multiplier, and y2Multiplier attributes do not need to be supplied.

Figure 3-18 - diagram WiresNotes

Figure 3-19 shows the diagram.

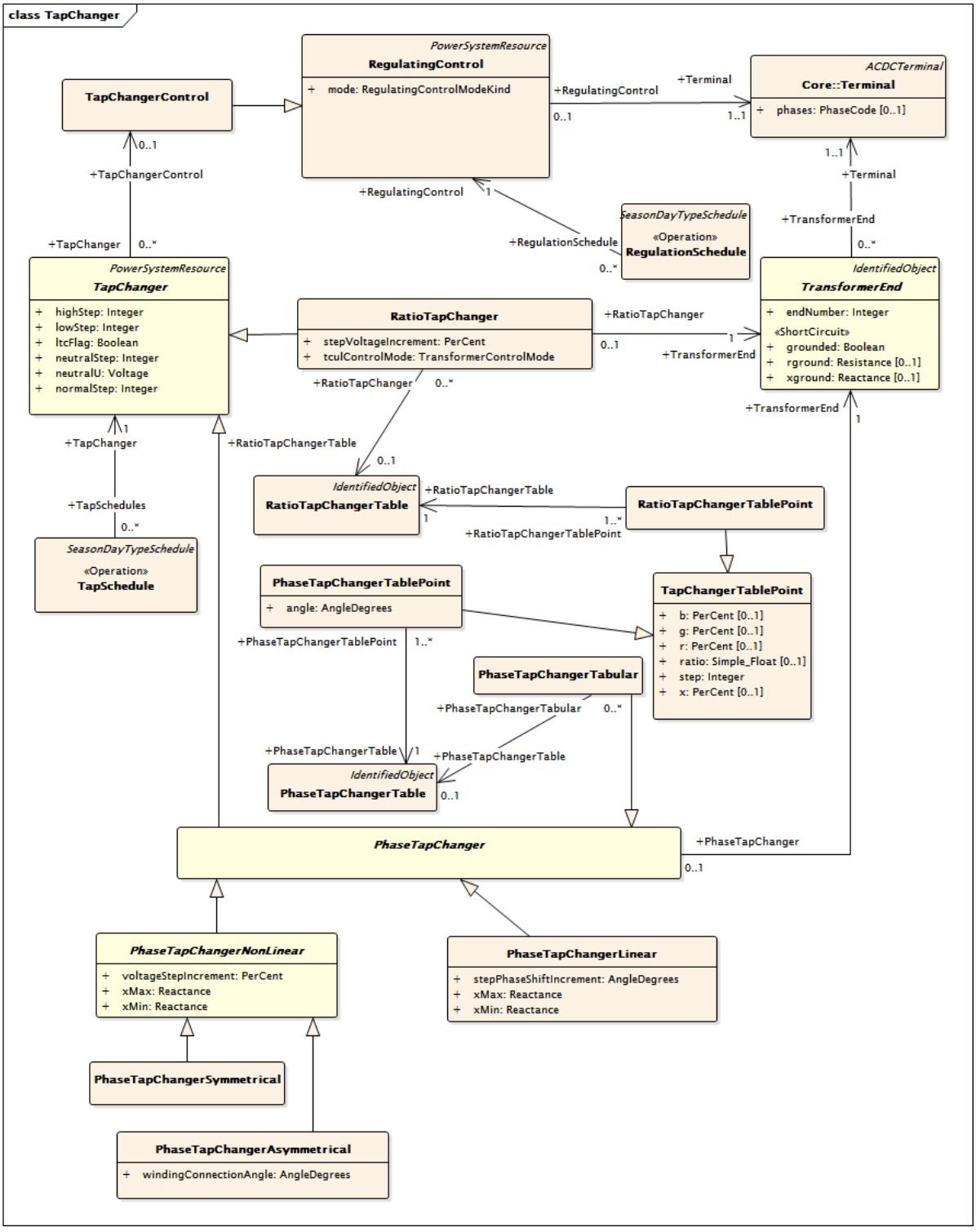


Figure 3-19 - diagram TapChanger

Figure 3-20 shows the diagram.

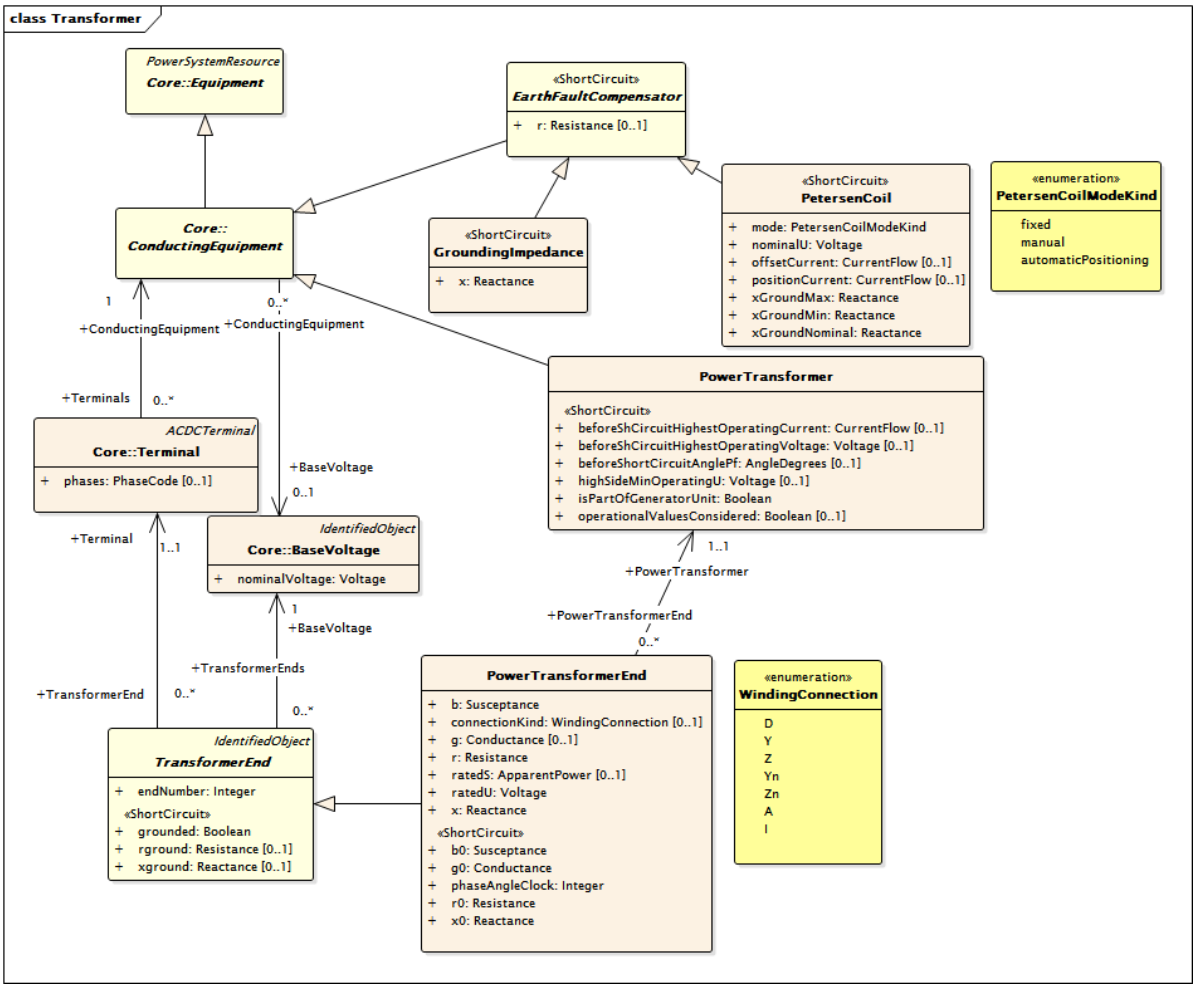


Figure 3-20 - diagram Transformer

3.9.2 ACLineSegment

A wire or combination of wires, with consistent electrical characteristics, building a single electrical system, used to carry alternating current between points in the power system. For symmetrical, transposed 3ph lines, it is sufficient to use attributes of the line segment, which describe impedances and admittances for the entire length of the segment. Additionally impedances can be computed by using length and associated per length impedances. The BaseVoltage at the two ends of ACLineSegments in a Line shall have the same BaseVoltage.nominalVoltage. However, boundary lines may have slightly different BaseVoltage.nominalVoltages and variation is allowed. Larger voltage difference in general requires use of an equivalent branch.

OCL constraint: An ACLineSegment must have a BaseVoltage.

Each ACLineSegment is required to have an association to a BaseVoltage. The association to Line is not required.

The positive sequence resistance (r) and reactance (x) are the same as negative sequence. Therefore negative sequence data is not modeled.
ENTSO-E exchanges allow 10 % difference of the BaseVoltage.nominalVoltage at the two ends of an ACLineSegment representing a complete tie-line or connecting to a boundary node.
Using the “EquipmentContainer” association, an ACLineSegment can only be contained by a Line, but the association to Line is not required.

Inheritance path: ->Conductor->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-161 shows all attributes of ACLineSegment.

Table 3-161 - Attributes of Wires::ACLineSegment

Attribute name	mult.	Attribute type	Description
b0ch (ShortCircuit)	1..1	Susceptance	Zero sequence shunt (charging) susceptance, uniformly distributed, of the entire line section.
bch	1..1	Susceptance	Positive sequence shunt (charging) susceptance, uniformly distributed, of the entire line section. This value represents the full charging over the full length of the line.
g0ch (ShortCircuit)	1..1	Conductance	Zero sequence shunt (charging) conductance, uniformly distributed, of the entire line section.
gch	0..1	Conductance	Positive sequence shunt (charging) conductance, uniformly distributed, of the entire line section.
r	1..1	Resistance	Positive sequence series resistance of the entire line section.
r0 (ShortCircuit)	1..1	Resistance	Zero sequence series resistance of the entire line section.
shortCircuitEndTemperature (ShortCircuit)	1..1	Temperature	Maximum permitted temperature at the end of SC for the calculation of minimum short-circuit currents. Used for short circuit data exchange according to IEC 60909
x	1..1	Reactance	Positive sequence series reactance of the entire line section.
x0 (ShortCircuit)	1..1	Reactance	Zero sequence series reactance of the entire line section.
length	0..1	Length	see Conductor
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-162 shows all association ends of ConductingEquipment with other classes.

Table 3-162 - Association ends of Core::ConductingEquipment with other classes

mult.	mult.	Class type name	Description
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.9.3 AsynchronousMachine

A rotating machine whose shaft rotates asynchronously with the electrical field. Also known as an induction machine with no external connection to the rotor windings, e.g squirrel-cage induction machine.

The attribute rxLockedRotorRatio is an optional attribute even if short circuit data is exchanged because IEC 60909 defines default values depending on motor size.

Inheritance path: ->RotatingMachine->RegulatingCondEq->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-163 shows all attributes of AsynchronousMachine.

Table 3-163 - Attributes of Wires::AsynchronousMachine

Attribute name	mult.	Attribute type	Description
converterFedDrive (ShortCircuit)	1..1	Boolean	Indicates whether the machine is a converter fed drive. Used for short circuit data exchange according to IEC 60909
efficiency (ShortCircuit)	1..1	PerCent	Efficiency of the asynchronous machine at nominal operation in percent. Indicator for converter drive motors. Used for short circuit data exchange according to IEC 60909
iaIrRatio (ShortCircuit)	1..1	Simple_Float	Ratio of locked-rotor current to the rated current of the motor (Ia/Ir). Used for short circuit data exchange according to IEC 60909
nominalFrequency	0..1	Frequency	Nameplate data indicates if the machine is 50 or 60 Hz.
nominalSpeed	0..1	RotationSpeed	Nameplate data. Depends on the slip

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			and number of pole pairs.
polePairNumber (ShortCircuit)	1..1	Integer	Number of pole pairs of stator. Used for short circuit data exchange according to IEC 60909
ratedMechanicalPower (ShortCircuit)	1..1	ActivePower	Rated mechanical power (Pr in the IEC 60909-0). Used for short circuit data exchange according to IEC 60909.
reversible (ShortCircuit)	1..1	Boolean	Indicates for converter drive motors if the power can be reversible. Used for short circuit data exchange according to IEC 60909
rxLockedRotorRatio (ShortCircuit)	0..1	Simple_Float	Locked rotor ratio (R/X). Used for short circuit data exchange according to IEC 60909
ratedPowerFactor	0..1	Simple_Float	see RotatingMachine
ratedS	0..1	ApparentPower	see RotatingMachine
ratedU	0..1	Voltage	see RotatingMachine
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-164 shows all association ends of RotatingMachine with other classes.

Table 3-164 - Association ends of Wires::RotatingMachine with other classes

mult.	mult.	Class type name	Description
GeneratingUnit	0..1	GeneratingUnit	Inherited Association Based On : Production:: GeneratingUnit.GeneratingUnit[0..1] ----- Wires:: RotatingMachine.RotatingMachine[1..*]
RegulatingControl	0..1	RegulatingControl	Inherited Association Based On : Wires:: RegulatingCondEq.RegulatingCondEq[0..*] ----- Wires:: RegulatingControl.RegulatingControl[0..1]
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] -----

			Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.9.4 Breaker

A mechanical switching device capable of making, carrying, and breaking currents under normal circuit conditions and also making, carrying for a specified time, and breaking currents under specified abnormal circuit conditions e.g. those of short circuit.

For switching Devices, Naming Convention and main containership hierarchy is : GeographicalRegion/SubGeographicalRegion/Substation/VoltageLevel or GeographicalRegion/SubGeographicalRegion/SubstationàVoltageLevel/Bay

Inheritance path: ->ProtectedSwitch->Switch->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-165 shows all attributes of Breaker.

Table 3-165 - Attributes of Wires::Breaker

Attribute name	mult.	Attribute type	Description
normalOpen	1..1	Boolean	see Switch
ratedCurrent	0..1	CurrentFlow	see Switch
retained	1..1	Boolean	see Switch
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-166 shows all association ends of ConductingEquipment with other classes.

Table 3-166 - Association ends of Core::ConductingEquipment with other classes

mult.	mult.	Class type name	Description
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core::

			Equipment.Equipments[0..*]

3.9.5 BusbarSection

A conductor, or group of conductors, with negligible impedance, that serve to connect other conducting equipment within a single substation. Voltage measurements are typically obtained from VoltageTransformers that are connected to busbar sections. A bus bar section may have many physical terminals but for analysis is modelled with exactly one logical terminal.

BusBarSection could be associated to only 1 Terminal.

The attribute ipMax is an optional attribute even if short circuit data is exchanged, as not always entered by the user (e.g. the IEC 60909-4 example test model described in Chapter 6.2 of the standard does not include these values.
Naming Convention and main containership hierarchy is : GeographicalRegion/SubGeographicalRegion/Substation/VoltageLevel

Inheritance path: ->Connector->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-167 shows all attributes of BusbarSection.

Table 3-167 - Attributes of Wires::BusbarSection

Attribute name	mult.	Attribute type	Description
ipMax (ShortCircuit)	0..1	CurrentFlow	Maximum allowable peak short-circuit current of busbar (Ipmax in the IEC 60909-0). Mechanical limit of the busbar in the substation itself. Used for short circuit data exchange according to IEC 60909
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-168 shows all association ends of ConductingEquipment with other classes.

Table 3-168 - Association ends of Core::ConductingEquipment with other classes

mult.	mult.	Class type name	Description
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1]

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			----- Core:: Equipment.Equipments[0..*]

3.9.6 Conductor(Abstract)

Combination of conducting material with consistent electrical characteristics, building a single electrical system, used to carry current between points in the power system.

Inheritance path: ->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-169 shows all attributes of Conductor.

Table 3-169 - Attributes of Wires::Conductor

Attribute name	mult.	Attribute type	Description
length	0..1	Length	Segment length for calculating line section capabilities
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-170 shows all association ends of ConductingEquipment with other classes.

Table 3-170 - Association ends of Core::ConductingEquipment with other classes

mult.	mult.	Class type name	Description
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.9.7 Connector(Abstract)

A conductor, or group of conductors, with negligible impedance, that serve to connect other conducting equipment within a single substation and are modelled with a single logical terminal.

Inheritance path: ->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-171 shows all attributes of Connector.

Table 3-171 - Attributes of Wires::Connector

Attribute name	mult.	Attribute type	Description

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-172 shows all association ends of ConductingEquipment with other classes.

Table 3-172 - Association ends of Core::ConductingEquipment with other classes

mult.	mult.	Class type name	Description
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.9.8 Disconnector

A manually operated or motor operated mechanical switching device used for changing the connections in a circuit, or for isolating a circuit or equipment from a source of power. It is required to open or close circuits when negligible current is broken or made.

For switching Devices, Naming Convention and main containership hierarchy is : GeographicalRegion/SubGeographicalRegion/Substation/VoltageLevel or GeographicalRegion/SubGeographicalRegion/SubstationàVoltageLevel/Bay

Inheritance path: ->Switch->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-173 shows all attributes of Disconnector.

Table 3-173 - Attributes of Wires::Disconnector

Attribute name	mult.	Attribute type	Description
normalOpen	1..1	Boolean	see Switch
ratedCurrent	0..1	CurrentFlow	see Switch
retained	1..1	Boolean	see Switch
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject

shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-174 shows all association ends of ConductingEquipment with other classes.

Table 3-174 - Association ends of Core::ConductingEquipment with other classes

mult.	mult.	Class type name	Description
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.9.9 (ShortCircuit) EarthFaultCompensator(Abstract)

A conducting equipment used to represent a connection to ground which is typically used to compensate earth faults.. An earth fault compensator device modeled with a single terminal implies a second terminal solidly connected to ground. If two terminals are modeled, the ground is not assumed and normal connection rules apply.

The GroundingImpedance and PetersenCoil connect the same way as other devices using Terminal. The phases attribute on Terminal indicates the connection is at the neutral phase. If a PowerTransformer has a Y connection on one terminal (secondary) and in case you wish to connect a PetersenCoil. The PowerTransformer's secondary Terminal where the coil is connected should be specified with phases attribute equal to ABCN and the PetersenCoil's Terminal should be specified with phase N. This allows the ACLineSegment to connect with phases ABC (the default) and the PetersenCoil to connect to the same TransformerTerminal but only on the N phase. It is expected all EarthFaultCompensators (GroundingImpedance and PetersenCoil) to have only one Terminal and be solidly connected to ground on the other unmodeled terminal. It is expected all PetersenCoil terminals to have only phase N. Basically a normal modeling of non-neutral network is done then add to the Petersen coil model which includes: 1) Adding the phasing of ABCN to the transformer secondary terminal so you have a place to connect neutral 2) Adding the PetersenCoil object with its associated Terminal which is phase N, since only one Terminal is modeled the PetersenCoil is assumed solidly grounded 3) Associate the PetersenCoil's single Terminal to the ConnectivityNode at transformer secondary.

Inheritance path: ->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-175 shows all attributes of EarthFaultCompensator.

Table 3-175 - Attributes of Wires::EarthFaultCompensator

Attribute name	mult.	Attribute type	Description
r (ShortCircuit)	0..1	Resistance	Nominal resistance of device.
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-176 shows all association ends of ConductingEquipment with other classes.

Table 3-176 - Association ends of Core::ConductingEquipment with other classes

mult.	mult.	Class type name	Description
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.9.10 EnergyConsumer

Generic user of energy - a point of consumption on the power system model.

Attributes (pfixed, qfixed, pfixedPct and qfixedPct) are used for load allocation. Attributes (pfixed and qfixed) represent base load, while attributes (pfixedPct and qfixedPct) represent the time-varying components

- The definition of the real and reactive power injections for an EnergyConsumer can be done using different sets of attributes. In the simplest case, the injections can be defined directly using only the attributes “pfixed” and “qfixed”. - To specify conforming and nonconforming loads, the classes ConformLoad, NonConformLoad, or their subtypes should be used. - The attributes defining the affect of voltage and frequency on the injection defined by an associated LoadResponseCharacteristic should be supplied, if they are available, but are not required.
Naming Convention and main containership hierarchy is : GeographicalRegion/SubGeographicalRegion/Substation/VoltageLevel
If LoadResponseCharacteristic is missing, this load is assumed to be constant power

Inheritance path: ->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-177 shows all attributes of EnergyConsumer.

Table 3-177 - Attributes of Wires::EnergyConsumer

Attribute name	mult.	Attribute type	Description
pfixed (Operation)	0..1	ActivePower	Active power of the load that is a fixed quantity. Load sign convention is used, i.e. positive sign means flow out from a node.
pfixedPct (Operation)	0..1	PerCent	Fixed active power as per cent of load group fixed active power. Load sign convention is used, i.e. positive sign means flow out from a node.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

qfixed (Operation)	0..1	ReactivePower	Reactive power of the load that is a fixed quantity. Load sign convention is used, i.e. positive sign means flow out from a node.
qfixedPct (Operation)	0..1	PerCent	Fixed reactive power as per cent of load group fixed reactive power. Load sign convention is used, i.e. positive sign means flow out from a node.
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-178 shows all association ends of EnergyConsumer with other classes.

Table 3-178 - Association ends of Wires::EnergyConsumer with other classes

mult.	mult.	Class type name	Description
LoadResponse	[0..1]	LoadResponseCharacteristic	The load response characteristic of this load. If missing, this load is assumed to be constant power. Association Based On : Wires:: EnergyConsumer.EnergyConsumer[0..*] ----- LoadModel:: LoadResponseCharacteristic.LoadResponse[0..1]
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.9.11 ExternalNetworkInjection

This class represents external network and it is used for IEC 60909 calculations.

The attributes ikSecond and voltageFactor are optional attributes even if short circuit data is exchanged. These attributes are used only if short circuit calculations are done according to superposition method.

Inheritance path: ->RegulatingCondEq->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-179 shows all attributes of ExternalNetworkInjection.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-179 - Attributes of Wires::ExternalNetworkInjection

Attribute name	mult.	Attribute type	Description
governorSCD	1..1	ActivePowerPerFrequency	Power Frequency Bias. This is the change in power injection divided by the change in frequency and negated. A positive value of the power frequency bias provides additional power injection upon a drop in frequency.
ikSecond (ShortCircuit)	0..1	Boolean	Indicates whether initial symmetrical short-circuit current and power have been calculated according to IEC (Ik").
maxInitialSymShCCurrent (ShortCircuit)	1..1	CurrentFlow	Maximum initial symmetrical short-circuit currents (Ik" max) in A (Ik" = Sk"/(SQRT(3) Un)). Used for short circuit data exchange according to IEC 60909
maxP	1..1	ActivePower	Maximum active power of the injection.
maxQ	1..1	ReactivePower	Not for short circuit modelling; It is used for modelling of infeed for load flow exchange. If maxQ and minQ are not used ReactiveCapabilityCurve can be used
maxR0ToX0Ratio (ShortCircuit)	1..1	Simple_Float	Maximum ratio of zero sequence resistance of Network Feeder to its zero sequence reactance (R(0)/X(0) max). Used for short circuit data exchange according to IEC 60909
maxR1ToX1Ratio (ShortCircuit)	1..1	Simple_Float	Maximum ratio of positive sequence resistance of Network Feeder to its positive sequence reactance (R(1)/X(1) max). Used for short circuit data exchange according to IEC 60909
maxZ0ToZ1Ratio (ShortCircuit)	1..1	Simple_Float	Maximum ratio of zero sequence impedance to its positive sequence impedance (Z(0)/Z(1) max). Used for short circuit data exchange according to IEC 60909
minInitialSymShCCurrent (ShortCircuit)	1..1	CurrentFlow	Minimum initial symmetrical short-circuit currents (Ik" min) in A

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			(Ik" = Sk"/(SQRT(3) Un)). Used for short circuit data exchange according to IEC 60909
minP	1..1	ActivePower	Minimum active power of the injection.
minQ	1..1	ReactivePower	Not for short circuit modelling; It is used for modelling of infeed for load flow exchange. If maxQ and minQ are not used ReactiveCapabilityCurve can be used
minR0ToX0Ratio (ShortCircuit)	1..1	Simple_Float	Indicates whether initial symmetrical short-circuit current and power have been calculated according to IEC (Ik"). Used for short circuit data exchange according to IEC 6090
minR1ToX1Ratio (ShortCircuit)	1..1	Simple_Float	Minimum ratio of positive sequence resistance of Network Feeder to its positive sequence reactance (R(1)/X(1) min). Used for short circuit data exchange according to IEC 60909
minZ0ToZ1Ratio (ShortCircuit)	1..1	Simple_Float	Minimum ratio of zero sequence impedance to its positive sequence impedance (Z(0)/Z(1) min). Used for short circuit data exchange according to IEC 60909
voltageFactor (ShortCircuit)	0..1	PU	Voltage factor in pu, which was used to calculate short-circuit current Ik" and power Sk".
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-180 shows all association ends of RegulatingCondEq with other classes.

Table 3-180 - Association ends of Wires::RegulatingCondEq with other classes

mult.	mult.	Class type name	Description
RegulatingControl	0..1	RegulatingControl	Inherited Association Based On : Wires::

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			RegulatingCondEq.RegulatingCondEq[0..*] --- Wires:: RegulatingControl.RegulatingControl[0..1]
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] --- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] --- Core:: Equipment.Equipments[0..*]

3.9.12 Ground

A point where the system is grounded used for connecting conducting equipment to ground. The power system model can have any number of grounds.

Inheritance path: ->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-181 shows all attributes of Ground.

Table 3-181 - Attributes of Wires::Ground

Attribute name	mult.	Attribute type	Description
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-182 shows all association ends of ConductingEquipment with other classes.

Table 3-182 - Association ends of Core::ConductingEquipment with other classes

mult.	mult.	Class type name	Description
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] --- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] --- Core:: Equipment.Equipments[0..*]

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

3.9.13 GroundDisconnector

A manually operated or motor operated mechanical switching device used for isolating a circuit or equipment from ground.

Inheritance path: ->Switch->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-183 shows all attributes of GroundDisconnector.

Table 3-183 - Attributes of Wires::GroundDisconnector

Attribute name	mult.	Attribute type	Description
normalOpen	1..1	Boolean	see Switch
ratedCurrent	0..1	CurrentFlow	see Switch
retained	1..1	Boolean	see Switch
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-184 shows all association ends of ConductingEquipment with other classes.

Table 3-184 - Association ends of Core::ConductingEquipment with other classes

mult.	mult.	Class type name	Description
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.9.14 GroundingImpedance

A fixed impedance device used for grounding.

Inheritance path: ->EarthFaultCompensator->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-185 shows all attributes of GroundingImpedance.

Table 3-185 - Attributes of Wires::GroundingImpedance

Attribute name	mult.	Attribute type	Description
x (ShortCircuit)	1..1	Reactance	Reactance of device.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

r	0..1	Resistance	see EarthFaultCompensator
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-186 shows all association ends of ConductingEquipment with other classes.

Table 3-186 - Association ends of Core::ConductingEquipment with other classes

mult.	mult.	Class type name	Description
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.9.15 Junction

A point where one or more conducting equipments are connected with zero resistance.

Inheritance path: ->Connector->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-187 shows all attributes of Junction.

Table 3-187 - Attributes of Wires::Junction

Attribute name	mult.	Attribute type	Description
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-188 shows all association ends of ConductingEquipment with other classes.

Table 3-188 - Association ends of Core::ConductingEquipment with other classes

mult.	mult.	Class type name	Description
BaseVoltage	0..1	BaseVoltage	

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.9.16 Line

Contains equipment beyond a substation belonging to a power transmission line.

Use of the Line class is not required. If used, it can only be used as a container for ACLineSegments and SeriesCompensators.

A Line is not required to be associated with a SubGeographicalRegion.

Inheritance path: ->EquipmentContainer->ConnectivityNodeContainer->PowerSystemResource->IdentifiedObject

Table 3-189 shows all attributes of Line.

Table 3-189 - Attributes of Wires::Line

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-190 shows all association ends of Line with other classes.

Table 3-190 - Association ends of Wires::Line with other classes

mult.	mult.	Class type name	Description
Region	[0..1]	SubGeographicalRegion	The sub-geographical region of the line. Association Based On : Core:: SubGeographicalRegion.Region[0..1] ----- Wires:: Line.Lines[0..*]

3.9.17 LinearShuntCompensator

A linear shunt compensator has banks or sections with equal admittance values.

Inheritance path: ->ShuntCompensator->RegulatingCondEq->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-191 shows all attributes of LinearShuntCompensator.

Table 3-191 - Attributes of Wires::LinearShuntCompensator

Attribute name	mult.	Attribute type	Description

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

b0PerSection (ShortCircuit)	1..1	Susceptance	Zero sequence shunt (charging) susceptance per section
bPerSection	1..1	Susceptance	Positive sequence shunt (charging) susceptance per section
g0PerSection (ShortCircuit)	1..1	Conductance	Zero sequence shunt (charging) conductance per section
gPerSection	1..1	Conductance	Positive sequence shunt (charging) conductance per section
aVRDelay	0..1	Seconds	see ShuntCompensator
grounded	0..1	Boolean	see ShuntCompensator
maximumSections	1..1	Integer	see ShuntCompensator
nomU	1..1	Voltage	see ShuntCompensator
normalSections	1..1	Integer	see ShuntCompensator
switchOnCount	0..1	Integer	see ShuntCompensator
switchOnDate	0..1	DateTime	see ShuntCompensator
voltageSensitivity	0..1	VoltagePerReactivePower	see ShuntCompensator
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-192 shows all association ends of RegulatingCondEq with other classes.

Table 3-192 - Association ends of Wires::RegulatingCondEq with other classes

mult.	mult.	Class type name	Description
RegulatingControl	0..1	RegulatingControl	Inherited Association Based On : Wires:: RegulatingCondEq.RegulatingCondEq[0..*] ----- Wires:: RegulatingControl.RegulatingControl[0..1]
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] -----

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			Core:: Equipment.Equipments[0..*]

3.9.18 LoadBreakSwitch

A mechanical switching device capable of making, carrying, and breaking currents under normal operating conditions.

For switching Devices, Naming Convention and main containership hierarchy is : GeographicalRegion/SubGeographicalRegion/Substation/VoltageLevel or GeographicalRegion/SubGeographicalRegion/SubstationàVoltageLevel/Bay

Inheritance path: ->ProtectedSwitch->Switch->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-193 shows all attributes of LoadBreakSwitch.

Table 3-193 - Attributes of Wires::LoadBreakSwitch

Attribute name	mult.	Attribute type	Description
normalOpen	1..1	Boolean	see Switch
ratedCurrent	0..1	CurrentFlow	see Switch
retained	1..1	Boolean	see Switch
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-194 shows all association ends of ConductingEquipment with other classes.

Table 3-194 - Association ends of Core::ConductingEquipment with other classes

mult.	mult.	Class type name	Description
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.9.19 MutualCoupling

This class represents the zero sequence line mutual coupling.

OCL constraint: First terminal of a mutual coupling must have a sequence number set
OCL constraint: Second terminal that of a mutual coupling must have a sequence number set

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

OCL constraint: The starting terminal for the calculation of distances along the first branch of the mutual coupling. Normally MutualCoupling would only be used for terminals of AC line segments. The terminals of a mutual coupling should point to different segments
OCL constraint: The starting terminal for the calculation of distances along the second branch of the mutual coupling. This should point to an ACLineSegment

Inheritance path: ->IdentifiedObject

Table 3-195 shows all attributes of MutualCoupling.

Table 3-195 - Attributes of Wires::MutualCoupling

Attribute name	mult.	Attribute type	Description
b0ch (ShortCircuit)	1..1	Susceptance	Zero sequence mutual coupling shunt (charging) susceptance, uniformly distributed, of the entire line section.
distance11 (ShortCircuit)	1..1	Length	Distance to the start of the coupled region from the first line's terminal having sequence number equal to 1.
distance12 (ShortCircuit)	1..1	Length	Distance to the end of the coupled region from the first line's terminal with sequence number equal to 1.
distance21 (ShortCircuit)	1..1	Length	Distance to the start of coupled region from the second line's terminal with sequence number equal to 1.
distance22 (ShortCircuit)	1..1	Length	Distance to the end of coupled region from the second line's terminal with sequence number equal to 1.
g0ch (ShortCircuit)	1..1	Conductance	Zero sequence mutual coupling shunt (charging) conductance, uniformly distributed, of the entire line section.
r0 (ShortCircuit)	1..1	Resistance	Zero sequence branch-to-branch mutual impedance coupling, resistance.
x0 (ShortCircuit)	1..1	Reactance	Zero sequence branch-to-branch mutual impedance coupling, reactance.
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-196 shows all association ends of MutualCoupling with other classes.

Table 3-196 - Association ends of Wires::MutualCoupling with other classes

mult.	mult.	Class type name	Description
First_Terminal (ShortCircuit)	[1..1]	Terminal	The starting terminal for the calculation of distances along the first branch of the mutual coupling. Normally MutualCoupling would only be used for terminals of AC line segments. The first and second terminals of a mutual coupling should point to different AC line segments. Association Based On : Wires:: MutualCoupling.HasFirstMutualCoupling[0..*] ----- Core:: Terminal.First_Terminal[1]
Second_Terminal (ShortCircuit)	[1..1]	Terminal	The starting terminal for the calculation of distances along the second branch of the mutual coupling. Association Based On : Wires:: MutualCoupling.HasSecondMutualCoupling[0..*] ----- Core:: Terminal.Second_Terminal[1]

3.9.20 (ShortCircuit) NonlinearShuntCompensator

A non linear shunt compensator has bank or section admittance values that differs.

Inheritance path: ->ShuntCompensator->RegulatingCondEq->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-197 shows all attributes of NonlinearShuntCompensator.

Table 3-197 - Attributes of Wires::NonlinearShuntCompensator

Attribute name	mult.	Attribute type	Description
aVRDelay	0..1	Seconds	see ShuntCompensator
grounded	0..1	Boolean	see ShuntCompensator
maximumSections	1..1	Integer	see ShuntCompensator
nomU	1..1	Voltage	see ShuntCompensator
normalSections	1..1	Integer	see ShuntCompensator
switchOnCount	0..1	Integer	see ShuntCompensator
switchOnDate	0..1	DateTime	see ShuntCompensator
voltageSensitivity	0..1	VoltagePerReactivePower	see ShuntCompensator
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-198 shows all association ends of RegulatingCondEq with other classes.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-198 - Association ends of Wires::RegulatingCondEq with other classes

mult.	mult.	Class type name	Description
RegulatingControl	0..1	RegulatingControl	Inherited Association Based On : Wires:: RegulatingCondEq.RegulatingCondEq[0..*] ----- Wires:: RegulatingControl.RegulatingControl[0..1]
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.9.21 NonlinearShuntCompensatorPoint

A non linear shunt compensator bank or section admittance value.

Table 3-199 shows all attributes of NonlinearShuntCompensatorPoint.

Table 3-199 - Attributes of Wires::NonlinearShuntCompensatorPoint

Attribute name	mult.	Attribute type	Description
b	1..1	Susceptance	Positive sequence shunt (charging) susceptance per section
b0 (ShortCircuit)	1..1	Susceptance	Zero sequence shunt (charging) susceptance per section
g	1..1	Conductance	Positive sequence shunt (charging) conductance per section
g0 (ShortCircuit)	1..1	Conductance	Zero sequence shunt (charging) conductance per section
sectionNumber	1..1	Integer	The number of the section.

Table 3-200 shows all association ends of NonlinearShuntCompensatorPoint with other classes.

Table 3-200 - Association ends of Wires::NonlinearShuntCompensatorPoint with other classes

mult.	mult.	Class type name	Description
NonlinearShuntCompensator	[1..1]	NonlinearShuntCompensator	Non-linear shunt compensator owning this point. Association Based On :

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			Wires:: NonlinearShuntCompensatorPoint.NonlinearShuntCompensatorPoints[1..*] ----- Wires:: NonlinearShuntCompensator.NonlinearShuntCompensator[1]

3.9.22 PetersenCoil

A tunable impedance device normally used to offset line charging during single line faults in an ungrounded section of network.

Inheritance path: ->EarthFaultCompensator->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-201 shows all attributes of PetersenCoil.

Table 3-201 - Attributes of Wires::PetersenCoil

Attribute name	mult.	Attribute type	Description
mode (ShortCircuit)	1..1	PetersenCoilModeKind	The mode of operation of the Petersen coil.
nominalU (ShortCircuit)	1..1	Voltage	The nominal voltage for which the coil is designed.
offsetCurrent (ShortCircuit)	0..1	CurrentFlow	The offset current that the Petersen coil controller is operating from the resonant point. This is normally a fixed amount for which the controller is configured and could be positive or negative. Typically 0 to 60 Amperes depending on voltage and resonance conditions.
positionCurrent (ShortCircuit)	0..1	CurrentFlow	The control current used to control the Petersen coil also known as the position current. Typically in the range of 20-200mA.
xGroundMax (ShortCircuit)	1..1	Reactance	The maximum reactance.
xGroundMin (ShortCircuit)	1..1	Reactance	The minimum reactance.
xGroundNominal (ShortCircuit)	1..1	Reactance	The nominal reactance. This is the operating point (normally over compensation) that is defined based on the resonance point in the healthy network condition. The impedance is calculated based on nominal voltage divided by position current.
r	0..1	Resistance	see EarthFaultCompensator

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-202 shows all association ends of ConductingEquipment with other classes.

Table 3-202 - Association ends of Core::ConductingEquipment with other classes

mult.	mult.	Class type name	Description
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.9.23 PhaseTapChanger(Abstract)

A transformer phase shifting tap model that controls the phase angle difference across the power transformer and potentially the active power flow through the power transformer. This phase tap model may also impact the voltage magnitude.

Inheritance path: ->TapChanger->PowerSystemResource->IdentifiedObject

Table 3-203 shows all attributes of PhaseTapChanger.

Table 3-203 - Attributes of Wires::PhaseTapChanger

Attribute name	mult.	Attribute type	Description
highStep	1..1	Integer	see TapChanger
lowStep	1..1	Integer	see TapChanger
ltcFlag	1..1	Boolean	see TapChanger
neutralStep	1..1	Integer	see TapChanger
neutralU	1..1	Voltage	see TapChanger
normalStep	1..1	Integer	see TapChanger
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-204 shows all association ends of PhaseTapChanger with other classes.

Table 3-204 - Association ends of Wires::PhaseTapChanger with other classes

mult.	mult.	Class type name	Description
TransformerEnd	[1..1]	TransformerEnd	Phase tap changer associated with this transformer end. Association Based On : Wires:: TransformerEnd.TransformerEnd[1] ----- Wires:: PhaseTapChanger.PhaseTapChanger[0..1]
TapChangerControl	0..1	TapChangerControl	Inherited Association Based On : Wires:: TapChanger.TapChanger[0..*] ----- Wires:: TapChangerControl.TapChangerControl[0..1]

3.9.24 PhaseTapChangerAsymmetrical

Describes the tap model for an asymmetrical phase shifting transformer in which the difference voltage vector adds to the primary side voltage. The angle between the primary side voltage and the difference voltage is named the winding connection angle. The phase shift depends on both the difference voltage magnitude and the winding connection angle.

Inheritance path: ->PhaseTapChangerNonLinear->PhaseTapChanger->TapChanger->PowerSystemResource->IdentifiedObject

Table 3-205 shows all attributes of PhaseTapChangerAsymmetrical.

Table 3-205 - Attributes of Wires::PhaseTapChangerAsymmetrical

Attribute name	mult.	Attribute type	Description
windingConnectionAngle	1..1	AngleDegrees	The phase angle between the in-phase winding and the out-of -phase winding used for creating phase shift. The out-of-phase winding produces what is known as the difference voltage. Setting this angle to 90 degrees is not the same as a symmemtrical transformer.
voltageStepIncrement	1..1	PerCent	see PhaseTapChangerNonLinear
xMax	1..1	Reactance	see PhaseTapChangerNonLinear
xMin	1..1	Reactance	see PhaseTapChangerNonLinear
highStep	1..1	Integer	see TapChanger
lowStep	1..1	Integer	see TapChanger
ltcFlag	1..1	Boolean	see TapChanger
neutralStep	1..1	Integer	see TapChanger
neutralU	1..1	Voltage	see TapChanger
normalStep	1..1	Integer	see TapChanger

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-206 shows all association ends of PhaseTapChanger with other classes.

Table 3-206 - Association ends of Wires::PhaseTapChanger with other classes

mult.	mult.	Class type name	Description
TransformerEnd	1..1	TransformerEnd	Inherited Association Based On : Wires:: TransformerEnd.TransformerEnd[1] ----- Wires:: PhaseTapChanger.PhaseTapChanger[0..1]
TapChangerControl	0..1	TapChangerControl	Inherited Association Based On : Wires:: TapChanger.TapChanger[0..*] ----- Wires:: TapChangerControl.TapChangerControl[0..1]

3.9.25 PhaseTapChangerLinear

Describes a tap changer with a linear relation between the tap step and the phase angle difference across the transformer. This is a mathematical model that is an approximation of a real phase tap changer.
The phase angle is computed as stepPhaseShitfIncrement times the tap position. The secondary side voltage magnitude is the same as at the primary side.

Inheritance path: ->PhaseTapChanger->TapChanger->PowerSystemResource->IdentifiedObject

Table 3-207 shows all attributes of PhaseTapChangerLinear.

Table 3-207 - Attributes of Wires::PhaseTapChangerLinear

Attribute name	mult.	Attribute type	Description
stepPhaseShiftIncrement	1..1	AngleDegrees	Phase shift per step position. A positive value indicates a positive phase shift from the winding where the tap is located to the other winding (for a two-winding transformer). The actual phase shift increment might be more accurately computed from the symmetrical or asymmetrical models or a tap step table lookup if those are available.
xMax	1..1	Reactance	The reactance depend on the tap position according to a "u" shaped curve. The

			maximum reactance (xMax) appear at the low and high tap positions.
xMin	1..1	Reactance	The reactance depend on the tap position according to a "u" shaped curve. The minimum reactance (xMin) appear at the mid tap position.
highStep	1..1	Integer	see TapChanger
lowStep	1..1	Integer	see TapChanger
ltcFlag	1..1	Boolean	see TapChanger
neutralStep	1..1	Integer	see TapChanger
neutralU	1..1	Voltage	see TapChanger
normalStep	1..1	Integer	see TapChanger
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-208 shows all association ends of PhaseTapChanger with other classes.

Table 3-208 - Association ends of Wires::PhaseTapChanger with other classes

mult.	mult.	Class type name	Description
TransformerEnd	1..1	TransformerEnd	Inherited Association Based On : Wires:: TransformerEnd.TransformerEnd[1] ----- Wires:: PhaseTapChanger.PhaseTapChanger[0..1]
TapChangerControl	0..1	TapChangerControl	Inherited Association Based On : Wires:: TapChanger.TapChanger[0..*] ----- Wires:: TapChangerControl.TapChangerControl[0..1]

3.9.26 PhaseTapChangerNonLinear(Abstract)

The non-linear phase tap changer describes the non-linear behavior of a phase tap changer. This is a base class for the symmetrical and asymmetrical phase tap changer models. The details of these models can be found in the IEC 61970-301 document.

Inheritance path: ->PhaseTapChanger->TapChanger->PowerSystemResource->IdentifiedObject

Table 3-209 shows all attributes of PhaseTapChangerNonLinear.

Table 3-209 - Attributes of Wires::PhaseTapChangerNonLinear

Attribute name	mult.	Attribute type	Description

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

voltageStepIncrement	1..1	PerCent	The voltage step increment on the out of phase winding specified in percent of nominal voltage of the transformer end.
xMax	1..1	Reactance	The reactance depend on the tap position according to a "u" shaped curve. The maximum reactance (xMax) appear at the low and high tap positions.
xMin	1..1	Reactance	The reactance depend on the tap position according to a "u" shaped curve. The minimum reactance (xMin) appear at the mid tap position.
highStep	1..1	Integer	see TapChanger
lowStep	1..1	Integer	see TapChanger
ltcFlag	1..1	Boolean	see TapChanger
neutralStep	1..1	Integer	see TapChanger
neutralU	1..1	Voltage	see TapChanger
normalStep	1..1	Integer	see TapChanger
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-210 shows all association ends of PhaseTapChanger with other classes.

Table 3-210 - Association ends of Wires::PhaseTapChanger with other classes

mult.	mult.	Class type name	Description
TransformerEnd	1..1	TransformerEnd	Inherited Association Based On : Wires:: TransformerEnd.TransformerEnd[1] ----- Wires:: PhaseTapChanger.PhaseTapChanger[0..1]
TapChangerControl	0..1	TapChangerControl	Inherited Association Based On : Wires:: TapChanger.TapChanger[0..*] ----- Wires:: TapChangerControl.TapChangerControl[0..1]

3.9.27 PhaseTapChangerSymmetrical

Describes a symmetrical phase shifting transformer tap model in which the secondary side voltage magnitude is the same as at the primary side. The difference voltage magnitude is the base in an equal-sided triangle where the sides corresponds to the primary and secondary voltages. The

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

phase angle difference corresponds to the top angle and can be expressed as twice the arctangent of half the total difference voltage.

Inheritance path: ->PhaseTapChangerNonLinear->PhaseTapChanger->TapChanger->PowerSystemResource->IdentifiedObject

Table 3-211 shows all attributes of PhaseTapChangerSymmetrical.

Table 3-211 - Attributes of Wires::PhaseTapChangerSymmetrical

Attribute name	mult.	Attribute type	Description
voltageStepIncrement	1..1	PerCent	see PhaseTapChangerNonLinear
xMax	1..1	Reactance	see PhaseTapChangerNonLinear
xMin	1..1	Reactance	see PhaseTapChangerNonLinear
highStep	1..1	Integer	see TapChanger
lowStep	1..1	Integer	see TapChanger
ltcFlag	1..1	Boolean	see TapChanger
neutralStep	1..1	Integer	see TapChanger
neutralU	1..1	Voltage	see TapChanger
normalStep	1..1	Integer	see TapChanger
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-212 shows all association ends of PhaseTapChanger with other classes.

Table 3-212 - Association ends of Wires::PhaseTapChanger with other classes

mult.	mult.	Class type name	Description
TransformerEnd	1..1	TransformerEnd	Inherited Association Based On : Wires:: TransformerEnd.TransformerEnd[1] ----- Wires:: PhaseTapChanger.PhaseTapChanger[0..1]
TapChangerControl	0..1	TapChangerControl	Inherited Association Based On : Wires:: TapChanger.TapChanger[0..*] ----- Wires:: TapChangerControl.TapChangerControl[0..1]

3.9.28 PhaseTapChangerTable

Describes a tabular curve for how the phase angle difference and impedance varies with the tap step.

Inheritance path: ->IdentifiedObject

Table 3-213 shows all attributes of PhaseTapChangerTable.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-213 - Attributes of Wires::PhaseTapChangerTable

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

3.9.29 PhaseTapChangerTablePoint

Describes each tap step in the phase tap changer tabular curve.

Inheritance path: ->TapChangerTablePoint

Table 3-214 shows all attributes of PhaseTapChangerTablePoint.

Table 3-214 - Attributes of Wires::PhaseTapChangerTablePoint

Attribute name	mult.	Attribute type	Description
angle	1..1	AngleDegrees	The angle difference in degrees.
b	0..1	PerCent	see TapChangerTablePoint
g	0..1	PerCent	see TapChangerTablePoint
r	0..1	PerCent	see TapChangerTablePoint
ratio	0..1	Simple_Float	see TapChangerTablePoint
step	1..1	Integer	see TapChangerTablePoint
x	0..1	PerCent	see TapChangerTablePoint

Table 3-215 shows all association ends of PhaseTapChangerTablePoint with other classes.

Table 3-215 - Association ends of Wires::PhaseTapChangerTablePoint with other classes

mult.	mult.	Class type name	Description
PhaseTapChangerTable	[1..1]	PhaseTapChangerTable	The table of this point. Association Based On : Wires:: PhaseTapChangerTablePoint.PhaseTapChangerTablePoint[1..*] ----- Wires:: PhaseTapChangerTable.PhaseTapChangerTable[1]

3.9.30 PhaseTapChangerTabular

Inheritance path: ->PhaseTapChanger->TapChanger->PowerSystemResource->IdentifiedObject

Table 3-216 shows all attributes of PhaseTapChangerTabular.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-216 - Attributes of Wires::PhaseTapChangerTabular

Attribute name	mult.	Attribute type	Description
highStep	1..1	Integer	see TapChanger
lowStep	1..1	Integer	see TapChanger
ltcFlag	1..1	Boolean	see TapChanger
neutralStep	1..1	Integer	see TapChanger
neutralU	1..1	Voltage	see TapChanger
normalStep	1..1	Integer	see TapChanger
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-217 shows all association ends of PhaseTapChangerTabular with other classes.

Table 3-217 - Association ends of Wires::PhaseTapChangerTabular with other classes

mult.	mult.	Class type name	Description
PhaseTapChangerTable	[0..1]	PhaseTapChangerTable	The phase tap changer table for this phase tap changer. Association Based On : Wires:: PhaseTapChangerTable.PhaseTapChangerTable[0..1] --- -- Wires:: PhaseTapChangerTabular.PhaseTapChangerTabular[0..*]
TransformerEnd	1..1	TransformerEnd	Inherited Association Based On : Wires:: TransformerEnd.TransformerEnd[1] ----- Wires:: PhaseTapChanger.PhaseTapChanger[0..1]
TapChangerControl	0..1	TapChangerControl	Inherited Association Based On : Wires:: TapChanger.TapChanger[0..*] ----- Wires:: TapChangerControl.TapChangerControl[0..1]

3.9.31 PowerTransformer

An electrical device consisting of two or more coupled windings, with or without a magnetic core, for introducing mutual coupling between electric circuits. Transformers can be used to control voltage and phase shift (active power flow).
A power transformer may be composed of separate transformer tanks that need not be identical.
A power transformer can be modeled with or without tanks and is intended for use in both balanced and unbalanced representations. A power transformer typically has two terminals, but may have one (grounding), three or more terminals.
The inherited association ConductingEquipment.BaseVoltage should not be used. The association from TransformerEnd to BaseVoltage should be used instead.

a PowerTransformer is contained in one Substation but it can connect a Terminal to another Substation

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Association between PowerTransformer and BaseVoltage is not exchanged
PowerTransformer now inherits from ConductingEquipment instead of Equipment. The terminals that were formerly attached to TransformerWinding are now attached to PowerTransformer. The Terminal.sequenceNumber distinquishes the terminals much as previously done by TransformerWinding.windingType:WindingType. A PowerTransformer may be balanced or unbalanced and may optionally model unbalanced tank level detail. The Terminal to PowerTransformerEnd association in addition to the Terminal to ConductingEquipment (to PowerTransformer instances) association is used. There is one terminal per PoweTransformerEnd, i.e. for a 2-winding transformer in total two terminals. Each terminal would have an association to the same PowerTransformer instance and one association to the individual PowerTransformerEnd instances. TransformerEnd.endNumber attribute is used to define high and lower sides of the PowerTransformer.
A PowerTransformer could be associated to 2 or 3 Terminals
A PowerTransformer can be either two winding or three winding. - A two winding transformer has two PowerTransformerEnds - A three winding transformer has three PowerTransformerEnds
Naming Convention and main containership hierarchy is : GeographicalRegion/SubGeographicalRegion/Substation

Inheritance path: ->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-218 shows all attributes of PowerTransformer.

Table 3-218 - Attributes of Wires::PowerTransformer

Attribute name	mult.	Attribute type	Description
beforeShCircuitHighestOperatingCurrent (ShortCircuit)	0..1	CurrentFlow	The highest operating current (Ib in the IEC 60909-0) before short circuit (depends on network configuration and relevant reliability philosophy). It is used for calculation of the impedance correction factor KT defined in IEC 60909-0.
beforeShCircuitHighestOperatingVoltage (ShortCircuit)	0..1	Voltage	The highest operating voltage (Ub in the IEC 60909-0) before short circuit. It is used for calculation of the impedance correction factor KT defined in IEC 60909-0. This is worst case voltage on the low side winding (Section 3.7.1 in the standard). Used to define operating conditions.
beforeShortCircuitAnglePf (ShortCircuit)	0..1	AngleDegrees	The angle of power factor before short circuit (phib in the IEC 60909-0). It is used for calculation of the impedance correction factor KT defined in IEC 60909-0. This is the worst case power factor. Used to define operating conditions.
highSideMinOperatingU (ShortCircuit)	0..1	Voltage	The minimum operating voltage (uQmin in the

			IEC 60909-0) at the high voltage side (Q side) of the unit transformer of the power station unit. A value well established from long-term operating experience of the system. It is used for calculation of the impedance correction factor KG defined in IEC 60909-0
isPartOfGeneratorUnit (ShortCircuit)	1..1	Boolean	Indicates whether the machine is part of a power station unit. Used for short circuit data exchange according to IEC 60909
operationalValuesConsidered (ShortCircuit)	0..1	Boolean	It is used to define if the data (other attributes related to short circuit data exchange) defines long term operational conditions or not. Used for short circuit data exchange according to IEC 60909.
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-219 shows all association ends of ConductingEquipment with other classes.

Table 3-219 - Association ends of Core::ConductingEquipment with other classes

mult.	mult.	Class type name	Description
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.9.32 PowerTransformerEnd

A PowerTransformerEnd is associated with each Terminal of a PowerTransformer. The impedance values r, r0, x, and x0 of a PowerTransformerEnd represents a star equivalent as follows

1) for a two Terminal PowerTransformer the high voltage PowerTransformerEnd has non zero

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

values on r, r0, x, and x0 while the low voltage PowerTransformerEnd has zero values for r, r0, x, and x0.
2) for a three Terminal PowerTransformer the three PowerTransformerEnds represents a star equivalent with each leg in the star represented by r, r0, x, and x0 values.
3) for a PowerTransformer with more than three Terminals the PowerTransformerEnd impedance values cannot be used. Instead use the TransformerMeshImpedance or split the transformer into multiple PowerTransformers.

- Each PowerTransformerEnd must be contained by a PowerTransformer. Because a PowerTransformerEnd (or any other object) can not be contained by more than one parent, a PowerTransformerEnd can not have an association to an EquipmentContainer (Substation, VoltageLevel, etc).

Inheritance path: ->TransformerEnd->IdentifiedObject

Table 3-220 shows all attributes of PowerTransformerEnd.

Table 3-220 - Attributes of Wires::PowerTransformerEnd

Attribute name	mult.	Attribute type	Description
b	1..1	Susceptance	Magnetizing branch susceptance (B mag). The value can be positive or negative.
connectionKind	0..1	WindingConnection	Kind of connection.
b0 (ShortCircuit)	1..1	Susceptance	Zero sequence magnetizing branch susceptance.
phaseAngleClock (ShortCircuit)	1..1	Integer	Terminal voltage phase angle displacement where 360 degrees are represented with clock hours. The valid values are 0 to 11. For example, for the secondary side end of a transformer with vector group code of 'Dyn11', specify the connection kind as wye with neutral and specify the phase angle of the clock as 11. The clock value of the transformer end number specified as 1, is assumed to be zero. Note the transformer end number is not assumed to be the same as the terminal sequence number.
ratedS	0..1	ApparentPower	Normal apparent power rating. The attribute shall be a positive value. For a two-winding transformer the values for the high and low voltage sides shall be identical.
g	0..1	Conductance	Magnetizing branch conductance.
ratedU	1..1	Voltage	Rated voltage: phase-phase for three-phase windings, and either phase-phase or

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			phase-neutral for single-phase windings. A high voltage side, as given by TransformerEnd.endNumber, shall have a ratedU that is greater or equal than ratedU for the lower voltage sides.
g0 (ShortCircuit)	1..1	Conductance	Zero sequence magnetizing branch conductance (star-model).
r	1..1	Resistance	Resistance (star-model) of the transformer end. The attribute shall be equal or greater than zero for non-equivalent transformers.
r0 (ShortCircuit)	1..1	Resistance	Zero sequence series resistance (star-model) of the transformer end.
x	1..1	Reactance	Positive sequence series reactance (star-model) of the transformer end.
x0 (ShortCircuit)	1..1	Reactance	Zero sequence series reactance of the transformer end.
rground (ShortCircuit)	0..1	Resistance	see TransformerEnd
endNumber	1..1	Integer	see TransformerEnd
grounded (ShortCircuit)	1..1	Boolean	see TransformerEnd
xground (ShortCircuit)	0..1	Reactance	see TransformerEnd
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-221 shows all association ends of PowerTransformerEnd with other classes.

Table 3-221 - Association ends of Wires::PowerTransformerEnd with other classes

mult.	mult.	Class type name	Description
PowerTransformer	[1..1]	PowerTransformer	The ends of this power transformer. Association Based On : Wires:: PowerTransformer.PowerTransformer[0..1] ----- Wires:: PowerTransformerEnd.PowerTransformerEnd[0..*]
Terminal	1..1	Terminal	Inherited Association Based On : Wires:: TransformerEnd.TransformerEnd[0..*] ----- Core:: Terminal.Terminal[0..1]
BaseVoltage	1..1	BaseVoltage	Inherited Association Based On : Wires:: TransformerEnd.TransformerEnds[0..*]

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			----- Core:: BaseVoltage.BaseVoltage[0..1]

3.9.33 ProtectedSwitch(Abstract)

A ProtectedSwitch is a switching device that can be operated by ProtectionEquipment.

Inheritance path: ->Switch->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-222 shows all attributes of ProtectedSwitch.

Table 3-222 - Attributes of Wires::ProtectedSwitch

Attribute name	mult.	Attribute type	Description
normalOpen	1..1	Boolean	see Switch
ratedCurrent	0..1	CurrentFlow	see Switch
retained	1..1	Boolean	see Switch
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-223 shows all association ends of ConductingEquipment with other classes.

Table 3-223 - Association ends of Core::ConductingEquipment with other classes

mult.	mult.	Class type name	Description
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.9.34 RatioTapChanger

A tap changer that changes the voltage ratio impacting the voltage magnitude but not the phase angle across the transformer.

- The attribute TapChanger.ltcflag specifies whether or not a TapChanger has load tap changing capabilities. If the ltcFlag is true, the attribute "stepVoltageIncrement" is required.

Inheritance path: ->TapChanger->PowerSystemResource->IdentifiedObject

Table 3-224 shows all attributes of RatioTapChanger.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-224 - Attributes of Wires::RatioTapChanger

Attribute name	mult.	Attribute type	Description
tculControlMode	1..1	TransformerControlMode	Specifies the regulation control mode (voltage or reactive) of the RatioTapChanger.
stepVoltageIncrement	1..1	PerCent	Tap step increment, in per cent of nominal voltage, per step position.
highStep	1..1	Integer	see TapChanger
lowStep	1..1	Integer	see TapChanger
ltcFlag	1..1	Boolean	see TapChanger
neutralStep	1..1	Integer	see TapChanger
neutralU	1..1	Voltage	see TapChanger
normalStep	1..1	Integer	see TapChanger
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-225 shows all association ends of RatioTapChanger with other classes.

Table 3-225 - Association ends of Wires::RatioTapChanger with other classes

mult.	mult.	Class type name	Description
RatioTapChangerTable	[0..1]	RatioTapChangerTable	The ratio tap changer of this tap ratio table. Association Based On : Wires:: RatioTapChanger.RatioTapChanger[0..*] ----- Wires:: RatioTapChangerTable.RatioTapChangerTable[0..1]
TransformerEnd	[1..1]	TransformerEnd	Ratio tap changer associated with this transformer end. Association Based On : Wires:: TransformerEnd.TransformerEnd[1] ----- Wires:: RatioTapChanger.RatioTapChanger[0..1]
TapChangerControl	0..1	TapChangerControl	Inherited Association Based On : Wires:: TapChanger.TapChanger[0..*] ----- Wires:: TapChangerControl.TapChangerControl[0..1]

3.9.35 RatioTapChangerTable

Describes a curve for how the voltage magnitude and impedance varies with the tap step.

Inheritance path: ->IdentifiedObject

Table 3-226 shows all attributes of RatioTapChangerTable.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-226 - Attributes of Wires::RatioTapChangerTable

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

3.9.36 RatioTapChangerTablePoint

Describes each tap step in the ratio tap changer tabular curve.

Inheritance path: ->TapChangerTablePoint

Table 3-227 shows all attributes of RatioTapChangerTablePoint.

Table 3-227 - Attributes of Wires::RatioTapChangerTablePoint

Attribute name	mult.	Attribute type	Description
b	0..1	PerCent	see TapChangerTablePoint
g	0..1	PerCent	see TapChangerTablePoint
r	0..1	PerCent	see TapChangerTablePoint
ratio	0..1	Simple_Float	see TapChangerTablePoint
step	1..1	Integer	see TapChangerTablePoint
x	0..1	PerCent	see TapChangerTablePoint

Table 3-228 shows all association ends of RatioTapChangerTablePoint with other classes.

Table 3-228 - Association ends of Wires::RatioTapChangerTablePoint with other classes

mult.	mult.	Class type name	Description
RatioTapChangerTable	[1..1]	RatioTapChangerTable	Points of this table. Association Based On : Wires:: RatioTapChangerTablePoint.RatioTapChangerTablePoint[1..*] ----- Wires:: RatioTapChangerTable.RatioTapChangerTable[1]

3.9.37 ReactiveCapabilityCurve

Reactive power rating envelope versus the synchronous machine's active power, in both the generating and motoring modes. For each active power value there is a corresponding high and low reactive power limit value. Typically there will be a separate curve for each coolant condition, such as hydrogen pressure. The Y1 axis values represent reactive minimum and the Y2 axis values represent reactive maximum.

- ReactiveCapabilityCurves are not required if the reactive power limits of the SynchronousMachine do not vary with real power output. - By convention, the Y1 axis values represent reactive minimum and the Y2 axis values represent reactive maximum. - Because the x value will always be specified in MW and the y values will always be specified in MVAr, the xMultiplier, y1Multiplier, and y2Multiplier attributes do not need to be supplied.

Inheritance path: ->Curve->IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-229 shows all attributes of ReactiveCapabilityCurve.

Table 3-229 - Attributes of Wires::ReactiveCapabilityCurve

Attribute name	mult.	Attribute type	Description
curveStyle	1..1	CurveStyle	see Curve
xUnit	1..1	UnitSymbol	see Curve
y1Unit	1..1	UnitSymbol	see Curve
y2Unit	0..1	UnitSymbol	see Curve
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

3.9.38 RegulatingCondEq(Abstract)

A type of conducting equipment that can regulate a quantity (i.e. voltage or flow) at a specific point in the network.

Inheritance path: ->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-230 shows all attributes of RegulatingCondEq.

Table 3-230 - Attributes of Wires::RegulatingCondEq

Attribute name	mult.	Attribute type	Description
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-231 shows all association ends of RegulatingCondEq with other classes.

Table 3-231 - Association ends of Wires::RegulatingCondEq with other classes

mult.	mult.	Class type name	Description
RegulatingControl	[0..1]	RegulatingControl	The regulating control scheme in which this equipment participates. Association Based On : Wires:: RegulatingCondEq.RegulatingCondEq[0..*] ----- Wires:: RegulatingControl.RegulatingControl[0..1]
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core::

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.9.39 RegulatingControl

Specifies a set of equipment that works together to control a power system quantity such as voltage or flow.
Remote bus voltage control is possible by specifying the controlled terminal located at some place remote from the controlling equipment.
In case multiple equipment, possibly of different types, control same terminal there must be only one RegulatingControl at that terminal. The most specific subtype of RegulatingControl shall be used in case such equipment participate in the control, e.g. TapChangerControl for tap changers. For flow control load sign convention is used, i.e. positive sign means flow out from a TopologicalNode (bus) into the conducting equipment.

RegulatingControl targetrange and targetvalue are required if a RegulationSchedule is not provided.

Inheritance path: ->PowerSystemResource->IdentifiedObject

Table 3-232 shows all attributes of RegulatingControl.

Table 3-232 - Attributes of Wires::RegulatingControl

Attribute name	mult.	Attribute type	Description
mode	1..1	RegulatingControlModeKind	The regulating control mode presently available. This specification allows for determining the kind of regulation without need for obtaining the units from a schedule.
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-233 shows all association ends of RegulatingControl with other classes.

Table 3-233 - Association ends of Wires::RegulatingControl with other classes

mult.	mult.	Class type name	Description
Terminal	[1..1]	Terminal	The controls regulating this terminal. Association Based On : Core:: Terminal.Terminal[0..1] ----- Wires:: RegulatingControl.RegulatingControl[0..*]

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

3.9.40 RegulationSchedule

A pre-established pattern over time for a controlled variable, e.g., busbar voltage.

- By convention, ”value1” represents the target voltage or real power. “value2” is the deviation. A value1 of 100 and value2 of 1 means regulating to 100KV plus or minus 1KV. The range would be from 99 KV to 101 KV. Because the regulation values will be specified in either kV for voltage or MW for real power, the value1Multiplier and value2Multiplier attributes do not need to be specified.

Inheritance path: ->SeasonDayTypeSchedule->RegularIntervalSchedule->BasicIntervalSchedule->IdentifiedObject

Table 3-234 shows all attributes of RegulationSchedule.

Table 3-234 - Attributes of Wires::RegulationSchedule

Attribute name	mult.	Attribute type	Description
timeStep	1..1	Seconds	see RegularIntervalSchedule
endTime	1..1	DateTime	see RegularIntervalSchedule
startTime	1..1	DateTime	see BasicIntervalSchedule
value1Unit	1..1	UnitSymbol	see BasicIntervalSchedule
value2Unit	0..1	UnitSymbol	see BasicIntervalSchedule
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-235 shows all association ends of RegulationSchedule with other classes.

Table 3-235 - Association ends of Wires::RegulationSchedule with other classes

mult.	mult.	Class type name	Description
RegulatingControl (Operation)	[1..1]	RegulatingControl	Regulating controls that have this Schedule. Association Based On : Wires:: RegulationSchedule.RegulationSchedule[0..*] ----- Wires:: RegulatingControl.RegulatingControl[1]
Season (Operation)	1..1	Season	Inherited Association Based On : LoadModel:: Season.Season[0..1] ----- LoadModel:: SeasonDayTypeSchedule.SeasonDayTypeSchedules[0..*]
DayType (Operation)	1..1	DayType	Inherited Association Based On : LoadModel:: DayType.DayType[0..1] ----- LoadModel:: SeasonDayTypeSchedule.SeasonDayTypeSchedules[0..*]

3.9.41 RotatingMachine(Abstract)

A rotating machine which may be used as a generator or motor.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

RotatingMachine.ratedS is required in case of dynamic data exchange.

If one RotatingMachine is associated with one GeneratingUnit the flag Equipment.aggregate should be consistent in case it is provided at both RotatingMachine and GeneratingUnit.

Inheritance path: ->RegulatingCondEq->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-236 shows all attributes of RotatingMachine.

Table 3-236 - Attributes of Wires::RotatingMachine

Attribute name	mult.	Attribute type	Description
ratedPowerFactor	0..1	Simple_Float	Power factor (nameplate data). It is primarily used for short circuit data exchange according to IEC 60909.
ratedS	0..1	ApparentPower	Nameplate apparent power rating for the unit. The attribute shall have a positive value.
ratedU	0..1	Voltage	Rated voltage (nameplate data, Ur in IEC 60909-0). It is primarily used for short circuit data exchange according to IEC 60909.
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-237 shows all association ends of RotatingMachine with other classes.

Table 3-237 - Association ends of Wires::RotatingMachine with other classes

mult.	mult.	Class type name	Description
GeneratingUnit	[0..1]	GeneratingUnit	A synchronous machine may operate as a generator and as such becomes a member of a generating unit. Association Based On : Production:: GeneratingUnit.GeneratingUnit[0..1] ----- Wires:: RotatingMachine.RotatingMachine[1..*]
RegulatingControl	0..1	RegulatingControl	Inherited Association Based On : Wires:: RegulatingCondEq.RegulatingCondEq[0..*] ----- Wires:: RegulatingControl.RegulatingControl[0..1]
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On :

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			Core:: BaseVoltage.BaseVoltage[0..1] --- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] --- Core:: Equipment.Equipments[0..*]

3.9.42 SeriesCompensator

A Series Compensator is a series capacitor or reactor or an AC transmission line without charging susceptance. It is a two terminal device.

Each SeriesCompensator is required to have an association to a BaseVoltage.

The positive sequence resistance (r) and reactance (x) are the same as negative sequence. Therefore negative sequence data is not modeled.

Inheritance path: ->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-238 shows all attributes of SeriesCompensator.

Table 3-238 - Attributes of Wires::SeriesCompensator

Attribute name	mult.	Attribute type	Description
r	1..1	Resistance	Positive sequence resistance.
r0 (ShortCircuit)	1..1	Resistance	Zero sequence resistance.
x	1..1	Reactance	Positive sequence reactance.
x0 (ShortCircuit)	1..1	Reactance	Zero sequence reactance.
varistorPresent	1..1	Boolean	Describe if a metal oxide varistor (mov) for over voltage protection is configured at the series compensator.
varistorRatedCurrent	1..1	CurrentFlow	The maximum current the varistor is designed to handle at specified duration.
varistorVoltageThreshold	1..1	Voltage	The dc voltage at which the varistor start conducting.
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-239 shows all association ends of ConductingEquipment with other classes.

Table 3-239 - Association ends of Core::ConductingEquipment with other classes

mult.	mult.	Class type name	Description
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.9.43 ShuntCompensator(Abstract)

A shunt capacitor or reactor or switchable bank of shunt capacitors or reactors. A section of a shunt compensator is an individual capacitor or reactor. A negative value for reactivePerSection indicates that the compensator is a reactor. ShuntCompensator is a single terminal device. Ground is implied.

Inheritance path: ->RegulatingCondEq->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-240 shows all attributes of ShuntCompensator.

Table 3-240 - Attributes of Wires::ShuntCompensator

Attribute name	mult.	Attribute type	Description
aVRDelay	0..1	Seconds	Time delay required for the device to be connected or disconnected by automatic voltage regulation (AVR).
grounded	0..1	Boolean	Used for Yn and Zn connections. True if the neutral is solidly grounded.
maximumSections	1..1	Integer	The maximum number of sections that may be switched in.
nomU	1..1	Voltage	The voltage at which the nominal reactive power may be calculated. This should normally be within 10% of the voltage at which the capacitor is connected to the network.
normalSections	1..1	Integer	The normal number of sections switched in.
switchOnCount	0..1	Integer	The switch on count since the capacitor

			count was last reset or initialized.
switchOnDate	0..1	DateTime	The date and time when the capacitor bank was last switched on.
voltageSensitivity	0..1	VoltagePerReactivePower	Voltage sensitivity required for the device to regulate the bus voltage, in voltage/reactive power.
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-241 shows all association ends of RegulatingCondEq with other classes.

Table 3-241 - Association ends of Wires::RegulatingCondEq with other classes

mult.	mult.	Class type name	Description
RegulatingControl	0..1	RegulatingControl	Inherited Association Based On : Wires:: RegulatingCondEq.RegulatingCondEq[0..*] ----- Wires:: RegulatingControl.RegulatingControl[0..1]
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.9.44 StaticVarCompensator

A facility for providing variable and controllable shunt reactive power. The SVC typically consists of a stepdown transformer, filter, thyristor-controlled reactor, and thyristor-switched capacitor arms.
The SVC may operate in fixed MVar output mode or in voltage control mode. When in voltage control mode, the output of the SVC will be proportional to the deviation of voltage at the controlled bus from the voltage setpoint. The SVC characteristic slope defines the proportion. If the voltage at the controlled bus is equal to the voltage setpoint, the SVC MVar output is zero.

Naming Convention and main containership hierarchy is : GeographicalRegion/SubGeographicalRegion/Substation/VoltageLevel

- The value of the “inductiveRating” attribute must always be negative. - The value of the “capactiveRating” attribute must always be positive. - It is required to use RegulatingControl for SVC.

Inheritance path: ->RegulatingCondEq->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-242 shows all attributes of StaticVarCompensator.

Table 3-242 - Attributes of Wires::StaticVarCompensator

Attribute name	mult.	Attribute type	Description
capacitiveRating	1..1	Reactance	Maximum available capacitive reactance.
inductiveRating	1..1	Reactance	Maximum available inductive reactance.
slope	1..1	VoltagePerReactivePower	The characteristics slope of an SVC defines how the reactive power output changes in proportion to the difference between the regulated bus voltage and the voltage setpoint.
sVCControlMode	1..1	SVCControlMode	SVC control mode.
voltageSetPoint	1..1	Voltage	The reactive power output of the SVC is proportional to the difference between the voltage at the regulated bus and the voltage setpoint. When the regulated bus voltage is equal to the voltage setpoint, the reactive power output is zero.
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-243 shows all association ends of RegulatingCondEq with other classes.

Table 3-243 - Association ends of Wires::RegulatingCondEq with other classes

mult.	mult.	Class type name	Description
RegulatingControl	0..1	RegulatingControl	Inherited Association Based On : Wires:: RegulatingCondEq.RegulatingCondEq[0..*] ----- Wires:: RegulatingControl.RegulatingControl[0..1]

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.9.45 Switch

A generic device designed to close, or open, or both, one or more electric circuits. All switches are two terminal devices including grounding switches.

For switching Devices, Naming Convention and main containership hierarchy is : GeographicalRegion/SubGeographicalRegion/Substation/VoltageLevel or GeographicalRegion/SubGeographicalRegion/SubstationàVoltageLevel/Bay

Instance data to be exchanged MUST make use of the most detailed class possible. The class Switch should only be used if the information to determine the more detailed class (Breaker, Disconnector, etc.) is not available.

Inheritance path: ->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-244 shows all attributes of Switch.

Table 3-244 - Attributes of Wires::Switch

Attribute name	mult.	Attribute type	Description
normalOpen	1..1	Boolean	The attribute is used in cases when no Measurement for the status value is present. If the Switch has a status measurement the Discrete.normalValue is expected to match with the Switch.normalOpen.
ratedCurrent	0..1	CurrentFlow	The maximum continuous current carrying capacity in amps governed by the device material and construction.
retained	1..1	Boolean	Branch is retained in a bus branch model. The flow through retained switches will normally be calculated in power flow.
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-245 shows all association ends of ConductingEquipment with other classes.

Table 3-245 - Association ends of Core::ConductingEquipment with other classes

mult.	mult.	Class type name	Description
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.9.46 SwitchSchedule

A schedule of switch positions. If RegularTimePoint.value1 is 0, the switch is open. If 1, the switch is closed.

Inheritance path: ->SeasonDayTypeSchedule->RegularIntervalSchedule->BasicIntervalSchedule->IdentifiedObject

Table 3-246 shows all attributes of SwitchSchedule.

Table 3-246 - Attributes of Wires::SwitchSchedule

Attribute name	mult.	Attribute type	Description
timeStep	1..1	Seconds	see RegularIntervalSchedule
endTime	1..1	DateTime	see RegularIntervalSchedule
startTime	1..1	DateTime	see BasicIntervalSchedule
value1Unit	1..1	UnitSymbol	see BasicIntervalSchedule
value2Unit	0..1	UnitSymbol	see BasicIntervalSchedule
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-247 shows all association ends of SwitchSchedule with other classes.

Table 3-247 - Association ends of Wires::SwitchSchedule with other classes

mult.	mult.	Class type name	Description
Switch (Operation)	[1..1]	Switch	A Switch can be associated with SwitchSchedules. Association Based On : Wires::

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			Switch.Switch[1] ----- Wires:: SwitchSchedule.SwitchSchedules[0..*]
Season (Operation)	1..1	Season	Inherited Association Based On : LoadModel:: Season.Season[0..1] ----- LoadModel:: SeasonDayTypeSchedule.SeasonDayTypeSchedules[0..*]
DayType (Operation)	1..1	DayType	Inherited Association Based On : LoadModel:: DayType.DayType[0..1] ----- LoadModel:: SeasonDayTypeSchedule.SeasonDayTypeSchedules[0..*]

3.9.47 SynchronousMachine

An electromechanical device that operates with shaft rotating synchronously with the network. It is a single machine operating either as a generator or synchronous condenser or pump.

OCL constraint: If a synchronous condenser is being modelled so that there is no capability for real power output, the SynchronousMachine is not required to be associated with a GeneratingUnit. In this case, the type and operatingMode attributes must both be set to "cond

The attribute satDirectTransX is not required for short circuit data exchange according to IEC 60909.

Naming Convention and main containership hierarchy is : GeographicalRegion/SubGeographicalRegion/Substation/VoltageLevel
If SynchronousMachine.earthing is true, then SynchronousMachine.earthingStarPointR and SynchronousMachine.earthingStarPointX are required.
- If a SynchronousMachine is not associated with a ReactiveCapabilityCurve, then the minQ and maxQ attributes will be used. - If a ReactiveCapabilityCurve is supplied, then the minQ and maxQ attributes are not required. - If a synchronous condenser is being modeled so that there is no capability for real power output, the SynchronousMachine is not required to be associated with a GeneratingUnit. In this case, the type and operatingMode attributes must both be set to “condenser”.

Inheritance path: ->RotatingMachine->RegulatingCondEq->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-248 shows all attributes of SynchronousMachine.

Table 3-248 - Attributes of Wires::SynchronousMachine

Attribute name	mult.	Attribute type	Description
earthing (ShortCircuit)	1..1	Boolean	Indicates whether or not the generator is earthed. Used for short circuit data exchange according to IEC 60909
earthingStarPointR (ShortCircuit)	0..1	Resistance	Generator star point earthing resistance (Re). Used for short circuit data exchange according to IEC 60909
earthingStarPointX (ShortCircuit)	0..1	Reactance	Generator star point earthing reactance (Xe). Used for short circuit data exchange according to IEC 60909

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

ikk (ShortCircuit)	0..1	CurrentFlow	Steady-state short-circuit current (in A for the profile) of generator with compound excitation during 3-phase short circuit. - Ikk=0: Generator with no compound excitation. - Ikk?0: Generator with compound excitation. Ikk is used to calculate the minimum steady-state short-circuit current for generators with compound excitation (Section 4.6.1.2 in the IEC 60909-0) Used only for single fed short circuit on a generator. (Section 4.3.4.2. in the IEC 60909-0)
maxQ	0..1	ReactivePower	Maximum reactive power limit. This is the maximum (nameplate) limit for the unit.
minQ	0..1	ReactivePower	Minimum reactive power limit for the unit.
mu (ShortCircuit)	0..1	Simple_Float	Factor to calculate the breaking current (Section 4.5.2.1 in the IEC 60909-0). Used only for single fed short circuit on a generator (Section 4.3.4.2. in the IEC 60909-0).
qPercent	0..1	PerCent	Percent of the coordinated reactive control that comes from this machine.
r0 (ShortCircuit)	1..1	PU	Zero sequence resistance of the synchronous machine.
r2 (ShortCircuit)	1..1	PU	Negative sequence resistance.
satDirectSubtransX (ShortCircuit)	1..1	PU	Direct-axis subtransient reactance saturated, also known as Xd"sat.
satDirectSyncX (ShortCircuit)	0..1	PU	Direct-axes saturated synchronous reactance (xdsat); reciprocal of short-circuit ration. Used for short circuit data exchange, only for single fed short circuit on a generator. (Section 4.3.4.2. in the IEC 60909-0).
satDirectTransX (ShortCircuit)	0..1	PU	Saturated Direct-axis transient reactance.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			The attribute is primarily used for short circuit calculations according to ANSI.
shortCircuitRotorType (ShortCircuit)	0..1	ShortCircuitRotorKind	Type of rotor, used by short circuit applications, only for single fed short circuit according to IEC 60909.
type	1..1	SynchronousMachineKind	Modes that this synchronous machine can operate in.
voltageRegulationRange (ShortCircuit)	0..1	PerCent	Range of generator voltage regulation (PG in the IEC 60909-0) used for calculation of the impedance correction factor KG defined in IEC 60909-0 This attribute is used to describe the operating voltage of the generating unit.
r (ShortCircuit)	1..1	Resistance	Equivalent resistance (RG) of generator. RG is considered for the calculation of all currents, except for the calculation of the peak current ip. Used for short circuit data exchange according to IEC 60909
x0 (ShortCircuit)	1..1	PU	Zero sequence reactance of the synchronous machine.
x2 (ShortCircuit)	1..1	PU	Negative sequence reactance.
ratedPowerFactor	0..1	Simple_Float	see RotatingMachine
ratedS	0..1	ApparentPower	see RotatingMachine
ratedU	0..1	Voltage	see RotatingMachine
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-249 shows all association ends of SynchronousMachine with other classes.

Table 3-249 - Association ends of Wires::SynchronousMachine with other classes

mult.	mult.	Class type name	Description
InitialReactiveCapabilityCurve	[0..1]	ReactiveCapabilityCurve	Synchronous machines using this curve as default. Association Based On :

			Wires:: ReactiveCapabilityCurve.InitialReactiveCapabilityCurve[0..1] - ---- Wires:: SynchronousMachine.InitiallyUsedBySynchronousMachines[1. .*]
GeneratingUnit	0..1	GeneratingUnit	Inherited Association Based On : Production:: GeneratingUnit.GeneratingUnit[0..1] ----- Wires:: RotatingMachine.RotatingMachine[1..*]
RegulatingControl	0..1	RegulatingControl	Inherited Association Based On : Wires:: RegulatingCondEq.RegulatingCondEq[0..*] ----- Wires:: RegulatingControl.RegulatingControl[0..1]
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.9.48 TapChanger(Abstract)

Mechanism for changing transformer winding tap positions.

Inheritance path: ->PowerSystemResource->IdentifiedObject

Table 3-250 shows all attributes of TapChanger.

Table 3-250 - Attributes of Wires::TapChanger

Attribute name	mult.	Attribute type	Description
highStep	1..1	Integer	Highest possible tap step position, advance from neutral. The attribute shall be greater than lowStep.
lowStep	1..1	Integer	Lowest possible tap step position, retard from neutral
ltcFlag	1..1	Boolean	Specifies whether or not a TapChanger has load tap changing capabilities.
neutralStep	1..1	Integer	The neutral tap step position for this winding. The attribute shall be equal or greater than lowStep and equal or less than highStep.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

neutralU	1..1	Voltage	Voltage at which the winding operates at the neutral tap setting.
normalStep	1..1	Integer	The tap step position used in "normal" network operation for this winding. For a "Fixed" tap changer indicates the current physical tap setting. The attribute shall be equal or greater than lowStep and equal or less than highStep.
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-251 shows all association ends of TapChanger with other classes.

Table 3-251 - Association ends of Wires::TapChanger with other classes

mult.	mult.	Class type name	Description
TapChangerControl	[0..1]	TapChangerControl	The tap changers that participates in this regulating tap control scheme. Association Based On : Wires:: TapChanger.TapChanger[0..*] ----- Wires:: TapChangerControl.TapChangerControl[0..1]

3.9.49 TapChangerControl

Describes behavior specific to tap changers, e.g. how the voltage at the end of a line varies with the load level and compensation of the voltage drop by tap adjustment.

Inheritance path: ->RegulatingControl->PowerSystemResource->IdentifiedObject

Table 3-252 shows all attributes of TapChangerControl.

Table 3-252 - Attributes of Wires::TapChangerControl

Attribute name	mult.	Attribute type	Description
mode	1..1	RegulatingControlModeKind	see RegulatingControl
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-253 shows all association ends of RegulatingControl with other classes.

Table 3-253 - Association ends of Wires::RegulatingControl with other classes

mult.	mult.	Class type name	Description
Terminal	1..1	Terminal	Inherited Association Based On : Core:: Terminal.Terminal[0..1] ----- Wires:: RegulatingControl.RegulatingControl[0..*]

3.9.50 TapChangerTablePoint

Table 3-254 shows all attributes of TapChangerTablePoint.

Table 3-254 - Attributes of Wires::TapChangerTablePoint

Attribute name	mult.	Attribute type	Description
b	0..1	PerCent	The magnetizing branch susceptance deviation in percent of nominal value. The actual susceptance is calculated as follows: calculated magnetizing susceptance = b(nominal) * (1 + b(from this class)/100). The b(nominal) is defined as the static magnetizing susceptance on the associated power transformer end or ends. This model assumes the star impedance (pi model) form.
g	0..1	PerCent	The magnetizing branch conductance deviation in percent of nominal value. The actual conductance is calculated as follows: calculated magnetizing conductance = g(nominal) * (1 + g(from this class)/100). The g(nominal) is defined as the static magnetizing conductance on the associated power transformer end or ends. This model assumes the star impedance (pi model) form.
r	0..1	PerCent	The resistance deviation in percent of nominal value. The actual reactance is calculated as follows: calculated resistance = r(nominal) * (1 + r(from this class)/100). The r(nominal) is defined as the static resistance on the associated power

			transformer end or ends. This model assumes the star impedance (pi model) form.
ratio	0..1	Simple_Float	The voltage ratio in per unit. Hence this is a value close to one.
step	1..1	Integer	The tap step.
x	0..1	PerCent	The series reactance deviation in percent of nominal value. The actual reactance is calculated as follows: calculated reactance = x(nominal) * (1 + x(from this class)/100). The x(nominal) is defined as the static series reactance on the associated power transformer end or ends. This model assumes the star impedance (pi model) form.

3.9.51 TapSchedule

A pre-established pattern over time for a tap step.

Inheritance path: ->SeasonDayTypeSchedule->RegularIntervalSchedule->BasicIntervalSchedule->IdentifiedObject

Table 3-255 shows all attributes of TapSchedule.

Table 3-255 - Attributes of Wires::TapSchedule

Attribute name	mult.	Attribute type	Description
timeStep	1..1	Seconds	see RegularIntervalSchedule
endTime	1..1	DateTime	see RegularIntervalSchedule
startTime	1..1	DateTime	see BasicIntervalSchedule
value1Unit	1..1	UnitSymbol	see BasicIntervalSchedule
value2Unit	0..1	UnitSymbol	see BasicIntervalSchedule
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-256 shows all association ends of TapSchedule with other classes.

Table 3-256 - Association ends of Wires::TapSchedule with other classes

mult.	mult.	Class type name	Description

TapChanger (Operation)	[1..1]	TapChanger	A TapChanger can have TapSchedules. Association Based On : Wires:: TapSchedule.TapSchedules[0..*] ----- Wires:: TapChanger.TapChanger[1]
Season (Operation)	1..1	Season	Inherited Association Based On : LoadModel:: Season.Season[0..1] ----- LoadModel:: SeasonDayTypeSchedule.SeasonDayTypeSchedules[0..*]
DayType (Operation)	1..1	DayType	Inherited Association Based On : LoadModel:: DayType.DayType[0..1] ----- LoadModel:: SeasonDayTypeSchedule.SeasonDayTypeSchedules[0..*]

3.9.52 TransformerEnd(Abstract)

A conducting connection point of a power transformer. It corresponds to a physical transformer winding terminal. In earlier CIM versions, the TransformerWinding class served a similar purpose, but this class is more flexible because it associates to terminal but is not a specialization of ConductingEquipment.

If TransformerEnd.grounded is true, then TransformerEnd.rground and TransformerEnd.xground are required

Inheritance path: ->IdentifiedObject

Table 3-257 shows all attributes of TransformerEnd.

Table 3-257 - Attributes of Wires::TransformerEnd

Attribute name	mult.	Attribute type	Description
rground (ShortCircuit)	0..1	Resistance	(for Yn and Zn connections) Resistance part of neutral impedance where 'grounded' is true.
endNumber	1..1	Integer	Number for this transformer end, corresponding to the end's order in the power transformer vector group or phase angle clock number. Highest voltage winding should be 1. Each end within a power transformer should have a unique subsequent end number. Note the transformer end number need not match the terminal sequence number.
grounded (ShortCircuit)	1..1	Boolean	(for Yn and Zn connections) True if the neutral is solidly grounded.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

xground (ShortCircuit)	0..1	Reactance	(for Yn and Zn connections) Reactive part of neutral impedance where 'grounded' is true.
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-258 shows all association ends of TransformerEnd with other classes.

Table 3-258 - Association ends of Wires::TransformerEnd with other classes

mult.	mult.	Class type name	Description
Terminal	[1..1]	Terminal	Terminal of the power transformer to which this transformer end belongs. Association Based On : Wires:: TransformerEnd.TransformerEnd[0..*] ----- Core:: Terminal.Terminal[0..1]
BaseVoltage	[1..1]	BaseVoltage	Base voltage of the transformer end. This is essential for PU calculation. Association Based On : Wires:: TransformerEnd.TransformerEnds[0..*] ----- Core:: BaseVoltage.BaseVoltage[0..1]

3.9.53 Enumerations

3.9.53.1 PetersenCoilModeKind

The mode of operation for a Petersen coil.

Literal	Description	Code
fixed	Fixed position.	
manual	Manual positioning.	
automaticPositioning	Automatic positioning.	

3.9.53.2 RegulatingControlModeKind

The kind of regulation model. For example regulating voltage, reactive power, active power, etc.

Literal	Description	Code
voltage	Voltage is specified.	
activePower	Active power is specified.	
reactivePower	Reactive power is specified.	
currentFlow	Current flow is specified.	
admittance	Admittance is specified.	
timeScheduled	Control switches on/off by time of day. The times may change on the weekend, or in different seasons.	

temperature	Control switches on/off based on the local temperature (i.e., a thermostat).	
powerFactor	Power factor is specified.	

3.9.53.3 ShortCircuitRotorKind

Type of rotor, used by short circuit applications.

Literal	Description	Code
salientPole1	Salient pole 1 in the IEC 60909	
salientPole2	Salient pole 2 in IEC 60909	
turboSeries1	Turbo Series 1 in the IEC 60909	
turboSeries2	Turbo series 2 in IEC 60909	

3.9.53.4 SVCControlMode

Static VAr Compensator control mode.

Literal	Description	Code
reactivePower		
voltage		

3.9.53.5 SynchronousMachineKind

Synchronous machine type.

Literal	Description	Code
generator		
condenser		
generatorOrCondenser		
motor		
generatorOrMotor		
motorOrCondenser		
generatorOrCondenserOrMotor		

3.9.53.6 TransformerControlMode

Control modes for a transformer.

Literal	Description	Code
volt	Voltage control	
reactive	Reactive power flow control	

3.9.53.7 WindingConnection

Winding connection type.

Literal	Description	Code
D	Delta	
Y	Wye	

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Z	ZigZag	
Yn	Wye, with neutral brought out for grounding.	
Zn	ZigZag, with neutral brought out for grounding.	
A	Autotransformer common winding	
I	Independent winding, for single-phase connections	

3.10 LoadModel

3.10.1 General

This package is responsible for modeling the energy consumers and the system load as curves and associated curve data. Special circumstances that may affect the load, such as seasons and daytypes, are also included here.
This information is used by Load Forecasting and Load Management.

Figure 3-21 shows the diagram.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

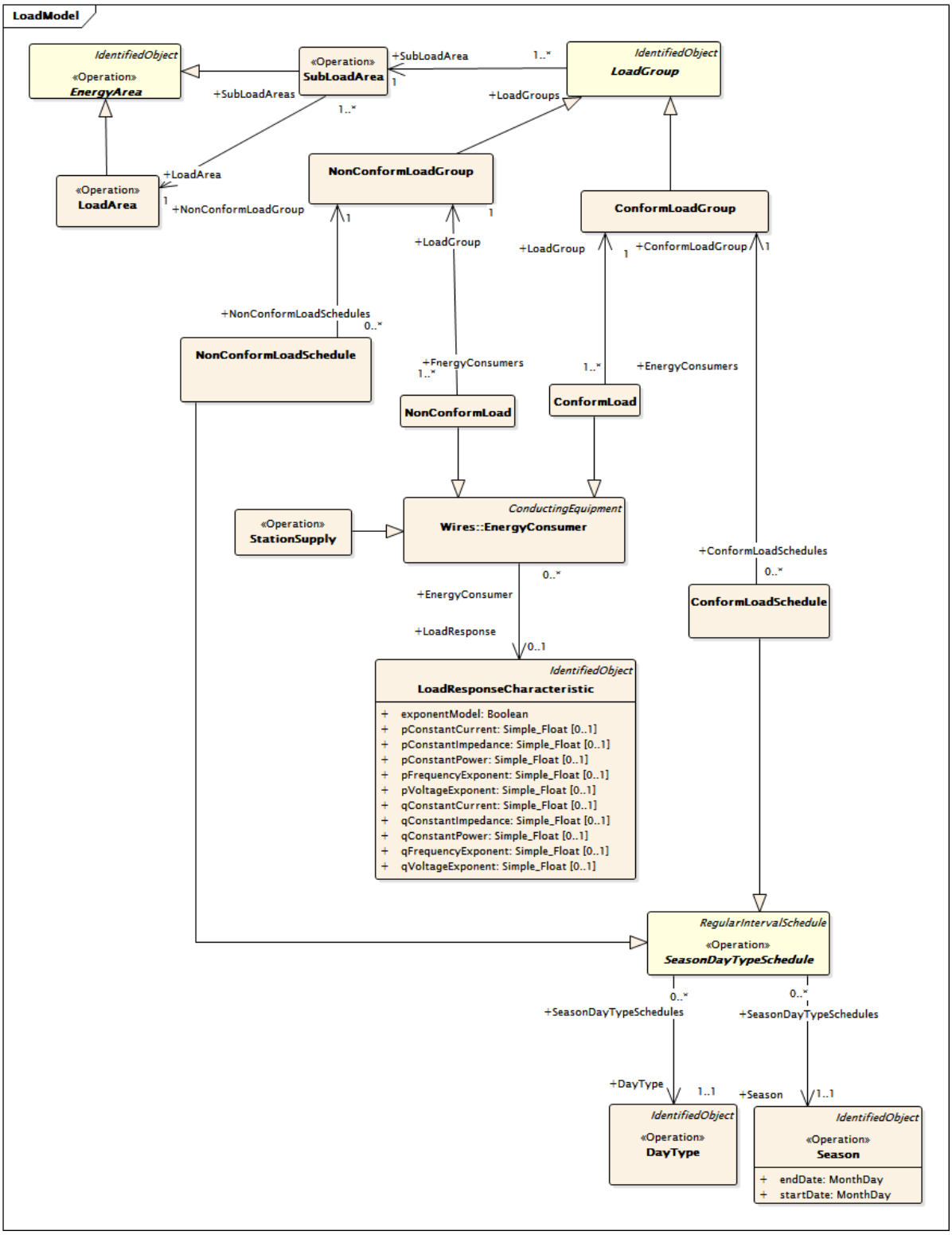


Figure 3-21 - diagram LoadModel

Figure 3-22 shows the diagram.

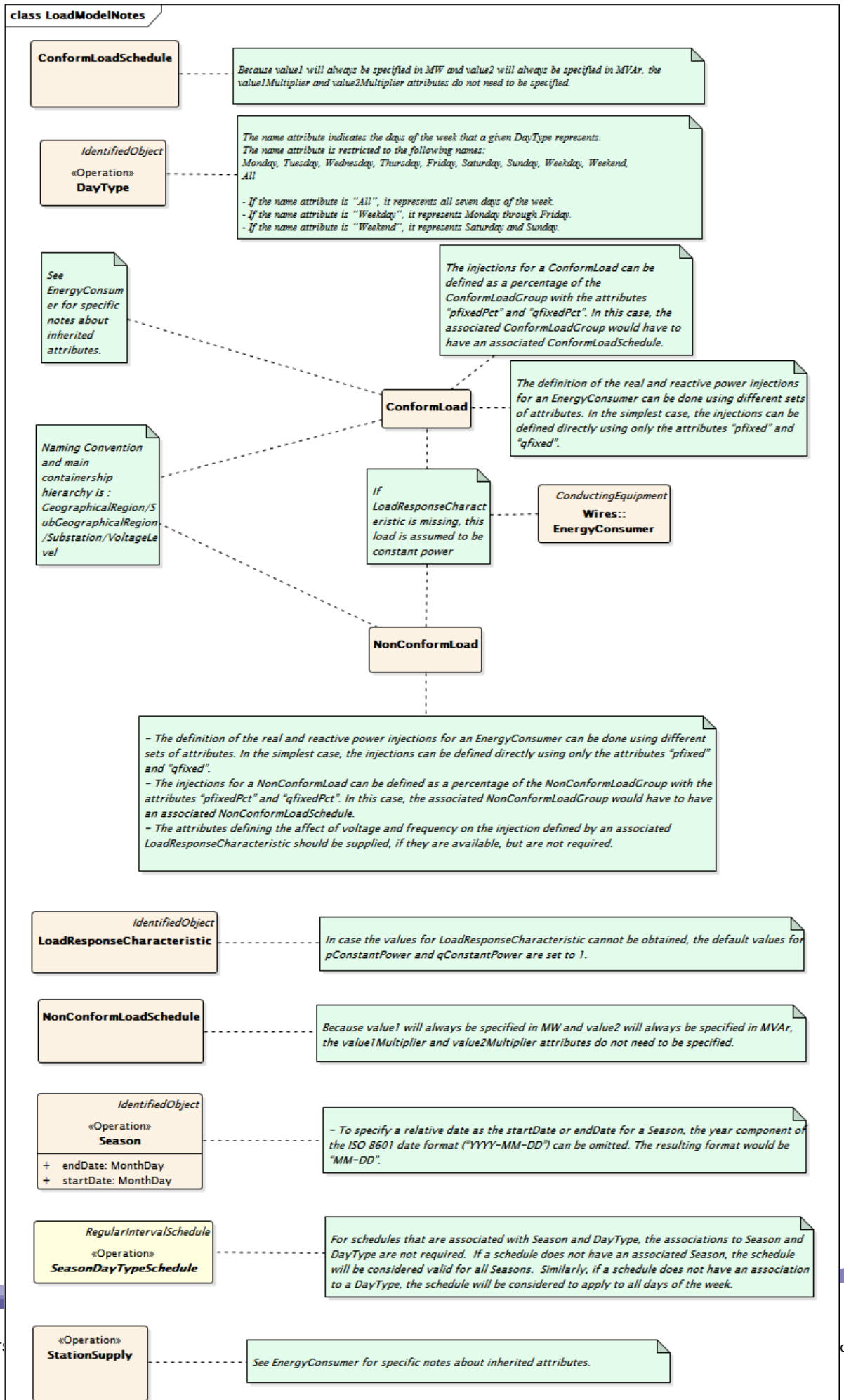

class LoadModelNotes
ConformLoadSchedule
Because value1 will always be specified in MW and value2 will always be specified in MVAr, the value1Multiplier and value2Multiplier attributes do not need to be specified.
IdentifiedObject
«Operation»
DayType
The name attribute indicates the days of the week that a given DayType represents.
The name attribute is restricted to the following names:
Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday, Weekday, Weekend, All
- If the name attribute is "All", it represents all seven days of the week.
- If the name attribute is "Weekday", it represents Monday through Friday.
- If the name attribute is "Weekend", it represents Saturday and Sunday.
See EnergyConsumer for specific notes about inherited attributes.
The injections for a ConformLoad can be defined as a percentage of the ConformLoadGroup with the attributes "pfixedPct" and "qfixedPct". In this case, the associated ConformLoadGroup would have to have an associated ConformLoadSchedule.
ConformLoad
The definition of the real and reactive power injections for an EnergyConsumer can be done using different sets of attributes. In the simplest case, the injections can be defined directly using only the attributes "pfixed" and "qfixed".
Naming Convention and main containership hierarchy is : GeographicalRegion/SubGeographicalRegion/Substation/VoltageLevel
If LoadResponseCharacteristic is missing, this load is assumed to be constant power
ConductingEquipment
Wires::
EnergyConsumer
NonConformLoad
- The definition of the real and reactive power injections for an EnergyConsumer can be done using different sets of attributes. In the simplest case, the injections can be defined directly using only the attributes "pfixed" and "qfixed".
- The injections for a NonConformLoad can be defined as a percentage of the NonConformLoadGroup with the attributes "pfixedPct" and "qfixedPct". In this case, the associated NonConformLoadGroup would have to have an associated NonConformLoadSchedule.
- The attributes defining the affect of voltage and frequency on the injection defined by an associated LoadResponseCharacteristic should be supplied, if they are available, but are not required.
IdentifiedObject
LoadResponseCharacteristic
In case the values for LoadResponseCharacteristic cannot be obtained, the default values for pConstantPower and qConstantPower are set to 1.
NonConformLoadSchedule
Because value1 will always be specified in MW and value2 will always be specified in MVAr, the value1Multiplier and value2Multiplier attributes do not need to be specified.
IdentifiedObject
«Operation»
Season
+ endDate: MonthDay
+ startDate: MonthDay
- To specify a relative date as the startDate or endDate for a Season, the year component of the ISO 8601 date format ("YYYY-MM-DD") can be omitted. The resulting format would be "MM-DD".
RegularIntervalSchedule
«Operation»
SeasonDayTypeSchedule
For schedules that are associated with Season and DayType, the associations to Season and DayType are not required. If a schedule does not have an associated Season, the schedule will be considered valid for all Seasons. Similarly, if a schedule does not have an association to a DayType, the schedule will be considered to apply to all days of the week.
«Operation»
StationSupply
See EnergyConsumer for specific notes about inherited attributes.

Figure 3-22 - diagram LoadModelNotes

3.10.2 ConformLoad

ConformLoad represent loads that follow a daily load change pattern where the pattern can be used to scale the load with a system load.

The injections for a ConformLoad can be defined as a percentage of the ConformLoadGroup with the attributes “pfixedPct” and “qfixedPct”. In this case, the associated ConformLoadGroup would have to have an associated ConformLoadSchedule.

The definition of the real and reactive power injections for an EnergyConsumer can be done using different sets of attributes. In the simplest case, the injections can be defined directly using only the attributes “pfixed” and “qfixed”.
Naming Convention and main containership hierarchy is : GeographicalRegion/SubGeographicalRegion/Substation/VoltageLevel
See EnergyConsumer for specific notes about inherited attributes.
If LoadResponseCharacteristic is missing, this load is assumed to be constant power

Inheritance path: ->EnergyConsumer->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-259 shows all attributes of ConformLoad.

Table 3-259 - Attributes of LoadModel::ConformLoad

Attribute name	mult.	Attribute type	Description
pfixed (Operation)	0..1	ActivePower	see EnergyConsumer
pfixedPct (Operation)	0..1	PerCent	see EnergyConsumer
qfixed (Operation)	0..1	ReactivePower	see EnergyConsumer
qfixedPct (Operation)	0..1	PerCent	see EnergyConsumer
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-260 shows all association ends of ConformLoad with other classes.

Table 3-260 - Association ends of LoadModel::ConformLoad with other classes

mult.	mult.	Class type name	Description
LoadGroup	[1..1]	ConformLoadGroup	Group of this ConformLoad. Association Based On : LoadModel:: ConformLoadGroup.LoadGroup[0..1] ----- LoadModel:: ConformLoad.EnergyConsumers[0..*]
LoadResponse	0..1	LoadResponseCharacteristic	Inherited Association Based On : Wires:: EnergyConsumer.EnergyConsumer[0..*] ----- LoadModel::

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			LoadResponseCharacteristic.LoadResponse[0..1]
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.10.3 ConformLoadGroup

A group of loads conforming to an allocation pattern.

Inheritance path: ->LoadGroup->IdentifiedObject

Table 3-261 shows all attributes of ConformLoadGroup.

Table 3-261 - Attributes of LoadModel::ConformLoadGroup

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-262 shows all association ends of LoadGroup with other classes.

Table 3-262 - Association ends of LoadModel::LoadGroup with other classes

mult.	mult.	Class type name	Description
SubLoadArea	1..1	SubLoadArea	Inherited Association Based On : LoadModel:: SubLoadArea.SubLoadArea[1] ----- LoadModel:: LoadGroup.LoadGroups[1..*]

3.10.4 ConformLoadSchedule

A curve of load versus time (X-axis) showing the active power values (Y1-axis) and reactive power (Y2-axis) for each unit of the period covered. This curve represents a typical pattern of load over the time period for a given day type and season.

Because value1 will always be specified in MW and value2 will always be specified in MVAr, the value1Multiplier and value2Multiplier attributes do not need to be specified.

Inheritance path: ->SeasonDayTypeSchedule->RegularIntervalSchedule->BasicIntervalSchedule->IdentifiedObject

Table 3-263 shows all attributes of ConformLoadSchedule.

Table 3-263 - Attributes of LoadModel::ConformLoadSchedule

Attribute name	mult.	Attribute type	Description
timeStep	1..1	Seconds	see RegularIntervalSchedule
endTime	1..1	DateTime	see RegularIntervalSchedule
startTime	1..1	DateTime	see BasicIntervalSchedule
value1Unit	1..1	UnitSymbol	see BasicIntervalSchedule
value2Unit	0..1	UnitSymbol	see BasicIntervalSchedule
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-264 shows all association ends of ConformLoadSchedule with other classes.

Table 3-264 - Association ends of LoadModel::ConformLoadSchedule with other classes

mult.	mult.	Class type name	Description
ConformLoadGroup	[1..1]	ConformLoadGroup	The ConformLoadGroup where the ConformLoadSchedule belongs. Association Based On : LoadModel:: ConformLoadGroup.ConformLoadGroup[1] ----- LoadModel:: ConformLoadSchedule.ConformLoadSchedules[1..*]
Season (Operation)	1..1	Season	Inherited Association Based On : LoadModel:: Season.Season[0..1] ----- LoadModel:: SeasonDayTypeSchedule.SeasonDayTypeSchedules[0..*]
DayType (Operation)	1..1	DayType	Inherited Association Based On : LoadModel:: DayType.DayType[0..1] ----- LoadModel:: SeasonDayTypeSchedule.SeasonDayTypeSchedules[0..*]

3.10.5 DayType

Group of similar days. For example it could be used to represent weekdays, weekend, or holidays.

The name attribute indicates the days of the week that a given DayType represents. The name attribute is restricted to the following names: Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday, Weekday, Weekend, All - If the name attribute is “All”, it represents all seven days of the week. - If the name attribute is “Weekday”, it represents Monday through Friday. - If the name attribute is “Weekend”, it represents Saturday and Sunday.

Inheritance path: ->IdentifiedObject

Table 3-265 shows all attributes of DayType.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-265 - Attributes of LoadModel::DayType

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

3.10.6 (Operation) EnergyArea(Abstract)

Describes an area having energy production or consumption. Specializations are intended to support the load allocation function as typically required in energy management systems or planning studies to allocate hypothesized load levels to individual load points for power flow analysis. Often the energy area can be linked to both measured and forecast load levels.

Inheritance path: ->IdentifiedObject

Table 3-266 shows all attributes of EnergyArea.

Table 3-266 - Attributes of LoadModel::EnergyArea

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

3.10.7 (Operation) LoadArea

The class is the root or first level in a hierarchical structure for grouping of loads for the purpose of load flow load scaling.

Inheritance path: ->EnergyArea->IdentifiedObject

Table 3-267 shows all attributes of LoadArea.

Table 3-267 - Attributes of LoadModel::LoadArea

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

3.10.8 LoadGroup(Abstract)

The class is the third level in a hierarchical structure for grouping of loads for the purpose of load flow load scaling.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Inheritance path: ->IdentifiedObject

Table 3-268 shows all attributes of LoadGroup.

Table 3-268 - Attributes of LoadModel::LoadGroup

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-269 shows all association ends of LoadGroup with other classes.

Table 3-269 - Association ends of LoadModel::LoadGroup with other classes

mult.	mult.	Class type name	Description
SubLoadArea	[1..1]	SubLoadArea	The SubLoadArea where the Loadgroup belongs. Association Based On : LoadModel:: SubLoadArea.SubLoadArea[1] ----- LoadModel:: LoadGroup.LoadGroups[1..*]

3.10.9 LoadResponseCharacteristic

Models the characteristic response of the load demand due to changes in system conditions such as voltage and frequency. This is not related to demand response.
If LoadResponseCharacteristic.exponentModel is True, the voltage exponents are specified and used as to calculate:
Active power component = Pnominal * (Voltage/cim:BaseVoltage.nominalVoltage) ** cim:LoadResponseCharacteristic.pVoltageExponent
Reactive power component = Qnominal * (Voltage/cim:BaseVoltage.nominalVoltage)** cim:LoadResponseCharacteristic.qVoltageExponent
Where * means "multiply" and ** is "raised to power of".

OCL constraint: If LoadResponseCharacteristic.exponentModel is false, the portion of reactive power modelled as constant current must be specified
OCL constraint: If LoadResponseCharacteristic.exponentModel is false, the portion of reactive power modelled as constant impedance must be specified
OCL constraint: If LoadResponseCharacteristic.exponentModel is false, the portion of reactive power modelled as constant power must be specified
OCL constraint: If LoadResponseCharacteristic.exponentModel is false, the portion of real power modelled as constant current must be specified
OCL constraint: If LoadResponseCharacteristic.exponentModel is false, the portion of real power modelled as constant impedance must be specified
OCL constraint: If LoadResponseCharacteristic.exponentModel is false, the portion of real power modelled as constant power must be specified
OCL constraint: If LoadResponseCharacteristic.exponentModel is true the Exponent of per unit voltage effecting reactive power must be specified
OCL constraint: If LoadResponseCharacteristic.exponentModel is true the Exponent of per unit voltage effecting real power must be specified

In case the values for LoadResponseCharacteristic cannot be obtained, the default values for pConstantPower and qConstantPower are set to 1.

Inheritance path: ->IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-270 shows all attributes of LoadResponseCharacteristic.

Table 3-270 - Attributes of LoadModel::LoadResponseCharacteristic

Attribute name	mult.	Attribute type	Description
exponentModel	1..1	Boolean	Indicates the exponential voltage dependency model is to be used. If false, the coefficient model is to be used. The exponential voltage dependency model consist of the attributes - pVoltageExponent - qVoltageExponent. The coefficient model consist of the attributes - pConstantImpedance - pConstantCurrent - pConstantPower - qConstantImpedance - qConstantCurrent - qConstantPower. The sum of pConstantImpedance, pConstantCurrent and pConstantPower shall equal 1. The sum of qConstantImpedance, qConstantCurrent and qConstantPower shall equal 1.
pConstantCurrent	0..1	Simple_Float	Portion of active power load modeled as constant current.
pConstantImpedance	0..1	Simple_Float	Portion of active power load modeled as constant impedance.
pConstantPower	0..1	Simple_Float	Portion of active power load modeled as constant power.
pFrequencyExponent	0..1	Simple_Float	Exponent of per unit frequency effecting active power.
pVoltageExponent	0..1	Simple_Float	Exponent of per unit voltage effecting real power.
qConstantCurrent	0..1	Simple_Float	Portion of reactive power load modeled as constant current.
qConstantImpedance	0..1	Simple_Float	Portion of reactive power load modeled as constant impedance.
qConstantPower	0..1	Simple_Float	Portion of reactive power load modeled as constant power.
qFrequencyExponent	0..1	Simple_Float	Exponent of per unit frequency effecting reactive power.

qVoltageExponent	0..1	Simple_Float	Exponent of per unit voltage effecting reactive power.
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

3.10.10 NonConformLoad

NonConformLoad represent loads that do not follow a daily load change pattern and changes are not correlated with the daily load change pattern.

- The definition of the real and reactive power injections for an EnergyConsumer can be done using different sets of attributes. In the simplest case, the injections can be defined directly using only the attributes “pfixed” and “qfixed”. - The injections for a NonConformLoad can be defined as a percentage of the NonConformLoadGroup with the attributes “pfixedPct” and “qfixedPct”. In this case, the associated NonConformLoadGroup would have to have an associated NonConformLoadSchedule. - The attributes defining the affect of voltage and frequency on the injection defined by an associated LoadResponseCharacteristic should be supplied, if they are available, but are not required.

Naming Convention and main containership hierarchy is : GeographicalRegion/SubGeographicalRegion/Substation/VoltageLevel
If LoadResponseCharacteristic is missing, this load is assumed to be constant power

Inheritance path: ->EnergyConsumer->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-271 shows all attributes of NonConformLoad.

Table 3-271 - Attributes of LoadModel::NonConformLoad

Attribute name	mult.	Attribute type	Description
pfixed (Operation)	0..1	ActivePower	see EnergyConsumer
pfixedPct (Operation)	0..1	PerCent	see EnergyConsumer
qfixed (Operation)	0..1	ReactivePower	see EnergyConsumer
qfixedPct (Operation)	0..1	PerCent	see EnergyConsumer
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-272 shows all association ends of NonConformLoad with other classes.

Table 3-272 - Association ends of LoadModel::NonConformLoad with other classes

mult.	mult.	Class type name	Description

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

LoadGroup	[1..1]	NonConformLoadGroup	Conform loads assigned to this ConformLoadGroup. Association Based On : LoadModel:: NonConformLoadGroup.LoadGroup[0..1] ----- LoadModel:: NonConformLoad.EnergyConsumers[0..*]
LoadResponse	0..1	LoadResponseCharacteristic	Inherited Association Based On : Wires:: EnergyConsumer.EnergyConsumer[0..*] ----- LoadModel:: LoadResponseCharacteristic.LoadResponse[0..1]
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.10.11 NonConformLoadGroup

Loads that do not follow a daily and seasonal load variation pattern.

Inheritance path: ->LoadGroup->IdentifiedObject

Table 3-273 shows all attributes of NonConformLoadGroup.

Table 3-273 - Attributes of LoadModel::NonConformLoadGroup

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-274 shows all association ends of LoadGroup with other classes.

Table 3-274 - Association ends of LoadModel::LoadGroup with other classes

mult.	mult.	Class type name	Description
SubLoadArea	1..1	SubLoadArea	Inherited Association Based On : LoadModel:: SubLoadArea.SubLoadArea[1] ----- LoadModel:: LoadGroup.LoadGroups[1..*]

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

3.10.12 NonConformLoadSchedule

An active power (Y1-axis) and reactive power (Y2-axis) schedule (curves) versus time (X-axis) for non-conforming loads, e.g., large industrial load or power station service (where modeled).

Because value1 will always be specified in MW and value2 will always be specified in MVAr, the value1Multiplier and value2Multiplier attributes do not need to be specified.

Inheritance path: ->SeasonDayTypeSchedule->RegularIntervalSchedule->BasicIntervalSchedule->IdentifiedObject

Table 3-275 shows all attributes of NonConformLoadSchedule.

Table 3-275 - Attributes of LoadModel::NonConformLoadSchedule

Attribute name	mult.	Attribute type	Description
timeStep	1..1	Seconds	see RegularIntervalSchedule
endTime	1..1	DateTime	see RegularIntervalSchedule
startTime	1..1	DateTime	see BasicIntervalSchedule
value1Unit	1..1	UnitSymbol	see BasicIntervalSchedule
value2Unit	0..1	UnitSymbol	see BasicIntervalSchedule
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-276 shows all association ends of NonConformLoadSchedule with other classes.

Table 3-276 - Association ends of LoadModel::NonConformLoadSchedule with other classes

mult.	mult.	Class type name	Description
NonConformLoadGroup	[1..1]	NonConformLoadGroup	The NonConformLoadGroup where the NonConformLoadSchedule belongs. Association Based On : LoadModel:: NonConformLoadGroup.NonConformLoadGroup[1] ----- LoadModel:: NonConformLoadSchedule.NonConformLoadSchedules[1..*]
Season (Operation)	1..1	Season	Inherited Association Based On : LoadModel:: Season.Season[0..1] ----- LoadModel:: SeasonDayTypeSchedule.SeasonDayTypeSchedules[0..*]
DayType (Operation)	1..1	DayType	Inherited Association Based On : LoadModel:: DayType.DayType[0..1] ----- LoadModel:: SeasonDayTypeSchedule.SeasonDayTypeSchedules[0..*]

3.10.13 Season

A specified time period of the year.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

- To specify a relative date as the startDate or endDate for a Season, the year component of the ISO 8601 date format (“YYYY-MM-DD”) can be omitted. The resulting format would be “MM-DD”.

Inheritance path: ->IdentifiedObject

Table 3-277 shows all attributes of Season.

Table 3-277 - Attributes of LoadModel::Season

Attribute name	mult.	Attribute type	Description
endDate (Operation)	1..1	MonthDay	Date season ends.
startDate (Operation)	1..1	MonthDay	Date season starts.
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

3.10.14 (Operation) SeasonDayTypeSchedule(Abstract)

A time schedule covering a 24 hour period, with curve data for a specific type of season and day.

For schedules that are associated with Season and DayType, the associations to Season and DayType are not required. If a schedule does not have an associated Season, the schedule will be considered valid for all Seasons. Similarly, if a schedule does not have an association to a DayType, the schedule will be considered to apply to all days of the week.

Inheritance path: ->RegularIntervalSchedule->BasicIntervalSchedule->IdentifiedObject

Table 3-278 shows all attributes of SeasonDayTypeSchedule.

Table 3-278 - Attributes of LoadModel::SeasonDayTypeSchedule

Attribute name	mult.	Attribute type	Description
timeStep	1..1	Seconds	see RegularIntervalSchedule
endTime	1..1	DateTime	see RegularIntervalSchedule
startTime	1..1	DateTime	see BasicIntervalSchedule
value1Unit	1..1	UnitSymbol	see BasicIntervalSchedule
value2Unit	0..1	UnitSymbol	see BasicIntervalSchedule
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-279 shows all association ends of SeasonDayTypeSchedule with other classes.

Table 3-279 - Association ends of LoadModel::SeasonDayTypeSchedule with other classes

mult.	mult.	Class	Description

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

		type name	
Season (Operation)	[1..1]	Season	Schedules that use this Season. Association Based On : LoadModel:: Season.Season[0..1] ----- LoadModel:: SeasonDayTypeSchedule.SeasonDayTypeSchedules[0..*]
DayType (Operation)	[1..1]	DayType	Schedules that use this DayType. Association Based On : LoadModel:: DayType.DayType[0..1] ----- LoadModel:: SeasonDayTypeSchedule.SeasonDayTypeSchedules[0..*]

3.10.15 (Operation) StationSupply

Station supply with load derived from the station output.

See EnergyConsumer for specific notes about inherited attributes.

Inheritance path: ->EnergyConsumer->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-280 shows all attributes of StationSupply.

Table 3-280 - Attributes of LoadModel::StationSupply

Attribute name	mult.	Attribute type	Description
pfixed (Operation)	0..1	ActivePower	see EnergyConsumer
pfixedPct (Operation)	0..1	PerCent	see EnergyConsumer
qfixed (Operation)	0..1	ReactivePower	see EnergyConsumer
qfixedPct (Operation)	0..1	PerCent	see EnergyConsumer
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-281 shows all association ends of EnergyConsumer with other classes.

Table 3-281 - Association ends of Wires::EnergyConsumer with other classes

mult.	mult.	Class type name	Description
LoadResponse	0..1	LoadResponseCharacteristic	Inherited Association Based On : Wires:: EnergyConsumer.EnergyConsumer[0..*] ----- LoadModel:: LoadResponseCharacteristic.LoadResponse[0..1]
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] -----

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.10.16 SubLoadArea

The class is the second level in a hierarchical structure for grouping of loads for the purpose of load flow load scaling.

Inheritance path: ->EnergyArea->IdentifiedObject

Table 3-282 shows all attributes of SubLoadArea.

Table 3-282 - Attributes of LoadModel::SubLoadArea

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-283 shows all association ends of SubLoadArea with other classes.

Table 3-283 - Association ends of LoadModel::SubLoadArea with other classes

mult.	mult.	Class type name	Description
LoadArea (Operation)	[1..1]	LoadArea	The LoadArea where the SubLoadArea belongs. Association Based On : LoadModel:: LoadArea.LoadArea[1] ----- LoadModel:: SubLoadArea.SubLoadAreas[1..*]

3.11 Equivalents

3.11.1 General

The equivalents package models equivalent networks.

Figure 3-23 shows the diagram.

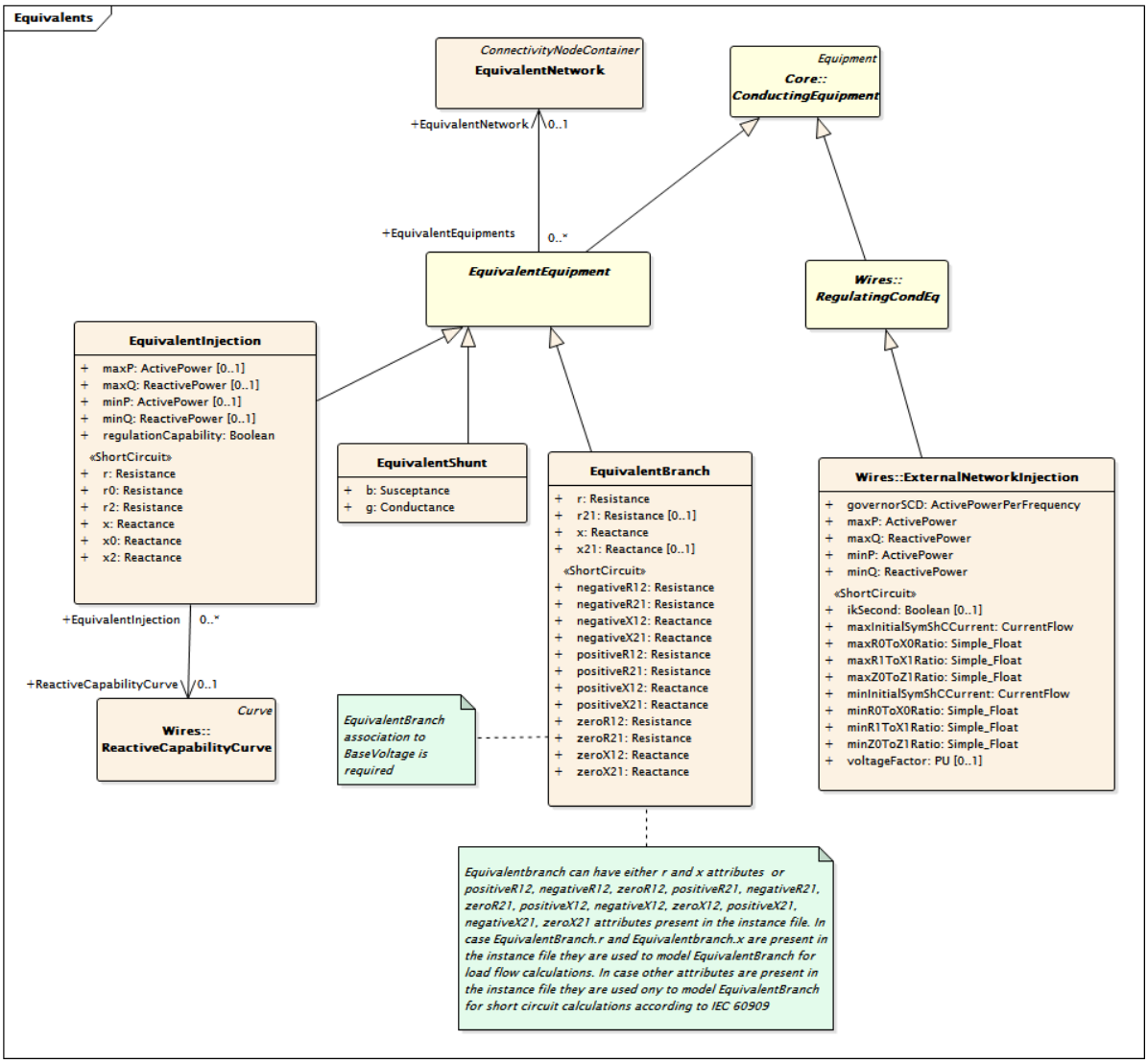


Figure 3-23 - diagram Equivalents

3.11.2 (Operation) EquivalentBranch

The class represents equivalent branches.

OCL constraint: If a device has more than one terminal it must have a sequenceNumber

Equivalentbranch can have either r and x attributes or positiveR12, negativeR12, zeroR12, positiveR21, negativeR21, zeroR21, positiveX12, negativeX12, zeroX12, positiveX21, negativeX21, zeroX21 attributes present in the instance file. In case EquivalentBranch.r and Equivalentbranch.x are present in the instance file they are used to model EquivalentBranch for load flow calculations. In case other attributes are present in the instance file they are used ony to model EquivalentBranch for short circuit calculations according to IEC 60909

EquivalentBranch association to BaseVoltage is required

Inheritance path: ->EquivalentEquipment->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-284 shows all attributes of EquivalentBranch.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-284 - Attributes of Equivalents::EquivalentBranch

Attribute name	mult.	Attribute type	Description
negativeR12 (ShortCircuit)	1..1	Resistance	Negative sequence series resistance from terminal sequence 1 to terminal sequence 2. Used for short circuit data exchange according to IEC 60909 EquivalentBranch is a result of network reduction prior to the data exchange
negativeR21 (ShortCircuit)	1..1	Resistance	Negative sequence series resistance from terminal sequence 2 to terminal sequence 1. Used for short circuit data exchange according to IEC 60909 EquivalentBranch is a result of network reduction prior to the data exchange
negativeX12 (ShortCircuit)	1..1	Reactance	Negative sequence series reactance from terminal sequence 1 to terminal sequence 2. Used for short circuit data exchange according to IEC 60909 Usage : EquivalentBranch is a result of network reduction prior to the data exchange
negativeX21 (ShortCircuit)	1..1	Reactance	Negative sequence series reactance from terminal sequence 2 to terminal sequence 1. Used for short circuit data exchange according to IEC 60909. Usage: EquivalentBranch is a result of network reduction prior to the data exchange
positiveR12 (ShortCircuit)	1..1	Resistance	Positive sequence series resistance from terminal sequence 1 to terminal sequence 2 . Used for short circuit data exchange according to IEC 60909. EquivalentBranch is a result of network reduction prior to the data exchange.
positiveR21 (ShortCircuit)	1..1	Resistance	Positive sequence series resistance from terminal sequence 2 to terminal sequence 1. Used for short circuit

			data exchange according to IEC 60909 EquivalentBranch is a result of network reduction prior to the data exchange
positiveX12 (ShortCircuit)	1..1	Reactance	Positive sequence series reactance from terminal sequence 1 to terminal sequence 2. Used for short circuit data exchange according to IEC 60909 Usage : EquivalentBranch is a result of network reduction prior to the data exchange
positiveX21 (ShortCircuit)	1..1	Reactance	Positive sequence series reactance from terminal sequence 2 to terminal sequence 1. Used for short circuit data exchange according to IEC 60909 Usage : EquivalentBranch is a result of network reduction prior to the data exchange
r	1..1	Resistance	Positive sequence series resistance of the reduced branch.
r21	0..1	Resistance	Resistance from terminal sequence 2 to terminal sequence 1 .Used for steady state power flow. This attribute is optional and represent unbalanced network such as off-nominal phase shifter. If only EquivalentBranch.r is given, then EquivalentBranch.r21 is assumed equal to EquivalentBranch.r. Usage rule : EquivalentBranch is a result of network reduction prior to the data exchange.
x	1..1	Reactance	Positive sequence series reactance of the reduced branch.
x21	0..1	Reactance	Reactance from terminal sequence 2 to terminal sequence 1 .Used for steady state power flow. This attribute is optional and represent unbalanced network such as off-nominal phase shifter. If only EquivalentBranch.x is given, then

			EquivalentBranch.x21 is assumed equal to EquivalentBranch.x. Usage rule : EquivalentBranch is a result of network reduction prior to the data exchange.
zeroR12 (ShortCircuit)	1..1	Resistance	Zero sequence series resistance from terminal sequence 1 to terminal sequence 2. Used for short circuit data exchange according to IEC 60909 EquivalentBranch is a result of network reduction prior to the data exchange
zeroR21 (ShortCircuit)	1..1	Resistance	Zero sequence series resistance from terminal sequence 2 to terminal sequence 1. Used for short circuit data exchange according to IEC 60909 Usage : EquivalentBranch is a result of network reduction prior to the data exchange
zeroX12 (ShortCircuit)	1..1	Reactance	Zero sequence series reactance from terminal sequence 1 to terminal sequence 2. Used for short circuit data exchange according to IEC 60909 Usage : EquivalentBranch is a result of network reduction prior to the data exchange
zeroX21 (ShortCircuit)	1..1	Reactance	Zero sequence series reactance from terminal sequence 2 to terminal sequence 1. Used for short circuit data exchange according to IEC 60909 Usage : EquivalentBranch is a result of network reduction prior to the data exchange
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-285 shows all association ends of EquivalentEquipment with other classes.

Table 3-285 - Association ends of Equivalents::EquivalentEquipment with other classes

mult.	mult.	Class type name	Description
EquivalentNetwork	0..1	EquivalentNetwork	Inherited Association Based On : Equivalents:: EquivalentEquipment.EquivalentEquipments[0..*] ----- Equivalents:: EquivalentNetwork.EquivalentNetwork[0..1]
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.11.3 EquivalentEquipment(Abstract)

The class represents equivalent objects that are the result of a network reduction. The class is the base for equivalent objects of different types.

Inheritance path: ->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-286 shows all attributes of EquivalentEquipment.

Table 3-286 - Attributes of Equivalents::EquivalentEquipment

Attribute name	mult.	Attribute type	Description
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-287 shows all association ends of EquivalentEquipment with other classes.

Table 3-287 - Association ends of Equivalents::EquivalentEquipment with other classes

mult.	mult.	Class type name	Description
EquivalentNetwork	[0..1]	EquivalentNetwork	The associated reduced equivalents. Association Based On : Equivalents:: EquivalentEquipment.EquivalentEquipments[0..*] ----- Equivalents:: EquivalentNetwork.EquivalentNetwork[0..1]

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.11.4 EquivalentInjection

This class represents equivalent injections (generation or load). Voltage regulation is allowed only at the point of connection.

Inheritance path: ->EquivalentEquipment->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-288 shows all attributes of EquivalentInjection.

Table 3-288 - Attributes of Equivalents::EquivalentInjection

Attribute name	mult.	Attribute type	Description
maxP	0..1	ActivePower	Maximum active power of the injection.
maxQ	0..1	ReactivePower	Used for modeling of infeed for load flow exchange. Not used for short circuit modeling. If maxQ and minQ are not used ReactiveCapabilityCurve can be used.
minP	0..1	ActivePower	Minimum active power of the injection.
minQ	0..1	ReactivePower	Used for modeling of infeed for load flow exchange. Not used for short circuit modeling. If maxQ and minQ are not used ReactiveCapabilityCurve can be used.
r (ShortCircuit)	1..1	Resistance	Positive sequence resistance. Used to represent Extended-Ward (IEC 60909). Usage : Extended-Ward is a result of network reduction prior to the data exchange.
r0 (ShortCircuit)	1..1	Resistance	Zero sequence resistance. Used to represent Extended-Ward (IEC 60909). Usage : Extended-Ward is a result of network reduction prior to the

			data exchange.
r2 (ShortCircuit)	1..1	Resistance	Negative sequence resistance. Used to represent Extended-Ward (IEC 60909). Usage : Extended-Ward is a result of network reduction prior to the data exchange.
regulationCapability	1..1	Boolean	Specifies whether or not the EquivalentInjection has the capability to regulate the local voltage.
x (ShortCircuit)	1..1	Reactance	Positive sequence reactance. Used to represent Extended-Ward (IEC 60909). Usage : Extended-Ward is a result of network reduction prior to the data exchange.
x0 (ShortCircuit)	1..1	Reactance	Zero sequence reactance. Used to represent Extended-Ward (IEC 60909). Usage : Extended-Ward is a result of network reduction prior to the data exchange.
x2 (ShortCircuit)	1..1	Reactance	Negative sequence reactance. Used to represent Extended-Ward (IEC 60909). Usage : Extended-Ward is a result of network reduction prior to the data exchange.
aggregate	0..1	Boolean	see Equipment
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-289 shows all association ends of EquivalentInjection with other classes.

Table 3-289 - Association ends of Equivalents::EquivalentInjection with other classes

mult.	mult.	Class type name	Description
ReactiveCapabilityCurve	[0..1]	ReactiveCapabilityCurve	The equivalent injection using this reactive capability curve. Association Based On : Wires:: ReactiveCapabilityCurve.ReactiveCapabilityCurve[0..1] ----- Equivalents:: EquivalentInjection.EquivalentInjection[0..*]

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

EquivalentNetwork	0..1	EquivalentNetwork	Inherited Association Based On : Equivalents:: EquivalentEquipment.EquivalentEquipments[0..*] ----- Equivalents:: EquivalentNetwork.EquivalentNetwork[0..1]
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.11.5 EquivalentNetwork

A class that represents an external meshed network that has been reduced to an electrically equivalent model. The ConnectivityNodes contained in the equivalent are intended to reflect internal nodes of the equivalent. The boundary Connectivity nodes where the equivalent connects outside itself are NOT contained by the equivalent.

Inheritance path: ->ConnectivityNodeContainer->PowerSystemResource->IdentifiedObject

Table 3-290 shows all attributes of EquivalentNetwork.

Table 3-290 - Attributes of Equivalents::EquivalentNetwork

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

3.11.6 EquivalentShunt

The class represents equivalent shunts.

Inheritance path: ->EquivalentEquipment->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 3-291 shows all attributes of EquivalentShunt.

Table 3-291 - Attributes of Equivalents::EquivalentShunt

Attribute name	mult.	Attribute type	Description
b	1..1	Susceptance	Positive sequence shunt susceptance.
g	1..1	Conductance	Positive sequence shunt conductance.
aggregate	0..1	Boolean	see Equipment

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-292 shows all association ends of EquivalentEquipment with other classes.

Table 3-292 - Association ends of Equivalents::EquivalentEquipment with other classes

mult.	mult.	Class type name	Description
EquivalentNetwork	0..1	EquivalentNetwork	Inherited Association Based On : Equivalents:: EquivalentEquipment.EquivalentEquipments[0..*] ----- Equivalents:: EquivalentNetwork.EquivalentNetwork[0..1]
BaseVoltage	0..1	BaseVoltage	Inherited Association Based On : Core:: BaseVoltage.BaseVoltage[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[0..*]
EquipmentContainer	0..1	EquipmentContainer	Inherited Association Based On : Core:: EquipmentContainer.EquipmentContainer[0..1] ----- Core:: Equipment.Equipments[0..*]

3.12 ControlArea

3.12.1 General

The ControlArea package models area specifications which can be used for a variety of purposes. The package as a whole models potentially overlapping control area specifications for the purpose of actual generation control, load forecast area load capture, or powerflow based analysis.

Figure 3-24 shows the diagram.

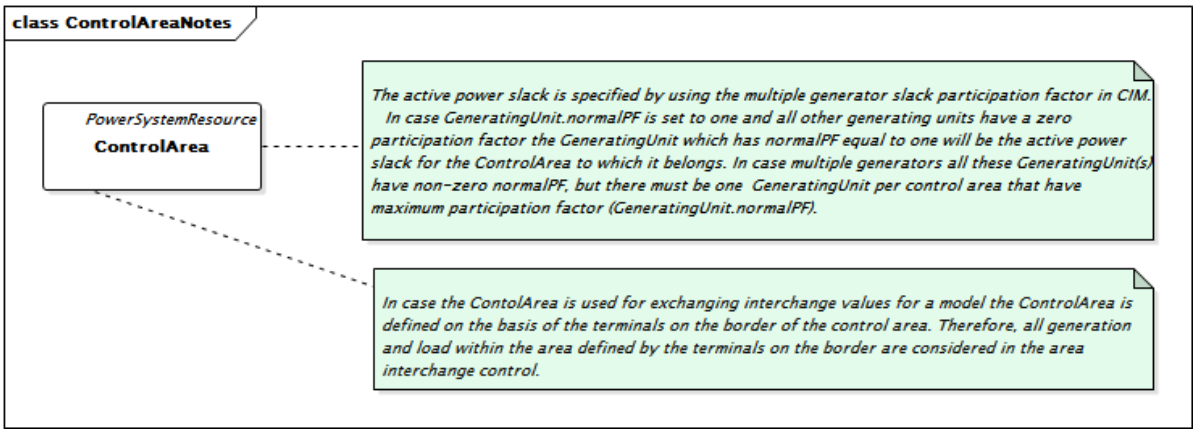


Figure 3-24 - diagram ControlAreaNotes

Figure 3-25 shows the diagram.

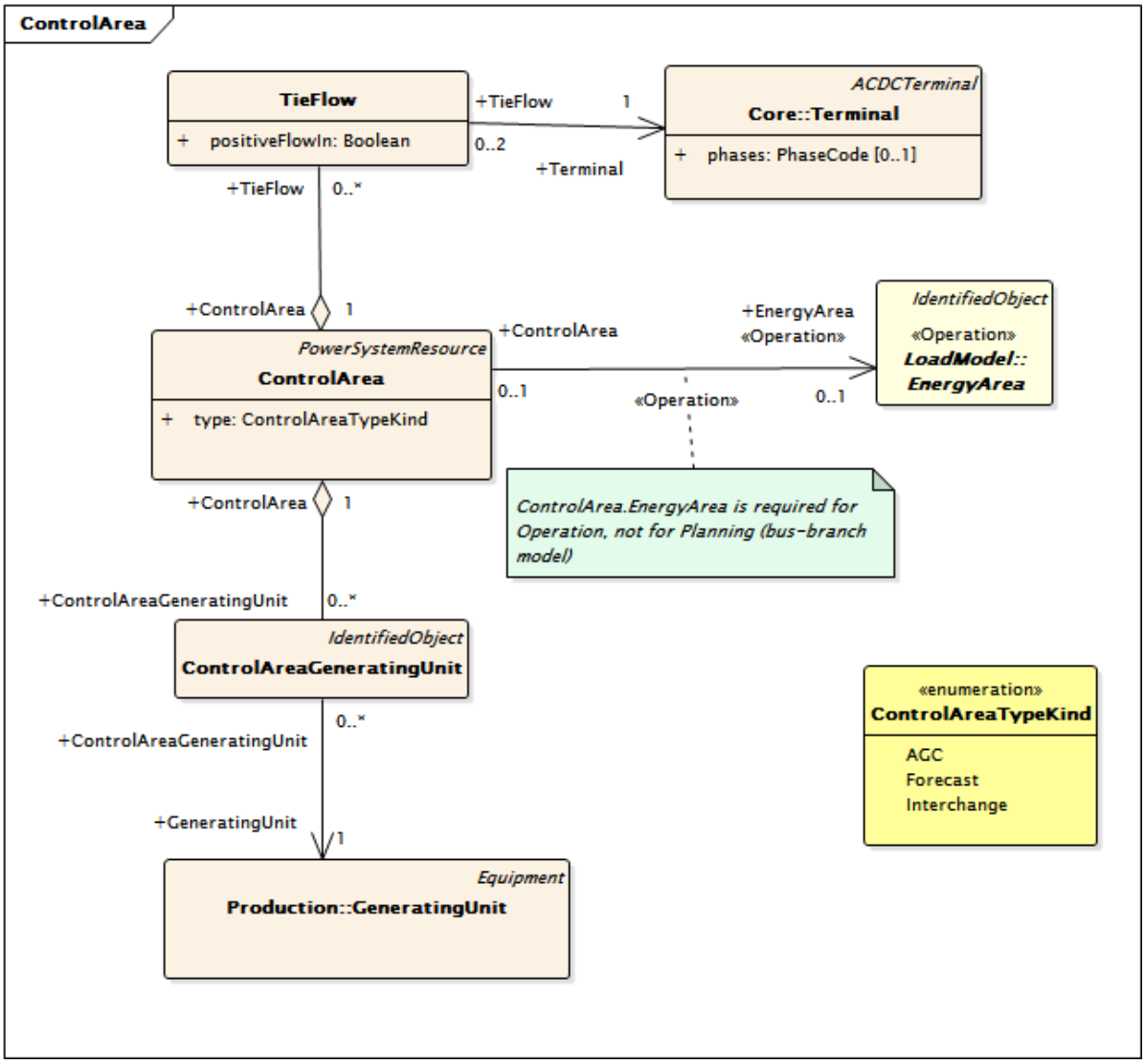


Figure 3-25 - diagram ControlArea

3.12.2 ControlArea

A control area is a grouping of generating units and/or loads and a cutset of tie lines (as terminals) which may be used for a variety of purposes including automatic generation control, powerflow solution area interchange control specification, and input to load forecasting. Note that any number of overlapping control area specifications can be superimposed on the physical model.

In case the ContolArea is used for exchanging interchange values for a model the ControlArea is defined on the basis of the terminals on the border of the control area. Therefore, all generation and load within the area defined by the terminals on the border are considered in the area interchange control.

The active power slack is specified by using the multiple generator slack participation factor in CIM. In case GeneratingUnit.normalPF is set to one and all other generating units have a zero participation factor the GeneratingUnit which has normalPF equal to one will be the active power slack for the ControlArea to which it belongs. In case multiple generators all these

GeneratingUnit(s) have non-zero normalPF, but there must be one GeneratingUnit per control area that have maximum participation factor (GeneratingUnit.normalPF).

Inheritance path: ->PowerSystemResource->IdentifiedObject

Table 3-293 shows all attributes of ControlArea.

Table 3-293 - Attributes of ControlArea::ControlArea

Attribute name	mult.	Attribute type	Description
type	1..1	ControlAreaTypeKind	The primary type of control area definition used to determine if this is used for automatic generation control, for planning interchange control, or other purposes. A control area specified with primary type of automatic generation control could still be forecast and used as an interchange area in power flow analysis.
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-294 shows all association ends of ControlArea with other classes.

Table 3-294 - Association ends of ControlArea::ControlArea with other classes

mult.	mult.	Class type name	Description
EnergyArea (Operation)	[0..1]	EnergyArea	The energy area that is forecast from this control area specification. Association Based On : ControlArea:: ControlArea.ControlArea[0..1] ----- LoadModel:: EnergyArea.EnergyArea[0..1]

3.12.3 (Operation) ControlAreaGeneratingUnit

A control area generating unit. This class is needed so that alternate control area definitions may include the same generating unit. Note only one instance within a control area should reference a specific generating unit.

Inheritance path: ->IdentifiedObject

Table 3-295 shows all attributes of ControlAreaGeneratingUnit.

Table 3-295 - Attributes of ControlArea::ControlAreaGeneratingUnit

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 3-296 shows all association ends of ControlAreaGeneratingUnit with other classes.

Table 3-296 - Association ends of ControlArea::ControlAreaGeneratingUnit with other classes

mult.	mult.	Class type name	Description
ControlArea	[1..1]	ControlArea	The parent control area for the generating unit specifications. Association Based On : ControlArea:: ControlAreaGeneratingUnit.ControlAreaGeneratingUnit[0..*] ----- ControlArea:: ControlArea.ControlArea[1]
GeneratingUnit	[1..1]	GeneratingUnit	The generating unit specified for this control area. Note that a control area should include a GeneratingUnit only once. Association Based On : ControlArea:: ControlAreaGeneratingUnit.ControlAreaGeneratingUnit[0..*] ----- Production:: GeneratingUnit.GeneratingUnit[1]

3.12.4 TieFlow

A flow specification in terms of location and direction for a control area.

Table 3-297 shows all attributes of TieFlow.

Table 3-297 - Attributes of ControlArea::TieFlow

Attribute name	mult.	Attribute type	Description
positiveFlowIn	1..1	Boolean	True if the flow into the terminal (load convention) is also flow into the control area. For example, this attribute should be true if using the tie line terminal further away from the control area. For example to represent a tie to a shunt component (like a load or generator) in another area, this is the near end of a branch and this attribute would be specified as false.

Table 3-298 shows all association ends of TieFlow with other classes.

Table 3-298 - Association ends of ControlArea::TieFlow with other classes

mult.	mult.	Class type name	Description

ControlArea	[1..1]	ControlArea	The control area of the tie flows. Association Based On : ControlArea:: TieFlow.TieFlow[0..*] ----- ControlArea:: ControlArea.ControlArea[1]
Terminal	[1..1]	Terminal	The terminal to which this tie flow belongs. Association Based On : ControlArea:: TieFlow.TieFlow[0..2] ----- Core:: Terminal.Terminal[1]

3.12.5 Enumerations

3.12.5.1 ControlAreaTypeKind

The type of control area.

Literal	Description	Code
AGC	Used for automatic generation control.	
Forecast	Used for load forecast.	
Interchange	Used for interchange specification or control.	

3.13 Global Used Datatypes

3.13.1 Datatypes

3.13.1.1 ApparentPower

Product of the RMS value of the voltage and the RMS value of the current.

Value type is IEEE 754 simple precision floating point

Table 3-299 shows all attributes of ApparentPower.

Table 3-299 - Attributes of DomainProfile::ApparentPower.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= VA
multiplier	0..1	UnitMultiplier	ConstantValue= M

3.13.1.2 ActivePower

Product of RMS value of the voltage and the RMS value of the in-phase component of the current.

Value type is IEEE 754 simple precision floating point

Table 3-300 shows all attributes of ActivePower.

Table 3-300 - Attributes of DomainProfile::ActivePower.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= W
multiplier	0..1	UnitMultiplier	ConstantValue= M

3.13.1.3 Voltage

Electrical voltage, can be both AC and DC.

Value type is IEEE 754 simple precision floating point

Table 3-301 shows all attributes of Voltage.

Table 3-301 - Attributes of DomainProfile::Voltage.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= V
multiplier	0..1	UnitMultiplier	ConstantValue= k

3.13.1.4 Resistance

Resistance (real part of impedance).

Value type is IEEE 754 simple precision floating point

Table 3-302 shows all attributes of Resistance.

Table 3-302 - Attributes of DomainProfile::Resistance.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= ohm
multiplier	0..1	UnitMultiplier	ConstantValue= none

3.13.1.5 ActivePowerPerCurrentFlow

Table 3-303 shows all attributes of ActivePowerPerCurrentFlow.

Table 3-303 - Attributes of DomainProfile::ActivePowerPerCurrentFlow.

Attribute name	mult.	Attribute type	Description
denominatorMultiplier	0..1	UnitMultiplier	ConstantValue= none
denominatorUnit	0..1	UnitSymbol	ConstantValue= A
multiplier	0..1	UnitMultiplier	ConstantValue= M
unit	0..1	UnitSymbol	ConstantValue= W
value	0..1	Float	

3.13.1.6 AngleDegrees

Measurement of angle in degrees.

Value type is IEEE 754 simple precision floating point

Table 3-304 shows all attributes of AngleDegrees.

Table 3-304 - Attributes of DomainProfile::AngleDegrees.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= deg
multiplier	0..1	UnitMultiplier	ConstantValue= none

3.13.1.7 CurrentFlow

Electrical current with sign convention: positive flow is out of the conducting equipment into the connectivity node. Can be both AC and DC.

Value type is IEEE 754 simple precision floating point

Table 3-305 shows all attributes of CurrentFlow.

Table 3-305 - Attributes of DomainProfile::CurrentFlow.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= A
multiplier	0..1	UnitMultiplier	ConstantValue= none

3.13.1.8 Inductance

Inductive part of reactance (imaginary part of impedance), at rated frequency.

Table 3-306 shows all attributes of Inductance.

Table 3-306 - Attributes of DomainProfile::Inductance.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= H
multiplier	0..1	UnitMultiplier	ConstantValue= none

3.13.1.9 Capacitance

Capacitive part of reactance (imaginary part of impedance), at rated frequency.

Table 3-307 shows all attributes of Capacitance.

Table 3-307 - Attributes of DomainProfile::Capacitance.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= F
multiplier	0..1	UnitMultiplier	ConstantValue= none

3.13.1.10 Length

Unit of length. Never negative.

Value type is IEEE 754 simple precision floating point

Table 3-308 shows all attributes of Length.

Table 3-308 - Attributes of DomainProfile::Length.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= m

multiplier	0..1	UnitMultiplier	ConstantValue= k

3.13.1.11 CapacitancePerLength

Capacitance per unit of length.

Table 3-309 shows all attributes of CapacitancePerLength.

Table 3-309 - Attributes of DomainProfile::CapacitancePerLength.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= F
multiplier	0..1	UnitMultiplier	ConstantValue= none
denominatorUnit	0..1	UnitSymbol	ConstantValue= m
denominatorMultiplier	0..1	UnitMultiplier	ConstantValue= none

3.13.1.12 InductancePerLength

Inductance per unit of length.

Table 3-310 shows all attributes of InductancePerLength.

Table 3-310 - Attributes of DomainProfile::InductancePerLength.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= H
multiplier	0..1	UnitMultiplier	ConstantValue= none
denominatorUnit	0..1	UnitSymbol	ConstantValue= m
denominatorMultiplier	0..1	UnitMultiplier	ConstantValue= none

3.13.1.13 ResistancePerLength

Resistance (real part of impedance) per unit of length.

Table 3-311 shows all attributes of ResistancePerLength.

Table 3-311 - Attributes of DomainProfile::ResistancePerLength.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= ohm
multiplier	0..1	UnitMultiplier	ConstantValue= none
denominatorUnit	0..1	UnitSymbol	ConstantValue= m
denominatorMultiplier	0..1	UnitMultiplier	ConstantValue= none

3.13.1.14 Simple_Float

A floating point number. The range is unspecified and not limited.

In ENTSO-E profile, Simple_Float range is the IEEE754 simple precision floating point one. It correspond to xs:float datatype

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-312 shows all attributes of Simple_Float.

Table 3-312 - Attributes of DomainProfile::Simple_Float.

Attribute name	mult.	Attribute type	Description
value	1..1	Float	

3.13.1.15 PerCent

Percentage on a defined base. For example, specify as 100 to indicate at the defined base.

Value type is IEEE 754 simple precision floating point

Table 3-313 shows all attributes of PerCent.

Table 3-313 - Attributes of DomainProfile::PerCent.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	Normally 0 - 100 on a defined base
unit	0..1	UnitSymbol	ConstantValue= none
multiplier	0..1	UnitMultiplier	ConstantValue= none

3.13.1.16 AngleRadians

Phase angle in radians.

Table 3-314 shows all attributes of AngleRadians.

Table 3-314 - Attributes of DomainProfile::AngleRadians.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= rad
multiplier	0..1	UnitMultiplier	ConstantValue= none

3.13.1.17 Reactance

Reactance (imaginary part of impedance), at rated frequency.

Value type is IEEE 754 simple precision floating point

Table 3-315 shows all attributes of Reactance.

Table 3-315 - Attributes of DomainProfile::Reactance.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= ohm
multiplier	0..1	UnitMultiplier	ConstantValue= none

3.13.1.18 Money

Amount of money.

Value type is IEEE 754 simple precision floating point

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-316 shows all attributes of Money.

Table 3-316 - Attributes of DomainProfile::Money.

Attribute name	mult.	Attribute type	Description
unit	0..1	Currency	ConstantValue= EUR
multiplier	0..1	UnitMultiplier	ConstantValue= none
value	0..1	Decimal	

3.13.1.19 Seconds

Time, in seconds.

Value type is IEEE 754 simple precision floating point

Table 3-317 shows all attributes of Seconds.

Table 3-317 - Attributes of DomainProfile::Seconds.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	Time, in seconds
unit	0..1	UnitSymbol	ConstantValue= s
multiplier	0..1	UnitMultiplier	ConstantValue= none

3.13.1.20 Susceptance

Imaginary part of admittance.

Value type is IEEE 754 simple precision floating point

Table 3-318 shows all attributes of Susceptance.

Table 3-318 - Attributes of DomainProfile::Susceptance.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= S
multiplier	0..1	UnitMultiplier	ConstantValue= none

3.13.1.21 Conductance

Factor by which voltage must be multiplied to give corresponding power lost from a circuit. Real part of admittance.

Value type is IEEE 754 simple precision floating point

Table 3-319 shows all attributes of Conductance.

Table 3-319 - Attributes of DomainProfile::Conductance.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= S
multiplier	0..1	UnitMultiplier	ConstantValue= none

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

3.13.1.22 Temperature

Value of temperature in degrees Celsius.

Value type is IEEE 754 simple precision floating point

Table 3-320 shows all attributes of Temperature.

Table 3-320 - Attributes of DomainProfile::Temperature.

Attribute name	mult.	Attribute type	Description
multiplier	0..1	UnitMultiplier	ConstantValue= none
unit	0..1	UnitSymbol	ConstantValue= degC
value	0..1	Float	

3.13.1.23 Frequency

Cycles per second.

Value type is IEEE 754 simple precision floating point

Table 3-321 shows all attributes of Frequency.

Table 3-321 - Attributes of DomainProfile::Frequency.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= Hz
multiplier	0..1	UnitMultiplier	DefaultValue= none

3.13.1.24 RotationSpeed

Number of revolutions per second.

Value type is IEEE 754 simple precision floating point

Table 3-322 shows all attributes of RotationSpeed.

Table 3-322 - Attributes of DomainProfile::RotationSpeed.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= none
multiplier	0..1	UnitMultiplier	ConstantValue= none
denominatorUnit	0..1	UnitSymbol	ConstantValue= s
denominatorMultiplier	0..1	UnitMultiplier	ConstantValue= none

3.13.1.25 ReactivePower

Product of RMS value of the voltage and the RMS value of the quadrature component of the current.

Value type is IEEE 754 simple precision floating point

Table 3-323 shows all attributes of ReactivePower.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 3-323 - Attributes of DomainProfile::ReactivePower.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= VAr
multiplier	0..1	UnitMultiplier	ConstantValue= M

3.13.1.26 ActivePowerPerFrequency

Active power variation with frequency.

Table 3-324 shows all attributes of ActivePowerPerFrequency.

Table 3-324 - Attributes of DomainProfile::ActivePowerPerFrequency.

Attribute name	mult.	Attribute type	Description
denominatorMultiplier	0..1	UnitMultiplier	ConstantValue= none
denominatorUnit	0..1	UnitSymbol	ConstantValue= Hz
multiplier	0..1	UnitMultiplier	ConstantValue= M
unit	0..1	UnitSymbol	ConstantValue= W
value	0..1	Float	

3.13.1.27 PU

Per Unit - a positive or negative value referred to a defined base. Values typically range from -10 to +10.

Value type is IEEE 754 simple precision floating point

Table 3-325 shows all attributes of PU.

Table 3-325 - Attributes of DomainProfile::PU.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= none
multiplier	0..1	UnitMultiplier	ConstantValue= none

3.13.1.28 VoltagePerReactivePower

Voltage variation with reactive power.

Value type is IEEE 754 simple precision floating point

Table 3-326 shows all attributes of VoltagePerReactivePower.

Table 3-326 - Attributes of DomainProfile::VoltagePerReactivePower.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= V
denominatorMultiplier	0..1	UnitMultiplier	ConstantValue= M
multiplier	0..1	UnitMultiplier	ConstantValue= k

denominatorUnit	0..1	UnitSymbol	ConstantValue= VAr

3.13.2 Enumerations

3.13.2.1 UnitSymbol

The units defined for usage in the CIM.

Literal	Description	Code
VA	Apparent power in volt ampere.	
W	Active power in watt.	
VAr	Reactive power in volt ampere reactive.	
VAh	Apparent energy in volt ampere hours.	
Wh	Real energy in what hours.	
VArh	Reactive energy in volt ampere reactive hours.	
V	Voltage in volt.	
ohm	Resistance in ohm.	
A	Current in ampere.	
F	Capacitance in farad.	
H	Inductance in henry.	
degC	Relative temperature in degrees Celsius. In the SI unit system the symbol is ºC. Electric charge is measured in coulomb that has the unit symbol C. To distinguish degree Celsius form coulomb the symbol used in the UML is degC. Reason for not using ºC is the special character º is difficult to manage in software.	
s	Time in seconds.	
min	Time in minutes.	
h	Time in hours.	
deg	Plane angle in degrees.	
rad	Plane angle in radians.	
J	Energy in joule.	
N	Force in newton.	
S	Conductance in siemens.	
none	Dimension less quantity, e.g. count, per unit, etc.	
Hz	Frequency in hertz.	
g	Mass in gram.	
Pa	Pressure in pascal (n/m2).	
m	Length in meter.	
m2	Area in square meters.	
m3	Volume in cubic meters.	

3.13.2.2 UnitMultiplier

The unit multipliers defined for the CIM.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Literal	Description	Code
p	Pico 10**-12.	
n	Nano 10**-9.	
micro	Micro 10**-6.	
m	Milli 10**-3.	
c	Centi 10**-2.	
d	Deci 10**-1.	
k	Kilo 10**3.	
M	Mega 10**6.	
G	Giga 10**9.	
T	Tera 10**12.	
none	No multiplier or equivalently multiply by 1.	

3.13.2.3 UnitSymbol

The units defined for usage in the CIM.

Literal	Description	Code
VA	Apparent power in volt ampere.	
W	Active power in watt.	
VAr	Reactive power in volt ampere reactive.	
VAh	Apparent energy in volt ampere hours.	
Wh	Real energy in what hours.	
VArh	Reactive energy in volt ampere reactive hours.	
V	Voltage in volt.	
ohm	Resistance in ohm.	
A	Current in ampere.	
F	Capacitance in farad.	
H	Inductance in henry.	
degC	Relative temperature in degrees Celsius. In the SI unit system the symbol is ºC. Electric charge is measured in coulomb that has the unit symbol C. To distinguish degree Celsius form coulomb the symbol used in the UML is degC. Reason for not using ºC is the special character º is difficult to manage in software.	
s	Time in seconds.	
min	Time in minutes.	
h	Time in hours.	
deg	Plane angle in degrees.	
rad	Plane angle in radians.	
J	Energy in joule.	
N	Force in newton.	
S	Conductance in siemens.	

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

none	Dimension less quantity, e.g. count, per unit, etc.	
Hz	Frequency in hertz.	
g	Mass in gram.	
Pa	Pressure in pascal (n/m2).	
m	Length in meter.	
m2	Area in square meters.	
m3	Volume in cubic meters.	

3.13.2.4 DCPolarityKind

Polarity for DC circuits.

Literal	Description	Code
positive	Positive pole.	
middle	Middle pole, potentially grounded.	
negative	Negative pole.	

3.13.2.5 DCConverterOperatingModeKind

The operating mode of an HVDC bipole.

Literal	Description	Code
bipolar	Bipolar operation.	
monopolarMetallicReturn	Monopolar operation with metallic return	
monopolarGroundReturn	Monopolar operation with ground return	

3.13.2.6 PhaseCode

Enumeration of phase identifiers. Allows designation of phases for both transmission and distribution equipment, circuits and loads. Residential and small commercial loads are often served from single-phase, or split-phase, secondary circuits. For example of s12N, phases 1 and 2 refer to hot wires that are 180 degrees out of phase, while N refers to the neutral wire. Through single-phase transformer connections, these secondary circuits may be served from one or two of the primary phases A, B, and C. For three-phase loads, use the A, B, C phase codes instead of s12N.

Literal	Description	Code
ABCN	Phases A, B, C, and N.	
ABC	Phases A, B, and C.	
ABN	Phases A, B, and neutral.	
ACN	Phases A, C and neutral.	
BCN	Phases B, C, and neutral.	
AB	Phases A and B.	
AC	Phases A and C.	
BC	Phases B and C.	
AN	Phases A and neutral.	
BN	Phases B and neutral.	

CN	Phases C and neutral.	
A	Phase A.	
B	Phase B.	
C	Phase C.	
N	Neutral phase.	
s1N	Secondary phase 1 and neutral.	
s2N	Secondary phase 2 and neutral.	
s12N	Secondary phases 1, 2, and neutral.	
s1	Secondary phase 1.	
s2	Secondary phase 2.	
s12	Secondary phase 1 and 2.	

3.13.2.7 Source

Source gives information related to the origin of a value.

Literal	Description	Code
PROCESS	The value is provided by input from the process I/O or being calculated from some function.	
DEFAULTED	The value contains a default value.	
SUBSTITUTED	The value is provided by input of an operator or by an automatic source.	

3.13.2.8 Validity

Validity for MeasurementValue.

Literal	Description	Code
GOOD	The value is marked good if no abnormal condition of the acquisition function or the information source is detected.	
QUESTIONABLE	The value is marked questionable if a supervision function detects an abnormal behaviour, however the value could still be valid. The client is responsible for determining whether or not values marked "questionable" should be used.	
INVALID	The value is marked invalid when a supervision function recognises abnormal conditions of the acquisition function or the information source (missing or non-operating updating devices). The value is not defined under this condition. The mark invalid is used to indicate to the client that the value may be incorrect and shall not be used.	

3.13.2.9 FuelType

Type of fuel.

Literal	Description	Code
coal	Generic coal, not including lignite type.	
oil	Oil.	
gas	Natural gas.	
lignite	The fuel is lignite coal. Note that this is a special type of coal, so the other enum of coal is reserved for hard coal types or if the exact type	

	of coal is not known.	
hardCoal	Hard coal	
oilShale	Oil Shale	

3.13.2.10 GeneratorControlSource

The source of controls for a generating unit.

Literal	Description	Code
unavailable	Not available.	
offAGC	Off of automatic generation control (AGC).	
onAGC	On automatic generation control (AGC).	
plantControl	Plant is controlling.	

3.13.2.11 Currency

Monetary currencies. Apologies for this list not being exhaustive.

Literal	Description	Code
USD	US dollar	
EUR	European euro	
AUD	Australian dollar	
CAD	Canadian dollar	
CHF	Swiss francs	
CNY	Chinese yuan renminbi	
DKK	Danish crown	
GBP	British pound	
JPY	Japanese yen	
NOK	Norwegian crown	
RUR	Russian ruble	
SEK	Swedish crown	
INR	India rupees	
other	Another type of currency.	

3.13.2.12 HydroEnergyConversionKind

Specifies the capability of the hydro generating unit to convert energy as a generator or pump.

Literal	Description	Code
generator	Able to generate power, but not able to pump water for energy storage.	
pumpAndGenerator	Able to both generate power and pump water for energy storage.	

3.13.2.13 HydroPlantStorageKind

The type of hydro power plant.

Literal	Description	Code

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

runOfRiver	Run of river.	
pumpedStorage	Pumped storage.	
storage	Storage.	

3.13.2.14 WindGenUnitKind

Kind of wind generating unit.

Literal	Description	Code
offshore	The wind generating unit is located offshore.	
onshore	The wind generating unit is located onshore.	

3.13.2.15 CurveStyle

Style or shape of curve.

Literal	Description	Code
constantYValue	The Y-axis values are assumed constant until the next curve point and prior to the first curve point.	
straightLineYValues	The Y-axis values are assumed to be a straight line between values. Also known as linear interpolation.	

3.13.2.16 LimitTypeKind

The enumeration defines the kinds of the limit types.

Literal	Description	Code
patl	The Permanent Admissible Transmission Loading (PATL) is the loading in Amps, MVA or MW that can be accepted by a network branch for an unlimited duration without any risk for the material. The duration attribute is not used and shall be excluded for the PATL limit type. Hence only one limit value exists for the PATL type.	
patlt	Permanent Admissible Transmission Loading Threshold (PATLT) is a value in engineering units defined for PATL and calculated using percentage less than 100 of the PATL type intended to alert operators of an arising condition. The percentage should be given in the name of the OperationalLimitSet. The aceptableDuration is another way to express the severity of the limit.	
tatl	Temporarily Admissible Transmission Loading (TATL) which is the loading in Amps, MVA or MW that can be accepted by a branch for a certain limited duration. The TATL can be defined in different ways: as a fixed percentage of the PATL for a given time (for example, 115% of the PATL that can be accepted during 15 minutes), pairs of TATL type and Duration calculated for each line taking into account its particular configuration and conditions of functioning (for example, it can define a TATL acceptable during 20 minutes and another one acceptable during 10 minutes). Such a definition of TATL can depend on the initial operating conditions of the network element (sag situation of a line). The duration attribute can be used define several TATL limit types. Hence multiple TATL limit values may exist having different durations.	
tc	Tripping Current (TC) is the ultimate intensity without any delay. It is defined as the threshold the line will trip without any possible remedial actions. The tripping of the network element is ordered by protections against short circuits or by overload protections, but in any case, the activation delay of these protections is not compatible with the reaction delay of an operator (less than one minute).	

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

	The duration is always zero and the duration attribute may be left out. Hence only one limit value exists for the TC type.	
tct	Tripping Current Threshold (TCT) is a value in engineering units defined for TC and calculated using percentage less than 100 of the TC type intended to alert operators of an arising condition. The percentage should be given in the name of the OperationalLimitSet. The aceptableDuration is another way to express the severity of the limit.	
highVoltage	Referring to the rating of the equipments, a voltage too high can lead to accelerated ageing or the destruction of the equipment. This limit type may or may not have duration.	
lowVoltage	A too low voltage can disturb the normal operation of some protections and transformer equipped with on-load tap changers, electronic power devices or can affect the behaviour of the auxiliaries of generation units. This limit type may or may not have duration.	

3.13.2.17 OperationalLimitDirectionKind

The direction attribute describes the side of a limit that is a violation.

Literal	Description	Code
high	High means that a monitored value above the limit value is a violation. If applied to a terminal flow, the positive direction is into the terminal.	
low	Low means a monitored value below the limit is a violation. If applied to a terminal flow, the positive direction is into the terminal.	
absoluteValue	An absoluteValue limit means that a monitored absolute value above the limit value is a violation.	

3.13.2.18 PetersenCoilModeKind

The mode of operation for a Petersen coil.

Literal	Description	Code
fixed	Fixed position.	
manual	Manual positioning.	
automaticPositioning	Automatic positioning.	

3.13.2.19 WindingConnection

Winding connection type.

Literal	Description	Code
D	Delta	
Y	Wye	
Z	ZigZag	
Yn	Wye, with neutral brought out for grounding.	
Zn	ZigZag, with neutral brought out for grounding.	
A	Autotransformer common winding	
I	Independent winding, for single-phase connections	

3.13.2.20 TransformerControlMode

Control modes for a transformer.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Literal	Description	Code
volt	Voltage control	
reactive	Reactive power flow control	

3.13.2.21 RegulatingControlModeKind

The kind of regulation model. For example regulating voltage, reactive power, active power, etc.

Literal	Description	Code
voltage	Voltage is specified.	
activePower	Active power is specified.	
reactivePower	Reactive power is specified.	
currentFlow	Current flow is specified.	
admittance	Admittance is specified.	
timeScheduled	Control switches on/off by time of day. The times may change on the weekend, or in different seasons.	
temperature	Control switches on/off based on the local temperature (i.e., a thermostat).	
powerFactor	Power factor is specified.	

3.13.2.22 SVCControlMode

Static VAr Compensator control mode.

Literal	Description	Code
reactivePower		
voltage		

3.13.2.23 ShortCircuitRotorKind

Type of rotor, used by short circuit applications.

Literal	Description	Code
salientPole1	Salient pole 1 in the IEC 60909	
salientPole2	Salient pole 2 in IEC 60909	
turboSeries1	Turbo Series 1 in the IEC 60909	
turboSeries2	Turbo series 2 in IEC 60909	

3.13.2.24 SynchronousMachineKind

Synchronous machine type.

Literal	Description	Code
generator		
condenser		
generatorOrCondenser		
motor		
generatorOrMotor		

motorOrCondenser		
generatorOrCondenserOrMotor		

3.13.2.25 ControlAreaTypeKind

The type of control area.

Literal	Description	Code
AGC	Used for automatic generation control.	
Forecast	Used for load forecast.	
Interchange	Used for interchange specification or control.	

3.13.3 Primitives

3.13.3.1 String

A string consisting of a sequence of characters. The character encoding is UTF-8. The string length is unspecified and unlimited.

3.13.3.2 Date

Date as "yyyy-mm-dd", which conforms with ISO 8601. UTC time zone is specified as "yyyy-mm-ddZ". A local timezone relative UTC is specified as "yyyy-mm-dd(+/-)hh:mm".

3.13.3.3 Float

A floating point number. The range is unspecified and not limited.

3.13.3.4 Integer

An integer number. The range is unspecified and not limited.

3.13.3.5 Float

A floating point number. The range is unspecified and not limited.

3.13.3.6 Boolean

A type with the value space "true" and "false".

3.13.3.7 DateTime

Date and time as "yyyy-mm-ddThh:mm:ss.sss", which conforms with ISO 8601. UTC time zone is specified as "yyyy-mm-ddThh:mm:ss.sssZ". A local timezone relative UTC is specified as "yyyy-mm-ddThh:mm:ss.sss-hh:mm". The second component (shown here as "ss.sss") could have any number of digits in its fractional part to allow any kind of precision beyond seconds.

3.13.3.8 Decimal

Decimal is the base-10 notational system for representing real numbers.

3.13.3.9 MonthDay

MonthDay format as "--mm-dd", which conforms with XSD data type gMonthDay.

4 TopologyProfile

4.1 General

This profile has been built on the basis of the IEC 61970-456 document and adjusted to fit the purpose of the ENTSO-E CGMES.

Figure 4-1 shows the diagram.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

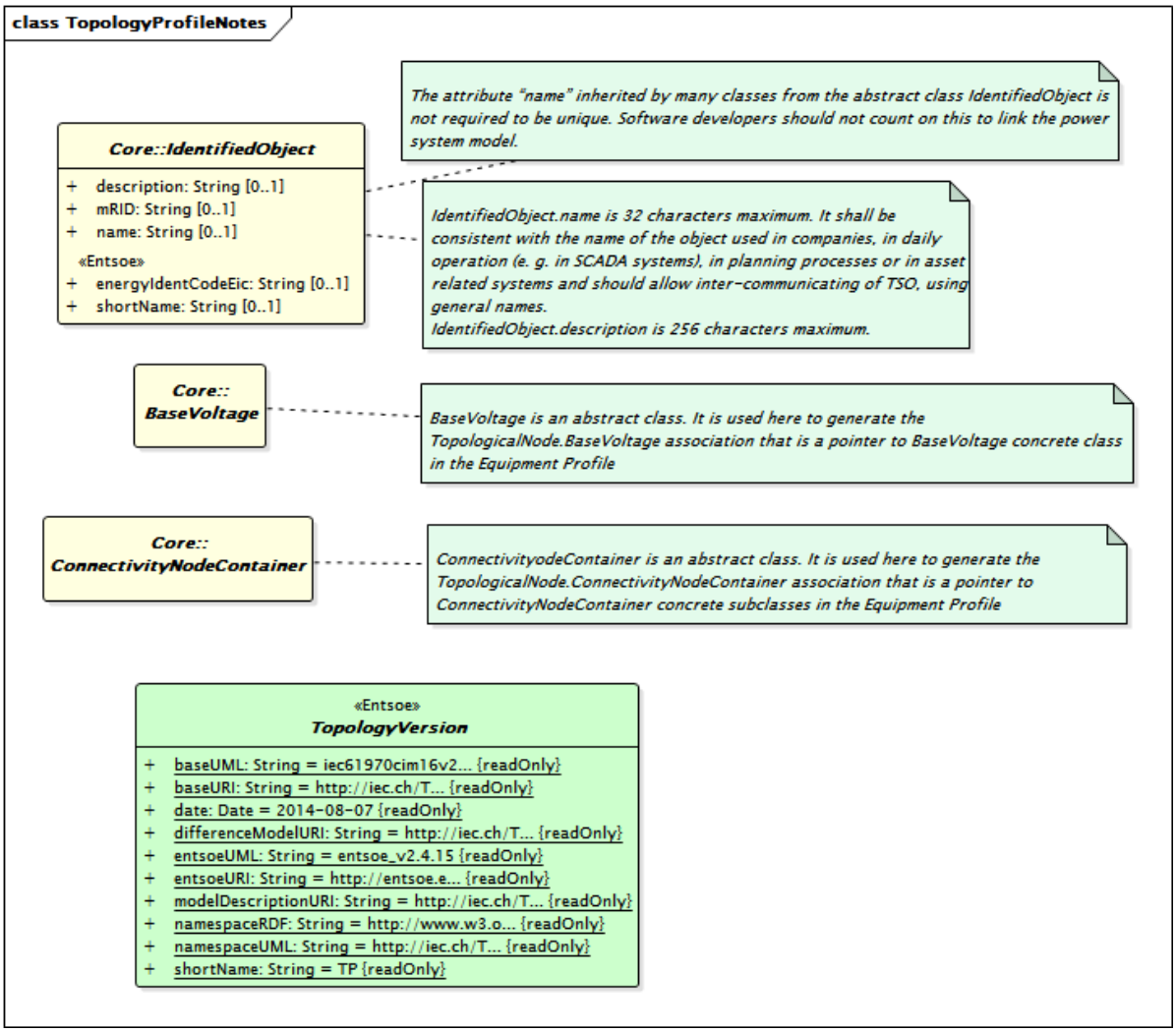


Figure 4-1 - diagram TopologyProfileNotes

Figure 4-2 shows the diagram.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

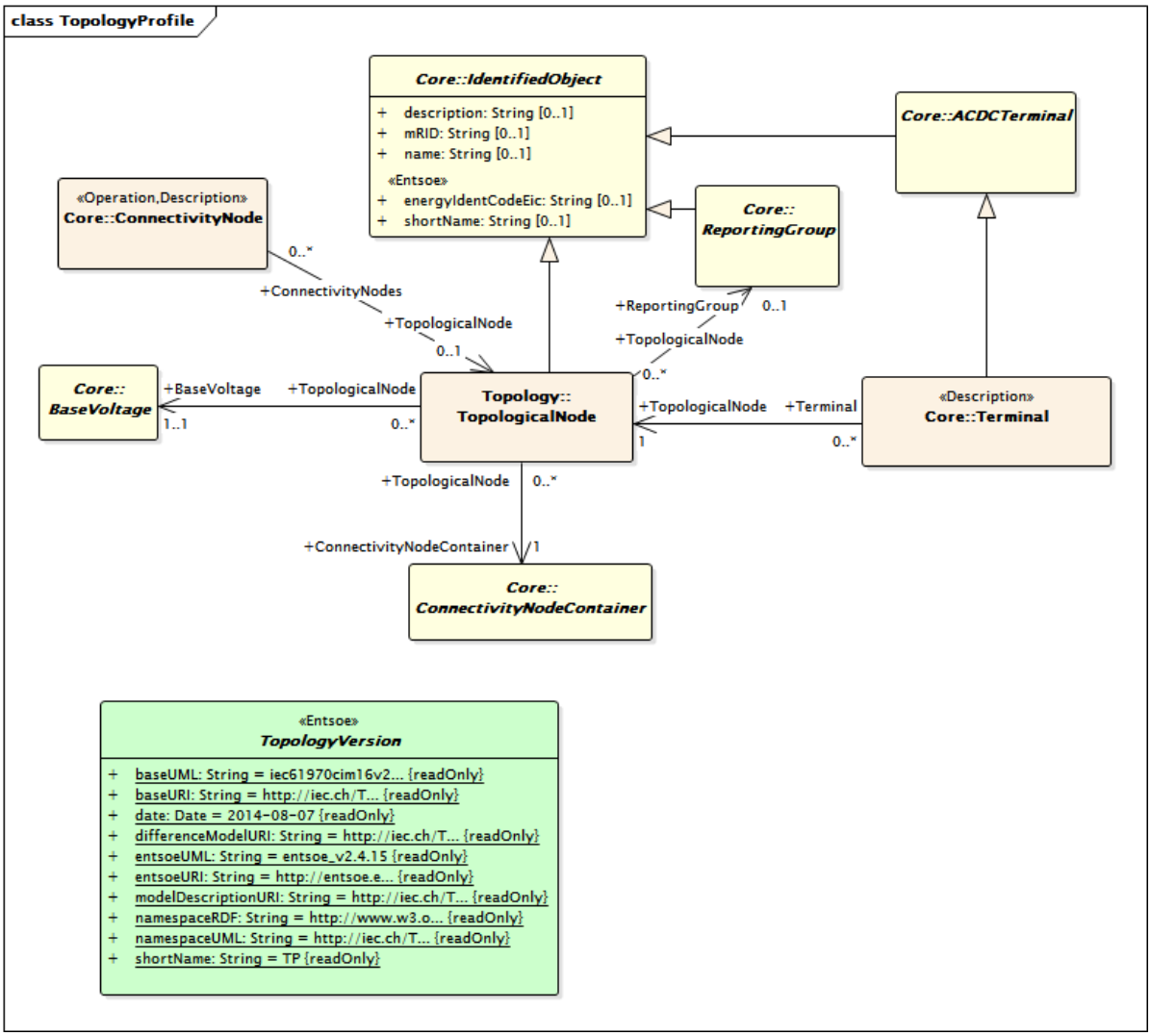


Figure 4-2 - diagram TopologyProfile

4.2 (Entsoe) TopologyVersion(Abstract)

Version details.

Table 4-1 shows all attributes of TopologyVersion.

Table 4-1 - Attributes of TopologyProfile::TopologyVersion

Attribute name	mult.	Attribute type	Description
baseUML (Entsoe)	1..1	String	ConstantValue= iec61970cim16v28_iec61968cim12v08_iec62325cim03v01a Base UML provided by CIM model manager.
baseURI (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/2013/61970-456/Topology/4 Profile URI used in the Model Exchange header and defined in IEC standards. It uniquely identifies the Profile and its version. It is given for information only and to identify the closest IEC profile to which this CGMES profile is based on.

date (Entsoe)	1..1	Date	ConstantValue= 2014-08-07 Profile creation date Form is YYYY-MM-DD for example for January 5, 2009 it is 2009-01-05.
differenceModelURI (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/61970-552/DifferenceModel/1# Difference model URI defined by IEC 61970-552.
entsoeUML (Entsoe)	1..1	String	ConstantValue= entsoe_v2.4.15 UML provided by ENTSO-E.
entsoeURI (Entsoe)	1..1	String	ConstantValue= http://entsoe.eu/CIM/Topology/4/1 Profile URI defined by ENTSO-E and used in the Model Exchange header. It uniquely identifies the Profile and its version. The last two elements in the URI (http://entsoe.eu/CIM/Topology/yy/zzz) indicate major and minor versions where: - yy - indicates a major version; - zzz - indicates a minor version.
modelDescriptionURI (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/61970-552/ModelDescription/1# Model Description URI defined by IEC 61970-552.
namespaceRDF (Entsoe)	1..1	String	ConstantValue= http://www.w3.org/1999/02/22-rdf-syntax-ns# RDF namespace.
namespaceUML (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/2013/CIM-schema-cim16# CIM UML namespace.
shortName (Entsoe)	1..1	String	ConstantValue= TP The short name of the profile used in profile documentation.

4.3 Core

4.3.1 General

Contains the core PowerSystemResource and ConductingEquipment entities

Figure 4-3 shows the diagram.

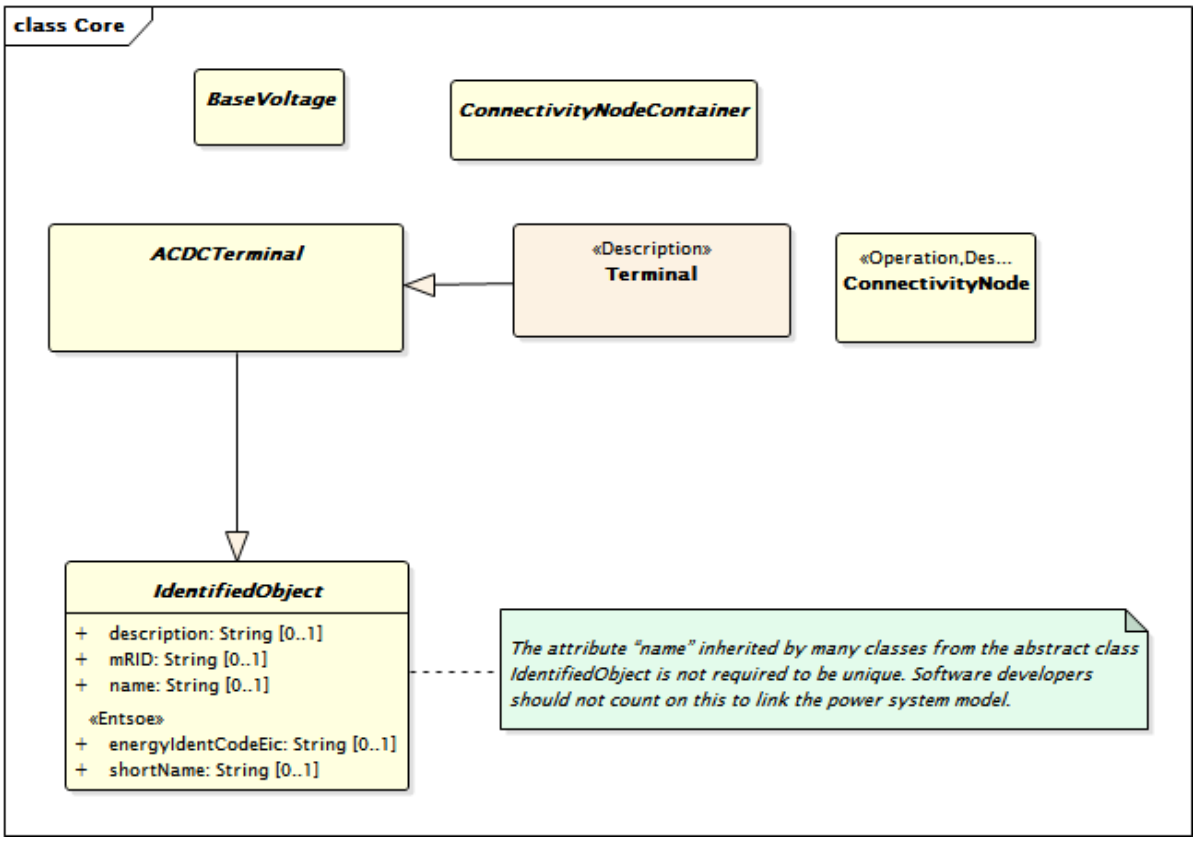


Figure 4-3 - diagram Core

4.3.2 ACDCTerminal(Abstract)

An electrical connection point (AC or DC) to a piece of conducting equipment. Terminals are connected at physical connection points called connectivity nodes.

Inheritance path: ->IdentifiedObject

Table 4-2 shows all attributes of ACDCTerminal.

Table 4-2 - Attributes of Core::ACDCTerminal

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

4.3.3 BaseVoltage(Abstract)

Defines a system base voltage which is referenced.

BaseVoltage is an abstract class. It is used here to generate the TopologicalNode.BaseVoltage association that is a pointer to BaseVoltage concrete class in the Equipment Profile

4.3.4 ConnectivityNode

Connectivity nodes are points where terminals of AC conducting equipment are connected together with zero impedance.

4.3.5 ConnectivityNodeContainer(Abstract)

A base class for all objects that may contain connectivity nodes or topological nodes.

ConnectivityodeContainer is an abstract class. It is used here to generate the TopologicalNode.ConnectivityNodeContainer association that is a pointer to ConnectivityNodeContainer concrete subclasses in the Equipment Profile

4.3.6 ReportingGroup(Abstract)

A reporting group is used for various ad-hoc groupings used for reporting.

Inheritance path: ->IdentifiedObject

Table 4-3 shows all attributes of ReportingGroup.

Table 4-3 - Attributes of Core::ReportingGroup

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

4.3.7 IdentifiedObject(Abstract)

This is a root class to provide common identification for all classes needing identification and naming attributes.

OCL constraint:R.4.10.11. Description length restriction (optional).
OCL constraint:R.4.10.11. Energy Ident Code length restriction (optional).
OCL constraint:R.4.10.11. Name length restriction.
OCL constraint:R.4.10.11. ShortName length restriction (optional).
OCL constraint:R.4.10.13. Name is required for all classes excepted ACDCTerminal.

IdentifiedObject.name is 32 characters maximum. It shall be consistent with the name of the object used in companies, in daily operation (e. g. in SCADA systems), in planning processes or in asset related systems and should allow inter-communicating of TSO, using general names. IdentifiedObject.description is 256 characters maximum.

The attribute “name” inherited by many classes from the abstract class IdentifiedObject is not required to be unique. Software developers should not count on this to link the power system model.

Table 4-4 shows all attributes of IdentifiedObject.

Table 4-4 - Attributes of Core::IdentifiedObject

Attribute name	mult.	Attribute type	Description
description	0..1	String	The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

energyIdentCodeEic (Entsoe)	0..1	String	The attribute is used for an exchange of the EIC code (Energy identification Code). The length of the string is 16 characters as defined by the EIC code. References: Local issuing offices for EIC: https://www.entsoe.eu/publications/edi-library/links-to-eic-websites/ EIC description: https://www.entsoe.eu/index.php?id=73&libCat=eic .
mRID	0..1	String	Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552 Edition 1, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name	0..1	String	The name is any free human readable and possibly non unique text naming the object.
shortName (Entsoe)	0..1	String	The attribute is used for an exchange of a human readable short name with length of the string 12 characters maximum.

4.3.8 (Entsoe) Terminal

An AC electrical connection point to a piece of conducting equipment. Terminals are connected at physical connection points called connectivity nodes.

Inheritance path: ->ACDCTerminal->IdentifiedObject

Table 4-5 shows all attributes of Terminal.

Table 4-5 - Attributes of Core::Terminal

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 4-6 shows all association ends of Terminal with other classes.

Table 4-6 - Association ends of Core::Terminal with other classes

mult.	mult.	Class type name	Description
TopologicalNode (Description)	[1..1]	TopologicalNode	The terminals associated with the topological node. This can be used as an alternative to the connectivity node path to terminal, thus making it unneccesary to model connectivity nodes in some cases. Note that if connectivity nodes are in the model, this association would probably not be used as an input specification. Association Based On : Core:: Terminal.Terminal[0..*] -----

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			Topology:: TopologicalNode.TopologicalNode[0..1]

4.4 DC

4.4.1 General

Figure 4-4 shows the diagram.

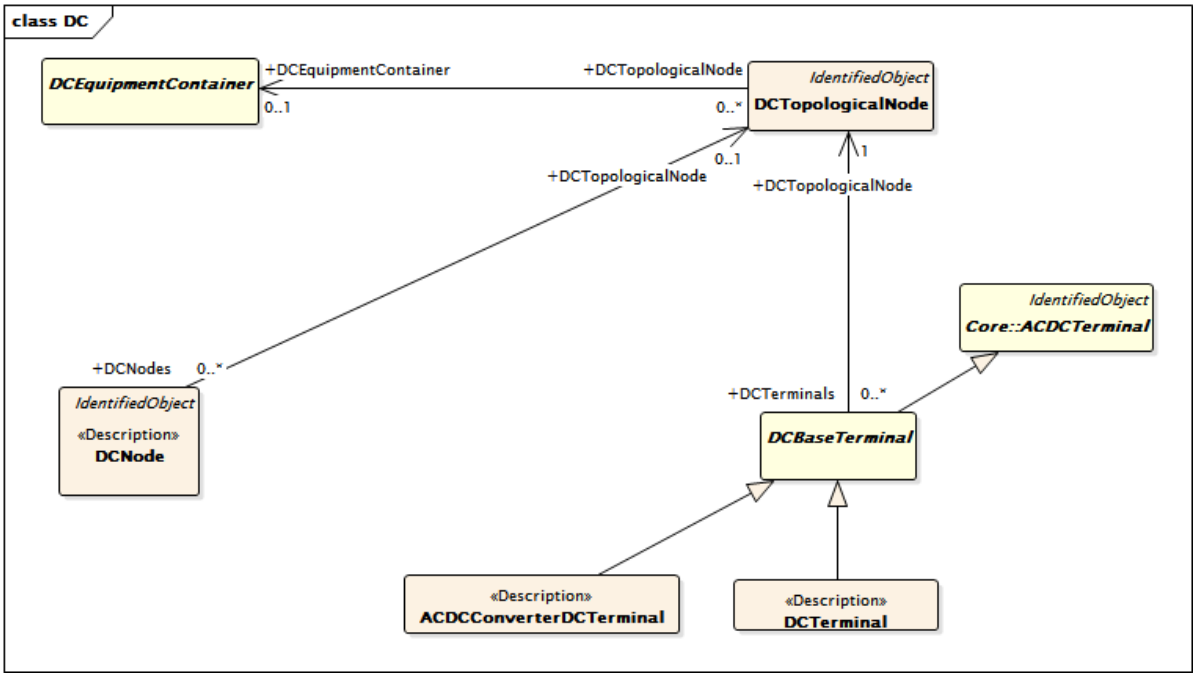


Figure 4-4 - diagram DC

4.4.2 (Description) ACDCConverterDCTerminal

A DC electrical connection point at the AC/DC converter. The AC/DC converter is electrically connected also to the AC side. The AC connection is inherited from the AC conducting equipment in the same way as any other AC equipment. The AC/DC converter DC terminal is separate from generic DC terminal to restrict the connection with the AC side to AC/DC converter and so that no other DC conducting equipment can be connected to the AC side.

Inheritance path: ->DCBaseTerminal->ACDCTerminal->IdentifiedObject

Table 4-7 shows all attributes of ACDCConverterDCTerminal.

Table 4-7 - Attributes of DC::ACDCConverterDCTerminal

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 4-8 shows all association ends of DCBaseTerminal with other classes.

Table 4-8 - Association ends of DC::DCBaseTerminal with other classes

mult.	mult.	Class type name	Description
DCTopologicalNode	1..1	DCTopologicalNode	Inherited Association Based On : DC:: DCBaseTerminal.DCTerminals[0..*] ----- Topology:: DCTopologicalNode.DCTopologicalNode[0..1]

4.4.3 DCBaseTerminal(Abstract)

An electrical connection point at a piece of DC conducting equipment. DC terminals are connected at one physical DC node that may have multiple DC terminals connected. A DC node is similar to an AC connectivity node. The model enforces that DC connections are distinct from AC connections.

Inheritance path: ->ACDCTerminal->IdentifiedObject

Table 4-9 shows all attributes of DCBaseTerminal.

Table 4-9 - Attributes of DC::DCBaseTerminal

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 4-10 shows all association ends of DCBaseTerminal with other classes.

Table 4-10 - Association ends of DC::DCBaseTerminal with other classes

mult.	mult.	Class type name	Description
DCTopologicalNode	[1..1]	DCTopologicalNode	See association end TopologicalNode.Terminal. Association Based On : DC:: DCBaseTerminal.DCTerminals[0..*] ----- Topology:: DCTopologicalNode.DCTopologicalNode[0..1]

4.4.4 DCEquipmentContainer(Abstract)

A modeling construct to provide a root class for containment of DC as well as AC equipment. The class differ from the EquipmentContaner for AC in that it may also contain DCNodes. Hence it can contain both AC and DC equipment.

4.4.5 DCNode

DC nodes are points where terminals of DC conducting equipment are connected together with zero impedance.

Inheritance path: ->IdentifiedObject

Table 4-11 shows all attributes of DCNode.

Table 4-11 - Attributes of DC::DCNode

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 4-12 shows all association ends of DCNode with other classes.

Table 4-12 - Association ends of DC::DCNode with other classes

mult.	mult.	Class type name	Description
DCTopologicalNode (Description)	[0..1]	DCTopologicalNode	See association end TopologicalNode.ConnectivityNodes. Association Based On : DC:: DCNode.DCNodes[0..*] ----- Topology:: DCTopologicalNode.DCTopologicalNode[0..1]

4.4.6 (Description) DCTerminal

An electrical connection point to generic DC conducting equipment.

Inheritance path: ->DCBaseTerminal->ACDCTerminal->IdentifiedObject

Table 4-13 shows all attributes of DCTerminal.

Table 4-13 - Attributes of DC::DCTerminal

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 4-14 shows all association ends of DCBaseTerminal with other classes.

Table 4-14 - Association ends of DC::DCBaseTerminal with other classes

mult.	mult.	Class type name	Description
DCTopologicalNode	1..1	DCTopologicalNode	Inherited Association Based On : DC:: DCBaseTerminal.DCTerminals[0..*] ----- Topology:: DCTopologicalNode.DCTopologicalNode[0..1]

4.4.7 DCTopologicalNode

DC bus.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Inheritance path: ->IdentifiedObject

Table 4-15 shows all attributes of DCTopologicalNode.

Table 4-15 - Attributes of DC::DCTopologicalNode

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 4-16 shows all association ends of DCTopologicalNode with other classes.

Table 4-16 - Association ends of DC::DCTopologicalNode with other classes

mult.	mult.	Class type name	Description
DCEquipmentContainer	[0..1]	DCEquipmentContainer	Association Based On : DC:: DCEquipmentContainer.DCEquipmentContainer[0..1] ----- Topology:: DCTopologicalNode.DCTopologicalNode[0..*]

4.5 Topology

4.5.1 General

4.5.2 TopologicalNode

For a detailed substation model a topological node is a set of connectivity nodes that, in the current network state, are connected together through any type of closed switches, including jumpers. Topological nodes change as the current network state changes (i.e., switches, breakers, etc. change state).
For a planning model, switch statuses are not used to form topological nodes. Instead they are manually created or deleted in a model builder tool. Topological nodes maintained this way are also called "busses".

Inheritance path: ->IdentifiedObject

Table 4-17 shows all attributes of TopologicalNode.

Table 4-17 - Attributes of Topology::TopologicalNode

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
energyIdentCodeEic (Entsoe)	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject
shortName (Entsoe)	0..1	String	see IdentifiedObject

Table 4-18 shows all association ends of TopologicalNode with other classes.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 4-18 - Association ends of Topology::TopologicalNode with other classes

mult.	mult.	Class type name	Description
ReportingGroup	[0..1]	ReportingGroup	The topological nodes that belong to the reporting group. Association Based On : Topology:: TopologicalNode.TopologicalNode[0..*] ----- Core:: ReportingGroup.ReportingGroup[0..1]
ConnectivityNodeContainer	[1..1]	ConnectivityNodeContainer	The connectivity node container to which the toplogical node belongs. Association Based On : Topology:: TopologicalNode.TopologicalNode[0..*] ----- Core:: ConnectivityNodeContainer.ConnectivityNodeContainer[0..1]
BaseVoltage	[1..1]	BaseVoltage	The base voltage of the topologocial node. Association Based On : Topology:: TopologicalNode.TopologicalNode[0..*] ----- Core:: BaseVoltage.BaseVoltage[0..1]

4.6 Global Used Datatypes

4.6.1 Primitives

4.6.1.1 String

A string consisting of a sequence of characters. The character encoding is UTF-8. The string length is unspecified and unlimited.

4.6.1.2 Date

Date as "yyyy-mm-dd", which conforms with ISO 8601. UTC time zone is specified as "yyyy-mm-ddZ". A local timezone relative UTC is specified as "yyyy-mm-dd(+/-)hh:mm".

5 SteadyStateHypothesisProfile

5.1 General

This profile has been built on the basis of the IEC 61970-456 document and adjusted to fit the purpose of the ENTSO-E CGMES.

Figure 5-1 shows the diagram.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

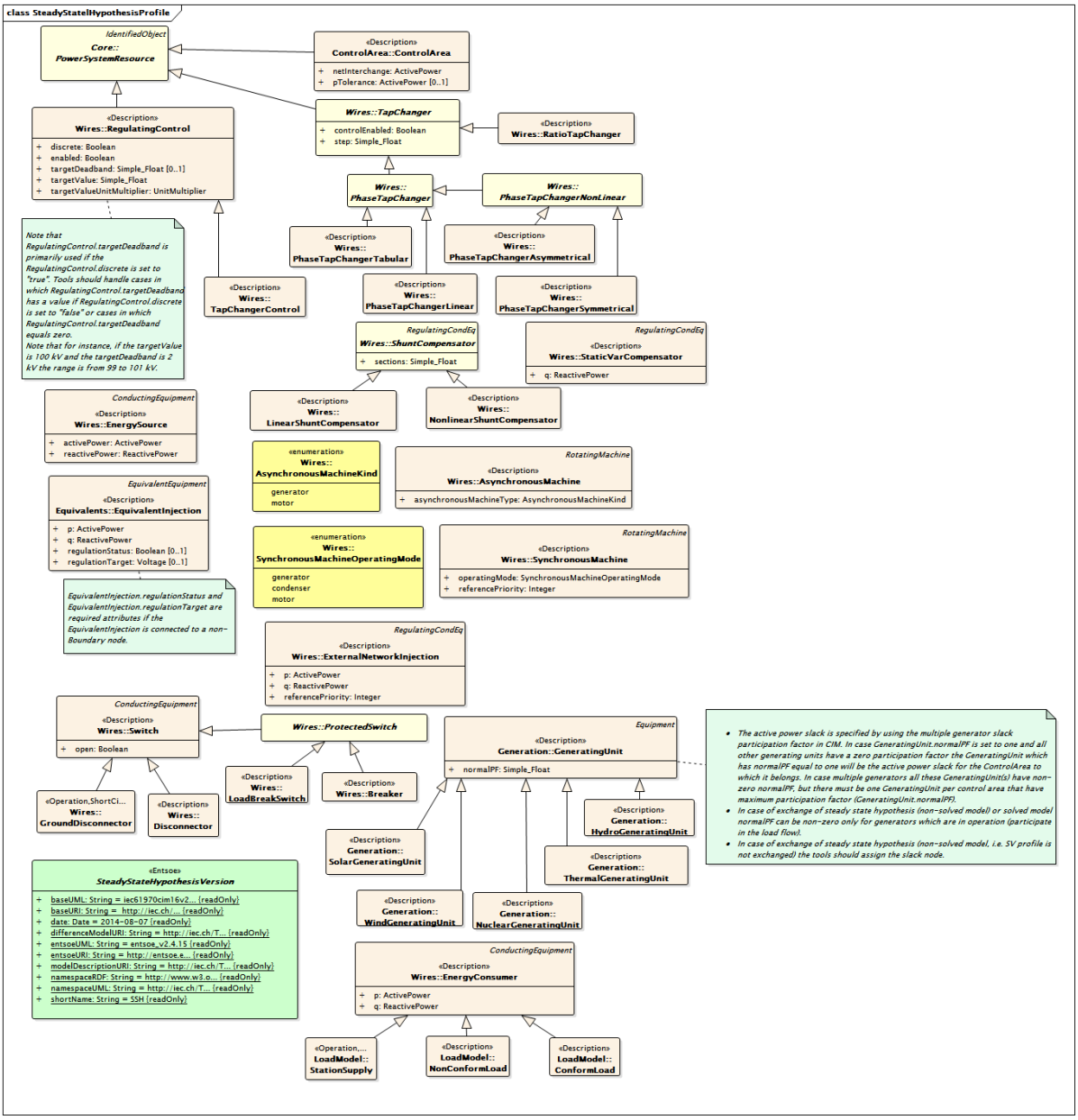


Figure 5-1 - diagram SteadyStatelHypothesisProfile

5.2 (Entsoe) SteadyStateHypothesisVersion(Abstract)

Version details.

Table 5-1 shows all attributes of SteadyStateHypothesisVersion.

Table 5-1 - Attributes of SteadyStateHypothesisProfile::SteadyStateHypothesisVersion

Attribute name	mult.	Attribute type	Description
baseUML (Entsoe)	1..1	String	ConstantValue= iec61970cim16v28_iec61968cim12v08_iec62325cim03v01a Base UML provided by CIM model manager.
baseURI (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/2013/61970-456/SteadyStateHypothesis/1

			Profile URI used in the Model Exchange header and defined in IEC standards. It uniquely identifies the Profile and its version. It is given for information only and to identify the closest IEC profile to which this CGMES profile is based on.
date (Entsoe)	1..1	Date	ConstantValue= 2014-08-07 Profile creation date Form is YYYY-MM-DD for example for January 5, 2009 it is 2009-01-05.
differenceModelURI (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/61970-552/DifferenceModel/1# Difference model URI defined by IEC 61970-552.
entsoeUML (Entsoe)	1..1	String	ConstantValue= entsoe_v2.4.15 UML provided by ENTSO-E.
entsoeURI (Entsoe)	1..1	String	ConstantValue= http://entsoe.eu/CIM/SteadyStateHypothesis/1/1 Profile URI defined by ENTSO-E and used in the Model Exchange header. It uniquely identifies the Profile and its version. The last two elements in the URI (http://entsoe.eu/CIM/SteadyStateHypothesis/yy/zzz) indicate major and minor versions where: - yy - indicates a major version; - zzz - indicates a minor version.
modelDescriptionURI (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/61970-552/ModelDescription/1# Model Description URI defined by IEC 61970-552.
namespaceRDF (Entsoe)	1..1	String	ConstantValue= http://www.w3.org/1999/02/22-rdf-syntax-ns# RDF namespace.
namespaceUML (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/2013/CIM-schema-cim16# CIM UML namespace.
shortName (Entsoe)	1..1	String	ConstantValue= SSH The short name of the profile used in profile documentation.

5.3 Core

5.3.1 General

Figure 5-2 shows the diagram.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

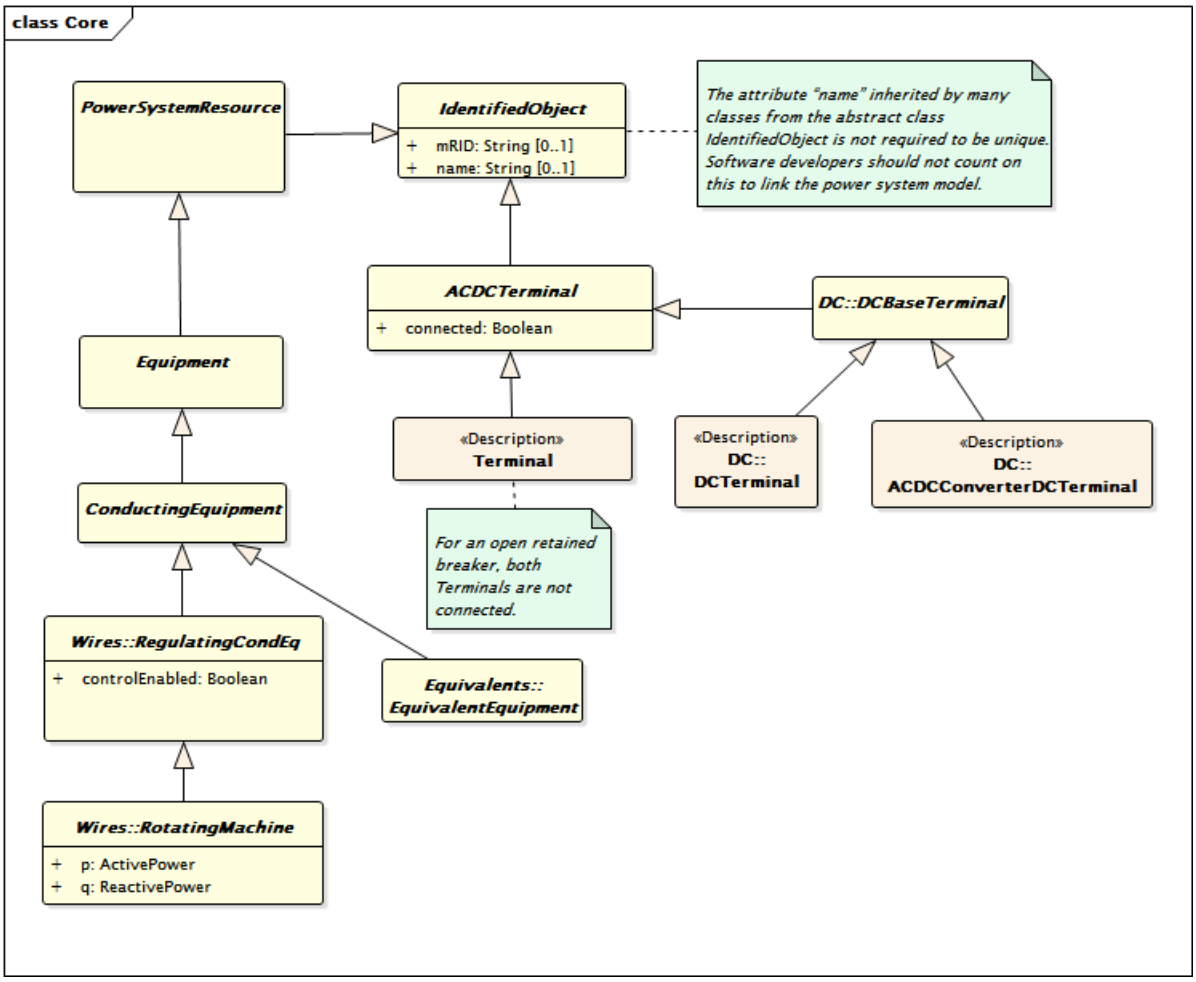


Figure 5-2 - diagram Core

5.3.2 ACDCTerminal(Abstract)

An electrical connection point (AC or DC) to a piece of conducting equipment. Terminals are connected at physical connection points called connectivity nodes.

Inheritance path: ->IdentifiedObject

Table 5-2 shows all attributes of ACDCTerminal.

Table 5-2 - Attributes of Core::ACDCTerminal

Attribute name	mult.	Attribute type	Description
connected	1..1	Boolean	The connected status is related to a bus-branch model and the topological node to terminal relation. True implies the terminal is connected to the related topological node and false implies it is not. In a bus-branch model, the connected status is used to tell if equipment

			is disconnected without having to change the connectivity described by the topological node to terminal relation. A valid case is that conducting equipment can be connected in one end and open in the other. In particular for an AC line segment, where the reactive line charging can be significant, this is a relevant case.
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.3.3 ConductingEquipment(Abstract)

The parts of the AC power system that are designed to carry current or that are conductively connected through terminals.

Inheritance path: ->Equipment->PowerSystemResource->IdentifiedObject

Table 5-3 shows all attributes of ConductingEquipment.

Table 5-3 - Attributes of Core::ConductingEquipment

Attribute name	mult.	Attribute type	Description
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.3.4 Equipment(Abstract)

The parts of a power system that are physical devices, electronic or mechanical.

Inheritance path: ->PowerSystemResource->IdentifiedObject

Table 5-4 shows all attributes of Equipment.

Table 5-4 - Attributes of Core::Equipment

Attribute name	mult.	Attribute type	Description
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.3.5 IdentifiedObject(Abstract)

This is a root class to provide common identification for all classes needing identification and naming attributes.

OCL constraint:R.4.10.11. Name length restriction (optional)

The attribute “name” inherited by many classes from the abstract class IdentifiedObject is not required to be unique. Software developers should not count on this to link the power system model.

Table 5-5 shows all attributes of IdentifiedObject.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 5-5 - Attributes of Core::IdentifiedObject

Attribute name	mult.	Attribute type	Description
mRID	0..1	String	Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552 Edition 1, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name	0..1	String	The name is any free human readable and possibly non unique text naming the object.

5.3.6 PowerSystemResource(Abstract)

A power system resource can be an item of equipment such as a switch, an equipment container containing many individual items of equipment such as a substation, or an organisational entity such as sub-control area. Power system resources can have measurements associated.

Inheritance path: ->IdentifiedObject

Table 5-6 shows all attributes of PowerSystemResource.

Table 5-6 - Attributes of Core::PowerSystemResource

Attribute name	mult.	Attribute type	Description
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.3.7 Terminal

An AC electrical connection point to a piece of conducting equipment. Terminals are connected at physical connection points called connectivity nodes.

For an open retained breaker, both Terminals are not connected.

Inheritance path: ->ACDCTerminal->IdentifiedObject

Table 5-7 shows all attributes of Terminal.

Table 5-7 - Attributes of Core::Terminal

Attribute name	mult.	Attribute type	Description
connected	1..1	Boolean	see ACDCTerminal

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.4 ControlArea

5.4.1 General

5.4.2 ControlArea

A control area is a grouping of generating units and/or loads and a cutset of tie lines (as terminals) which may be used for a variety of purposes including automatic generation control, powerflow solution area interchange control specification, and input to load forecasting. Note that any number of overlapping control area specifications can be superimposed on the physical model.

Inheritance path: ->PowerSystemResource->IdentifiedObject

Table 5-8 shows all attributes of ControlArea.

Table 5-8 - Attributes of ControlArea::ControlArea

Attribute name	mult.	Attribute type	Description
netInterchange (Description)	1..1	ActivePower	The specified positive net interchange into the control area, i.e. positive sign means flow in to the area.
pTolerance (Description)	0..1	ActivePower	Active power net interchange tolerance
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.5 DC

5.5.1 General

Figure 5-3 shows the diagram.

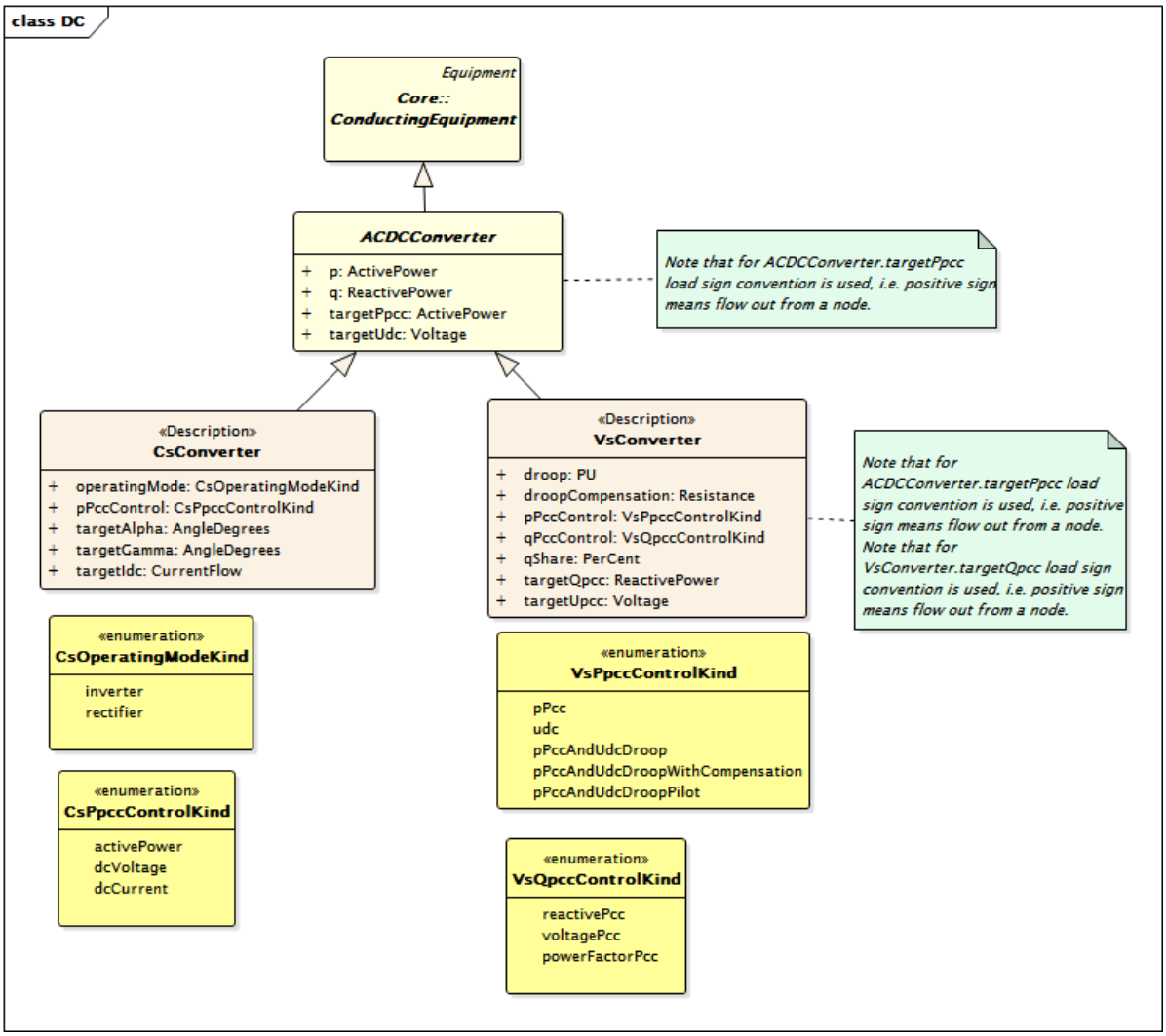


Figure 5-3 - diagram DC

5.5.2 ACDCConverter(Abstract)

A unit with valves for three phases, together with unit control equipment, essential protective and switching devices, DC storage capacitors, phase reactors and auxiliaries, if any, used for conversion.

Note that for ACDCConverter.targetPpcc load sign convention is used, i.e. positive sign means flow out from a node.

Inheritance path: ->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 5-9 shows all attributes of ACDCConverter.

Table 5-9 - Attributes of DC::ACDCConverter

Attribute name	mult.	Attribute type	Description
p	1..1	ActivePower	Active power at the point of common coupling. Load sign convention is used, i.e. positive sign means

			flow out from a node. Starting value for a steady state solution in the case a simplified power flow model is used.
q	1..1	ReactivePower	Reactive power at the point of common coupling. Load sign convention is used, i.e. positive sign means flow out from a node. Starting value for a steady state solution in the case a simplified power flow model is used.
targetPpcc	1..1	ActivePower	Real power injection target in AC grid, at point of common coupling.
targetUdc	1..1	Voltage	Target value for DC voltage magnitude.
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.5.3 ACDCConverterDCTerminal

A DC electrical connection point at the AC/DC converter. The AC/DC converter is electrically connected also to the AC side. The AC connection is inherited from the AC conducting equipment in the same way as any other AC equipment. The AC/DC converter DC terminal is separate from generic DC terminal to restrict the connection with the AC side to AC/DC converter and so that no other DC conducting equipment can be connected to the AC side.

Inheritance path: ->DCBaseTerminal->ACDCTerminal->IdentifiedObject

Table 5-10 shows all attributes of ACDCConverterDCTerminal.

Table 5-10 - Attributes of DC::ACDCConverterDCTerminal

Attribute name	mult.	Attribute type	Description
connected	1..1	Boolean	see ACDCTerminal
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.5.4 CsConverter

DC side of the current source converter (CSC).

Inheritance path: ->ACDCConverter->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 5-11 shows all attributes of CsConverter.

Table 5-11 - Attributes of DC::CsConverter

Attribute name	mult.	Attribute type	Description
operatingMode (Description)	1..1	CsOperatingModeKind	Indicates whether the

			DC pole is operating as an inverter or as a rectifier. CSC control variable used in power flow.
pPccControl (Description)	1..1	CsPpccControlKind	
targetAlpha (Description)	1..1	AngleDegrees	Target firing angle. CSC control variable used in power flow.
targetGamma (Description)	1..1	AngleDegrees	Target extinction angle. CSC control variable used in power flow.
targetIdc (Description)	1..1	CurrentFlow	DC current target value. CSC control variable used in power flow.
p	1..1	ActivePower	see ACDCConverter
q	1..1	ReactivePower	see ACDCConverter
targetPpcc	1..1	ActivePower	see ACDCConverter
targetUdc	1..1	Voltage	see ACDCConverter
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.5.5 DCBaseTerminal(Abstract)

An electrical connection point at a piece of DC conducting equipment. DC terminals are connected at one physical DC node that may have multiple DC terminals connected. A DC node is similar to an AC connectivity node. The model enforces that DC connections are distinct from AC connections.

Inheritance path: ->ACDCTerminal->IdentifiedObject

Table 5-12 shows all attributes of DCBaseTerminal.

Table 5-12 - Attributes of DC::DCBaseTerminal

Attribute name	mult.	Attribute type	Description
connected	1..1	Boolean	see ACDCTerminal
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.5.6 DCTerminal

An electrical connection point to generic DC conducting equipment.

Inheritance path: ->DCBaseTerminal->ACDCTerminal->IdentifiedObject

Table 5-13 shows all attributes of DCTerminal.

Table 5-13 - Attributes of DC::DCTerminal

Attribute name	mult.	Attribute type	Description
connected	1..1	Boolean	see ACDCTerminal
mRID	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

name	0..1	String	see IdentifiedObject

5.5.7 VsConverter

DC side of the voltage source converter (VSC).

Note that for ACDCConverter.targetPpcc load sign convention is used, i.e. positive sign means flow out from a node. Note that for VsConverter.targetQpcc load sign convention is used, i.e. positive sign means flow out from a node.

Inheritance path: ->ACDCConverter->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 5-14 shows all attributes of VsConverter.

Table 5-14 - Attributes of DC::VsConverter

Attribute name	mult.	Attribute type	Description
droop (Description)	1..1	PU	Droop constant; pu value is obtained as D [kV^2 / MW] x Sb / Ubdc^2.
droopCompensation (Description)	1..1	Resistance	Compensation (resistance) constant. Used to compensate for voltage drop when controlling voltage at a distant bus.
pPccControl (Description)	1..1	VsPpccControlKind	Kind of control of real power and/or DC voltage.
qPccControl (Description)	1..1	VsQpccControlKind	
qShare (Description)	1..1	PerCent	Reactive power sharing factor among parallel converters on Uac control.
targetQpcc (Description)	1..1	ReactivePower	Reactive power injection target in AC grid, at point of common coupling.
targetUpcc (Description)	1..1	Voltage	Voltage target in AC grid, at point of common coupling.
p	1..1	ActivePower	see ACDCConverter
q	1..1	ReactivePower	see ACDCConverter
targetPpcc	1..1	ActivePower	see ACDCConverter
targetUdc	1..1	Voltage	see ACDCConverter
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.5.8 Enumerations

5.5.8.1 CsOperatingModeKind

Operating mode for HVDC line operating as Current Source Converter.

Literal	Description	Code
inverter	Operating as inverter	
rectifier	Operating as rectifier.	

5.5.8.2 CsPpccControlKind

Active power control modes for HVDC line operating as Current Source Converter.

Literal	Description	Code
activePower	Active power control at AC side.	
dcVoltage	DC voltage control.	
dcCurrent	DC current control	

5.5.8.3 VsPpccControlKind

Types applicable to the control of real power and/or DC voltage by voltage source converter.

Literal	Description	Code
pPcc	Control variable (target) is real power at PCC bus.	
udc	Control variable (target) is DC voltage and real power at PCC bus is derived.	
pPccAndUdcDroop	Control variables (targets) are both active power at point of common coupling and local DC voltage, with the droop.	
pPccAndUdcDroopWithCompensation	Control variables (targets) are both active power at point of common coupling and compensated DC voltage, with the droop; compensation factor is the resistance, as an approximation of the DC voltage of a common (real or virtual) node in the DC network.	
pPccAndUdcDroopPilot	Control variables (targets) are both active power at point of common coupling and the pilot DC voltage, with the droop.	

5.5.8.4 VsQpccControlKind

Literal	Description	Code
reactivePcc		
voltagePcc		
powerFactorPcc		

5.6 Equivalents

5.6.1 General

5.6.2 EquivalentEquipment(Abstract)

The class represents equivalent objects that are the result of a network reduction. The class is the base for equivalent objects of different types.

Inheritance path: ->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 5-15 shows all attributes of EquivalentEquipment.

Table 5-15 - Attributes of Equivalents::EquivalentEquipment

Attribute name	mult.	Attribute type	Description
mRID	0..1	String	see IdentifiedObject

name	0..1	String	see IdentifiedObject

5.6.3 EquivalentInjection

This class represents equivalent injections (generation or load). Voltage regulation is allowed only at the point of connection.

EquivalentInjection.regulationStatus and EquivalentInjection.regulationTarget are required attributes if the EquivalentInjection is connected to a non-Boundary node.

Inheritance path: ->EquivalentEquipment->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 5-16 shows all attributes of EquivalentInjection.

Table 5-16 - Attributes of Equivalents::EquivalentInjection

Attribute name	mult.	Attribute type	Description
regulationStatus (Description)	0..1	Boolean	Specifies the default regulation status of the EquivalentInjection. True is regulating. False is not regulating.
regulationTarget (Description)	0..1	Voltage	The target voltage for voltage regulation.
p (Description)	1..1	ActivePower	Equivalent active power injection. Load sign convention is used, i.e. positive sign means flow out from a node. Starting value for steady state solutions.
q (Description)	1..1	ReactivePower	Equivalent reactive power injection. Load sign convention is used, i.e. positive sign means flow out from a node. Starting value for steady state solutions.
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.7 Generation

5.7.1 General

5.7.2 (Description) GeneratingUnit

A single or set of synchronous machines for converting mechanical power into alternating-current power. For example, individual machines within a set may be defined for scheduling purposes while a single control signal is derived for the set. In this case there would be a GeneratingUnit for each member of the set and an additional GeneratingUnit corresponding to the set.

- The active power slack is specified by using the multiple generator slack participation factor in CIM. In case GeneratingUnit.normalPF is set to one and all other generating units have a zero participation factor the GeneratingUnit which has normalPF equal to one will be the active power slack for the ControlArea to which it belongs. In case multiple generators all these GeneratingUnit(s) have non-zero normalPF, but there must be one GeneratingUnit per control area that have maximum participation factor (GeneratingUnit.normalPF).

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

- In case of exchange of steady state hypothesis (non-solved model) or solved model normalPF can be non-zero only for generators which are in operation (participate in the load flow).
- In case of exchange of steady state hypothesis (non-solved model, i.e. SV profile is not exchanged) the tools should assign the slack node.

Inheritance path: ->Equipment->PowerSystemResource->IdentifiedObject

Table 5-17 shows all attributes of GeneratingUnit.

Table 5-17 - Attributes of Generation::GeneratingUnit

Attribute name	mult.	Attribute type	Description
normalPF (Description)	1..1	Simple_Float	Generating unit economic participation factor.
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.7.3 (Description) HydroGeneratingUnit

A generating unit whose prime mover is a hydraulic turbine (e.g., Francis, Pelton, Kaplan).

Inheritance path: ->GeneratingUnit->Equipment->PowerSystemResource->IdentifiedObject

Table 5-18 shows all attributes of HydroGeneratingUnit.

Table 5-18 - Attributes of Generation::HydroGeneratingUnit

Attribute name	mult.	Attribute type	Description
normalPF	1..1	Simple_Float	see GeneratingUnit
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.7.4 (Description) NuclearGeneratingUnit

A nuclear generating unit.

Inheritance path: ->GeneratingUnit->Equipment->PowerSystemResource->IdentifiedObject

Table 5-19 shows all attributes of NuclearGeneratingUnit.

Table 5-19 - Attributes of Generation::NuclearGeneratingUnit

Attribute name	mult.	Attribute type	Description
normalPF	1..1	Simple_Float	see GeneratingUnit
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.7.5 (Description) SolarGeneratingUnit

A solar thermal generating unit.

Inheritance path: ->GeneratingUnit->Equipment->PowerSystemResource->IdentifiedObject

Table 5-20 shows all attributes of SolarGeneratingUnit.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 5-20 - Attributes of Generation::SolarGeneratingUnit

Attribute name	mult.	Attribute type	Description
normalPF	1..1	Simple_Float	see GeneratingUnit
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.7.6 (Description) ThermalGeneratingUnit

A generating unit whose prime mover could be a steam turbine, combustion turbine, or diesel engine.

Inheritance path: ->GeneratingUnit->Equipment->PowerSystemResource->IdentifiedObject

Table 5-21 shows all attributes of ThermalGeneratingUnit.

Table 5-21 - Attributes of Generation::ThermalGeneratingUnit

Attribute name	mult.	Attribute type	Description
normalPF	1..1	Simple_Float	see GeneratingUnit
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.7.7 (Description) WindGeneratingUnit

A wind driven generating unit. May be used to represent a single turbine or an aggregation.

Inheritance path: ->GeneratingUnit->Equipment->PowerSystemResource->IdentifiedObject

Table 5-22 shows all attributes of WindGeneratingUnit.

Table 5-22 - Attributes of Generation::WindGeneratingUnit

Attribute name	mult.	Attribute type	Description
normalPF	1..1	Simple_Float	see GeneratingUnit
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.8 Wires

5.8.1 General

5.8.2 (Description) AsynchronousMachine

A rotating machine whose shaft rotates asynchronously with the electrical field. Also known as an induction machine with no external connection to the rotor windings, e.g squirrel-cage induction machine.

Inheritance path: ->RotatingMachine->RegulatingCondEq->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 5-23 shows all attributes of AsynchronousMachine.

Table 5-23 - Attributes of Wires::AsynchronousMachine

Attribute name	mult.	Attribute type	Description

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

asynchronousMachineType (Description)	1..1	AsynchronousMachineKind	Indicates the type of Asynchronous Machine (motor or generator).
p	1..1	ActivePower	see RotatingMachine
q	1..1	ReactivePower	see RotatingMachine
controlEnabled	1..1	Boolean	see RegulatingCondEq
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.8.3 (Description) Breaker

A mechanical switching device capable of making, carrying, and breaking currents under normal circuit conditions and also making, carrying for a specified time, and breaking currents under specified abnormal circuit conditions e.g. those of short circuit.

Inheritance path: ->ProtectedSwitch->Switch->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 5-24 shows all attributes of Breaker.

Table 5-24 - Attributes of Wires::Breaker

Attribute name	mult.	Attribute type	Description
open	1..1	Boolean	see Switch
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.8.4 (Description) Disconnector

A manually operated or motor operated mechanical switching device used for changing the connections in a circuit, or for isolating a circuit or equipment from a source of power. It is required to open or close circuits when negligible current is broken or made.

Inheritance path: ->Switch->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 5-25 shows all attributes of Disconnector.

Table 5-25 - Attributes of Wires::Disconnector

Attribute name	mult.	Attribute type	Description
open	1..1	Boolean	see Switch
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.8.5 (Description) EnergyConsumer

Generic user of energy - a point of consumption on the power system model.

Inheritance path: ->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 5-26 shows all attributes of EnergyConsumer.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 5-26 - Attributes of Wires::EnergyConsumer

Attribute name	mult.	Attribute type	Description
p (Description)	1..1	ActivePower	Active power of the load. Load sign convention is used, i.e. positive sign means flow out from a node. For voltage dependent loads the value is at rated voltage. Starting value for a steady state solution.
q (Description)	1..1	ReactivePower	Reactive power of the load. Load sign convention is used, i.e. positive sign means flow out from a node. For voltage dependent loads the value is at rated voltage. Starting value for a steady state solution.
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.8.6 (Description) EnergySource

A generic equivalent for an energy supplier on a transmission or distribution voltage level.

Inheritance path: ->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 5-27 shows all attributes of EnergySource.

Table 5-27 - Attributes of Wires::EnergySource

Attribute name	mult.	Attribute type	Description
activePower (Description)	1..1	ActivePower	High voltage source active injection. Load sign convention is used, i.e. positive sign means flow out from a node. Starting value for steady state solutions.
reactivePower (Description)	1..1	ReactivePower	High voltage source reactive injection. Load sign convention is used, i.e. positive sign means flow out from a node. Starting value for steady state solutions.
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.8.7 (Description) ExternalNetworkInjection

This class represents external network and it is used for IEC 60909 calculations.

Inheritance path: ->RegulatingCondEq->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 5-28 shows all attributes of ExternalNetworkInjection.

Table 5-28 - Attributes of Wires::ExternalNetworkInjection

Attribute name	mult.	Attribute type	Description
referencePriority (Description)	1..1	Integer	Priority of unit for use as powerflow voltage phase angle reference bus selection. 0 = don t care (default) 1 = highest priority. 2 is less than 1 and so on.
p (Description)	1..1	ActivePower	Active power injection. Load sign convention is used, i.e. positive sign means flow out from a node. Starting value for steady state solutions.
q (Description)	1..1	ReactivePower	Reactive power injection. Load sign convention is used, i.e. positive sign means flow out from a node. Starting value for steady state solutions.
controlEnabled	1..1	Boolean	see RegulatingCondEq
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.8.8 (Description) GroundDisconnector

A manually operated or motor operated mechanical switching device used for isolating a circuit or equipment from ground.

Inheritance path: ->Switch->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 5-29 shows all attributes of GroundDisconnector.

Table 5-29 - Attributes of Wires::GroundDisconnector

Attribute name	mult.	Attribute type	Description
open	1..1	Boolean	see Switch
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.8.9 (Description) LoadBreakSwitch

A mechanical switching device capable of making, carrying, and breaking currents under normal operating conditions.

Inheritance path: ->ProtectedSwitch->Switch->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 5-30 shows all attributes of LoadBreakSwitch.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 5-30 - Attributes of Wires::LoadBreakSwitch

Attribute name	mult.	Attribute type	Description
open	1..1	Boolean	see Switch
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.8.10 PhaseTapChanger(Abstract)

A transformer phase shifting tap model that controls the phase angle difference across the power transformer and potentially the active power flow through the power transformer. This phase tap model may also impact the voltage magnitude.

Inheritance path: ->TapChanger->PowerSystemResource->IdentifiedObject

Table 5-31 shows all attributes of PhaseTapChanger.

Table 5-31 - Attributes of Wires::PhaseTapChanger

Attribute name	mult.	Attribute type	Description
controlEnabled	1..1	Boolean	see TapChanger
step	1..1	Simple_Float	see TapChanger
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.8.11 PhaseTapChangerAsymmetrical

Describes the tap model for an asymmetrical phase shifting transformer in which the difference voltage vector adds to the primary side voltage. The angle between the primary side voltage and the difference voltage is named the winding connection angle. The phase shift depends on both the difference voltage magnitude and the winding connection angle.

Inheritance path: ->PhaseTapChangerNonLinear->PhaseTapChanger->TapChanger->PowerSystemResource->IdentifiedObject

Table 5-32 shows all attributes of PhaseTapChangerAsymmetrical.

Table 5-32 - Attributes of Wires::PhaseTapChangerAsymmetrical

Attribute name	mult.	Attribute type	Description
controlEnabled	1..1	Boolean	see TapChanger
step	1..1	Simple_Float	see TapChanger
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.8.12 PhaseTapChangerLinear

Describes a tap changer with a linear relation between the tap step and the phase angle difference across the transformer. This is a mathematical model that is an approximation of a real phase tap changer.
The phase angle is computed as stepPhaseShitfIncrement times the tap position. The secondary side voltage magnitude is the same as at the primary side.

Inheritance path: ->PhaseTapChanger->TapChanger->PowerSystemResource->IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 5-33 shows all attributes of PhaseTapChangerLinear.

Table 5-33 - Attributes of Wires::PhaseTapChangerLinear

Attribute name	mult.	Attribute type	Description
controlEnabled	1..1	Boolean	see TapChanger
step	1..1	Simple_Float	see TapChanger
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.8.13 PhaseTapChangerNonLinear(Abstract)

The non-linear phase tap changer describes the non-linear behavior of a phase tap changer. This is a base class for the symmetrical and asymmetrical phase tap changer models. The details of these models can be found in the IEC 61970-301 document.

Inheritance path: ->PhaseTapChanger->TapChanger->PowerSystemResource->IdentifiedObject

Table 5-34 shows all attributes of PhaseTapChangerNonLinear.

Table 5-34 - Attributes of Wires::PhaseTapChangerNonLinear

Attribute name	mult.	Attribute type	Description
controlEnabled	1..1	Boolean	see TapChanger
step	1..1	Simple_Float	see TapChanger
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.8.14 PhaseTapChangerSymmetrical

Describes a symmetrical phase shifting transformer tap model in which the secondary side voltage magnitude is the same as at the primary side. The difference voltage magnitude is the base in an equal-sided triangle where the sides corresponds to the primary and secondary voltages. The phase angle difference corresponds to the top angle and can be expressed as twice the arctangent of half the total difference voltage.

Inheritance path: ->PhaseTapChangerNonLinear->PhaseTapChanger->TapChanger->PowerSystemResource->IdentifiedObject

Table 5-35 shows all attributes of PhaseTapChangerSymmetrical.

Table 5-35 - Attributes of Wires::PhaseTapChangerSymmetrical

Attribute name	mult.	Attribute type	Description
controlEnabled	1..1	Boolean	see TapChanger
step	1..1	Simple_Float	see TapChanger
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.8.15 PhaseTapChangerTabular

Inheritance path: ->PhaseTapChanger->TapChanger->PowerSystemResource->IdentifiedObject

Table 5-36 shows all attributes of PhaseTapChangerTabular.

Table 5-36 - Attributes of Wires::PhaseTapChangerTabular

Attribute name	mult.	Attribute type	Description
controlEnabled	1..1	Boolean	see TapChanger
step	1..1	Simple_Float	see TapChanger
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.8.16 ProtectedSwitch(Abstract)

A ProtectedSwitch is a switching device that can be operated by ProtectionEquipment.

Inheritance path: ->Switch->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 5-37 shows all attributes of ProtectedSwitch.

Table 5-37 - Attributes of Wires::ProtectedSwitch

Attribute name	mult.	Attribute type	Description
open	1..1	Boolean	see Switch
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.8.17 RatioTapChanger

A tap changer that changes the voltage ratio impacting the voltage magnitude but not the phase angle across the transformer.

Inheritance path: ->TapChanger->PowerSystemResource->IdentifiedObject

Table 5-38 shows all attributes of RatioTapChanger.

Table 5-38 - Attributes of Wires::RatioTapChanger

Attribute name	mult.	Attribute type	Description
controlEnabled	1..1	Boolean	see TapChanger
step	1..1	Simple_Float	see TapChanger
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.8.18 RegulatingCondEq(Abstract)

A type of conducting equipment that can regulate a quantity (i.e. voltage or flow) at a specific point in the network.

Inheritance path: ->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 5-39 shows all attributes of RegulatingCondEq.

Table 5-39 - Attributes of Wires::RegulatingCondEq

Attribute name	mult.	Attribute type	Description

controlEnabled	1..1	Boolean	Specifies the regulation status of the equipment. True is regulating, false is not regulating.
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.8.19 RotatingMachine(Abstract)

A rotating machine which may be used as a generator or motor.

Inheritance path: ->RegulatingCondEq->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 5-40 shows all attributes of RotatingMachine.

Table 5-40 - Attributes of Wires::RotatingMachine

Attribute name	mult.	Attribute type	Description
p	1..1	ActivePower	Active power injection. Load sign convention is used, i.e. positive sign means flow out from a node. Starting value for a steady state solution.
q	1..1	ReactivePower	Reactive power injection. Load sign convention is used, i.e. positive sign means flow out from a node. Starting value for a steady state solution.
controlEnabled	1..1	Boolean	see RegulatingCondEq
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.8.20 StaticVarCompensator

A facility for providing variable and controllable shunt reactive power. The SVC typically consists of a stepdown transformer, filter, thyristor-controlled reactor, and thyristor-switched capacitor arms.
The SVC may operate in fixed MVar output mode or in voltage control mode. When in voltage control mode, the output of the SVC will be proportional to the deviation of voltage at the controlled bus from the voltage setpoint. The SVC characteristic slope defines the proportion. If the voltage at the controlled bus is equal to the voltage setpoint, the SVC MVar output is zero.

Inheritance path: ->RegulatingCondEq->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 5-41 shows all attributes of StaticVarCompensator.

Table 5-41 - Attributes of Wires::StaticVarCompensator

Attribute name	mult.	Attribute type	Description
q (Description)	1..1	ReactivePower	Reactive power injection. Load sign convention is used, i.e.

			positive sign means flow out from a node. Starting value for a steady state solution.
controlEnabled	1..1	Boolean	see RegulatingCondEq
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.8.21 (Description) Switch

A generic device designed to close, or open, or both, one or more electric circuits. All switches are two terminal devices including grounding switches.

Inheritance path: ->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 5-42 shows all attributes of Switch.

Table 5-42 - Attributes of Wires::Switch

Attribute name	mult.	Attribute type	Description
open (Description)	1..1	Boolean	The attribute tells if the switch is considered open when used as input to topology processing.
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.8.22 (Description) SynchronousMachine

An electromechanical device that operates with shaft rotating synchronously with the network. It is a single machine operating either as a generator or synchronous condenser or pump.

Inheritance path: ->RotatingMachine->RegulatingCondEq->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 5-43 shows all attributes of SynchronousMachine.

Table 5-43 - Attributes of Wires::SynchronousMachine

Attribute name	mult.	Attribute type	Description
operatingMode (Description)	1..1	SynchronousMachineOperatingMode	Current mode of operation.
referencePriority (Description)	1..1	Integer	Priority of unit for use as powerflow voltage phase angle reference bus selection. 0 = don t care (default) 1 = highest priority. 2 is less than 1 and so on.
p	1..1	ActivePower	see RotatingMachine
q	1..1	ReactivePower	see RotatingMachine
controlEnabled	1..1	Boolean	see RegulatingCondEq
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

5.8.23 (Description) TapChangerControl

Describes behavior specific to tap changers, e.g. how the voltage at the end of a line varies with the load level and compensation of the voltage drop by tap adjustment.

Inheritance path: ->RegulatingControl->PowerSystemResource->IdentifiedObject

Table 5-44 shows all attributes of TapChangerControl.

Table 5-44 - Attributes of Wires::TapChangerControl

Attribute name	mult.	Attribute type	Description
discrete	1..1	Boolean	see RegulatingControl
enabled	1..1	Boolean	see RegulatingControl
targetDeadband	0..1	Simple_Float	see RegulatingControl
targetValue	1..1	Simple_Float	see RegulatingControl
targetValueUnitMultiplier	1..1	UnitMultiplier	see RegulatingControl
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.8.24 (Description) LinearShuntCompensator

A linear shunt compensator has banks or sections with equal admittance values.

Inheritance path: ->ShuntCompensator->RegulatingCondEq->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 5-45 shows all attributes of LinearShuntCompensator.

Table 5-45 - Attributes of Wires::LinearShuntCompensator

Attribute name	mult.	Attribute type	Description
sections	1..1	Simple_Float	see ShuntCompensator
controlEnabled	1..1	Boolean	see RegulatingCondEq
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.8.25 (Description) NonlinearShuntCompensator

A non linear shunt compensator has bank or section admittance values that differs.

Inheritance path: ->ShuntCompensator->RegulatingCondEq->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 5-46 shows all attributes of NonlinearShuntCompensator.

Table 5-46 - Attributes of Wires::NonlinearShuntCompensator

Attribute name	mult.	Attribute type	Description
sections	1..1	Simple_Float	see ShuntCompensator
controlEnabled	1..1	Boolean	see RegulatingCondEq
mRID	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

name	0..1	String	see IdentifiedObject

5.8.26 (Description) RegulatingControl

Specifies a set of equipment that works together to control a power system quantity such as voltage or flow.
Remote bus voltage control is possible by specifying the controlled terminal located at some place remote from the controlling equipment.
In case multiple equipment, possibly of different types, control same terminal there must be only one RegulatingControl at that terminal. The most specific subtype of RegulatingControl shall be used in case such equipment participate in the control, e.g. TapChangerControl for tap changers. For flow control load sign convention is used, i.e. positive sign means flow out from a TopologicalNode (bus) into the conducting equipment.

Note that RegulatingControl.targetDeadband is primarily used if the RegulatingControl.discrete is set to "true". Tools should handle cases in which RegulatingControl.targetDeadband has a value if RegulatingControl.discrete is set to "false" or cases in which RegulatingControl.targetDeadband equals zero. Note that for instance, if the targetValue is 100 kV and the targetDeadband is 2 kV the range is from 99 to 101 kV.

Inheritance path: ->PowerSystemResource->IdentifiedObject

Table 5-47 shows all attributes of RegulatingControl.

Table 5-47 - Attributes of Wires::RegulatingControl

Attribute name	mult.	Attribute type	Description
discrete (Description)	1..1	Boolean	The regulation is performed in a discrete mode. This applies to equipment with discrete controls, e.g. tap changers and shunt compensators.
enabled (Description)	1..1	Boolean	The flag tells if regulation is enabled.
targetDeadband (Description)	0..1	Simple_Float	This is a deadband used with discrete control to avoid excessive update of controls like tap changers and shunt compensator banks while regulating. The units of those appropriate for the mode.
targetValue (Description)	1..1	Simple_Float	The target value specified for case input. This value can be used for the target value without the use of schedules. The value has the units appropriate to the mode attribute.
targetValueUnitMultiplier (Description)	1..1	UnitMultiplier	Specify the multiplier for used for the targetValue.
mRID	0..1	String	see IdentifiedObject

name	0..1	String	see IdentifiedObject

5.8.27 ShuntCompensator(Abstract)

A shunt capacitor or reactor or switchable bank of shunt capacitors or reactors. A section of a shunt compensator is an individual capacitor or reactor. A negative value for reactivePerSection indicates that the compensator is a reactor. ShuntCompensator is a single terminal device. Ground is implied.

Inheritance path: ->RegulatingCondEq->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 5-48 shows all attributes of ShuntCompensator.

Table 5-48 - Attributes of Wires::ShuntCompensator

Attribute name	mult.	Attribute type	Description
sections	1..1	Simple_Float	Shunt compensator sections in use. Starting value for steady state solution. Non integer values are allowed to support continuous variables. The reasons for continuous value are to support study cases where no discrete shunt compensators has yet been designed, a solutions where a narrow voltage band force the sections to oscillate or accommodate for a continuous solution as input.
controlEnabled	1..1	Boolean	see RegulatingCondEq
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.8.28 TapChanger(Abstract)

Mechanism for changing transformer winding tap positions.

Inheritance path: ->PowerSystemResource->IdentifiedObject

Table 5-49 shows all attributes of TapChanger.

Table 5-49 - Attributes of Wires::TapChanger

Attribute name	mult.	Attribute type	Description
controlEnabled	1..1	Boolean	Specifies the regulation status of the equipment. True is regulating, false is not regulating.
step	1..1	Simple_Float	Tap changer position. Starting step for a steady state solution. Non integer values are allowed to support

			continuous tap variables. The reasons for continuous value are to support study cases where no discrete tap changers has yet been designed, a solutions where a narrow voltage band force the tap step to oscillate or accommodate for a continuous solution as input. The attribute shall be equal or greater than lowStep and equal or less than highStep.
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.8.29 Enumerations

5.8.29.1 AsynchronousMachineKind

Kind of Asynchronous Machine.

Literal	Description	Code
generator	The Asynchronous Machine is a generator.	
motor	The Asynchronous Machine is a motor.	

5.8.29.2 SynchronousMachineOperatingMode

Synchronous machine operating mode.

Literal	Description	Code
generator		
condenser		
motor		

5.9 LoadModel

5.9.1 General

5.9.2 ConformLoad

ConformLoad represent loads that follow a daily load change pattern where the pattern can be used to scale the load with a system load.

Inheritance path: ->EnergyConsumer->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 5-50 shows all attributes of ConformLoad.

Table 5-50 - Attributes of LoadModel::ConformLoad

Attribute name	mult.	Attribute type	Description
p	1..1	ActivePower	see EnergyConsumer
q	1..1	ReactivePower	see EnergyConsumer
mRID	0..1	String	see IdentifiedObject

name	0..1	String	see IdentifiedObject

5.9.3 NonConformLoad

NonConformLoad represent loads that do not follow a daily load change pattern and changes are not correlated with the daily load change pattern.

Inheritance path: ->EnergyConsumer->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 5-51 shows all attributes of NonConformLoad.

Table 5-51 - Attributes of LoadModel::NonConformLoad

Attribute name	mult.	Attribute type	Description
p	1..1	ActivePower	see EnergyConsumer
q	1..1	ReactivePower	see EnergyConsumer
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.9.4 StationSupply

Station supply with load derived from the station output.

Inheritance path: ->EnergyConsumer->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 5-52 shows all attributes of StationSupply.

Table 5-52 - Attributes of LoadModel::StationSupply

Attribute name	mult.	Attribute type	Description
p	1..1	ActivePower	see EnergyConsumer
q	1..1	ReactivePower	see EnergyConsumer
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

5.10 Global Used Datatypes

5.10.1 Datatypes

5.10.1.1 ActivePower

Product of RMS value of the voltage and the RMS value of the in-phase component of the current.

Value type is IEEE 754 simple precision floating point

Table 5-53 shows all attributes of ActivePower.

Table 5-53 - Attributes of DomainProfile::ActivePower.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= W
multiplier	0..1	UnitMultiplier	ConstantValue= M

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

5.10.1.2 ReactivePower

Product of RMS value of the voltage and the RMS value of the quadrature component of the current.

Value type is IEEE 754 simple precision floating point

Table 5-54 shows all attributes of ReactivePower.

Table 5-54 - Attributes of DomainProfile::ReactivePower.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= VAr
multiplier	0..1	UnitMultiplier	ConstantValue= M

5.10.1.3 Voltage

Electrical voltage, can be both AC and DC.

Value type is IEEE 754 simple precision floating point

Table 5-55 shows all attributes of Voltage.

Table 5-55 - Attributes of DomainProfile::Voltage.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= V
multiplier	0..1	UnitMultiplier	ConstantValue= k

5.10.1.4 AngleDegrees

Measurement of angle in degrees.

Value type is IEEE 754 simple precision floating point

Table 5-56 shows all attributes of AngleDegrees.

Table 5-56 - Attributes of DomainProfile::AngleDegrees.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= deg
multiplier	0..1	UnitMultiplier	ConstantValue= none

5.10.1.5 CurrentFlow

Electrical current with sign convention: positive flow is out of the conducting equipment into the connectivity node. Can be both AC and DC.

Value type is IEEE 754 simple precision floating point

Table 5-57 shows all attributes of CurrentFlow.

Table 5-57 - Attributes of DomainProfile::CurrentFlow.

Attribute name	mult.	Attribute type	Description

value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= A
multiplier	0..1	UnitMultiplier	ConstantValue= none

5.10.1.6 PU

Per Unit - a positive or negative value referred to a defined base. Values typically range from -10 to +10.

Value type is IEEE 754 simple precision floating point

Table 5-58 shows all attributes of PU.

Table 5-58 - Attributes of DomainProfile::PU.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= none
multiplier	0..1	UnitMultiplier	ConstantValue= none

5.10.1.7 Resistance

Resistance (real part of impedance).

Value type is IEEE 754 simple precision floating point

Table 5-59 shows all attributes of Resistance.

Table 5-59 - Attributes of DomainProfile::Resistance.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= ohm
multiplier	0..1	UnitMultiplier	ConstantValue= none

5.10.1.8 PerCent

Percentage on a defined base. For example, specify as 100 to indicate at the defined base.

Value type is IEEE 754 simple precision floating point

Table 5-60 shows all attributes of PerCent.

Table 5-60 - Attributes of DomainProfile::PerCent.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	Normally 0 - 100 on a defined base
unit	0..1	UnitSymbol	ConstantValue= none
multiplier	0..1	UnitMultiplier	ConstantValue= none

5.10.1.9 Simple_Float

A floating point number. The range is unspecified and not limited.

In ENTSO-E profile, Simple_Float range is the IEEE754 simple precision floating point one. It correspond to xs:float datatype

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 5-61 shows all attributes of Simple_Float.

Table 5-61 - Attributes of DomainProfile::Simple_Float.

Attribute name	mult.	Attribute type	Description
value	1..1	Float	

5.10.2 Enumerations

5.10.2.1 UnitSymbol

The units defined for usage in the CIM.

Literal	Description	Code
VA	Apparent power in volt ampere.	
W	Active power in watt.	
VAr	Reactive power in volt ampere reactive.	
VAh	Apparent energy in volt ampere hours.	
Wh	Real energy in what hours.	
VArh	Reactive energy in volt ampere reactive hours.	
V	Voltage in volt.	
ohm	Resistance in ohm.	
A	Current in ampere.	
F	Capacitance in farad.	
H	Inductance in henry.	
degC	Relative temperature in degrees Celsius. In the SI unit system the symbol is ºC. Electric charge is measured in coulomb that has the unit symbol C. To distinguish degree Celsius form coulomb the symbol used in the UML is degC. Reason for not using ºC is the special character º is difficult to manage in software.	
s	Time in seconds.	
min	Time in minutes.	
h	Time in hours.	
deg	Plane angle in degrees.	
rad	Plane angle in radians.	
J	Energy in joule.	
N	Force in newton.	
S	Conductance in siemens.	
none	Dimension less quantity, e.g. count, per unit, etc.	
Hz	Frequency in hertz.	
g	Mass in gram.	
Pa	Pressure in pascal (n/m2).	
m	Length in meter.	
m2	Area in square meters.	

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

m3	Volume in cubic meters.	

5.10.2.2 UnitMultiplier

The unit multipliers defined for the CIM.

Literal	Description	Code
p	Pico 10**-12.	
n	Nano 10**-9.	
micro	Micro 10**-6.	
m	Milli 10**-3.	
c	Centi 10**-2.	
d	Deci 10**-1.	
k	Kilo 10**3.	
M	Mega 10**6.	
G	Giga 10**9.	
T	Tera 10**12.	
none	No multiplier or equivalently multiply by 1.	

5.10.2.3 CsOperatingModeKind

Operating mode for HVDC line operating as Current Source Converter.

Literal	Description	Code
inverter	Operating as inverter	
rectifier	Operating as rectifier.	

5.10.2.4 CsPpccControlKind

Active power control modes for HVDC line operating as Current Source Converter.

Literal	Description	Code
activePower	Active power control at AC side.	
dcVoltage	DC voltage control.	
dcCurrent	DC current control	

5.10.2.5 VsPpccControlKind

Types applicable to the control of real power and/or DC voltage by voltage source converter.

Literal	Description	Code
pPcc	Control variable (target) is real power at PCC bus.	
udc	Control variable (target) is DC voltage and real power at PCC bus is derived.	
pPccAndUdcDroop	Control variables (targets) are both active power at point of common coupling and local DC voltage, with the droop.	
pPccAndUdcDroopWithCompensation	Control variables (targets) are both active power at point of common coupling and compensated DC voltage, with the droop; compensation factor is the resistance, as an approximation of the DC voltage of a common (real or virtual) node in the DC network.	

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

pPccAndUdcDroopPilot	Control variables (targets) are both active power at point of common coupling and the pilot DC voltage, with the droop.	

5.10.2.6 VsQpccControlKind

Literal	Description	Code
reactivePcc		
voltagePcc		
powerFactorPcc		

5.10.2.7 AsynchronousMachineKind

Kind of Asynchronous Machine.

Literal	Description	Code
generator	The Asynchronous Machine is a generator.	
motor	The Asynchronous Machine is a motor.	

5.10.2.8 SynchronousMachineOperatingMode

Synchronous machine operating mode.

Literal	Description	Code
generator		
condenser		
motor		

5.10.3 Primitives

5.10.3.1 String

A string consisting of a sequence of characters. The character encoding is UTF-8. The string length is unspecified and unlimited.

5.10.3.2 Date

Date as "yyyy-mm-dd", which conforms with ISO 8601. UTC time zone is specified as "yyyy-mm-ddZ". A local timezone relative UTC is specified as "yyyy-mm-dd(+/-)hh:mm".

5.10.3.3 Boolean

A type with the value space "true" and "false".

5.10.3.4 Float

A floating point number. The range is unspecified and not limited.

5.10.3.5 Integer

An integer number. The range is unspecified and not limited.

6 StateVariablesProfile

6.1 General

This profile has been built on the basis of the IEC 61970-456 document and adjusted to fit the purpose of the ENTSO-E CGMES.

Figure 6-1 shows the diagram.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

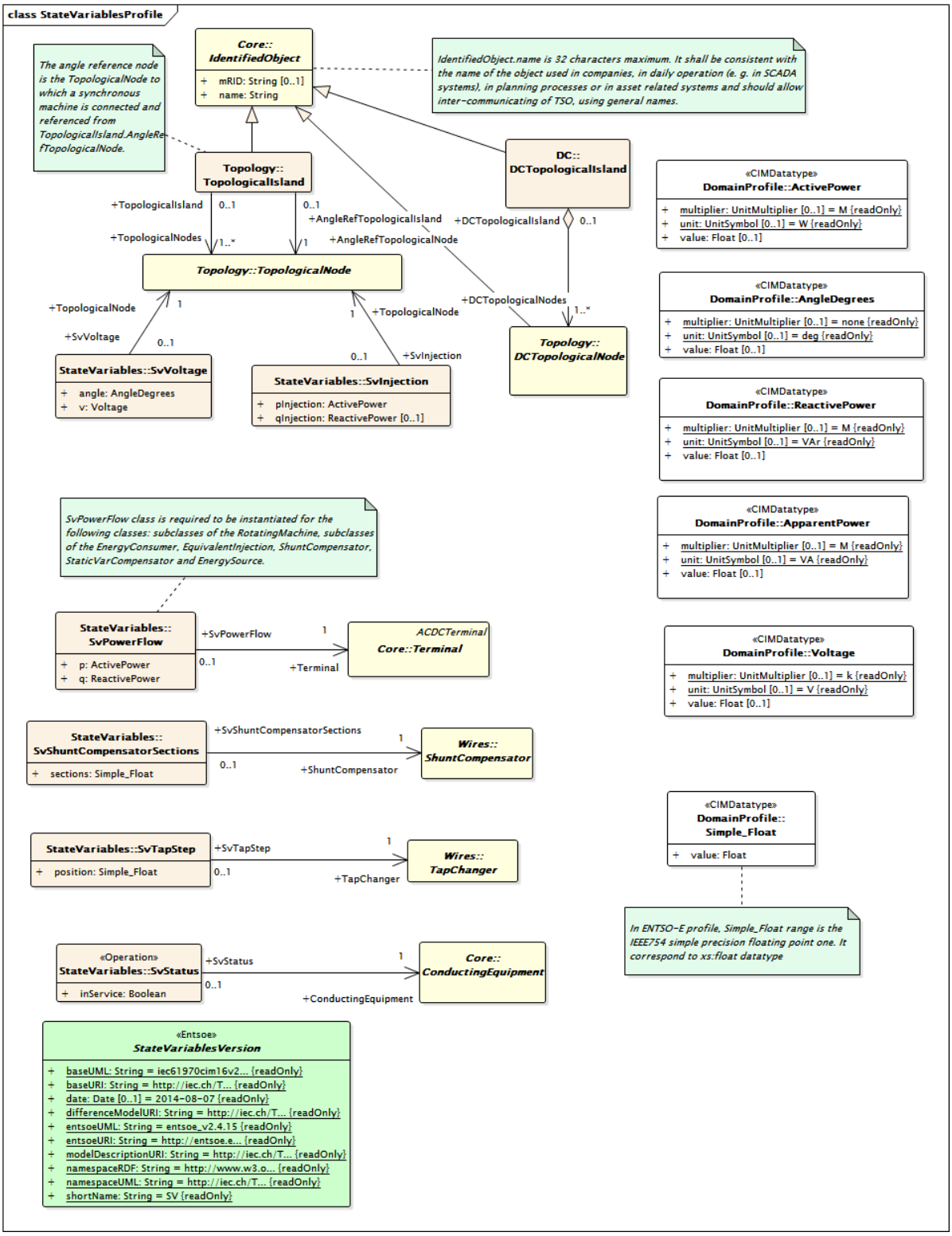


Figure 6-1 - diagram StateVariablesProfile

6.2 (Entsoe) StateVariablesVersion(Abstract)

Version details.

Table 6-1 shows all attributes of StateVariablesVersion.

Table 6-1 - Attributes of StateVariablesProfile::StateVariablesVersion

Attribute name	mult.	Attribute type	Description
baseUML (Entsoe)	1..1	String	ConstantValue= iec61970cim16v28_iec61968cim12v08_iec62325cim03v01a Base UML provided by CIM model manager.
baseURI (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/2013/61970-456/StateVariables/4 Profile URI used in the Model Exchange header and defined in IEC standards. It uniquely identifies the Profile and its version. It is given for information only and to identify the closest IEC profile to which this CGMES profile is based on.
date (Entsoe)	0..1	Date	ConstantValue= 2014-08-07 Profile creation date Form is YYYY-MM-DD for example for January 5, 2009 it is 2009-01-05.
differenceModelURI (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/61970-552/DifferenceModel/1# Difference model URI defined by IEC 61970-552.
entsoeUML (Entsoe)	1..1	String	ConstantValue= entsoe_v2.4.15 UML provided by ENTSO-E.
entsoeURI (Entsoe)	1..1	String	ConstantValue= http://entsoe.eu/CIM/StateVariables/4/1 Profile URI defined by ENTSO-E and used in the Model Exchange header. It uniquely identifies the Profile and its version. The last two elements in the URI (http://entsoe.eu/CIM/StateVariables/yy/zzz) indicate major and minor versions where: - yy - indicates a major version; - zzz - indicates a minor version.
modelDescriptionURI (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/61970-552/ModelDescription/1# Model Description URI defined by IEC 61970-552.
namespaceRDF (Entsoe)	1..1	String	ConstantValue= http://www.w3.org/1999/02/22-rdf-syntax-ns# RDF namespace.
namespaceUML (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/2013/CIM-schema-cim16# CIM UML namespace.
shortName (Entsoe)	1..1	String	ConstantValue= SV The short name of the profile used in profile documentation.

6.3 Core

6.3.1 General

Figure 6-2 shows the diagram.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

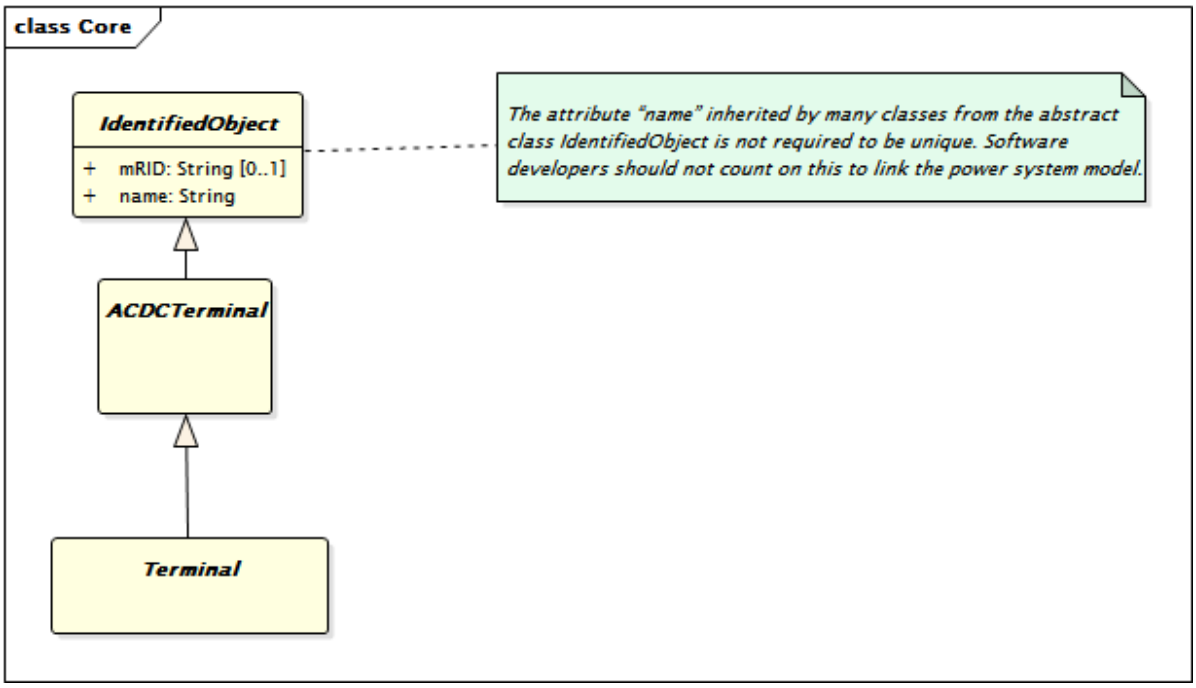


Figure 6-2 - diagram Core

6.3.2 ACDCTerminal(Abstract)

An electrical connection point (AC or DC) to a piece of conducting equipment. Terminals are connected at physical connection points called connectivity nodes.

Inheritance path: ->IdentifiedObject

Table 6-2 shows all attributes of ACDCTerminal.

Table 6-2 - Attributes of Core::ACDCTerminal

Attribute name	mult.	Attribute type	Description
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject

6.3.3 ConductingEquipment(Abstract)

The parts of the AC power system that are designed to carry current or that are conductively connected through terminals.

6.3.4 IdentifiedObject(Abstract)

This is a root class to provide common identification for all classes needing identification and naming attributes.

OCL constraint:R.4.10.11. Name length restriction

IdentifiedObject.name is 32 characters maximum. It shall be consistent with the name of the object used in companies, in daily operation (e. g. in SCADA systems), in planning processes or in asset related systems and should allow inter-communicating of TSO, using general names.

The attribute “name” inherited by many classes from the abstract class IdentifiedObject is not required to be unique. Software developers should not count on this to link the power system model.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 6-3 shows all attributes of IdentifiedObject.

Table 6-3 - Attributes of Core::IdentifiedObject

Attribute name	mult.	Attribute type	Description
mRID	0..1	String	Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552 Edition 1, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name	1..1	String	The name is any free human readable and possibly non unique text naming the object.

6.3.5 Terminal(Abstract)

An AC electrical connection point to a piece of conducting equipment. Terminals are connected at physical connection points called connectivity nodes.

Inheritance path: ->ACDCTerminal->IdentifiedObject

Table 6-4 shows all attributes of Terminal.

Table 6-4 - Attributes of Core::Terminal

Attribute name	mult.	Attribute type	Description
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject

6.4 DC

6.4.1 General

Figure 6-3 shows the diagram.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

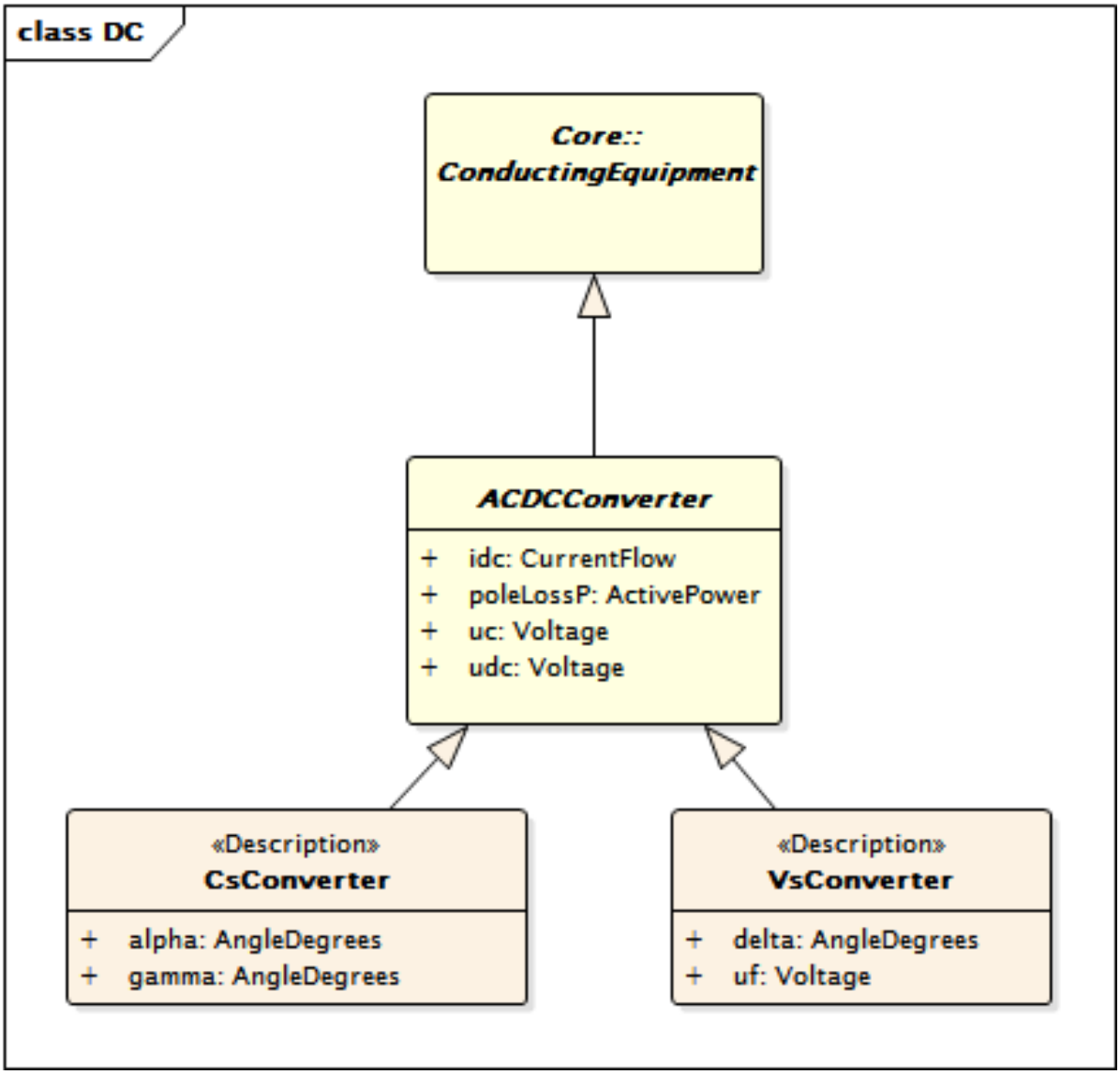


Figure 6-3 - diagram DC

6.4.2 ACDCConverter(Abstract)

A unit with valves for three phases, together with unit control equipment, essential protective and switching devices, DC storage capacitors, phase reactors and auxiliaries, if any, used for conversion.

Inheritance path: ->ConductingEquipment

Table 6-5 shows all attributes of ACDCConverter.

Table 6-5 - Attributes of DC::ACDCConverter

Attribute name	mult.	Attribute type	Description
idc	1..1	CurrentFlow	Converter DC current, also called Id. Converter state variable, result from

			power flow.
poleLossP	1..1	ActivePower	The active power loss at a DC Pole = idleLoss + switchingLoss*|Idc| + resitiveLoss*Idc^2 For lossless operation Pdc=Pac For rectifier operation with losses Pdc=Pac-lossP For inverter operation with losses Pdc=Pac+lossP Converter state variable used in power flow.
uc	1..1	Voltage	Converter voltage, the voltage at the AC side of the bridge. Converter state variable, result from power flow.
udc	1..1	Voltage	Converter voltage at the DC side, also called Ud. Converter state variable, result from power flow.

6.4.3 CsConverter

DC side of the current source converter (CSC).

Inheritance path: ->ACDCConverter->ConductingEquipment

Table 6-6 shows all attributes of CsConverter.

Table 6-6 - Attributes of DC::CsConverter

Attribute name	mult.	Attribute type	Description
alpha (Description)	1..1	AngleDegrees	Firing angle, typical value between 10 and 18 degrees for a rectifier. CSC state variable, result from power flow.
gamma (Description)	1..1	AngleDegrees	Extinction angle. CSC state variable, result from power flow.
idc	1..1	CurrentFlow	see ACDCConverter
poleLossP	1..1	ActivePower	see ACDCConverter
uc	1..1	Voltage	see ACDCConverter
udc	1..1	Voltage	see ACDCConverter

6.4.4 (Description) DCTopologicalIsland

An electrically connected subset of the network. DC topological islands can change as the current network state changes: e.g. due to
- disconnect switches or breakers change state in a SCADA/EMS
- manual creation, change or deletion of topological nodes in a planning tool.

Inheritance path: ->IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 6-7 shows all attributes of DCTopologicalIsland.

Table 6-7 - Attributes of DC::DCTopologicalIsland

Attribute name	mult.	Attribute type	Description
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject

Table 6-8 shows all association ends of DCTopologicalIsland with other classes.

Table 6-8 - Association ends of DC::DCTopologicalIsland with other classes

mult.	mult.	Class type name	Description
DCTopologicalNodes	[1..*]	DCTopologicalNode	Association Based On : Topology:: DCTopologicalNode.DCTopologicalNodes[1..*] ---- DC:: DCTopologicalIsland.DCTopologicalIsland[0..1]

6.4.5 VsConverter

DC side of the voltage source converter (VSC).

Inheritance path: ->ACDCConverter->ConductingEquipment

Table 6-9 shows all attributes of VsConverter.

Table 6-9 - Attributes of DC::VsConverter

Attribute name	mult.	Attribute type	Description
delta (Description)	1..1	AngleDegrees	Angle between uf and uc. Converter state variable used in power flow.
uf (Description)	1..1	Voltage	Filter bus voltage. Converter state variable, result from power flow.
idc	1..1	CurrentFlow	see ACDCConverter
poleLossP	1..1	ActivePower	see ACDCConverter
uc	1..1	Voltage	see ACDCConverter
udc	1..1	Voltage	see ACDCConverter

6.5 StateVariables

6.5.1 General

6.5.2 (Description) SvStatus

State variable for status.

Table 6-10 shows all attributes of SvStatus.

Table 6-10 - Attributes of StateVariables::SvStatus

Attribute name	mult.	Attribute type	Description

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

inService (Operation)	1..1	Boolean	The in service status as a result of topology processing.

Table 6-11 shows all association ends of SvStatus with other classes.

Table 6-11 - Association ends of StateVariables::SvStatus with other classes

mult.	mult.	Class type name	Description
ConductingEquipment (Operation)	[1..1]	ConductingEquipment	The conducting equipment associated with the status state variable. Association Based On : StateVariables:: SvStatus.SvStatus[0..1] ----- Core:: ConductingEquipment.ConductingEquipment[1]

6.5.3 (Operation) SvInjection

The SvInjection is reporting the calculated bus injection minus the sum of the terminal flows. The terminal flow is positive out from the bus (load sign convention) and bus injection has positive flow into the bus. SvInjection may have the remainder after state estimation or slack after power flow calculation.

Table 6-12 shows all attributes of SvInjection.

Table 6-12 - Attributes of StateVariables::SvInjection

Attribute name	mult.	Attribute type	Description
pInjection	1..1	ActivePower	The active power injected into the bus in addition to injections from equipment terminals. Positive sign means injection into the TopologicalNode (bus).
qInjection	0..1	ReactivePower	The reactive power injected into the bus in addition to injections from equipment terminals. Positive sign means injection into the TopologicalNode (bus).

Table 6-13 shows all association ends of SvInjection with other classes.

Table 6-13 - Association ends of StateVariables::SvInjection with other classes

mult.	mult.	Class type name	Description
TopologicalNode	[1..1]	TopologicalNode	The injection flows state variables associated with the topological node. Association Based On : StateVariables:: SvInjection.SvInjection[0..1] ----- Topology:: TopologicalNode.TopologicalNode[1]

6.5.4 SvPowerFlow

State variable for power flow. Load convention is used for flow direction. This means flow out from the TopologicalNode into the equipment is positive.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

OCL constraint: The active power flow into the ConductingEquipment. If the associated Terminal.connected status is "false", the flow specified in the SvPowerFlow.p should be zero.
OCL constraint: The reactive power flow into the ConductingEquipment. If the associated Terminal.connected status is "false", the flow specified in the SvPowerFlow.q should be zero.
SvPowerFlow class is required to be instantiated for the following classes: subclasses of the RotatingMachine, subclasses of the EnergyConsumer, EquivalentInjection, ShuntCompensator, StaticVarCompensator and EnergySource.

Table 6-14 shows all attributes of SvPowerFlow.

Table 6-14 - Attributes of StateVariables::SvPowerFlow

Attribute name	mult.	Attribute type	Description
p	1..1	ActivePower	The active power flow. Load sign convention is used, i.e. positive sign means flow out from a TopologicalNode (bus) into the conducting equipment.
q	1..1	ReactivePower	The reactive power flow. Load sign convention is used, i.e. positive sign means flow out from a TopologicalNode (bus) into the conducting equipment.

Table 6-15 shows all association ends of SvPowerFlow with other classes.

Table 6-15 - Association ends of StateVariables::SvPowerFlow with other classes

mult.	mult.	Class type name	Description
Terminal	[1..1]	Terminal	The terminal associated with the power flow state variable. Association Based On : StateVariables:: SvPowerFlow.SvPowerFlow[0..1] ----- Core:: Terminal.Terminal[1]

6.5.5 SvShuntCompensatorSections

State variable for the number of sections in service for a shunt compensator.

Table 6-16 shows all attributes of SvShuntCompensatorSections.

Table 6-16 - Attributes of StateVariables::SvShuntCompensatorSections

Attribute name	mult.	Attribute type	Description
sections	1..1	Simple_Float	The number of sections in service as a continous variable. To get integer value scale with ShuntCompensator.bPerSection.

Table 6-17 shows all association ends of SvShuntCompensatorSections with other classes.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 6-17 - Association ends of StateVariables::SvShuntCompensatorSections with other classes

mult.	mult.	Class type name	Description
ShuntCompensator	[1..1]	ShuntCompensator	The shunt compensator for which the state applies. Association Based On : StateVariables:: SvShuntCompensatorSections.SvShuntCompensatorSections[0..1] ----- Wires:: ShuntCompensator.ShuntCompensator[1]

6.5.6 SvTapStep

State variable for transformer tap step. This class is to be used for taps of LTC (load tap changing) transformers, not fixed tap transformers.

Table 6-18 shows all attributes of SvTapStep.

Table 6-18 - Attributes of StateVariables::SvTapStep

Attribute name	mult.	Attribute type	Description
position	1..1	Simple_Float	The floating point tap position. This is not the tap ratio, but rather the tap step position as defined by the related tap changer model and normally is constrained to be within the range of minimum and maximum tap positions.

Table 6-19 shows all association ends of SvTapStep with other classes.

Table 6-19 - Association ends of StateVariables::SvTapStep with other classes

mult.	mult.	Class type name	Description
TapChanger	[1..1]	TapChanger	The tap changer associated with the tap step state. Association Based On : StateVariables:: SvTapStep.SvTapStep[0..1] ----- Wires:: TapChanger.TapChanger[1]

6.5.7 SvVoltage

State variable for voltage.

Table 6-20 shows all attributes of SvVoltage.

Table 6-20 - Attributes of StateVariables::SvVoltage

Attribute name	mult.	Attribute type	Description
angle	1..1	AngleDegrees	The voltage angle of the topological node complex voltage with respect to system reference.
v	1..1	Voltage	The voltage magnitude of the topological node.

Table 6-21 shows all association ends of SvVoltage with other classes.

Table 6-21 - Association ends of StateVariables::SvVoltage with other classes

mult.	mult.	Class type name	Description
TopologicalNode	[1..1]	TopologicalNode	The state voltage associated with the topological node. Association Based On : StateVariables:: SvVoltage.SvVoltage[0..1] ----- Topology:: TopologicalNode.TopologicalNode[1]

6.6 Topology

6.6.1 General

Figure 6-4 shows the diagram.

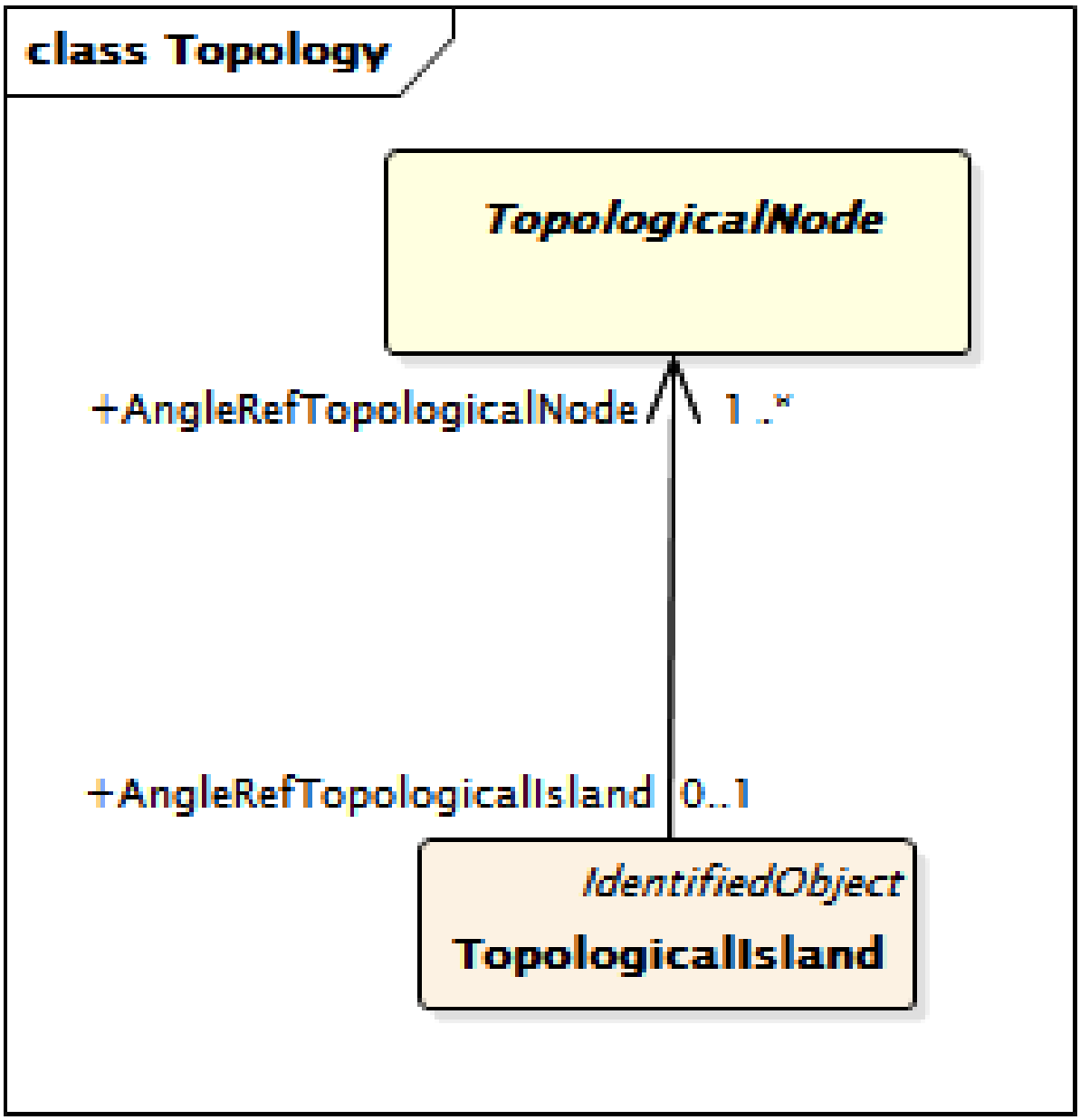


Figure 6-4 - diagram Topology

6.6.2 DCTopologicalNode(Abstract)

DC bus.

Inheritance path: ->IdentifiedObject

Table 6-22 shows all attributes of DCTopologicalNode.

Table 6-22 - Attributes of Topology::DCTopologicalNode

Attribute name	mult.	Attribute type	Description
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject

6.6.3 TopologicalNode(Abstract)

For a detailed substation model a topological node is a set of connectivity nodes that, in the current network state, are connected together through any type of closed switches, including jumpers. Topological nodes change as the current network state changes (i.e., switches, breakers, etc. change state).
For a planning model, switch statuses are not used to form topological nodes. Instead they are manually created or deleted in a model builder tool. Topological nodes maintained this way are also called "busses".

6.6.4 TopologicalIsland

An electrically connected subset of the network. Topological islands can change as the current network state changes: e.g. due to
- disconnect switches or breakers change state in a SCADA/EMS
- manual creation, change or deletion of topological nodes in a planning tool.

The angle reference node is the TopologicalNode to which a synchronous machine is connected and referenced from TopologicalIsland.AngleRefTopologicalNode.

Inheritance path: ->IdentifiedObject

Table 6-23 shows all attributes of TopologicalIsland.

Table 6-23 - Attributes of Topology::TopologicalIsland

Attribute name	mult.	Attribute type	Description
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject

Table 6-24 shows all association ends of TopologicalIsland with other classes.

Table 6-24 - Association ends of Topology::TopologicalIsland with other classes

mult.	mult.	Class type name	Description
AngleRefTopologicalNode	[1..1]	TopologicalNode	The angle reference for the island. Normally there is one TopologicalNode that is selected as the angle reference for each island. Other reference schemes exist, so the association is typically optional. Association Based On : Topology:: TopologicalIsland.AngleRefTopologicalIsland[0..1] ---- Topology:: TopologicalNode.AngleRefTopologicalNode[0..1]

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

TopologicalNodes	[1..*]	TopologicalNode	A topological node belongs to a topological island. Association Based On : Topology:: TopologicalNode.TopologicalNodes[1..*] ----- Topology:: TopologicalIsland.TopologicalIsland[0..1]

6.7 Wires

6.7.1 General

Figure 6-5 shows the diagram.

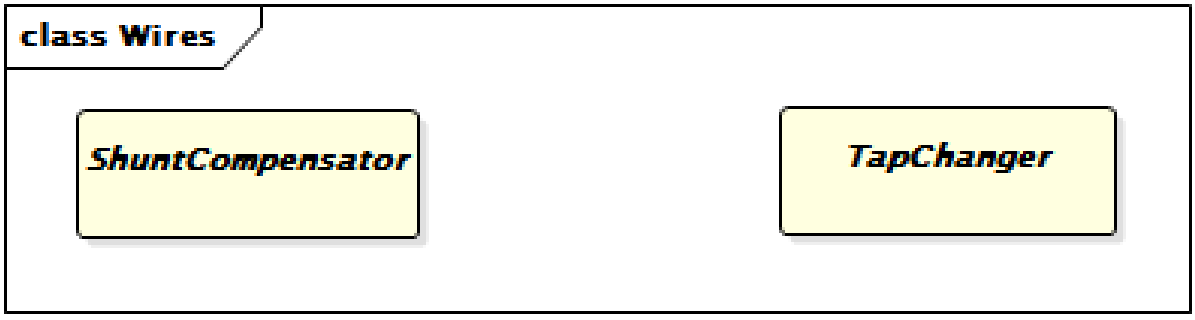


Figure 6-5 - diagram Wires

6.7.2 ShuntCompensator(Abstract)

A shunt capacitor or reactor or switchable bank of shunt capacitors or reactors. A section of a shunt compensator is an individual capacitor or reactor. A negative value for reactivePerSection indicates that the compensator is a reactor. ShuntCompensator is a single terminal device. Ground is implied.

6.7.3 TapChanger(Abstract)

Mechanism for changing transformer winding tap positions.

6.8 Global Used Datatypes

6.8.1 Datatypes

6.8.1.1 CurrentFlow

Electrical current with sign convention: positive flow is out of the conducting equipment into the connectivity node. Can be both AC and DC.

Value type is IEEE 754 simple precision floating point

Table 6-25 shows all attributes of CurrentFlow.

Table 6-25 - Attributes of DomainProfile::CurrentFlow.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= A
multiplier	0..1	UnitMultiplier	ConstantValue= none

6.8.1.2 ActivePower

Product of RMS value of the voltage and the RMS value of the in-phase component of the current.

Value type is IEEE 754 simple precision floating point

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 6-26 shows all attributes of ActivePower.

Table 6-26 - Attributes of DomainProfile::ActivePower.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= W
multiplier	0..1	UnitMultiplier	ConstantValue= M

6.8.1.3 Voltage

Electrical voltage, can be both AC and DC.

Value type is IEEE 754 simple precision floating point

Table 6-27 shows all attributes of Voltage.

Table 6-27 - Attributes of DomainProfile::Voltage.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= V
multiplier	0..1	UnitMultiplier	ConstantValue= k

6.8.1.4 AngleDegrees

Measurement of angle in degrees.

Value type is IEEE 754 simple precision floating point

Table 6-28 shows all attributes of AngleDegrees.

Table 6-28 - Attributes of DomainProfile::AngleDegrees.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= deg
multiplier	0..1	UnitMultiplier	ConstantValue= none

6.8.1.5 ReactivePower

Product of RMS value of the voltage and the RMS value of the quadrature component of the current.

Value type is IEEE 754 simple precision floating point

Table 6-29 shows all attributes of ReactivePower.

Table 6-29 - Attributes of DomainProfile::ReactivePower.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= VAr
multiplier	0..1	UnitMultiplier	ConstantValue= M

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

6.8.1.6 Simple_Float

A floating point number. The range is unspecified and not limited.

In ENTSO-E profile, Simple_Float range is the IEEE754 simple precision floating point one. It correspond to xs:float datatype

Table 6-30 shows all attributes of Simple_Float.

Table 6-30 - Attributes of DomainProfile::Simple_Float.

Attribute name	mult.	Attribute type	Description
value	1..1	Float	

6.8.2 Enumerations

6.8.2.1 UnitSymbol

The units defined for usage in the CIM.

Literal	Description	Code
VA	Apparent power in volt ampere.	
W	Active power in watt.	
VAr	Reactive power in volt ampere reactive.	
VAh	Apparent energy in volt ampere hours.	
Wh	Real energy in what hours.	
VArh	Reactive energy in volt ampere reactive hours.	
V	Voltage in volt.	
ohm	Resistance in ohm.	
A	Current in ampere.	
F	Capacitance in farad.	
H	Inductance in henry.	
degC	Relative temperature in degrees Celsius. In the SI unit system the symbol is ºC. Electric charge is measured in coulomb that has the unit symbol C. To distinguish degree Celsius form coulomb the symbol used in the UML is degC. Reason for not using ºC is the special character º is difficult to manage in software.	
s	Time in seconds.	
min	Time in minutes.	
h	Time in hours.	
deg	Plane angle in degrees.	
rad	Plane angle in radians.	
J	Energy in joule.	
N	Force in newton.	
S	Conductance in siemens.	
none	Dimension less quantity, e.g. count, per unit, etc.	
Hz	Frequency in hertz.	

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

g	Mass in gram.	
Pa	Pressure in pascal (n/m2).	
m	Length in meter.	
m2	Area in square meters.	
m3	Volume in cubic meters.	

6.8.2.2 UnitMultiplier

The unit multipliers defined for the CIM.

Literal	Description	Code
p	Pico 10**-12.	
n	Nano 10**-9.	
micro	Micro 10**-6.	
m	Milli 10**-3.	
c	Centi 10**-2.	
d	Deci 10**-1.	
k	Kilo 10**3.	
M	Mega 10**6.	
G	Giga 10**9.	
T	Tera 10**12.	
none	No multiplier or equivalently multiply by 1.	

6.8.3 Primitives

6.8.3.1 String

A string consisting of a sequence of characters. The character encoding is UTF-8. The string length is unspecified and unlimited.

6.8.3.2 Date

Date as "yyyy-mm-dd", which conforms with ISO 8601. UTC time zone is specified as "yyyy-mm-ddZ". A local timezone relative UTC is specified as "yyyy-mm-dd(+/-)hh:mm".

6.8.3.3 Float

A floating point number. The range is unspecified and not limited.

6.8.3.4 Boolean

A type with the value space "true" and "false".

7 DiagramLayoutProfile

7.1 General

This profile has been built on the basis of the IEC 61970-453 document and adjusted to fit the purpose of the ENTSO-E CGMES.

Figure 7-1 shows the diagram.

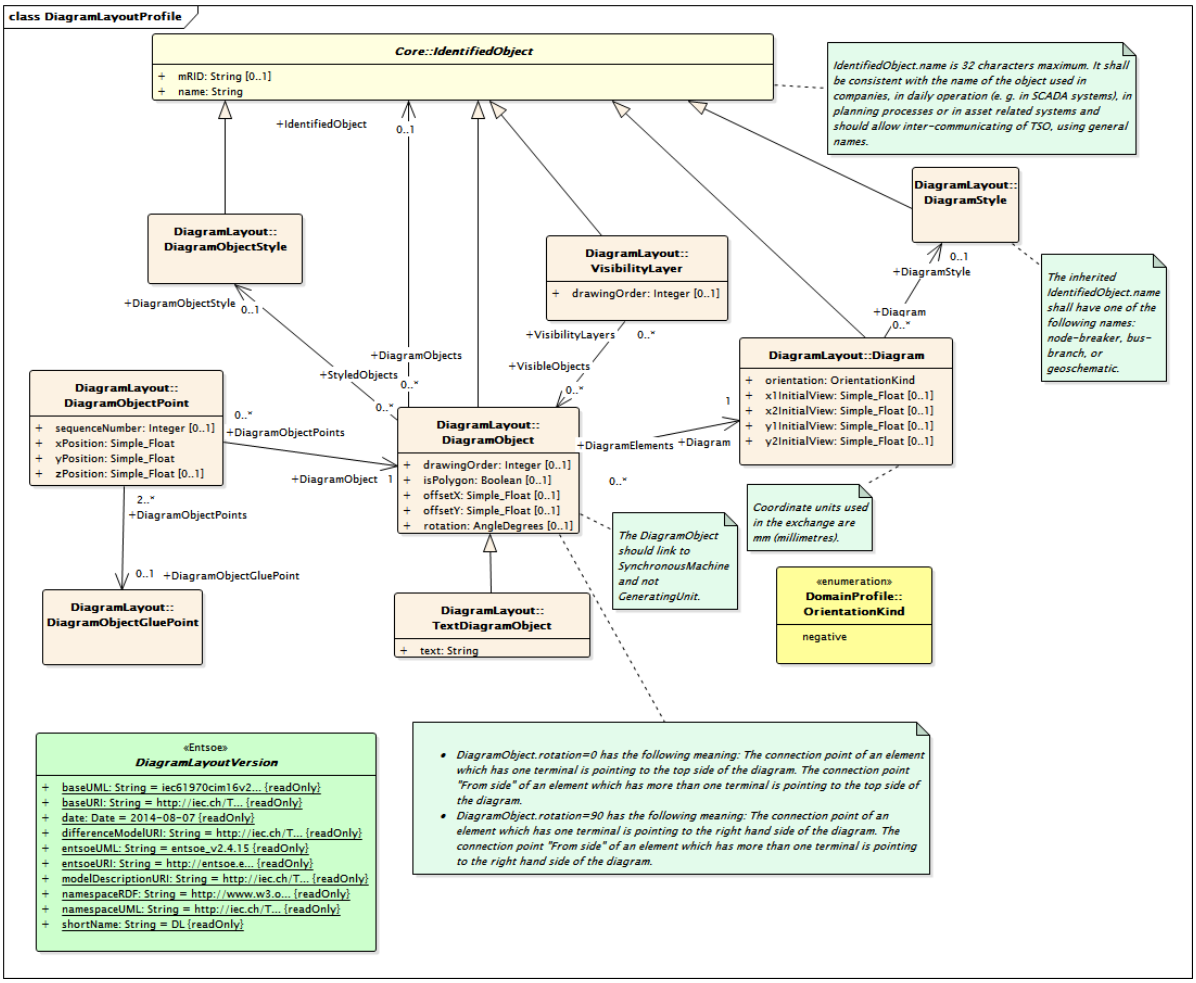


Figure 7-1 - diagram DiagramLayoutProfile

7.2 (Entsoe) DiagramLayoutVersion(Abstract)

Version details.

Table 7-1 shows all attributes of DiagramLayoutVersion.

Table 7-1 - Attributes of DiagramLayoutProfile::DiagramLayoutVersion

Attribute name	mult.	Attribute type	Description
baseUML (Entsoe)	1..1	String	ConstantValue= iec61970cim16v28_iec61968cim12v08_iec62325cim03v01a Base UML provided by CIM model manager.
baseURI (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/2013/61970-453/DiagramLayout/3 Profile URI used in the Model Exchange header and defined in IEC standards. It uniquely identifies the Profile and its version. It is given for information only and to identify the closest IEC profile to which this CGMES profile is based on.
date (Entsoe)	1..1	Date	ConstantValue= 2014-08-07 Profile creation date Form is YYYY-MM-DD for example for January 5, 2009 it is 2009-01-05.

differenceModelURI (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/61970-552/DifferenceModel/1# Difference model URI defined by IEC 61970-552.
entsoeUML (Entsoe)	1..1	String	ConstantValue= entsoe_v2.4.15 UML provided by ENTSO-E.
entsoeURI (Entsoe)	1..1	String	ConstantValue= http://entsoe.eu/CIM/DiagramLayout/3/1 Profile URI defined by ENTSO-E and used in the Model Exchange header. It uniquely identifies the Profile and its version. The last two elements in the URI (http://entsoe.eu/CIM/DiagramLayout/yy/zzz) indicate major and minor versions where: - yy - indicates a major version; - zzz - indicates a minor version.
modelDescriptionURI (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/61970-552/ModelDescription/1# Model Description URI defined by IEC 61970-552.
namespaceRDF (Entsoe)	1..1	String	ConstantValue= http://www.w3.org/1999/02/22-rdf-syntax-ns# RDF namespace.
namespaceUML (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/2013/CIM-schema-cim16# CIM UML namespace.
shortName (Entsoe)	1..1	String	ConstantValue= DL The short name of the profile used in profile documentation.

7.3 DiagramLayout

7.3.1 General

7.3.2 (Entsoe) DiagramStyle

The diagram style refer to a style used by the originating system for a diagram. A diagram style describes information such as schematic, geographic, bus-branch etc.

The inherited IdentifiedObject.name shall have one of the following names: node-breaker, bus-branch, or geoschematic.

Inheritance path: ->IdentifiedObject

Table 7-2 shows all attributes of DiagramStyle.

Table 7-2 - Attributes of DiagramLayout::DiagramStyle

Attribute name	mult.	Attribute type	Description
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject

7.3.3 (Entsoe) Diagram

The diagram being exchanged. The coordinate system is a standard Cartesian coordinate system and the orientation attribute defines the orientation.

Coordinate units used in the exchange are mm (millimetres).

Inheritance path: ->IdentifiedObject

Table 7-3 shows all attributes of Diagram.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 7-3 - Attributes of DiagramLayout::Diagram

Attribute name	mult.	Attribute type	Description
orientation	1..1	OrientationKind	Coordinate system orientation of the diagram.
x1InitialView	0..1	Simple_Float	X coordinate of the first corner of the initial view.
x2InitialView	0..1	Simple_Float	X coordinate of the second corner of the initial view.
y1InitialView	0..1	Simple_Float	Y coordinate of the first corner of the initial view.
y2InitialView	0..1	Simple_Float	Y coordinate of the second corner of the initial view.
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject

Table 7-4 shows all association ends of Diagram with other classes.

Table 7-4 - Association ends of DiagramLayout::Diagram with other classes

mult.	mult.	Class type name	Description
DiagramStyle	[0..1]	DiagramStyle	A Diagram may have a DiagramStyle. Association Based On : DiagramLayout:: DiagramStyle.DiagramStyle[0..1] ----- DiagramLayout:: Diagram.Diagram[0..*]

7.3.4 DiagramObject

An object that defines one or more points in a given space. This object can be associated with anything that specializes IdentifiedObject. For single line diagrams such objects typically include such items as analog values, breakers, disconnectors, power transformers, and transmission lines.

- DiagramObject.rotation=0 has the following meaning: The connection point of an element which has one terminal is pointing to the top side of the diagram. The connection point "From side" of an element which has more than one terminal is pointing to the top side of the diagram.
- DiagramObject.rotation=90 has the following meaning: The connection point of an element which has one terminal is pointing to the right hand side of the diagram. The connection point "From side" of an element which has more than one terminal is pointing to the right hand side of the diagram.

The DiagramObject should link to SynchronousMachine and not GeneratingUnit.

Inheritance path: ->IdentifiedObject

Table 7-5 shows all attributes of DiagramObject.

Table 7-5 - Attributes of DiagramLayout::DiagramObject

Attribute name	mult.	Attribute type	Description

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

drawingOrder	0..1	Integer	The drawing order of this element. The higher the number, the later the element is drawn in sequence. This is used to ensure that elements that overlap are rendered in the correct order.
isPolygon	0..1	Boolean	Defines whether or not the diagram objects points define the boundaries of a polygon or the routing of a polyline. If this value is true then a receiving application should consider the first and last points to be connected.
offsetX	0..1	Simple_Float	The offset in the X direction. This is used for defining the offset from centre for rendering an icon (the default is that a single point specifies the centre of the icon). The offset is in per-unit with 0 indicating there is no offset from the horizontal centre of the icon. -0.5 indicates it is offset by 50% to the left and 0.5 indicates an offset of 50% to the right.
offsetY	0..1	Simple_Float	The offset in the Y direction. This is used for defining the offset from centre for rendering an icon (the default is that a single point specifies the centre of the icon). The offset is in per-unit with 0 indicating there is no offset from the vertical centre of the icon. The offset direction is dependent on the orientation of the diagram, with -0.5 and 0.5 indicating an offset of +/- 50% on the vertical axis.
rotation	0..1	AngleDegrees	Sets the angle of rotation of the diagram object. Zero degrees is pointing to the top of the diagram. Rotation is clockwise.
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 7-6 shows all association ends of DiagramObject with other classes.

Table 7-6 - Association ends of DiagramLayout::DiagramObject with other classes

mult.	mult.	Class type name	Description
IdentifiedObject	[0..1]	IdentifiedObject	The diagram objects that are associated with the domain object. Association Based On : Core:: IdentifiedObject.IdentifiedObject[0..1] ----- DiagramLayout:: DiagramObject.DiagramObjects[0..*]
DiagramObjectStyle	[0..1]	DiagramObjectStyle	A diagram object has a style associated that provides a reference for the style used in the originating system. Association Based On : DiagramLayout:: DiagramObject.StyledObjects[0..*] ----- DiagramLayout:: DiagramObjectStyle.DiagramObjectStyle[0..1]
Diagram	[1..1]	Diagram	A diagram object is part of a diagram. Association Based On : DiagramLayout:: DiagramObject.DiagramElements[0..*] ----- DiagramLayout:: Diagram.Diagram[0..1]

7.3.5 DiagramObjectGluePoint

This is used for grouping diagram object points from different diagram objects that are considered to be glued together in a diagram even if they are not at the exact same coordinates.

7.3.6 DiagramObjectPoint

A point in a given space defined by 3 coordinates and associated to a diagram object. The coordinates may be positive or negative as the origin does not have to be in the corner of a diagram.

Table 7-7 shows all attributes of DiagramObjectPoint.

Table 7-7 - Attributes of DiagramLayout::DiagramObjectPoint

Attribute name	mult.	Attribute type	Description
sequenceNumber	0..1	Integer	The sequence position of the point, used for defining the order of points for diagram objects acting as a polyline or polygon with more than one point.
xPosition	1..1	Simple_Float	The X coordinate of this point.
yPosition	1..1	Simple_Float	The Y coordinate of this point.
zPosition	0..1	Simple_Float	The Z coordinate of this point.

Table 7-8 shows all association ends of DiagramObjectPoint with other classes.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 7-8 - Association ends of DiagramLayout::DiagramObjectPoint with other classes

mult.	mult.	Class type name	Description
DiagramObjectGluePoint	[0..1]	DiagramObjectGluePoint	A diagram object glue point is associated with 2 or more object points that are considered to be 'glued' together. Association Based On : DiagramLayout:: DiagramObjectPoint.DiagramObjectPoints[2..*] ----- DiagramLayout:: DiagramObjectGluePoint.DiagramObjectGluePoint[0..1]
DiagramObject	[1..1]	DiagramObject	The diagram object with which the points are associated. Association Based On : DiagramLayout:: DiagramObject.DiagramObject[1] ----- DiagramLayout:: DiagramObjectPoint.DiagramObjectPoints[0..*]

7.3.7 DiagramObjectStyle

A reference to a style used by the originating system for a diagram object. A diagram object style describes information such as line thickness, shape such as circle or rectangle etc, and color.

Inheritance path: ->IdentifiedObject

Table 7-9 shows all attributes of DiagramObjectStyle.

Table 7-9 - Attributes of DiagramLayout::DiagramObjectStyle

Attribute name	mult.	Attribute type	Description
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject

7.3.8 TextDiagramObject

A diagram object for placing free-text or text derived from an associated domain object.

Inheritance path: ->DiagramObject->IdentifiedObject

Table 7-10 shows all attributes of TextDiagramObject.

Table 7-10 - Attributes of DiagramLayout::TextDiagramObject

Attribute name	mult.	Attribute type	Description
text	1..1	String	The text that is displayed by this text diagram object.
drawingOrder	0..1	Integer	see DiagramObject
isPolygon	0..1	Boolean	see DiagramObject
offsetX	0..1	Simple_Float	see DiagramObject
offsetY	0..1	Simple_Float	see DiagramObject
rotation	0..1	AngleDegrees	see DiagramObject
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 7-11 shows all association ends of DiagramObject with other classes.

Table 7-11 - Association ends of DiagramLayout::DiagramObject with other classes

mult.	mult.	Class type name	Description
IdentifiedObject	0..1	IdentifiedObject	Inherited Association Based On : Core:: IdentifiedObject.IdentifiedObject[0..1] ----- DiagramLayout:: DiagramObject.DiagramObjects[0..*]
DiagramObjectStyle	0..1	DiagramObjectStyle	Inherited Association Based On : DiagramLayout:: DiagramObject.StyledObjects[0..*] ----- DiagramLayout:: DiagramObjectStyle.DiagramObjectStyle[0..1]
Diagram	1..1	Diagram	Inherited Association Based On : DiagramLayout:: DiagramObject.DiagramElements[0..*] ----- DiagramLayout:: Diagram.Diagram[0..1]

7.3.9 VisibilityLayer

Layers are typically used for grouping diagram objects according to themes and scales. Themes are used to display or hide certain information (e.g., lakes, borders), while scales are used for hiding or displaying information depending on the current zoom level (hide text when it is too small to be read, or when it exceeds the screen size). This is also called de-cluttering. CIM based graphics exchange will support an m:n relationship between diagram objects and layers. It will be the task of the importing system to convert an m:n case into an appropriate 1:n representation if the importing system does not support m:n.

Inheritance path: ->IdentifiedObject

Table 7-12 shows all attributes of VisibilityLayer.

Table 7-12 - Attributes of DiagramLayout::VisibilityLayer

Attribute name	mult.	Attribute type	Description
drawingOrder	0..1	Integer	The drawing order for this layer. The higher the number, the later the layer and the objects within it are rendered.
mRID	0..1	String	see IdentifiedObject
name	1..1	String	see IdentifiedObject

Table 7-13 shows all association ends of VisibilityLayer with other classes.

Table 7-13 - Association ends of DiagramLayout::VisibilityLayer with other classes

mult.	mult.	Class type name	Description
VisibleObjects	[0..*]	DiagramObject	A visibility layer can contain one or more diagram

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			objects. Association Based On : DiagramLayout:: DiagramObject.VisibleObjects[0..*] ----- DiagramLayout:: VisibilityLayer.VisibilityLayers[0..*]

7.4 Core

7.4.1 General

7.4.2 IdentifiedObject(Abstract)

This is a root class to provide common identification for all classes needing identification and naming attributes.

OCL constraint:R.4.10.11. Name length restriction

IdentifiedObject.name is 32 characters maximum. It shall be consistent with the name of the object used in companies, in daily operation (e. g. in SCADA systems), in planning processes or in asset related systems and should allow inter-communicating of TSO, using general names.

Table 7-14 shows all attributes of IdentifiedObject.

Table 7-14 - Attributes of Core::IdentifiedObject

Attribute name	mult.	Attribute type	Description
mRID	0..1	String	Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552 Edition 1, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name	1..1	String	The name is any free human readable and possibly non unique text naming the object.

7.5 Global Used Datatypes

7.5.1 Datatypes

7.5.1.1 Simple_Float

A floating point number. The range is unspecified and not limited.

In ENTSO-E profile, Simple_Float range is the IEEE754 simple precision floating point one. It correspond to xs:float datatype

Table 7-15 shows all attributes of Simple_Float.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 7-15 - Attributes of DomainProfile::Simple_Float.

Attribute name	mult.	Attribute type	Description
value	1..1	Float	

7.5.1.2 AngleDegrees

Measurement of angle in degrees.

Value type is IEEE 754 simple precision floating point

Table 7-16 shows all attributes of AngleDegrees.

Table 7-16 - Attributes of DomainProfile::AngleDegrees.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= deg
multiplier	0..1	UnitMultiplier	ConstantValue= none

7.5.2 Enumerations

7.5.2.1 OrientationKind

The orientation of the coordinate system with respect to top, left, and the coordinate number system.

Literal	Description	Code
negative	For 2D diagrams, a negative orientation gives the left-hand orientation (favoured by computer graphics displays) with X values increasing from left to right and Y values increasing from top to bottom. This is also known as a left hand orientation.	

7.5.2.2 UnitSymbol

The units defined for usage in the CIM.

Literal	Description	Code
VA	Apparent power in volt ampere.	
W	Active power in watt.	
VAr	Reactive power in volt ampere reactive.	
VAh	Apparent energy in volt ampere hours.	
Wh	Real energy in what hours.	
VArh	Reactive energy in volt ampere reactive hours.	
V	Voltage in volt.	
ohm	Resistance in ohm.	
A	Current in ampere.	
F	Capacitance in farad.	
H	Inductance in henry.	
degC	Relative temperature in degrees Celsius. In the SI unit system the symbol is ºC. Electric charge is measured in coulomb that has the unit symbol C. To distinguish degree Celsius form coulomb the symbol used in the UML is degC. Reason for not using ºC is the	

	special character º is difficult to manage in software.	
s	Time in seconds.	
min	Time in minutes.	
h	Time in hours.	
deg	Plane angle in degrees.	
rad	Plane angle in radians.	
J	Energy in joule.	
N	Force in newton.	
S	Conductance in siemens.	
none	Dimension less quantity, e.g. count, per unit, etc.	
Hz	Frequency in hertz.	
g	Mass in gram.	
Pa	Pressure in pascal (n/m2).	
m	Length in meter.	
m2	Area in square meters.	
m3	Volume in cubic meters.	

7.5.2.3 UnitMultiplier

The unit multipliers defined for the CIM.

Literal	Description	Code
p	Pico 10**-12.	
n	Nano 10**-9.	
micro	Micro 10**-6.	
m	Milli 10**-3.	
c	Centi 10**-2.	
d	Deci 10**-1.	
k	Kilo 10**3.	
M	Mega 10**6.	
G	Giga 10**9.	
T	Tera 10**12.	
none	No multiplier or equivalently multiply by 1.	

7.5.3 Primitives

7.5.3.1 String

A string consisting of a sequence of characters. The character encoding is UTF-8. The string length is unspecified and unlimited.

7.5.3.2 Date

Date as "yyyy-mm-dd", which conforms with ISO 8601. UTC time zone is specified as "yyyy-mm-ddZ". A local timezone relative UTC is specified as "yyyy-mm-dd(+/-)hh:mm".

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

7.5.3.3 Float

A floating point number. The range is unspecified and not limited.

7.5.3.4 Integer

An integer number. The range is unspecified and not limited.

7.5.3.5 Boolean

A type with the value space "true" and "false".

8 GeographicalLocationProfile

8.1 General

This geographical location profile is built on the basis of the IEC profile. There is small difference in one association which is specific for exchanges between TSOs.

Figure 8-1 shows the diagram.

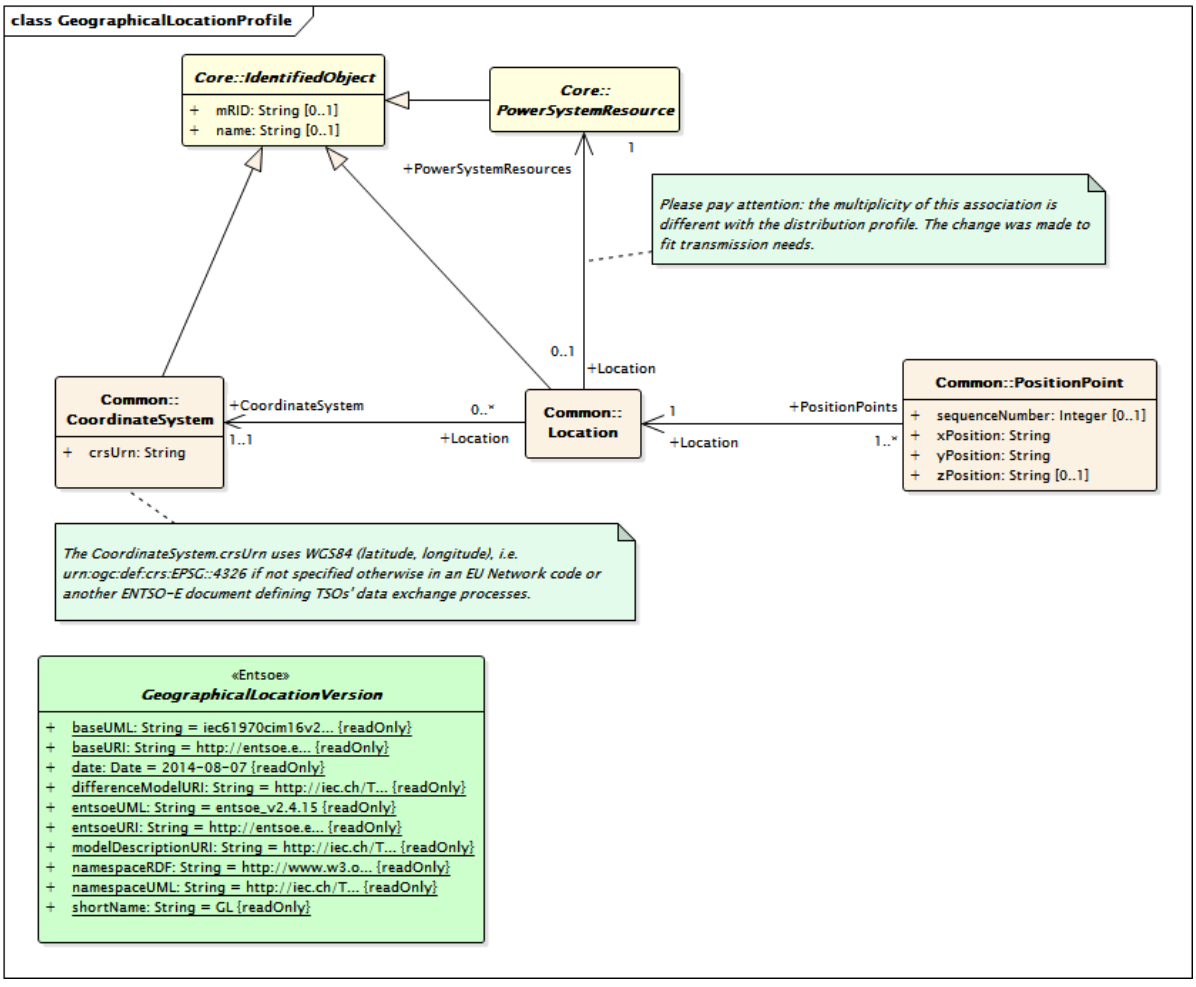


Figure 8-1 - diagram GeographicalLocationProfile

8.2 (Entsoe) GeographicalLocationVersion(Abstract)

Version details.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 8-1 shows all attributes of GeographicalLocationVersion.

Table 8-1 - Attributes of GeographicalLocationProfile::GeographicalLocationVersion

Attribute name	mult.	Attribute type	Description
baseUML (Entsoe)	1..1	String	ConstantValue= iec61970cim16v28_iec61968cim12v08_iec62325cim03v01a Base UML provided by CIM model manager.
baseURI (Entsoe)	1..1	String	ConstantValue= http://entsoe.eu/Secretariat/2013/Geographical/2 Profile URI used in the Model Exchange header and defined in IEC standards. It uniquely identifies the Profile and its version. It is given for information only and to identify the closest IEC profile to which this CGMES profile is based on.
date (Entsoe)	1..1	Date	ConstantValue= 2014-08-07 Profile creation date Form is YYYY-MM-DD for example for January 5, 2009 it is 2009-01-05.
differenceModelURI (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/61970-552/DifferenceModel/1# Difference model URI defined by IEC 61970-552.
entsoeUML (Entsoe)	1..1	String	ConstantValue= entsoe_v2.4.15 UML provided by ENTSO-E.
entsoeURI (Entsoe)	1..1	String	ConstantValue= http://entsoe.eu/CIM/GeographicalLocation/2/1 Profile URI defined by ENTSO-E and used in the Model Exchange header. It uniquely identifies the Profile and its version. The last two elements in the URI (http://entsoe.eu/CIM/GeographicalLocation/yy/zzz) indicate major and minor versions where: - yy - indicates a major version; - zzz - indicates a minor version.
modelDescriptionURI (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/61970-552/ModelDescription/1# Model Description URI defined by IEC 61970-552.
namespaceRDF (Entsoe)	1..1	String	ConstantValue= http://www.w3.org/1999/02/22-rdf-syntax-ns# RDF namespace.
namespaceUML (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/2013/CIM-schema-cim16# CIM UML namespace.
shortName (Entsoe)	1..1	String	ConstantValue= GL The short name of the profile used in profile documentation.

8.3 Common

8.3.1 General

8.3.2 (Entsoe) CoordinateSystem

Coordinate reference system.

The CoordinateSystem.crsUrn uses WGS84 (latitude, longitude), i.e. urn:ogc:def:crs:EPSG::4326 if not specified otherwise in an EU Network code or another ENTSO-E document defining TSOs' data exchange processes.

Inheritance path: ->IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 8-2 shows all attributes of CoordinateSystem.

Table 8-2 - Attributes of Common::CoordinateSystem

Attribute name	mult.	Attribute type	Description
crsUrn	1..1	String	A Uniform Resource Name (URN) for the coordinate reference system (crs) used to define 'Location.PositionPoints'. An example would be the European Petroleum Survey Group (EPSG) code for a coordinate reference system, defined in URN under the Open Geospatial Consortium (OGC) namespace as: urn:ogc:def:uom:EPSG::XXXX, where XXXX is an EPSG code (a full list of codes can be found at the EPSG Registry web site http://www.epsg-registry.org/). To define the coordinate system as being WGS84 (latitude, longitude) using an EPSG OGC, this attribute would be urn:ogc:def:uom:EPSG::4236. A profile should limit this code to a set of allowed URNs agreed to by all sending and receiving parties.
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

8.3.3 Location

The place, scene, or point of something where someone or something has been, is, and/or will be at a given moment in time. It can be defined with one or more postition points (coordinates) in a given coordinate system.

Inheritance path: ->IdentifiedObject

Table 8-3 shows all attributes of Location.

Table 8-3 - Attributes of Common::Location

Attribute name	mult.	Attribute type	Description
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 8-4 shows all association ends of Location with other classes.

Table 8-4 - Association ends of Common::Location with other classes

mult.	mult.	Class type name	Description
PowerSystemResources	[1..1]	PowerSystemResource	All power system resources at this location. Association Based On : Common:: Location.Location[0..1] ----- Core::

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			PowerSystemResource.PowerSystemResources[0..*]
CoordinateSystem	[1..1]	CoordinateSystem	Coordinate system used to describe position points of this location. Association Based On : Common:: Location.Locations[0..*] ----- Common:: CoordinateSystem.CoordinateSystem[0..1]

8.3.4 PositionPoint

Set of spatial coordinates that determine a point, defined in the coordinate system specified in 'Location.CoordinateSystem'. Use a single position point instance to desribe a point-oriented location. Use a sequence of position points to describe a line-oriented object (physical location of non-point oriented objects like cables or lines), or area of an object (like a substation or a geographical zone - in this case, have first and last position point with the same values).

Table 8-5 shows all attributes of PositionPoint.

Table 8-5 - Attributes of Common::PositionPoint

Attribute name	mult.	Attribute type	Description
sequenceNumber	0..1	Integer	Zero-relative sequence number of this point within a series of points.
xPosition	1..1	String	X axis position.
yPosition	1..1	String	Y axis position.
zPosition	0..1	String	(if applicable) Z axis position.

Table 8-6 shows all association ends of PositionPoint with other classes.

Table 8-6 - Association ends of Common::PositionPoint with other classes

mult.	mult.	Class type name	Description
Location	[1..1]	Location	Location described by this position point. Association Based On : Common:: Location.Location[1] ----- Common:: PositionPoint.PositionPoints[0..*]

8.4 Core

8.4.1 General

Figure 8-2 shows the diagram.

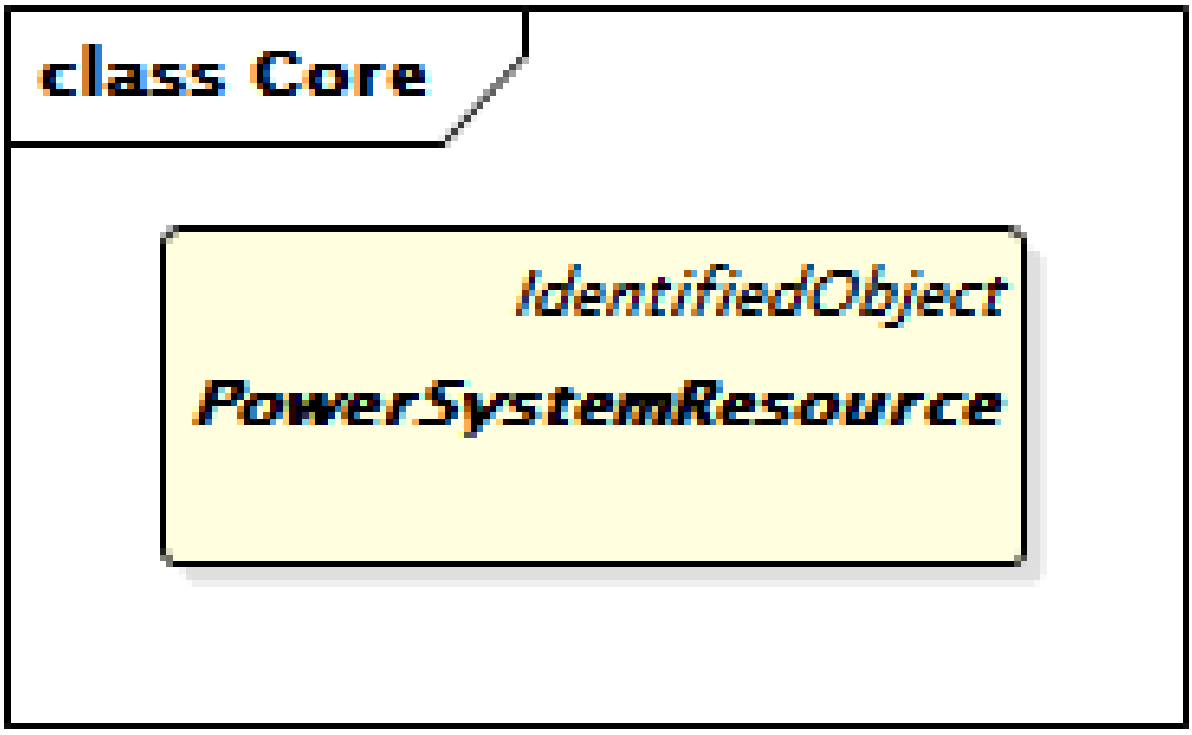


Figure 8-2 - diagram Core

8.4.2 IdentifiedObject(Abstract)

This is a root class to provide common identification for all classes needing identification and naming attributes.

Table 8-7 shows all attributes of IdentifiedObject.

Table 8-7 - Attributes of Core::IdentifiedObject

Attribute name	mult.	Attribute type	Description
mRID	0..1	String	Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552 Edition 1, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name	0..1	String	The name is any free human readable and possibly non unique

			text naming the object.

8.4.3 PowerSystemResource(Abstract)

A power system resource can be an item of equipment such as a switch, an equipment container containing many individual items of equipment such as a substation, or an organisational entity such as sub-control area. Power system resources can have measurements associated.

Inheritance path: ->IdentifiedObject

Table 8-8 shows all attributes of PowerSystemResource.

Table 8-8 - Attributes of Core::PowerSystemResource

Attribute name	mult.	Attribute type	Description
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

8.5 Global Used Datatypes

8.5.1 Primitives

8.5.1.1 String

A string consisting of a sequence of characters. The character encoding is UTF-8. The string length is unspecified and unlimited.

8.5.1.2 Date

Date as "yyyy-mm-dd", which conforms with ISO 8601. UTC time zone is specified as "yyyy-mm-ddZ". A local timezone relative UTC is specified as "yyyy-mm-dd(+/-)hh:mm".

8.5.1.3 Integer

An integer number. The range is unspecified and not limited.

9 DynamicsProfile

9.1 General

The CIM dynamic model definitions reflect the most common IEEE or, in the case of wind models, IEC, representations of models as well as models included in some of the transient stability software widely used by utilities.
These dynamic models are intended to ensure interoperability between different vendors' software products currently in use by electric utility energy companies, utilities, TSOs and RTO/ISOs. It is important to note that each vendor is free to select its own internal implementation of these models. Differences in vendor results, as long as they are within accepted engineering practice, caused by different internal representations, are acceptable.
Notes:
1. Wind models package is defined in accordance with IEC 61400-27-1 version CDV 2013-08-15.

9.2 (Entsoe) DynamicsVersion(Abstract)

Version details.

Table 9-1 shows all attributes of DynamicsVersion.

Table 9-1 - Attributes of DynamicsProfile::DynamicsVersion

Attribute name	mult.	Attribute type	Description
baseUML (Entsoe)	1..1	String	ConstantValue=

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			iec61970cim16v28_iec61968cim12v08_iec62325cim03v01a Base UML provided by CIM model manager.
baseURI (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/2013/61970-457/Dynamics/3 Profile URI used in the Model Exchange header and defined in IEC standards. It uniquely identifies the Profile and its version. It is given for information only and to identify the closest IEC profile to which this CGMES profile is based on.
date (Entsoe)	1..1	Date	ConstantValue= 2014-08-07 Profile creation date Form is YYYY-MM-DD for example for January 5, 2009 it is 2009-01-05.
differenceModelURI (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/61970-552/DifferenceModel/1# Difference model URI defined by IEC 61970-552.
entsoeUML (Entsoe)	1..1	String	ConstantValue= entsoe_v2.4.15 UML provided by ENTSO-E.
entsoeURI (Entsoe)	1..1	String	ConstantValue= http://entsoe.eu/CIM/Dynamics/3/1 Profile URI defined by ENTSO-E and used in the Model Exchange header. It uniquely identifies the Profile and its version. The last two elements in the URI (http://entsoe.eu/CIM/Dynamics/yy/zzz) indicate major and minor versions where: - yy - indicates a major version; - zzz - indicates a minor version.
modelDescriptionURI (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/61970-552/ModelDescription/1# Model Description URI defined by IEC 61970-552.
namespaceRDF (Entsoe)	1..1	String	ConstantValue= http://www.w3.org/1999/02/22-rdf-syntax-ns# RDF namespace.
namespaceUML (Entsoe)	1..1	String	ConstantValue= http://iec.ch/TC57/2013/CIM-schema-cim16# CIM UML namespace.
shortName (Entsoe)	1..1	String	ConstantValue= DY The short name of the profile used in profile documentation.

9.3 Core

9.3.1 General

Figure 9-1 shows the diagram.

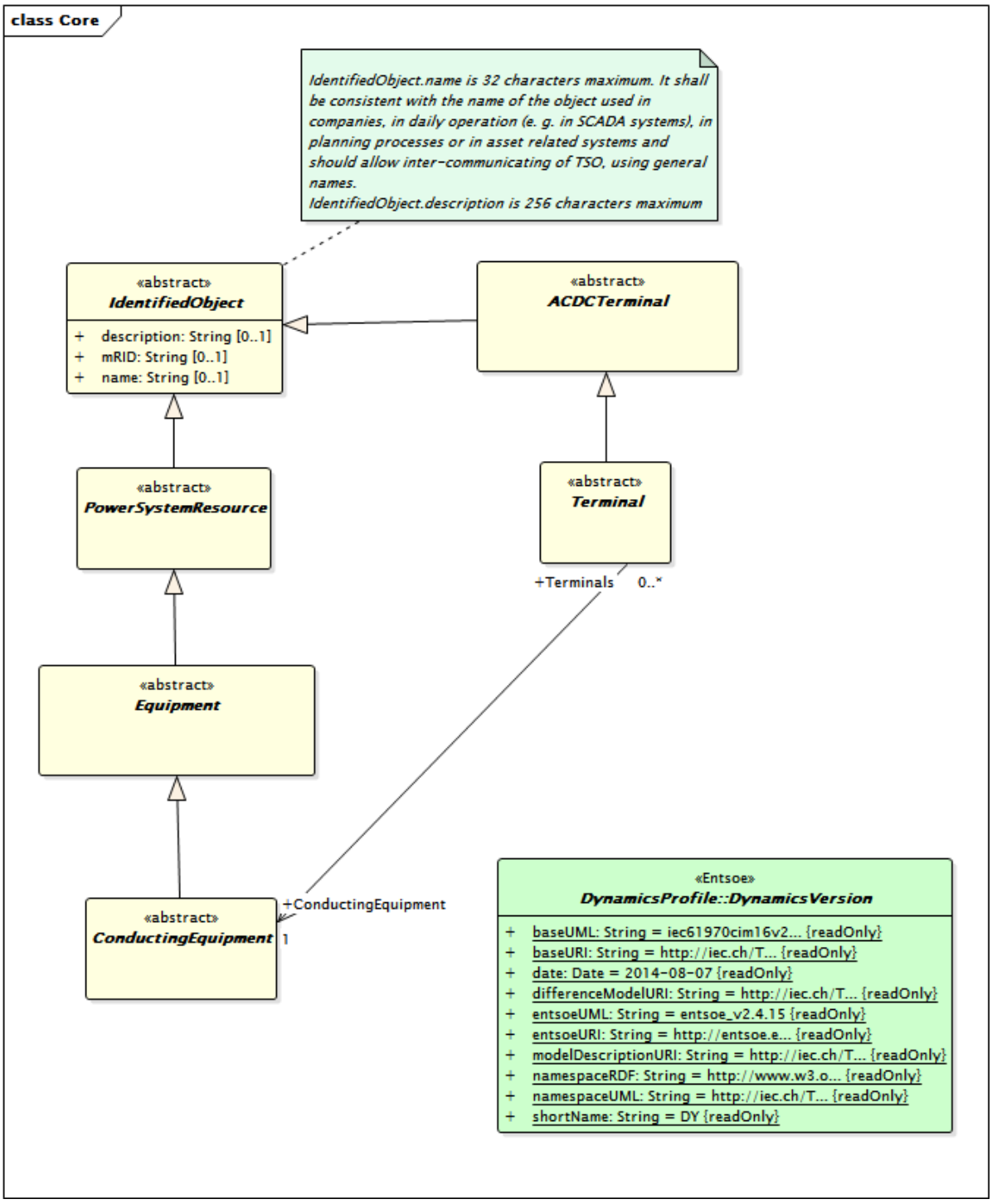


Figure 9-1 - diagram Core

9.3.2 ACDCTerminal(Abstract)

An electrical connection point (AC or DC) to a piece of conducting equipment. Terminals are connected at physical connection points called connectivity nodes.

Inheritance path: ->IdentifiedObject

Table 9-2 shows all attributes of ACDCTerminal.

Table 9-2 - Attributes of Core::ACDCTerminal

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

9.3.3 ConductingEquipment(Abstract)

The parts of the AC power system that are designed to carry current or that are conductively connected through terminals.

Inheritance path: ->Equipment->PowerSystemResource->IdentifiedObject

Table 9-3 shows all attributes of ConductingEquipment.

Table 9-3 - Attributes of Core::ConductingEquipment

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

9.3.4 Equipment(Abstract)

The parts of a power system that are physical devices, electronic or mechanical.

Inheritance path: ->PowerSystemResource->IdentifiedObject

Table 9-4 shows all attributes of Equipment.

Table 9-4 - Attributes of Core::Equipment

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

9.3.5 IdentifiedObject(Abstract)

This is a root class to provide common identification for all classes needing identification and naming attributes.

OCL constraint:R.4.10.11. Description length restriction (optional)
OCL constraint:R.4.10.11. Name length restriction

IdentifiedObject.name is 32 characters maximum. It shall be consistent with the name of the object used in companies, in daily operation (e. g. in SCADA systems), in planning processes or in asset related systems and should allow inter-communicating of TSO, using general names. IdentifiedObject.description is 256 characters maximum

Table 9-5 shows all attributes of IdentifiedObject.

Table 9-5 - Attributes of Core::IdentifiedObject

Attribute name	mult.	Attribute type	Description

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

description	0..1	String	The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy.
mRID	0..1	String	Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552 Edition 1, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name	0..1	String	The name is any free human readable and possibly non unique text naming the object.

9.3.6 PowerSystemResource(Abstract)

A power system resource can be an item of equipment such as a switch, an equipment container containing many individual items of equipment such as a substation, or an organisational entity such as sub-control area. Power system resources can have measurements associated.

Inheritance path: ->IdentifiedObject

Table 9-6 shows all attributes of PowerSystemResource.

Table 9-6 - Attributes of Core::PowerSystemResource

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

9.3.7 Terminal(Abstract)

An AC electrical connection point to a piece of conducting equipment. Terminals are connected at physical connection points called connectivity nodes.

Inheritance path: ->ACDCTerminal->IdentifiedObject

Table 9-7 shows all attributes of Terminal.

Table 9-7 - Attributes of Core::Terminal

Attribute name	mult.	Attribute type	Description

description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-8 shows all association ends of Terminal with other classes.

Table 9-8 - Association ends of Core::Terminal with other classes

mult.	mult.	Class type name	Description
ConductingEquipment	[1..1]	ConductingEquipment	The conducting equipment of the terminal. Conducting equipment have terminals that may be connected to other conducting equipment terminals via connectivity nodes or topological nodes. Association Based On : Core:: Terminal.Terminals[0..*] ----- Core:: ConductingEquipment.ConductingEquipment[1]

9.4 Wires

9.4.1 General

Figure 9-2 shows the diagram.

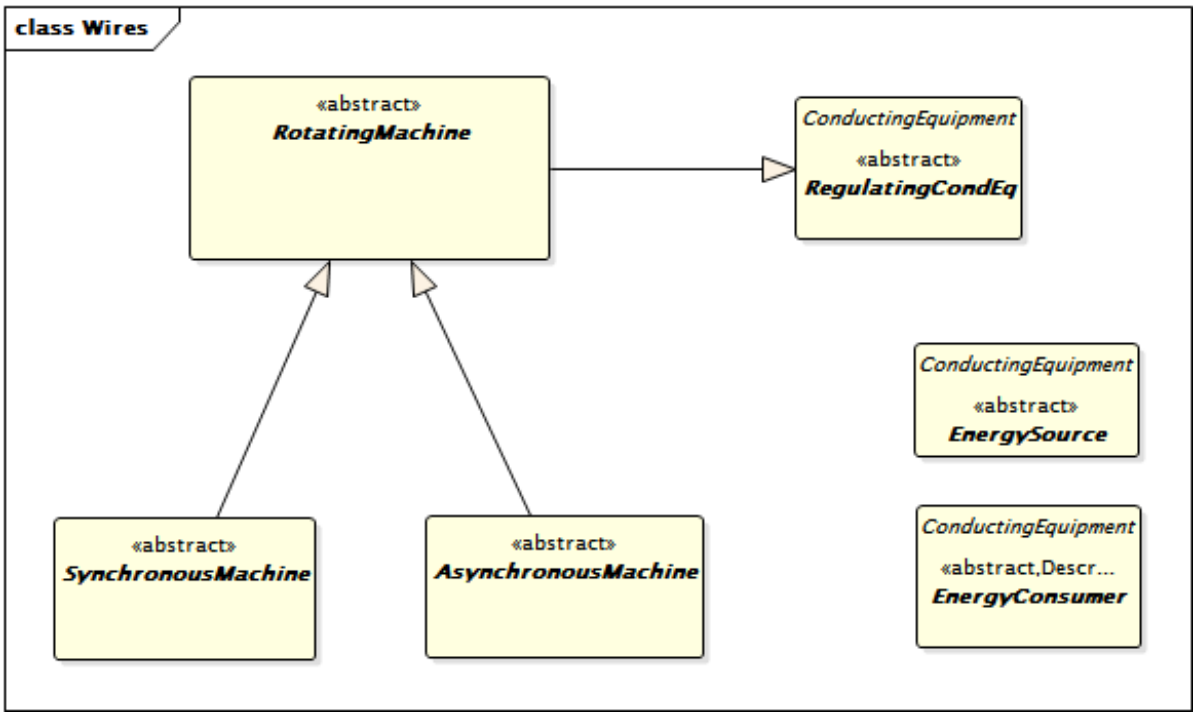


Figure 9-2 - diagram Wires

9.4.2 AsynchronousMachine(Abstract)

A rotating machine whose shaft rotates asynchronously with the electrical field. Also known as an induction machine with no external connection to the rotor windings, e.g squirrel-cage induction machine.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Inheritance path: ->RotatingMachine->RegulatingCondEq->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 9-9 shows all attributes of AsynchronousMachine.

Table 9-9 - Attributes of Wires::AsynchronousMachine

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

9.4.3 (Description) EnergyConsumer(Abstract)

Generic user of energy - a point of consumption on the power system model.

Inheritance path: ->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 9-10 shows all attributes of EnergyConsumer.

Table 9-10 - Attributes of Wires::EnergyConsumer

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-11 shows all association ends of EnergyConsumer with other classes.

Table 9-11 - Association ends of Wires::EnergyConsumer with other classes

mult.	mult.	Class type name	Description
LoadDynamics (Description)	[0..1]	LoadDynamics	Load dynamics model used to describe dynamic behavior of this energy consumer. Association Based On : LoadDynamics:: LoadDynamics.LoadDynamics[0..1] ----- Wires:: EnergyConsumer.EnergyConsumer[0..*]

9.4.4 EnergySource(Abstract)

A generic equivalent for an energy supplier on a transmission or distribution voltage level.

Inheritance path: ->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 9-12 shows all attributes of EnergySource.

Table 9-12 - Attributes of Wires::EnergySource

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

9.4.5 RegulatingCondEq(Abstract)

A type of conducting equipment that can regulate a quantity (i.e. voltage or flow) at a specific point in the network.

Inheritance path: ->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 9-13 shows all attributes of RegulatingCondEq.

Table 9-13 - Attributes of Wires::RegulatingCondEq

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

9.4.6 RotatingMachine(Abstract)

A rotating machine which may be used as a generator or motor.

Inheritance path: ->RegulatingCondEq->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 9-14 shows all attributes of RotatingMachine.

Table 9-14 - Attributes of Wires::RotatingMachine

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

9.4.7 SynchronousMachine(Abstract)

An electromechanical device that operates with shaft rotating synchronously with the network. It is a single machine operating either as a generator or synchronous condenser or pump.

Inheritance path: ->RotatingMachine->RegulatingCondEq->ConductingEquipment->Equipment->PowerSystemResource->IdentifiedObject

Table 9-15 shows all attributes of SynchronousMachine.

Table 9-15 - Attributes of Wires::SynchronousMachine

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

9.5 StandardInterconnections

9.5.1 General

This section describes the standard interconnections for various types of equipment. These interconnections are understood by the application programs and can be identified based on the presence of one of the key classes with a relationship to the static power flow model: SynchronousMachineDynamics, AsynchronousMachineDynamics, EnergyConsumerDynamics or

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

WindTurbineType3or4Dynamics.
The relationships between classes expressed in the interconnection diagrams are intended to support dynamic behaviour described by either standard models or user-defined models. In the interconnection diagrams, boxes which are black in colour represent function blocks whose functionality can be provided by one of many standard models or by a used-defined model. Blue boxes represent specific standard models. A dashed box means that the function block or specific standard model is optional.

Figure 9-3 shows the diagram.

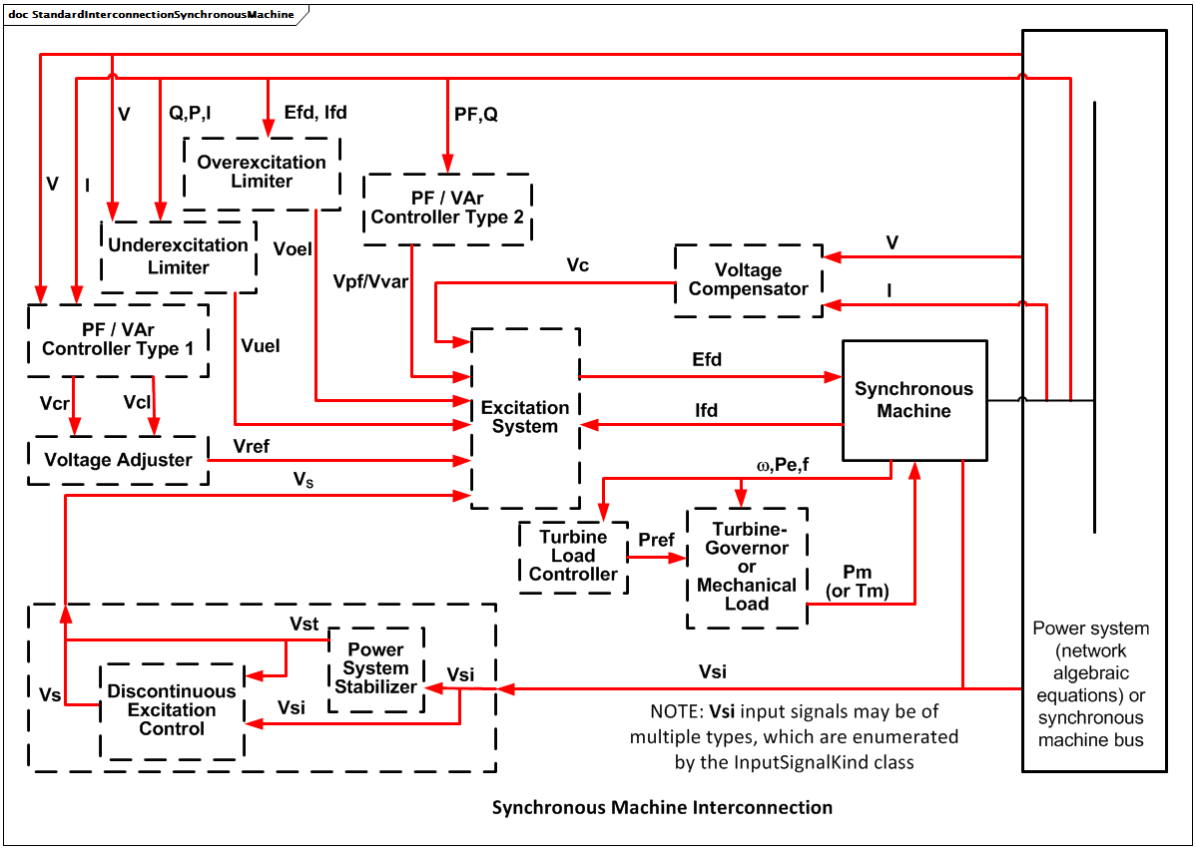


Figure 9-3 - diagram StandardInterconnectionSynchronousMachine

A dynamic model including a single synchronous machine (generator, motor or condenser) and its related equipment models (function blocks) is associated with a synchronous machine in the static (power flow) model and has the standard interconnection shown above.
The interface between the dynamic synchronous machine and the network algebraic equations is application dependent. The variables used for this interface do not need to be identified since they are internal to the application program, well-defined and not necessary to be exchanged between applications.
The interfaces between different blocks in the above figure are indicative of the variety of input and output signals that are defined by the individual standard models. The figure provides information on most of the alternative interfaces, for instance, the turbine-governor (for generators) or the mechanical load (for motors) and the dynamic synchronous machine may be mechanical torque (Tm) or mechanical power (Pm).

Figure 9-4 shows the diagram.

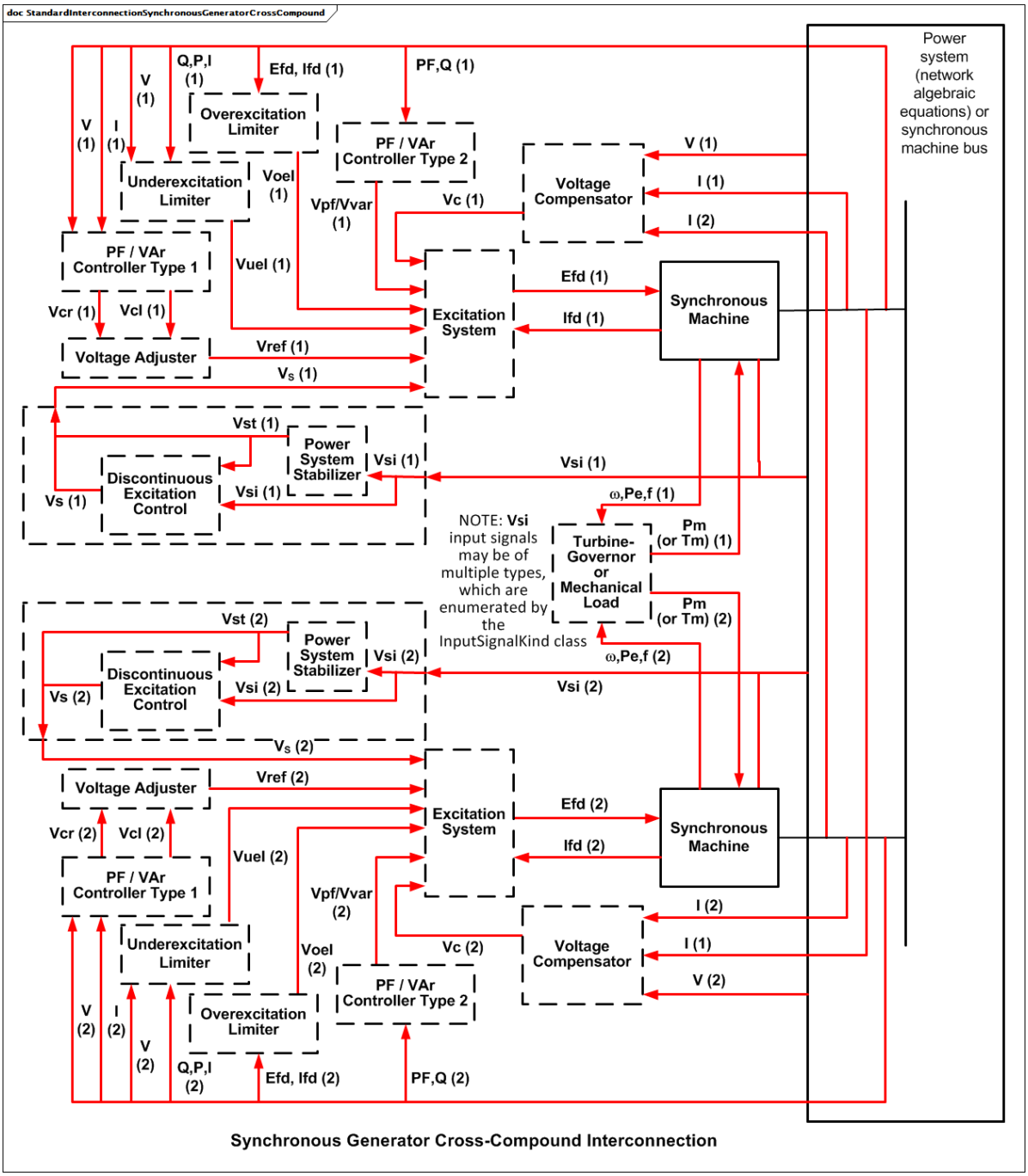


Figure 9-4 - diagram StandardInterconnectionSynchronousGeneratorCrossCompound

A cross-compound dynamic model includes multiple synchronous generators and their related equipment models (function blocks), is associated with multiple synchronous machines in the static (power flow) model and has the standard interconnection shown above. Generator 2 is the second unit of a cross-compound pair of generators and is usually connected to the same terminal bus. A single turbine-governor model determines the mechanical power for both units.

Figure 9-5 shows the diagram.

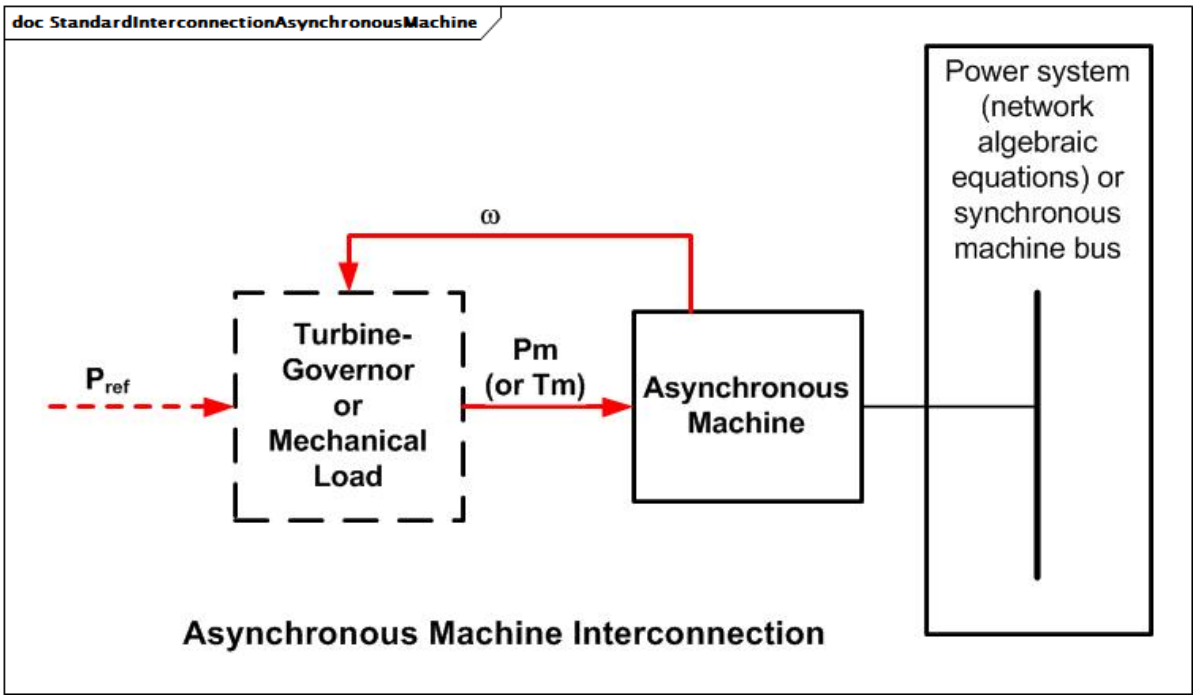


Figure 9-5 - diagram StandardInterconnectionAsynchronousMachine

A dynamic model including an asynchronous machine (either a generator or motor) and its related equipment models (function blocks) is associated with an asynchronous machine in the static (power flow) model and has the standard interconnection shown above. This interconnection is for a “squirrel-cage” induction machine or a wound-rotor induction machine with short-circuited field windings. Other models and a different interconnection are required for a wound-rotor machine with external connections to the field windings. The interface between the dynamic asynchronous machine and the network algebraic equations is application dependent.

Figure 9-6 shows the diagram.

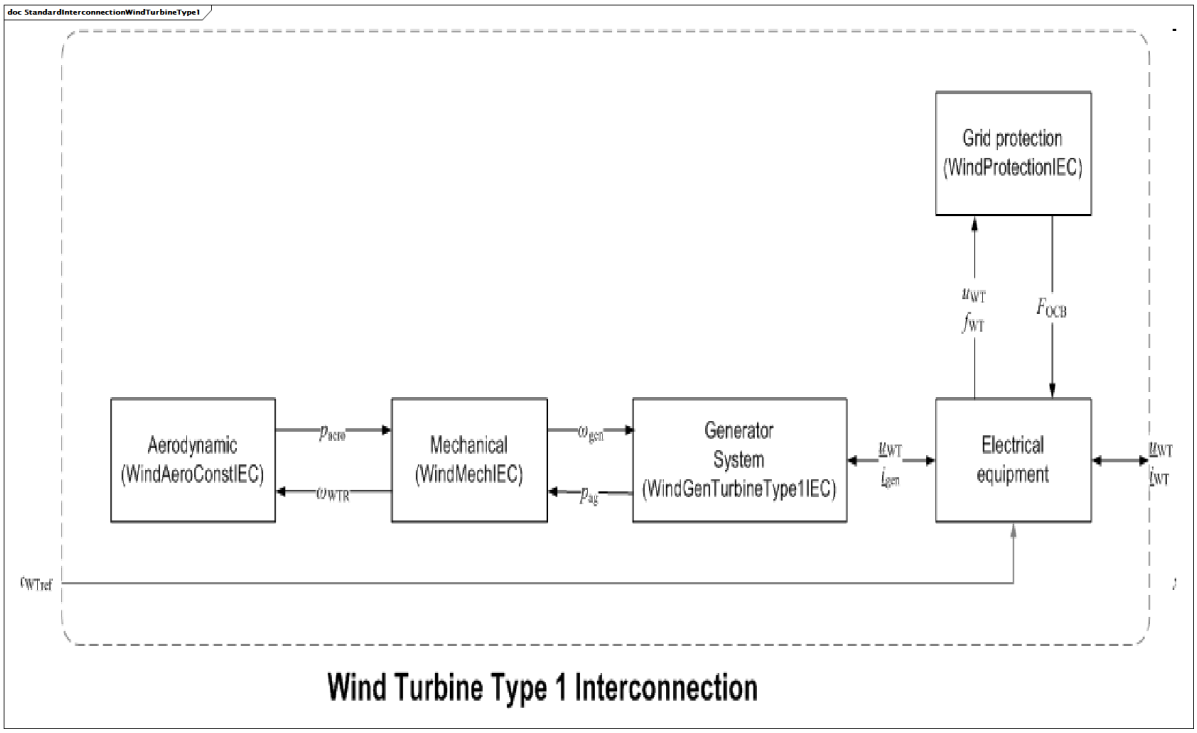


Figure 9-6 - diagram StandardInterconnectionWindTurbineType1

A dynamic model representing a Type 1 wind turbine and its related standard equipment models is associated with an asynchronous machine in the static (power flow) model and has the two the standard interconnection shown in the figure. Reference: IEC Standard 61400-27-1 Section 6.5.2.

Figure 9-7 shows the diagram.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

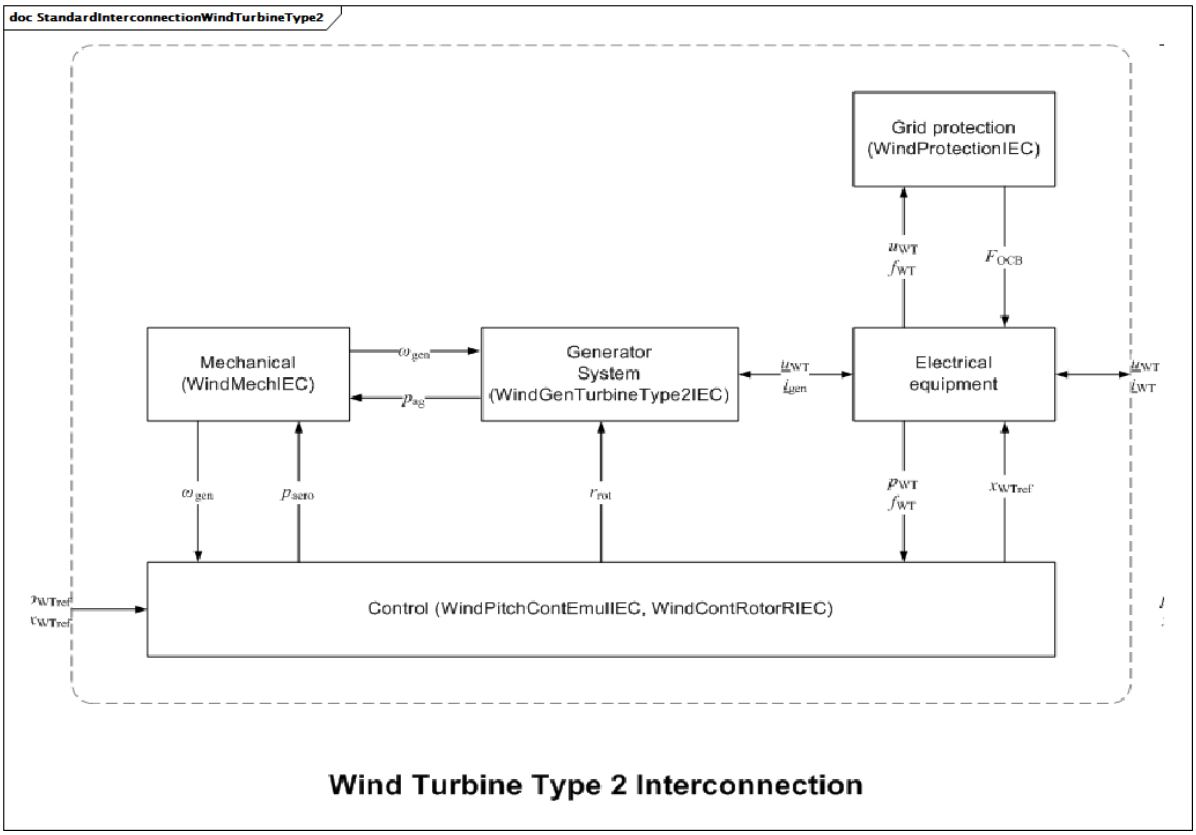


Figure 9-7 - diagram StandardInterconnectionWindTurbineType2

A dynamic model representing a Type 2 wind generator and its related standard equipment models is associated with an asynchronous machine in the static (power flow) model and has the standard interconnection shown above.
The Type 2 turbine is equipped with a variable rotor resistance (VRR) and therefore uses a wound rotor asynchronous generator (WRAG). Type 2 wind turbines are also normally equipped with blade angle control.
More information is contained in IEC Standard 61400-27-1 Section 6.4.2.

Figure 9-8 shows the diagram.

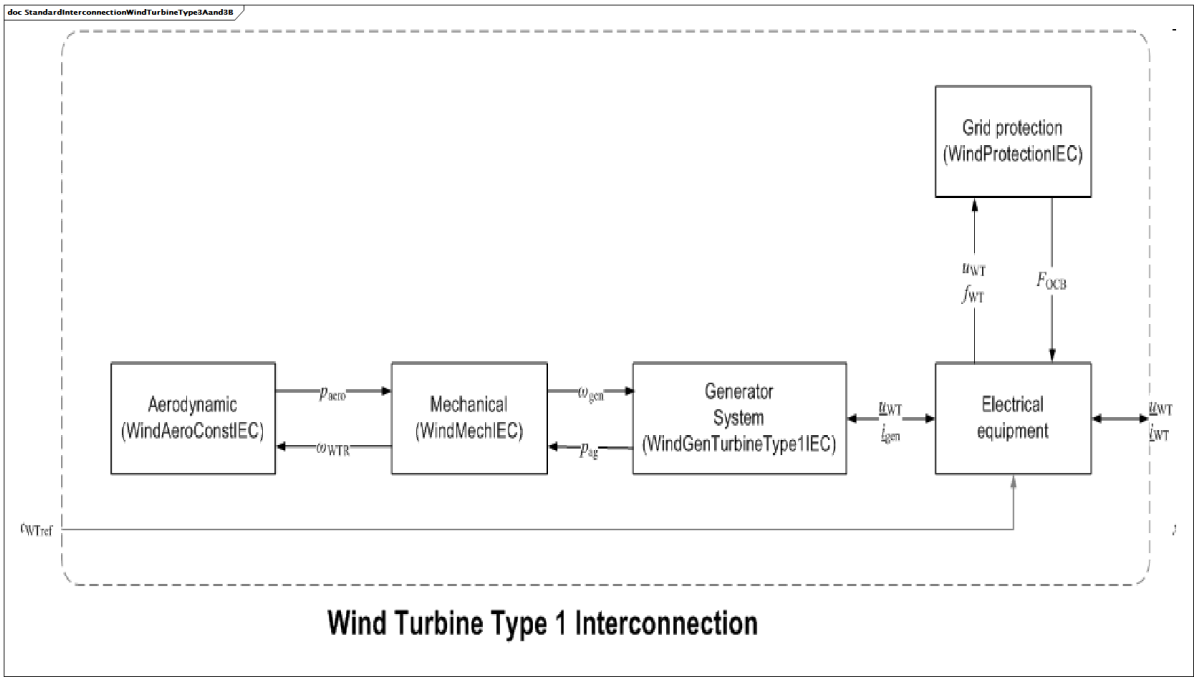


Figure 9-8 - diagram StandardInterconnectionWindTurbineType3Aand3B

A dynamic model representing a Type 3 wind turbine and its related standard equipment models is associated with a current source model in the static (power flow) model and has the standard interconnection shown in the figure. Reference: IEC Standard 61400-27-1 Section 6.5.4.

Figure 9-9 shows the diagram.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

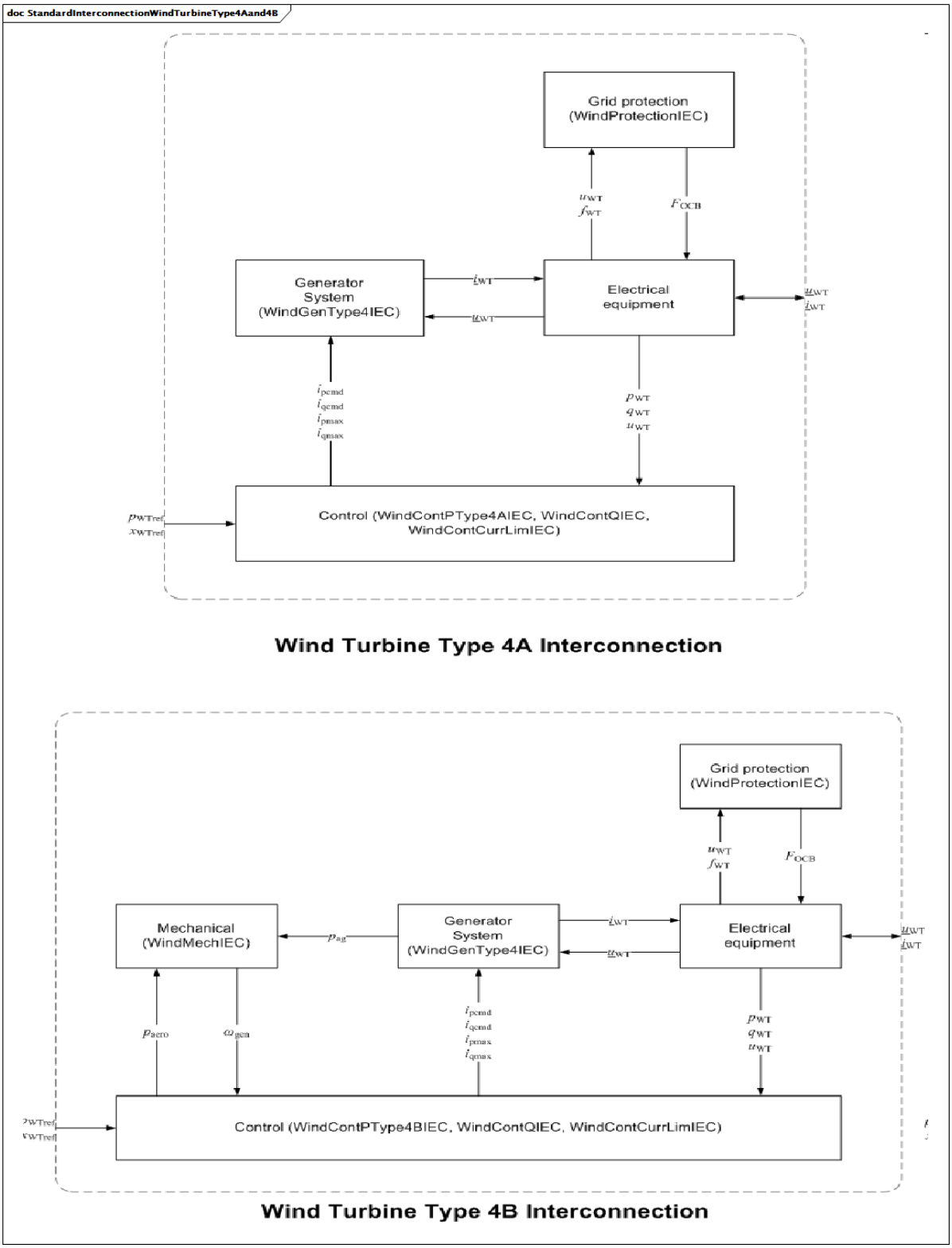


Figure 9-9 - diagram StandardInterconnectionWindTurbineType4Aand4B

A dynamic model representing a Type 4 wind turbine and its related standard equipment models is associated with a current source model in the static (power flow) model and has one of the two the

standard interconnections shown in the figure.
Reference: IEC Standard 61400-27-1 Section 6.5.5.

Figure 9-10 shows the diagram.

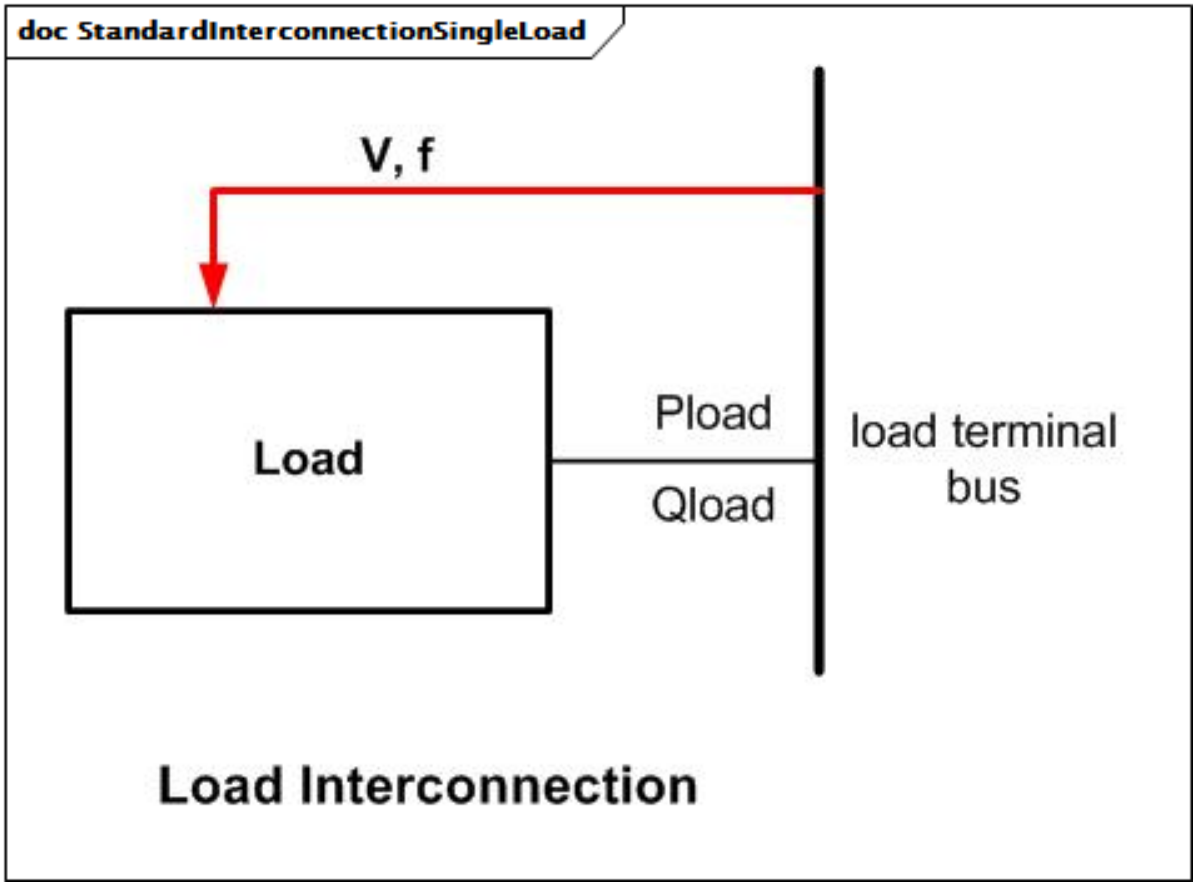


Figure 9-10 - diagram StandardInterconnectionSingleLoad

A load model is associated with an energy consumer in the power flow data and has no standard interconnection with other models.
The P and Q consumption of the load are determined as a function of the magnitude and frequency of the voltage at the terminal bus as shown above.
The interface between the model and the network algebraic equations is application dependent. Dynamic (motor) load models generally interface with the network equations in a similar manner to motor models.

Figure 9-11 shows the diagram.

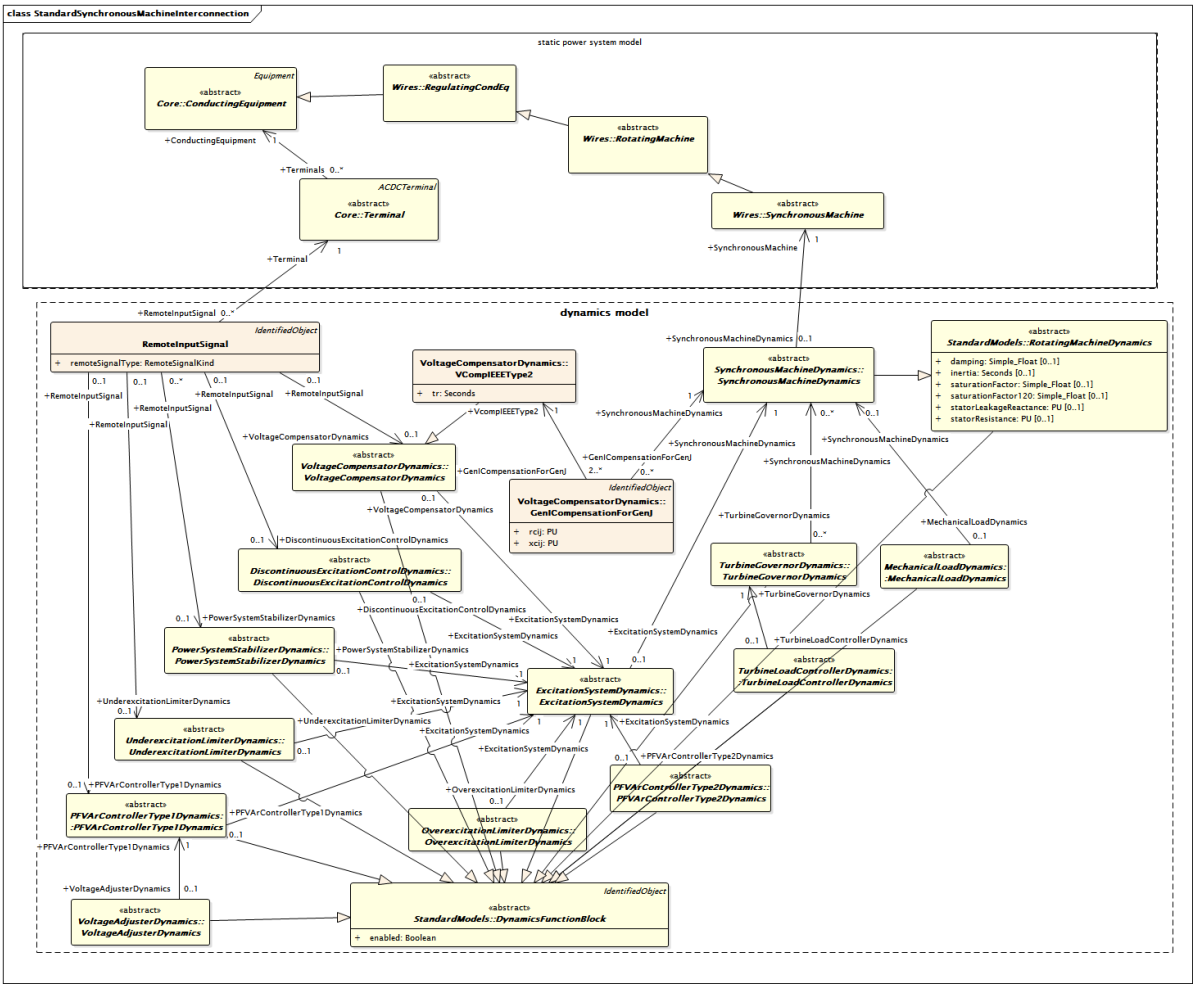


Figure 9-11 - diagram StandardSynchronousMachineInterconnection

This diagram shows the classes necessary for both of the two types of synchronous machine interconnection models: standard synchronous machine and standard synchronous generator cross-compound.
A standard synchronous machine interconnection has a single instance of a SynchronousMachineDynamics child class whereas a generator cross-compound connection has multiple instances of SynchronousMachineDynamics child classes related by a single instance of a TurbineGovernorDynamics child class.
A valid standard synchronous machine interconnection model reflects the following rules. It may have:

- 1 instance of TurbineGovernorDynamics child class, but only if SynchronousMachine.operatingMode = generator
- 1 instance of MechanicalLoadDynamics child class, but only if SynchronousMachine.operatingMode = motor or SynchronousMachine.operatingMode = condenser

It may not have

- any instance of GenICompensationForGenJ

A generator cross-compound interconnection model reflects the following rules. It may have:

- multiple instances of SynchronousMachineDynamics child classes, each of which has its related SynchronousMachine.operatingMode = generator
- 1 instance of TurbineGovernorDynamics child class

It must have

- 2 or more instances of GenICompensationForGenJ associated with each instance of a SynchronousMachineDynamics child class.

Figure 9-12 shows the diagram.

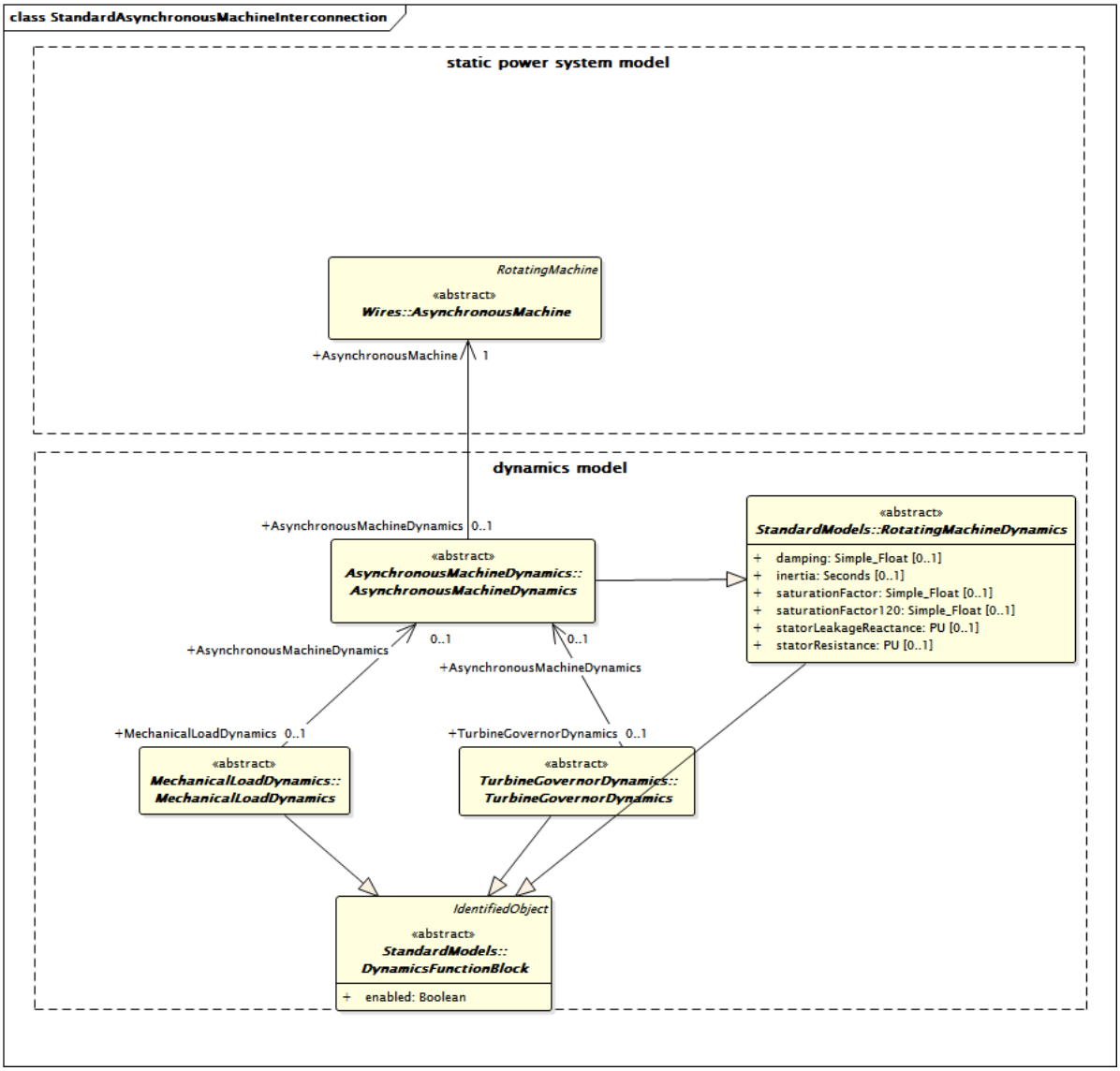


Figure 9-12 - diagram StandardAsynchronousMachineInterconnection

This diagram shows the classes necessary for the asynchronous machine interconnection model. A standard asynchronous machine interconnection has a single instance of an AsynchronousMachineDynamics child class.
A valid standard asynchronous machine interconnection model also reflects the following rules. It may have:

- 1 instance of TurbineGovernorDynamics child class, but only if AsynchronousMachine.asynchronousMachineType = generator
- 1 instance of MechanicalLoadDynamics child class, but only if AsynchronousMachine.asynchronousMachineType = motor.

Figure 9-13 shows the diagram.

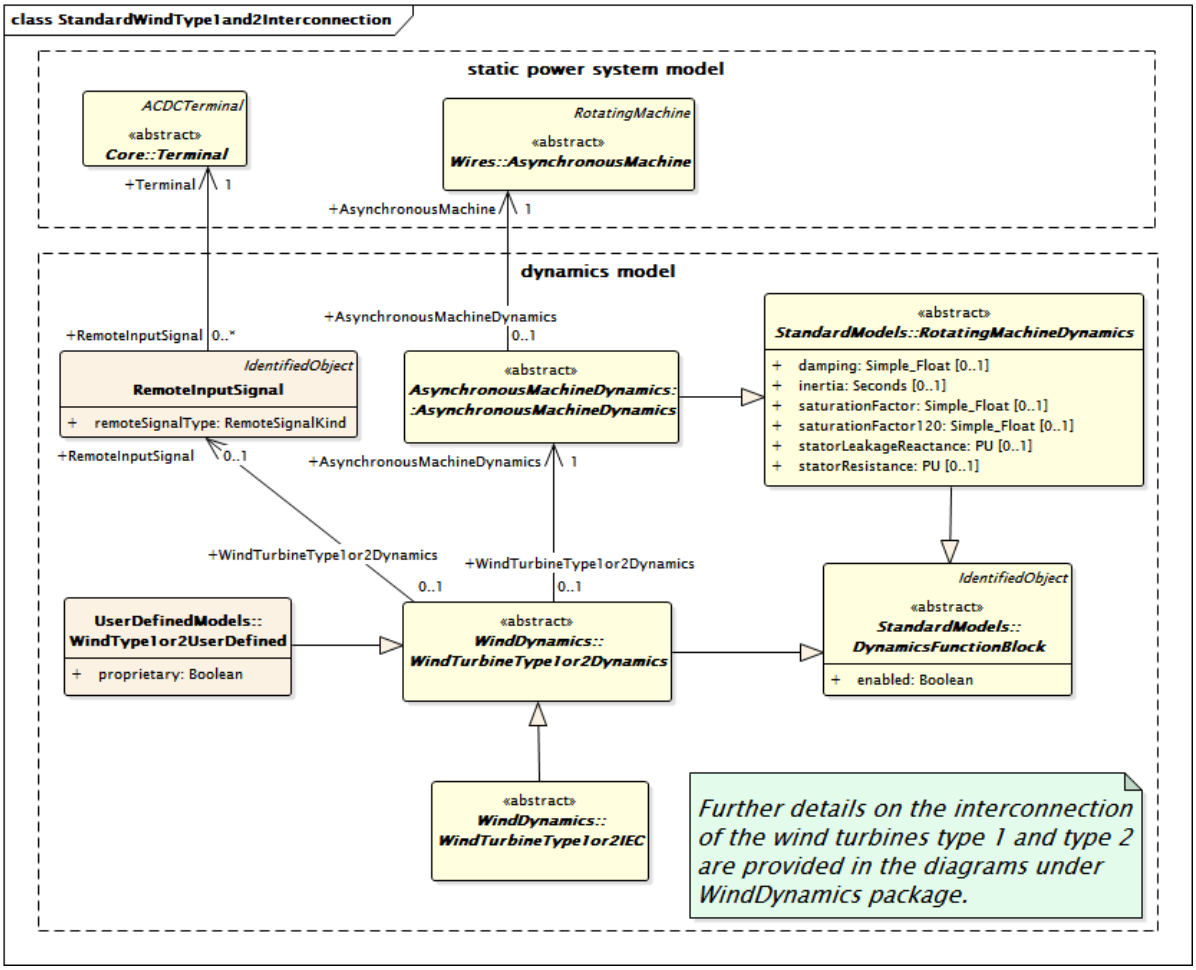


Figure 9-13 - diagram StandardWindType1and2Interconnection

This diagram shows the classes necessary for wind Type 1 and Type 2 interconnection models. Standard IEC wind turbines Type 1 and 2 interconnection models all have a single instance of an AsynchronousMachineDynamics child class and a single instance of the WindGenTurbineType1IEC class or the WindGenTurbineType2IEC class.
A valid IEC wind Type 1 interconnection model also must have an instance of the following classes (and no others):

- WindMechIEC

- WindAeroConstIEC
- WindProtectionIEC.

A valid IEC wind Type 2 interconnection model also must have an instance of the following classes (and no others):

- WindMechIEC

- WindContRotorRIEC
- WindProtectionIEC
- WindPitchContEmulIEC
- WindDynamicsLookupTable.

Figure 9-14 shows the diagram.

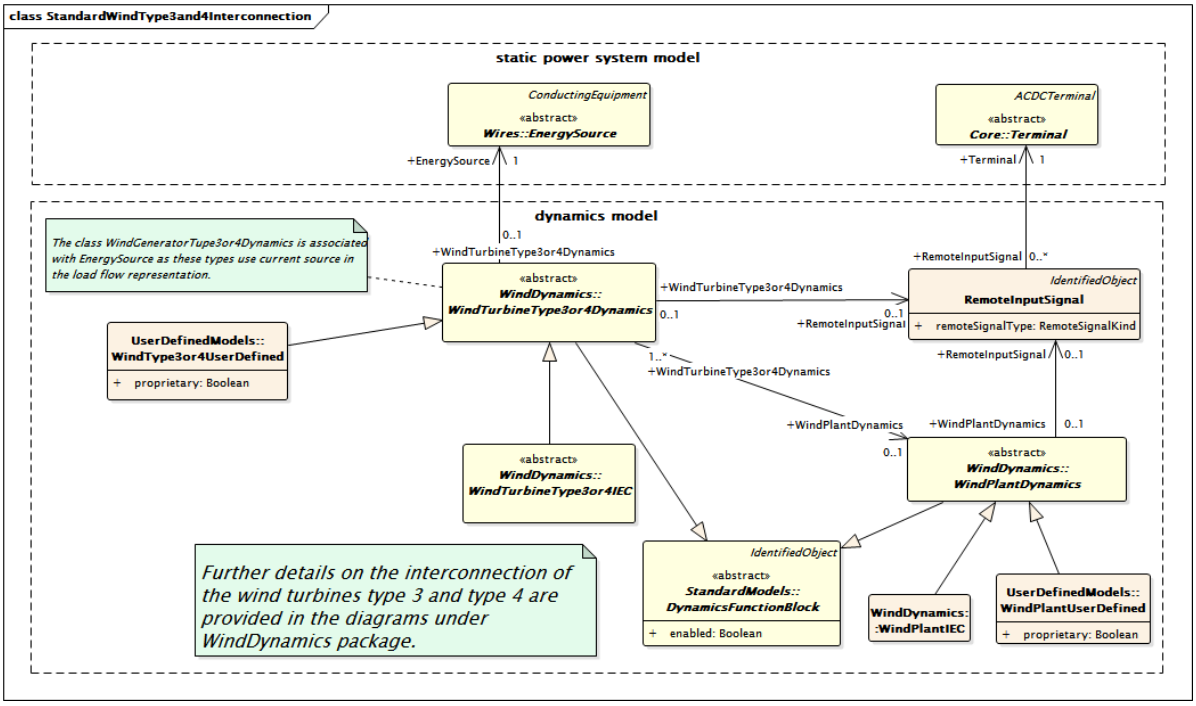


Figure 9-14 - diagram StandardWindType3and4Interconnection

This diagram shows the classes necessary for wind Type 3 and Type 4 interconnection models, including the wind plant model.
Standard IEC wind Type 3A, 3B, 4A, 4B and plant interconnection models all have a single instances as follows:
IEC wind Type 3A:

- WindGenTurbineType3aIEC
- WindProtectionIEC
- WindContQIEC
- WindContCurrLimIEC
- WindMechIEC

- WindAeroLinearIEC
- WindContPType3IEC
- WindContPitchAngleIEC
- WindDynamicsLookupTable.

IEC wind Type 3B:

- WindGenTurbineType3bIEC
- WindProtectionIEC
- WindContQIEC
- WindContCurrLimIEC
- WindMechIEC

- WindAeroLinearIEC
- WindContPType3IEC
- WindContPitchAngleIEC
- WindDynamicsLookupTable.

IEC wind Type 4A:

- WindTurbineType4AIEC
- WindProtectionIEC
- WindContQIEC
- WindContCurrLimIEC

- WindContPType4AIEC.

IEC wind Type 4B:

- WindTurbineType4BIEC
- WindMechIEC
- WindProtectionIEC
- WindContQIEC
- WindContCurrLimIEC

- WindContPType4BIEC.

Figure 9-15 shows the diagram.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

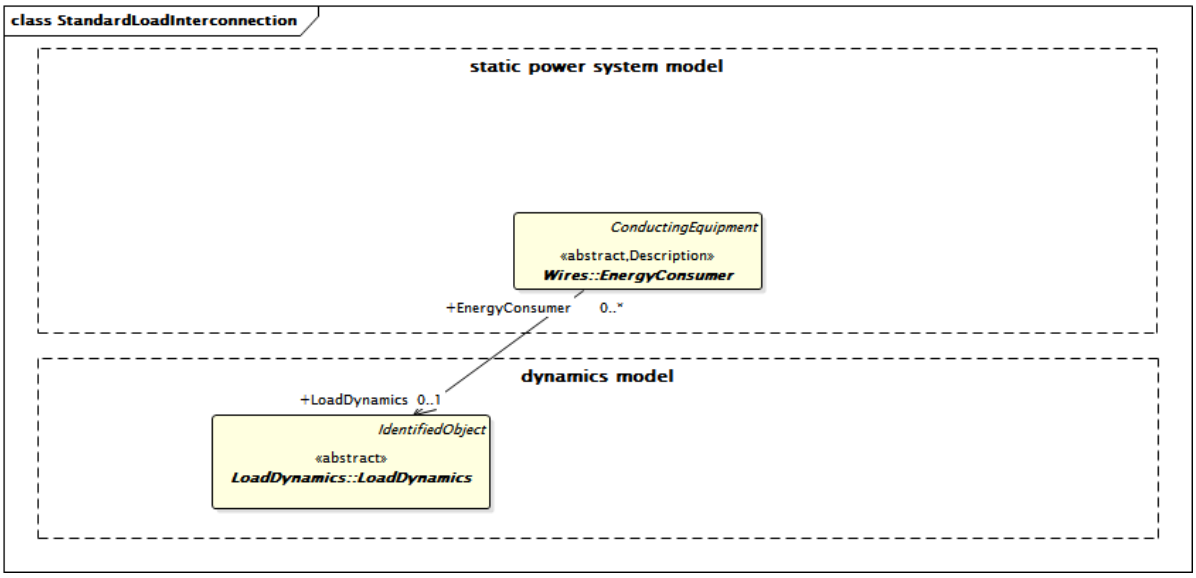


Figure 9-15 - diagram StandardLoadInterconnection

This diagram shows the classes necessary for a load interconnection model. A standard load interconnection model has a single instance of an EnergyConsumerDynamics child class plus a related instance of a LoadDynamics child class.

9.5.2 RemoteInputSignal

Supports connection to a terminal associated with a remote bus from which an input signal of a specific type is coming.

Inheritance path: ->IdentifiedObject

Table 9-16 shows all attributes of RemoteInputSignal.

Table 9-16 - Attributes of StandardInterconnections::RemoteInputSignal

Attribute name	mult.	Attribute type	Description
remoteSignalType	1..1	RemoteSignalKind	Type of input signal.
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-17 shows all association ends of RemoteInputSignal with other classes.

Table 9-17 - Association ends of StandardInterconnections::RemoteInputSignal with other classes

mult.	mult.	Class type name	Description
PFVArControllerType1Dynamics	[0..1]	PFVArControllerType1Dynamics	Power Factor or VAr controller Type I model using this remote input signal. Association Based On : PFVArControllerType1Dynamics:: PFVArControllerType1Dynamics.PFVArControllerType1Dynamics[0..1] ----- StandardInterconnections:: RemoteInputSignal.RemoteInputSignal[0..1]

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

UnderexcitationLimiterDynamics	[0..1]	UnderexcitationLimiterDynamics	Underexcitation limiter model using this remote input signal. Association Based On : UnderexcitationLimiterDynamics:: UnderexcitationLimiterDynamics.UnderexcitationLimiterDynamics[0..1] ----- StandardInterconnections:: RemoteInputSignal.RemoteInputSignal[0..1]
VoltageCompensatorDynamics	[0..1]	VoltageCompensatorDynamics	Voltage compensator model using this remote input signal. Association Based On : VoltageCompensatorDynamics:: VoltageCompensatorDynamics.VoltageCompensatorDynamics[0..1] ----- StandardInterconnections:: RemoteInputSignal.RemoteInputSignal[0..1]
PowerSystemStabilizerDynamics	[0..1]	PowerSystemStabilizerDynamics	Power system stabilizer model using this remote input signal. Association Based On : PowerSystemStabilizerDynamics:: PowerSystemStabilizerDynamics.PowerSystemStabilizerDynamics[0..1] ----- StandardInterconnections:: RemoteInputSignal.RemoteInputSignal[0..*]
DiscontinuousExcitationControlDynamics	[0..1]	DiscontinuousExcitationControlDynamics	Discontinuous excitation control model using this remote input signal. Association Based On : DiscontinuousExcitationControlDynamics:: DiscontinuousExcitationControlDynamics.DiscontinuousExcitationControlDynamics[0..1] ----- StandardInterconnections:: RemoteInputSignal.RemoteInputSignal[0..1]
Terminal	[1..1]	Terminal	Remote terminal with which this input signal is associated. Association Based On : StandardInterconnections:: RemoteInputSignal.RemoteInputSignal[0..*] ----- Core:: Terminal.Terminal[1]

9.5.3 Enumerations

9.5.3.1 RemoteSignalKind

Type of input signal coming from remote bus.

Literal	Description	Code
remoteBusVoltageFrequency	Input is voltage frequency from remote terminal bus.	
remoteBusVoltageFrequencyDeviation	Input is voltage frequency deviation from remote terminal bus.	
remoteBusFrequency	Input is frequency from remote terminal bus.	
remoteBusFrequencyDeviation	Input is frequency deviation from remote terminal bus.	
remoteBusVoltageAmplitude	Input is voltage amplitude from remote terminal bus.	
remoteBusVoltage	Input is voltage from remote terminal bus.	
remoteBranchCurrentAmplitude	Input is branch current amplitude from remote terminal bus.	
remoteBusVoltageAmplitudeDerivative	Input is branch current amplitude derivative from remote terminal bus.	
remotePuBusVoltageDerivative	Input is PU voltage derivative from remote terminal bus.	

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

9.6 StandardModels

9.6.1 General

This section contains standard dynamic model specifications grouped into packages by standard function block (type of equipment being modelled). In the CIM, standard dynamic models are expressed by means of a class named with the standard model name and attributes reflecting each of the parameters necessary to describe the behaviour of an instance of the standard model.

9.6.2 DynamicsFunctionBlock(Abstract)

Abstract parent class for all Dynamics function blocks.

Inheritance path: ->IdentifiedObject

Table 9-18 shows all attributes of DynamicsFunctionBlock.

Table 9-18 - Attributes of StandardModels::DynamicsFunctionBlock

Attribute name	mult.	Attribute type	Description
enabled	1..1	Boolean	Function block used indicator. true = use of function block is enabled false = use of function block is disabled.
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

9.6.3 RotatingMachineDynamics(Abstract)

Abstract parent class for all synchronous and asynchronous machine standard models.

Inheritance path: ->DynamicsFunctionBlock->IdentifiedObject

Table 9-19 shows all attributes of RotatingMachineDynamics.

Table 9-19 - Attributes of StandardModels::RotatingMachineDynamics

Attribute name	mult.	Attribute type	Description
damping	0..1	Simple_Float	Damping torque coefficient (D). A proportionality constant that, when multiplied by the angular velocity of the rotor poles with respect to the magnetic field (frequency), results in the damping torque. This value is often zero when the sources of damping torques (generator damper windings, load damping effects, etc.) are modelled in detail. Typical Value = 0.
inertia	0..1	Seconds	Inertia constant of generator or motor and mechanical load (H) (>0). This is the specification for the stored energy in the rotating mass when operating at rated speed. For a generator, this includes the generator plus all other elements (turbine, exciter) on the same shaft and has units of MW*sec. For a motor, it includes the motor plus its mechanical load. Conventional units are per unit on the generator MVA

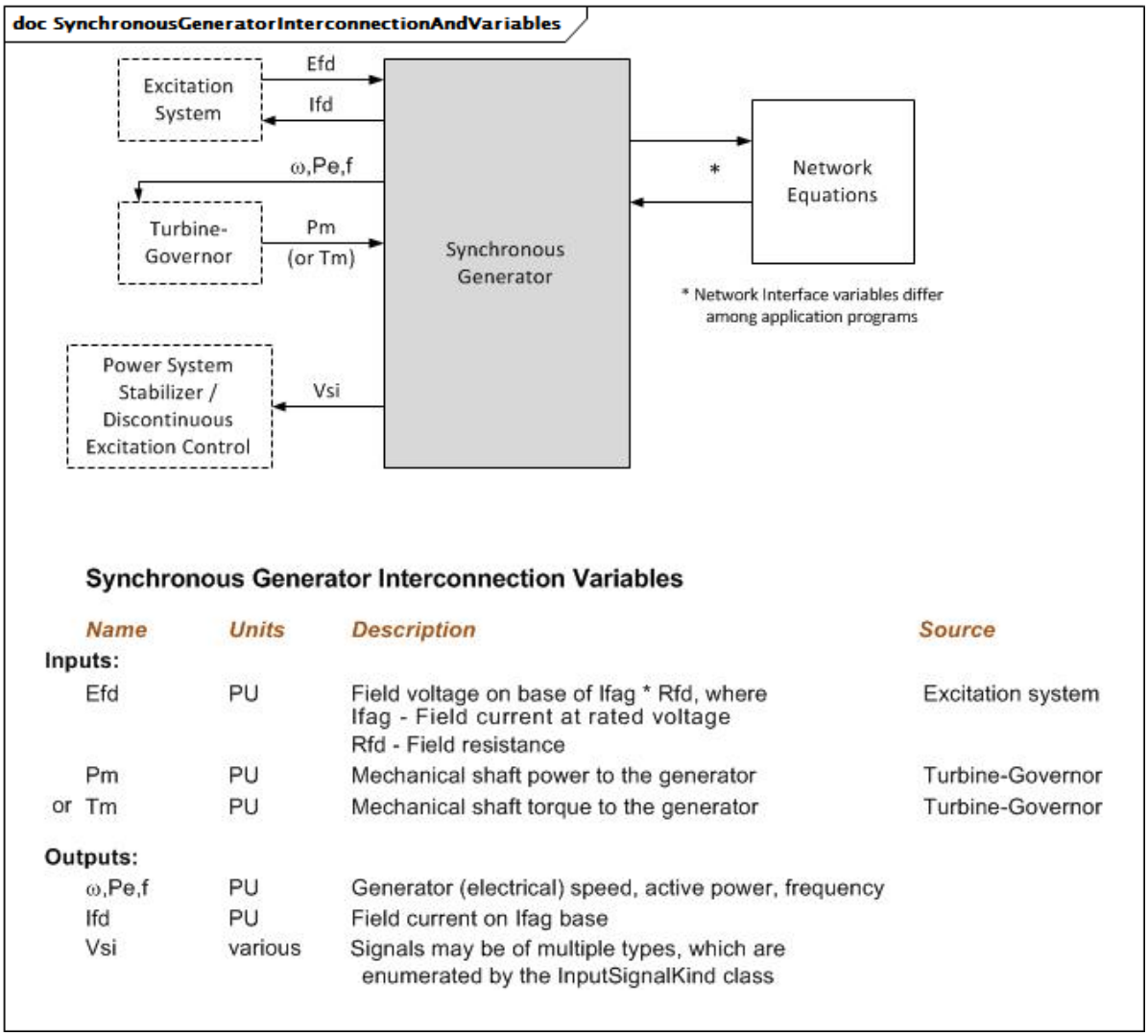
			base, usually expressed as MW*second/MVA or just second. This value is used in the accelerating power reference frame for operator training simulator solutions. Typical Value = 3.
saturationFactor	0..1	Simple_Float	Saturation factor at rated terminal voltage (S1) (> or =0). Not used by simplified model. Defined by defined by S(E1) in the SynchronousMachineSaturationParameters diagram. Typical Value = 0.02.
saturationFactor120	0..1	Simple_Float	Saturation factor at 120% of rated terminal voltage (S12) (> or =S1). Not used by the simplified model, defined by S(E2) in the SynchronousMachineSaturationParameters diagram. Typical Value = 0.12.
statorLeakageReactance	0..1	PU	Stator leakage reactance (Xl) (> or =0). Typical Value = 0.15.
statorResistance	0..1	PU	Stator (armature) resistance (Rs) (> or =0). Typical Value = 0.005.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

9.6.4 SynchronousMachineDynamics

9.6.4.1 General

For conventional power generating units (e.g., thermal, hydro, combustion turbine), a synchronous machine model represents the electrical characteristics of the generator and the mechanical characteristics of the turbine-generator rotational inertia. Large industrial motors or groups of similar motors may be represented by individual motor models which are represented as **generators with negative active power** in the static (power flow) data. The interconnection with the electrical network equations may differ among simulation tools. The tool only needs to know the synchronous machine to establish the correct interconnection. The interconnection with motor's equipment could also differ due to input and output signals required by standard models.

Figure 9-16 shows the diagram.



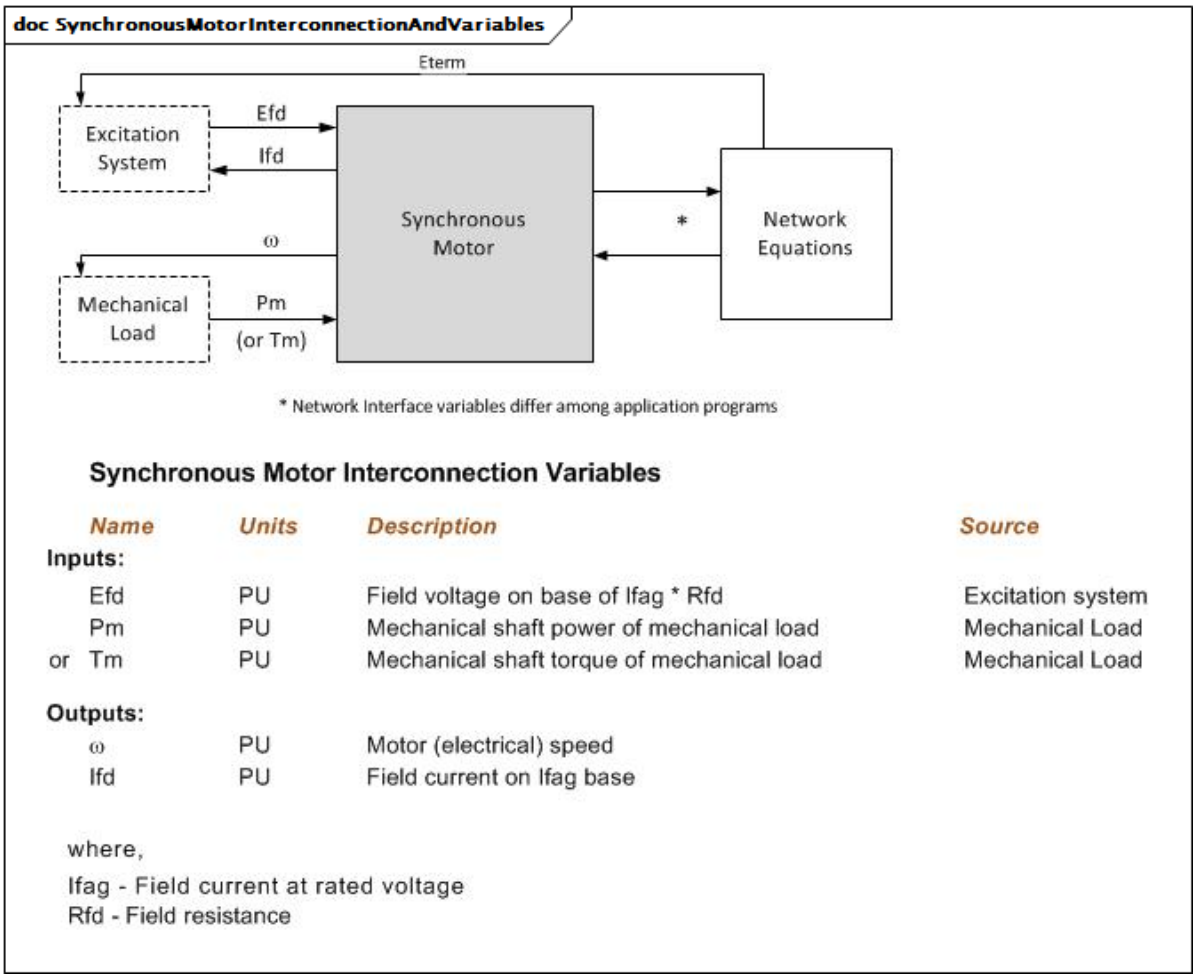
Synchronous Generator Interconnection Variables

Name	Units	Description	Source
Inputs:			
Efd	PU	Field voltage on base of Ifag * Rfd, where Ifag - Field current at rated voltage Rfd - Field resistance	Excitation system
Pm	PU	Mechanical shaft power to the generator	Turbine-Governor
or Tm	PU	Mechanical shaft torque to the generator	Turbine-Governor
Outputs:			
ω,Pe,f	PU	Generator (electrical) speed, active power, frequency	
Ifd	PU	Field current on Ifag base	
Vsi	various	Signals may be of multiple types, which are enumerated by the InputSignalKind class	

Figure 9-16 - diagram SynchronousGeneratorInterconnectionAndVariables

This diagram illustrates the inputs and outputs for a synchronous generator.

Figure 9-17 shows the diagram.



Synchronous Motor Interconnection Variables

	Name	Units	Description	Source
Inputs:				
	Efd	PU	Field voltage on base of Ifag * Rfd	Excitation system
	Pm	PU	Mechanical shaft power of mechanical load	Mechanical Load
or	Tm	PU	Mechanical shaft torque of mechanical load	Mechanical Load
Outputs:				
	ω	PU	Motor (electrical) speed	
	Ifd	PU	Field current on Ifag base	

where,
Ifag - Field current at rated voltage
Rfd - Field resistance

Figure 9-17 - diagram SynchronousMotorInterconnectionAndVariables

This diagram illustrates the inputs and outputs for a synchronous motor.

Figure 9-18 shows the diagram.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

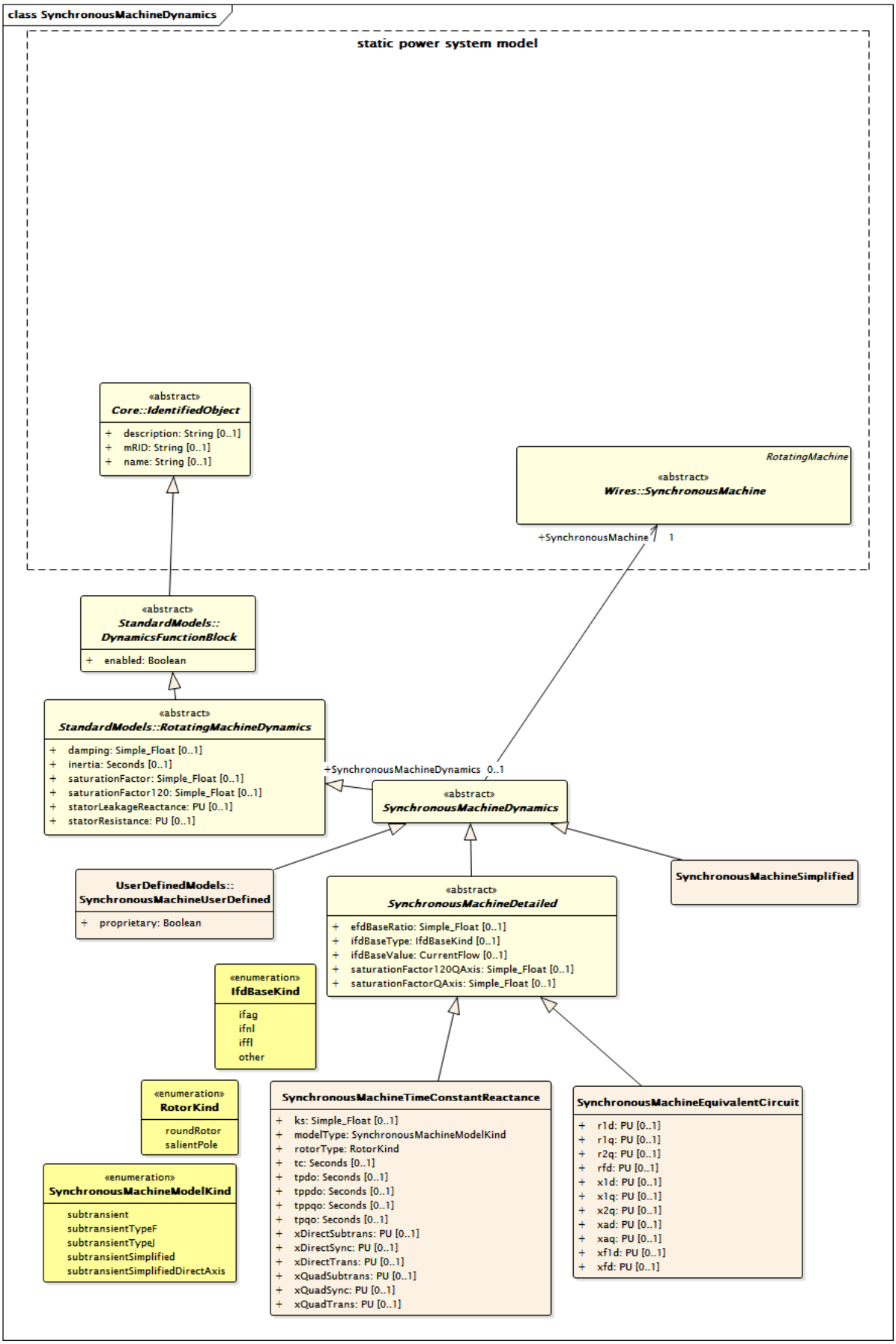

class SynchronousMachineDynamics
static power system model
«abstract»
Core::IdentifiedObject
+ description: String [0..1]
+ mRID: String [0..1]
+ name: String [0..1]
RotatingMachine
«abstract»
Wires::SynchronousMachine
+SynchronousMachine
1
«abstract»
StandardModels::
DynamicsFunctionBlock
+ enabled: Boolean
«abstract»
StandardModels::RotatingMachineDynamics
+ damping: Simple_Float [0..1]
+ inertia: Seconds [0..1]
+ saturationFactor: Simple_Float [0..1]
+ saturationFactor120: Simple_Float [0..1]
+ statorLeakageReactance: PU [0..1]
+ statorResistance: PU [0..1]
+SynchronousMachineDynamics
0..1
«abstract»
SynchronousMachineDynamics
UserDefinedModels::
SynchronousMachineUserDefined
+ proprietary: Boolean
«abstract»
SynchronousMachineDetailed
+ efdBaseRatio: Simple_Float [0..1]
+ ifdBaseType: IfdBaseKind [0..1]
+ ifdBaseValue: CurrentFlow [0..1]
+ saturationFactor120QAxis: Simple_Float [0..1]
+ saturationFactorQAxis: Simple_Float [0..1]
SynchronousMachineSimplified
«enumeration»
IfdBaseKind
ifag
ifnl
iffl
other
«enumeration»
RotorKind
roundRotor
salientPole
«enumeration»
SynchronousMachineModelKind
subtransient
subtransientTypeF
subtransientTypeJ
subtransientSimplified
subtransientSimplifiedDirectAxis
SynchronousMachineTimeConstantReactance
+ ks: Simple_Float [0..1]
+ modelType: SynchronousMachineModelKind
+ rotorType: RotorKind
+ tc: Seconds [0..1]
+ tpdo: Seconds [0..1]
+ tppdo: Seconds [0..1]
+ tppqo: Seconds [0..1]
+ tpqo: Seconds [0..1]
+ xDirectSubtrans: PU [0..1]
+ xDirectSync: PU [0..1]
+ xDirectTrans: PU [0..1]
+ xQuadSubtrans: PU [0..1]
+ xQuadSync: PU [0..1]
+ xQuadTrans: PU [0..1]
SynchronousMachineEquivalentCircuit
+ r1d: PU [0..1]
+ r1q: PU [0..1]
+ r2q: PU [0..1]
+ rfd: PU [0..1]
+ x1d: PU [0..1]
+ x1q: PU [0..1]
+ x2q: PU [0..1]
+ xad: PU [0..1]
+ xaq: PU [0..1]
+ xf1d: PU [0..1]
+ xfd: PU [0..1]

Figure 9-18 - diagram SynchronousMachineDynamics

This diagram shows static power system model classes and dynamics model classes related to the synchronous machine dynamics model. An association between the SynchronousMachineDynamics class and the SynchronousMachine class provides a means for relating the dynamics model to the static model.

Figure 9-19 shows the diagram.

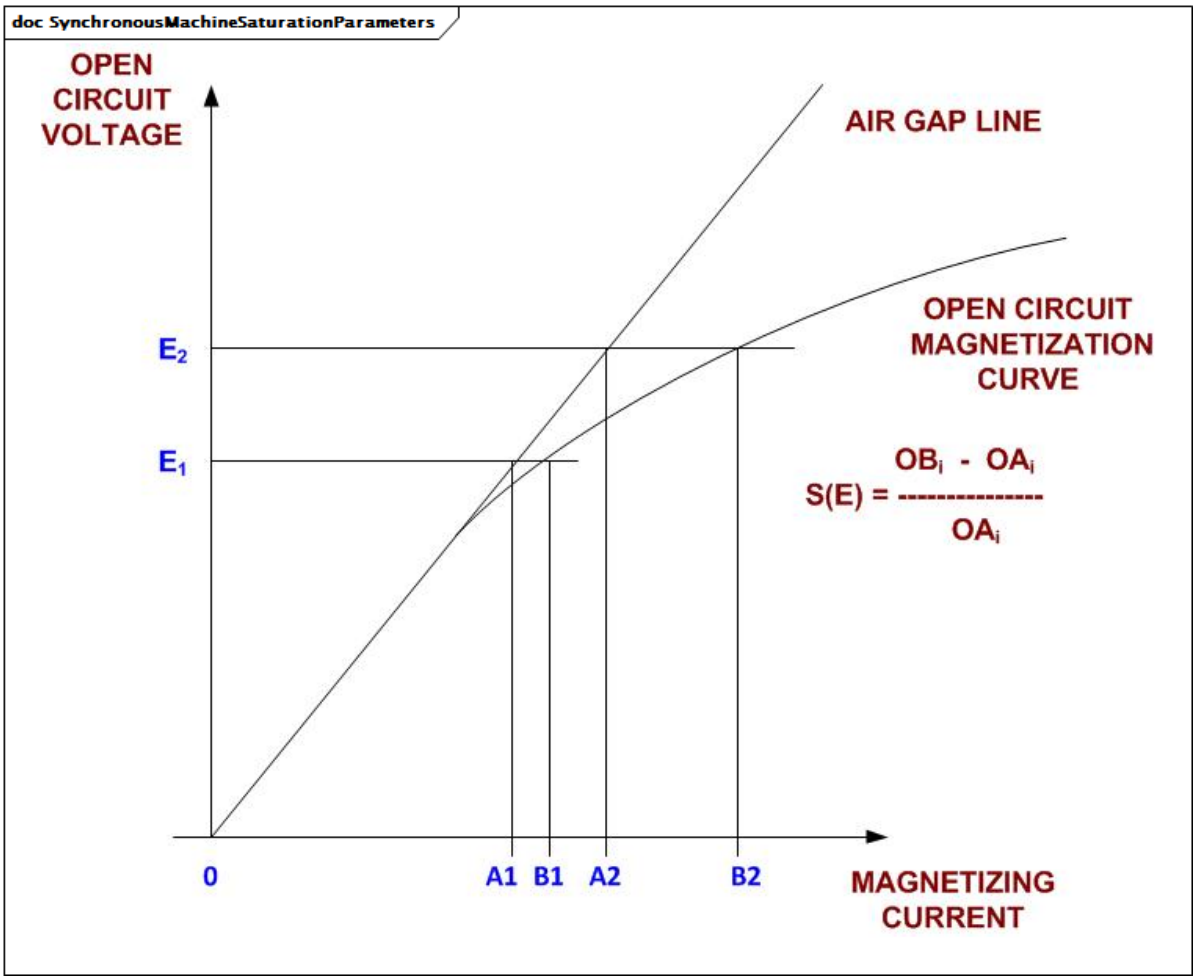


Figure 9-19 - diagram SynchronousMachineSaturationParameters

Note: The quantity OA_i in amperes is normally called **Ifag** -- Field current at rated voltage, open circuit on the air gap (no saturation) line. OB_i is **Ifnl** – Field current at rated voltage with no load on the machine.
Saturation factors for generators are given by RotatingMachineDynamics.saturationFactor and RotatingMachineDynamics.saturationFactor120.
For exciters, the values are given in the attributes for each standard excitation system model.

Figure 9-20 shows the diagram.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

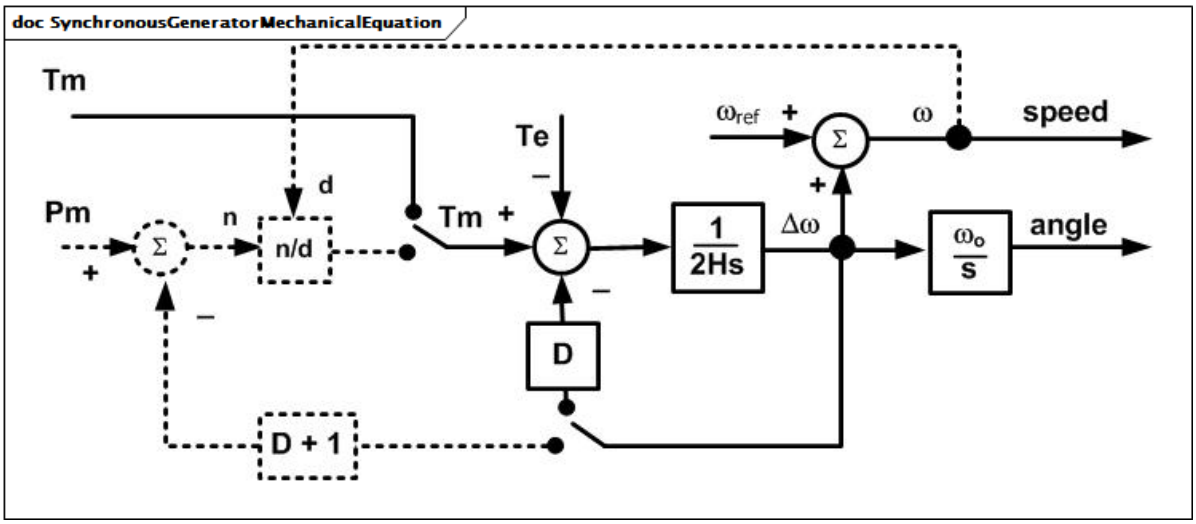


Figure 9-20 - diagram SynchronousGeneratorMechanicalEquation

The mechanical equations for all the variations of the synchronous generator model are the same and can be represented by the SynchronousGeneratorMechanicalEquation block diagram. All variables are per unit on generator MVA base except angle, which is in radians. **omega$_0$** is the system synchronous frequency in radians per second. **omega$_{ref}$** is the per unit system frequency (or isolated area frequency), which may be constant at the initial frequency or vary during a simulation depending on the application program.
The input to the mechanical equations may be either **Pm** or **Tm** depending on the application program. Application programs using **Pm** as the input with the damping feedback subtracted from **Pm** will need to increase **D** by 1 as shown. In that case, the damping feedback to the torque summing point is not used.

Figure 9-21 shows the diagram.

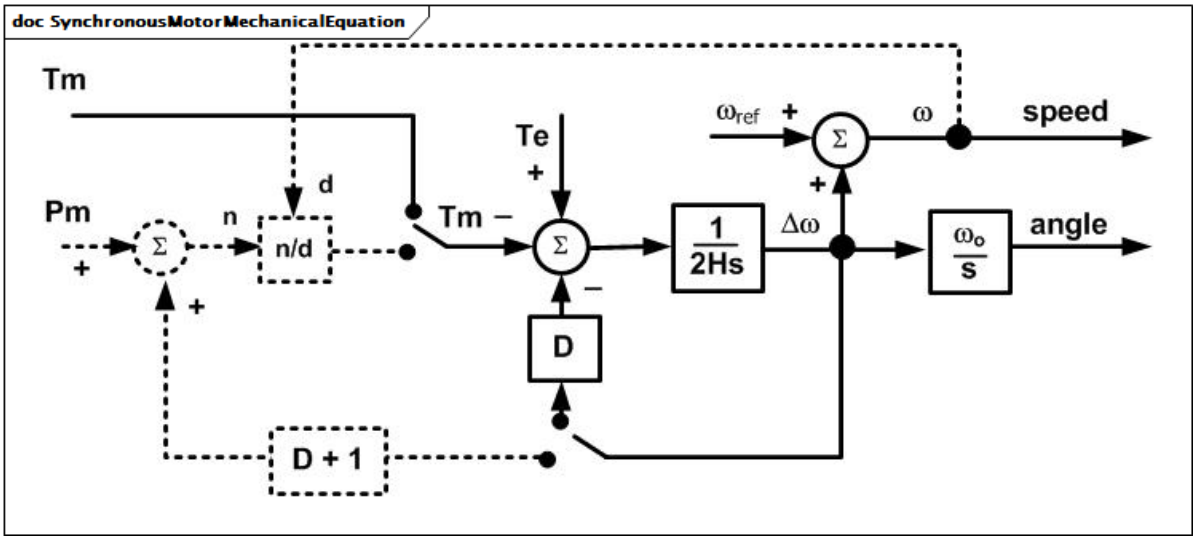


Figure 9-21 - diagram SynchronousMotorMechanicalEquation

The mechanical equations for all variations of the synchronous motor model are the same and can be represented by the SynchronousMotorMechanicalEquation diagram.
All variables are per unit on motor MVA base except angle, which is in radians. **omega$_0$** is the

system synchronous frequency in radians per second. **omega$_{ref}$** is the per unit system frequency (or isolated area frequency), which may be constant at the initial frequency or vary during a simulation depending on the application program.

Figure 9-22 shows the diagram.

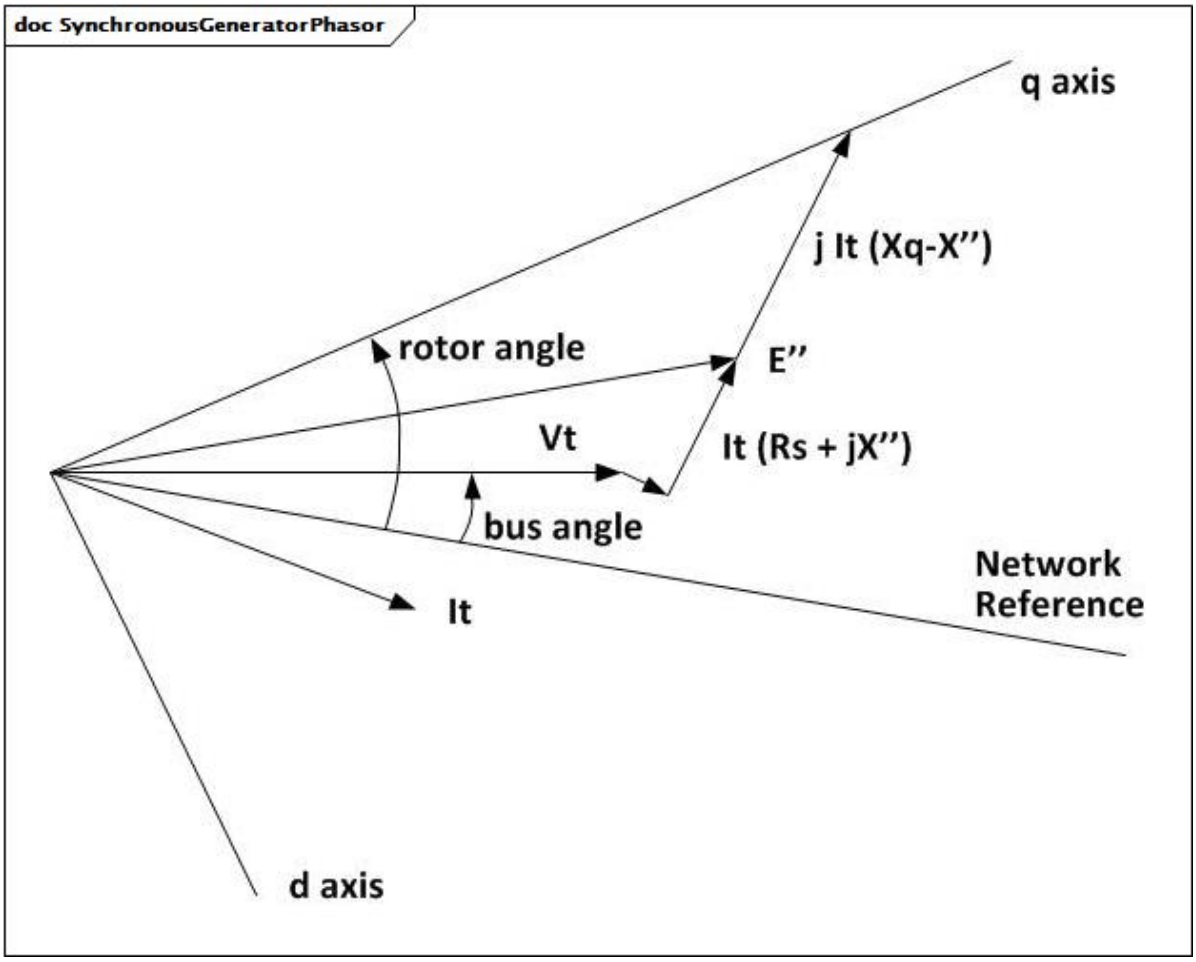


Figure 9-22 - diagram SynchronousGeneratorPhasor

The definition of d and q axis variables is based on the SynchronousGeneratorPhasor diagram (counter-clockwise rotation), for the case of an overexcited generator (generating Q).

Figure 9-23 shows the diagram.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

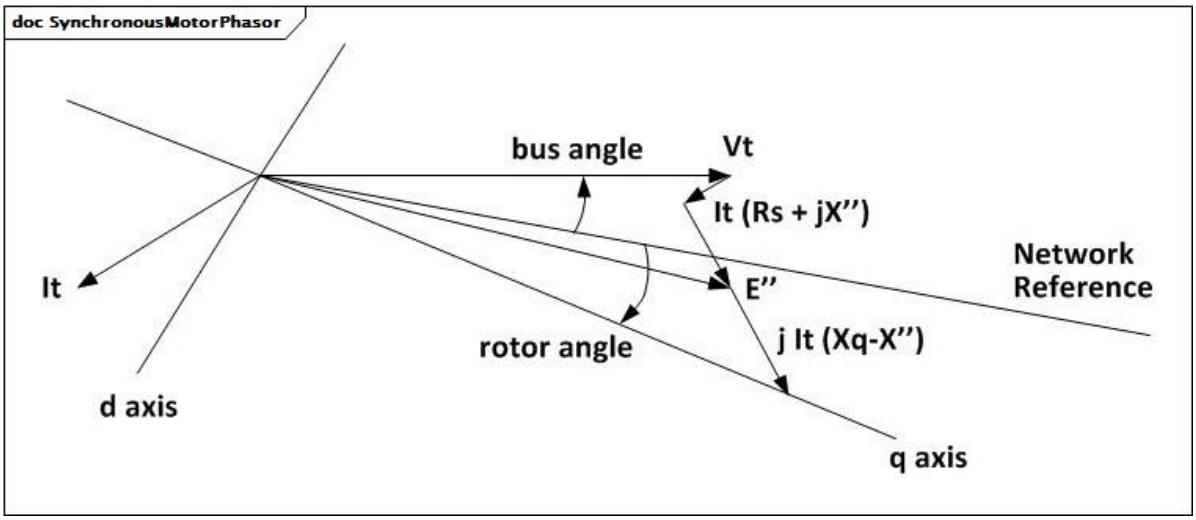


Figure 9-23 - diagram SynchronousMotorPhasor

The definition of d and q axis variables is based on the SynchronousMotorPhasor diagram (counter-clockwise rotation), for the case of a motor consuming active power and overexcited (generating reactive power).

9.6.4.2 SynchronousMachineSimplified

The simplified model represents a synchronous generator as a constant internal voltage behind an impedance (**Rs** + **jXp**) as shown in the Simplified diagram.
Since internal voltage is held constant, there is no **Efd** input and any excitation system model will be ignored. There is also no **Ifd** output.
This model should not be used for representing a real generator except, perhaps, small generators whose response is insignificant.
The parameters used for the Simplified model include:

- RotatingMachineDynamics.damping (D)
- RotatingMachineDynamics.inertia (H)
- RotatingMachineDynamics.statorLeakageReactance (used to exchange jXp for SynchronousMachineSimplified)
- RotatingMachineDynamics.statorResistance (Rs).

Figure 9-24 class diagram

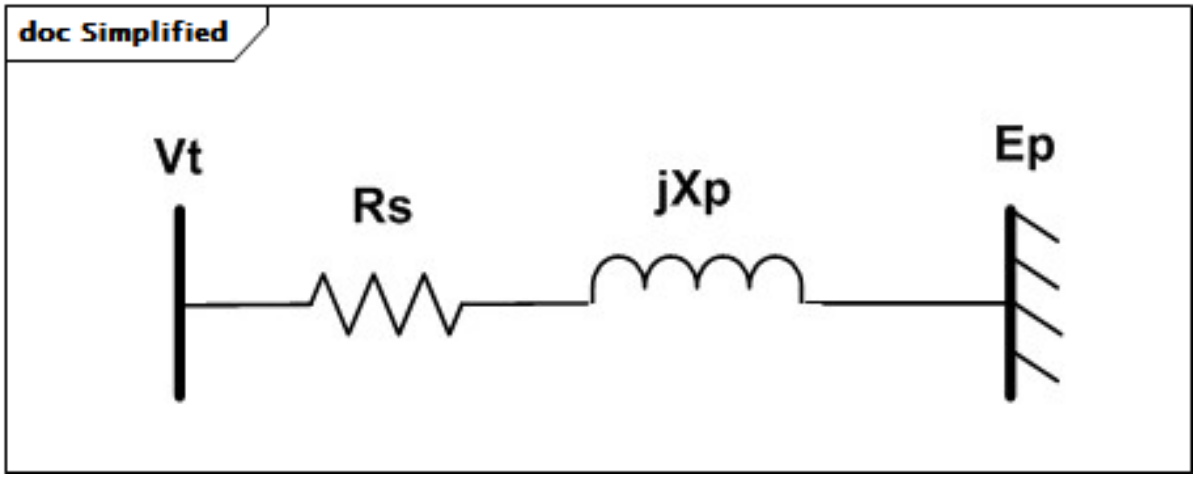


Figure 9-24 - figure Simplified

Parameter Notes:

1. The model is often used for gross equivalents of parts of a system that are not represented in detail. In this case the MVA rating would be the combined rating of all generators in the equivalenced area.
2. **Rs** + **jXp** would be the short circuit equivalent impedance at the location of the equivalent generator on the generator MVA rating base. Inertia constant (**H)** would be typical or average values for the generators in the equivalenced area.
3. The internal reactance (**Xp**) may be labelled in different ways (X', X", X'd, X"d) by different simulation tools.

Inheritance path: ->SynchronousMachineDynamics->RotatingMachineDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-20 shows all attributes of SynchronousMachineSimplified.

Table 9-20 - Attributes of SynchronousMachineDynamics::SynchronousMachineSimplified

Attribute name	mult.	Attribute type	Description
damping	0..1	Simple_Float	see RotatingMachineDynamics
inertia	0..1	Seconds	see RotatingMachineDynamics
saturationFactor	0..1	Simple_Float	see RotatingMachineDynamics
saturationFactor120	0..1	Simple_Float	see RotatingMachineDynamics
statorLeakageReactance	0..1	PU	see RotatingMachineDynamics
statorResistance	0..1	PU	see RotatingMachineDynamics
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-21 shows all association ends of SynchronousMachineDynamics with other classes.

Table 9-21 - Association ends of SynchronousMachineDynamics::SynchronousMachineDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachine	1..1	SynchronousMachine	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..1] ----- Wires:: SynchronousMachine.SynchronousMachine[1]

9.6.4.3 SynchronousMachineDynamics(Abstract)

Synchronous machine whose behaviour is described by reference to a standard model expressed in one of the following forms:

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

- simplified (or classical), where a group of generators or motors is not modelled in detail

- detailed, in equivalent circuit form
- detailed, in time constant reactance form

or by definition of a user-defined model. **Note:** It is a common practice to represent small generators by a negative load rather than by a dynamic generator model when performing dynamics simulations. In this case a SynchronousMachine in the static model is not represented by anything in the dynamics model, instead it is treated as ordinary load. **Parameter Notes:**

1. Synchronous machine parameters such as **Xl, Xd**, **Xp** etc. are actually used as inductances (L) in the models, but are commonly referred to as reactances since, at nominal frequency, the per unit values are the same. However, some references use the symbol L instead of X.

Inheritance path: ->RotatingMachineDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-22 shows all attributes of SynchronousMachineDynamics.

Table 9-22 - Attributes of SynchronousMachineDynamics::SynchronousMachineDynamics

Attribute name	mult.	Attribute type	Description
damping	0..1	Simple_Float	see RotatingMachineDynamics
inertia	0..1	Seconds	see RotatingMachineDynamics
saturationFactor	0..1	Simple_Float	see RotatingMachineDynamics
saturationFactor120	0..1	Simple_Float	see RotatingMachineDynamics
statorLeakageReactance	0..1	PU	see RotatingMachineDynamics
statorResistance	0..1	PU	see RotatingMachineDynamics
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-23 shows all association ends of SynchronousMachineDynamics with other classes.

Table 9-23 - Association ends of SynchronousMachineDynamics::SynchronousMachineDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachine	[1..1]	SynchronousMachine	Synchronous machine to which synchronous machine dynamics model applies. Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..1] ----- Wires:: SynchronousMachine.SynchronousMachine[1]

9.6.4.4 SynchronousMachineDetailed(Abstract)

All synchronous machine detailed types use a subset of the same data parameters and input/output variables.
The several variations differ in the following ways:

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

- The number of equivalent windings that are included
- The way in which saturation is incorporated into the model.
- Whether or not “subtransient saliency” (**X''q** not = **X''d**) is represented.

Note: It is not necessary for each simulation tool to have separate models for each of the model types. The same model can often be used for several types by alternative logic within the model. Also, differences in saturation representation may not result in significant model performance differences so model substitutions are often acceptable.

Inheritance path: ->SynchronousMachineDynamics->RotatingMachineDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-24 shows all attributes of SynchronousMachineDetailed.

Table 9-24 - Attributes of SynchronousMachineDynamics::SynchronousMachineDetailed

Attribute name	mult.	Attribute type	Description
saturationFactorQAxis	0..1	Simple_Float	Q-axis saturation factor at rated terminal voltage (S1q) (>= 0). Typical Value = 0.02.
saturationFactor120QAxis	0..1	Simple_Float	Q-axis saturation factor at 120% of rated terminal voltage (S12q) (>=S1q). Typical Value = 0.12.
efdBaseRatio	0..1	Simple_Float	Ratio of Efd bases of exciter and generator models. Typical Value = 1.
ifdBaseType	0..1	IfdBaseKind	Excitation base system mode. Typical Value = ifag.
ifdBaseValue	0..1	CurrentFlow	Ifd base current if .ifdBaseType = other. Not needed if .ifdBaseType not = other. Unit = A. Typical Value = 0.
damping	0..1	Simple_Float	see RotatingMachineDynamics
inertia	0..1	Seconds	see RotatingMachineDynamics
saturationFactor	0..1	Simple_Float	see RotatingMachineDynamics
saturationFactor120	0..1	Simple_Float	see RotatingMachineDynamics
statorLeakageReactance	0..1	PU	see RotatingMachineDynamics
statorResistance	0..1	PU	see RotatingMachineDynamics
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-25 shows all association ends of SynchronousMachineDynamics with other classes.

Table 9-25 - Association ends of SynchronousMachineDynamics::SynchronousMachineDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachine	1..1	SynchronousMachine	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..1] ----- Wires:: SynchronousMachine.SynchronousMachine[1]

9.6.4.5 SynchronousMachineTimeConstantReactance

Synchronous machine detailed modelling types are defined by the combination of the attributes SynchronousMachineTimeConstantReactance.modelType and SynchronousMachineTimeConstantReactance.rotorType.

Parameter notes:

1. The “p” in the time-related attribute names is a substitution for a “prime” in the usual parameter notation, e.g. tpdo refers to **T'do**.

The parameters used for models expressed in time constant reactance form include:

- RotatingMachine.ratedS (MVAbase)
- RotatingMachineDynamics.damping (D)
- RotatingMachineDynamics.inertia (H)
- RotatingMachineDynamics.saturationFactor (S1)
- RotatingMachineDynamics.saturationFactor120 (S12)
- RotatingMachineDynamics.statorLeakageReactance (Xl)
- RotatingMachineDynamics.statorResistance (Rs)
- SynchronousMachineTimeConstantReactance.ks (Ks)
- SynchronousMachineDetailed.saturationFactorQAxis (S1q)
- SynchronousMachineDetailed.saturationFactor120QAxis (S12q)
- SynchronousMachineDetailed.efdBaseRatio
- SynchronousMachineDetailed.ifdBaseType
- SynchronousMachineDetailed.ifdBaseValue, if present
- .xDirectSync (Xd)
- .xDirectTrans (X'd)
- .xDirectSubtrans (X''d)
- .xQuadSync (Xq)
- .xQuadTrans (X'q)
- .xQuadSubtrans (X''q)
- .tpdo (T'do)
- .tppdo (T''do)
- .tpqo (T'qo)
- .tppqo (T''qo)
- .tc.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Figure 9-25 class diagram

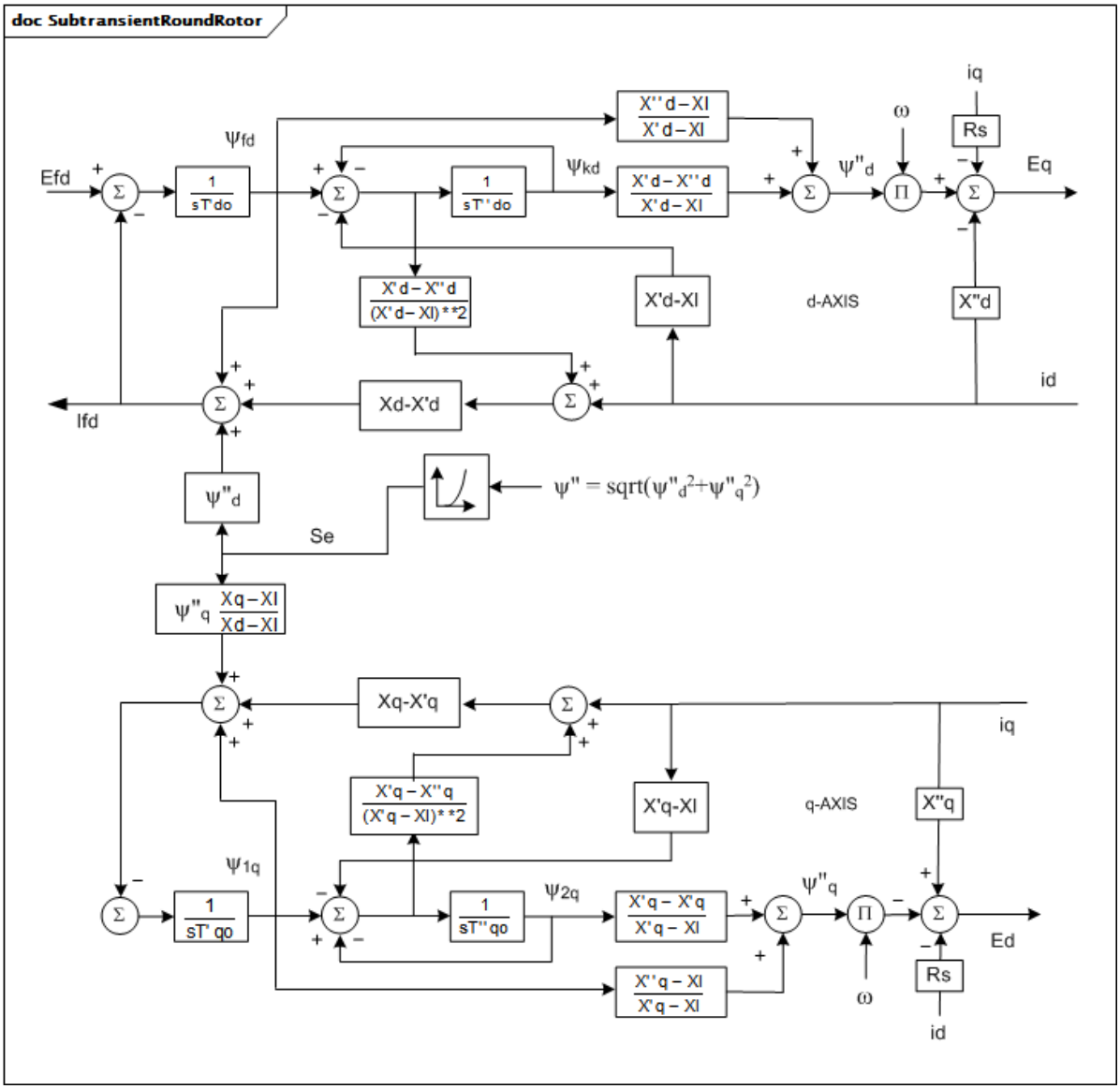


Figure 9-25 - figure SubtransientRoundRotor

The Subtransient Round Rotor Type is modelled using one of the concrete child classes of the SynchronousMachineDetailed class. If the SynchronousMachineTimeConstantReactance class is used, .modelType = subtransient and .rotorType = roundRotor. The complete equivalent circuit is used with two rotor windings in each axis. Ifd represents LadIfd (XadIfd) and is a voltage proportional to field current in the per unit system (IEEE 421.5-2005, Annex B). It is equal to Efd in steady-state.
Parameter Notes:

1. **X''q** may be assumed to be equal to **X''d** (no subtransient saliency).
2. Saturation is modeled in both the d and q axes.
3. For generators, the following parameters may be omitted with defaults shown: **X''q** (=**X''d**), **Ks** (=0), **S1q** (=**S1**), **S12q** (=**S12**), **Tc** (=0).

4. For motors, the following parameter is not used: **X''q**.

Figure 9-26 class diagram

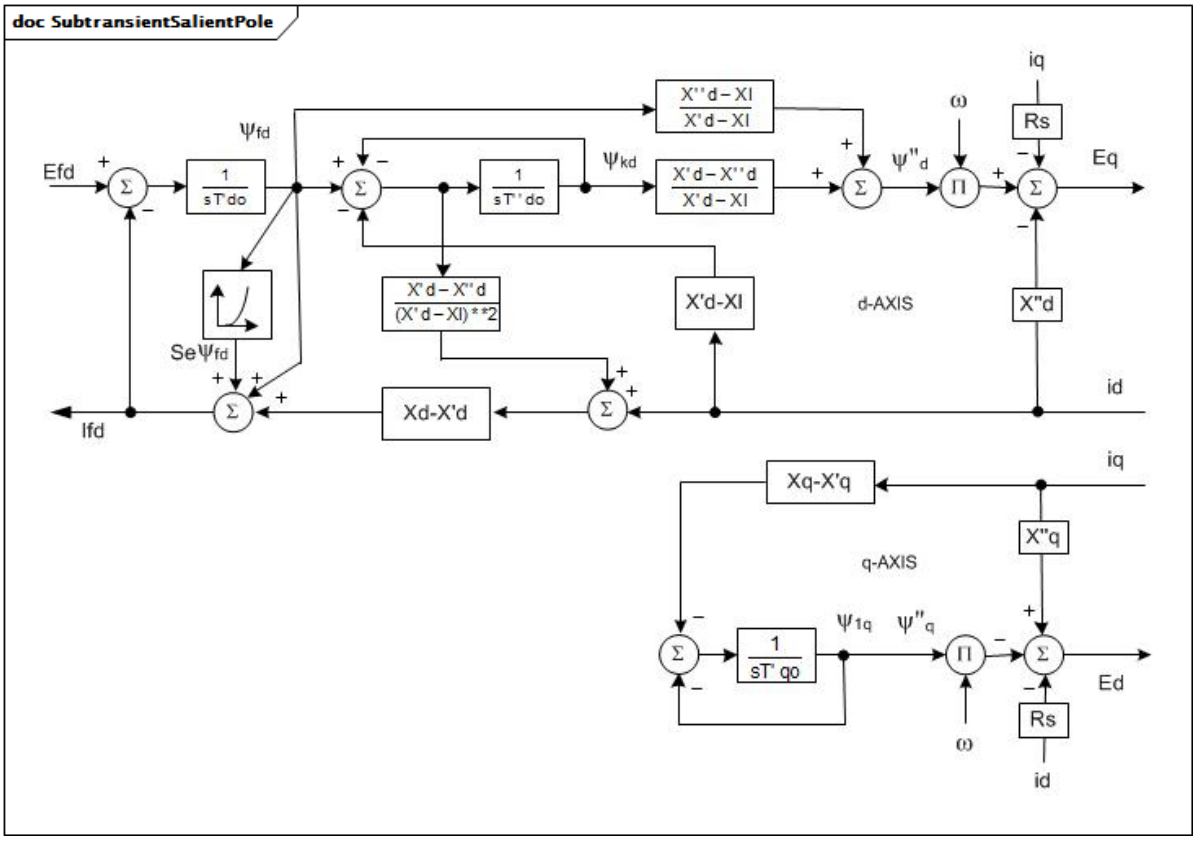


Figure 9-26 - figure SubtransientSalientPole

The Subtransient Salient Pole Type is modelled using one of the concrete child classes of the SynchronousMachineDetailed class. If the SynchronousMachineTimeConstantReactance class is used, .modelType = subtransient and .rotorKind = salientPole.
The d-axis equivalent circuit is the same as for the Subtransient Round Rotor Type. The q-axis has only one equivalent rotor winding, which may be labelled as transient (**X'q**) or subtransient (**X''q**); **X'q** is used for the CIM description.
Ifd represents LadIfd (XadIfd) and is a voltage proportional to field current in the per unit system (IEEE 421.5-2005, Annex B). It is equal to Efd in steady-state.
Parameter Notes:

1. **X''q** (=**X'q**) is assumed to be equal to **X''d** (no subtransient saliency).
2. Saturation is modelled in the d axis only.
3. For generators, the following input parameters may be omitted with default shown: **X'q** (=**X''d**), **X''q** (=**X''d**), **T''qo**, **Ks** (=0), **S1q** (=0), **S12q** (=0), **Tc** (=0).
4. For motors, the following input parameters are not used: **X'q**, **X''q**, **T''qo**.

Figure 9-27 class diagram

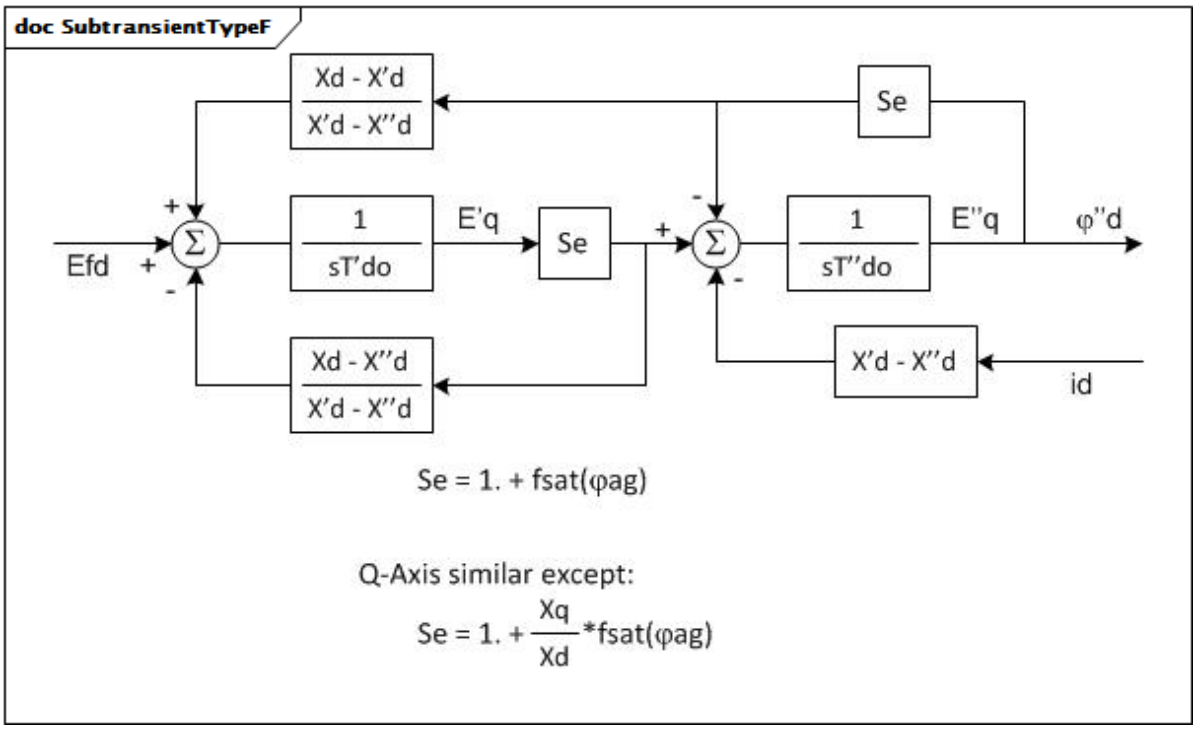


Figure 9-27 - figure SubtransientTypeF

The Subtransient Type F Type is modelled using one of the concrete child classes of the SynchronousMachineDetailed class. If the SynchronousMachineTimeConstantReactance class is used, .modelType = subtransientTypeF.
This model is the WECC type F model with uniform inductance rotor modelling; shaft speed effects are neglected. It has a similar level of detail to the Subtransient Round Rotor Type but permits subtransient saliency (**X''q** not = **X''d**) and models saturation differently. The Subtransient Round Rotor Type can usually be substituted without significant loss of accuracy. Saturation is modelled in both the d and q axes.

Figure 9-28 class diagram

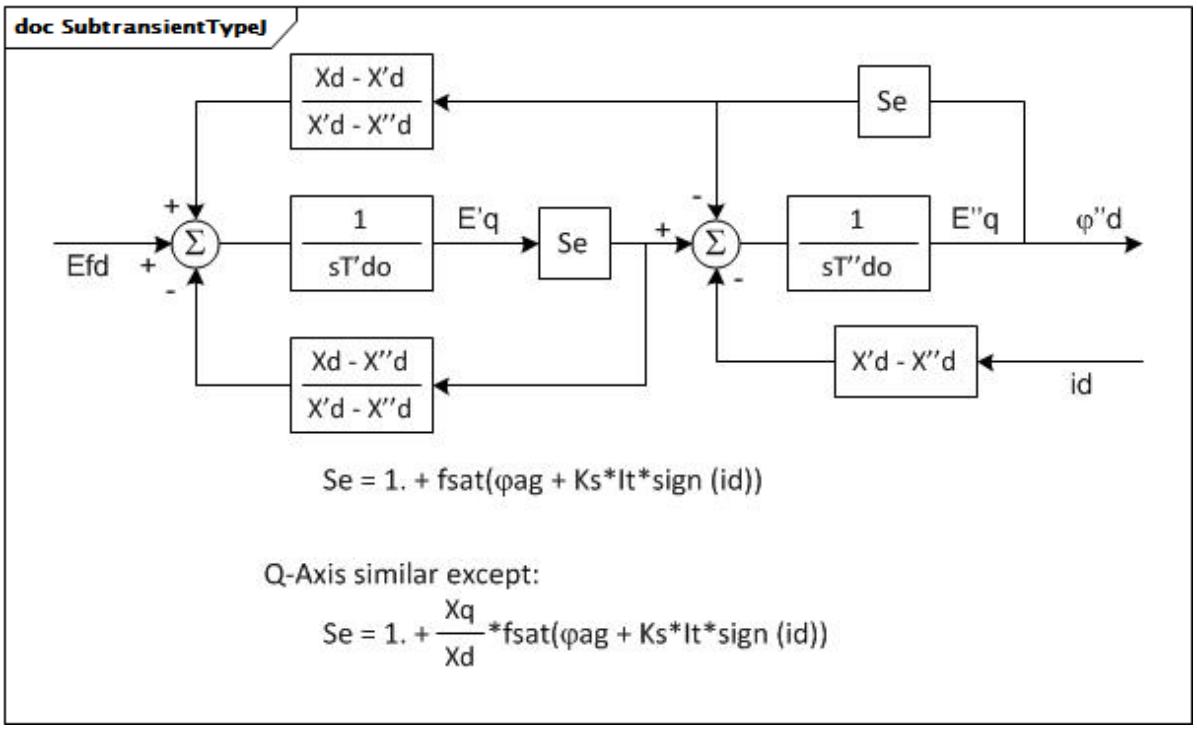
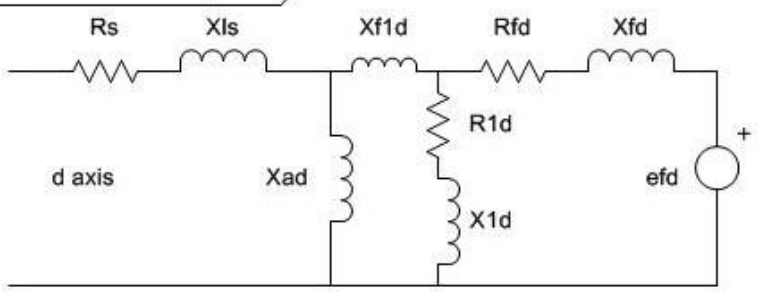


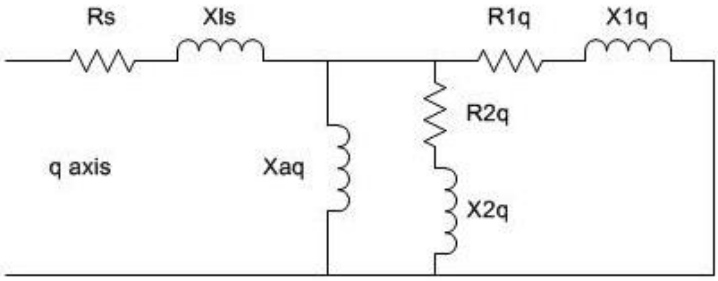
Figure 9-28 - figure SubtransientTypeJ

The Subtransient Type J Type is modelled using one of the concrete child classes of the SynchronousMachineDetailed class. If the SynchronousMachineTimeConstantReactance class is used, .modelType = subtransientTypeJ.
This model is the WECC type J model with uniform inductance ratios rotor modelling with modified saturation model; shaft speed effects are neglected. This model is the same as Subtransient Type F but includes the effect of generator loading on saturation. With the usage of the saturation constant (**Ks**) parameter, it allows reproduction of the measured "V curves" over the whole range of reactive power, particularly on some hydro (salient pole) machines. Saturation is modelled in both the d and q axes.

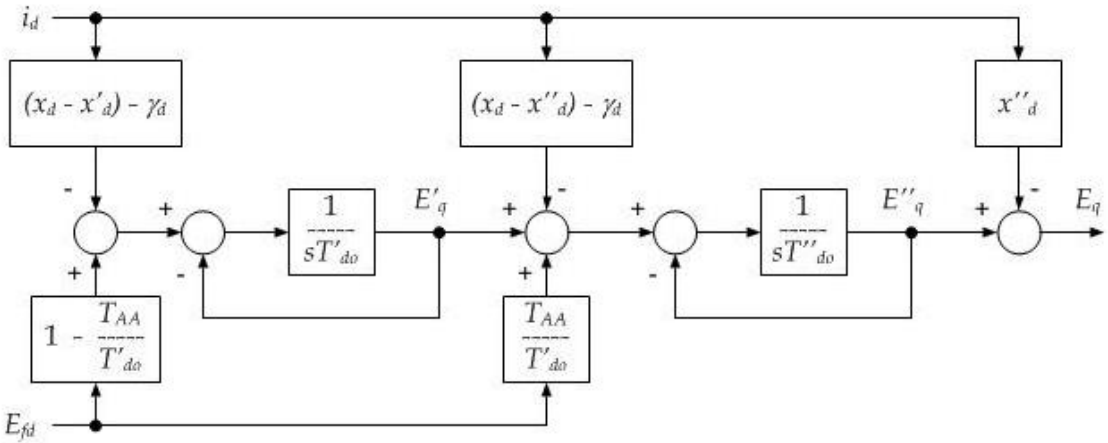
Figure 9-29 class diagram

doc SubtransientRoundRotorSimplified





Equivalent Circuit

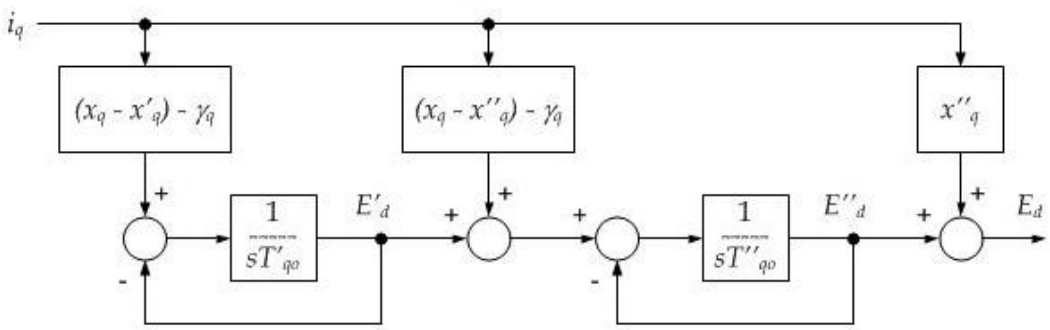


$$\gamma_d = \frac{T''_d}{T'_{d0}} \cdot (x_d - x'_d)$$

$$T''_d = \frac{1}{r_{1d}} \cdot \left((x_{1d} + x_{ad}) - \frac{x_{ad}^2 \cdot (x_{1d} + x_{fd})}{(x_{fd} + x_{ad}) \cdot (x_{1d} + x_{ad}) - x_{ad}^2} \right)$$

$$T_{AA} = \frac{x_{1d}}{r_{1d}}$$

d-axis Block Diagram



$$\gamma_q = \frac{T''_q}{T'_{q0}} \cdot (x_q - x'_q)$$

Figure 9-29 - figure SubtransientRoundRotorSimplified

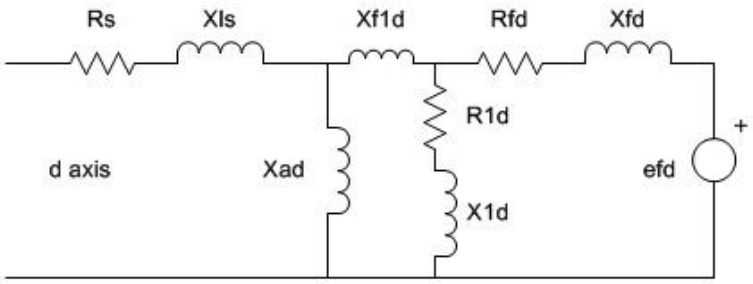
The Subtransient Round Rotor Simplified Type is modelled using one of the concrete child classes of the SynchronousMachineDetailed class. If the SynchronousMachineTimeConstantReactance class is used, .modelType = subtransientSimplified and .rotorKind = roundRotor. It is a simplified version of Subtransient Round Rotor Type, magnetic coupling between the direct and quadrature axes is ignored.
Parameter Notes:

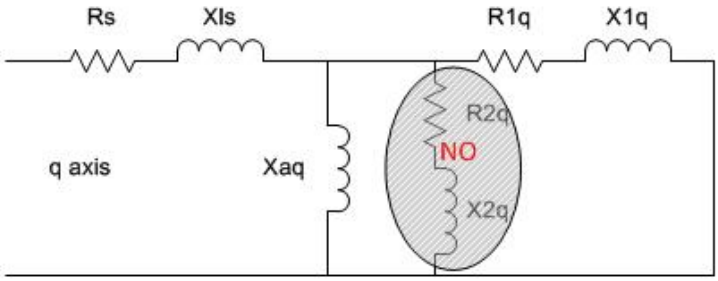
1. The effects of speed variation are neglected.
2. No armature resistance.
3. No saturation effects.
4. No subtransient saliency.
5. Direct Axis with field circuit and one equivalent damper circuit, **X'd** not = **X''d**.
6. Quadrature Axis with two equivalent damper circuit, **X''q** not = **X'q**.
7. For generators and motors the following input parameters are needed: **Xd**, **X'd**, **X''d**, **T'do**, **T''do** and **Xq**, **X'q**, **X''q**, **T'qo**, **T''qo**.

Figure 9-30 class diagram

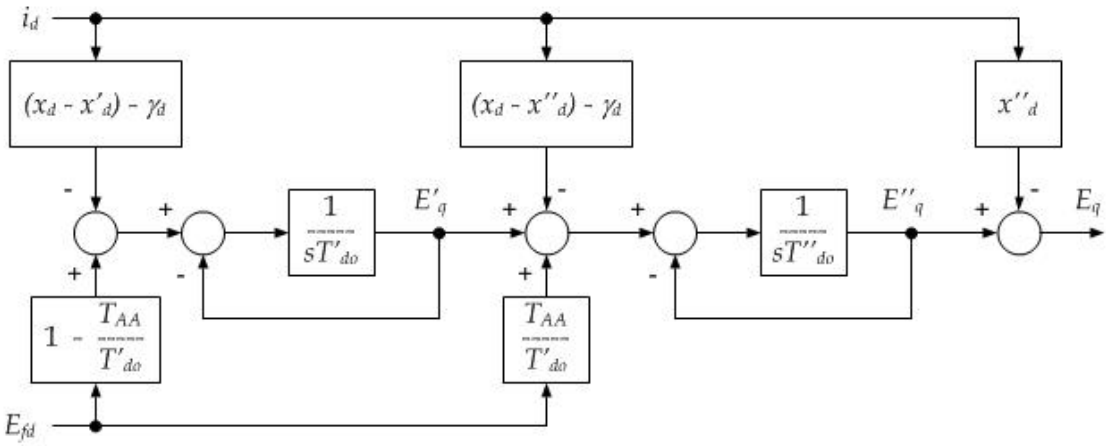
ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

doc SubtransientSalientPoleSimplified





Equivalent Circuit

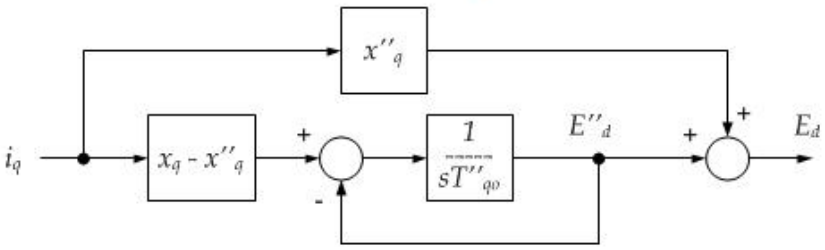


$$\gamma_d = \frac{T''_d}{T'_{d0}} \cdot (x_d - x'_d)$$

$$T''_d = \frac{1}{r_{1d}} \cdot \left((x_{1d} + x_{ad}) - \frac{x_{ad}^2 \cdot (x_{1d} + x_{fd})}{(x_{fd} + x_{ad}) \cdot (x_{1d} + x_{ad}) - x_{ad}^2} \right)$$

$$T_{AA} = \frac{x_{1d}}{r_{1d}}$$

d-axis Block Diagram



q-axis Block Diagram

Figure 9-30 - figure SubtransientSalientPoleSimplified

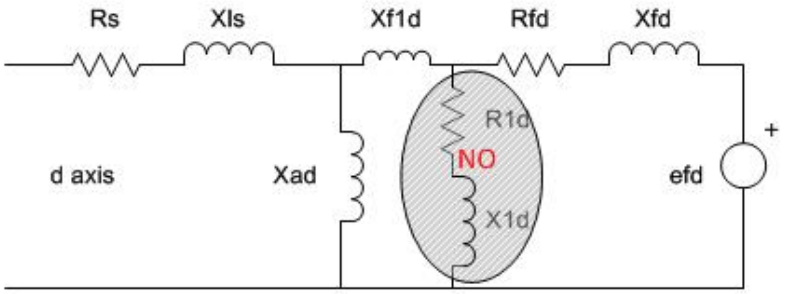
The Subtransient Salient Pole Simplified Type is modelled using one of the concrete child classes of the SynchronousMachineDetailed class. If the SynchronousMachineTimeConstantReactance class is used, .modelType = subtransientSimplified and .rotorKind = salientPole. It is a simplified version of Subtransient Salient Pole Type, magnetic coupling between the direct and quadrature axes is ignored.
Parameter Notes:

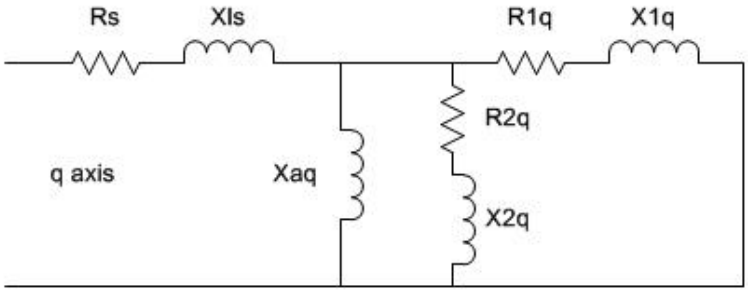
1. The effects of speed variation are neglected.
2. No armature resistance.
3. No saturation effects.
4. No subtransient saliency.
5. Direct Axis with field circuit and one equivalent damper circuit, **X'd** not = **X''d**.
6. Quadrature Axis with only one equivalent damper circuit, which is labeled as subtransient (**X''q**), **X''q** =**X'q**.
7. For generators and motors the following input parameters are needed: **Xd**, **X'd**, **X''d**, **T'do**, **T''do** and **Xq**, **X''q**, **T''qo**.

Figure 9-31 class diagram

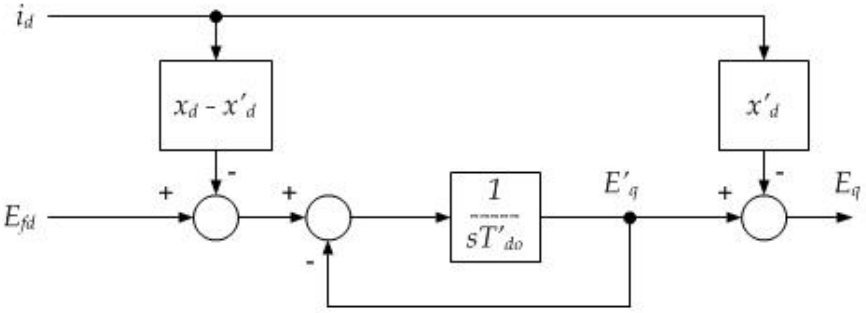
ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

doc SubtransientRoundRotorSimplifiedDirectAxis

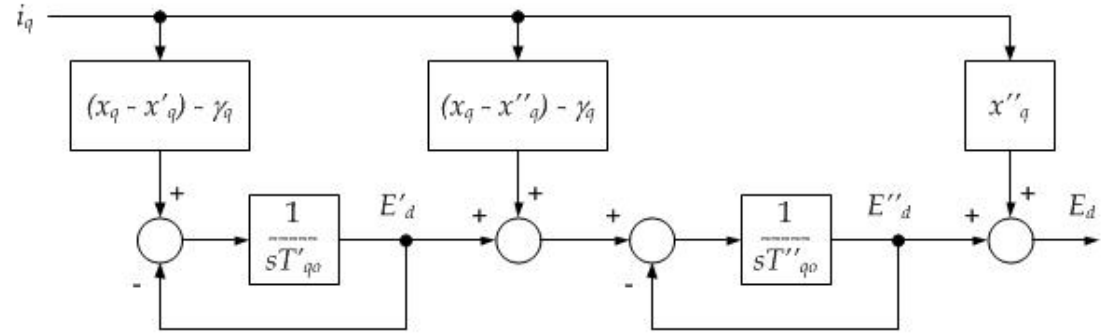




Equivalent Circuit



d-axis Block Diagram



$$\gamma_q = \frac{T''_q}{T'_{q0}} \cdot (x_q - x'_q)$$

$$T''_q = \frac{(x_{1q} + x_{aq})}{r_{1q}} \cdot \left(1 - \frac{x^2_{aq}}{(x_{ls} + x_{aq}) \cdot (x_{1q} + x_{aq})}\right)$$

q-axis Block Diagram

Figure 9-31 - figure SubtransientRoundRotorSimplifiedDirectAxis

The Subtransient Round Rotor Simplified Direct Axis Type is modelled using one of the concrete child classes of the SynchronousMachineDetailed class. If the SynchronousMachineTimeConstantReactance class is used, .modelType = subtransientSimplifiedDirectAxis and .rotorKind = roundRotor.
It is a simplified version of a Subtransient Round Rotor Type with no damper circuit on d-axis. This simplification contributes to a reduction in computational effort by reducing the order of the differential equation system. The q-axis has two equivalent rotor windings. Magnetic coupling between the direct and quadrature axes is ignored.
Parameter Notes:

1. The effects of speed variation are neglected.
2. No armature resistance.
3. No saturation effects.
4. No subtransient saliency.
5. Direct Axis with field circuit only, no equivalent damper circuit. Approximation: **X'd** = **X''d**.
6. Quadrature Axis with two equivalent damper circuit, **X''q** not = **X'q**.
7. For generators and motors the following input parameters are needed: **Xd**, **X'd**, **T'do** and **Xq**, **X'q**, **X''q**, **T'qo**, **T''qo**.

Figure 9-32 class diagram

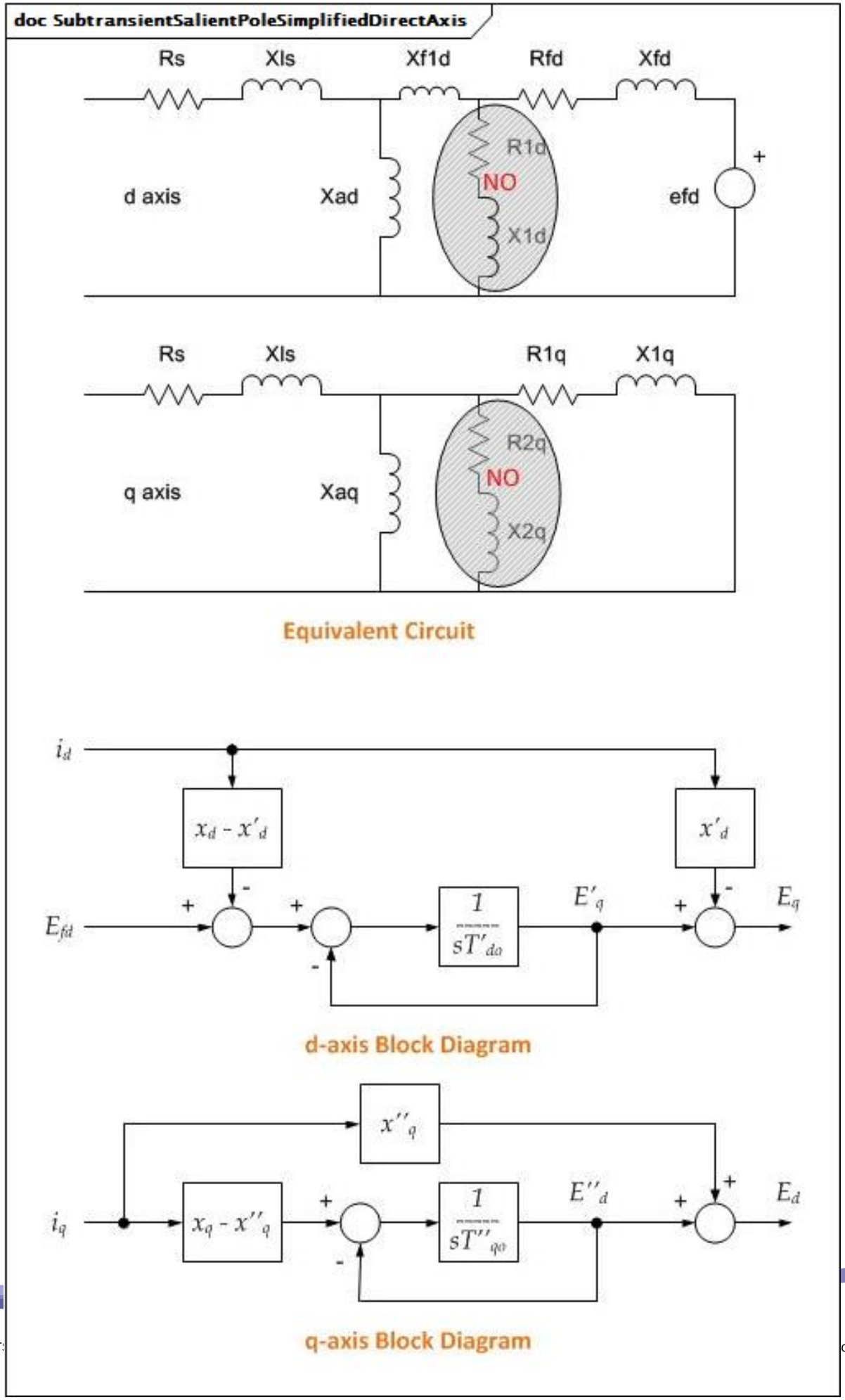

doc SubtransientSalientPoleSimplifiedDirectAxis
Rs
Xls
Xf1d
Rfd
Xfd
R1d
NO
X1d
d axis
Xad
efd
+
Rs
Xls
R1q
X1q
R2q
NO
X2q
q axis
Xaq
Equivalent Circuit
i_d
$x_d - x'_d$
x'_d
E_{fd}
+
-
+
-
$\frac{1}{sT'_{do}}$
E'_q
+
-
E_q
d-axis Block Diagram
x''_q
i_q
$x_q - x''_q$
+
-
$\frac{1}{sT''_{qo}}$
E''_d
+
+
E_d
q-axis Block Diagram

Figure 9-32 - figure SubtransientSalientPoleSimplifiedDirectAxis

The Subtransient Salient Pole Simplified Direct Axis Type is modelled using one of the concrete child classes of the SynchronousMachineDetailed class. If the SynchronousMachineTimeConstantReactance class is used, .modelType = subtransientSimplifiedDirectAxis and .rotorKind = salientPole. It is a simplified version of a Subtransient Salient Pole Type with no damper circuit on d-axis. This simplification contributes to a reduction in computational effort by reducing the order of the differential equation system. The q-axis has only one equivalent rotor winding. Magnetic coupling between the direct and quadrature axes is ignored.
Parameter Notes:

1. The effects of speed variation are neglected.
2. No armature resistance.
3. No saturation effects.
4. No subtransient saliency.
5. Direct Axis with field circuit only, no equivalent damper circuit. Approximation: **X'd** = **X''d**.
6. Quadrature Axis with only one equivalent damper circuit, which is labelled as subtransient (**X''q**), **X''q** =**X'q**.
7. For generators and motors the following input parameters are needed: **Xd**, **X'd**, **T'do** and **Xq**, **X''q**, **T''qo**.

Inheritance path: ->SynchronousMachineDetailed->SynchronousMachineDynamics->RotatingMachineDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-26 shows all attributes of SynchronousMachineTimeConstantReactance.

Table 9-26 - Attributes of SynchronousMachineDynamics::SynchronousMachineTimeConstantReactance

Attribute name	mult.	Attribute type	Description
rotorType	1..1	RotorKind	Type of rotor on physical machine.
modelType	1..1	SynchronousMachineModelKind	Type of synchronous machine model used in Dynamic simulation applications.
ks	0..1	Simple_Float	Saturation loading correction factor (Ks) (>= 0). Used only by Type J model. Typical Value = 0.
xDirectSync	0..1	PU	Direct-axis synchronous reactance (Xd) (>= X'd). The quotient of a sustained value of that AC component of armature voltage that is produced by the total direct-axis flux due to direct-axis armature current and the value of the AC component of this current, the machine running at rated speed. Typical Value = 1.8.
xDirectTrans	0..1	PU	Direct-axis transient reactance (unsaturated) (X'd) (> =X''d). Typical Value =

			0.5.
xDirectSubtrans	0..1	PU	Direct-axis subtransient reactance (unsaturated) (X''d) (> Xl). Typical Value = 0.2.
xQuadSync	0..1	PU	Quadrature-axis synchronous reactance (Xq) (> =X'q). The ratio of the component of reactive armature voltage, due to the quadrature-axis component of armature current, to this component of current, under steady state conditions and at rated frequency. Typical Value = 1.6.
xQuadTrans	0..1	PU	Quadrature-axis transient reactance (X'q) (> =X''q). Typical Value = 0.3.
xQuadSubtrans	0..1	PU	Quadrature-axis subtransient reactance (X''q) (> Xl). Typical Value = 0.2.
tpdo	0..1	Seconds	Direct-axis transient rotor time constant (T'do) (> T''do). Typical Value = 5.
tppdo	0..1	Seconds	Direct-axis subtransient rotor time constant (T''do) (> 0). Typical Value = 0.03.
tpqo	0..1	Seconds	Quadrature-axis transient rotor time constant (T'qo) (> T''qo). Typical Value = 0.5.
tppqo	0..1	Seconds	Quadrature-axis subtransient rotor time constant (T''qo) (> 0). Typical Value = 0.03.
tc	0..1	Seconds	Damping time constant for “Canay” reactance. Typical Value = 0.
saturationFactorQAxis	0..1	Simple_Float	see SynchronousMachineDetailed
saturationFactor120QAxis	0..1	Simple_Float	see SynchronousMachineDetailed
efdBaseRatio	0..1	Simple_Float	see SynchronousMachineDetailed
ifdBaseType	0..1	IfdBaseKind	see SynchronousMachineDetailed
ifdBaseValue	0..1	CurrentFlow	see SynchronousMachineDetailed
damping	0..1	Simple_Float	see RotatingMachineDynamics
inertia	0..1	Seconds	see RotatingMachineDynamics
saturationFactor	0..1	Simple_Float	see RotatingMachineDynamics
saturationFactor120	0..1	Simple_Float	see

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			RotatingMachineDynamics
statorLeakageReactance	0..1	PU	see RotatingMachineDynamics
statorResistance	0..1	PU	see RotatingMachineDynamics
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-27 shows all association ends of SynchronousMachineDynamics with other classes.

Table 9-27 - Association ends of SynchronousMachineDynamics::SynchronousMachineDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachine	1..1	SynchronousMachine	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..1] ----- Wires:: SynchronousMachine.SynchronousMachine[1]

9.6.4.6 SynchronousMachineEquivalentCircuit

The electrical equations for all variations of the synchronous models are based on the SynchronousEquivalentCircuit diagram for the direct and quadrature axes.
Equations for conversion between Equivalent Circuit and Time Constant Reactance forms:
Xd = Xad + Xl
X'd = Xl + Xad * Xfd / (Xad + Xfd)
X"d = Xl + Xad * Xfd * X1d / (Xad * Xfd + Xad * X1d + Xfd * X1d)
Xq = Xaq + Xl
X'q = Xl + Xaq * X1q / (Xaq+ X1q)
X"q = Xl + Xaq * X1q* X2q / (Xaq * X1q + Xaq * X2q + X1q * X2q)
T'do = (Xad + Xfd) / (omega$_0$ * Rfd)
T"do = (Xad * Xfd + Xad * X1d + Xfd * X1d) / (omega$_0$ * R1d * (Xad + Xfd)
T'qo = (Xaq + X1q) / (omega$_0$ * R1q)
T"qo = (Xaq * X1q + Xaq * X2q + X1q * X2q)/ (omega$_0$ * R2q * (Xaq + X1q)
Same equations using CIM attributes from SynchronousMachineTimeConstantReactance class on left of = sign and SynchronousMachineEquivalentCircuit class on right (except as noted):
xDirectSync = xad + RotatingMachineDynamics.statorLeakageReactance
xDirectTrans = RotatingMachineDynamics.statorLeakageReactance + xad * xfd / (xad + xfd)
xDirectSubtrans = RotatingMachineDynamics.statorLeakageReactance + xad * xfd * x1d / (xad * xfd + xad * x1d + xfd * x1d)
xQuadSync = xaq + RotatingMachineDynamics.statorLeakageReactance
xQuadTrans = RotatingMachineDynamics.statorLeakageReactance + xaq * x1q / (xaq+ x1q)
xQuadSubtrans = RotatingMachineDynamics.statorLeakageReactance + xaq * x1q* x2q / (xaq * x1q + xaq * x2q + x1q * x2q)
tpdo = (xad + xfd) / (2*pi*nominal frequency * rfd)
tppdo = (xad * xfd + xad * x1d + xfd * x1d) / (2*pi*nominal frequency * r1d * (xad + xfd)
tpqo = (xaq + x1q) / (2*pi*nominal frequency * r1q)
tppqo = (xaq * x1q + xaq * x2q + x1q * x2q)/ (2*pi*nominal frequency * r2q * (xaq + x1q).
Are only valid for a simplified model where "Canay" reactance is zero.

Figure 9-33 class diagram

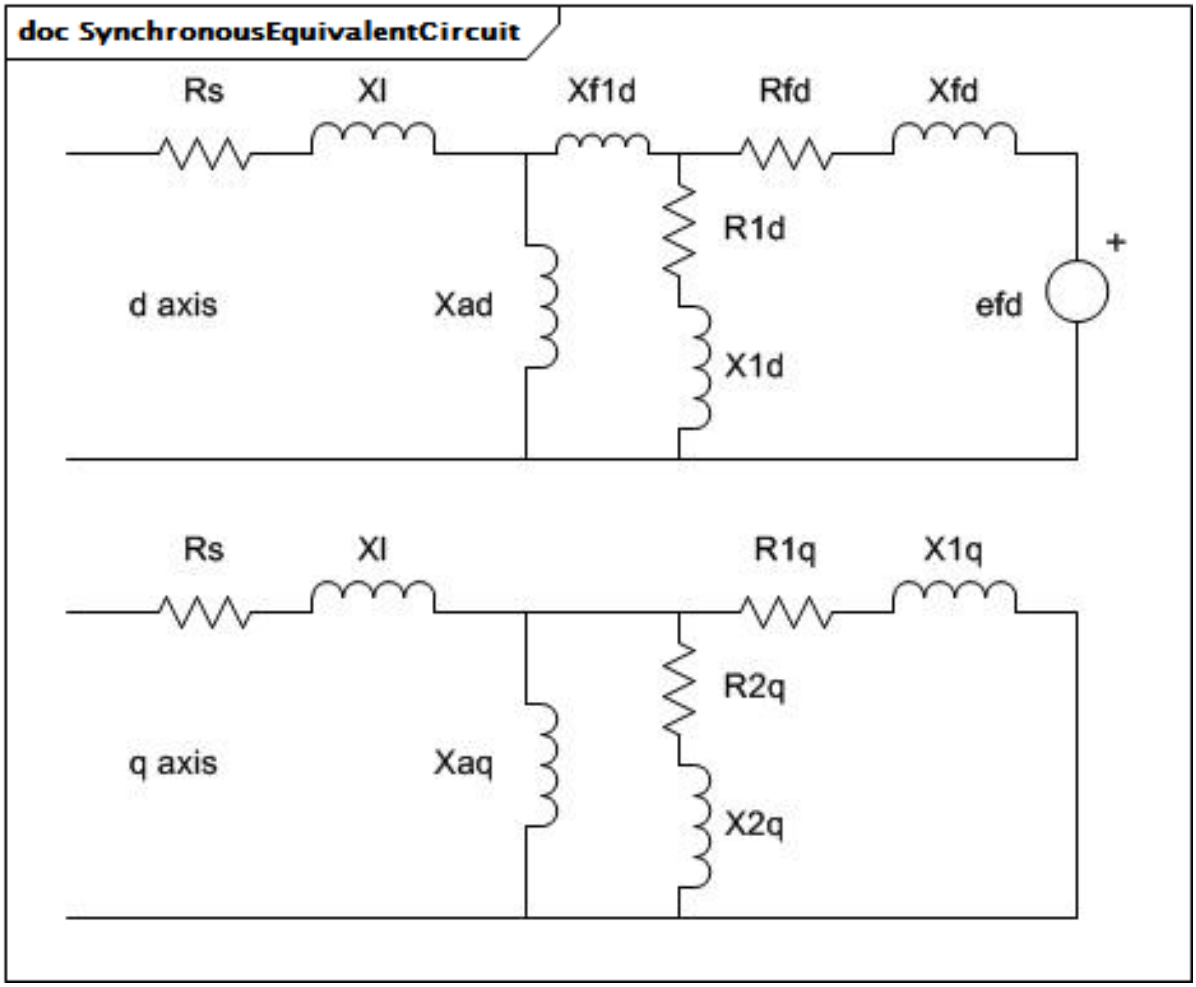


Figure 9-33 - figure SynchronousEquivalentCircuit

In each axis, the branches represent the stator leakage reactance (**Xl**) and resistance (**Rs**), the magnetizing reactance (**Xad**, **Xaq**), the physical field winding (**Rfd**, **Xfd**) on the rotor, and equivalent windings for eddy current flow in the rotor iron.

Inheritance path: ->SynchronousMachineDetailed->SynchronousMachineDynamics->RotatingMachineDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-28 shows all attributes of SynchronousMachineEquivalentCircuit.

Table 9-28 - Attributes of SynchronousMachineDynamics::SynchronousMachineEquivalentCircuit

Attribute name	mult.	Attribute type	Description
xad	0..1	PU	D-axis mutual reactance.
rfd	0..1	PU	Field winding resistance.
xfd	0..1	PU	Field winding leakage reactance.
r1d	0..1	PU	D-axis damper 1 winding resistance.
x1d	0..1	PU	D-axis damper 1 winding

			leakage reactance.
xf1d	0..1	PU	Differential mutual (“Canay”) reactance.
xaq	0..1	PU	Q-axis mutual reactance.
r1q	0..1	PU	Q-axis damper 1 winding resistance.
x1q	0..1	PU	Q-axis damper 1 winding leakage reactance.
r2q	0..1	PU	Q-axis damper 2 winding resistance.
x2q	0..1	PU	Q-axis damper 2 winding leakage reactance.
saturationFactorQAxis	0..1	Simple_Float	see SynchronousMachineDetailed
saturationFactor120QAxis	0..1	Simple_Float	see SynchronousMachineDetailed
efdBaseRatio	0..1	Simple_Float	see SynchronousMachineDetailed
ifdBaseType	0..1	IfdBaseKind	see SynchronousMachineDetailed
ifdBaseValue	0..1	CurrentFlow	see SynchronousMachineDetailed
damping	0..1	Simple_Float	see RotatingMachineDynamics
inertia	0..1	Seconds	see RotatingMachineDynamics
saturationFactor	0..1	Simple_Float	see RotatingMachineDynamics
saturationFactor120	0..1	Simple_Float	see RotatingMachineDynamics
statorLeakageReactance	0..1	PU	see RotatingMachineDynamics
statorResistance	0..1	PU	see RotatingMachineDynamics
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-29 shows all association ends of SynchronousMachineDynamics with other classes.

Table 9-29 - Association ends of SynchronousMachineDynamics::SynchronousMachineDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachine	1..1	SynchronousMachine	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..1] -----

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			Wires:: SynchronousMachine.SynchronousMachine[1]

9.6.4.7 Enumerations

9.6.4.7.1 IfdBaseKind

Excitation base system mode.

Literal	Description	Code
ifag	Air gap line mode. ifdBaseValue is computed, not defined by the user, in this mode.	
ifnl	No load system with saturation mode. ifdBaseValue is computed, not defined by the user, in this mode.	
iffl	Full load system mode. ifdBaseValue is computed, not defined by the user, in this mode.	
other	Free mode. ifdBaseValue is defined by the user in this mode.	

9.6.4.7.2 SynchronousMachineModelKind

Type of synchronous machine model used in Dynamic simulation applications.

Literal	Description	Code
subtransient	Subtransient synchronous machine model.	
subtransientTypeF	WECC Type F variant of subtransient synchronous machine model.	
subtransientTypeJ	WECC Type J variant of subtransient synchronous machine model.	
subtransientSimplified	Simplified version of subtransient synchronous machine model where magnetic coupling between the direct and quadrature axes is ignored.	
subtransientSimplifiedDirectAxis	Simplified version of a subtransient synchronous machine model with no damper circuit on d-axis.	

9.6.4.7.3 RotorKind

Type of rotor on physical machine.

Literal	Description	Code
roundRotor	Round rotor type of synchronous machine.	
salientPole	Salient pole type of synchronous machine.	

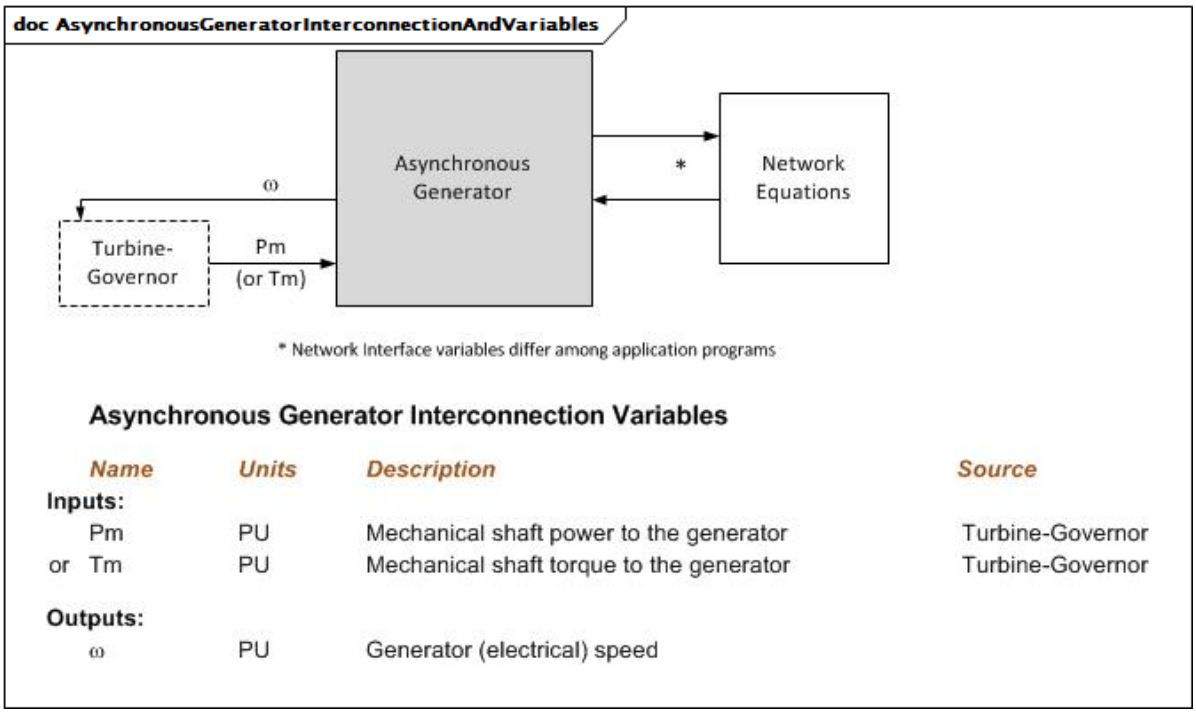
9.6.5 AsynchronousMachineDynamics

9.6.5.1 General

An asynchronous machine model represents a (induction) generator or motor with no external connection to the rotor windings, e.g., squirrel-cage induction machine. The interconnection with the electrical network equations may differ among simulation tools. The program only needs to know the terminal to which this asynchronous machine is connected in order to establish the correct interconnection. The interconnection with motor's equipment could also differ due to input and output signals required by standard models. The asynchronous machine model is used to model wind generators Type 1 and Type 2. For these, normal practice is to include the rotor flux transients and neglect the stator flux transients.

Figure 9-34 shows the diagram.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu



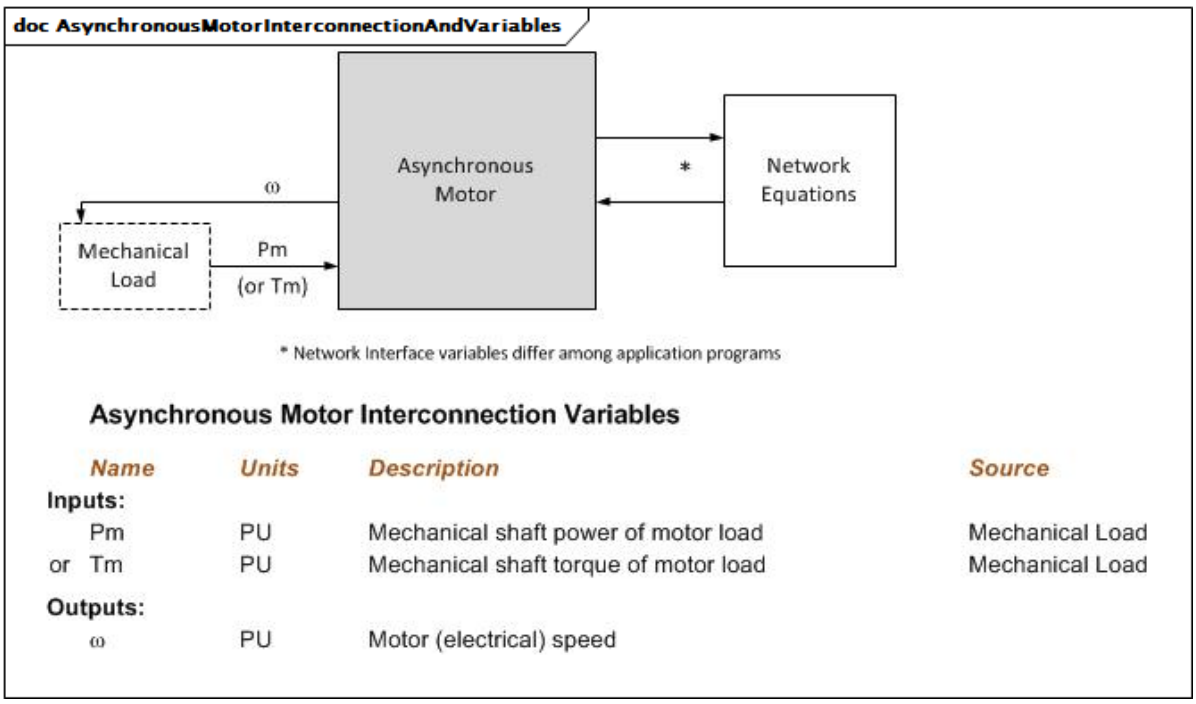
Asynchronous Generator Interconnection Variables

Name	Units	Description	Source
Inputs:			
Pm	PU	Mechanical shaft power to the generator	Turbine-Governor
or Tm	PU	Mechanical shaft torque to the generator	Turbine-Governor
Outputs:			
ω	PU	Generator (electrical) speed	

Figure 9-34 - diagram AsynchronousGeneratorInterconnectionAndVariables

This diagram illustrates the inputs and outputs for an asynchronous generator.

Figure 9-35 shows the diagram.



Asynchronous Motor Interconnection Variables

Name	Units	Description	Source
Inputs:			
Pm	PU	Mechanical shaft power of motor load	Mechanical Load
or Tm	PU	Mechanical shaft torque of motor load	Mechanical Load
Outputs:			
ω	PU	Motor (electrical) speed	

Figure 9-35 - diagram AsynchronousMotorInterconnectionAndVariables

This diagram illustrates the inputs and outputs for an asynchronous motor.

Figure 9-36 shows the diagram.

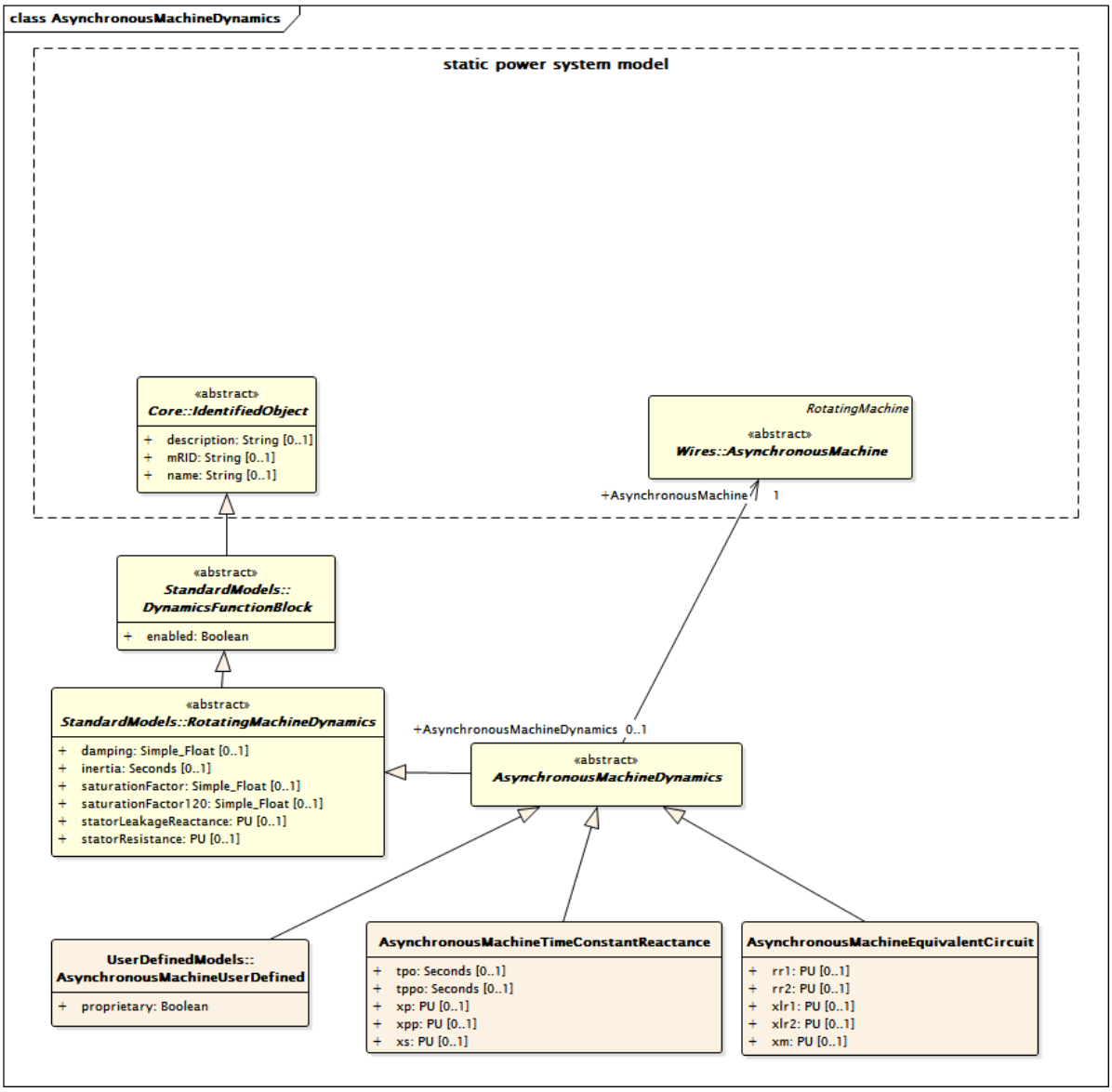


Figure 9-36 - diagram AsynchronousMachineDynamics

This diagram shows static power system model classes and dynamics model classes related to the asynchronous machine dynamics model. An association between the AsynchronousMachineDynamics class and the AsynchronousMachine class provides a means for relating the dynamics model to the static model.

Figure 9-37 shows the diagram.

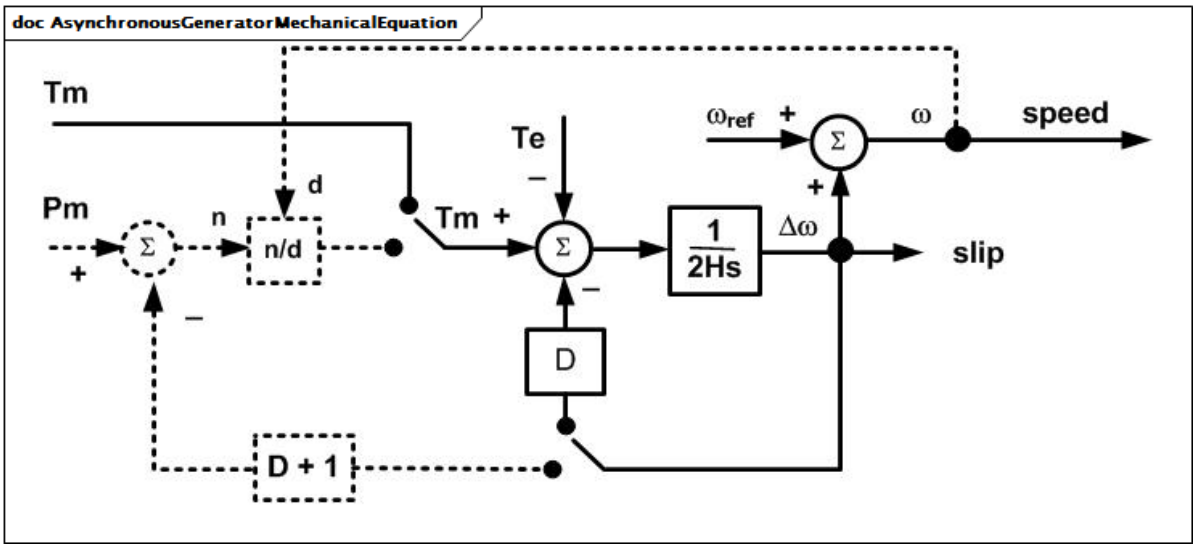


Figure 9-37 - diagram AsynchronousGeneratorMechanicalEquation

All variables are per unit on motor MVA base except angle, which is in radians. **omega$_0$** is the system synchronous frequency in radians per second. **omega$_{ref}$** is the per unit system frequency (or isolated area frequency), which may be constant at the initial frequency or vary during a simulation depending on the application program.

Figure 9-38 shows the diagram.

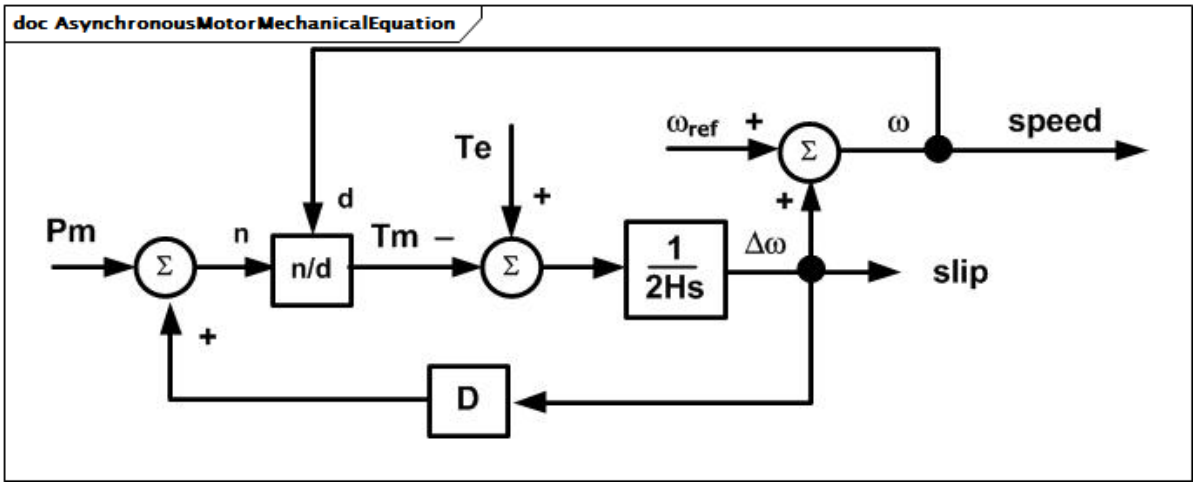


Figure 9-38 - diagram AsynchronousMotorMechanicalEquation

All variables are per unit on motor MVA base except angle, which is in radians. **omega$_0$** is the system synchronous frequency in radians per second. **omega$_{ref}$** is the per unit system frequency (or isolated area frequency), which may be constant at the initial frequency or vary during a simulation depending on the application program.

9.6.5.2 AsynchronousMachineDynamics(Abstract)

Asynchronous machine whose behaviour is described by reference to a standard model expressed in either time constant reactance form or equivalent circuit form or by definition of a user-defined model.

Parameter Notes:

1. Asynchronous machine parameters such as **Xl, Xs** etc. are actually used as inductances (L) in the model, but are commonly referred to as reactances since, at nominal frequency, the per unit values are the same. However, some references use the symbol L instead of X.

Inheritance path: ->RotatingMachineDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-30 shows all attributes of AsynchronousMachineDynamics.

Table 9-30 - Attributes of AsynchronousMachineDynamics::AsynchronousMachineDynamics

Attribute name	mult.	Attribute type	Description
damping	0..1	Simple_Float	see RotatingMachineDynamics
inertia	0..1	Seconds	see RotatingMachineDynamics
saturationFactor	0..1	Simple_Float	see RotatingMachineDynamics
saturationFactor120	0..1	Simple_Float	see RotatingMachineDynamics
statorLeakageReactance	0..1	PU	see RotatingMachineDynamics
statorResistance	0..1	PU	see RotatingMachineDynamics
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-31 shows all association ends of AsynchronousMachineDynamics with other classes.

Table 9-31 - Association ends of AsynchronousMachineDynamics::AsynchronousMachineDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachine	[1..1]	AsynchronousMachine	Asynchronous machine to which this asynchronous machine dynamics model applies. Association Based On : AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1] ----- Wires:: AsynchronousMachine.AsynchronousMachine[1]

9.6.5.3 AsynchronousMachineTimeConstantReactance

Parameter Notes:

1. If **X''** = **X'**, a single cage (one equivalent rotor winding per axis) is modelled.
2. The “p” in the attribute names is a substitution for a “prime” in the usual parameter notation, e.g. tpo refers to T'o.

The parameters used for models expressed in time constant reactance form include:

- RotatingMachine.ratedS (MVAbase)
- RotatingMachineDynamics.damping (D)

- RotatingMachineDynamics.inertia (H)
- RotatingMachineDynamics.saturationFactor (S1)
- RotatingMachineDynamics.saturationFactor120 (S12)
- RotatingMachineDynamics.statorLeakageReactance (Xl)
- RotatingMachineDynamics.statorResistance (Rs)
- .xs (Xs)
- .xp (X')
- .xpp (X'')
- .tpo (T'o)
- .tppo (T''o).

Inheritance path: ->AsynchronousMachineDynamics->RotatingMachineDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-32 shows all attributes of AsynchronousMachineTimeConstantReactance.

Table 9-32 - Attributes of AsynchronousMachineDynamics::AsynchronousMachineTimeConstantReactance

Attribute name	mult.	Attribute type	Description
xs	0..1	PU	Synchronous reactance (Xs) (>= X'). Typical Value = 1.8.
xp	0..1	PU	Transient reactance (unsaturated) (X') (>=X''). Typical Value = 0.5.
xpp	0..1	PU	Subtransient reactance (unsaturated) (X'') (> Xl). Typical Value = 0.2.
tpo	0..1	Seconds	Transient rotor time constant (T'o) (> T''o). Typical Value = 5.
tppo	0..1	Seconds	Subtransient rotor time constant (T''o) (> 0). Typical Value = 0.03.
damping	0..1	Simple_Float	see RotatingMachineDynamics
inertia	0..1	Seconds	see RotatingMachineDynamics
saturationFactor	0..1	Simple_Float	see RotatingMachineDynamics
saturationFactor120	0..1	Simple_Float	see RotatingMachineDynamics
statorLeakageReactance	0..1	PU	see RotatingMachineDynamics
statorResistance	0..1	PU	see RotatingMachineDynamics
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 9-33 shows all association ends of AsynchronousMachineDynamics with other classes.

Table 9-33 - Association ends of AsynchronousMachineDynamics::AsynchronousMachineDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachine	1..1	AsynchronousMachine	Inherited Association Based On : AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1] --- Wires:: AsynchronousMachine.AsynchronousMachine[1]

9.6.5.4 AsynchronousMachineEquivalentCircuit

The electrical equations of all variations of the asynchronous model are based on the AsynchronousEquivalentCircuit diagram for the direct and quadrature axes, with two equivalent rotor windings in each axis.

Equations for conversion between Equivalent Circuit and Time Constant Reactance forms:

Xs = Xm + Xl

X' = Xl + Xm * Xlr1 / (Xm + Xlr1)

X'' = Xl + Xm * Xlr1* Xlr2 / (Xm * Xlr1 + Xm * Xlr2 + Xlr1 * Xlr2)

T'o = (Xm + Xlr1) / (omega$_0$ * Rr1)

T''o = (Xm * Xlr1 + Xm * Xlr2 + Xlr1 * Xlr2) / (omega$_0$ * Rr2 * (Xm + Xlr1)

Same equations using CIM attributes from AsynchronousMachineTimeConstantReactance class on left of = sign and AsynchronousMachineEquivalentCircuit class on right (except as noted):

xs = xm + RotatingMachineDynamics.statorLeakageReactance

xp = RotatingMachineDynamics.statorLeakageReactance + xm * xlr1 / (xm + xlr1)

xpp = RotatingMachineDynamics.statorLeakageReactance + xm * xlr1* xlr2 / (xm * xlr1 + xm * xlr2 + xlr1 * xlr2)

tpo = (xm + xlr1) / (2*pi*nominal frequency * rr1)

tppo = (xm * xlr1 + xm * xlr2 + xlr1 * xlr2) / (2*pi*nominal frequency * rr2 * (xm + xlr1).

Figure 9-39 class diagram

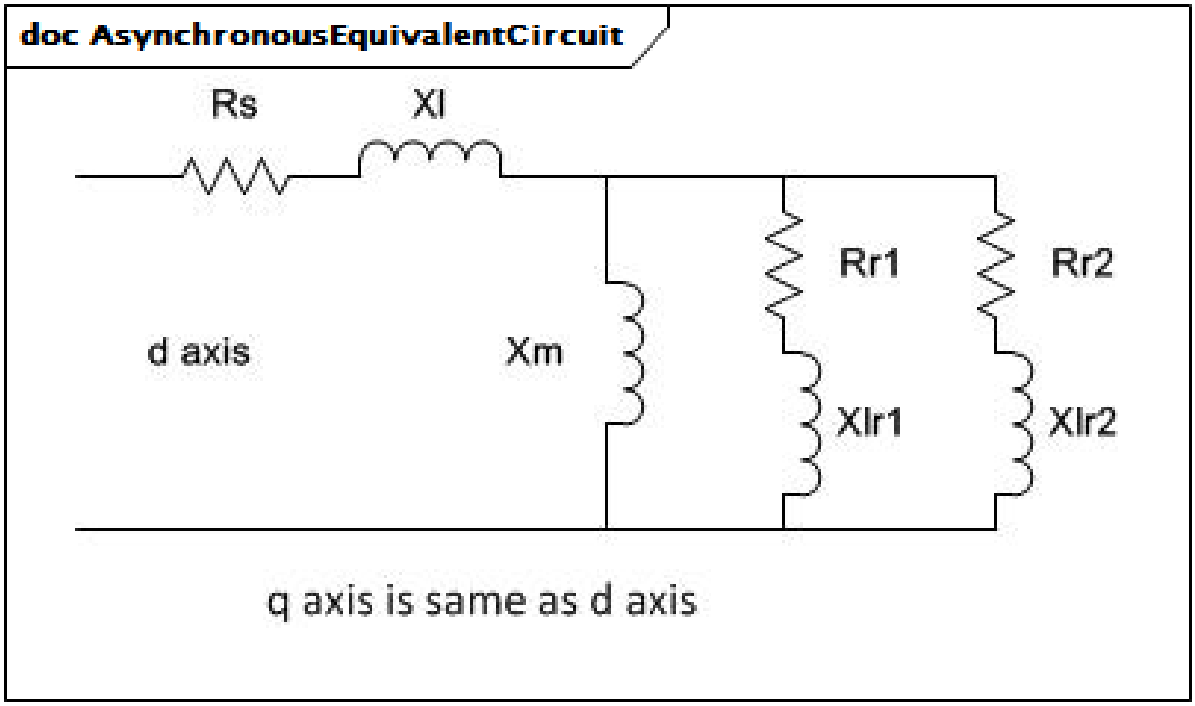


Figure 9-39 - figure AsynchronousEquivalentCircuit

In each axis, the branches represent the stator leakage reactance (**Xl**) and resistance (**Rs**), the magnetizing reactance (**Xm**), and the resistance and leakage reactance of equivalent windings (**Rr1**, **Xlr1**, etc.) on the rotor.

Inheritance path: ->AsynchronousMachineDynamics->RotatingMachineDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-34 shows all attributes of AsynchronousMachineEquivalentCircuit.

Table 9-34 - Attributes of AsynchronousMachineDynamics::AsynchronousMachineEquivalentCircuit

Attribute name	mult.	Attribute type	Description
xm	0..1	PU	Magnetizing reactance.
rr1	0..1	PU	Damper 1 winding resistance.
xlr1	0..1	PU	Damper 1 winding leakage reactance.
rr2	0..1	PU	Damper 2 winding resistance.
xlr2	0..1	PU	Damper 2 winding leakage reactance.
damping	0..1	Simple_Float	see RotatingMachineDynamics
inertia	0..1	Seconds	see RotatingMachineDynamics
saturationFactor	0..1	Simple_Float	see RotatingMachineDynamics

saturationFactor120	0..1	Simple_Float	see RotatingMachineDynamics
statorLeakageReactance	0..1	PU	see RotatingMachineDynamics
statorResistance	0..1	PU	see RotatingMachineDynamics
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-35 shows all association ends of AsynchronousMachineDynamics with other classes.

Table 9-35 - Association ends of AsynchronousMachineDynamics::AsynchronousMachineDynamics with other classes

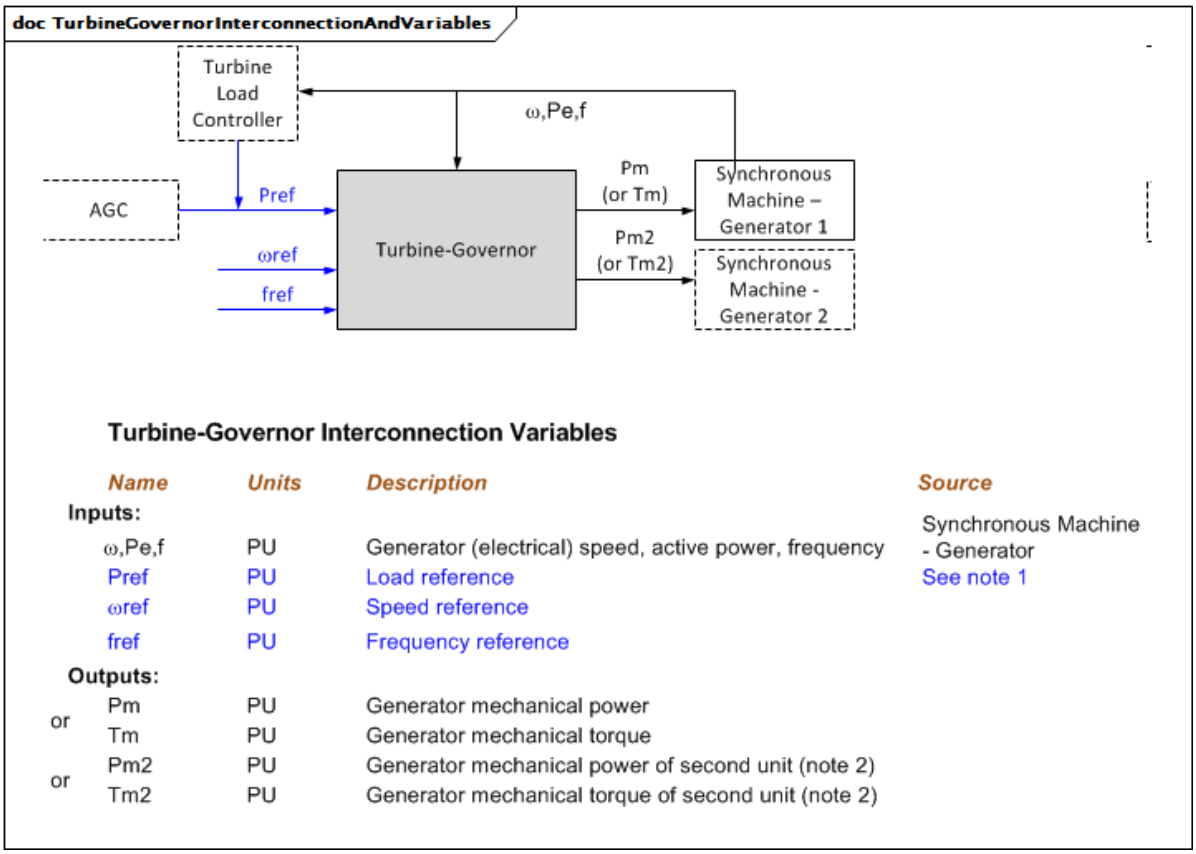
mult.	mult.	Class type name	Description
AsynchronousMachine	1..1	AsynchronousMachine	Inherited Association Based On : AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1] --- Wires:: AsynchronousMachine.AsynchronousMachine[1]

9.6.6 TurbineGovernorDynamics

9.6.6.1 General

The turbine-governor model is linked to one or two synchronous generators and determines the shaft mechanical power (Pm) or torque (Tm) for the generator model. Unlike IEEE standard models for other function blocks, the three IEEE turbine-governor standard models (GovHydroIEEE0, GovHydroIEEE2, GovSteamIEEE1) are documented in IEEE Transactions not in IEEE standards. For that reason, diagrams are supplied for those models. A 2012 IEEE report, *Dynamic Models for Turbine-Governors in Power System Studies*, provides updated information on a variety of models including IEEE, vendor and reliability authority models. Fully incorporating the results of that report into the CIM Dynamics model is a future effort.

Figure 9-40 shows the diagram.



Turbine-Governor Interconnection Variables

	Name	Units	Description	Source
Inputs:				
	ω,Pe,f	PU	Generator (electrical) speed, active power, frequency	Synchronous Machine - Generator
	Pref	PU	Load reference	See note 1
	ωref	PU	Speed reference	
	fref	PU	Frequency reference	
Outputs:				
	Pm	PU	Generator mechanical power	
or	Tm	PU	Generator mechanical torque	
	Pm2	PU	Generator mechanical power of second unit (note 2)	
or	Tm2	PU	Generator mechanical torque of second unit (note 2)	

Figure 9-40 - diagram TurbineGovernorInterconnectionAndVariables

This diagram illustrates the inputs and outputs for a turbine-governor.
Notes:

1. **Pref** is usually held constant for stability analysis, but may be determined by a user-defined model or, for long-term dynamics, an area-wide automatic generation control (AGC) model. For some models, **Pref** has units of PU speed instead of PU power.
2. Some turbine-governor models are used in cross-compound interconnections, e.g. GovSteam1, where they are connected to two generators (synchronous machines).
3. The vast majority of turbine-governor models have mwbase as a parameter. This parameter corresponds to GeneratingUnit.ratedGrossMaxP from the Production package, but is specified explicitly in the Dynamics package to support use of a different value by transient analysis tools, if needed.

Figure 9-41 shows the diagram.

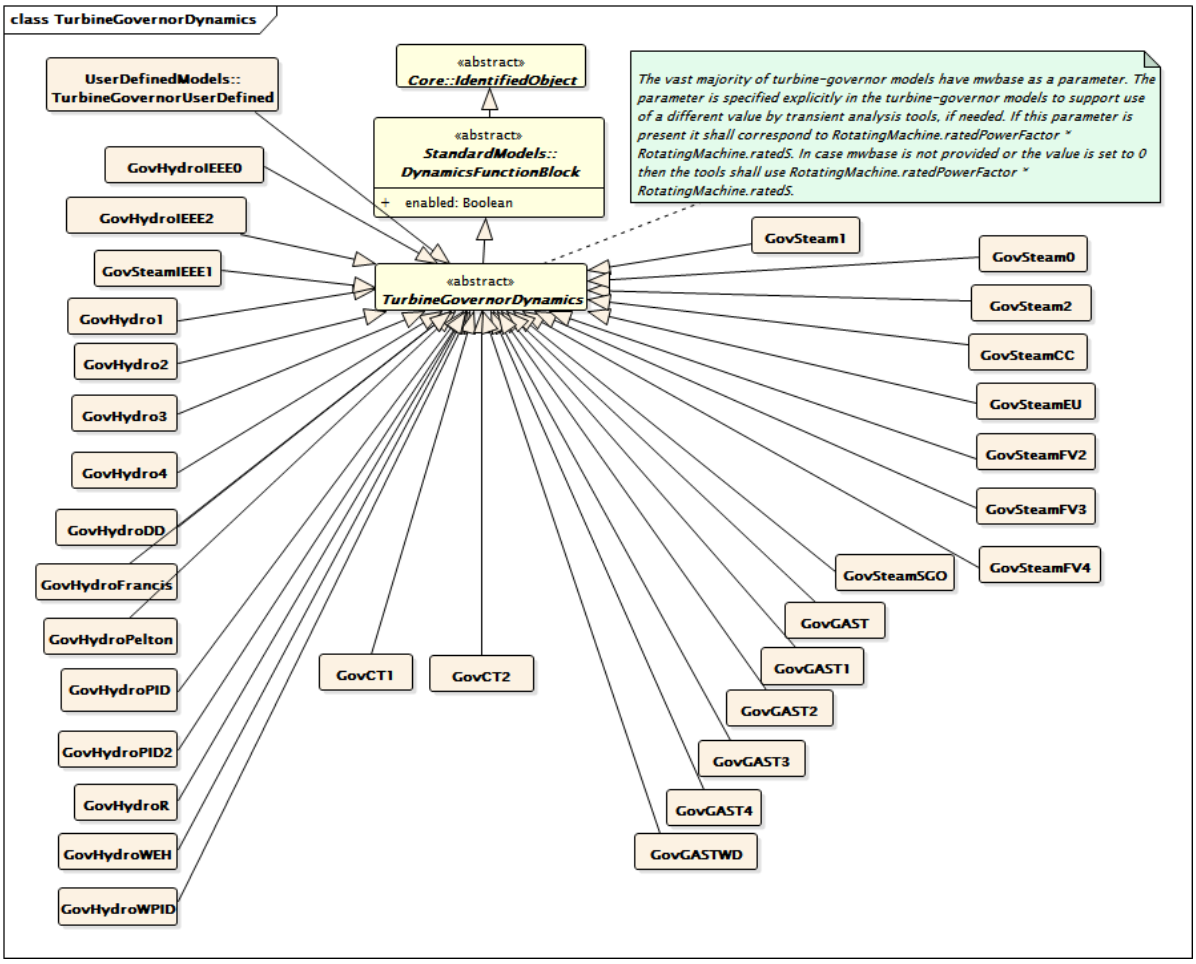


Figure 9-41 - diagram TurbineGovernorDynamics

This diagram shows dynamic standard model turbine-governor classes and their inheritance from the TurbineGovernorDynamics class.

9.6.6.2 TurbineGovernorDynamics(Abstract)

Turbine-governor function block whose behavior is described by reference to a standard model or by definition of a user-defined model.

The vast majority of turbine-governor models have mwbase as a parameter. The parameter is specified explicitly in the turbine-governor models to support use of a different value by transient analysis tools, if needed. If this parameter is present it shall correspond to RotatingMachine.ratedPowerFactor * RotatingMachine.ratedS. In case mwbase is not provided or the value is set to 0 then the tools shall use RotatingMachine.ratedPowerFactor * RotatingMachine.ratedS.

Inheritance path: ->DynamicsFunctionBlock->IdentifiedObject

Table 9-36 shows all attributes of TurbineGovernorDynamics.

Table 9-36 - Attributes of TurbineGovernorDynamics::TurbineGovernorDynamics

Attribute name	mult.	Attribute type	Description
enabled	1..1	Boolean	see DynamicsFunctionBlock

description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-37 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-37 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	[0..1]	AsynchronousMachineDynamics	Asynchronous machine model with which this turbine-governor model is associated. Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] ----- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	[0..*]	SynchronousMachineDynamics	Turbine-governor model associated with this synchronous machine model. Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] ----- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.6.6.3 GovHydroIEEE0

IEEE Simplified Hydro Governor-Turbine Model. Used for Mechanical-Hydraulic and Electro-Hydraulic turbine governors, with our without steam feedback. Typical values given are for Mechanical-Hydraulic.
Reference: IEEE Transactions on Power Apparatus and Systems November/December 1973, Volume PAS-92, Number 6 *Dynamic Models for Steam and Hydro Turbines in Power System Studies*, Page 1904.

Figure 9-42 class diagram

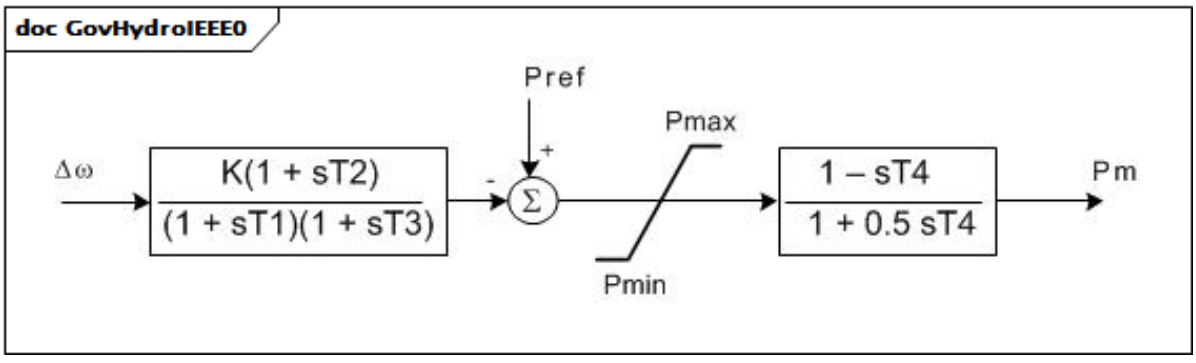


Figure 9-42 - figure GovHydroIEEE0

This diagram is supplied to document the CIM model interpretation of the IEEE Transactions on Power Apparatus and Systems November/December 1973, Volume PAS-92, Number 6 *Dynamic Models for Steam and Hydro Turbines in Power System Studies*, Page 1904.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-38 shows all attributes of GovHydroIEEE0.

Table 9-38 - Attributes of TurbineGovernorDynamics::GovHydroIEEE0

Attribute name	mult.	Attribute type	Description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
k	1..1	PU	Governor gain (K).
t1	1..1	Seconds	Governor lag time constant (T1). Typical Value = 0.25.
t2	1..1	Seconds	Governor lead time constant (T2). Typical Value = 0.
t3	1..1	Seconds	Gate actuator time constant (T3). Typical Value = 0.1.
t4	1..1	Seconds	Water starting time (T4).
pmax	1..1	PU	Gate maximum (Pmax).
pmin	1..1	PU	Gate minimum (Pmin).
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-39 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-39 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] --- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] --- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

9.6.6.4 GovHydroIEEE2

IEEE hydro turbine governor model represents plants with straightforward penstock configurations and hydraulic-dashpot governors. Reference: IEEE Transactions on Power Apparatus and Systems November/December 1973, Volume PAS-92, Number 6 *Dynamic Models for Steam and Hydro Turbines in Power System Studies*, Page 1904.

Figure 9-43 class diagram

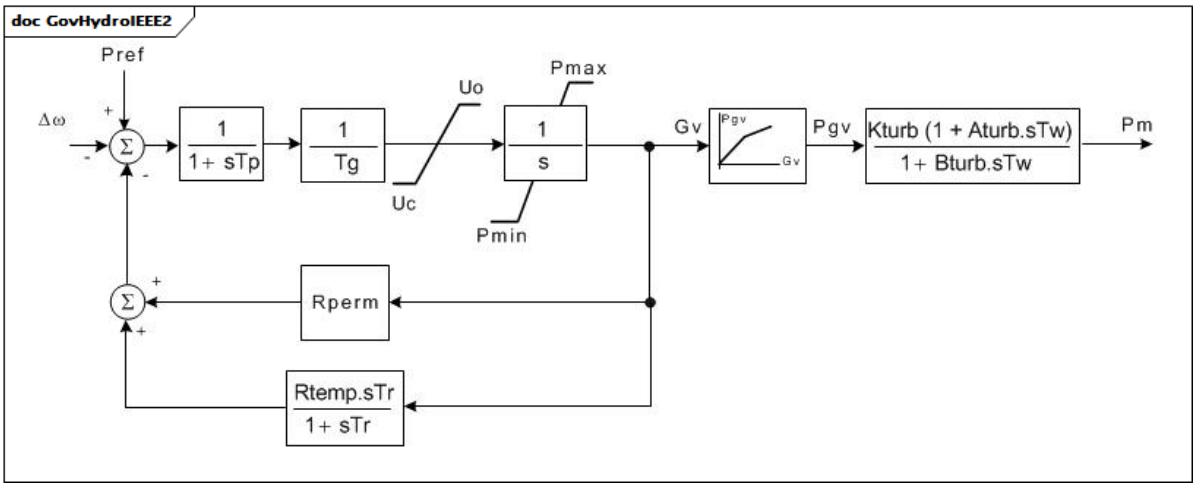


Figure 9-43 - figure GovHydroIEEE2

This diagram is supplied to document the CIM model interpretation of the IEEE Transactions on Power Apparatus and Systems November/December 1973, Volume PAS-92, Number 6 *Dynamic Models for Steam and Hydro Turbines in Power System Studies*, Page 1904.
Parameter Notes:

1. Per unit parameters are on base of **MWbase**, which is normally the MW capability of the turbine.
2. **Tr**, **Tg**, **Tw**, **Kturb**, and **Bturb** must be greater than zero.
3. This turbine governor model is a reasonable approximation to the performance of simple hydro plants for small motions of the gates about a given load, but it represents the variation of water inertia effect with load only if the values of the turbine parameters are properly adjusted by the user as load is varied. For idealized turbine model, set **Kturb** = 1., **Aturb** = -1., **Bturb** = 0.5.
4. The nonlinear gain between gate position and power may be input with up to 6 points. The (0.,0.) and (1.,1.) points are assumed and therefore not included as attributes in the class. The output is not allowed to go beyond 0. and 1. However, if **Pmax** > 1., the input and output are scaled by **Pmax**.

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-40 shows all attributes of GovHydroIEEE2.

Table 9-40 - Attributes of TurbineGovernorDynamics::GovHydroIEEE2

Attribute name	mult.	Attribute type	Description

mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
tg	1..1	Seconds	Gate servo time constant (Tg). Typical Value = 0.5.
tp	1..1	Seconds	Pilot servo valve time constant (Tp). Typical Value = 0.03.
uo	1..1	Simple_Float	Maximum gate opening velocity (Uo). Unit = PU/sec. Typical Value = 0.1.
uc	1..1	Simple_Float	Maximum gate closing velocity (Uc) (<0). Typical Value = -0.1.
pmax	1..1	PU	Maximum gate opening (Pmax). Typical Value = 1.
pmin	1..1	PU	Minimum gate opening (Pmin). Typical Value = 0.
rperm	1..1	PU	Permanent droop (Rperm). Typical Value = 0.05.
rtemp	1..1	PU	Temporary droop (Rtemp). Typical Value = 0.5.
tr	1..1	Seconds	Dashpot time constant (Tr). Typical Value = 12.
tw	1..1	Seconds	Water inertia time constant (Tw). Typical Value = 2.
kturb	1..1	PU	Turbine gain (Kturb). Typical Value = 1.
aturb	1..1	PU	Turbine numerator multiplier (Aturb). Typical Value = -1.
bturb	1..1	PU	Turbine denominator multiplier (Bturb). Typical Value = 0.5.
gv1	1..1	PU	Nonlinear gain point 1, PU gv (Gv1). Typical Value = 0.
pgv1	1..1	PU	Nonlinear gain point 1, PU power (Pgv1). Typical Value = 0.
gv2	1..1	PU	Nonlinear gain point 2, PU gv (Gv2). Typical Value = 0.
pgv2	1..1	PU	Nonlinear gain point 2, PU power (Pgv2). Typical Value = 0.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

gv3	1..1	PU	Nonlinear gain point 3, PU gv (Gv3). Typical Value = 0.
pgv3	1..1	PU	Nonlinear gain point 3, PU power (Pgv3). Typical Value = 0.
gv4	1..1	PU	Nonlinear gain point 4, PU gv (Gv4). Typical Value = 0.
pgv4	1..1	PU	Nonlinear gain point 4, PU power (Pgv4). Typical Value = 0.
gv5	1..1	PU	Nonlinear gain point 5, PU gv (Gv5). Typical Value = 0.
pgv5	1..1	PU	Nonlinear gain point 5, PU power (Pgv5). Typical Value = 0.
gv6	1..1	PU	Nonlinear gain point 6, PU gv (Gv6). Typical Value = 0.
pgv6	1..1	PU	Nonlinear gain point 6, PU power (Pgv6). Typical Value = 0.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-41 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-41 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] --- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] --- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.6.6.5 GovSteamIEEE1

IEEE steam turbine governor model. Reference: IEEE Transactions on Power Apparatus and Systems November/December 1973, Volume PAS-92, Number 6 *Dynamic Models for Steam and Hydro Turbines in Power System Studies*, Page 1904.
Parameter Notes:

1. Per unit parameters are on base of **MWbase**, which is normally the MW capability of the turbine.
2. **T3** must be greater than zero. All other time constants may be zero.
3. For a tandem-compound turbine the parameters **K2**, **K4**, **K6**, and **K8** are ignored. For a cross-compound turbine, two generators are connected to this turbine-governor model.
4. Each generator must be represented in the load flow by data on its own MVA base. The values of **K1**, **K3**, **K5**, **K7** must be specified to describe the proportionate development of power on the first turbine shaft. **K2**, **K4**, **K6**, **K8** must describe the second turbine shaft. Normally **K1** + **K3** + **K5** + **K7** = 1.0 and **K2** + **K4** + **K6** + **K8** = 1.0 (if second generator is present).
5. The division of power between the two shafts is in proportion to the values of MVA bases of the two generators. The initial condition load flow should, therefore, have the two generators loaded to the same fraction of each one's MVA base.

Figure 9-44 class diagram

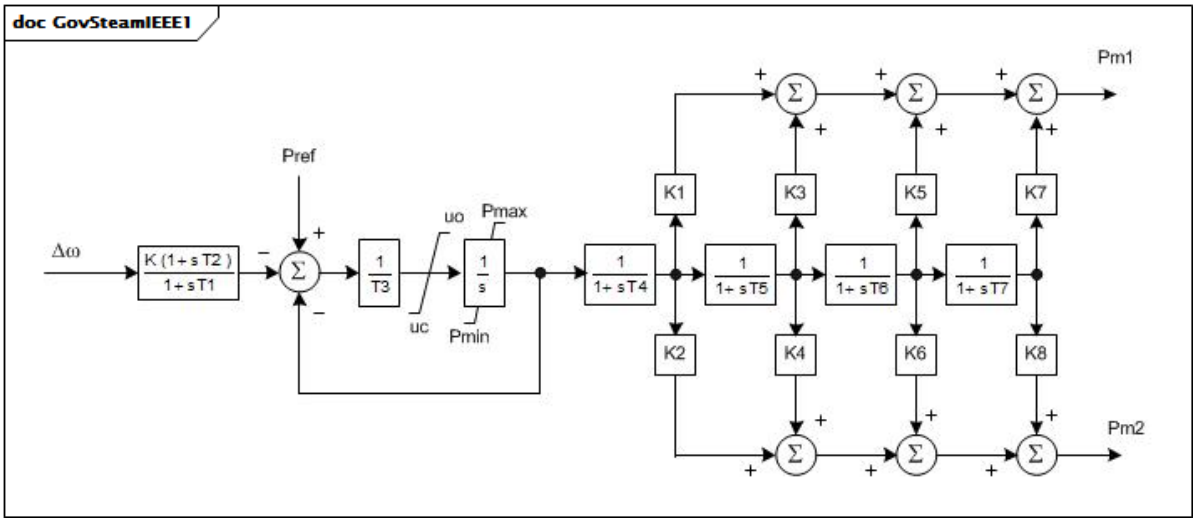


Figure 9-44 - figure GovSteamIEEE1

This diagram is supplied to document the CIM model interpretation of the IEEE Transactions on Power Apparatus and Systems November/December 1973, Volume PAS-92, Number 6 *Dynamic Models for Steam and Hydro Turbines in Power System Studies*, Page 1904.

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-42 shows all attributes of GovSteamIEEE1.

Table 9-42 - Attributes of TurbineGovernorDynamics::GovSteamIEEE1

Attribute name	mult.	Attribute type	Description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0).
k	1..1	PU	Governor gain (reciprocal of droop) (K) (> 0). Typical Value = 25.
t1	1..1	Seconds	Governor lag time constant (T1). Typical Value = 0.
t2	1..1	Seconds	Governor lead time constant (T2). Typical Value = 0.
t3	1..1	Seconds	Valve positioner time constant (T3) (> 0). Typical Value = 0.1.
uo	1..1	Simple_Float	Maximum valve opening velocity (Uo) (> 0). Unit = PU/sec. Typical Value = 1.
uc	1..1	Simple_Float	Maximum valve closing velocity (Uc) (< 0). Unit = PU/sec. Typical Value = -10.
pmax	1..1	PU	Maximum valve opening (Pmax) (> Pmin). Typical Value = 1.
pmin	1..1	PU	Minimum valve opening (Pmin) (>= 0). Typical Value = 0.
t4	1..1	Seconds	Inlet piping/steam bowl time constant (T4). Typical Value = 0.3.
k1	1..1	Simple_Float	Fraction of HP shaft power after first boiler pass (K1). Typical Value = 0.2.
k2	1..1	Simple_Float	Fraction of LP shaft power after first boiler pass (K2). Typical Value = 0.
t5	1..1	Seconds	Time constant of second boiler pass (T5). Typical Value = 5.
k3	1..1	Simple_Float	Fraction of HP shaft power after second boiler pass (K3). Typical Value = 0.3.
k4	1..1	Simple_Float	Fraction of LP shaft power after second boiler pass (K4). Typical Value = 0.
t6	1..1	Seconds	Time constant of third boiler pass (T6).

			Typical Value = 0.5.
k5	1..1	Simple_Float	Fraction of HP shaft power after third boiler pass (K5). Typical Value = 0.5.
k6	1..1	Simple_Float	Fraction of LP shaft power after third boiler pass (K6). Typical Value = 0.
t7	1..1	Seconds	Time constant of fourth boiler pass (T7). Typical Value = 0.
k7	1..1	Simple_Float	Fraction of HP shaft power after fourth boiler pass (K7). Typical Value = 0.
k8	1..1	Simple_Float	Fraction of LP shaft power after fourth boiler pass (K8). Typical Value = 0.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-43 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-43 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] ----- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] ----- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.6.6.6 GovCT1

General model for any prime mover with a PID governor, used primarily for combustion turbine and combined cycle units.
This model can be used to represent a variety of prime movers controlled by PID governors. It is suitable, for example, for representation of

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

- gas turbine and single shaft combined cycle turbines
- diesel engines with modern electronic or digital governors
- steam turbines where steam is supplied from a large boiler drum or a large header whose pressure is substantially constant over the period under study
- simple hydro turbines in dam configurations where the water column length is short and water inertia effects are minimal.

Additional information on this model is available in the 2012 IEEE report, *Dynamic Models for Turbine-Governors in Power System Studies*, section 3.1.2.3 page 3-4 (GGOV1).

Figure 9-45 class diagram

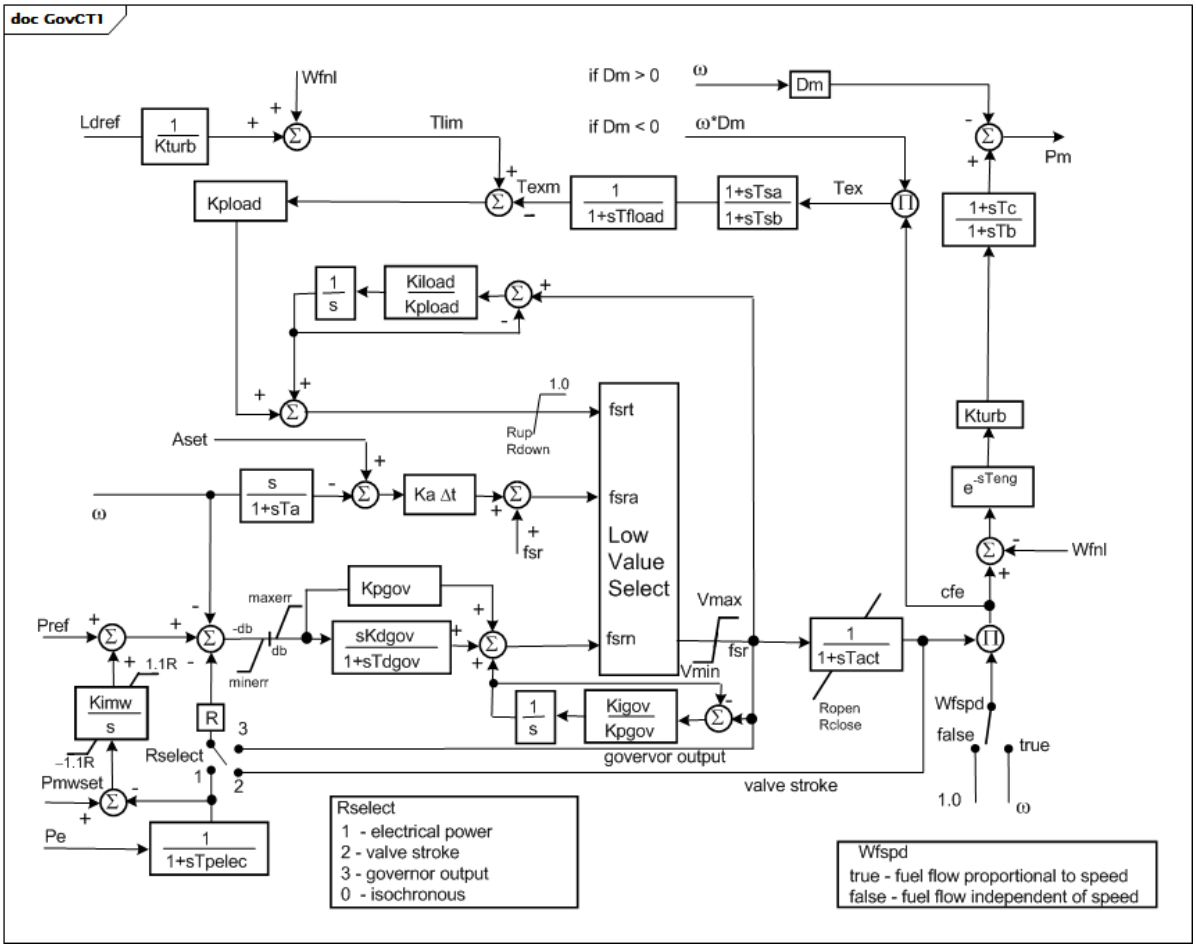


Figure 9-45 - figure GovCT1

Parameter Notes:

1. Per unit parameters are on base of **MWbase**, which is normally the MW capability of the turbine.
2. The range of fuel valve travel and of fuel flow is unity. Thus the largest possible value of **Vmax** is 1.0 and the smallest possible value of **Vmin** is zero. **Vmax** may, however, be reduced below unity to represent a loading limit that may be imposed by the operator or a

supervisory control system. For gas turbines **Vmin** should normally be greater than zero and less than **Wfnl** to represent a minimum firing limit. The value of the fuel flow at maximum output must be less than, or equal to unity, depending on the value of **Kturb**.

3. The load limiter module may be used to impose a maximum output limit such as an exhaust temperature limit. To do this the time constant **Tfload** should be set to represent the time constant in the measurement of temperature (or other signal), and the gains of the limiter, **Kpload**, **Kiload**, should be set to give prompt stable control when on limit. The load limit can be deactivated by setting the parameter **Ldref** to a high value.
4. This model includes a simple representation of a supervisory load controller. This controller is active if the parameter **Kimw** is non-zero. The load controller is a slow acting reset loop that adjusts the speed/load reference of the turbine governor to hold the electrical power output of the unit at its initial condition value. This value is stored in the parameter **Pmwset** (which equals Pe, initial generated power) when the model is initialized, and can be changed thereafter. The load controller must be adjusted to respond gently relative to the speed governor.
5. The parameters **Aset**, **Ka**, and **Ta** describe an acceleration limiter. **Ta** must be non-zero, but the acceleration limiter can be disabled by setting **Aset** to a large value, such as 1.
6. The parameters **Tsa** and **Tsb**, are provided to augment the exhaust gas temperature measurement subsystem in gas turbines. For example, they may be set to values such as 4., 5., to represent the 'radiation shield' element of large gas turbines.
7. The parameters **Rup** and **Rdown** specify the maximum rate of increase and decrease of the output of the load limit controller (**Kpload**/**Kiload**). These parameters should normally be set, or defaulted to 99/-99, but may be given particular values to represent the temperature limit controls of some GE heavy-duty engine controls.
8. The fuel flow command **fsr** is determined by whichever is lowest of **fsrt**, **fsra**, and **fsrn**. Although not explicitly shown in the GovCT1 diagram, the signals that are not in control track **fsr** so that they do not "windup" beyond that value. This represents GE gas turbine control practice but may not be true for other controller designs.
9. As shown in the GovCT1 diagram, when **Kpgov** is non-zero, the governor PI control is implemented to "track" **fsr** to prevent windup when **fsr** is limited by another signal (**fsrt**, **fsra**) or **Vmax**/**Vmin**. If **Kpgov** is zero, the integral path is implemented directly. The same applies to the load limiter PI control with regard to **Kpload**.

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-44 shows all attributes of GovCT1.

Table 9-44 - Attributes of TurbineGovernorDynamics::GovCT1

Attribute name	mult.	Attribute type	Description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
r	1..1	PU	Permanent droop (R). Typical Value = 0.04.
rselect	1..1	DroopSignalFeedbackKind	Feedback signal for droop (Rselect). Typical Value = electricalPower.
tpelec	1..1	Seconds	Electrical power transducer time constant (Tpelec) (>0). Typical Value = 1.
maxerr	1..1	PU	Maximum value for speed error signal (maxerr). Typical Value

			= 0.05.
minerr	1..1	PU	Minimum value for speed error signal (minerr). Typical Value = -0.05.
kpgov	1..1	PU	Governor proportional gain (Kpgov). Typical Value = 10.
kigov	1..1	PU	Governor integral gain (Kigov). Typical Value = 2.
kdgov	1..1	PU	Governor derivative gain (Kdgov). Typical Value = 0.
tdgov	1..1	Seconds	Governor derivative controller time constant (Tdgov). Typical Value = 1.
vmax	1..1	PU	Maximum valve position limit (Vmax). Typical Value = 1.
vmin	1..1	PU	Minimum valve position limit (Vmin). Typical Value = 0.15.
tact	1..1	Seconds	Actuator time constant (Tact). Typical Value = 0.5.
kturb	1..1	PU	Turbine gain (Kturb) (>0). Typical Value = 1.5.
wfnl	1..1	PU	No load fuel flow (Wfnl). Typical Value = 0.2.
tb	1..1	Seconds	Turbine lag time constant (Tb) (>0). Typical Value = 0.5.
tc	1..1	Seconds	Turbine lead time constant (Tc). Typical Value = 0.
wfspd	1..1	Boolean	Switch for fuel source characteristic to recognize that fuel flow, for a given fuel valve stroke, can be proportional to engine speed (Wfspd). true = fuel flow proportional to speed (for some gas turbines and diesel engines with positive displacement fuel injectors) false = fuel control system keeps fuel flow independent of engine speed. Typical Value = true.
teng	1..1	Seconds	Transport time delay for diesel engine used in

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			representing diesel engines where there is a small but measurable transport delay between a change in fuel flow setting and the development of torque (Teng). Teng should be zero in all but special cases where this transport delay is of particular concern. Typical Value = 0.
tfload	1..1	Seconds	Load Limiter time constant (Tfload) (>0). Typical Value = 3.
kpload	1..1	PU	Load limiter proportional gain for PI controller (Kpload). Typical Value = 2.
kiload	1..1	PU	Load limiter integral gain for PI controller (Kiload). Typical Value = 0.67.
ldref	1..1	PU	Load limiter reference value (Ldref). Typical Value = 1.
dm	1..1	PU	Speed sensitivity coefficient (Dm). Dm can represent either the variation of the engine power with the shaft speed or the variation of maximum power capability with shaft speed. If it is positive it describes the falling slope of the engine speed verses power characteristic as speed increases. A slightly falling characteristic is typical for reciprocating engines and some aero-derivative turbines. If it is negative the engine power is assumed to be unaffected by the shaft speed, but the maximum permissible fuel flow is taken to fall with falling shaft speed. This is characteristic of single-shaft industrial turbines due to exhaust temperature limits. Typical Value = 0.
ropen	1..1	Simple_Float	Maximum valve opening rate (Ropen). Unit = PU/sec. Typical Value = 0.10.
rclose	1..1	Simple_Float	Minimum valve closing rate (Rclose). Unit = PU/sec. Typical Value =

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			-0.1.
kimw	1..1	PU	Power controller (reset) gain (Kimw). The default value of 0.01 corresponds to a reset time of 100 seconds. A value of 0.001 corresponds to a relatively slow acting load controller. Typical Value = 0.01.
aset	1..1	Simple_Float	Acceleration limiter setpoint (Aset). Unit = PU/sec. Typical Value = 0.01.
ka	1..1	PU	Acceleration limiter gain (Ka). Typical Value = 10.
ta	1..1	Seconds	Acceleration limiter time constant (Ta) (>0). Typical Value = 0.1.
db	1..1	PU	Speed governor dead band in per unit speed (db). In the majority of applications, it is recommended that this value be set to zero. Typical Value = 0.
tsa	1..1	Seconds	Temperature detection lead time constant (Tsa). Typical Value = 4.
tsb	1..1	Seconds	Temperature detection lag time constant (Tsb). Typical Value = 5.
rup	1..1	PU	Maximum rate of load limit increase (Rup). Typical Value = 99.
rdown	1..1	PU	Maximum rate of load limit decrease (Rdown). Typical Value = -99.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-45 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-45 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics::

			TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] --- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] --- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.6.6.7 GovCT2

General governor model with frequency-dependent fuel flow limit. This model is a modification of the GovCT1 model in order to represent the frequency-dependent fuel flow limit of a specific gas turbine manufacturer.

Figure 9-46 class diagram

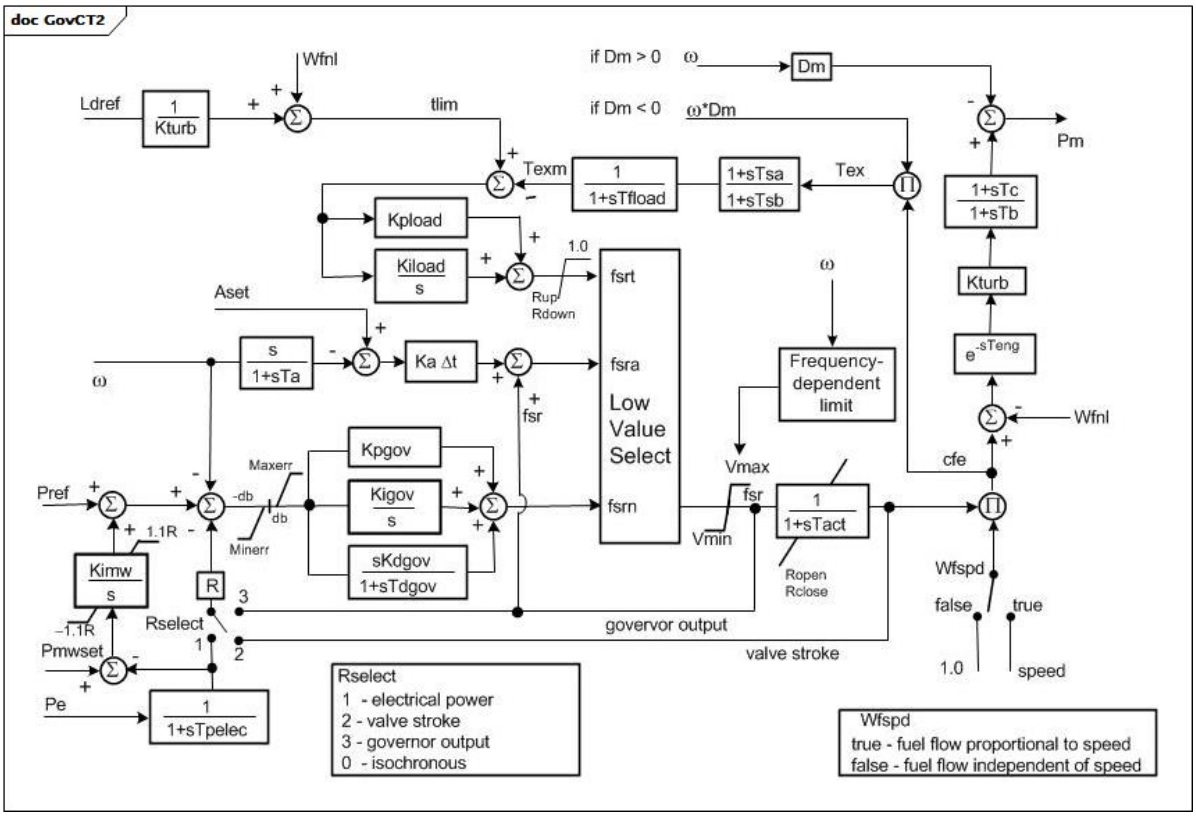


Figure 9-46 - figure GovCT2

Parameter Notes:

1. The frequency-dependent limit reduces the **Vmax** limit on fuel flow signal (**fsr**). For normal operation, the limiter performs no action. When frequency (generator speed) drops below **Flim1**, the highest frequency data point, the desired value for the power limit (**Plim**) is

determined by linear interpolation between associated data pairs. The maximum value of the limiter (**Vmax**) will ramp to the **fsr** value corresponding to **Plim** (**Plim** / **Kturb** + **Wfnl**) at the **Prate** ramp rate. **Plim** will be updated as frequency changes. If frequency subsequently rises back above **Flim1**, the value of the limit will ramp up at the rate **Prate** back to the normal value of **Vmax**.

2. The frequency-dependent limit is defined by a set of up to 10 pairs of points relating frequency (generator speed in Hz.) and power. The points must be monotonically decreasing in both the magnitude of frequency and power. If fewer than 10 points are needed to define the relationship, values of zero must be entered for the remaining frequencies and power limits. The value of the last non-zero data pair are used as a lower limit, that is, the last two values are not extrapolated to calculate lower power limits. If there is only one set of frequency and power limit, those values will be used as a single limit of power applied at that frequency and below.
3. Refer to GovCT1 model for other notes. Aside from the frequency-dependent limt, GovCT2 is identical to GovCT1, except that the temperature fuel command **fsrt** does not track fsr when it is not in control. Instead, it stays at its upper limit (1.).
4. The **Kpgov**/**Kigov** and **Kpload**/**Kiload** controllers include tracking logic to ensure smooth transfer between active controllers. This logic is not shown on the GovCT2 diagram.
5. **Pmwset** equals Pe, initial generated power.

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-46 shows all attributes of GovCT2.

Table 9-46 - Attributes of TurbineGovernorDynamics::GovCT2

Attribute name	mult.	Attribute type	Description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
r	1..1	PU	Permanent droop (R). Typical Value = 0.05.
rselect	1..1	DroopSignalFeedbackKind	Feedback signal for droop (Rselect). Typical Value = electricalPower.
tpelec	1..1	Seconds	Electrical power transducer time constant (Tpelec). Typical Value = 2.5.
maxerr	1..1	PU	Maximum value for speed error signal (Maxerr). Typical Value = 1.
minerr	1..1	PU	Minimum value for speed error signal (Minerr). Typical Value = -1.
kpgov	1..1	PU	Governor proportional gain (Kpgov). Typical Value = 4.
kigov	1..1	PU	Governor integral gain (Kigov). Typical Value = 0.45.
kdgov	1..1	PU	Governor derivative gain (Kdgov). Typical

			Value = 0.
tdgov	1..1	Seconds	Governor derivative controller time constant (Tdgov). Typical Value = 1.
vmax	1..1	PU	Maximum valve position limit (Vmax). Typical Value = 1.
vmin	1..1	PU	Minimum valve position limit (Vmin). Typical Value = 0.175.
tact	1..1	Seconds	Actuator time constant (Tact). Typical Value = 0.4.
kturb	1..1	PU	Turbine gain (Kturb). Typical Value = 1.9168.
wfnl	1..1	PU	No load fuel flow (Wfnl). Typical Value = 0.187.
tb	1..1	Seconds	Turbine lag time constant (Tb). Typical Value = 0.1.
tc	1..1	Seconds	Turbine lead time constant (Tc). Typical Value = 0.
wfspd	1..1	Boolean	Switch for fuel source characteristic to recognize that fuel flow, for a given fuel valve stroke, can be proportional to engine speed (Wfspd). true = fuel flow proportional to speed (for some gas turbines and diesel engines with positive displacement fuel injectors) false = fuel control system keeps fuel flow independent of engine speed. Typical Value = false.
teng	1..1	Seconds	Transport time delay for diesel engine used in representing diesel engines where there is a small but measurable transport delay between a change in fuel flow setting and the development of torque (Teng). Teng should be zero in all but special cases where this transport delay is of particular concern. Typical Value = 0.
tfload	1..1	Seconds	Load Limiter time constant (Tfload). Typical Value = 3.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

kpload	1..1	PU	Load limiter proportional gain for PI controller (Kpload). Typical Value = 1.
kiload	1..1	PU	Load limiter integral gain for PI controller (Kiload). Typical Value = 1.
ldref	1..1	PU	Load limiter reference value (Ldref). Typical Value = 1.
dm	1..1	PU	Speed sensitivity coefficient (Dm). Dm can represent either the variation of the engine power with the shaft speed or the variation of maximum power capability with shaft speed. If it is positive it describes the falling slope of the engine speed verses power characteristic as speed increases. A slightly falling characteristic is typical for reciprocating engines and some aero-derivative turbines. If it is negative the engine power is assumed to be unaffected by the shaft speed, but the maximum permissible fuel flow is taken to fall with falling shaft speed. This is characteristic of single-shaft industrial turbines due to exhaust temperature limits. Typical Value = 0.
ropen	1..1	Simple_Float	Maximum valve opening rate (Ropen). Unit = PU/sec. Typical Value = 99.
rclose	1..1	Simple_Float	Minimum valve closing rate (Rclose). Unit = PU/sec. Typical Value = -99.
kimw	1..1	PU	Power controller (reset) gain (Kimw). The default value of 0.01 corresponds to a reset time of 100 seconds. A value of 0.001 corresponds to a relatively slow acting load controller. Typical Value = 0.
aset	1..1	Simple_Float	Acceleration limiter setpoint (Aset). Unit = PU/sec. Typical Value = 10.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

ka	1..1	PU	Acceleration limiter Gain (Ka). Typical Value = 10.
ta	1..1	Seconds	Acceleration limiter time constant (Ta). Typical Value = 1.
db	1..1	PU	Speed governor dead band in per unit speed (db). In the majority of applications, it is recommended that this value be set to zero. Typical Value = 0.
tsa	1..1	Seconds	Temperature detection lead time constant (Tsa). Typical Value = 0.
tsb	1..1	Seconds	Temperature detection lag time constant (Tsb). Typical Value = 50.
rup	1..1	PU	Maximum rate of load limit increase (Rup). Typical Value = 99.
rdown	1..1	PU	Maximum rate of load limit decrease (Rdown). Typical Value = -99.
prate	1..1	PU	Ramp rate for frequency-dependent power limit (Prate). Typical Value = 0.017.
flim1	1..1	Frequency	Frequency threshold 1 (Flim1). Unit = Hz. Typical Value = 59.
plim1	1..1	PU	Power limit 1 (Plim1). Typical Value = 0.8325.
flim2	1..1	Frequency	Frequency threshold 2 (Flim2). Unit = Hz. Typical Value = 0.
plim2	1..1	PU	Power limit 2 (Plim2). Typical Value = 0.
flim3	1..1	Frequency	Frequency threshold 3 (Flim3). Unit = Hz. Typical Value = 0.
plim3	1..1	PU	Power limit 3 (Plim3). Typical Value = 0.
flim4	1..1	Frequency	Frequency threshold 4 (Flim4). Unit = Hz. Typical Value = 0.
plim4	1..1	PU	Power limit 4 (Plim4). Typical Value = 0.
flim5	1..1	Frequency	Frequency threshold 5 (Flim5). Unit = Hz. Typical Value = 0.
plim5	1..1	PU	Power limit 5 (Plim5). Typical Value = 0.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

flim6	1..1	Frequency	Frequency threshold 6 (Flim6). Unit = Hz. Typical Value = 0.
plim6	1..1	PU	Power limit 6 (Plim6). Typical Value = 0.
flim7	1..1	Frequency	Frequency threshold 7 (Flim7). Unit = Hz. Typical Value = 0.
plim7	1..1	PU	Power limit 7 (Plim7). Typical Value = 0.
flim8	1..1	Frequency	Frequency threshold 8 (Flim8). Unit = Hz. Typical Value = 0.
plim8	1..1	PU	Power limit 8 (Plim8). Typical Value = 0.
flim9	1..1	Frequency	Frequency threshold 9 (Flim9). Unit = Hz. Typical Value = 0.
plim9	1..1	PU	Power Limit 9 (Plim9). Typical Value = 0.
flim10	1..1	Frequency	Frequency threshold 10 (Flim10). Unit = Hz. Typical Value = 0.
plim10	1..1	PU	Power limit 10 (Plim10). Typical Value = 0.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-47 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-47 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] --- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] --- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

9.6.6.8 GovGAST

Single shaft gas turbine.

Figure 9-47 class diagram

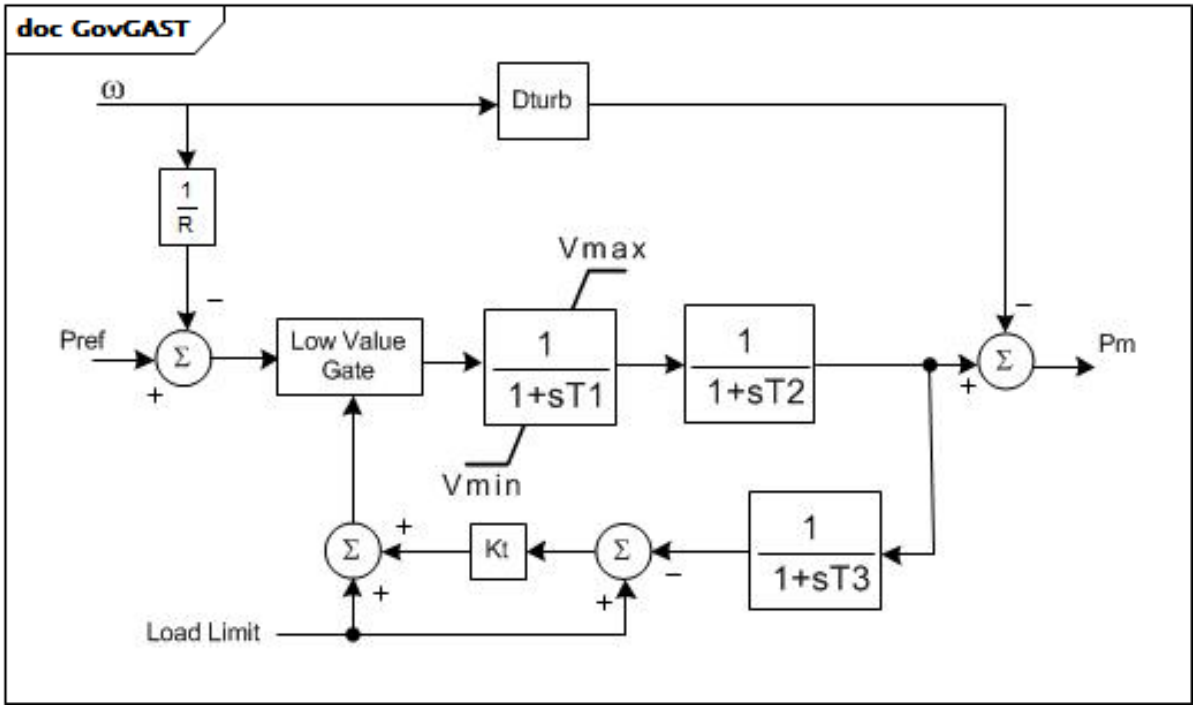


Figure 9-47 - figure GovGAST

Parameter **Notes:**
T1, **T2**, **T3**,and **R** must be greater than zero.

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-48 shows all attributes of GovGAST.

Table 9-48 - Attributes of TurbineGovernorDynamics::GovGAST

Attribute name	mult.	Attribute type	Description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0).
r	1..1	PU	Permanent droop (R). Typical Value = 0.04.
t1	1..1	Seconds	Governor mechanism time constant (T1). T1 represents the natural valve positioning time constant of the governor for small disturbances, as seen when rate limiting is not in effect. Typical Value = 0.5.
t2	1..1	Seconds	Turbine power time constant (T2). T2

			represents delay due to internal energy storage of the gas turbine engine. T2 can be used to give a rough approximation to the delay associated with acceleration of the compressor spool of a multi-shaft engine, or with the compressibility of gas in the plenum of a the free power turbine of an aero-derivative unit, for example. Typical Value = 0.5.
t3	1..1	Seconds	Turbine exhaust temperature time constant (T3). Typical Value = 3.
at	1..1	PU	Ambient temperature load limit (Load Limit). Typical Value = 1.
kt	1..1	PU	Temperature limiter gain (Kt). Typical Value = 3.
vmax	1..1	PU	Maximum turbine power, PU of MWbase (Vmax). Typical Value = 1.
vmin	1..1	PU	Minimum turbine power, PU of MWbase (Vmin). Typical Value = 0.
dturb	1..1	PU	Turbine damping factor (Dturb). Typical Value = 0.18.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-49 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-49 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] --- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] --- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.6.6.9 GovGAST1

Modified single shaft gas turbine.

Figure 9-48 class diagram

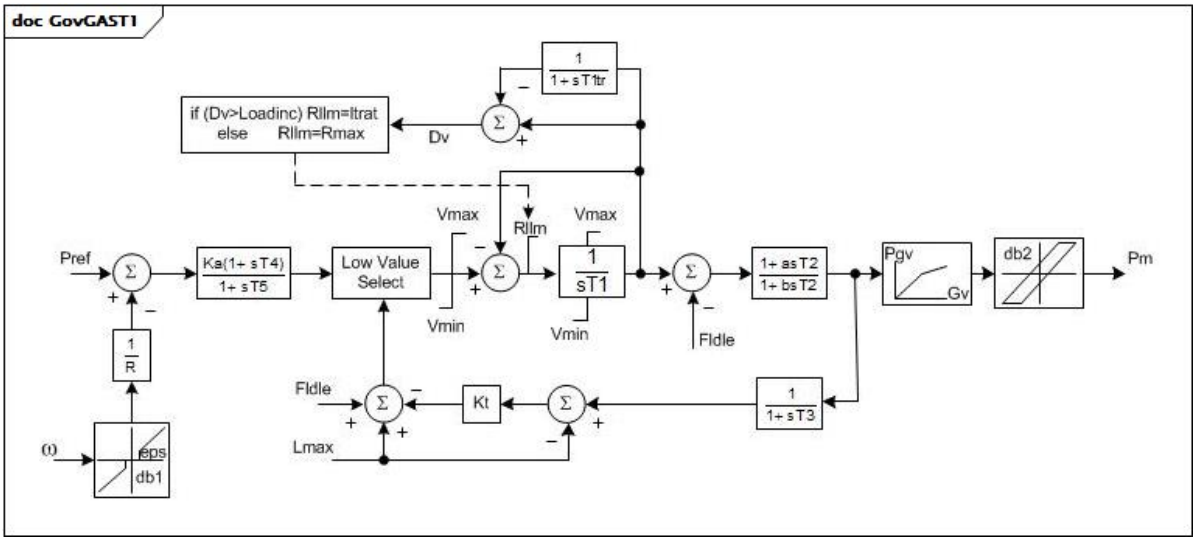


Figure 9-48 - figure GovGAST1

Parameter Notes:

1. **T1**, **T2**, **T3**, **Tltr**, **R**, and **b** must be greater than zero.
2. **Rmax**, **Linc**, **Tltr**, **Ltrate** represent a loading rate limit subsystem. For small excursions about a steady load the fuel valve can open at a rate of **Rmax** PU/sec. The time constant, **Tltr**, is used to obtain a long term average valve position. When a large output increase is called for, the fuel valve is allowed to open at the full rate only until the valve opening has moved ahead of the long term average position by **Linc** PU. After this the valve is permitted to open only at the rate, **Ltrat**, until the long term average position catches up.
3. The nonlinear gain on the output may be input with up to 6 points. The (0.,0.) and (1.,1.) points are assumed and need not be exchanged. The output is not allowed to go beyond 0. and 1. However, if **Vmax** > 1., the input and output are scaled by this value. If the values are set to zero, a straight line is used.
4. If **T5** is zero, the entire gain and lead-lag block is bypassed.

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-50 shows all attributes of GovGAST1.

Table 9-50 - Attributes of TurbineGovernorDynamics::GovGAST1

Attribute name	mult.	Attribute type	Description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
r	1..1	PU	Permanent droop (R). Typical Value = 0.04.
t1	1..1	Seconds	Governor mechanism time constant (T1). T1 represents the natural valve positioning time constant of the governor for small disturbances, as seen when rate limiting is not in effect. Typical Value = 0.5.
t2	1..1	Seconds	Turbine power time constant (T2). T2 represents delay due to internal energy storage of the gas turbine engine. T2 can be used to give a rough approximation to the delay associated with acceleration of the compressor spool of a multi-shaft engine, or with the compressibility of gas in the plenum of the free power turbine of an aero-derivative unit, for example. Typical Value = 0.5.
t3	1..1	Seconds	Turbine exhaust temperature time constant (T3). T3 represents delay in the exhaust temperature and load limiting system. Typical Value = 3.
lmax	1..1	PU	Ambient temperature load limit (Lmax). Lmax is the turbine power output corresponding to the limiting exhaust gas temperature. Typical Value = 1.
kt	1..1	PU	Temperature limiter gain (Kt). Typical Value = 3.
vmax	1..1	PU	Maximum turbine power, PU of MWbase (Vmax). Typical Value = 1.
vmin	1..1	PU	Minimum turbine power, PU of MWbase (Vmin). Typical Value = 0.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

fidle	1..1	PU	Fuel flow at zero power output (Fidle). Typical Value = 0.18.
rmax	1..1	Simple_Float	Maximum fuel valve opening rate (Rmax). Unit = PU/sec. Typical Value = 1.
loadinc	1..1	PU	Valve position change allowed at fast rate (Loadinc). Typical Value = 0.05.
tltr	1..1	Seconds	Valve position averaging time constant (Tltr). Typical Value = 10.
ltrate	1..1	Simple_Float	Maximum long term fuel valve opening rate (Ltrate). Typical Value = 0.02.
a	1..1	Simple_Float	Turbine power time constant numerator scale factor (a). Typical Value = 0.8.
b	1..1	Simple_Float	Turbine power time constant denominator scale factor (b). Typical Value = 1.
db1	1..1	Frequency	Intentional dead-band width (db1). Unit = Hz. Typical Value = 0.
eps	1..1	Frequency	Intentional db hysteresis (eps). Unit = Hz. Typical Value = 0.
db2	1..1	ActivePower	Unintentional dead-band (db2). Unit = MW. Typical Value = 0.
gv1	1..1	PU	Nonlinear gain point 1, PU gv (Gv1). Typical Value = 0.
pgv1	1..1	PU	Nonlinear gain point 1, PU power (Pgv1). Typical Value = 0.
gv2	1..1	PU	Nonlinear gain point 2,PU gv (Gv2). Typical Value = 0.
pgv2	1..1	PU	Nonlinear gain point 2, PU power (Pgv2). Typical Value = 0.
gv3	1..1	PU	Nonlinear gain point 3, PU gv (Gv3). Typical Value = 0.
pgv3	1..1	PU	Nonlinear gain point 3, PU power (Pgv3). Typical Value = 0.
gv4	1..1	PU	Nonlinear gain point 4, PU gv (Gv4). Typical

			Value = 0.
pgv4	1..1	PU	Nonlinear gain point 4, PU power (Pgv4). Typical Value = 0.
gv5	1..1	PU	Nonlinear gain point 5, PU gv (Gv5). Typical Value = 0.
pgv5	1..1	PU	Nonlinear gain point 5, PU power (Pgv5). Typical Value = 0.
gv6	1..1	PU	Nonlinear gain point 6, PU gv (Gv6). Typical Value = 0.
pgv6	1..1	PU	Nonlinear gain point 6, PU power (Pgv6). Typical Value = 0.
ka	1..1	PU	Governor gain (Ka). Typical Value = 0.
t4	1..1	Seconds	Governor lead time constant (T4). Typical Value = 0.
t5	1..1	Seconds	Governor lag time constant (T5). Typical Value = 0.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-51 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-51 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] ----- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] ----- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

9.6.6.10 GovGAST2

Gas turbine model.

Figure 9-49 class diagram

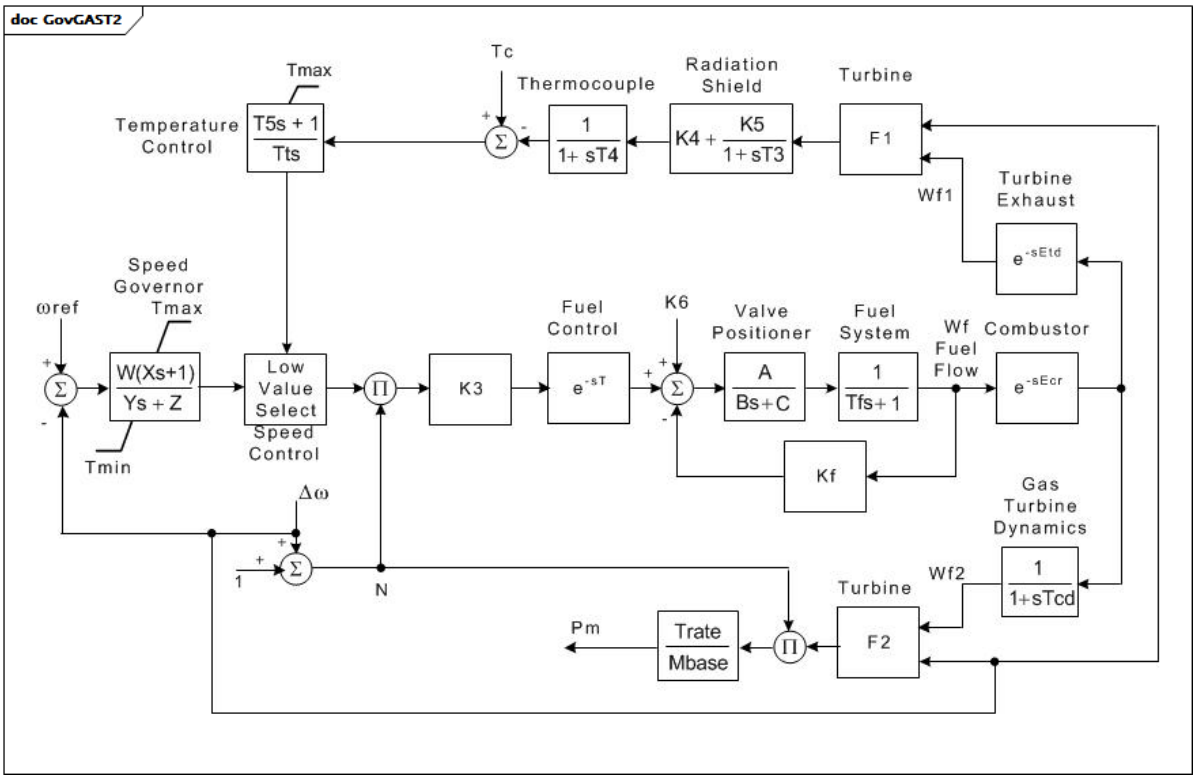


Figure 9-49 - figure GovGAST2

Parameter Notes:

1. Temperature control output is set to output of speed governor when temperature control input changes from positive to negative.
2. **F1** = **Tr** – **Af1**(1 – **Wf1**) – **Bf1**(**delta omega**).
3. **F2** = **Af2** + **Bf2**(**Wf2**) – **Cf2**(**delta omega**).

F2 = K_{hhv}(**Wf2** - 0.23) +0.5(1 - **delta omega**) **F2** = 0.5(1 - **delta omega**) +K_{hhv} x **Wf2** - 0.23 x K_{hhv} **Af2** = 0.5(1 - **delta omega**) **Bf2** = K_{hhv} **Cf2** = 0.23 x K_{hhv}.

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-52 shows all attributes of GovGAST2.

Table 9-52 - Attributes of TurbineGovernorDynamics::GovGAST2

Attribute name	mult.	Attribute type	Description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
w	1..1	PU	Governor gain (1/droop)

			on turbine rating (W).
x	1..1	Seconds	Governor lead time constant (X).
y	1..1	Seconds	Governor lag time constant (Y) (>0).
z	1..1	Boolean	Governor mode (Z). true = Droop false = ISO.
etd	1..1	Seconds	Turbine and exhaust delay (Etd).
tcd	1..1	Seconds	Compressor discharge time constant (Tcd).
trate	1..1	ActivePower	Turbine rating (Trate). Unit = MW.
t	1..1	Seconds	Fuel Control Time Constant (T).
tmax	1..1	PU	Maximum Turbine limit (Tmax).
tmin	1..1	PU	Minimum Turbine limit (Tmin).
ecr	1..1	Seconds	Combustion reaction time delay (Ecr).
k3	1..1	PU	Ratio of Fuel Adjustment (K3).
a	1..1	Simple_Float	Valve positioner (A).
b	1..1	Simple_Float	Valve positioner (B).
c	1..1	Simple_Float	Valve positioner (C).
tf	1..1	Seconds	Fuel system time constant (Tf).
kf	1..1	PU	Fuel system feedback (Kf).
k5	1..1	PU	Gain of radiation shield (K5).
k4	1..1	PU	Gain of radiation shield (K4).
t3	1..1	Seconds	Radiation shield time constant (T3).
t4	1..1	Seconds	Thermocouple time constant (T4).
tt	1..1	Seconds	Temperature controller integration rate (Tt).
t5	1..1	Seconds	Temperature control time constant (T5).
af1	1..1	PU	Exhaust temperature Parameter (Af1). Unit = per unit temperature. Based on temperature in degrees C.
bf1	1..1	PU	(Bf1). Bf1 = E(1-w)

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			where E (speed sensitivity coefficient) is 0.55 to 0.65 x Tr. Unit = per unit temperature. Based on temperature in degrees C.
af2	1..1	PU	Coefficient equal to 0.5(1-speed) (Af2).
bf2	1..1	PU	Turbine Torque Coefficient K_{hhv} (depends on heating value of fuel stream in combustion chamber) (Bf2).
cf2	1..1	PU	Coefficient defining fuel flow where power output is 0% (Cf2). Synchronous but no output. Typically 0.23 x K_{hhv} (23% fuel flow).
tr	1..1	Temperature	Rated temperature (Tr). Unit = °C depending on parameters Af1 and Bf1.
k6	1..1	PU	Minimum fuel flow (K6).
tc	1..1	Temperature	Temperature control (Tc). Unit = °F or °C depending on constants Af1 and Bf1.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-53 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-53 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] ----- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] ----- TurbineGovernorDynamics::

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.6.6.11 GovGAST3

Generic turbogas with acceleration and temperature controller.

Figure 9-50 class diagram

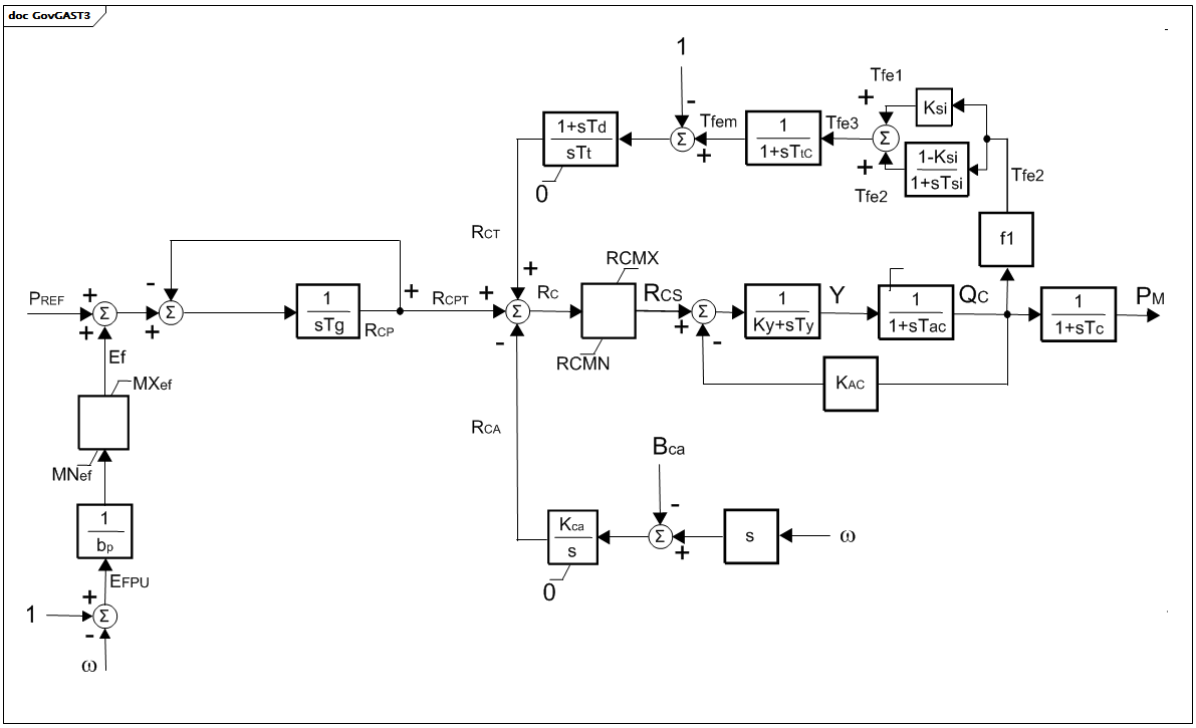


Figure 9-50 - figure GovGAST3

Parameter Notes:

1. Per unit parameters are on base of MW capability of the turbine.
2. omega has units of per unit speed.
3. Function for exhaust temperature calculation shown in Gov33TGTExhaustTemperature diagram.

Figure 9-51 class diagram

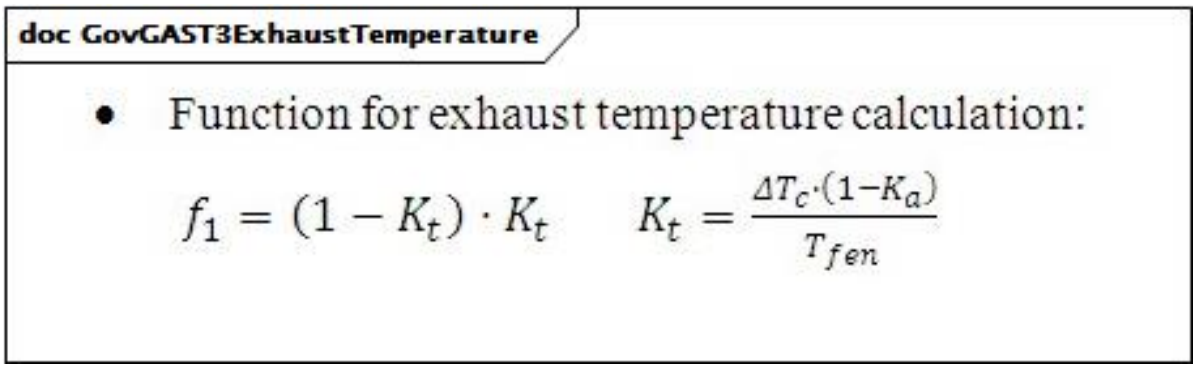


Figure 9-51 - figure GovGAST3ExhaustTemperature

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-54 shows all attributes of GovGAST3.

Table 9-54 - Attributes of TurbineGovernorDynamics::GovGAST3

Attribute name	mult.	Attribute type	Description
bp	1..1	PU	Droop (bp). Typical Value = 0.05.
tg	1..1	Seconds	Time constant of speed governor (Tg). Typical Value = 0.05.
rcmx	1..1	PU	Maximum fuel flow (RCMX). Typical Value = 1.
rcmn	1..1	PU	Minimum fuel flow (RCMN). Typical Value = -0.1.
ky	1..1	Simple_Float	Coefficient of transfer function of fuel valve positioner (Ky). Typical Value = 1.
ty	1..1	Seconds	Time constant of fuel valve positioner (Ty). Typical Value = 0.2.
tac	1..1	Seconds	Fuel control time constant (Tac). Typical Value = 0.1.
kac	1..1	Simple_Float	Fuel system feedback (K_{AC}). Typical Value = 0.
tc	1..1	Seconds	Compressor discharge volume time constant (Tc). Typical Value = 0.2.
bca	1..1	Simple_Float	Acceleration limit set-point (Bca). Unit = 1/s. Typical Value = 0.01.
kca	1..1	Simple_Float	Acceleration control integral gain (Kca). Unit = 1/s. Typical Value = 100.
dtc	1..1	Temperature	Exhaust temperature variation due to fuel flow increasing from 0 to 1 PU (deltaTc). Typical Value = 390.
ka	1..1	PU	Minimum fuel flow (Ka). Typical Value = 0.23.
tsi	1..1	Seconds	Time constant of radiation shield (Tsi). Typical Value = 15.
ksi	1..1	Simple_Float	Gain of radiation shield (Ksi). Typical Value =

			0.8.
ttc	1..1	Seconds	Time constant of thermocouple (Ttc). Typical Value = 2.5.
tfen	1..1	Temperature	Turbine rated exhaust temperature correspondent to Pm=1 PU (Tfen). Typical Value = 540.
td	1..1	Seconds	Temperature controller derivative gain (Td). Typical Value = 3.3.
tt	1..1	Temperature	Temperature controller integration rate (Tt). Typical Value = 250.
mxef	1..1	PU	Fuel flow maximum positive error value (MX_{EF}). Typical Value = 0.05.
mnef	1..1	PU	Fuel flow maximum negative error value (MN_{EF}). Typical Value = -0.05.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-55 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-55 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] --- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] --- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.6.6.12 GovGAST4

Generic turbogas.

Figure 9-52 class diagram

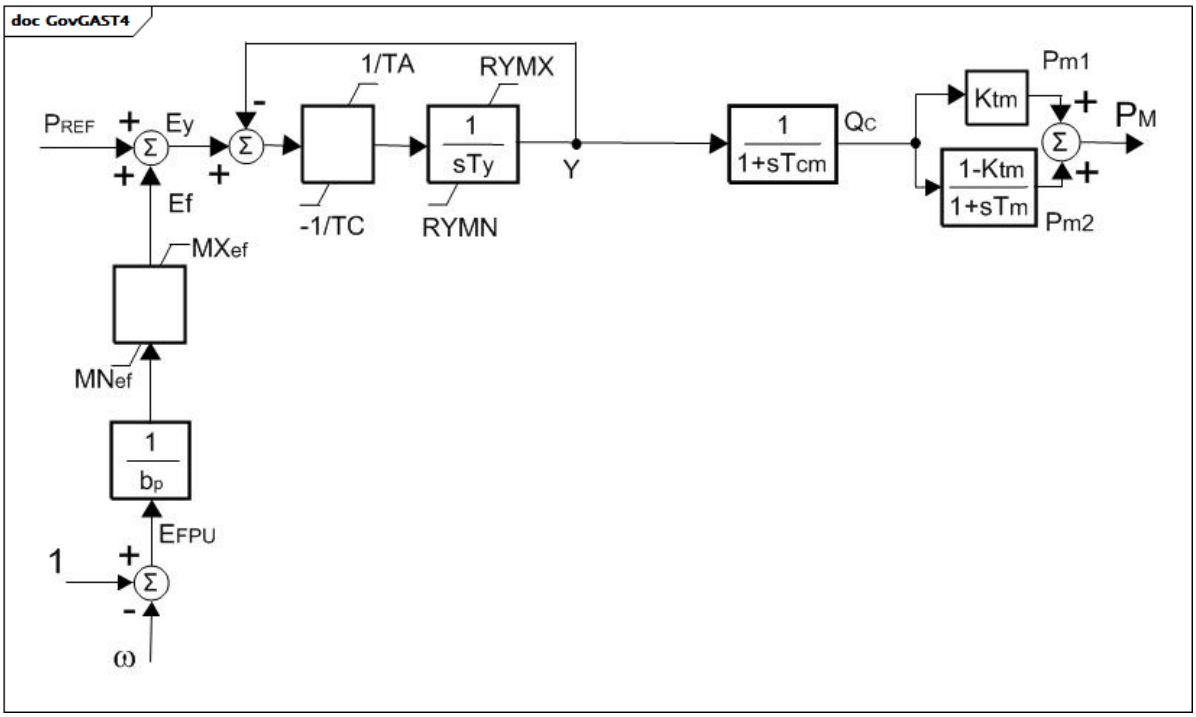


Figure 9-52 - figure GovGAST4

Parameter Notes:

1. Per unit parameters are on base of MW capability of the turbine.
2. omega has units of per unit speed.

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-56 shows all attributes of GovGAST4.

Table 9-56 - Attributes of TurbineGovernorDynamics::GovGAST4

Attribute name	mult.	Attribute type	Description
bp	1..1	PU	Droop (bp). Typical Value = 0.05.
tv	1..1	Seconds	Time constant of fuel valve positioner (T_y). Typical Value = 0.1.
ta	1..1	Seconds	Maximum gate opening velocity (T_A). Typical Value = 3.
tc	1..1	Seconds	Maximum gate closing velocity (T_c). Typical Value = 0.5.
tcm	1..1	Seconds	Fuel control time constant (T_{cm}). Typical Value = 0.1.
ktm	1..1	PU	Compressor gain (K_{tm}).

			Typical Value = 0.
tm	1..1	Seconds	Compressor discharge volume time constant (T_m). Typical Value = 0.2.
rymx	1..1	PU	Maximum valve opening (RYMX). Typical Value = 1.1.
rymn	1..1	PU	Minimum valve opening (RYMN). Typical Value = 0.
mxef	1..1	PU	Fuel flow maximum positive error value (MX_{EF}). Typical Value = 0.05.
mnef	1..1	PU	Fuel flow maximum negative error value (MN_{EF}). Typical Value = -0.05.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-57 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-57 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] --- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] --- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.6.6.13 GovGASTWD

Woodward Gas turbine governor model.

Figure 9-53 class diagram

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

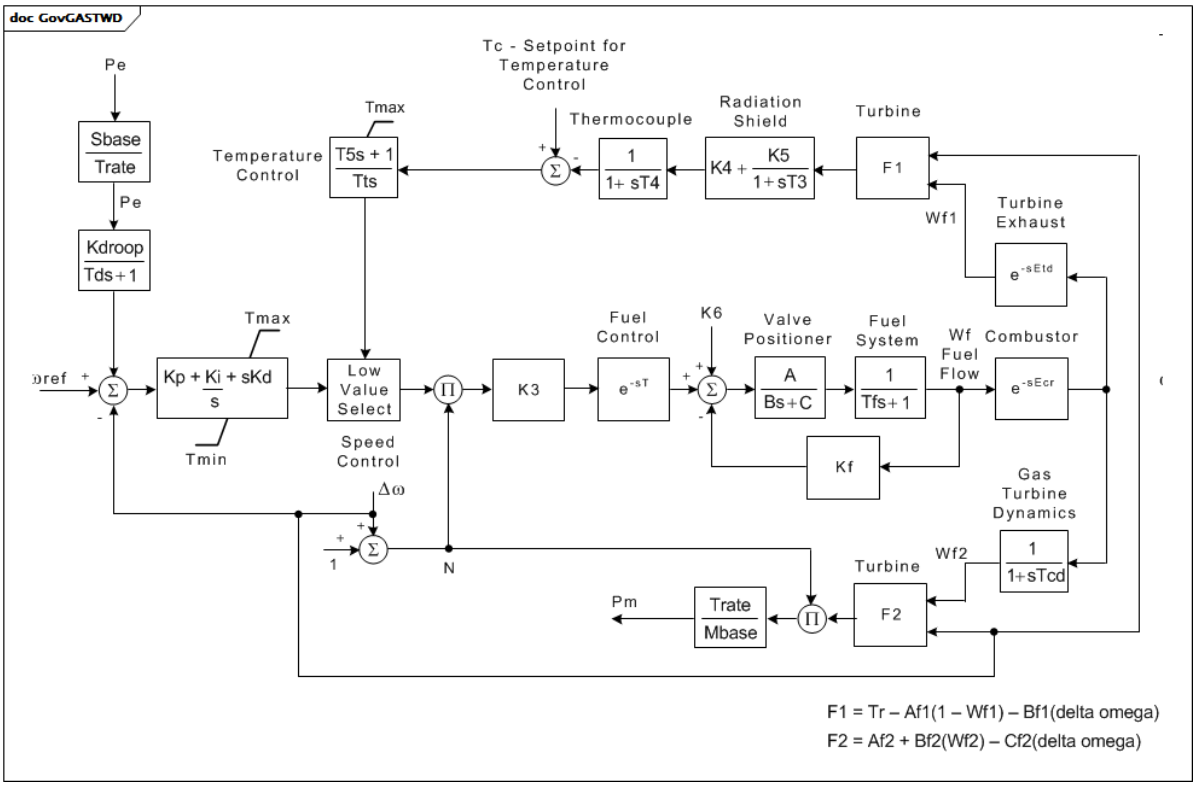


Figure 9-53 - figure GovGASTWD

Parameter Notes:

1. Temperature control output is set to output of speed governor when temperature control input changes from positive to negative.
2. **F1** = **Tr** – **Af1**(1 – **Wf1**) – **Bf1**(**delta omega**)
3. **F2** = **Af2** + **Bf2**(**Wf2**) – **Cf2**(**delta omega**).

F2 = K_{hhv}(**Wf2** - 0.23) +0.5(1 - **delta omega**) **F2** = 0.5(1 - **delta omega**) +K_{hhv} x **Wf2** - 0.23 x K_{hhv} **Af2** = 0.5(1 - **delta omega**) **Bf2** = K_{hhv} **Cf2** = 0.23 x K_{hhv}.

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-58 shows all attributes of GovGASTWD.

Table 9-58 - Attributes of TurbineGovernorDynamics::GovGASTWD

Attribute name	mult.	Attribute type	Description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
kdroop	1..1	PU	(Kdroop).
kp	1..1	PU	PID Proportional gain (Kp).
ki	1..1	PU	Isochronous Governor Gain (Ki).

kd	1..1	PU	Drop Governor Gain (Kd).
etd	1..1	Seconds	Turbine and exhaust delay (Etd).
tcd	1..1	Seconds	Compressor discharge time constant (Tcd).
trate	1..1	ActivePower	Turbine rating (Trate). Unit = MW.
t	1..1	Seconds	Fuel Control Time Constant (T).
tmax	1..1	PU	Maximum Turbine limit (Tmax).
tmin	1..1	PU	Minimum Turbine limit (Tmin).
ecr	1..1	Seconds	Combustion reaction time delay (Ecr).
k3	1..1	PU	Ratio of Fuel Adjustment (K3).
a	1..1	Simple_Float	Valve positioner (*A*).
b	1..1	Simple_Float	Valve positioner (*B*).
c	1..1	Simple_Float	Valve positioner (*C*).
tf	1..1	Seconds	Fuel system time constant (Tf).
kf	1..1	PU	Fuel system feedback (Kf).
k5	1..1	PU	Gain of radiation shield (K5).
k4	1..1	PU	Gain of radiation shield (K4).
t3	1..1	Seconds	Radiation shield time constant (T3).
t4	1..1	Seconds	Thermocouple time constant (T4).
tt	1..1	Seconds	Temperature controller integration rate (Tt).
t5	1..1	Seconds	Temperature control time constant (T5).
af1	1..1	PU	Exhaust temperature Parameter (Af1).
bf1	1..1	PU	(Bf1). Bf1 = E(1-w) where E (speed sensitivity coefficient) is 0.55 to 0.65 x Tr.
af2	1..1	PU	Coefficient equal to 0.5(1-speed) (Af2).
bf2	1..1	PU	Turbine Torque Coefficient K_{hhv} (depends on heating value of fuel stream in combustion chamber)

			(Bf2).
cf2	1..1	PU	Coefficient defining fuel flow where power output is 0% (Cf2). Synchronous but no output. Typically 0.23 x K_{hhv} (23% fuel flow).
tr	1..1	Temperature	Rated temperature (Tr).
k6	1..1	PU	Minimum fuel flow (K6).
tc	1..1	Temperature	Temperature control (Tc).
td	1..1	Seconds	Power transducer time constant (Td).
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-59 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-59 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] --- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] --- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.6.6.14 GovHydro1

Basic Hydro turbine governor model.

Figure 9-54 class diagram

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

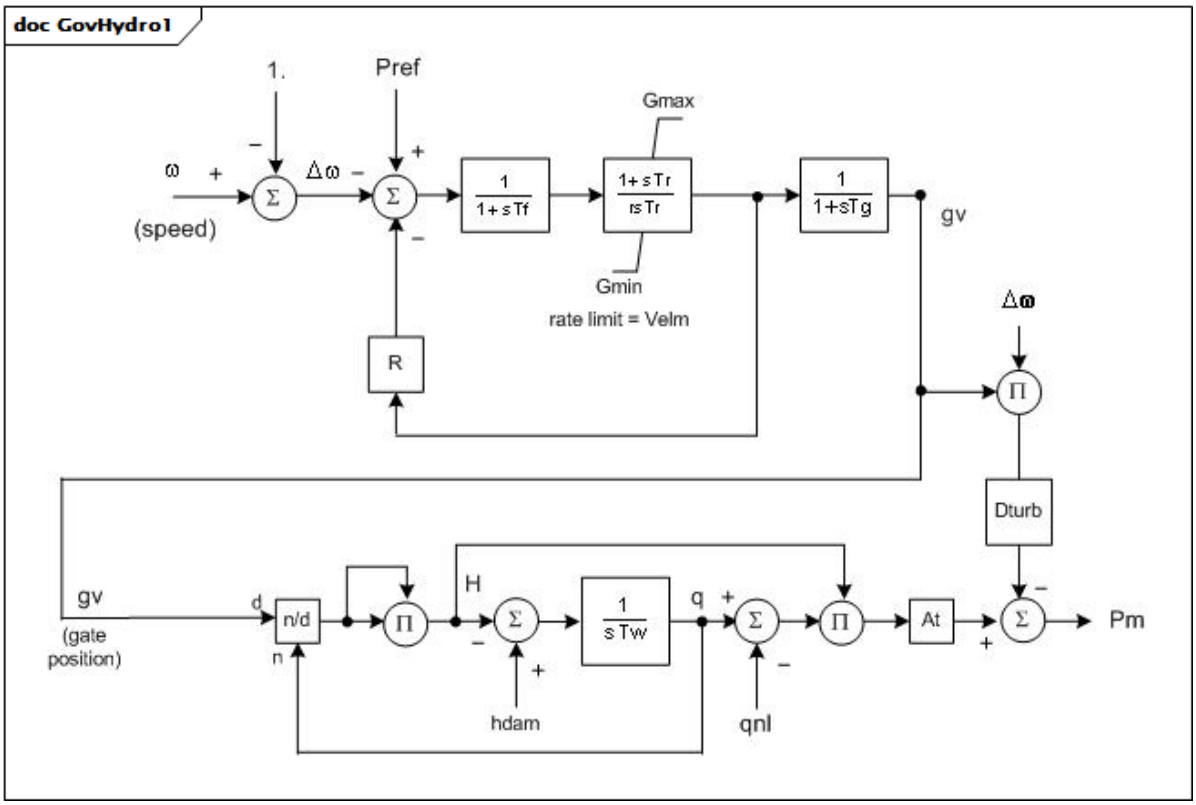


Figure 9-54 - figure GovHydro1

Parameter Notes:

1. Per unit parameters are on base of **MWbase**, which is normally the MW capability of the turbine.
2. The gates travel over a range of 1.0 per unit from fully closed to fully opened. The gate position is normally greater than zero at zero power and normally less than 1.0 when power is 1.0 PU. **Gmax** and **Gmin** are operating limits.
3. **Tr**, **Tf**, **Tg**, **Tw** must be greater than zero.
4. **Dturb** has the dimensions delta P(PU of mwcap) /delta speed(PU).

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-60 shows all attributes of GovHydro1.

Table 9-60 - Attributes of TurbineGovernorDynamics::GovHydro1

Attribute name	mult.	Attribute type	Description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
rperm	1..1	PU	Permanent droop (R) (>0). Typical Value = 0.04.
rtemp	1..1	PU	Temporary droop (r) (>R). Typical Value =

			0.3.
tr	1..1	Seconds	Washout time constant (Tr) (>0). Typical Value = 5.
tf	1..1	Seconds	Filter time constant (*Tf*) (>0). Typical Value = 0.05.
tg	1..1	Seconds	Gate servo time constant (Tg) (>0). Typical Value = 0.5.
velm	1..1	Simple_Float	Maximum gate velocity (Vlem) (>0). Typical Value = 0.2.
gmax	1..1	PU	Maximum gate opening (Gmax) (>0). Typical Value = 1.
gmin	1..1	PU	Minimum gate opening (Gmin) (>=0). Typical Value = 0.
tw	1..1	Seconds	Water inertia time constant (Tw) (>0). Typical Value = 1.
at	1..1	PU	Turbine gain (At) (>0). Typical Value = 1.2.
dturb	1..1	PU	Turbine damping factor (Dturb) (>=0). Typical Value = 0.5.
qnl	1..1	PU	No-load flow at nominal head (qnl) (>=0). Typical Value = 0.08.
hdam	1..1	PU	Turbine nominal head (hdam). Typical Value = 1.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-61 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-61 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] --- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] ----- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.6.6.15 GovHydro2

IEEE hydro turbine governor model represents plants with straightforward penstock configurations and hydraulic-dashpot governors.

Figure 9-55 class diagram

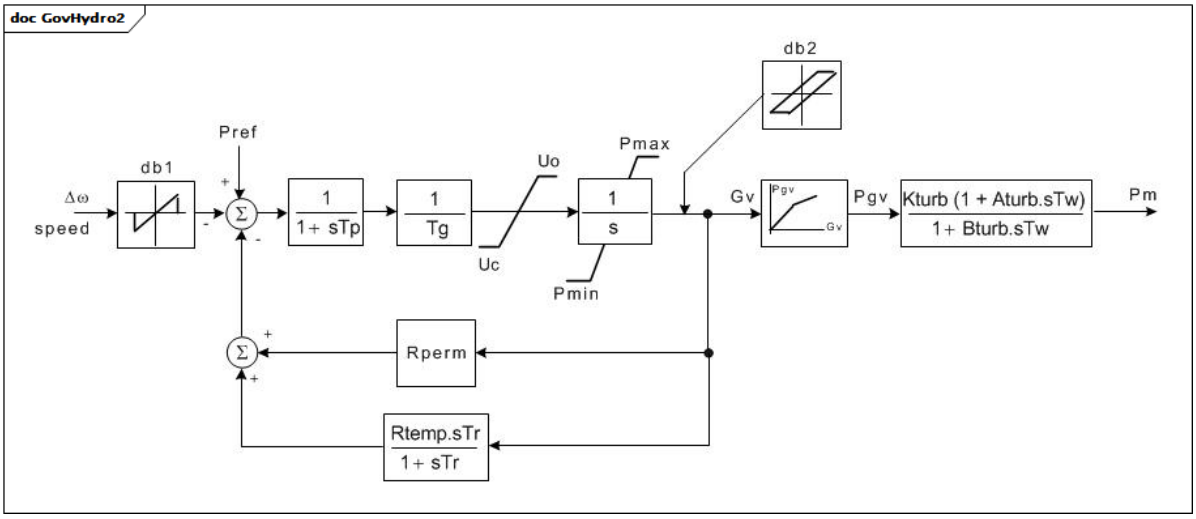


Figure 9-55 - figure GovHydro2

Parameter Notes:

1. Per unit parameters are on base of **MWbase**, which is normally the MW capability of the turbine.
2. **Tr**, **Tg**, **Tw**, **Kturb**, and **Bturb** must be greater than zero
3. This turbine governor model is a reasonable approximation to the performance of simple hydro plants for small motions of the gates about a given load, but it represents the variation of water inertia effect with load only if the values of the turbine parameters are properly adjusted by the user as load is varied. For idealized turbine model, set **Kturb** = 1., **Aturb** = -1., **Bturb** = 0.5.
4. The nonlinear gain between gate position and power may be input with up to 6 points. The (0.,0.) and (1.,1.) points are assumed and need not be exchanged. The output is not allowed to go beyond 0. and 1. However, if **Pmax** > 1., the input and output are scaled by **Pmax**. If **Gv1** is a negative number, the default hydro curve will be used. If zero values are provided, a straight line is used.
5. For compatibility with older versions of this model: **Kturb** = a23, **Aturb** = a11 – a13*a21/a23, **Bturb** = a11.

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-62 shows all attributes of GovHydro2.

Table 9-62 - Attributes of TurbineGovernorDynamics::GovHydro2

Attribute name	mult.	Attribute type	Description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
tg	1..1	Seconds	Gate servo time constant (Tg). Typical Value = 0.5.
tp	1..1	Seconds	Pilot servo valve time constant (Tp). Typical Value = 0.03.
uo	1..1	Simple_Float	Maximum gate opening velocity (Uo). Unit = PU/sec. Typical Value = 0.1.
uc	1..1	Simple_Float	Maximum gate closing velocity (Uc) (<0). Unit = PU/sec. Typical Value = -0.1.
pmax	1..1	PU	Maximum gate opening (Pmax). Typical Value = 1.
pmin	1..1	PU	Minimum gate opening; (*Pmin*). Typical Value = 0.
rperm	1..1	PU	Permanent droop (Rperm). Typical Value = 0.05.
rtemp	1..1	PU	Temporary droop (Rtemp). Typical Value = 0.5.
tr	1..1	Seconds	Dashpot time constant (Tr). Typical Value = 12.
tw	1..1	Seconds	Water inertia time constant (Tw). Typical Value = 2.
kturb	1..1	PU	Turbine gain (Kturb). Typical Value = 1.
aturb	1..1	PU	Turbine numerator multiplier (Aturb). Typical Value = -1.
bturb	1..1	PU	Turbine denominator multiplier (Bturb). Typical Value = 0.5.
db1	1..1	Frequency	Intentional deadband width (db1). Unit = Hz. Typical Value = 0.
eps	1..1	Frequency	Intentional db hysteresis (eps). Unit = Hz. Typical Value = 0.
db2	1..1	ActivePower	Unintentional deadband (db2). Unit = MW. Typical Value = 0.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

gv1	1..1	PU	Nonlinear gain point 1, PU gv (Gv1). Typical Value = 0.
pgv1	1..1	PU	Nonlinear gain point 1, PU power (Pgv1). Typical Value = 0.
gv2	1..1	PU	Nonlinear gain point 2, PU gv (Gv2). Typical Value = 0.
pgv2	1..1	PU	Nonlinear gain point 2, PU power (Pgv2). Typical Value = 0.
gv3	1..1	PU	Nonlinear gain point 3, PU gv (Gv3). Typical Value = 0.
pgv3	1..1	PU	Nonlinear gain point 3, PU power (Pgv3). Typical Value = 0.
gv4	1..1	PU	Nonlinear gain point 4, PU gv (Gv4). Typical Value = 0.
pgv4	1..1	PU	Nonlinear gain point 4, PU power (Pgv4). Typical Value = 0.
gv5	1..1	PU	Nonlinear gain point 5, PU gv (Gv5). Typical Value = 0.
pgv5	1..1	PU	Nonlinear gain point 5, PU power (Pgv5). Typical Value = 0.
gv6	1..1	PU	Nonlinear gain point 6, PU gv (Gv6). Typical Value = 0.
pgv6	1..1	PU	Nonlinear gain point 6, PU power (Pgv6). Typical Value = 0.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-63 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-63 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] -----

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] --- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.6.6.16 GovHydro3

Modified IEEE Hydro Governor-Turbine Model. This model differs from that defined in the IEEE modeling guideline paper in that the limits on gate position and velocity do not permit "wind up" of the upstream signals.

Figure 9-56 class diagram

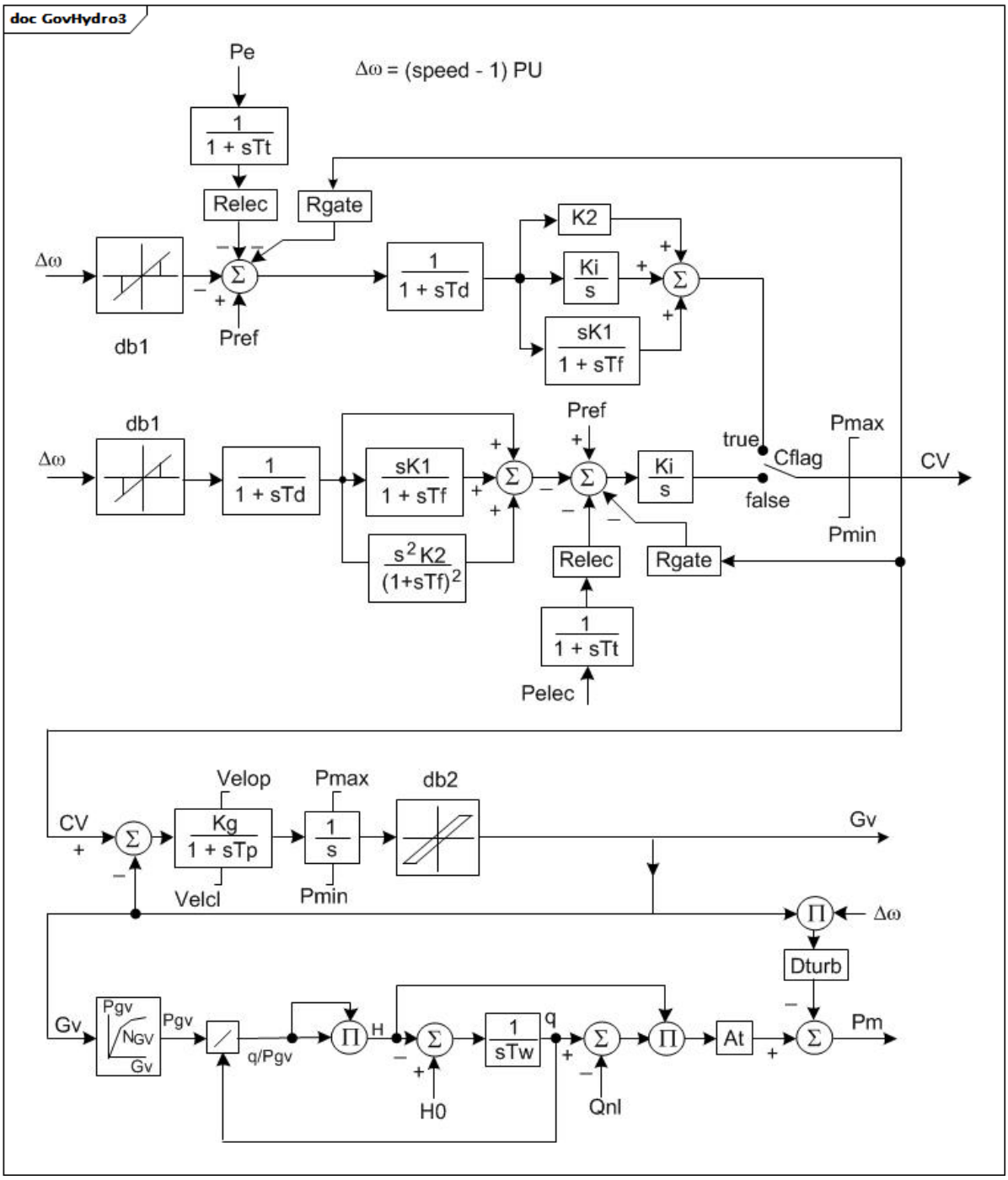


Figure 9-56 - figure GovHydro3

Parameter Notes:

1. **Tt**, **Tg**, **Tw**, and **At** must be greater than zero.
2. The nonlinear gain between gate position and power may be input with up to 6 points. The (0.,0.) and (1.,1.) points are assumed and need not be exchanged. The output is not allowed to go beyond 0. and 1. However, if **Pmax** > 1., the input and output are scaled by

Pmax. If **Gv1** is a negative number, the default hydro curve will be used. If all zero values are provided, a straight line is used.

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-64 shows all attributes of GovHydro3.

Table 9-64 - Attributes of TurbineGovernorDynamics::GovHydro3

Attribute name	mult.	Attribute type	Description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
pmax	1..1	PU	Maximum gate opening, PU of MWbase (Pmax). Typical Value = 1.
pmin	1..1	PU	Minimum gate opening, PU of MWbase (Pmin). Typical Value = 0.
governorControl	1..1	Boolean	Governor control flag (Cflag). true = PID control is active false = double derivative control is active. Typical Value = true.
rgate	1..1	PU	Steady-state droop, PU, for governor output feedback (Rgate). Typical Value = 0.
relec	1..1	PU	Steady-state droop, PU, for electrical power feedback (Relec). Typical Value = 0.05.
td	1..1	Seconds	Input filter time constant (Td). Typical Value = 0.05.
tf	1..1	Seconds	Washout time constant (Tf). Typical Value = 0.1.
tp	1..1	Seconds	Gate servo time constant (Tp). Typical Value = 0.05.
velop	1..1	Simple_Float	Maximum gate opening velocity (Velop). Unit = PU/sec. Typical Value = 0.2.
velcl	1..1	Simple_Float	Maximum gate closing velocity (Velcl). Unit = PU/sec. Typical Value = -0.2.
k1	1..1	PU	Derivative gain (K1). Typical Value = 0.01.
k2	1..1	PU	Double derivative gain, if Cflag = -1 (K2). Typical Value = 2.5.

ki	1..1	PU	Integral gain (Ki). Typical Value = 0.5.
kg	1..1	PU	Gate servo gain (Kg). Typical Value = 2.
tt	1..1	Seconds	Power feedback time constant (Tt). Typical Value = 0.2.
db1	1..1	Frequency	Intentional dead-band width (db1). Unit = Hz. Typical Value = 0.
eps	1..1	Frequency	Intentional db hysteresis (eps). Unit = Hz. Typical Value = 0.
db2	1..1	ActivePower	Unintentional dead-band (db2). Unit = MW. Typical Value = 0.
tw	1..1	Seconds	Water inertia time constant (Tw). Typical Value = 1.
at	1..1	PU	Turbine gain (At). Typical Value = 1.2.
dturb	1..1	PU	Turbine damping factor (Dturb). Typical Value = 0.2.
qnl	1..1	PU	No-load turbine flow at nominal head (Qnl). Typical Value = 0.08.
h0	1..1	PU	Turbine nominal head (H0). Typical Value = 1.
gv1	1..1	PU	Nonlinear gain point 1, PU gv (Gv1). Typical Value = 0.
pgv1	1..1	PU	Nonlinear gain point 1, PU power (Pgv1). Typical Value = 0.
gv2	1..1	PU	Nonlinear gain point 2, PU gv (Gv2). Typical Value = 0.
pgv2	1..1	PU	Nonlinear gain point 2, PU power (Pgv2). Typical Value = 0.
gv3	1..1	PU	Nonlinear gain point 3, PU gv (Gv3). Typical Value = 0.
pgv3	1..1	PU	Nonlinear gain point 3, PU power (Pgv3). Typical Value = 0.
gv4	1..1	PU	Nonlinear gain point 4, PU gv (Gv4). Typical Value = 0.
pgv4	1..1	PU	Nonlinear gain point 4, PU power (Pgv4). Typical Value = 0.

gv5	1..1	PU	Nonlinear gain point 5, PU gv (Gv5). Typical Value = 0.
pgv5	1..1	PU	Nonlinear gain point 5, PU power (Pgv5). Typical Value = 0.
gv6	1..1	PU	Nonlinear gain point 6, PU gv (Gv6). Typical Value = 0.
pgv6	1..1	PU	Nonlinear gain point 6, PU power (Pgv6). Typical Value = 0.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-65 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-65 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] ----- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] ----- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.6.6.17 GovHydro4

Hydro turbine and governor. Represents plants with straight-forward penstock configurations and hydraulic governors of traditional 'dashpot' type. This model can be used to represent simple, Francis, Pelton or Kaplan turbines.

Figure 9-57 class diagram

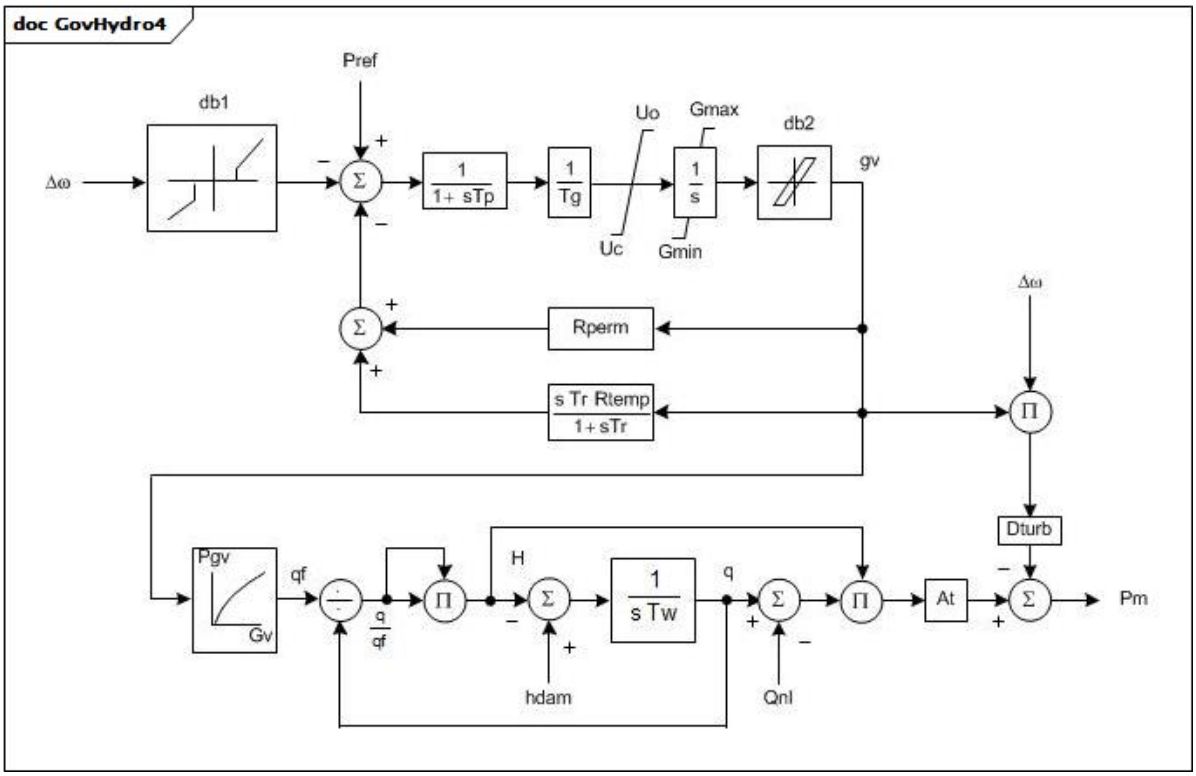


Figure 9-57 - figure GovHydro4

Parameter Notes:

1. The gates travel over a range of 1.0 per unit from fully closed to fully opened. The gates are at a position greater than zero at zero power and, normally, less than 1.0 when power is 1.0 PU **Gmax** and **Gmin** are operating limits.
2. The characteristics of the turbine can be represented by three different gate-head-flow-power characteristics: a) Simple General Purpose Model: a simple linear relationship given by **At, Qnl** as shown in the GovHydro4SimpleHydroTurbine diagram, b) Francis/Pelton Turbine Model: a nonlinear relationship between flow and gate opening as shown in the GovHydro4FrancisPeltonTurbine diagram and described by **Gv0, Pgv0**, thru **Pgv5** and c) Kaplan Turbine Model: a nonlinear relationship between flow, gate, and blade angle as shown in the GovHydro4KaplanTurbine diagram and described by **Gv0**, **Pgv0**, thru **Pgv5**, and **Bgv0** thru **Bgv5**. The Kaplan turbine model is the same as for the GovHydro1 model.
3. The simple turbine model is selected by omitting **Gv0** and all following parameters from the parameter list. (Note - you could set **Gv0** and all following parameters to zero, except **hdam** which should be set to the appropriate nonzero value to describe the available head). The simple model will normally have a value slightly greater than unity for **At** and a value in the range 0 - 0.15 for **Qnl**.
4. The Francis/Pelton turbine model is invoked by specifying values for **Gv0**, **Pgv0**, through **Gv5**, **Pgv5** as shown in the GovHydro4FrancisPeltonTurbine diagram. The values of **Gv0** - **Gv5** must be in increasing order and **Gv5** must be less than unity. The values of **Pgv0** must be in increasing order and the value of **Pgv5** must be less than unity. If **Gv0** is greater than 0., the first two data points will be used to extrapolate downward for gate positions less than **Gv0**. The curve described by **Gv0** - **Pgv5** describes the relationship of flow through the turbine to stroke of the gate actuating servomotor. This curve is proportional to turbine power, with the constant of proportionality being a function of available head. This constant must be specified by the turbine gain, **At**. If **Gv0** is a negative

number, the default hydro curve will be used. If all zero values are provided, a straight line is used. This option is provided for compatibility with older data bases of the now-discontinued WSCC dynamic simulation program and should be used only when needed for that compatibility.

Figure 9-58 class diagram

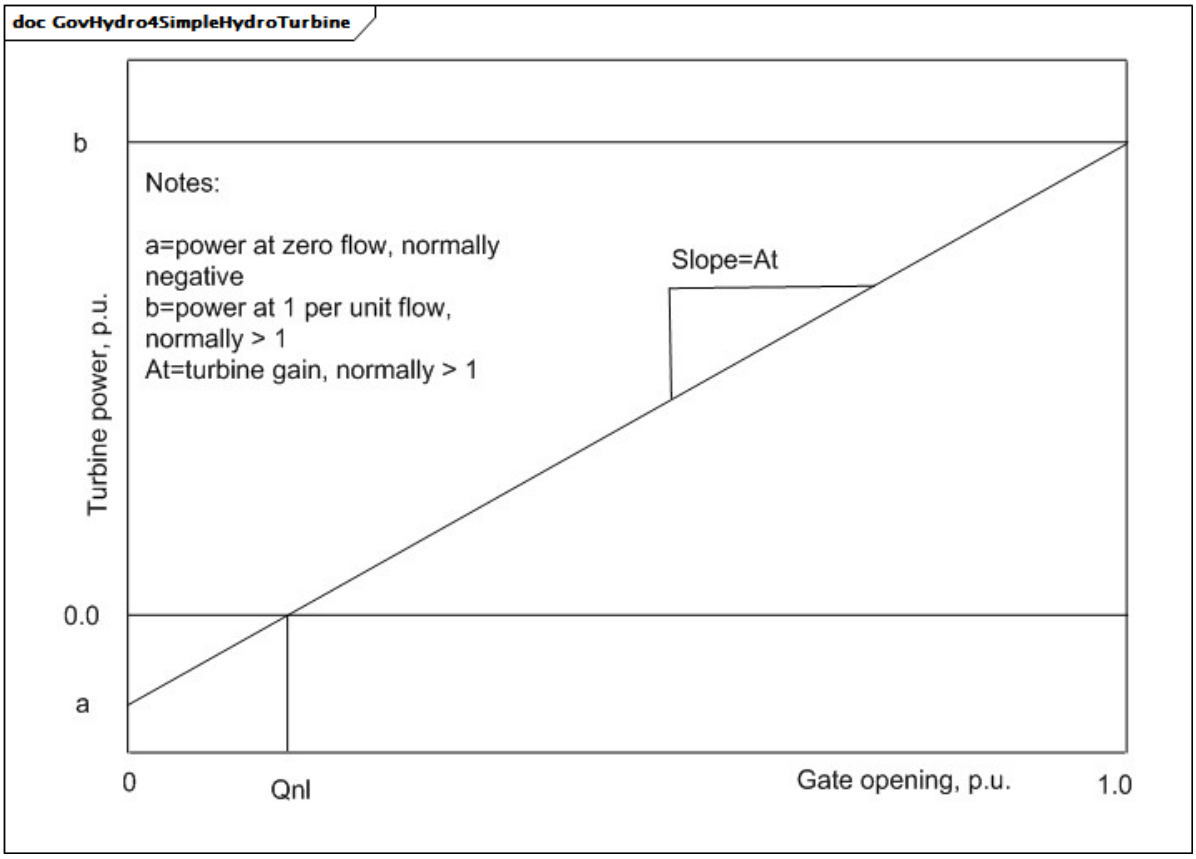


Figure 9-58 - figure GovHydro4SimpleHydroTurbine

Figure 9-59 class diagram

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

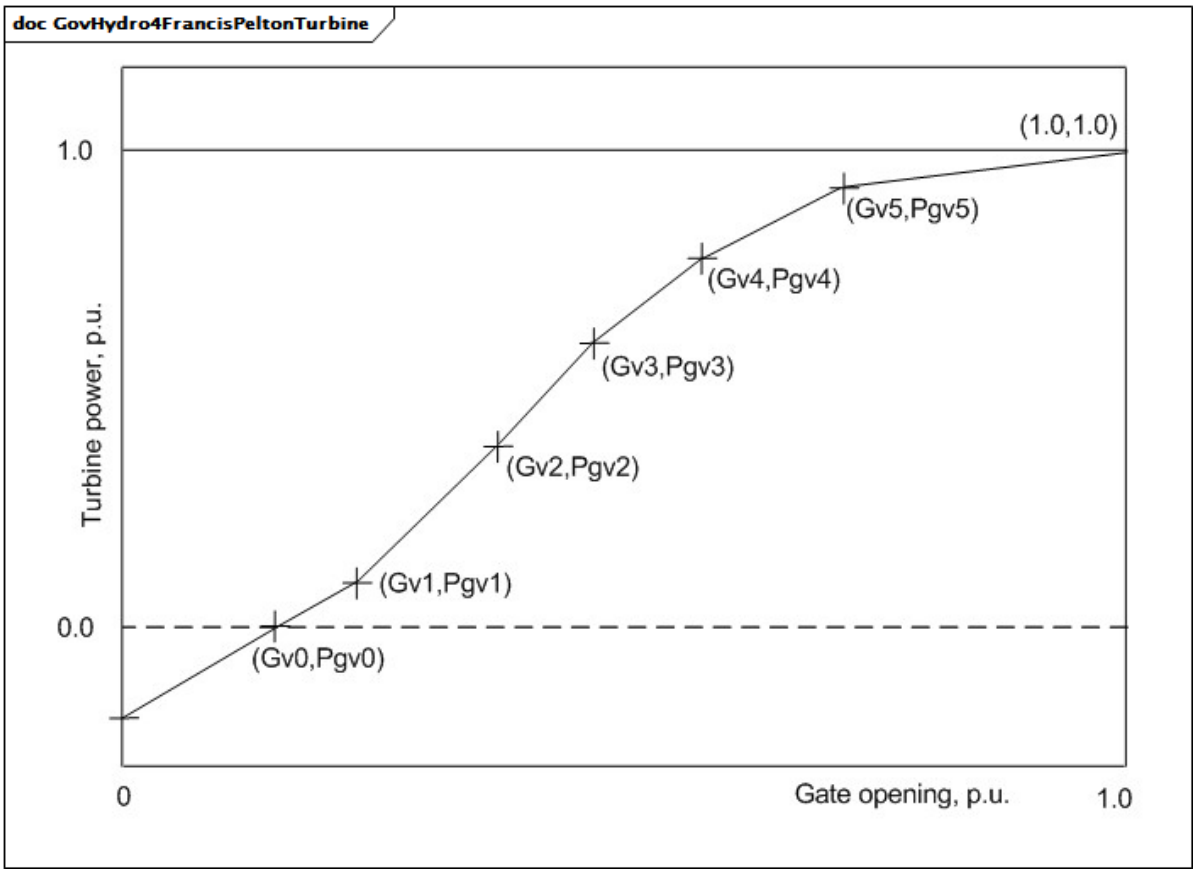


Figure 9-59 - figure GovHydro4FrancisPeltonTurbine

Figure 9-60 class diagram

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

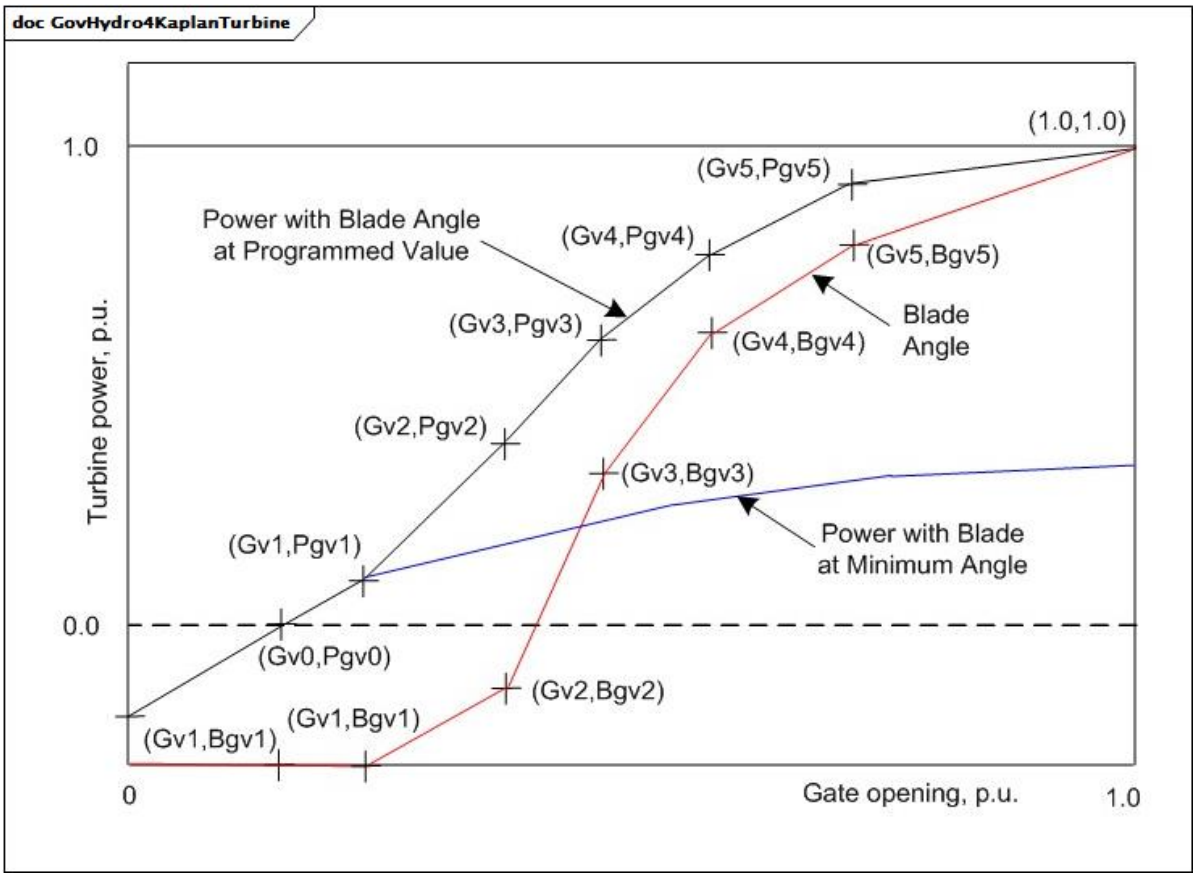


Figure 9-60 - figure GovHydro4KaplanTurbine

Figure 9-61 class diagram

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

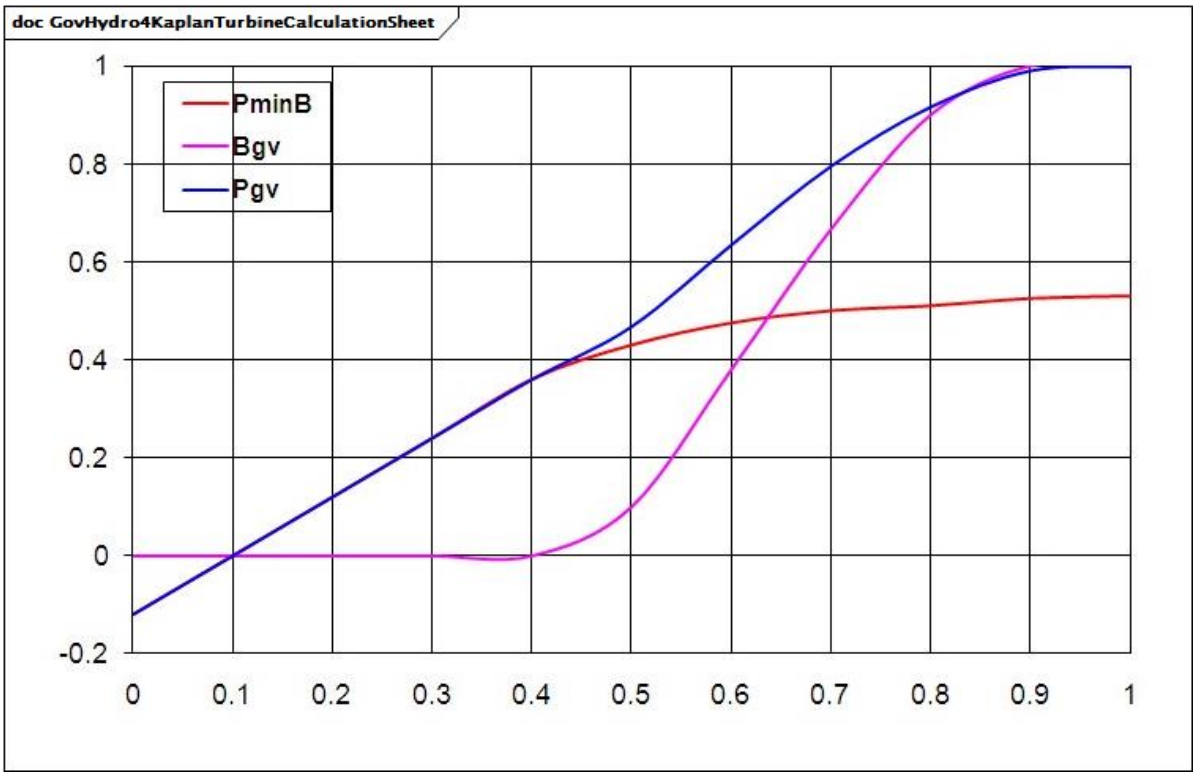


Figure 9-61 - figure GovHydro4KaplanTurbineCalculationSheet

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-66 shows all attributes of GovHydro4.

Table 9-66 - Attributes of TurbineGovernorDynamics::GovHydro4

Attribute name	mult.	Attribute type	Description
mwbase	1..1	ActivePower	Base for power values (MWbase) (>0). Unit = MW.
tg	1..1	Seconds	Gate servo time constant (Tg) (>0). Typical Value = 0.5.
tp	1..1	Seconds	Pilot servo time constant (Tp). Typical Value = 0.1.
uo	1..1	Simple_Float	Max gate opening velocity (Uo). Typical Vlaue = 0.2.
uc	1..1	Simple_Float	Max gate closing velocity (Uc). Typical Value = 0.2.
gmax	1..1	PU	Maximum gate opening, PU of MWbase (Gmax). Typical Value = 1.
gmin	1..1	PU	Minimum gate opening, PU of MWbase (Gmin). Typical Value = 0.

rperm	1..1	Seconds	Permanent droop (Rperm). Typical Value = 0.05.
rtemp	1..1	Seconds	Temporary droop (Rtemp). Typical Value = 0.3.
tr	1..1	Seconds	Dashpot time constant (Tr) (>0). Typical Value = 5.
tw	1..1	Seconds	Water inertia time constant (Tw) (>0). Typical Value = 1.
at	1..1	PU	Turbine gain (At). Typical Value = 1.2.
dturb	1..1	PU	Turbine damping factor (Dturb). Unit = delta P (PU of MWbase) / delta speed (PU). Typical Value = 0.5. Typical Value Francis = 1.1, Kaplan = 1.1.
hdam	1..1	PU	Head available at dam (hdam). Typical Value = 1.
qn1	1..1	PU	No-load flow at nominal head (Qnl). Typical Value = 0.08. Typical Value Francis = 0, Kaplan = 0.
db1	1..1	Frequency	Intentional deadband width (db1). Unit = Hz. Typical Value = 0.
eps	1..1	Frequency	Intentional db hysteresis (eps). Unit = Hz. Typical Value = 0.
db2	1..1	ActivePower	Unintentional dead-band (db2). Unit = MW. Typical Value = 0.
gv0	1..1	PU	Nonlinear gain point 0, PU gv (Gv0). Typical Value = 0. Typical Value Francis = 0.1, Kaplan = 0.1.
pgv0	1..1	PU	Nonlinear gain point 0, PU power (Pgv0). Typical Value = 0.
gv1	1..1	PU	Nonlinear gain point 1, PU gv (Gv1). Typical Value = 0. Typical Value Francis = 0.4, Kaplan = 0.4.
pgv1	1..1	PU	Nonlinear gain point 1, PU power (Pgv1). Typical Value = 0. Typical Value Francis = 0.42, Kaplan = 0.35.
gv2	1..1	PU	Nonlinear gain point 2,

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			PU gv (Gv2). Typical Value = 0. Typical Value Francis = 0.5, Kaplan = 0.5.
pgv2	1..1	PU	Nonlinear gain point 2, PU power (Pgv2). Typical Value = 0. Typical Value Francis = 0.56, Kaplan = 0.468.
gv3	1..1	PU	Nonlinear gain point 3, PU gv (Gv3). Typical Value = 0. Typical Value Francis = 0.7, Kaplan = 0.7.
pgv3	1..1	PU	Nonlinear gain point 3, PU power (Pgv3). Typical Value = 0. Typical Value Francis = 0.8, Kaplan = 0.796.
gv4	1..1	PU	Nonlinear gain point 4, PU gv (Gv4). Typical Value = 0. Typical Value Francis = 0.8, Kaplan = 0.8.
pgv4	1..1	PU	Nonlinear gain point 4, PU power (Pgv4). Typical Value = 0. Typical Value Francis = 0.9, Kaplan = 0.917.
gv5	1..1	PU	Nonlinear gain point 5, PU gv (Gv5). Typical Value = 0. Typical Value Francis = 0.9, Kaplan = 0.9.
pgv5	1..1	PU	Nonlinear gain point 5, PU power (Pgv5). Typical Value = 0. Typical Value Francis = 0.97, Kaplan = 0.99.
bgv0	1..1	PU	Kaplan blade servo point 0 (Bgv0). Typical Value = 0.
bgv1	1..1	PU	Kaplan blade servo point 1 (Bgv1). Typical Value = 0.
bgv2	1..1	PU	Kaplan blade servo point 2 (Bgv2). Typical Value = 0. Typical Value Francis = 0, Kaplan = 0.1.
bgv3	1..1	PU	Kaplan blade servo point 3 (Bgv3). Typical Value = 0. Typical Value Francis = 0, Kaplan = 0.667.
bgv4	1..1	PU	Kaplan blade servo point 4 (Bgv4). Typical Value = 0. Typical Value Francis = 0, Kaplan = 0.9.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

bgv5	1..1	PU	Kaplan blade servo point 5 (Bgv5). Typical Value = 0. Typical Value Francis = 0, Kaplan = 1.
bmax	1..1	Simple_Float	Maximum blade adjustment factor (Bmax). Typical Value = 0. Typical Value Francis = 0, Kaplan = 1.1276.
tblade	1..1	Seconds	Blade servo time constant (Tblade). Typical Value = 100.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-67 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-67 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] --- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] --- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.6.6.18 GovHydroDD

Double derivative hydro governor and turbine.

Figure 9-62 class diagram

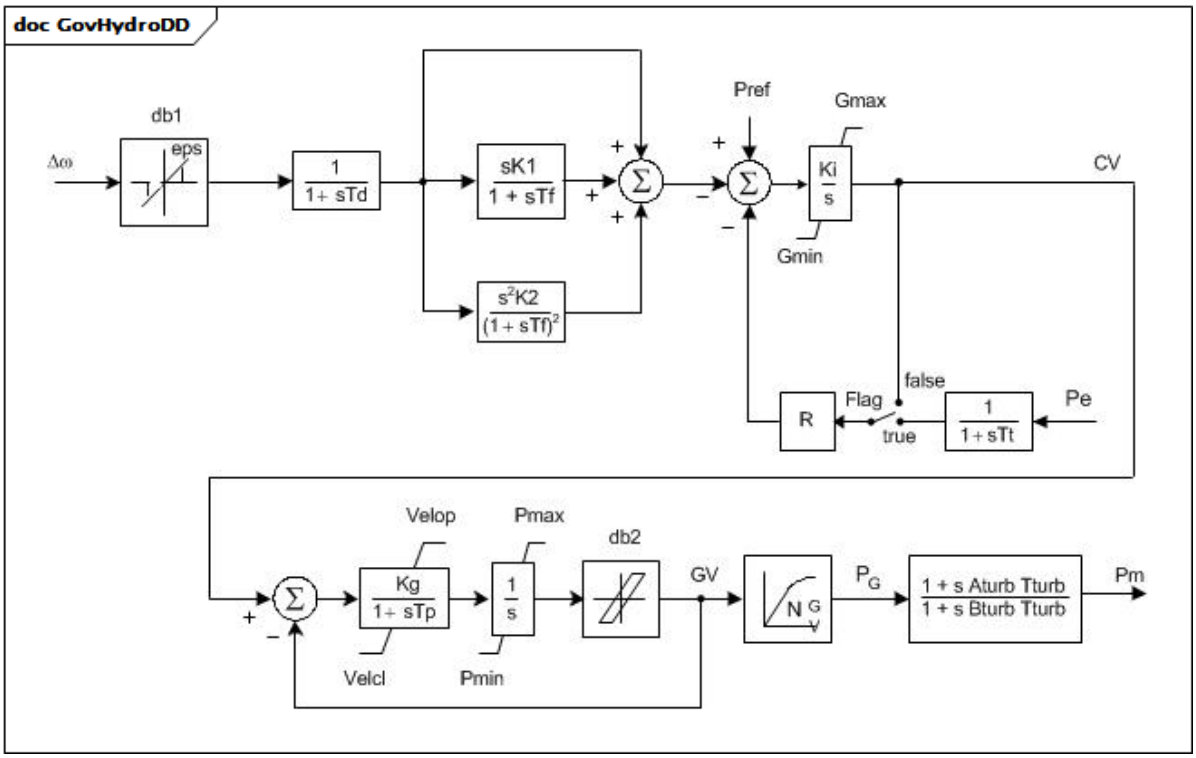


Figure 9-62 - figure GovHydroDD

Parameter Notes:

1. **Td**, **Tp**, **Tt**, and **Tturb** may be zero to bypass the corresponding transfer function blocks. If **Tf** is input as zero, but **K1** or **K2** is not zero, **Tf** is set to a small value (4 times integration time step). If **Bturb*****Tturb** is less than 4 times integration time step, **Tturb** is set to zero to bypass the block.
2. The nonlinear gain between gate position and power may be input with up to 6 points. The (0.,0.) and (1.,1.) points are assumed and need not be exchanged. The output is not allowed to go beyond 0. and 1. However, if **Pmax** > 1., the input and output are scaled by **Pmax**. If **Gv1** is a negative number, the default hydro curve will be used. If all zero values are provided, a straight line is used.
3. For a hydro turbine, set **Tturb** = **Tw** (water starting time), set **Aturb** = -1. and **Bturb** = 0.5. For a diesel engine or two-shaft gas turbine, set **Aturb** = 0.8 and **Bturb** = 1.0 and **Tturb** = 1 second. For a small steam turbine with a non-reheat steam supply or a single- shaft gas turbine, set **Aturb** = 0. and **Bturb** = 1.0 and **Tturb** = 0.2 second.

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-68 shows all attributes of GovHydroDD.

Table 9-68 - Attributes of TurbineGovernorDynamics::GovHydroDD

Attribute name	mult.	Attribute type	Description
mwbase	1..1	ActivePower	Base for power values (MWbase) (>0). Unit = MW.

pmax	1..1	PU	Maximum gate opening, PU of MWbase (Pmax). Typical Value = 1.
pmin	1..1	PU	Minimum gate opening, PU of MWbase (Pmin). Typical Value = 0.
r	1..1	PU	Steady state droop (R). Typical Value = 0.05.
td	1..1	Seconds	Input filter time constant (Td). Typical Value = 0.
tf	1..1	Seconds	Washout time constant (Tf). Typical Value = 0.1.
tp	1..1	Seconds	Gate servo time constant (Tp). Typical Value = 0.35.
velop	1..1	Simple_Float	Maximum gate opening velocity (Velop). Unit = PU/sec. Typical Value = 0.09.
velcl	1..1	Simple_Float	Maximum gate closing velocity (Velcl). Unit = PU/sec. Typical Value = -0.14.
k1	1..1	PU	Single derivative gain (K1). Typical Value = 3.6.
k2	1..1	PU	Double derivative gain (K2). Typical Value = 0.2.
ki	1..1	PU	Integral gain (Ki). Typical Value = 1.
kg	1..1	PU	Gate servo gain (Kg). Typical Value = 3.
tturb	1..1	Seconds	Turbine time constant (Tturb) (note 3). Typical Value = 0.8.
aturb	1..1	PU	Turbine numerator multiplier (Aturb) (note 3). Typical Value = -1.
bturb	1..1	PU	Turbine denominator multiplier (Bturb) (note 3). Typical Value = 0.5.
tt	1..1	Seconds	Power feedback time constant (Tt). Typical Value = 0.02.
db1	1..1	Frequency	Intentional dead-band width (db1). Unit = Hz. Typical Value = 0.
eps	1..1	Frequency	Intentional db hysteresis (eps). Unit = Hz. Typical Value = 0.
db2	1..1	ActivePower	Unintentional dead-band (db2). Unit = MW.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			Typical Value = 0.
gv1	1..1	PU	Nonlinear gain point 1, PU gv (Gv1). Typical Value = 0.
pgv1	1..1	PU	Nonlinear gain point 1, PU power (Pgv1). Typical Value = 0.
gv2	1..1	PU	Nonlinear gain point 2, PU gv (Gv2). Typical Value = 0.
pgv2	1..1	PU	Nonlinear gain point 2, PU power (Pgv2). Typical Value = 0.
gv3	1..1	PU	Nonlinear gain point 3, PU gv (Gv3). Typical Value = 0.
pgv3	1..1	PU	Nonlinear gain point 3, PU power (Pgv3). Typical Value = 0.
gv4	1..1	PU	Nonlinear gain point 4, PU gv (Gv4). Typical Value = 0.
pgv4	1..1	PU	Nonlinear gain point 4, PU power (Pgv4). Typical Value = 0.
gv5	1..1	PU	Nonlinear gain point 5, PU gv (Gv5). Typical Value = 0.
pgv5	1..1	PU	Nonlinear gain point 5, PU power (Pgv5). Typical Value = 0.
gv6	1..1	PU	Nonlinear gain point 6, PU gv (Gv6). Typical Value = 0.
pgv6	1..1	PU	Nonlinear gain point 6, PU power (Pgv6). Typical Value = 0.
gmax	1..1	PU	Maximum gate opening (Gmax). Typical Value = 0.
gmin	1..1	PU	Minimum gate opening (Gmin). Typical Value = 0.
inputSignal	1..1	Boolean	Input signal switch (Flag). true = Pe input is used false = feedback is received from CV. Flag is normally dependent on Tt. If Tf is zero, Flag is set to false. If Tf is not zero, Flag is set to true. Typical Value = true.
enabled	1..1	Boolean	see DynamicsFunctionBlock

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-69 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-69 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] --- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] --- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.6.6.19 GovHydroFrancis

Detailed hydro unit - Francis model. This model can be used to represent three types of governors. A schematic of the hydraulic system of detailed hydro unit models, like Francis and Pelton, is provided in the DetailedHydroModelHydraulicSystem diagram.

Figure 9-63 class diagram

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

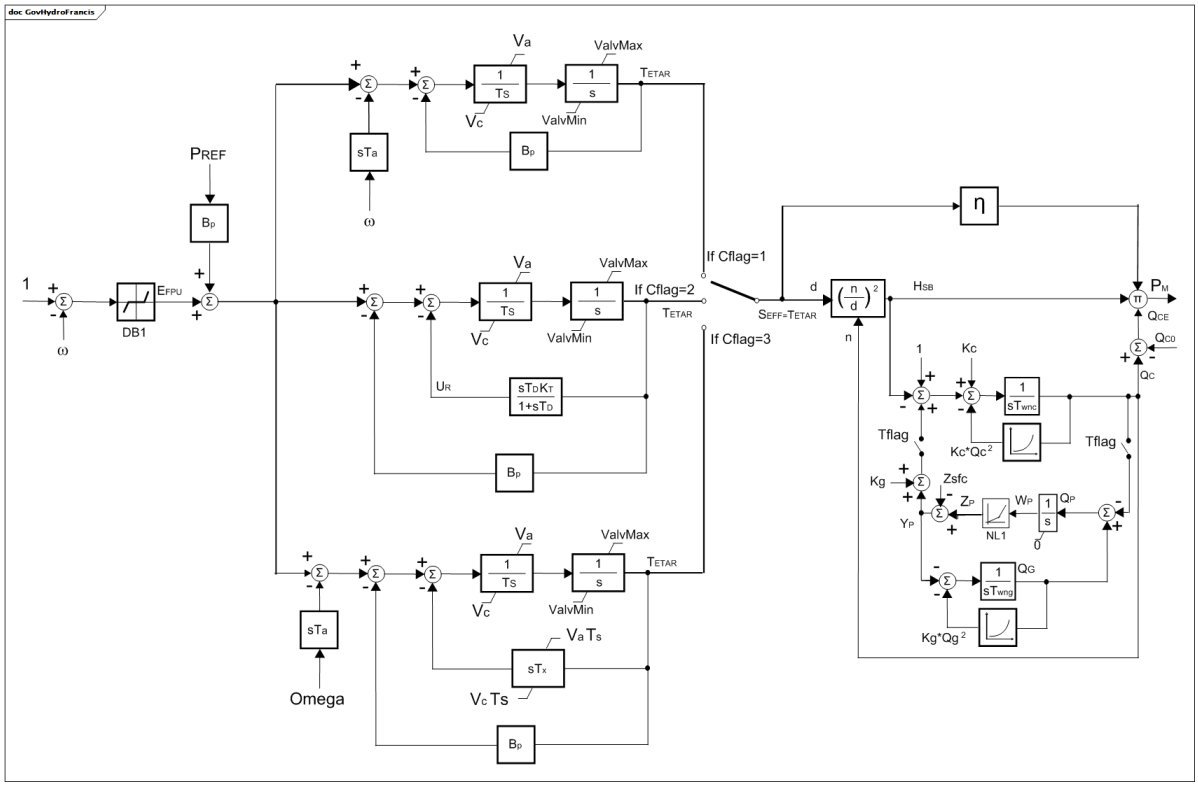


Figure 9-63 - figure GovHydroFrancis

Parameter Notes:

1. Governors per unit parameters are on base of MW capability of the turbine.
2. Rated hydraulic head **Hn** [m] and rated flow **Qn** [m3/s] are the design parameters of the hydraulic system capability.
3. Hydraulic system head attributes (**Zsfc**, **H1**, **H2**) are in meters but in the block diagram are in per unit on base of **Hn**.
4. Hydraulic area attributes (**Av0**, **Av1**) are in square meters but in the block diagram are in per unit on base of **Qn**/**Hn** [m2/s].
5. omega has units of per unit speed.
6. Non-linear gain and efficiency equations shown in GovHydroFrancisNonLinearGainAndEfficiency diagram.

Figure 9-64 class diagram

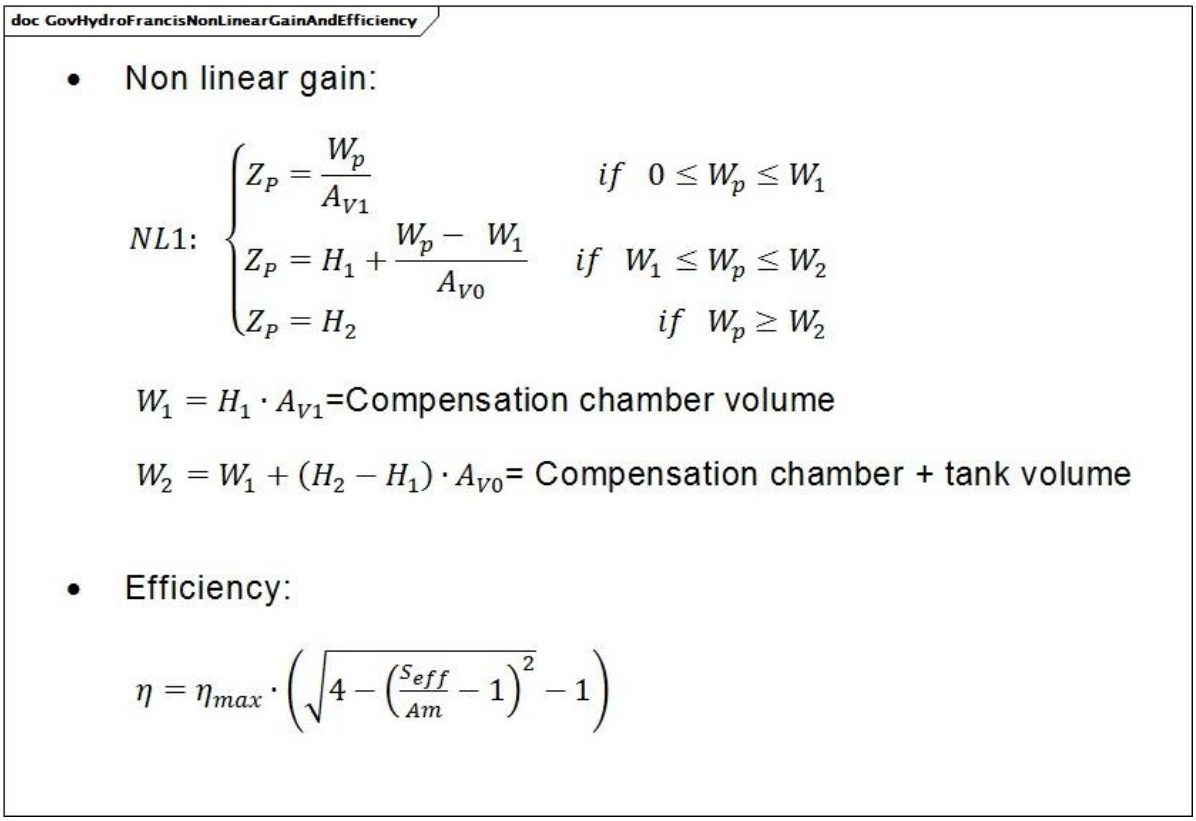


Figure 9-64 - figure GovHydroFrancisNonLinearGainAndEfficiency

Figure 9-65 class diagram

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

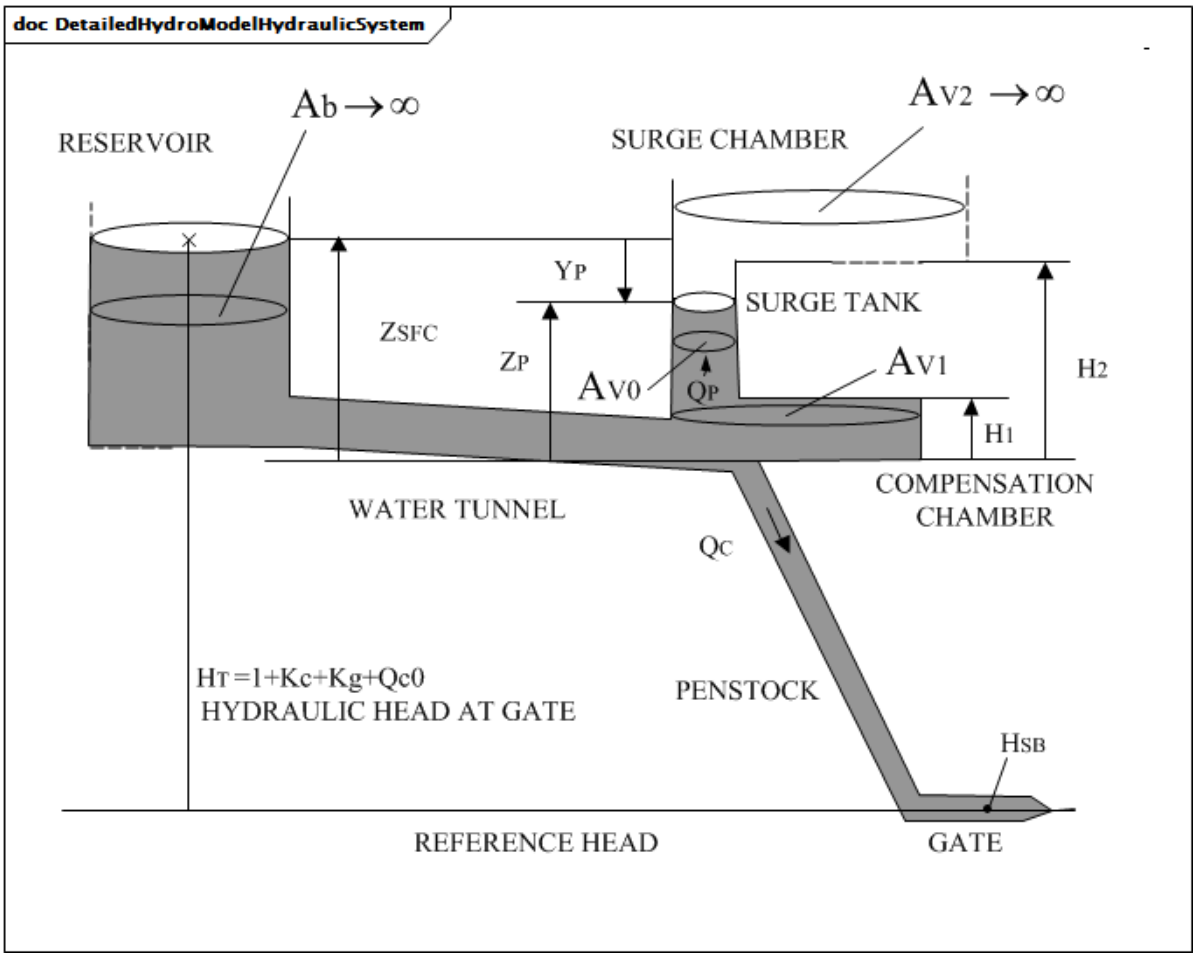


Figure 9-65 - figure DetailedHydroModelHydraulicSystem

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-70 shows all attributes of GovHydroFrancis.

Table 9-70 - Attributes of TurbineGovernorDynamics::GovHydroFrancis

Attribute name	mult.	Attribute type	Description
am	1..1	PU	Opening section S_{eff} at the maximum efficiency (Am). Typical Value = 0.7.
av0	1..1	Area	Area of the surge tank (A_{V0}). Unit = m^2. Typical Value = 30.
av1	1..1	Area	Area of the compensation tank (A_{V1}). Unit = m^2. Typical Value = 700.
bp	1..1	PU	Droop (Bp). Typical Value = 0.05.
db1	1..1	Frequency	Intentional dead-band width (DB1). Unit = Hz. Typical Value = 0.
etamax	1..1	PU	Maximum efficiency (EtaMax). Typical Value = 1.05.
governorControl	1..1	FrancisGovernorControlKind	Governor control flag (Cflag).

			Typical Value = mechanicHydrolicTachoAccelerator.
h1	1..1	Length	Head of compensation chamber water level with respect to the level of penstock (H_1). Unit = m. Typical Value = 4.
h2	1..1	Length	Head of surge tank water level with respect to the level of penstock (H_2). Unit = m. Typical Value = 40.
hn	1..1	Length	Rated hydraulic head (H_n). Unit = m. Typical Value = 250.
kc	1..1	PU	Penstock loss coefficient (due to friction) (Kc). Typical Value = 0.025.
kg	1..1	PU	Water tunnel and surge chamber loss coefficient (due to friction) (Kg). Typical Value = 0.025.
kt	1..1	PU	Washout gain (Kt). Typical Value = 0.25.
qc0	1..1	PU	No-load turbine flow at nominal head (Qc0). Typical Value = 0.21.
qn	1..1	VolumeFlowRate	Rated flow (Q_n). Unit = m^3/s. Typical Value = 40.
ta	1..1	Seconds	Derivative gain (Ta). Typical Value = 3.
td	1..1	Seconds	Washout time constant (Td). Typical Value = 3.
ts	1..1	Seconds	Gate servo time constant (Ts). Typical Value = 0.5.
twnc	1..1	Seconds	Water inertia time constant (Twnc). Typical Value = 1.
twng	1..1	Seconds	Water tunnel and surge chamber inertia time constant (Twng). Typical Value = 3.
tx	1..1	Seconds	Derivative feedback gain (Tx). Typical Value = 1.
va	1..1	Simple_Float	Maximum gate opening velocity (Va). Unit = PU/sec. Typical Value = 0.011.
valvmax	1..1	PU	Maximum gate opening (ValvMax). Typical Value = 1.
valvmin	1..1	PU	Minimum gate opening (ValvMin). Typical Value = 0.
vc	1..1	Simple_Float	Maximum gate closing velocity (Vc). Unit = PU/sec. Typical Value = -0.011.
waterTunnelSurgeChamberSimulation	1..1	Boolean	Water tunnel and surge chamber simulation (Tflag). true = enable of water tunnel and surge chamber simulation false = inhibit of water tunnel and surge chamber simulation. Typical Value = false.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

zsfc	1..1	Length	Head of upper water level with respect to the level of penstock (Zsfc). Unit = m. Typical Value = 25.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-71 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-71 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] --- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] --- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.6.6.20 GovHydroPelton

Detailed hydro unit - Pelton model. This model can be used to represent the dynamic related to water tunnel and surge chamber.
A schematic of the hydraulic system of detailed hydro unit models, like Francis and Pelton, is located under the GovHydroFrancis class.

Figure 9-66 class diagram

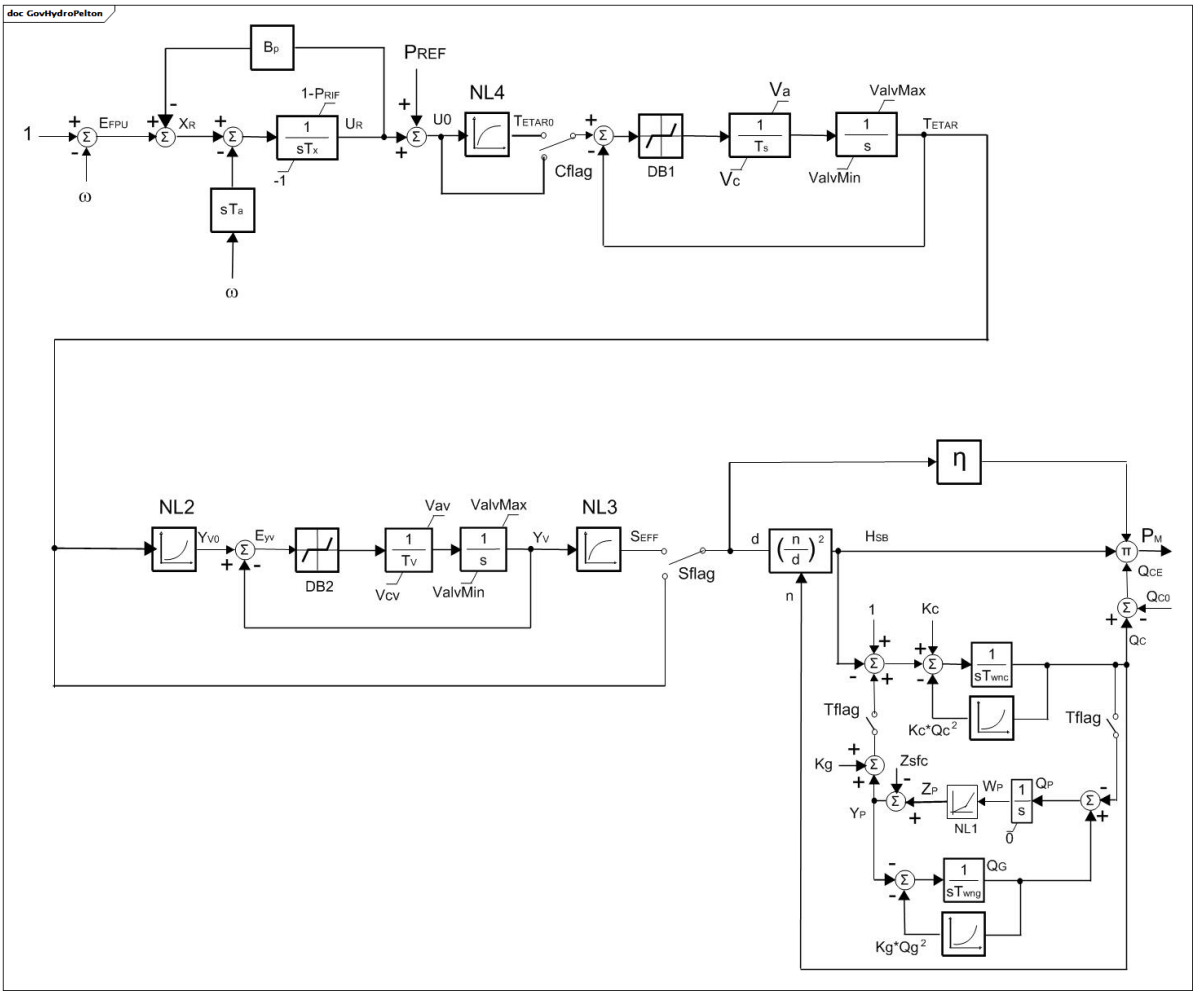


Figure 9-66 - figure GovHydroPelton

Parameter Notes:

1. Governors per unit parameters are on base of MW capability of the turbine.
2. Rated hydraulic head **Hn** [m] and rated flow **Qn** [m3/s] are the design parameters of the hydraulic system capability.
3. Hydraulic system head attributes (**Zsfc**, **H1**, **H2**) are in meters but in the block diagram are in per unit on base of **Hn**.
4. Hydraulic area attributes (**Av0**, **Av1**) are in square meters but in the block diagram are in per unit on base of **Qn**/**Hn** [m2/s].
5. omega has units of per unit speed.
6. Non-linear gains and efficiency shown in GovHydroPeltonNonLinearGainAndEfficiency diagram.

Figure 9-67 class diagram

doc GovHydroPeltonNonLinearGainAndEfficiency

- Non linear gains:

$$NL1: \begin{cases} Z_p = \frac{W_p}{A_{V1}} & if \quad 0 \le W_p \le W_1 \\ Z_p = H_1 + \frac{W_p - W_1}{A_{V0}} & if \quad W_1 \le W_p \le W_2 \\ Z_p = H_2 & if \quad W_p \ge W_2 \end{cases}$$

$$NL2: \quad Y_{V0} = (TetaR)^3$$

$$NL3: \quad S_{eff} = -1.1 + \sqrt{1.21 + Y_V \cdot (4.2 - Y_V)}$$

$$NL4: \quad TetaR0 = \left(2.1 - \sqrt{5.62 - (U_0 + 1.1)^2}\right)^{1/3}$$

$W_1 = H_1 \cdot A_{V1}$=Compensation chamber volume

$W_2 = W_1 + (H_2 - H_1) \cdot A_{V0}$= Compensation chamber + tank volume

- Efficiency:

$$\eta = \begin{cases} 1 & if \; S_{eff} \ge 0.3 \\ 0.7 + S_{eff} & if \; 0.13 \le S_{eff} < 0.3 \\ 13.83 \cdot (S_{eff} - 0.07) & if \; 0.07 \le S_{eff} < 0.13 \\ 0 & if \; S_{eff} < 0.7 \end{cases}$$

Figure 9-67 - figure GovHydroPeltonNonLinearGainAndEfficiency

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-72 shows all attributes of GovHydroPelton.

Table 9-72 - Attributes of TurbineGovernorDynamics::GovHydroPelton

Attribute name	mult.	Attribute type	Description
av0	1..1	Area	Area of the surge tank (A_{V0}). Unit = m^2. Typical Value = 30.
av1	1..1	Area	Area of the compensation tank (A_{V1}). Unit = m^2. Typical Value = 700.
bp	1..1	PU	Droop (bp). Typical Value = 0.05.

db1	1..1	Frequency	Intentional dead-band width (DB1). Unit = Hz. Typical Value = 0.
db2	1..1	Frequency	Intentional dead-band width of valve opening error (DB2). Unit = Hz. Typical Value = 0.01.
h1	1..1	Length	Head of compensation chamber water level with respect to the level of penstock (H_1). Unit = m. Typical Value = 4.
h2	1..1	Length	Head of surge tank water level with respect to the level of penstock (H_2). Unit = m. Typical Value = 40.
hn	1..1	Length	Rated hydraulic head (H_n). Unit = m. Typical Value = 250.
kc	1..1	PU	Penstock loss coefficient (due to friction) (Kc). Typical Value = 0.025.
kg	1..1	PU	Water tunnel and surge chamber loss coefficient (due to friction) (Kg). Typical Value = -0.025.
qc0	1..1	PU	No-load turbine flow at nominal head (Qc0). Typical Value = 0.05.
qn	1..1	VolumeFlowRate	Rated flow (Q_n). Unit = m^3/s. Typical Value = 40.
simplifiedPelton	1..1	Boolean	Simplified Pelton model simulation (Sflag). true = enable of simplified Pelton model simulation false = enable of complete Pelton model simulation (non linear gain). Typical Value = false.
staticCompensating	1..1	Boolean	Static compensating characteristic (Cflag). true = enable of static compensating characteristic false = inhibit of static compensating characteristic. Typical Value = false.
ta	1..1	Seconds	Derivative gain (accelerometer time constant) (Ta). Typical Value = 3.
ts	1..1	Seconds	Gate servo time constant (Ts). Typical

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			Value = 0.15.
tv	1..1	Seconds	Servomotor integrator time constant (TV). Typical Value = 0.3.
twnc	1..1	Seconds	Water inertia time constant (Twnc). Typical Value = 1.
twng	1..1	Seconds	Water tunnel and surge chamber inertia time constant (Twng). Typical Value = 3.
tx	1..1	Seconds	Electronic integrator time constant (Tx). Typical Value = 0.5.
va	1..1	Simple_Float	Maximum gate opening velocity (Va). Unit = PU/sec. Typical Value = 0.016.
valvmax	1..1	PU	Maximum gate opening (ValvMax). Typical Value = 1.
valvmin	1..1	PU	Minimum gate opening (ValvMin). Typical Value = 0.
vav	1..1	PU	Maximum servomotor valve opening velocity (Vav). Typical Value = 0.017.
vc	1..1	Simple_Float	Maximum gate closing velocity (Vc). Unit = PU/sec. Typical Value = -0.016.
vcv	1..1	PU	Maximum servomotor valve closing velocity (Vcv). Typical Value = -0.017.
waterTunnelSurgeChamberSimulation	1..1	Boolean	Water tunnel and surge chamber simulation (Tflag). true = enable of water tunnel and surge chamber simulation false = inhibit of water tunnel and surge chamber simulation. Typical Value = false.
zsfc	1..1	Length	Head of upper water level with respect to the level of penstock (Zsfc). Unit = m. Typical Value = 25.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-73 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-73 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] --- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] --- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.6.6.21 GovHydroPID

PID governor and turbine.

Figure 9-68 class diagram

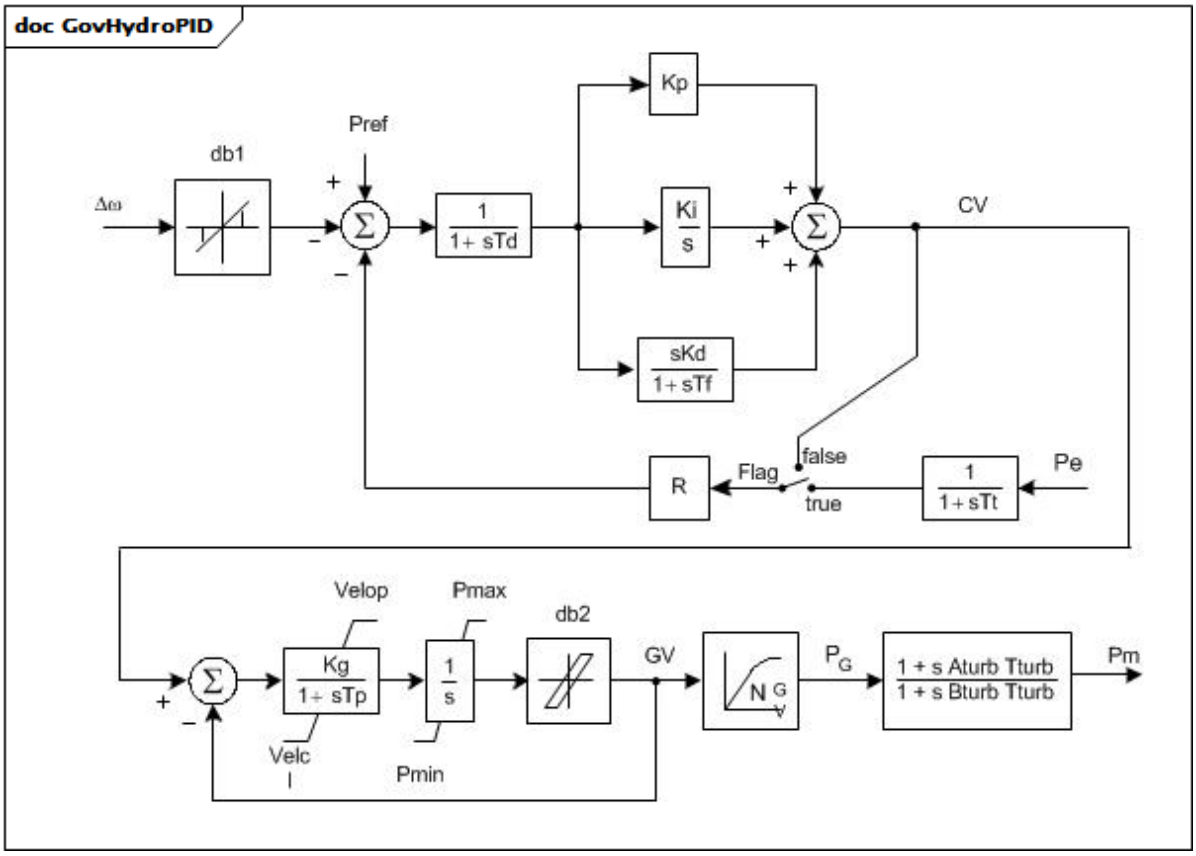


Figure 9-68 - figure GovHydroPID

Parameter Notes:

1. **Td**, **Tp**, **Tt**, and **Tturb** may be zero to bypass the corresponding transfer function blocks. If **Tf** is zero, but **Kd** is not zero, **Tf** is set to a small value (4 times integration time step).
2. The nonlinear gain between gate position and power may be input with up to 6 points. The (0.,0.) and (1.,1.) points are assumed and need not be input. The output is not allowed to go beyond 0. and 1. However, if **Pmax** > 1, the input and output are scaled by **Pmax**. If **Gv1** is a negative number, the default hydro curve will be used. If all zero values are provided, a straight line is used.
3. For a hydro turbine, set **Tturb** = **Tw** (water starting time), set **Aturb** = -1 and **Bturb** = 0.5. For a diesel engine or two-shaft gas turbine, set **Aturb** = 0.8 and **Bturb** = 1 and Tturb = 1 second. For a small steam turbine with a non-reheat steam supply or a single- shaft gas turbine, set **Aturb** = 0 and **Bturb** = 1 and **Tturb** = 0 second.

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-74 shows all attributes of GovHydroPID.

Table 9-74 - Attributes of TurbineGovernorDynamics::GovHydroPID

Attribute name	mult.	Attribute type	Description
mwbase	1..1	ActivePower	Base for power values (MWbase) (>0). Unit = MW.
pmax	1..1	PU	Maximum gate opening, PU of MWbase (Pmax). Typical Value = 1.
pmin	1..1	PU	Minimum gate opening, PU of MWbase (Pmin). Typical Value = 0.
r	1..1	PU	Steady state droop (R). Typical Value = 0.05.
td	1..1	Seconds	Input filter time constant (Td). Typical Value = 0.
tf	1..1	Seconds	Washout time constant (Tf). Typical Value = 0.1.
tp	1..1	Seconds	Gate servo time constant (Tp). Typical Value = 0.35.
velop	1..1	Simple_Float	Maximum gate opening velocity (Velop). Unit = PU/sec. Typical Value = 0.09.
velcl	1..1	Simple_Float	Maximum gate closing velocity (Velcl). Unit = PU/sec. Typical Value = -0.14.
kd	1..1	PU	Derivative gain (Kd). Typical Value = 1.11.
kp	1..1	PU	Proportional gain (Kp).

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			Typical Value = 0.1.
ki	1..1	PU	Integral gain (Ki). Typical Value = 0.36.
kg	1..1	PU	Gate servo gain (Kg). Typical Value = 2.5.
tturb	1..1	Seconds	Turbine time constant (Tturb) (note 3). Typical Value = 0.8.
aturb	1..1	PU	Turbine numerator multiplier (Aturb) (note 3). Typical Value -1.
bturb	1..1	PU	Turbine denominator multiplier (Bturb) (note 3). Typical Value = 0.5.
tt	1..1	Seconds	Power feedback time constant (Tt). Typical Value = 0.02.
db1	1..1	Frequency	Intentional dead-band width (db1). Unit = Hz. Typical Value = 0.
inputSignal	1..1	Boolean	Input signal switch (Flag). true = Pe input is used false = feedback is received from CV. Flag is normally dependent on Tt. If Tf is zero, Flag is set to false. If Tf is not zero, Flag is set to true. Typical Value = true.
eps	1..1	Frequency	Intentional db hysteresis (eps). Unit = Hz. Typical Value = 0.
db2	1..1	ActivePower	Unintentional dead-band (db2). Unit = MW. Typical Value = 0.
gv1	1..1	PU	Nonlinear gain point 1, PU gv (Gv1). Typical Value = 0.
pgv1	1..1	PU	Nonlinear gain point 1, PU power (Pgv1). Typical Value = 0.
gv2	1..1	PU	Nonlinear gain point 2, PU gv (Gv2). Typical Value = 0.
pgv2	1..1	PU	Nonlinear gain point 2, PU power (Pgv2). Typical Value = 0.
gv3	1..1	PU	Nonlinear gain point 3, PU gv (Gv3). Typical Value = 0.
pgv3	1..1	PU	Nonlinear gain point 3, PU power (Pgv3). Typical Value = 0.

gv4	1..1	PU	Nonlinear gain point 4, PU gv (Gv4). Typical Value = 0.
pgv4	1..1	PU	Nonlinear gain point 4, PU power (Pgv4). Typical Value = 0.
gv5	1..1	PU	Nonlinear gain point 5, PU gv (Gv5). Typical Value = 0.
pgv5	1..1	PU	Nonlinear gain point 5, PU power (Pgv5). Typical Value = 0.
gv6	1..1	PU	Nonlinear gain point 6, PU gv (Gv6). Typical Value = 0.
pgv6	1..1	PU	Nonlinear gain point 6, PU power (Pgv6). Typical Value = 0.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-75 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-75 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] ----- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] ----- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.6.6.22 GovHydroPID2

Hydro turbine and governor. Represents plants with straight forward penstock configurations and "three term" electro-hydraulic governors (i.e. Woodard electronic).

Figure 9-69 class diagram

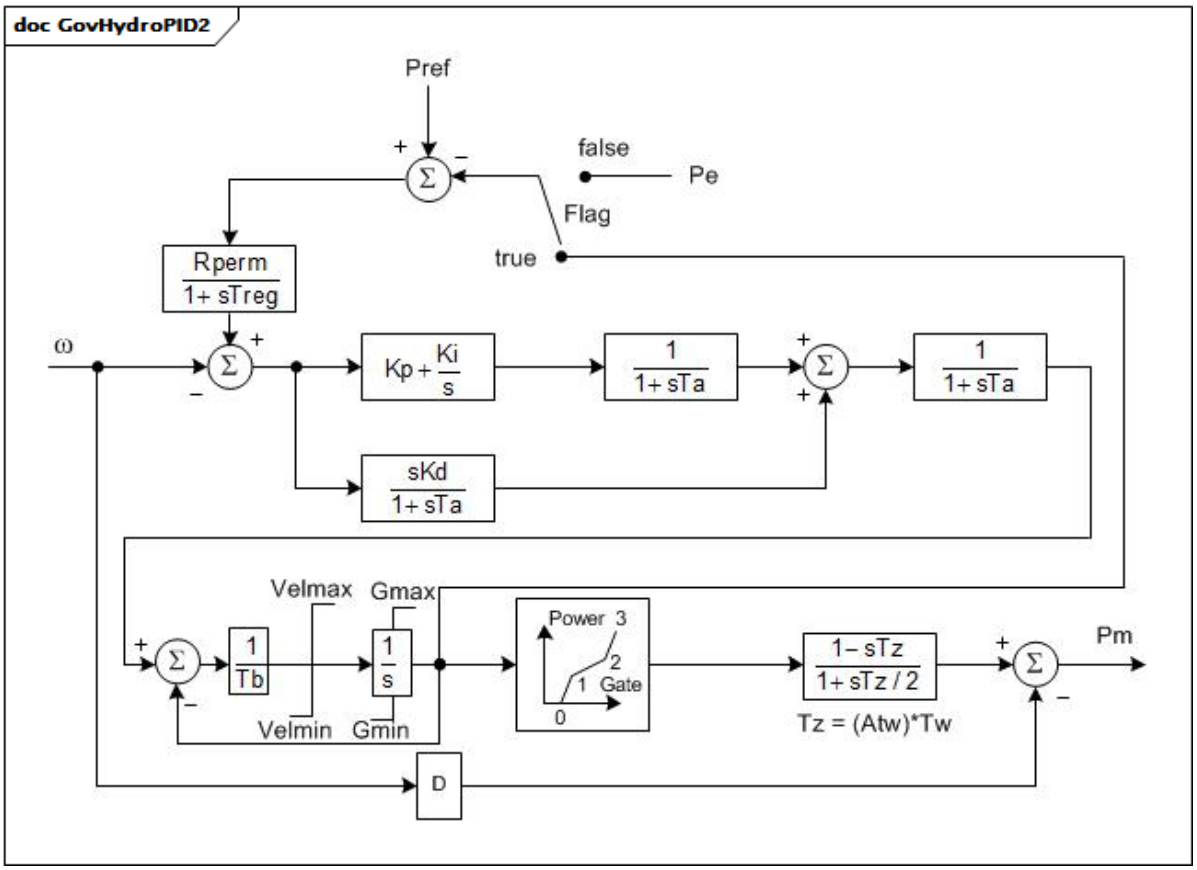


Figure 9-69 - figure GovHydroPID2

Parameter Notes:

1. The gates travel over a range of 1.0 per unit from full closed to full opened. The gates are at a position greater than zero at zero power and, normally, less than 1.0 when power is 1.0 PU.
2. **G0**, **G1**, **P1**, **G2**, **P2**, **P3** specify the variation of power with gate **G2**>**G1**>**G0.**
3. This model uses a simplified turbine-penstock model that does not account for variation of water inertia effect with gate opening. To make this model correspond to other models using the "classical" turbine-penstock model, set **Atw** to unity. To approximate water inertia effects at part load, **Atw** can be set to the per unit gate opening about which perturbations are centered.

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-76 shows all attributes of GovHydroPID2.

Table 9-76 - Attributes of TurbineGovernorDynamics::GovHydroPID2

Attribute name	mult.	Attribute type	Description
mwbase	1..1	ActivePower	Base for power values (MWbase) (>0). Unit = MW.
treg	1..1	Seconds	Speed detector time

			constant (Treg). Typical Value = 0.
rperm	1..1	PU	Permanent drop (Rperm). Typical Value = 0.
kp	1..1	PU	Proportional gain (Kp). Typical Value = 0.
ki	1..1	Simple_Float	Reset gain (Ki). Unit = PU/ sec. Typical Value = 0.
kd	1..1	PU	Derivative gain (Kd). Typical Value = 0.
ta	1..1	Seconds	Controller time constant (Ta) (>0). Typical Value = 0.
tb	1..1	Seconds	Gate servo time constant (Tb) (>0). Typical Value = 0.
velmax	1..1	Simple_Float	Maximum gate opening velocity (Velmax). Unit = PU/sec. Typical Value = 0.
velmin	1..1	Simple_Float	Maximum gate closing velocity (Velmin). Unit = PU/sec. Typical Value = 0.
gmax	1..1	PU	Maximum gate opening (Gmax). Typical Value = 0.
gmin	1..1	PU	Minimum gate opening (Gmin). Typical Value = 0.
tw	1..1	Seconds	Water inertia time constant (Tw) (>0). Typical Value = 0.
d	1..1	PU	Turbine damping factor (D). Unit = delta P / delta speed. Typical Value = 0.
g0	1..1	PU	Gate opening at speed no load (G0). Typical Value = 0.
g1	1..1	PU	Intermediate gate opening (G1). Typical Value = 0.
p1	1..1	PU	Power at gate opening G1 (P1). Typical Value = 0.
g2	1..1	PU	Intermediate gate opening (G2). Typical Value = 0.
p2	1..1	PU	Power at gate opening G2 (P2). Typical Value = 0.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

p3	1..1	PU	Power at full opened gate (P3). Typical Value = 0.
atw	1..1	PU	Factor multiplying Tw (Atw). Typical Value = 0.
feedbackSignal	1..1	Boolean	Feedback signal type flag (Flag). true = use gate position feedback signal false = use Pe.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-77 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-77 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] --- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] --- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.6.6.23 GovHydroR

Fourth order lead-lag governor and hydro turbine.

Figure 9-70 class diagram

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

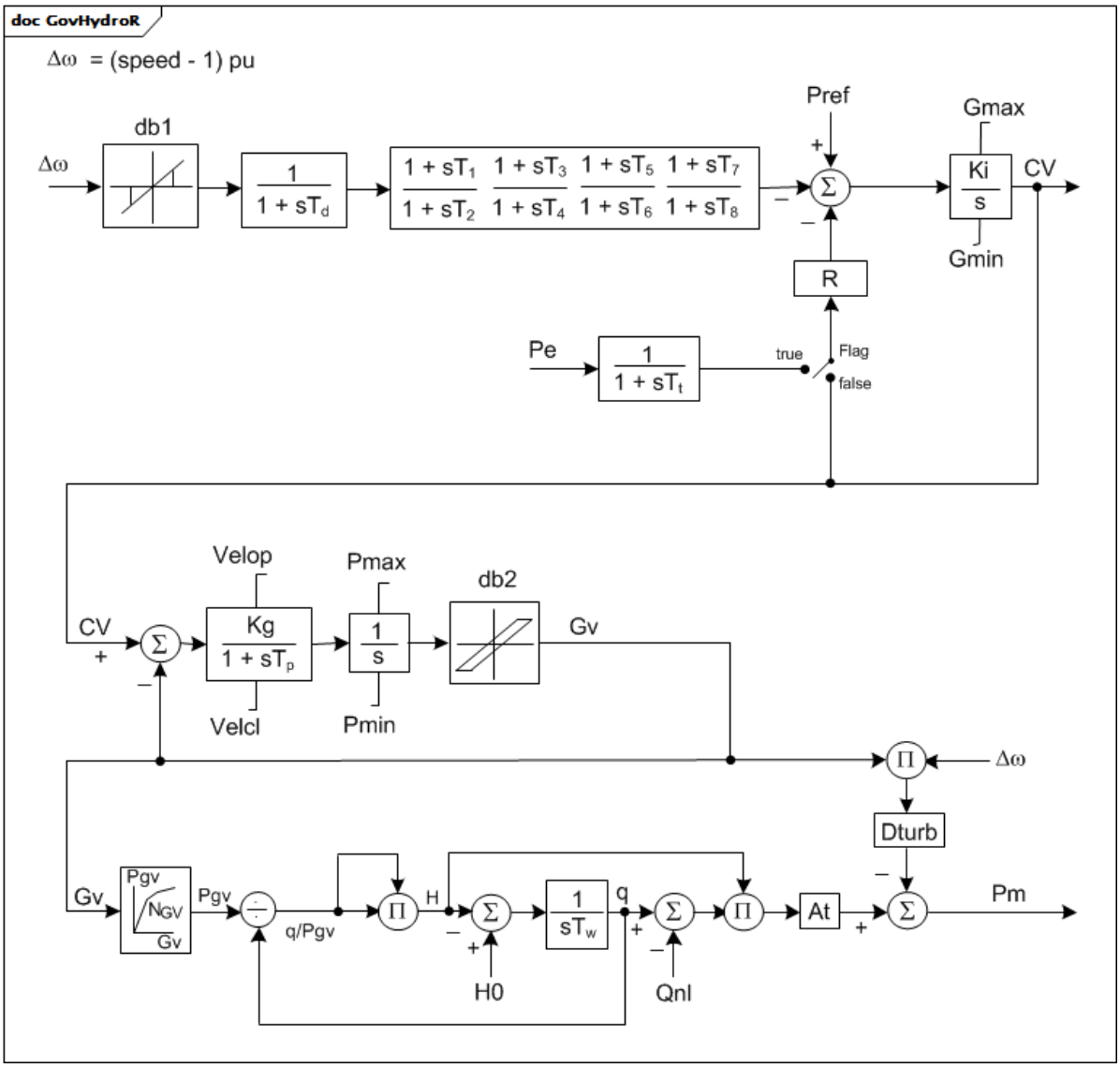


Figure 9-70 - figure GovHydroR

Parameter Notes:

1. **Tw** must be greater than zero.
2. The nonlinear gain between gate position and power may be input with up to 6 points. The (0.,0.) and (1.,1.) points are assumed and need not be exchanged. The output is not allowed to go beyond 0. and 1. However, if **Pmax** > 1., the input and output are scaled by **Pmax**. If **Gv1** is a negative number, the default hydro curve will be used. If all zero values are provided, a straight line is used.

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-78 shows all attributes of GovHydroR.

Table 9-78 - Attributes of TurbineGovernorDynamics::GovHydroR

Attribute name	mult.	Attribute type	Description
mwbase	1..1	ActivePower	Base for power values (MWbase) (>0). Unit = MW.
pmax	1..1	PU	Maximum gate opening, PU of MWbase (Pmax). Typical Value = 1.
pmin	1..1	PU	Minimum gate opening, PU of MWbase (Pmin). Typical Value = 0.
r	1..1	PU	Steady-state droop (R). Typical Value = 0.05.
td	1..1	Seconds	Input filter time constant (Td). Typical Value = 0.05.
t1	1..1	Seconds	Lead time constant 1 (T1). Typical Value = 1.5.
t2	1..1	Seconds	Lag time constant 1 (T2). Typical Value = 0.1.
t3	1..1	Seconds	Lead time constant 2 (T3). Typical Value = 1.5.
t4	1..1	Seconds	Lag time constant 2 (T4). Typical Value = 0.1.
t5	1..1	Seconds	Lead time constant 3 (T5). Typical Value = 0.
t6	1..1	Seconds	Lag time constant 3 (T6). Typical Value = 0.05.
t7	1..1	Seconds	Lead time constant 4 (T7). Typical Value = 0.
t8	1..1	Seconds	Lag time constant 4 (T8). Typical Value = 0.05.
tp	1..1	Seconds	Gate servo time constant (Tp). Typical Value = 0.05.
velop	1..1	Simple_Float	Maximum gate opening velocity (Velop). Unit = PU/sec. Typical Value = 0.2.
velcl	1..1	Simple_Float	Maximum gate closing velocity (Velcl). Unit = PU/sec. Typical Value = -0.2.
ki	1..1	PU	Integral gain (Ki). Typical Value = 0.5.
kg	1..1	PU	Gate servo gain (Kg). Typical Value = 2.

gmax	1..1	PU	Maximum governor output (Gmax). Typical Value = 1.05.
gmin	1..1	PU	Minimum governor output (Gmin). Typical Value = -0.05.
tt	1..1	Seconds	Power feedback time constant (Tt). Typical Value = 0.
inputSignal	1..1	Boolean	Input signal switch (Flag). true = Pe input is used false = feedback is received from CV. Flag is normally dependent on Tt. If Tf is zero, Flag is set to false. If Tf is not zero, Flag is set to true. Typical Value = true.
db1	1..1	Frequency	Intentional dead-band width (db1). Unit = Hz. Typical Value = 0.
eps	1..1	Frequency	Intentional db hysteresis (eps). Unit = Hz. Typical Value = 0.
db2	1..1	ActivePower	Unintentional dead-band (db2). Unit = MW. Typical Value = 0.
tw	1..1	Seconds	Water inertia time constant (Tw). Typical Value = 1.
at	1..1	PU	Turbine gain (At). Typical Value = 1.2.
dturb	1..1	PU	Turbine damping factor (Dturb). Typical Value = 0.2.
qnl	1..1	PU	No-load turbine flow at nominal head (Qnl). Typical Value = 0.08.
h0	1..1	PU	Turbine nominal head (H0). Typical Value = 1.
gv1	1..1	PU	Nonlinear gain point 1, PU gv (Gv1). Typical Value = 0.
pgv1	1..1	PU	Nonlinear gain point 1, PU power (Pgv1). Typical Value = 0.
gv2	1..1	PU	Nonlinear gain point 2, PU gv (Gv2). Typical Value = 0.
pgv2	1..1	PU	Nonlinear gain point 2, PU power (Pgv2). Typical Value = 0.
gv3	1..1	PU	Nonlinear gain point 3, PU gv (Gv3). Typical

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			Value = 0.
pgv3	1..1	PU	Nonlinear gain point 3, PU power (Pgv3). Typical Value = 0.
gv4	1..1	PU	Nonlinear gain point 4, PU gv (Gv4). Typical Value = 0.
pgv4	1..1	PU	Nonlinear gain point 4, PU power (Pgv4). Typical Value = 0.
gv5	1..1	PU	Nonlinear gain point 5, PU gv (Gv5). Typical Value = 0.
pgv5	1..1	PU	Nonlinear gain point 5, PU power (Pgv5). Typical Value = 0.
gv6	1..1	PU	Nonlinear gain point 6, PU gv (Gv6). Typical Value = 0.
pgv6	1..1	PU	Nonlinear gain point 6, PU power (Pgv6). Typical Value = 0.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-79 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-79 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] --- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] --- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.6.6.24 GovHydroWEH

Woodward Electric Hydro Governor Model.

Figure 9-71 class diagram

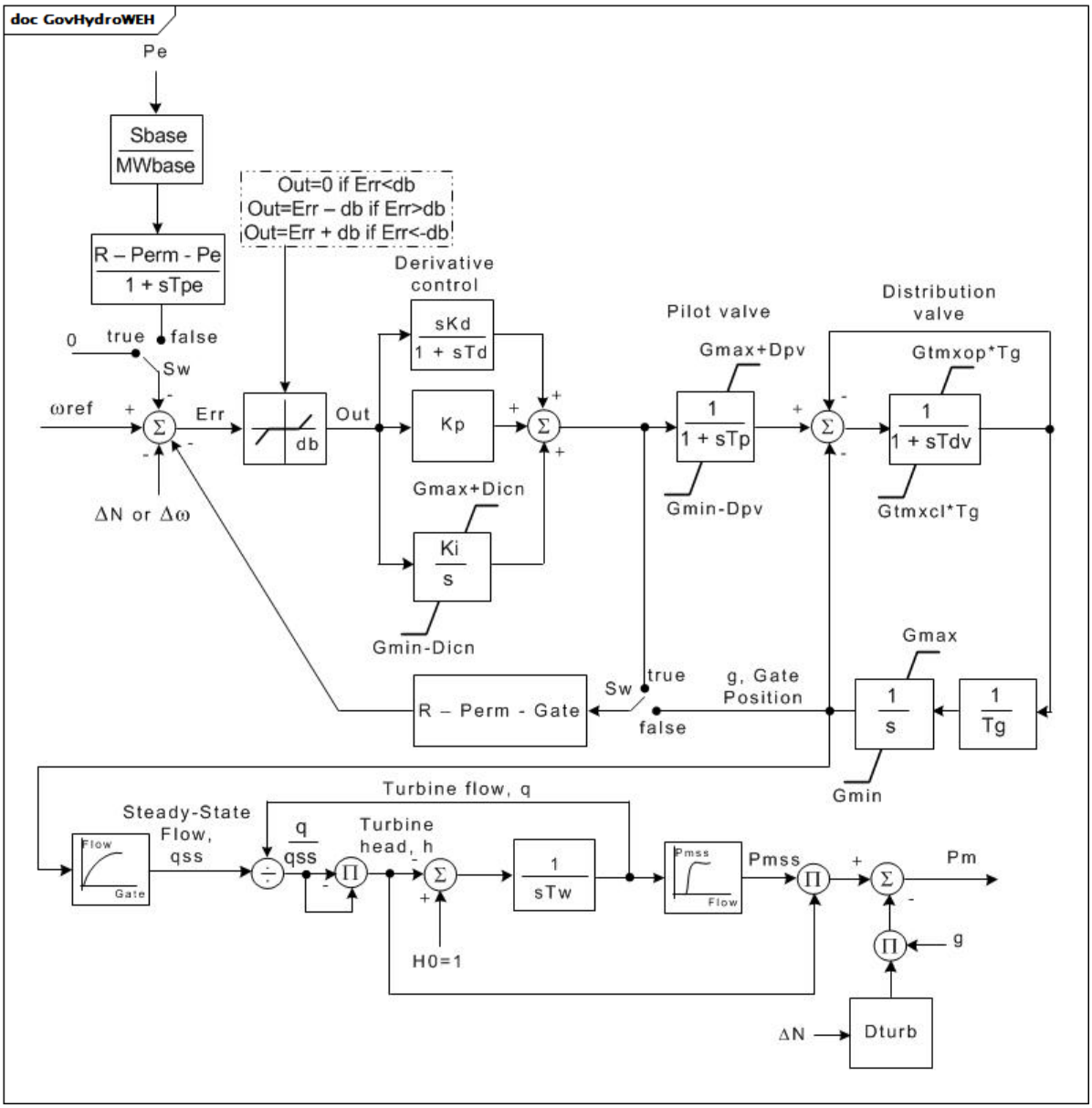


Figure 9-71 - figure GovHydroWEH

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-80 shows all attributes of GovHydroWEH.

Table 9-80 - Attributes of TurbineGovernorDynamics::GovHydroWEH

Attribute name	mult.	Attribute type	Description
mwbase	1..1	ActivePower	Base for power values (MWbase) (>0). Unit = MW.
rpg	1..1	Simple_Float	Permanent droop for governor output feedback (R-Perm-Gate).

rpp	1..1	Simple_Float	Permanent droop for electrical power feedback (R-Perm-Pe).
tpe	1..1	Seconds	Electrical power droop time constant (Tpe).
kp	1..1	PU	Derivative control gain (Kp).
ki	1..1	PU	Derivative controller Integral gain (Ki).
kd	1..1	PU	Derivative controller derivative gain (Kd).
td	1..1	Seconds	Derivative controller time constant to limit the derivative characteristic beyond a breakdown frequency to avoid amplification of high-frequency noise (Td).
tp	1..1	Seconds	Pilot Valve time lag time constant (Tp).
tdv	1..1	Seconds	Distributive Valve time lag time constant (Tdv).
tg	1..1	Seconds	Value to allow the Distribution valve controller to advance beyond the gate movement rate limit (Tg).
gtmxop	1..1	PU	Maximum gate opening rate (Gtmxop).
gtmxcl	1..1	PU	Maximum gate closing rate (Gtmxcl).
gmax	1..1	PU	Maximum Gate Position (Gmax).
gmin	1..1	PU	Minimum Gate Position (Gmin).
dturb	1..1	PU	Turbine damping factor (Dturb). Unit = delta P (PU of MWbase) / delta speed (PU).
tw	1..1	Seconds	Water inertia time constant (Tw) (>0).
db	1..1	PU	Speed Dead Band (db).
dpv	1..1	PU	Value to allow the Pilot valve controller to advance beyond the gate limits (Dpv).
dicn	1..1	PU	Value to allow the integral controller to advance beyond the gate limits (Dicn).
feedbackSignal	1..1	Boolean	Feedback signal selection (Sw).

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			true = PID Output (if R-Perm-Gate=droop and R-Perm-Pe=0) false = Electrical Power (if R-Perm-Gate=0 and R-Perm-Pe=droop) or false = Gate Position (if R-Perm-Gate=droop and R-Perm-Pe=0).
gv1	1..1	PU	Gate 1 (Gv1). Gate Position value for point 1 for lookup table representing water flow through the turbine as a function of gate position to produce steady state flow.
gv2	1..1	PU	Gate 2 (Gv2). Gate Position value for point 2 for lookup table representing water flow through the turbine as a function of gate position to produce steady state flow.
gv3	1..1	PU	Gate 3 (Gv3). Gate Position value for point 3 for lookup table representing water flow through the turbine as a function of gate position to produce steady state flow.
gv4	1..1	PU	Gate 4 (Gv4). Gate Position value for point 4 for lookup table representing water flow through the turbine as a function of gate position to produce steady state flow.
gv5	1..1	PU	Gate 5 (Gv5). Gate Position value for point 5 for lookup table representing water flow through the turbine as a function of gate position to produce steady state flow.
fl1	1..1	PU	Flow Gate 1 (Fl1). Flow value for gate position point 1 for lookup table representing water flow through the turbine as a function of gate position to produce steady state flow.
fl2	1..1	PU	Flow Gate 2 (Fl2). Flow value for gate position point 2 for lookup table representing water flow through the turbine as a function of gate position to produce steady state

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			flow.
fl3	1..1	PU	Flow Gate 3 (Fl3). Flow value for gate position point 3 for lookup table representing water flow through the turbine as a function of gate position to produce steady state flow.
fl4	1..1	PU	Flow Gate 4 (Fl4). Flow value for gate position point 4 for lookup table representing water flow through the turbine as a function of gate position to produce steady state flow.
fl5	1..1	PU	Flow Gate 5 (Fl5). Flow value for gate position point 5 for lookup table representing water flow through the turbine as a function of gate position to produce steady state flow.
fp1	1..1	PU	Flow P1 (Fp1). Turbine Flow value for point 1 for lookup table representing per unit mechanical power on machine MVA rating as a function of turbine flow.
fp2	1..1	PU	Flow P2 (Fp2). Turbine Flow value for point 2 for lookup table representing per unit mechanical power on machine MVA rating as a function of turbine flow.
fp3	1..1	PU	Flow P3 (Fp3). Turbine Flow value for point 3 for lookup table representing per unit mechanical power on machine MVA rating as a function of turbine flow.
fp4	1..1	PU	Flow P4 (Fp4). Turbine Flow value for point 4 for lookup table representing per unit mechanical power on machine MVA rating as a function of turbine flow.
fp5	1..1	PU	Flow P5 (Fp5). Turbine Flow value for point 5 for lookup table representing per unit mechanical power on machine MVA rating as a function of turbine

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			flow.
fp6	1..1	PU	Flow P6 (Fp6). Turbine Flow value for point 6 for lookup table representing per unit mechanical power on machine MVA rating as a function of turbine flow.
fp7	1..1	PU	Flow P7 (Fp7). Turbine Flow value for point 7 for lookup table representing per unit mechanical power on machine MVA rating as a function of turbine flow.
fp8	1..1	PU	Flow P8 (Fp8). Turbine Flow value for point 8 for lookup table representing per unit mechanical power on machine MVA rating as a function of turbine flow.
fp9	1..1	PU	Flow P9 (Fp9). Turbine Flow value for point 9 for lookup table representing per unit mechanical power on machine MVA rating as a function of turbine flow.
fp10	1..1	PU	Flow P10 (Fp10). Turbine Flow value for point 10 for lookup table representing per unit mechanical power on machine MVA rating as a function of turbine flow.
pmss1	1..1	PU	Pmss Flow P1 (Pmss1). Mechanical Power output Pmss for Turbine Flow point 1 for lookup table representing per unit mechanical power on machine MVA rating as a function of turbine flow.
pmss2	1..1	PU	Pmss Flow P2 (Pmss2). Mechanical Power output Pmss for Turbine Flow point 2 for lookup table representing per unit mechanical power on machine MVA rating as a function of turbine flow.
pmss3	1..1	PU	Pmss Flow P3 (Pmss3). Mechanical Power output Pmss for Turbine Flow point 3 for lookup table representing per

			unit mechanical power on machine MVA rating as a function of turbine flow.
pmss4	1..1	PU	Pmss Flow P4 (Pmss4). Mechanical Power output Pmss for Turbine Flow point 4 for lookup table representing per unit mechanical power on machine MVA rating as a function of turbine flow.
pmss5	1..1	PU	Pmss Flow P5 (Pmss5). Mechanical Power output Pmss for Turbine Flow point 5 for lookup table representing per unit mechanical power on machine MVA rating as a function of turbine flow.
pmss6	1..1	PU	Pmss Flow P6 (Pmss6). Mechanical Power output Pmss for Turbine Flow point 6 for lookup table representing per unit mechanical power on machine MVA rating as a function of turbine flow.
pmss7	1..1	PU	Pmss Flow P7 (Pmss7). Mechanical Power output Pmss for Turbine Flow point 7 for lookup table representing per unit mechanical power on machine MVA rating as a function of turbine flow.
pmss8	1..1	PU	Pmss Flow P8 (Pmss8). Mechanical Power output Pmss for Turbine Flow point 8 for lookup table representing per unit mechanical power on machine MVA rating as a function of turbine flow.
pmss9	1..1	PU	Pmss Flow P9 (Pmss9). Mechanical Power output Pmss for Turbine Flow point 9 for lookup table representing per unit mechanical power on machine MVA rating as a function of turbine flow.
pmss10	1..1	PU	Pmss Flow P10 (Pmss10). Mechanical Power output Pmss for Turbine Flow point 10 for lookup table representing per unit mechanical power on

			machine MVA rating as a function of turbine flow.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-81 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-81 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] --- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] --- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.6.6.25 GovHydroWPID

Woodward PID Hydro Governor.

Figure 9-72 class diagram

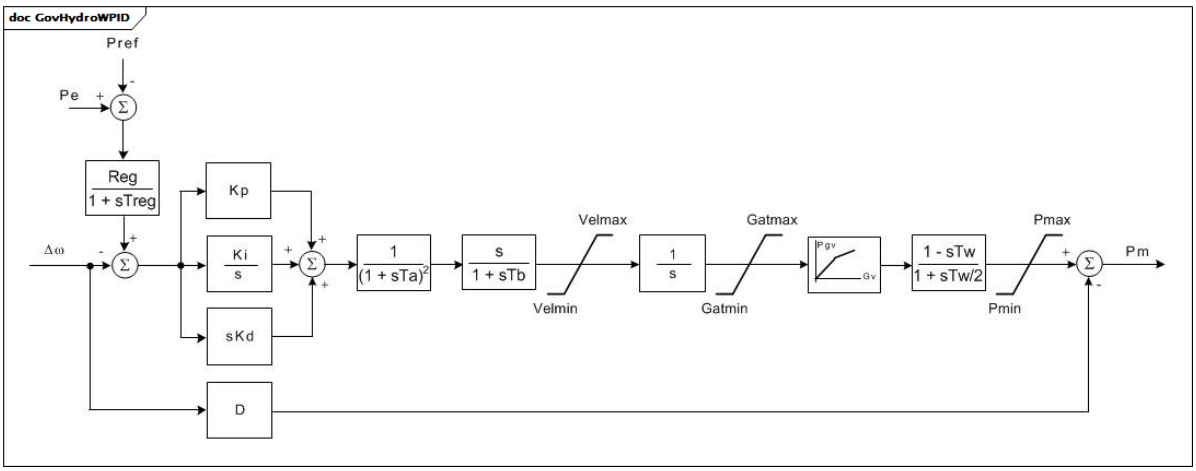


Figure 9-72 - figure GovHydroWPID

Parameter Notes:

1. The following attributes are using fixed values: **Pgv0**=0.0, **Gv3**=1.0.

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-82 shows all attributes of GovHydroWPID.

Table 9-82 - Attributes of TurbineGovernorDynamics::GovHydroWPID

Attribute name	mult.	Attribute type	Description
mwbase	1..1	ActivePower	Base for power values (MWbase) (>0). Unit = MW.
treg	1..1	Seconds	Speed detector time constant (Treg).
reg	1..1	PU	Permanent drop (Reg).
kp	1..1	PU	Proportional gain (Kp). Typical Value = 0.1.
ki	1..1	PU	Reset gain (Ki). Typical Value = 0.36.
kd	1..1	PU	Derivative gain (Kd). Typical Value = 1.11.
ta	1..1	Seconds	Controller time constant (Ta) (>0). Typical Value = 0.
tb	1..1	Seconds	Gate servo time constant (Tb) (>0). Typical Value = 0.
velmax	1..1	PU	Maximum gate opening velocity (Velmax). Unit = PU/sec. Typical Value = 0.
velmin	1..1	PU	Maximum gate closing velocity (Velmin). Unit = PU/sec. Typical Value = 0.
gatmax	1..1	PU	Gate opening Limit Maximum (Gatmax).
gatmin	1..1	PU	Gate opening Limit Minimum (Gatmin).
tw	1..1	Seconds	Water inertia time constant (Tw) (>0). Typical Value = 0.
pmax	1..1	PU	Maximum Power Output (Pmax).
pmin	1..1	PU	Minimum Power Output (Pmin).
d	1..1	PU	Turbine damping factor (D). Unit = delta P / delta speed.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

gv3	1..1	PU	Gate position 3 (Gv3).
gv1	1..1	PU	Gate position 1 (Gv1).
pgv1	1..1	PU	Output at Gv1 PU of MWbase (Pgv1).
gv2	1..1	PU	Gate position 2 (Gv2).
pgv2	1..1	PU	Output at Gv2 PU of MWbase (Pgv2).
pgv3	1..1	PU	Output at Gv3 PU of MWbase (Pgv3).
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-83 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-83 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] --- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] --- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.6.6.26 GovSteam0

A simplified steam turbine governor model.

Figure 9-73 class diagram

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

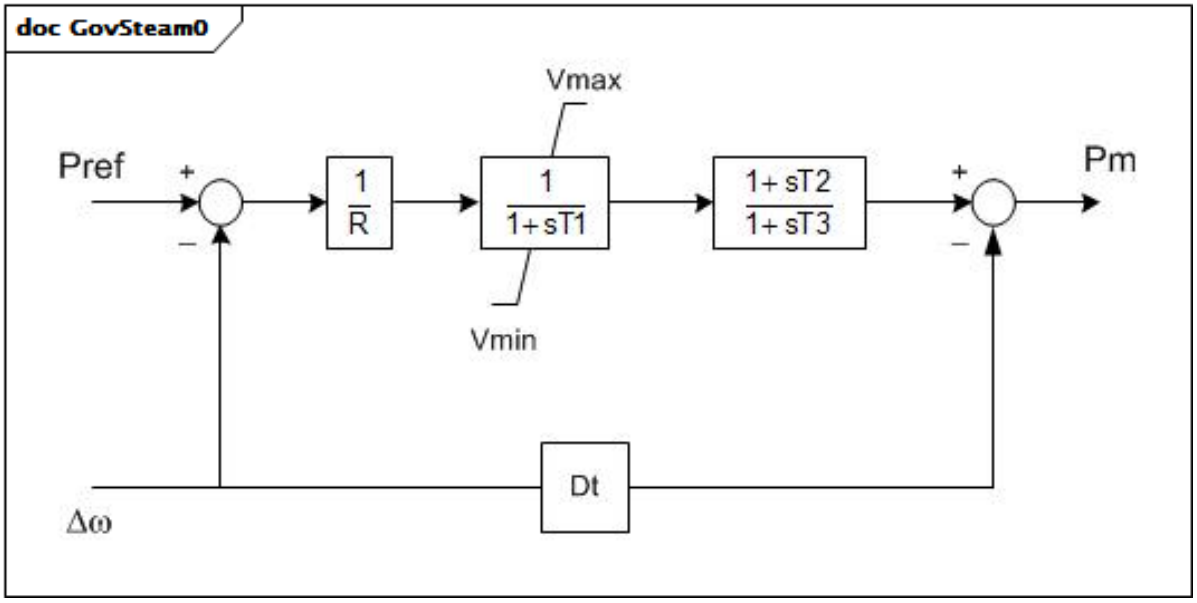


Figure 9-73 - figure GovSteam0

Parameter Notes:

1. **T1** and **T3** must be greater than zero.
2. **T2** may be zero. To represent a non-reheat turbine set **T2** = 0 and **T3** to about 0.5 sec. For reheat turbine, set **T2** = **T3** * (HP turbine fraction).

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-84 shows all attributes of GovSteam0.

Table 9-84 - Attributes of TurbineGovernorDynamics::GovSteam0

Attribute name	mult.	Attribute type	Description
mwbase	1..1	ActivePower	Base for power values (MWbase) (>0). Unit = MW.
r	1..1	PU	Permanent droop (R). Typical Value = 0.05.
t1	1..1	Seconds	Steam bowl time constant (T1). Typical Value = 0.5.
vmax	1..1	PU	Maximum valve position, PU of mwcap (Vmax). Typical Value = 1.
vmin	1..1	PU	Minimum valve position, PU of mwcap (Vmin). Typical Value = 0.
t2	1..1	Seconds	Numerator time constant of T2/T3 block (T2). Typical Value = 3.

t3	1..1	Seconds	Reheater time constant (T3). Typical Value = 10.
dt	1..1	PU	Turbine damping coefficient (Dt). Unit = delta P / delta speed. Typical Value = 0.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-85 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-85 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] --- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] --- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.6.6.27 GovSteam1

Steam turbine governor model, based on the GovSteamIEEE1 model (with optional deadband and nonlinear valve gain added).

Figure 9-74 class diagram

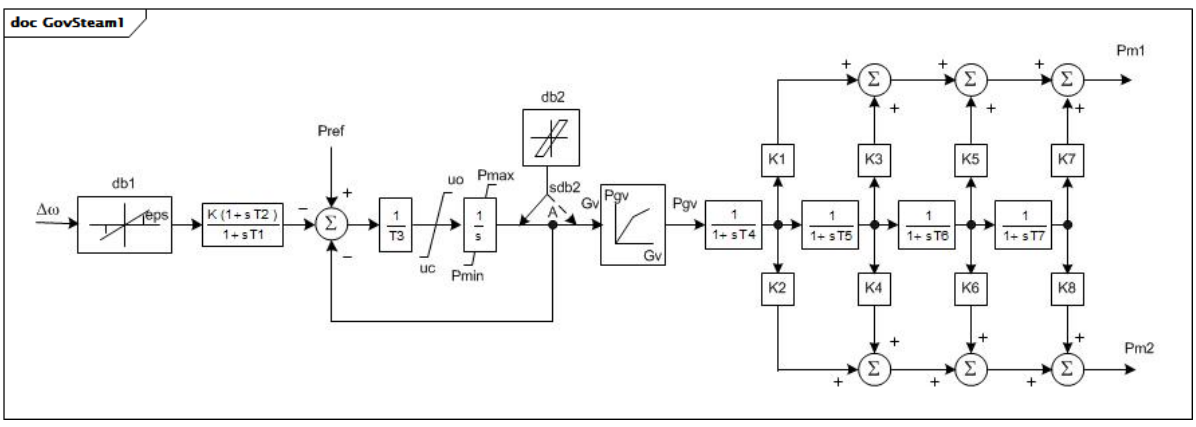


Figure 9-74 - figure GovSteam1

Parameter Notes:

1. Per unit parameters are on base of **MWbase**, which is normally the MW capability of the turbine.
2. **T3** must be greater than zero. All other time constants may be zero.
3. For a tandem-compound turbine the parameters **K2**, **K4**, **K6**, and **K8** are ignored. For a cross-compound turbine, two generators are connected to this turbine-governor model.
4. Each generator must be represented in the load flow by data on its own MVA base. The values of **K1**, **K3**, **K5**, **K7** must be specified to describe the proportionate development of power on the first turbine shaft. **K2**, **K4**, **K6**, **K8** must describe the second turbine shaft. Normally **K1** + **K3** + **K5** + **K7** = 1.0 and **K2** + **K4** + **K6** + **K8** = 1.0 (if second generator is present).
5. The division of power between the two shafts is in proportion to the values of MVA bases of the two generators. The initial condition load flow should, therefore, have the two generators loaded to the same fraction of each one's MVA base.
6. The intentional input speed deadband follows the relationship shown in the GovSteamIEEE1InputSpeedDeadband diagram.
7. The unintentional deadband can be used to represent backlash hysteresis in the control components by the relationship shown in the GovSteamIEEE1BacklashHysteresis diagram.
8. The nonlinear gain between gate position and power is represented with up to 6 points. The (0.,0.) and (1.,1.) points are assumed and need not be exchanged. The output is not allowed to go beyond 0. and 1. However, if **Pmax** > 1., the input and output are scaled by **Pmax**.
9. If **GV1** is a negative number, the default full-arc steam valve curve shown in the default will be used. If all zero values are provided, a straight line is used.

Figure 9-75 class diagram

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

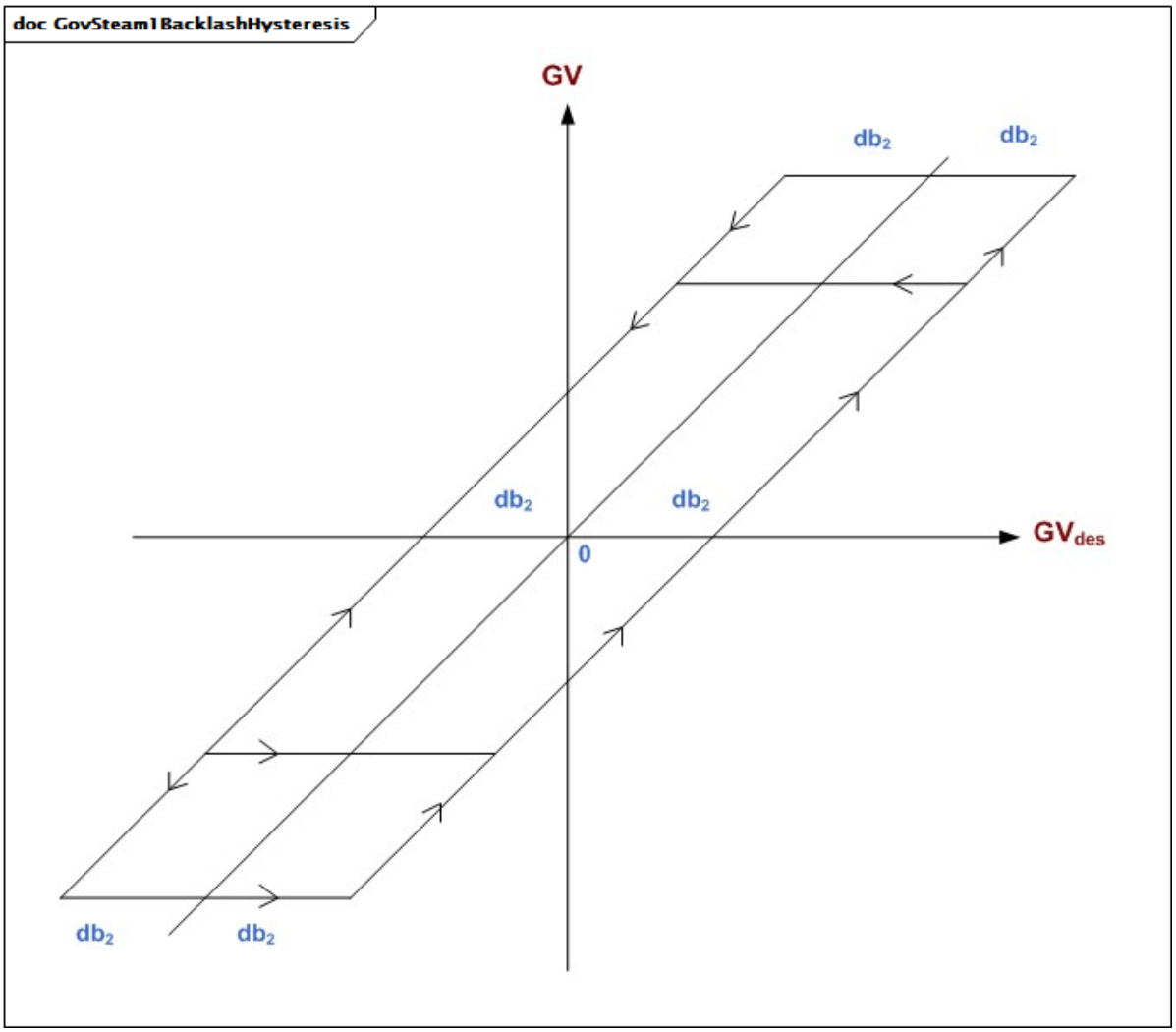


Figure 9-75 - figure GovSteam1BacklashHysteresis

Figure 9-76 class diagram

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

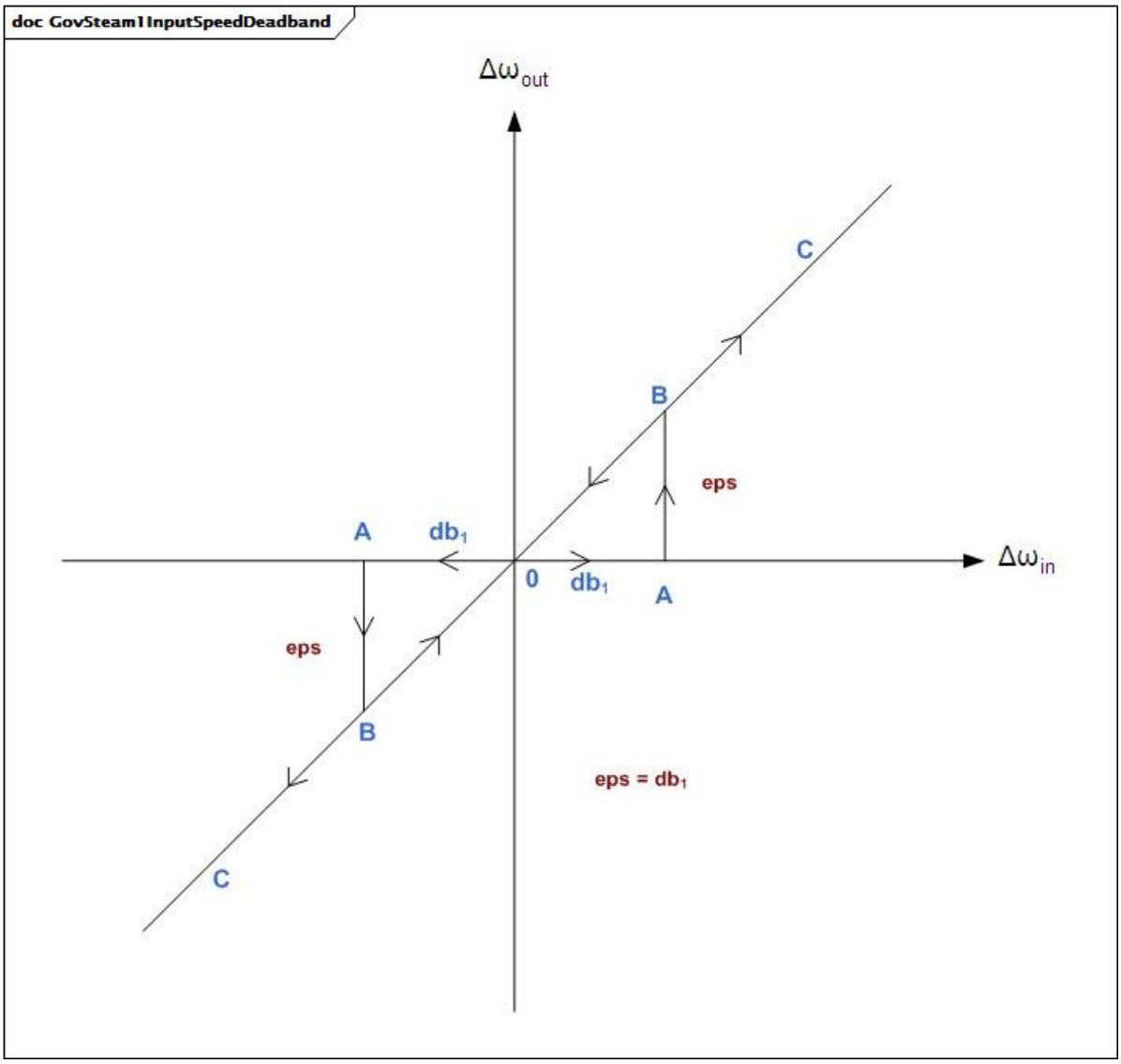


Figure 9-76 - figure GovSteam1InputSpeedDeadband

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-86 shows all attributes of GovSteam1.

Table 9-86 - Attributes of TurbineGovernorDynamics::GovSteam1

Attribute name	mult.	Attribute type	Description
mwbase	1..1	ActivePower	Base for power values (MWbase) (>0). Unit = MW.
k	1..1	PU	Governor gain (reciprocal of droop) (K) (>0). Typical Value = 25.
t1	1..1	Seconds	Governor lag time constant (T1). Typical Value = 0.

t2	1..1	Seconds	Governor lead time constant (T2). Typical Value = 0.
t3	1..1	Seconds	Valve positioner time constant (T3) (>0). Typical Value = 0.1.
uo	1..1	Simple_Float	Maximum valve opening velocity (Uo) (>0). Unit = PU/sec. Typical Value = 1.
uc	1..1	Simple_Float	Maximum valve closing velocity (Uc) (<0). Unit = PU/sec. Typical Value = -10.
pmax	1..1	PU	Maximum valve opening (Pmax) (> Pmin). Typical Value = 1.
pmin	1..1	PU	Minimum valve opening (Pmin) (>=0). Typical Value = 0.
t4	1..1	Seconds	Inlet piping/steam bowl time constant (T4). Typical Value = 0.3.
k1	1..1	Simple_Float	Fraction of HP shaft power after first boiler pass (K1). Typical Value = 0.2.
k2	1..1	Simple_Float	Fraction of LP shaft power after first boiler pass (K2). Typical Value = 0.
t5	1..1	Seconds	Time constant of second boiler pass (T5). Typical Value = 5.
k3	1..1	Simple_Float	Fraction of HP shaft power after second boiler pass (K3). Typical Value = 0.3.
k4	1..1	Simple_Float	Fraction of LP shaft power after second boiler pass (K4). Typical Value = 0.
t6	1..1	Seconds	Time constant of third boiler pass (T6). Typical Value = 0.5.
k5	1..1	Simple_Float	Fraction of HP shaft power after third boiler pass (K5). Typical Value = 0.5.
k6	1..1	Simple_Float	Fraction of LP shaft power after third boiler pass (K6). Typical Value = 0.
t7	1..1	Seconds	Time constant of fourth boiler pass (T7). Typical Value = 0.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

k7	1..1	Simple_Float	Fraction of HP shaft power after fourth boiler pass (K7). Typical Value = 0.
k8	1..1	Simple_Float	Fraction of LP shaft power after fourth boiler pass (K8). Typical Value = 0.
db1	1..1	Frequency	Intentional deadband width (db1). Unit = Hz. Typical Value = 0.
eps	1..1	Frequency	Intentional db hysteresis (eps). Unit = Hz. Typical Value = 0.
sdb1	1..1	Boolean	Intentional deadband indicator. true = intentional deadband is applied false = intentional deadband is not applied. Typical Value = true.
sdb2	1..1	Boolean	Unintentional deadband location. true = intentional deadband is applied before point "A" false = intentional deadband is applied after point "A". Typical Value = true.
db2	1..1	ActivePower	Unintentional deadband (db2). Unit = MW. Typical Value = 0.
valve	1..1	Boolean	Nonlinear valve characteristic. true = nonlinear valve characteristic is used false = nonlinear valve characteristic is not used. Typical Value = true.
gv1	1..1	PU	Nonlinear gain valve position point 1 (GV1). Typical Value = 0.
pgv1	1..1	PU	Nonlinear gain power value point 1 (Pgv1). Typical Value = 0.
gv2	1..1	PU	Nonlinear gain valve position point 2 (GV2). Typical Value = 0.4.
pgv2	1..1	PU	Nonlinear gain power value point 2 (Pgv2). Typical Value = 0.75.
gv3	1..1	PU	Nonlinear gain valve position point 3 (GV3). Typical Value = 0.5.
pgv3	1..1	PU	Nonlinear gain power value point 3 (Pgv3).

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			Typical Value = 0.91.
gv4	1..1	PU	Nonlinear gain valve position point 4 (GV4). Typical Value = 0.6.
pgv4	1..1	PU	Nonlinear gain power value point 4 (Pgv4). Typical Value = 0.98.
gv5	1..1	PU	Nonlinear gain valve position point 5 (GV5). Typical Value = 1.
pgv5	1..1	PU	Nonlinear gain power value point 5 (Pgv5). Typical Value = 1.
gv6	1..1	PU	Nonlinear gain valve position point 6 (GV6). Typical Value = 0.
pgv6	1..1	PU	Nonlinear gain power value point 6 (Pgv6). Typical Value = 0.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-87 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-87 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] --- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] --- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.6.6.28 GovSteam2

Simplified governor model.

Figure 9-77 class diagram

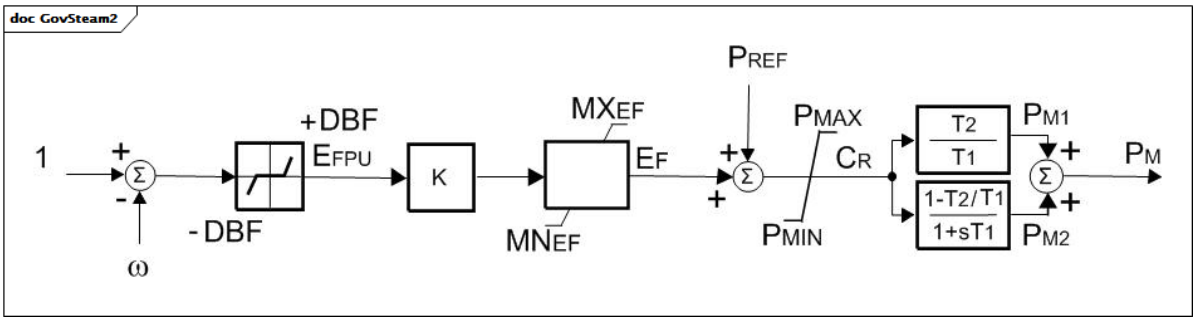


Figure 9-77 - figure GovSteam2

Parameter Notes:

1. Per unit parameters are on base of MW capability of the turbine.
2. This model can be used for both Thermal and Hydro power plants.
3. omega has units of per unit speed.

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-88 shows all attributes of GovSteam2.

Table 9-88 - Attributes of TurbineGovernorDynamics::GovSteam2

Attribute name	mult.	Attribute type	Description
k	1..1	Simple_Float	Governor gain (reciprocal of droop) (K). Typical Value = 20.
dbf	1..1	PU	Frequency dead band (DBF). Typical Value = 0.
t1	1..1	Seconds	Governor lag time constant (T_1) (>0). Typical Value = 0.45.
t2	1..1	Seconds	Governor lead time constant (T_2) (may be 0). Typical Value = 0.
pmax	1..1	PU	Maximum fuel flow (P_{MAX}). Typical Value = 1.
pmin	1..1	PU	Minimum fuel flow (P_{MIN}). Typical Value = 0.
mxef	1..1	PU	Fuel flow maximum positive error value (MX_{EF}). Typical Value = 1.
mnef	1..1	PU	Fuel flow maximum negative error value (MN_{EF}). Typical Value = -1.
enabled	1..1	Boolean	see

			DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-89 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-89 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] --- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] --- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.6.6.29 GovSteamCC

Cross compound turbine governor model.

Figure 9-78 class diagram

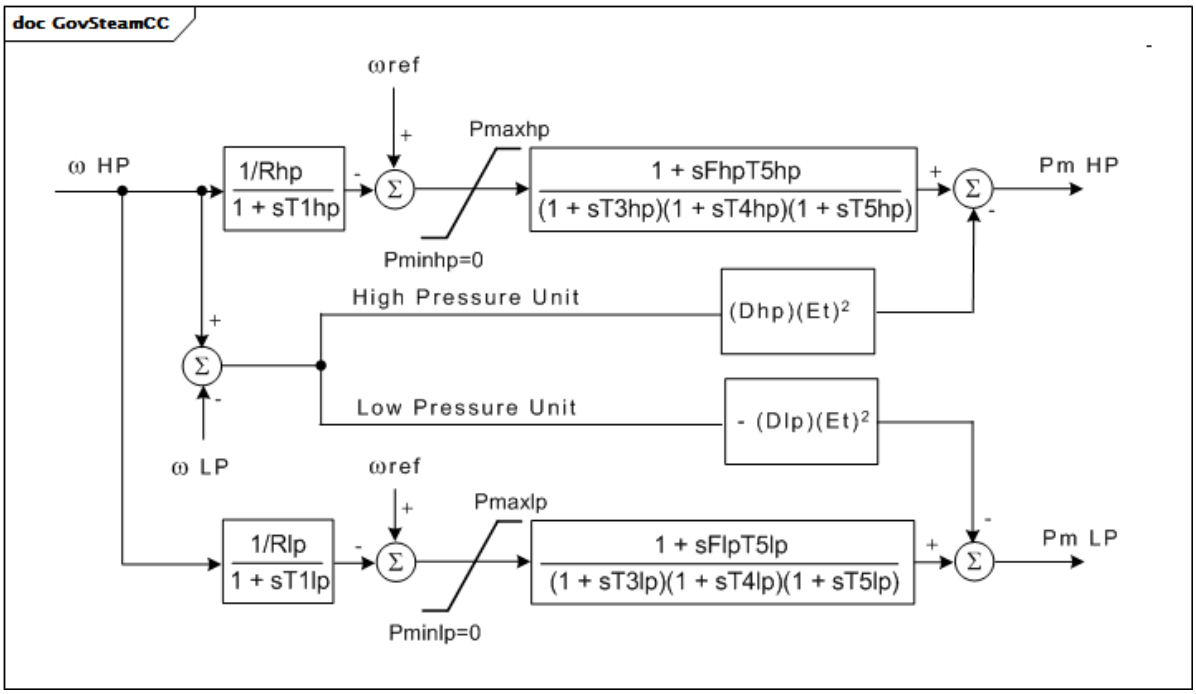


Figure 9-78 - figure GovSteamCC

Parameter Notes:

1. Both generators must be present and in service.
2. Each generator must be represented in the load flow by data stated on its own MVA base. The division of power between the two turbines is determined by the values of MWbase of the two generators, not by the values of **Fhp** and **Flp**.
3. **Fhp** and **Flp** may be used to represent the presence of dual reheat stages for a single turbine. If there is only one reheat stage, between the HP and LP turbines, **Fhp** and **Flp** should normally be unity and zero, respectively, and **T5lp** should be set to the reheater time constant.
4. **Dhp** and **Dlp** give a rough approximation of the damping of intermachine speed oscillations by electromagnetic effects. When the generators are represented by the round rotor model with proper values for subtransient time constants this damping is represented accurately in the generator electrical torques and **Dhp**, **Dlp** should be set to zero.
5. **Rhp** and **Rlp** must be greater than zero. If any of the time constants are zero, they are bypassed.

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-90 shows all attributes of GovSteamCC.

Table 9-90 - Attributes of TurbineGovernorDynamics::GovSteamCC

Attribute name	mult.	Attribute type	Description
mwbase	1..1	ActivePower	Base for power values (MWbase) (>0). Unit = MW.
pmaxhp	1..1	PU	Maximum HP value position (Pmaxhp). Typical Value = 1.
rhp	1..1	PU	HP governor droop (Rhp). Typical Value = 0.05.
t1hp	1..1	Seconds	HP governor time constant (T1hp). Typical Value = 0.1.
t3hp	1..1	Seconds	HP turbine time constant (T3hp). Typical Value = 0.1.
t4hp	1..1	Seconds	HP turbine time constant (T4hp). Typical Value = 0.1.
t5hp	1..1	Seconds	HP reheater time constant (T5hp). Typical Value = 10.
fhp	1..1	PU	Fraction of HP power ahead of reheater (Fhp). Typical Value = 0.3.
dhp	1..1	PU	HP damping factor (Dhp). Typical Value = 0.

pmaxlp	1..1	PU	Maximum LP value position (Pmaxlp). Typical Value = 1.
rlp	1..1	PU	LP governor droop (Rlp). Typical Value = 0.05.
t1lp	1..1	Seconds	LP governor time constant (T1lp). Typical Value = 0.1.
t3lp	1..1	Seconds	LP turbine time constant (T3lp). Typical Value = 0.1.
t4lp	1..1	Seconds	LP turbine time constant (T4lp). Typical Value = 0.1.
t5lp	1..1	Seconds	LP reheater time constant (T5lp). Typical Value = 10.
flp	1..1	PU	Fraction of LP power ahead of reheater (Flp). Typical Value = 0.7.
dlp	1..1	PU	LP damping factor (Dlp). Typical Value = 0.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-91 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-91 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] --- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] --- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.6.6.30 GovSteamEU

Simplified model of boiler and steam turbine with PID governor.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Figure 9-79 class diagram

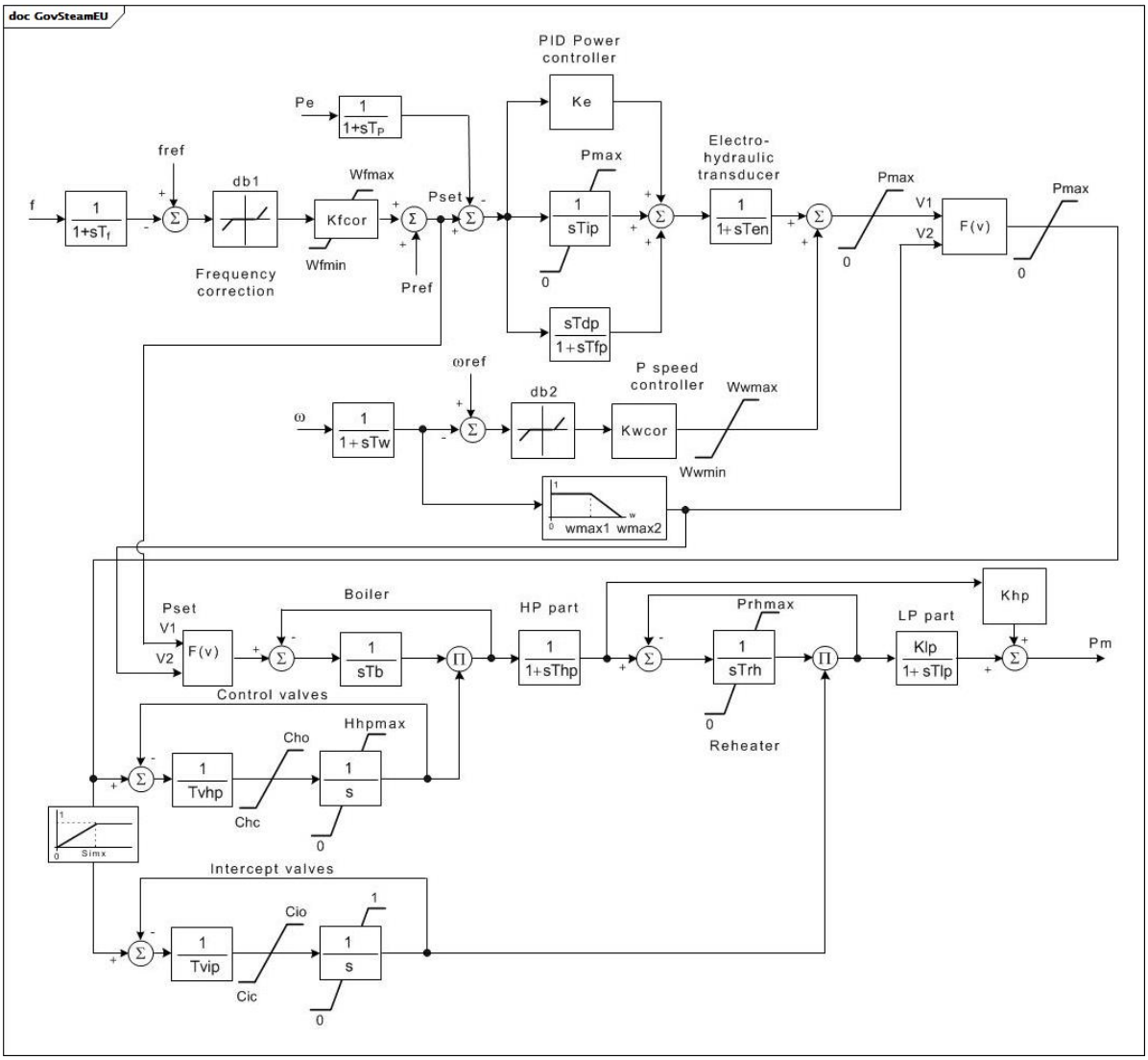


Figure 9-79 - figure GovSteamEU

Parameter Notes:

1. Per unit parameters are on base of turbine MW capability **MWbase**.
2. **Thp** and **Tlp** must be greater than zero.
3. **Trh** may be zero. To represent a non-reheat turbine set **Trh** = 0 and **Tlp** to about 0.5 sec.
4. **F(v)** is minimum value selection
5. Frequency (**f**) is local (nodal) frequency.

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-92 shows all attributes of GovSteamEU.

Table 9-92 - Attributes of TurbineGovernorDynamics::GovSteamEU

Attribute name	mult.	Attribute type	Description
mwbase	1..1	ActivePower	Base for power values (MWbase) (>0). Unit = MW.
tp	1..1	Seconds	Power transducer time constant (Tp). Typical Value = 0.07.
ke	1..1	PU	Gain of the power controller (Ke). Typical Value = 0.65.
tip	1..1	Seconds	Integral time constant of the power controller (Tip). Typical Value = 2.
tdp	1..1	Seconds	Derivative time constant of the power controller (Tdp). Typical Value = 0.
tfp	1..1	Seconds	Time constant of the power controller (Tfp). Typical Value = 0.
tf	1..1	Seconds	Frequency transducer time constant (Tf). Typical Value = 0.
kfcor	1..1	PU	Gain of the frequency corrector (Kfcor). Typical Value = 20.
db1	1..1	PU	Dead band of the frequency corrector (db1). Typical Value = 0.
wfmax	1..1	PU	Upper limit for frequency correction (Wfmax). Typical Value = 0.05.
wfmin	1..1	PU	Lower limit for frequency correction (Wfmin). Typical Value = -0.05.
pmax	1..1	PU	Maximal active power of the turbine (Pmax). Typical Value = 1.
ten	1..1	Seconds	Electro hydraulic transducer (Ten). Typical Value = 0.1.
tw	1..1	Seconds	Speed transducer time constant (Tw). Typical Value = 0.02.
kwcor	1..1	PU	Gain of the speed governor (Kwcor). Typical Value = 20.
db2	1..1	PU	Dead band of the speed governor (db2). Typical Value = 0.0004.

wwmax	1..1	PU	Upper limit for the speed governor (Wwmax). Typical Value = 0.1.
wwmin	1..1	PU	Lower limit for the speed governor frequency correction (Wwmin). Typical Value = -1.
wmax1	1..1	PU	Emergency speed control lower limit (wmax1). Typical Value = 1.025.
wmax2	1..1	PU	Emergency speed control upper limit (wmax2). Typical Value = 1.05.
tvhp	1..1	Seconds	Control valves servo time constant (Tvhp). Typical Value = 0.1.
cho	1..1	Simple_Float	Control valves rate opening limit (Cho). Unit = PU/sec. Typical Value = 0.17.
chc	1..1	Simple_Float	Control valves rate closing limit (Chc). Unit = PU/sec. Typical Value = -3.3.
hhpmax	1..1	PU	Maximum control valve position (Hhpmax). Typical Value = 1.
tvip	1..1	Seconds	Intercept valves servo time constant (Tvip). Typical Value = 0.15.
cio	1..1	PU	Intercept valves rate opening limit (Cio). Typical Value = 0.123.
cic	1..1	PU	Intercept valves rate closing limit (Cic). Typical Value = -2.2.
simx	1..1	PU	Intercept valves transfer limit (Simx). Typical Value = 0.425.
thp	1..1	Seconds	High pressure (HP) time constant of the turbine (Thp). Typical Value = 0.31.
trh	1..1	Seconds	Reheater time constant of the turbine (Trh). Typical Value = 8.
tlp	1..1	Seconds	Low pressure(LP) time constant of the turbine (Tlp). Typical Value = 0.45.
prhmax	1..1	PU	Maximum low pressure limit (Prhmax). Typical Value = 1.4.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

khp	1..1	PU	Fraction of total turbine output generated by HP part (Khp). Typical Value = 0.277.
klp	1..1	PU	Fraction of total turbine output generated by HP part (Klp). Typical Value = 0.723.
tb	1..1	Seconds	Boiler time constant (Tb). Typical Value = 100.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-93 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-93 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] ----- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] ----- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.6.6.31 GovSteamFV2

Steam turbine governor with reheat time constants and modeling of the effects of fast valve closing to reduce mechanical power.

Figure 9-80 class diagram

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

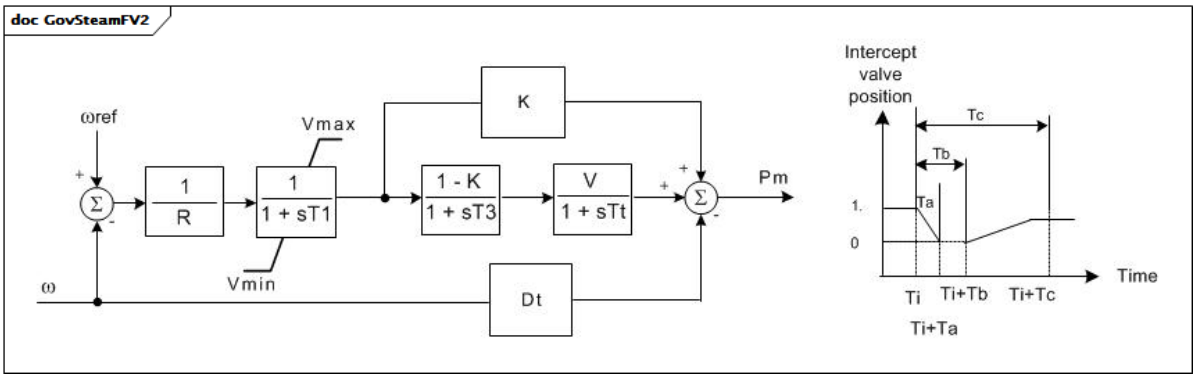


Figure 9-80 - figure GovSteamFV2

Valve position is determined by the time curve. The valve position is either 1 or 0.

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-94 shows all attributes of GovSteamFV2.

Table 9-94 - Attributes of TurbineGovernorDynamics::GovSteamFV2

Attribute name	mult.	Attribute type	Description
mwbase	1..1	ActivePower	Alternate Base used instead of Machine base in equipment model if necessary (MWbase) (>0). Unit = MW.
r	1..1	PU	(R).
t1	1..1	Seconds	Governor time constant (T1).
vmax	1..1	PU	(Vmax).
vmin	1..1	PU	(Vmin).
k	1..1	PU	Fraction of the turbine power developed by turbine sections not involved in fast valving (K).
t3	1..1	Seconds	Reheater time constant (T3).
dt	1..1	PU	(Dt).
tt	1..1	Seconds	Time constant with which power falls off after intercept valve closure (Tt).
ta	1..1	Seconds	Time after initial time for valve to close (Ta).
tb	1..1	Seconds	Time after initial time for valve to begin opening (Tb).
tc	1..1	Seconds	Time after initial time for valve to become

			fully open (Tc).
ti	1..1	Seconds	Initial time to begin fast valving (Ti).
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-95 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-95 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] --- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] --- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.6.6.32 GovSteamFV3

Simplified GovSteamIEEE1 Steam turbine governor model with Prmax limit and fast valving.

Figure 9-81 class diagram

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

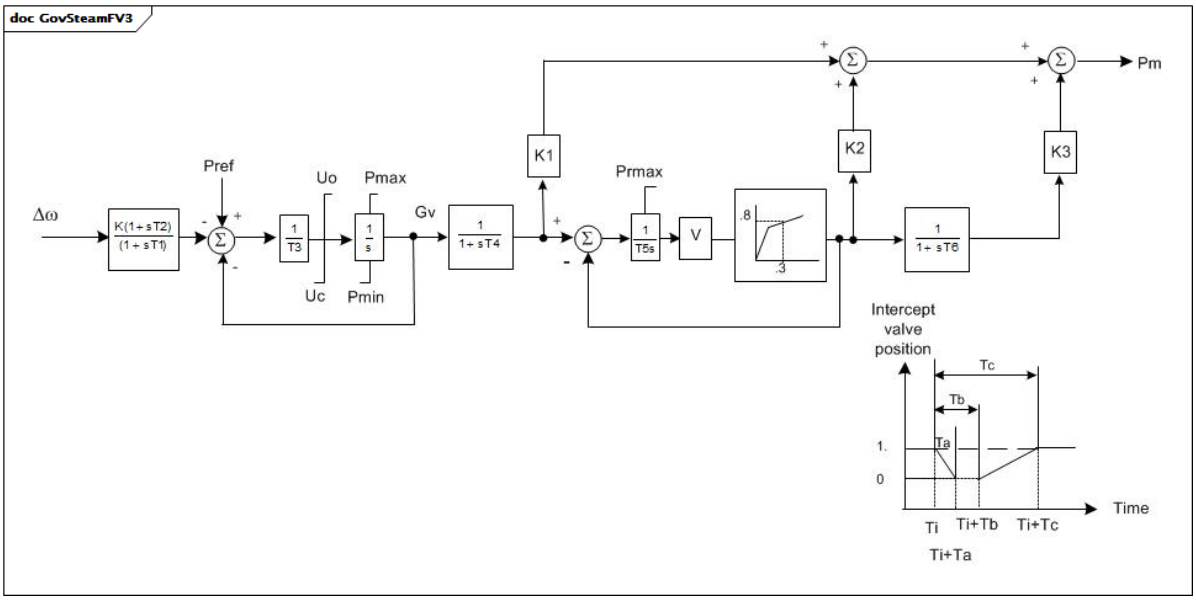


Figure 9-81 - figure GovSteamFV3

Parameter Notes:

1. **T3** must be greater than zero. **T5** must be greater than zero if fast valving is used. All other time constants may be zero.
2. The gains **K1, K2, K3** describe the division of power output among turbine stages. Normally, **K1**+ **K2** + **K3** = 1.
3. The times **Ta**, **Tb**, and **Tc** are all with respect to the initiation time.
4. The value **Prmax** is a limit on the pressure in the reheater. Normally, this value should be about 1.10 p.u.
5. A piece-wise linear characteristic for intercept valve flow vs. position is built in.

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-96 shows all attributes of GovSteamFV3.

Table 9-96 - Attributes of TurbineGovernorDynamics::GovSteamFV3

Attribute name	mult.	Attribute type	Description
mwbase	1..1	ActivePower	Base for power values (MWbase) (>0). Unit = MW.
k	1..1	PU	Governor gain, (reciprocal of droop) (K). Typical Value = 20.
t1	1..1	Seconds	Governor lead time constant (T1). Typical Value = 0.
t2	1..1	Seconds	Governor lag time constant (T2). Typical Value = 0.
t3	1..1	Seconds	Valve positioner time constant (T3). Typical

			Value = 0.
uo	1..1	Simple_Float	Maximum valve opening velocity (Uo). Unit = PU/sec. Typical Value = 0.1.
uc	1..1	Simple_Float	Maximum valve closing velocity (Uc). Unit = PU/sec. Typical Value = -1.
pmax	1..1	PU	Maximum valve opening, PU of MWbase (Pmax). Typical Value = 1.
pmin	1..1	PU	Minimum valve opening, PU of MWbase (Pmin). Typical Value = 0.
t4	1..1	Seconds	Inlet piping/steam bowl time constant (T4). Typical Value = 0.2.
k1	1..1	PU	Fraction of turbine power developed after first boiler pass (K1). Typical Value = 0.2.
t5	1..1	Seconds	Time constant of second boiler pass (i.e. reheater) (T5). Typical Value = 0.5.
k2	1..1	PU	Fraction of turbine power developed after second boiler pass (K2). Typical Value = 0.2.
t6	1..1	Seconds	Time constant of crossover or third boiler pass (T6). Typical Value = 10.
k3	1..1	PU	Fraction of hp turbine power developed after crossover or third boiler pass (K3). Typical Value = 0.6.
ta	1..1	Seconds	Time to close intercept valve (IV) (Ta). Typical Value = 0.97.
tb	1..1	Seconds	Time until IV starts to reopen (Tb). Typical Value = 0.98.
tc	1..1	Seconds	Time until IV is fully open (Tc). Typical Value = 0.99.
prmax	1..1	PU	Max. pressure in reheater (Prmax). Typical Value = 1.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-97 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-97 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] --- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] --- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.6.6.33 GovSteamFV4

Detailed electro-hydraulic governor for steam unit.

Figure 9-82 class diagram

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

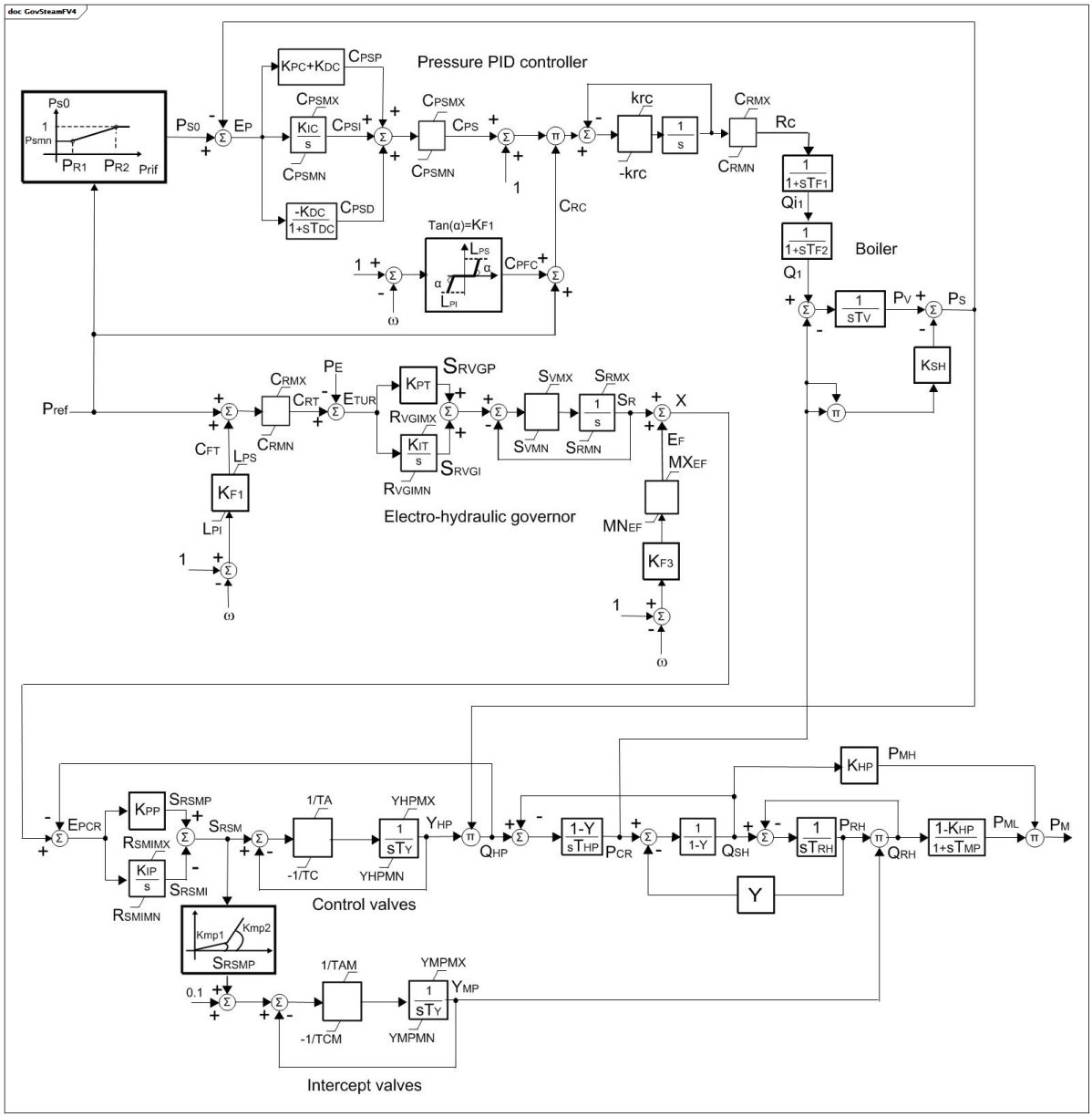


Figure 9-82 - figure GovSteamFV4

Parameter Notes:

1. Per unit parameters are on base of MW capability of the turbine.
2. This model can be used for steam units in coordinated control, not belonging to a combined-cycle power plant.
3. omega has units of per unit speed.

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-98 shows all attributes of GovSteamFV4.

Table 9-98 - Attributes of TurbineGovernorDynamics::GovSteamFV4

Attribute name	mult.	Attribute type	Description
kf1	1..1	PU	Frequency bias (reciprocal of droop) (Kf1). Typical Value = 20.
kf3	1..1	PU	Frequency control (reciprocal of droop) (Kf3). Typical Value = 20.
lps	1..1	PU	Maximum positive power error (Lps). Typical Value = 0.03.
lpi	1..1	PU	Maximum negative power error (Lpi). Typical Value = -0.15.
mxef	1..1	PU	Upper limit for frequency correction (MX_{EF}). Typical Value = 0.05.
mnef	1..1	PU	Lower limit for frequency correction (MN_{EF}). Typical Value = -0.05.
crmx	1..1	PU	Maximum value of regulator set-point (Crmx). Typical Value = 1.2.
crmn	1..1	PU	Minimum value of regulator set-point (Crmn). Typical Value = 0.
kpt	1..1	PU	Proportional gain of electro-hydraulic regulator (Kpt). Typical Value = 0.3.
kit	1..1	PU	Integral gain of electro-hydraulic regulator (Kit). Typical Value = 0.04.
rvgmx	1..1	PU	Maximum value of integral regulator (Rvgmx). Typical Value = 1.2.
rvgmn	1..1	PU	Minimum value of integral regulator (Rvgmn). Typical Value = 0.
svmx	1..1	Simple_Float	Maximum regulator gate opening velocity (Svmx). Typical Value = 0.0333.
svmn	1..1	Simple_Float	Maximum regulator gate closing velocity (Svmn). Typical Value = -0.0333.
srmx	1..1	PU	Maximum valve opening

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			(Srmx). Typical Value = 1.1.
srmn	1..1	PU	Minimum valve opening (Srmn). Typical Value = 0.
kpp	1..1	PU	Proportional gain of pressure feedback regulator (Kpp). Typical Value = 1.
kip	1..1	PU	Integral gain of pressure feedback regulator (Kip). Typical Value = 0.5.
rsmimx	1..1	PU	Maximum value of integral regulator (Rsmimx). Typical Value = 1.1.
rsmimn	1..1	PU	Minimum value of integral regulator (Rsmimn). Typical Value = 0.
kmp1	1..1	PU	First gain coefficient of intercept valves characteristic (Kmp1). Typical Value = 0.5.
kmp2	1..1	PU	Second gain coefficient of intercept valves characteristic (Kmp2). Typical Value = 3.5.
srsmp	1..1	PU	Intercept valves characteristic discontinuity point (Srsmp). Typical Value = 0.43.
ta	1..1	Seconds	Control valves rate opening time (Ta). Typical Value = 0.8.
tc	1..1	Seconds	Control valves rate closing time (Tc). Typical Value = 0.5.
ty	1..1	Seconds	Control valves servo time constant (Ty). Typical Value = 0.1.
yhpmx	1..1	PU	Maximum control valve position (Yhpmx). Typical Value = 1.1.
yhpmn	1..1	PU	Minimum control valve position (Yhpmn). Typical Value = 0.
tam	1..1	Seconds	Intercept valves rate opening time (Tam). Typical Value = 0.8.
tcm	1..1	Seconds	Intercept valves rate closing time (Tcm). Typical Value = 0.5.
ympmx	1..1	PU	Maximum intercept valve position (Ympmx).

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			Typical Value = 1.1.
ympmn	1..1	PU	Minimum intercept valve position (Ympmn). Typical Value = 0.
y	1..1	PU	Coefficient of linearized equations of turbine (Stodola formulation) (Y). Typical Value = 0.13.
thp	1..1	Seconds	High pressure (HP) time constant of the turbine (Thp). Typical Value = 0.15.
trh	1..1	Seconds	Reheater time constant of the turbine (Trh). Typical Value = 10.
tmp	1..1	Seconds	Low pressure (LP) time constant of the turbine (Tmp). Typical Value = 0.4.
khp	1..1	PU	Fraction of total turbine output generated by HP part (Khp). Typical Value = 0.35.
pr1	1..1	PU	First value of pressure set point static characteristic (Pr1). Typical Value = 0.2.
pr2	1..1	PU	Second value of pressure set point static characteristic, corresponding to Ps0 = 1.0 PU (Pr2). Typical Value = 0.75.
psmn	1..1	PU	Minimum value of pressure set point static characteristic (Psmn). Typical Value = 1.
kpc	1..1	PU	Proportional gain of pressure regulator (Kpc). Typical Value = 0.5.
kic	1..1	PU	Integral gain of pressure regulator (Kic). Typical Value = 0.0033.
kdc	1..1	PU	Derivative gain of pressure regulator (Kdc). Typical Value = 1.
tdc	1..1	Seconds	Derivative time constant of pressure regulator (Tdc). Typical Value = 90.
cpsmx	1..1	PU	Maximum value of pressure regulator output (Cpsmx). Typical Value = 1.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

cpsmn	1..1	PU	Minimum value of pressure regulator output (Cpsmn). Typical Value = -1.
krc	1..1	PU	Maximum variation of fuel flow (Krc). Typical Value = 0.05.
tf1	1..1	Seconds	Time constant of fuel regulation (Tf1). Typical Value = 10.
tf2	1..1	Seconds	Time constant of steam chest (Tf2). Typical Value = 10.
tv	1..1	Seconds	Boiler time constant (Tv). Typical Value = 60.
ksh	1..1	PU	Pressure loss due to flow friction in the boiler tubes (Ksh). Typical Value = 0.08.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-99 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-99 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] ----- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] ----- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.6.6.34 GovSteamSGO

Simplified Steam turbine governor model.

Figure 9-83 class diagram

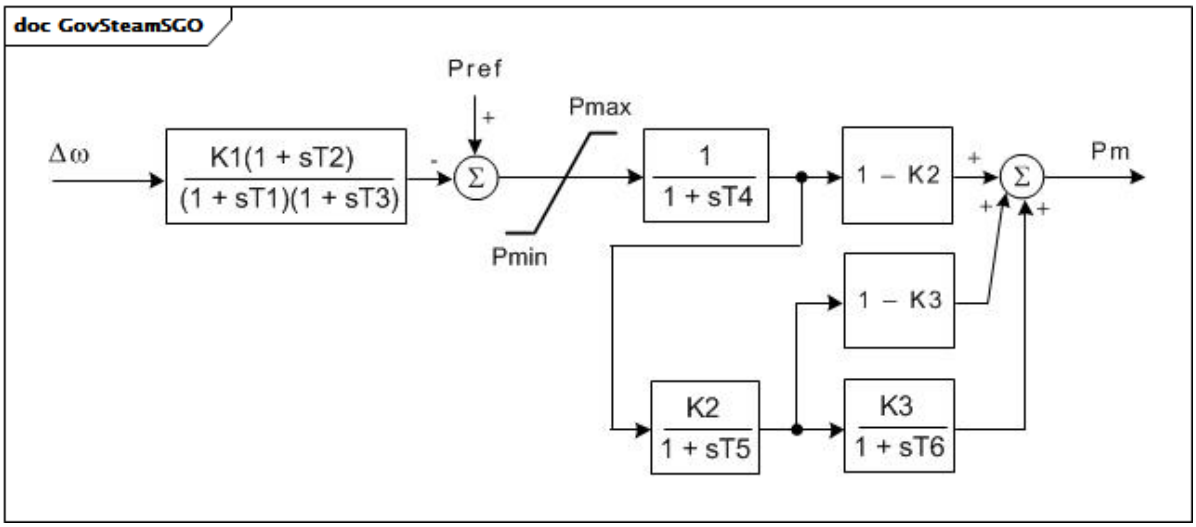


Figure 9-83 - figure GovSteamSGO

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-100 shows all attributes of GovSteamSGO.

Table 9-100 - Attributes of TurbineGovernorDynamics::GovSteamSGO

Attribute name	mult.	Attribute type	Description
mwbase	1..1	ActivePower	Base for power values (MWbase) (>0). Unit = MW.
t1	1..1	Seconds	Controller lag (T1).
t2	1..1	Seconds	Controller lead compensation (T2).
t3	1..1	Seconds	Governor lag (T3) (>0).
t4	1..1	Seconds	Delay due to steam inlet volumes associated with steam chest and inlet piping (T4).
t5	1..1	Seconds	Reheater delay including hot and cold leads (T5).
t6	1..1	Seconds	Delay due to IP-LP turbine, crossover pipes and LP end hoods (T6).
k1	1..1	PU	One/per unit regulation (K1).
k2	1..1	PU	Fraction (K2).
k3	1..1	PU	Fraction (K3).
pmax	1..1	PU	Upper power limit (Pmax).
pmin	1..1	Seconds	Lower power limit (Pmin).

enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-101 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-101 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] --- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] --- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.6.6.35 Enumerations

9.6.6.35.1 DroopSignalFeedbackKind

Governor droop signal feedback source.

Literal	Description	Code
electricalPower	Electrical power feedback (connection indicated as 1 in the block diagrams of models, e.g. GovCT1, GovCT2).	
none	No droop signal feedback, is isochronous governor.	
fuelValveStroke	Fuel valve stroke feedback (true stroke) (connection indicated as 2 in the block diagrams of model, e.g. GovCT1, GovCT2).	
governorOutput	Governor output feedback (requested stroke) (connection indicated as 3 in the block diagrams of models, e.g. GovCT1, GovCT2).	

9.6.6.35.2 FrancisGovernorControlKind

Governor control flag for Francis hydro model.

Literal	Description	Code
mechanicHydrolicTachoAccelerator	Mechanic-hydraulic regulator with tacho-accelerometer (Cflag = 1).	
mechanicHydraulicTransientFeedback	Mechanic-hydraulic regulator with transient feedback (Cflag=2).	
electromechanicalElectrohydraulic	Electromechanical and electrohydraulic regulator (Cflag=3).	

9.6.7 TurbineLoadControllerDynamics

9.6.7.1 General

A turbine load controller acts to maintain turbine power at a set value by continuous adjustment of the turbine governor speed-load reference.

Figure 9-84 shows the diagram.

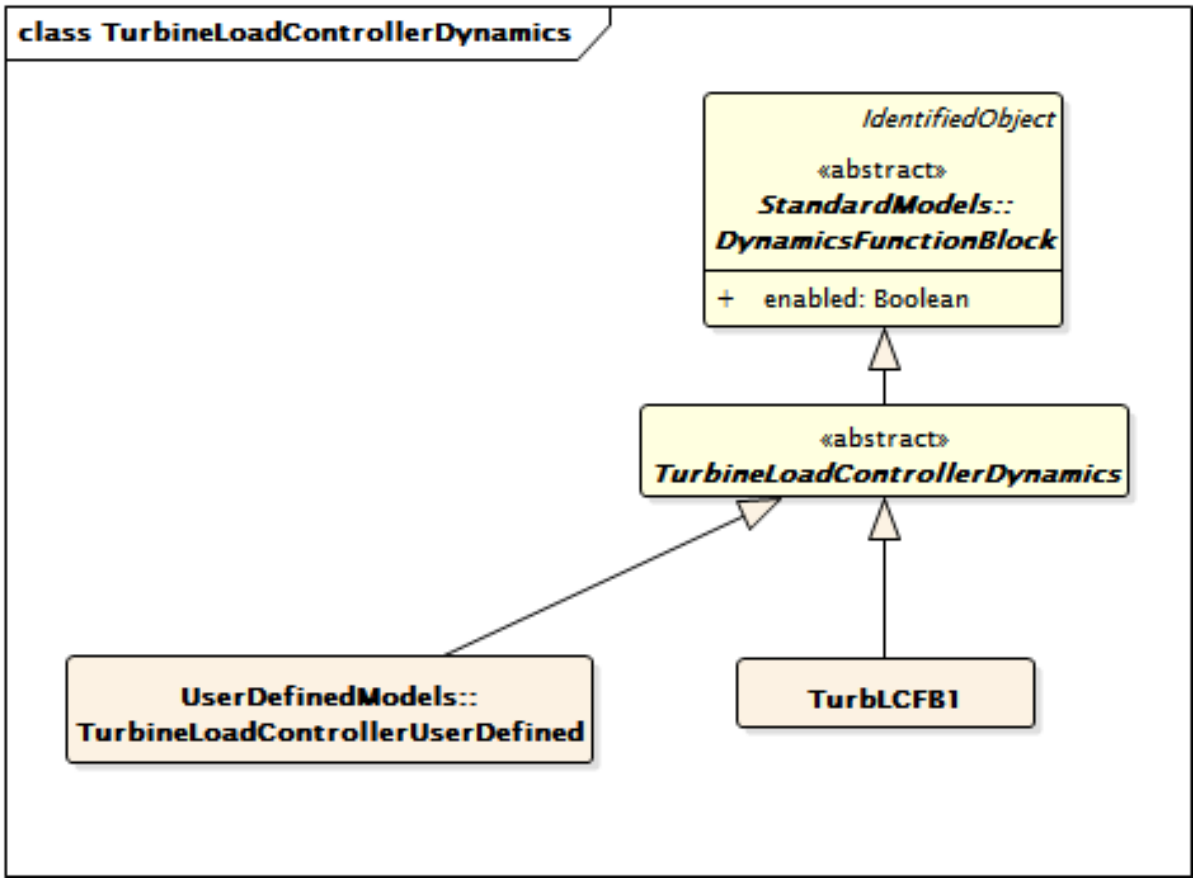


Figure 9-84 - diagram TurbineLoadControllerDynamics

This diagram shows dynamic standard model turbine load controller classes and their inheritance from the TurbineLoadControllerDynamics class.

9.6.7.2 TurbineLoadControllerDynamics(Abstract)

Turbine load controller function block whose behavior is described by reference to a standard model or by definition of a user-defined model.

Inheritance path: ->DynamicsFunctionBlock->IdentifiedObject

Table 9-102 shows all attributes of TurbineLoadControllerDynamics.

Table 9-102 - Attributes of TurbineLoadControllerDynamics::TurbineLoadControllerDynamics

Attribute name	mult.	Attribute type	Description
enabled	1..1	Boolean	see DynamicsFunctionBlock

description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-103 shows all association ends of TurbineLoadControllerDynamics with other classes.

Table 9-103 - Association ends of TurbineLoadControllerDynamics::TurbineLoadControllerDynamics with other classes

mult.	mult.	Class type name	Description
TurbineGovernorDynamics	[1..1]	TurbineGovernorDynamics	Turbine-governor controlled by this turbine load controller. Association Based On : TurbineLoadControllerDynamics:: TurbineLoadControllerDynamics.TurbineLoadControllerDynamics [0..1] ----- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[1]

9.6.7.3 TurbLCFB1

Turbine Load Controller model developed in the WECC. This model represents a supervisory turbine load controller that acts to maintain turbine power at a set value by continuous adjustment of the turbine governor speed-load reference. This model is intended to represent slow reset 'outer loop' controllers managing the action of the turbine governor.

Figure 9-85 class diagram

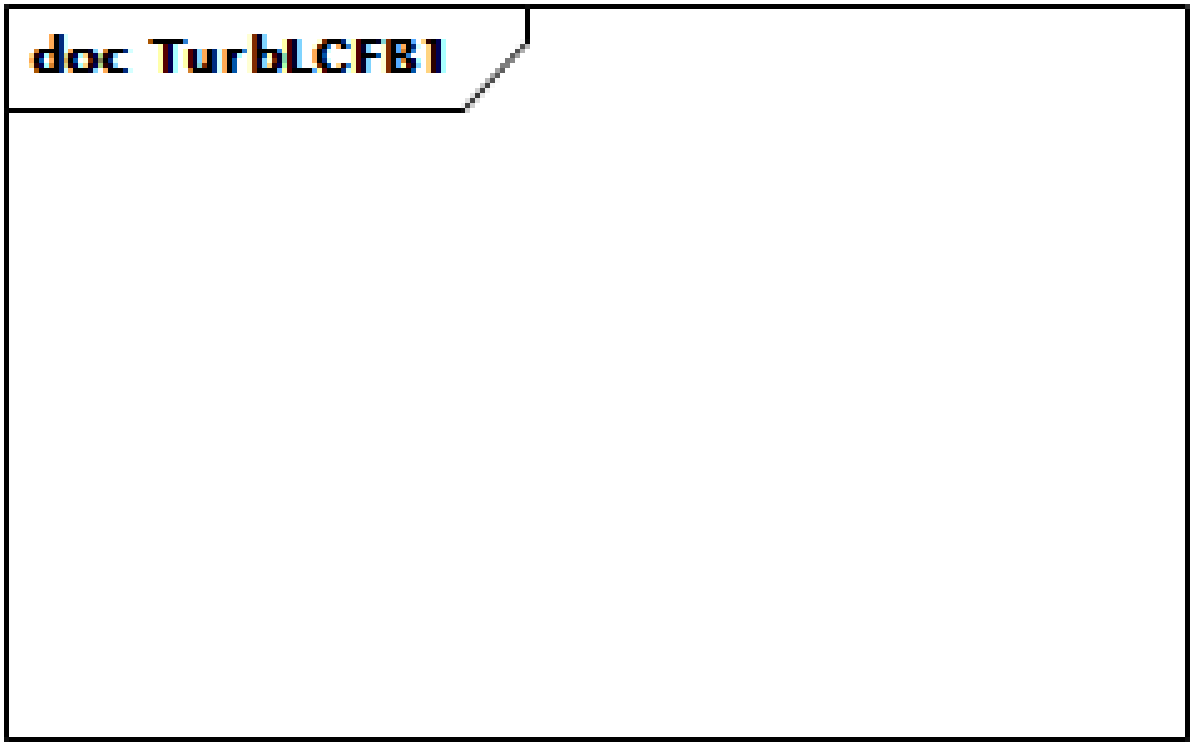


Figure 9-85 - figure TurbLCFB1

Parameter Notes:

1. The load reference of this supervisory load control loop is accessible as the parameter, **Pmwset**. **Pmwset** is given a value automatically when the model is initialized. This value overwrites any value entered prior to initialization.
2. The controller acts by applying a bias to the turbine-governor speed load reference, **Pref**. This reference is initialized in the normal manner by the turbine-governor model.
3. The output of the controller is limited to plus or minus **Irmax**. To prevent the controller from adjusting the turbine power excessively, **Irmax** should not exceed approximately one half of the governor droop or approximately 0.025 per unit.
4. The error limit, **Emax**, is intended to set a maximum rate of change of turbine power. If limitation of load change rate is desired, **Emax** should be set such that **Emax** = (max rate, PU sec)*(droop of governor in PU)/(**Ki** of TurbLCFB1). Note that governor droop used in this calculation must be in per unit relative to the base power, **MWbase**, of the turbine. Note, also, that some governor models (such as GovSteamIEEE1) have their droop specified by the value of a gain whose value is the reciprocal of the droop. For example, in GovSteamIEEE1 the droop is specified by the gain, **K**, whose typical value is 1/0.04 = 25.
5. This model recognizes the two alternative ways of specifying the turbine-governor load reference. In models such as GovCT1 and GovHydro1, the reference is a speed setpoint. In other models such as GovSteamIEEE1 the reference is a per unit load value. The references are generally as follows: 1) speed reference (1 + [initial power or valve position]*[droop]), 2) load reference (0 + [initial power or valve position]). The parameters of TurbLCFB1, other than **Type**, must always be set on the basis of a governor speed reference.

Inheritance path: ->TurbineLoadControllerDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-104 shows all attributes of TurbLCFB1.

Table 9-104 - Attributes of TurbineLoadControllerDynamics::TurbLCFB1

Attribute name	mult.	Attribute type	Description
mwbase	1..1	ActivePower	Base for power values (MWbase) (>0). Unit = MW.
speedReferenceGovernor	1..1	Boolean	Type of turbine governor reference (Type). true = speed reference governor false = load reference governor. Typical Value = true.
db	1..1	PU	Controller dead band (db). Typical Value = 0.
emax	1..1	PU	Maximum control error (Emax) (note 4). Typical Value = 0.02.
fb	1..1	PU	Frequency bias gain (Fb). Typical Value = 0.
kp	1..1	PU	Proportional gain (Kp). Typical Value = 0.
ki	1..1	PU	Integral gain (Ki). Typical Value = 0.
fbf	1..1	Boolean	Frequency bias flag (Fbf).

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			true = enable frequency bias false = disable frequency bias. Typical Value = false.
pbf	1..1	Boolean	Power controller flag (Pbf). true = enable load controller false = disable load controller. Typical Value = false.
tpelec	1..1	Seconds	Power transducer time constant (Tpelec). Typical Value = 0.
irmax	1..1	PU	Maximum turbine speed/load reference bias (Irmax) (note 3). Typical Value = 0.
pmwset	1..1	ActivePower	Power controller setpoint (Pmwset) (note 1). Unit = MW. Typical Value = 0.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-105 shows all association ends of TurbineLoadControllerDynamics with other classes.

Table 9-105 - Association ends of TurbineLoadControllerDynamics::TurbineLoadControllerDynamics with other classes

mult.	mult.	Class type name	Description
TurbineGovernorDynamics	1..1	TurbineGovernorDynamics	Inherited Association Based On : TurbineLoadControllerDynamics:: TurbineLoadControllerDynamics.TurbineLoadControllerDynamics[0..1] --- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[1]

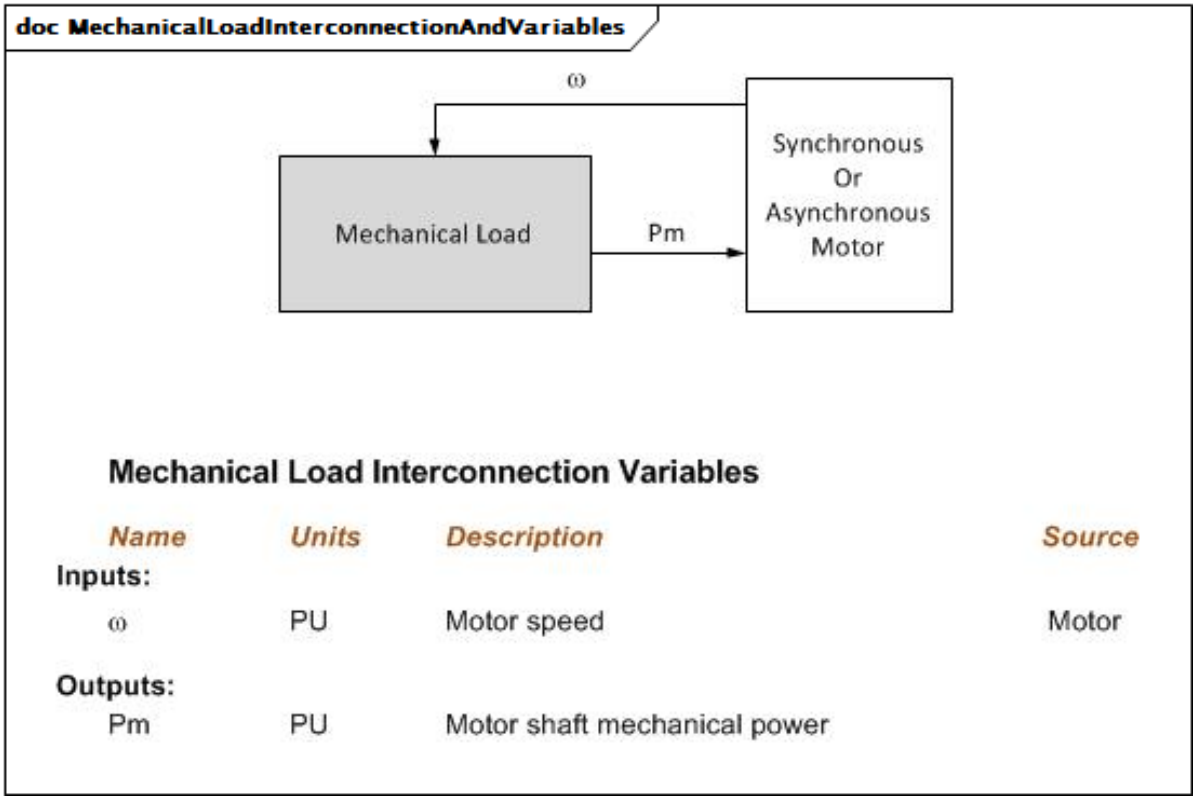
9.6.8 MechanicalLoadDynamics

9.6.8.1 General

A mechanical load represents the variation in a motor's shaft torque or power as a function of shaft speed.

Figure 9-86 shows the diagram.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu



Mechanical Load Interconnection Variables

Name	Units	Description	Source
Inputs:			
ω	PU	Motor speed	Motor
Outputs:			
Pm	PU	Motor shaft mechanical power	

Figure 9-86 - diagram MechanicalLoadInterconnectionAndVariables

This diagram illustrates the inputs and outputs for a mechanical load.

Figure 9-87 shows the diagram.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

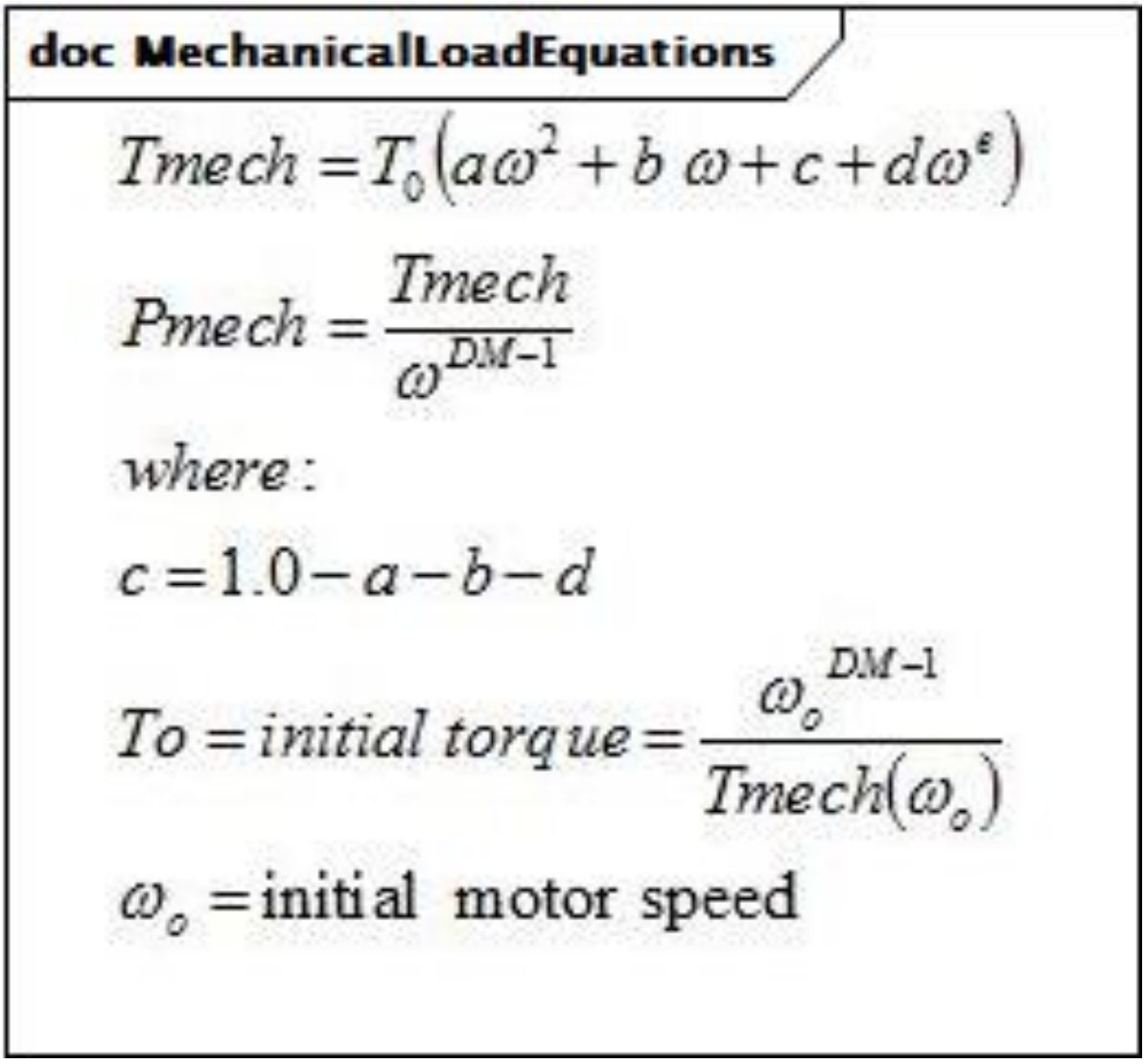


Figure 9-87 - diagram MechanicalLoadEquations

Figure 9-88 shows the diagram.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

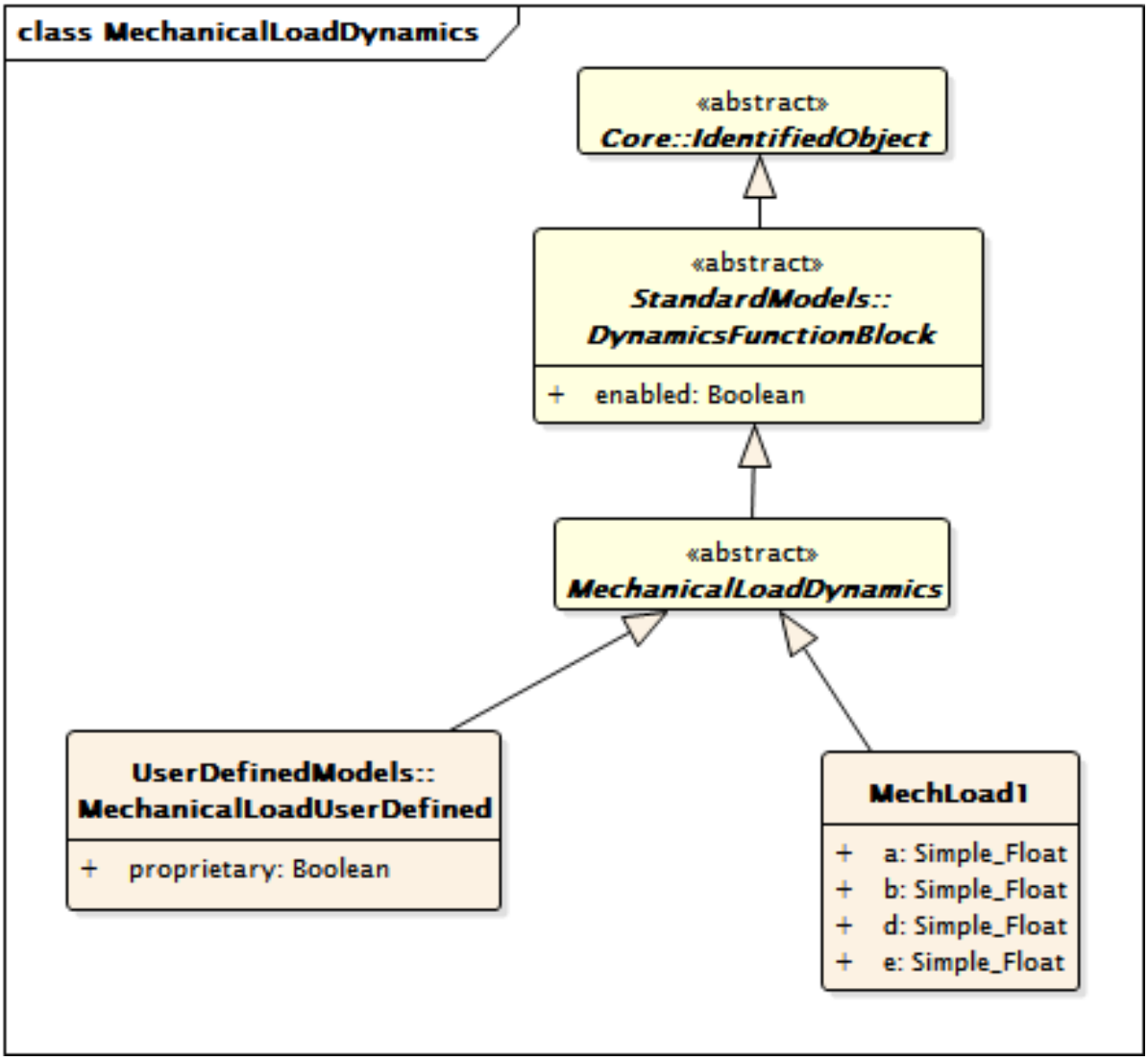


Figure 9-88 - diagram MechanicalLoadDynamics

This diagram shows dynamic standard model mechanical load classes and their inheritance from the MechanicalLoadDynamics class.

9.6.8.2 MechanicalLoadDynamics(Abstract)

Mechanical load function block whose behavior is described by reference to a standard model or by definition of a user-defined model.

Inheritance path: ->DynamicsFunctionBlock->IdentifiedObject

Table 9-106 shows all attributes of MechanicalLoadDynamics.

Table 9-106 - Attributes of MechanicalLoadDynamics::MechanicalLoadDynamics

Attribute name	mult.	Attribute type	Description
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-107 shows all association ends of MechanicalLoadDynamics with other classes.

Table 9-107 - Association ends of MechanicalLoadDynamics::MechanicalLoadDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	[0..1]	AsynchronousMachineDynamics	Asynchronous machine model with which this mechanical load model is associated. Association Based On : MechanicalLoadDynamics::MechanicalLoadDynamics.MechanicalLoadDynamics[0..1] --- AsynchronousMachineDynamics::AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	[0..1]	SynchronousMachineDynamics	Synchronous machine model with which this mechanical load model is associated. Association Based On : MechanicalLoadDynamics::MechanicalLoadDynamics.MechanicalLoadDynamics[0..1] --- SynchronousMachineDynamics::SynchronousMachineDynamics.SynchronousMachineDynamics[0..1]

9.6.8.3 MechLoad1

Mechanical load model type 1.

Inheritance path: ->MechanicalLoadDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-108 shows all attributes of MechLoad1.

Table 9-108 - Attributes of MechanicalLoadDynamics::MechLoad1

Attribute name	mult.	Attribute type	Description
a	1..1	Simple_Float	Speed squared coefficient (a).
b	1..1	Simple_Float	Speed coefficient (b).
d	1..1	Simple_Float	Speed to the exponent coefficient (d).
e	1..1	Simple_Float	Exponent (e).
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-109 shows all association ends of MechanicalLoadDynamics with other classes.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 9-109 - Association ends of MechanicalLoadDynamics::MechanicalLoadDynamics with other classes

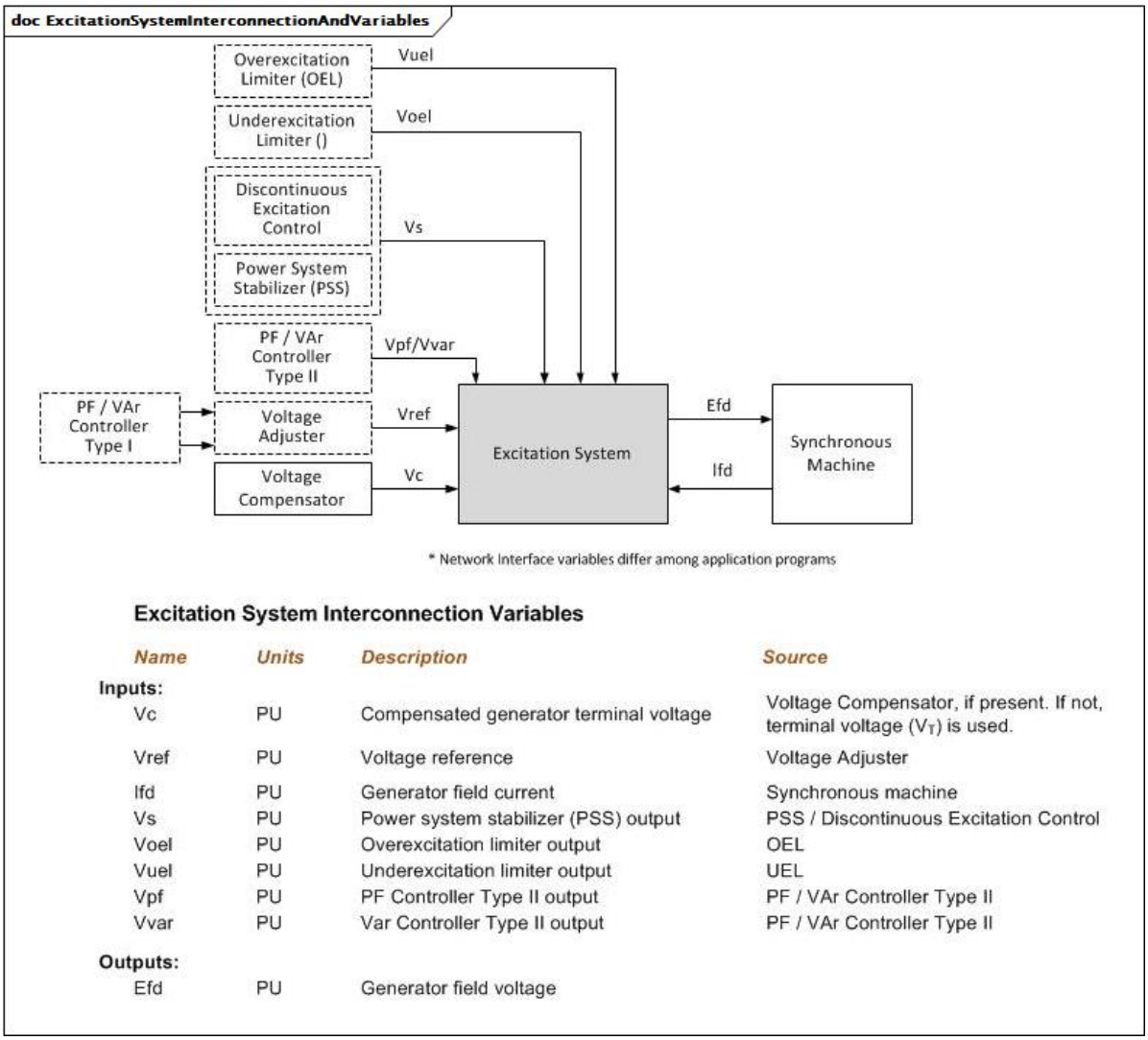
mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : MechanicalLoadDynamics:: MechanicalLoadDynamics.MechanicalLoadDynamics[0..1] --- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	Inherited Association Based On : MechanicalLoadDynamics:: MechanicalLoadDynamics.MechanicalLoadDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..1]

9.6.9 ExcitationSystemDynamics

9.6.9.1 General

The excitation system model provides the field voltage (Efd) for a synchronous machine model. It is linked to a specific generator (synchronous machine). The data parameters are different for each excitation system model; the same parameter name may have different meaning in different models.

Figure 9-89 shows the diagram.



Excitation System Interconnection Variables

Name	Units	Description	Source
Inputs:			
Vc	PU	Compensated generator terminal voltage	Voltage Compensator, if present. If not, terminal voltage (V_T) is used.
Vref	PU	Voltage reference	Voltage Adjuster
Ifd	PU	Generator field current	Synchronous machine
Vs	PU	Power system stabilizer (PSS) output	PSS / Discontinuous Excitation Control
Voel	PU	Overexcitation limiter output	OEL
Vuel	PU	Underexcitation limiter output	UEL
Vpf	PU	PF Controller Type II output	PF / VAr Controller Type II
Vvar	PU	Var Controller Type II output	PF / VAr Controller Type II
Outputs:			
Efd	PU	Generator field voltage	

Figure 9-89 - diagram ExcitationSystemInterconnectionAndVariables

This diagram illustrates the inputs and outputs for an excitation system.

Figure 9-90 shows the diagram.

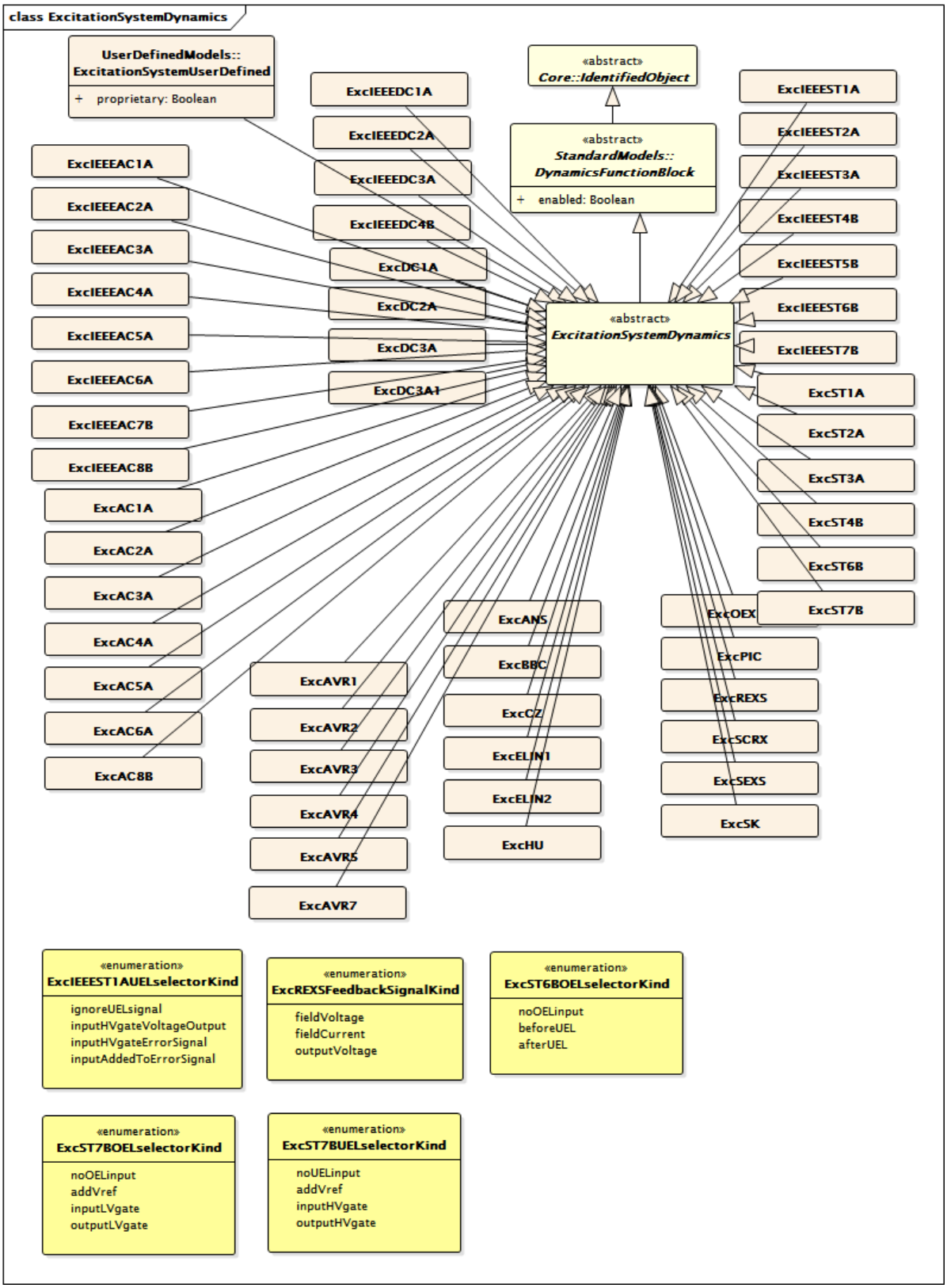


Figure 9-90 - diagram ExcitationSystemDynamics

This diagram shows dynamic standard model excitation system classes and their inheritance from the ExcitationSystemDynamics class.

9.6.9.2 ExcitationSystemDynamics(Abstract)

Excitation system function block whose behavior is described by reference to a standard model or by definition of a user-defined model.

Inheritance path: ->DynamicsFunctionBlock->IdentifiedObject

Table 9-110 shows all attributes of ExcitationSystemDynamics.

Table 9-110 - Attributes of ExcitationSystemDynamics::ExcitationSystemDynamics

Attribute name	mult.	Attribute type	Description
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-111 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-111 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	[1..1]	SynchronousMachineDynamics	Synchronous machine model with which this excitation system model is associated. Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] ----- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.3 ExcIEEEAC1A

The class represents IEEE Std 421.5-2005 type AC1A model. The model represents the field-controlled alternator-rectifier excitation systems designated Type AC1A. These excitation systems consist of an alternator main exciter with non-controlled rectifiers. Reference: IEEE Standard 421.5-2005 Section 6.1.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-112 shows all attributes of ExcIEEEAC1A.

Table 9-112 - Attributes of ExcitationSystemDynamics::ExcIEEEAC1A

Attribute name	mult.	Attribute type	Description
tb	1..1	Seconds	Voltage regulator time constant (T_B). Typical Value = 0.
tc	1..1	Seconds	Voltage regulator time constant (T_C). Typical Value = 0.

ka	1..1	PU	Voltage regulator gain (K_A). Typical Value = 400.
ta	1..1	Seconds	Voltage regulator time constant (T_A). Typical Value = 0.02.
vamax	1..1	PU	Maximum voltage regulator output (V_{AMAX}). Typical Value = 14.5.
vamin	1..1	PU	Minimum voltage regulator output (V_{AMIN}). Typical Value = -14.5.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (T_E). Typical Value = 0.8.
kf	1..1	PU	Excitation control system stabilizer gains (K_F). Typical Value = 0.03.
tf	1..1	Seconds	Excitation control system stabilizer time constant (T_F). Typical Value = 1.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (K_C). Typical Value = 0.2.
kd	1..1	PU	Demagnetizing factor, a function of exciter alternator reactances (K_D). Typical Value = 0.38.
ke	1..1	PU	Exciter constant related to self-excited field (K_E). Typical Value = 1.
ve1	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (V_{E1}). Typical Value = 4.18.
seve1	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, V_{E1}, back of commutating reactance ($S_E[V_{E1}]$). Typical Value = 0.1.
ve2	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (V_{E2}). Typical Value = 3.14.
seve2	1..1	Simple_Float	Exciter saturation function value at the

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			corresponding exciter voltage, V_{E2}, back of commutating reactance ($S_E[V_{E2}]$). Typical Value = 0.03.
vrmax	1..1	PU	Maximum voltage regulator outputs (V_{RMAX}). Typical Value = 6.03.
vrmin	1..1	PU	Minimum voltage regulator outputs (V_{RMIN}). Typical Value = -5.43.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-113 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-113 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] ----- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.4 ExcIEEEAC2A

The class represents IEEE Std 421.5-2005 type AC2A model. The model represents a high initial response field-controlled alternator-rectifier excitation system. The alternator main exciter is used with non-controlled rectifiers. The Type AC2A model is similar to that of Type AC1A except for the inclusion of exciter time constant compensation and exciter field current limiting elements. Reference: IEEE Standard 421.5-2005 Section 6.2.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-114 shows all attributes of ExcIEEEAC2A.

Table 9-114 - Attributes of ExcitationSystemDynamics::ExcIEEEAC2A

Attribute name	mult.	Attribute type	Description
tb	1..1	Seconds	Voltage regulator time constant (T_B). Typical Value = 0.
tc	1..1	Seconds	Voltage regulator time constant (T_C). Typical Value = 0.
ka	1..1	PU	Voltage regulator gain

			(K_A). Typical Value = 400.
ta	1..1	Seconds	Voltage regulator time constant (T_A). Typical Value = 0.02.
vamax	1..1	PU	Maximum voltage regulator output (V_{AMAX}). Typical Value = 8.
vamin	1..1	PU	Minimum voltage regulator output (V_{AMIN}). Typical Value = -8.
kb	1..1	PU	Second stage regulator gain (K_B). Typical Value = 25.
vrmax	1..1	PU	Maximum voltage regulator outputs (V_{RMAX}). Typical Value = 105.
vrmin	1..1	PU	Minimum voltage regulator outputs (V_{RMIN}). Typical Value = -95.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (T_E). Typical Value = 0.6.
vfemax	1..1	PU	Exciter field current limit reference (V_{FEMAX}). Typical Value = 4.4.
kh	1..1	PU	Exciter field current feedback gain (K_H). Typical Value = 1.
kf	1..1	PU	Excitation control system stabilizer gains (K_F). Typical Value = 0.03.
tf	1..1	Seconds	Excitation control system stabilizer time constant (T_F). Typical Value = 1.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (K_C). Typical Value = 0.28.
kd	1..1	PU	Demagnetizing factor, a function of exciter alternator reactances (K_D). Typical Value = 0.35.
ke	1..1	PU	Exciter constant related to self-excited field (K_E). Typical Value = 1.
ve1	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is

			defined (V_{E1}). Typical Value = 4.4.
seve1	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, V_{E1}, back of commutating reactance ($S_E[V_{E1}]$). Typical Value = 0.037.
ve2	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (V_{E2}). Typical Value = 3.3.
seve2	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, V_{E2}, back of commutating reactance ($S_E[V_{E2}]$). Typical Value = 0.012.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-115 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-115 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.5 ExcIEEEAC3A

The class represents IEEE Std 421.5-2005 type AC3A model. The model represents the field-controlled alternator-rectifier excitation systems designated Type AC3A. These excitation systems include an alternator main exciter with non-controlled rectifiers. The exciter employs self-excitation, and the voltage regulator power is derived from the exciter output voltage. Therefore, this system has an additional nonlinearity, simulated by the use of a multiplier whose inputs are the voltage regulator command signal, **Va**, and the exciter output voltage, **Efd**, times $\mathbf{K_R}$. This model is applicable to excitation systems employing static voltage regulators. Reference: IEEE Standard 421.5-2005 Section 6.3.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-116 shows all attributes of ExcIEEEAC3A.

Table 9-116 - Attributes of ExcitationSystemDynamics::ExcIEEEAC3A

Attribute name	mult.	Attribute type	Description
tb	1..1	Seconds	Voltage regulator time constant (T_B). Typical Value = 0.
tc	1..1	Seconds	Voltage regulator time constant (T_C). Typical Value = 0.
ka	1..1	PU	Voltage regulator gain (K_A). Typical Value = 45.62.
ta	1..1	Seconds	Voltage regulator time constant (T_A). Typical Value = 0.013.
vamax	1..1	PU	Maximum voltage regulator output (V_{AMAX}). Typical Value = 1.
vamin	1..1	PU	Minimum voltage regulator output (V_{AMIN}). Typical Value = -0.95.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (T_E). Typical Value = 1.17.
vemin	1..1	PU	Minimum exciter voltage output (V_{EMIN}). Typical Value = 0.1.
kr	1..1	PU	Constant associated with regulator and alternator field power supply (K_R). Typical Value = 3.77.
kf	1..1	PU	Excitation control system stabilizer gains (K_F). Typical Value = 0.143.
tf	1..1	Seconds	Excitation control system stabilizer time constant (T_F). Typical Value = 1.
kn	1..1	PU	Excitation control system stabilizer gain (K_N). Typical Value = 0.05.
efdn	1..1	PU	Value of *EFD* at which feedback gain changes (E_{FDN}). Typical Value = 2.36.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (K_C). Typical Value = 0.104.
kd	1..1	PU	Demagnetizing factor, a function of exciter

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			alternator reactances (K_D). Typical Value = 0.499.
ke	1..1	PU	Exciter constant related to self-excited field (K_E). Typical Value = 1.
vfemax	1..1	PU	Exciter field current limit reference (V_{FEMAX}). Typical Value = 16.
ve1	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (V_{E1}) equals V_{EMAX} (V_{E1}). Typical Value = 6.24.
seve1	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, V_{E1}, back of commutating reactance ($S_E[V_{E1}]$). Typical Value = 1.143.
ve2	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (V_{E2}). Typical Value = 4.68.
seve2	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, V_{E2}, back of commutating reactance ($S_E[V_{E2}]$). Typical Value = 0.1.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-117 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-117 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] ----- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.6 ExcIEEEAC4A

The class represents IEEE Std 421.5-2005 type AC4A model. The model represents type AC4A alternator-supplied controlled-rectifier excitation system which is quite different from the other type ac systems. This high initial response excitation system utilizes a full thyristor bridge in the exciter output circuit. The voltage regulator controls the firing of the thyristor bridges. The exciter alternator uses an independent voltage regulator to control its output voltage to a constant value. These effects are not modeled; however, transient loading effects on the exciter alternator are included.
Reference: IEEE Standard 421.5-2005 Section 6.4.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-118 shows all attributes of ExcIEEEAC4A.

Table 9-118 - Attributes of ExcitationSystemDynamics::ExcIEEEAC4A

Attribute name	mult.	Attribute type	Description
vimax	1..1	PU	Maximum voltage regulator input limit (V_{IMAX}). Typical Value = 10.
vimin	1..1	PU	Minimum voltage regulator input limit (V_{IMIN}). Typical Value = -10.
tc	1..1	Seconds	Voltage regulator time constant (T_C). Typical Value = 1.
tb	1..1	Seconds	Voltage regulator time constant (T_B). Typical Value = 10.
ka	1..1	PU	Voltage regulator gain (K_A). Typical Value = 200.
ta	1..1	Seconds	Voltage regulator time constant (T_A). Typical Value = 0.015.
vrmax	1..1	PU	Maximum voltage regulator output (V_{RMAX}). Typical Value = 5.64.
vrmin	1..1	PU	Minimum voltage regulator output (V_{RMIN}). Typical Value = -4.53.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (K_C). Typical Value = 0.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-119 shows all association ends of ExcitationSystemDynamics with other classes.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 9-119 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.7 ExcIEEEAC5A

The class represents IEEE Std 421.5-2005 type AC5A model. The model represents a simplified model for brushless excitation systems. The regulator is supplied from a source, such as a permanent magnet generator, which is not affected by system disturbances. Unlike other ac models, this model uses loaded rather than open circuit exciter saturation data in the same way as it is used for the dc models. Because the model has been widely implemented by the industry, it is sometimes used to represent other types of systems when either detailed data for them are not available or simplified models are required. Reference: IEEE Standard 421.5-2005 Section 6.5.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-120 shows all attributes of ExcIEEEAC5A.

Table 9-120 - Attributes of ExcitationSystemDynamics::ExcIEEEAC5A

Attribute name	mult.	Attribute type	Description
ka	1..1	PU	Voltage regulator gain (K_A). Typical Value = 400.
ta	1..1	Seconds	Voltage regulator time constant (T_A). Typical Value = 0.02.
vrmax	1..1	PU	Maximum voltage regulator output (V_{RMAX}). Typical Value = 7.3.
vrmin	1..1	PU	Minimum voltage regulator output (V_{RMIN}). Typical Value = -7.3.
ke	1..1	PU	Exciter constant related to self-excited field (K_E). Typical Value = 1.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (T_E). Typical Value = 0.8.
kf	1..1	PU	Excitation control system stabilizer gains (K_F). Typical Value = 0.03.

tf1	1..1	Seconds	Excitation control system stabilizer time constant (T_{F1}). Typical Value = 1.
tf2	1..1	Seconds	Excitation control system stabilizer time constant (T_{F2}). Typical Value = 1.
tf3	1..1	Seconds	Excitation control system stabilizer time constant (T_{F3}). Typical Value = 1.
efd1	1..1	PU	Exciter voltage at which exciter saturation is defined (E_{FD1}). Typical Value = 5.6.
seefd1	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, E_{FD1} ($S_E[E_{FD1}]$). Typical Value = 0.86.
efd2	1..1	PU	Exciter voltage at which exciter saturation is defined (E_{FD2}). Typical Value = 4.2.
seefd2	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, E_{FD2} ($S_E[E_{FD2}]$). Typical Value = 0.5.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-121 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-121 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.8 ExcIEEEAC6A

The class represents IEEE Std 421.5-2005 type AC6A model. The model represents field-controlled alternator-rectifier excitation systems with system-supplied electronic voltage regulators. The maximum output of the regulator, V_R, is a function of terminal voltage, V_T. The field current

limiter included in the original model AC6A remains in the 2005 update. Reference: IEEE Standard 421.5-2005 Section 6.6.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-122 shows all attributes of ExcIEEEAC6A.

Table 9-122 - Attributes of ExcitationSystemDynamics::ExcIEEEAC6A

Attribute name	mult.	Attribute type	Description
ka	1..1	PU	Voltage regulator gain (K_A). Typical Value = 536.
ta	1..1	Seconds	Voltage regulator time constant (T_A). Typical Value = 0.086.
tk	1..1	Seconds	Voltage regulator time constant (T_K). Typical Value = 0.18.
tb	1..1	Seconds	Voltage regulator time constant (T_B). Typical Value = 9.
tc	1..1	Seconds	Voltage regulator time constant (T_C). Typical Value = 3.
vamax	1..1	PU	Maximum voltage regulator output (V_{AMAX}). Typical Value = 75.
vamin	1..1	PU	Minimum voltage regulator output (V_{AMIN}). Typical Value = -75.
vrmax	1..1	PU	Maximum voltage regulator output (V_{RMAX}). Typical Value = 44.
vrmin	1..1	PU	Minimum voltage regulator output (V_{RMIN}). Typical Value = -36.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (T_E). Typical Value = 1.
kh	1..1	PU	Exciter field current limiter gain (K_H). Typical Value = 92.
tj	1..1	Seconds	Exciter field current limiter time constant (T_J). Typical Value = 0.02.
th	1..1	Seconds	Exciter field current limiter time constant (T_H). Typical Value = 0.08.
vfelim	1..1	PU	Exciter field current limit reference (V_{FELIM}).

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			Typical Value = 19.
vhmax	1..1	PU	Maximum field current limiter signal reference (V_{HMAX}). Typical Value = 75.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (K_C). Typical Value = 0.173.
kd	1..1	PU	Demagnetizing factor, a function of exciter alternator reactances (K_D). Typical Value = 1.91.
ke	1..1	PU	Exciter constant related to self-excited field (K_E). Typical Value = 1.6.
ve1	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (V_{E1}) equals V_{EMAX} (V_{E1}). Typical Value = 7.4.
seve1	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, V_{E1}, back of commutating reactance ($S_E[V_{E1}]$). Typical Value = 0.214.
ve2	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (V_{E2}). Typical Value = 5.55.
seve2	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, V_{E2}, back of commutating reactance ($S_E[V_{E2}]$). Typical Value = 0.044.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-123 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-123 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult	Class type name	Description

	.		
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.9 ExcIEEEAC7B

The class represents IEEE Std 421.5-2005 type AC7B model. The model represents excitation systems which consist of an ac alternator with either stationary or rotating rectifiers to produce the dc field requirements. It is an upgrade to earlier ac excitation systems, which replace only the controls but retain the ac alternator and diode rectifier bridge. Reference: IEEE Standard 421.5-2005 Section 6.7. **Note:** In the IEEE Standard 421.5 – 2005, the [1 / sT_E] block is shown as [1 / (1 + sT_E)], which is incorrect.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-124 shows all attributes of ExcIEEEAC7B.

Table 9-124 - Attributes of ExcitationSystemDynamics::ExcIEEEAC7B

Attribute name	mult.	Attribute type	Description
kpr	1..1	PU	Voltage regulator proportional gain (K_{PR}). Typical Value = 4.24.
kir	1..1	PU	Voltage regulator integral gain (K_{IR}). Typical Value = 4.24.
kdr	1..1	PU	Voltage regulator derivative gain (K_{DR}). Typical Value = 0.
tdr	1..1	Seconds	Lag time constant (T_{DR}). Typical Value = 0.
vrmax	1..1	PU	Maximum voltage regulator output (V_{RMAX}). Typical Value = 5.79.
vrmin	1..1	PU	Minimum voltage regulator output (V_{RMIN}). Typical Value = -5.79.
kpa	1..1	PU	Voltage regulator proportional gain (K_{PA}). Typical Value = 65.36.
kia	1..1	PU	Voltage regulator integral gain (K_{IA}). Typical Value = 59.69.
vamax	1..1	PU	Maximum voltage regulator output (V_{AMAX}). Typical Value = 1.
vamin	1..1	PU	Minimum voltage regulator output (V_{AMIN}). Typical Value = -0.95.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

kp	1..1	PU	Potential circuit gain coefficient (K_P). Typical Value = 4.96.
kl	1..1	PU	Exciter field voltage lower limit parameter (K_L). Typical Value = 10.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (T_E). Typical Value = 1.1.
vfemax	1..1	PU	Exciter field current limit reference (V_{FEMAX}). Typical Value = 6.9.
vemin	1..1	PU	Minimum exciter voltage output (V_{EMIN}). Typical Value = 0.
ke	1..1	PU	Exciter constant related to self-excited field (K_E). Typical Value = 1.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (K_C). Typical Value = 0.18.
kd	1..1	PU	Demagnetizing factor, a function of exciter alternator reactances (K_D). Typical Value = 0.02.
kf1	1..1	PU	Excitation control system stabilizer gain (K_{F1}). Typical Value = 0.212.
kf2	1..1	PU	Excitation control system stabilizer gain (K_{F2}). Typical Value = 0.
kf3	1..1	PU	Excitation control system stabilizer gain (K_{F3}). Typical Value = 0.
tf	1..1	Seconds	Excitation control system stabilizer time constant (T_F). Typical Value = 1.
ve1	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (V_{E1}) equals V_{EMAX} (V_{E1}). Typical Value = 6.3.
seve1	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, V_{E1}, back of commutating reactance ($S_E[V_{E1}]$). Typical Value = 0.44.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

ve2	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (V_{E2}). Typical Value = 3.02.
seve2	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, V_{E2}, back of commutating reactance ($S_E[V_{E2}]$). Typical Value = 0.075.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-125 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-125 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.10 ExcIEEEAC8B

The class represents IEEE Std 421.5-2005 type AC8B model. This model represents a PID voltage regulator with either a brushless exciter or dc exciter. The AVR in this model consists of PID control, with separate constants for the proportional ($\mathbf{K_{PR}}$), integral ($\mathbf{K_{IR}}$), and derivative ($\mathbf{K_{DR}}$) gains. The representation of the brushless exciter ($\mathbf{T_E}$, $\mathbf{K_E}$, $\mathbf{S_E}$, $\mathbf{K_C}$, $\mathbf{K_D}$) is similar to the model Type AC2A. The Type AC8B model can be used to represent static voltage regulators applied to brushless excitation systems. Digitally based voltage regulators feeding dc rotating main exciters can be represented with the AC Type AC8B model with the parameters $\mathbf{K_C}$ and $\mathbf{K_D}$ set to 0. For thyristor power stages fed from the generator terminals, the limits $\mathbf{V_{RMAX}}$ and $\mathbf{V_{RMIN}}$ should be a function of terminal voltage: $\mathbf{V_T}$ * $\mathbf{V_{RMAX}}$ and $\mathbf{V_T}$ * $\mathbf{V_{RMIN}}$. Reference: IEEE Standard 421.5-2005 Section 6.8.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-126 shows all attributes of ExcIEEEAC8B.

Table 9-126 - Attributes of ExcitationSystemDynamics::ExcIEEEAC8B

Attribute name	mult.	Attribute type	Description
kpr	1..1	PU	Voltage regulator proportional gain (K_{PR}). Typical Value = 80.

kir	1..1	PU	Voltage regulator integral gain (K_{IR}). Typical Value = 5.
kdr	1..1	PU	Voltage regulator derivative gain (K_{DR}). Typical Value = 10.
tdr	1..1	Seconds	Lag time constant (T_{DR}). Typical Value = 0.1.
vrmax	1..1	PU	Maximum voltage regulator output (V_{RMAX}). Typical Value = 35.
vrmin	1..1	PU	Minimum voltage regulator output (V_{RMIN}). Typical Value = 0.
ka	1..1	PU	Voltage regulator gain (K_A). Typical Value = 1.
ta	1..1	Seconds	Voltage regulator time constant (T_A). Typical Value = 0.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (T_E). Typical Value = 1.2.
vfemax	1..1	PU	Exciter field current limit reference (V_{FEMAX}). Typical Value = 6.
vemin	1..1	PU	Minimum exciter voltage output (V_{EMIN}). Typical Value = 0.
ke	1..1	PU	Exciter constant related to self-excited field (K_E). Typical Value = 1.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (K_C). Typical Value = 0.55.
kd	1..1	PU	Demagnetizing factor, a function of exciter alternator reactances (K_D). Typical Value = 1.1.
ve1	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (V_{E1}) equals V_{EMAX} (V_{E1}). Typical Value = 6.5.
seve1	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, V_{E1}, back of commutating reactance ($S_E[V_{E1}]$). Typical Value = 0.3.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

ve2	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (V_{E2}). Typical Value = 9.
seve2	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, V_{E2}, back of commutating reactance ($S_E[V_{E2}]$). Typical Value = 3.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-127 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-127 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.11 ExcIEEEDC1A

The class represents IEEE Std 421.5-2005 type DC1A model. This model represents field-controlled dc commutator exciters with continuously acting voltage regulators (especially the direct-acting rheostatic, rotating amplifier, and magnetic amplifier types). Because this model has been widely implemented by the industry, it is sometimes used to represent other types of systems when detailed data for them are not available or when a simplified model is required. Reference: IEEE Standard 421.5-2005 Section 5.1.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-128 shows all attributes of ExcIEEEDC1A.

Table 9-128 - Attributes of ExcitationSystemDynamics::ExcIEEEDC1A

Attribute name	mult.	Attribute type	Description
ka	1..1	PU	Voltage regulator gain (K_A). Typical Value = 46.
ta	1..1	Seconds	Voltage regulator time constant (T_A). Typical Value = 0.06.

tb	1..1	Seconds	Voltage regulator time constant (T_B). Typical Value = 0.
tc	1..1	Seconds	Voltage regulator time constant (T_C). Typical Value = 0.
vrmax	1..1	PU	Maximum voltage regulator output (V_{RMAX}). Typical Value = 1.
vrmin	1..1	PU	Minimum voltage regulator output (V_{RMIN}). Typical Value = -0.9.
ke	1..1	PU	Exciter constant related to self-excited field (K_E). Typical Value = 0.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (T_E). Typical Value = 0.46.
kf	1..1	PU	Excitation control system stabilizer gain (K_F). Typical Value = 0.1.
tf	1..1	Seconds	Excitation control system stabilizer time constant (T_F). Typical Value = 1.
efd1	1..1	PU	Exciter voltage at which exciter saturation is defined (E_{FD1}). Typical Value = 3.1.
seefd1	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, E_{FD1} ($S_E[E_{FD1}]$). Typical Value = 0.33.
efd2	1..1	PU	Exciter voltage at which exciter saturation is defined (E_{FD2}). Typical Value = 2.3.
seefd2	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, E_{FD2} ($S_E[E_{FD2}]$). Typical Value = 0.1.
uelin	1..1	Boolean	UEL input (uelin). true = input is connected to the HV gate false = input connects to the error signal. Typical Value = true.
exclim	1..1	Boolean	(exclim). IEEE standard is ambiguous about lower limit on exciter output. true = a lower limit of

			zero is applied to integrator output false = a lower limit of zero is not applied to integrator output. Typical Value = true.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-129 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-129 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] ----- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.12 ExcIEEEDC2A

The class represents IEEE Std 421.5-2005 type DC2A model. This model represents represent field-controlled dc commutator exciters with continuously acting voltage regulators having supplies obtained from the generator or auxiliary bus. It differs from the Type DC1A model only in the voltage regulator output limits, which are now proportional to terminal voltage $\mathbf{V_T}$. It is representative of solid-state replacements for various forms of older mechanical and rotating amplifier regulating equipment connected to dc commutator exciters. Reference: IEEE Standard 421.5-2005 Section 5.2.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-130 shows all attributes of ExcIEEEDC2A.

Table 9-130 - Attributes of ExcitationSystemDynamics::ExcIEEEDC2A

Attribute name	mult.	Attribute type	Description
efd1	1..1	PU	Exciter voltage at which exciter saturation is defined (E_{FD1}). Typical Value = 3.05.
efd2	1..1	PU	Exciter voltage at which exciter saturation is defined (E_{FD2}). Typical Value = 2.29.
exclim	1..1	PU	(exclim). IEEE standard is ambiguous about lower limit on exciter output. Typical Value = - 999 which means that

			there is no limit applied.
ka	1..1	PU	Voltage regulator gain (K_A). Typical Value = 300.
ke	1..1	PU	Exciter constant related to self-excited field (K_E). Typical Value = 1.
kf	1..1	PU	Excitation control system stabilizer gain (K_F). Typical Value = 0.1.
seefd1	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, E_{FD1} ($S_E[E_{FD1}]$). Typical Value = 0.279.
seefd2	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, E_{FD2} ($S_E[E_{FD2}]$). Typical Value = 0.117.
ta	1..1	Seconds	Voltage regulator time constant (T_A). Typical Value = 0.01.
tb	1..1	Seconds	Voltage regulator time constant (T_B). Typical Value = 0.
tc	1..1	Seconds	Voltage regulator time constant (T_C). Typical Value = 0.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (T_E). Typical Value = 1.33.
tf	1..1	Seconds	Excitation control system stabilizer time constant (T_F). Typical Value = 0.675.
uelin	1..1	Boolean	UEL input (uelin). true = input is connected to the HV gate false = input connects to the error signal. Typical Value = true.
vrmax	1..1	PU	Maximum voltage regulator output (V_{RMAX}). Typical Value = 4.95.
vrmin	1..1	PU	Minimum voltage regulator output (V_{RMIN}). Typical Value = -4.9.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-131 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-131 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.13 ExcIEEEDC3A

The class represents IEEE Std 421.5-2005 type DC3A model. This model represents represent older systems, in particular those dc commutator exciters with non-continuously acting regulators that were commonly used before the development of the continuously acting varieties. These systems respond at basically two different rates, depending upon the magnitude of voltage error. For small errors, adjustment is made periodically with a signal to a motor-operated rheostat. Larger errors cause resistors to be quickly shorted or inserted and a strong forcing signal applied to the exciter. Continuous motion of the motor-operated rheostat occurs for these larger error signals, even though it is bypassed by contactor action. Reference: IEEE Standard 421.5-2005 Section 5.3.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-132 shows all attributes of ExcIEEEDC3A.

Table 9-132 - Attributes of ExcitationSystemDynamics::ExcIEEEDC3A

Attribute name	mult.	Attribute type	Description
trh	1..1	Seconds	Rheostat travel time (T_{RH}). Typical Value = 20.
kv	1..1	PU	Fast raise/lower contact setting (K_V). Typical Value = 0.05.
vrmax	1..1	PU	Maximum voltage regulator output (V_{RMAX}). Typical Value = 1.
vrmin	1..1	PU	Minimum voltage regulator output (V_{RMIN}). Typical Value = 0.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (T_E). Typical Value = 0.5.
ke	1..1	PU	Exciter constant related

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			to self-excited field (K_E). Typical Value = 0.05.
efd1	1..1	PU	Exciter voltage at which exciter saturation is defined (E_{FD1}). Typical Value = 3.375.
seefd1	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, E_{FD1} ($S_E[E_{FD1}]$). Typical Value = 0.267.
efd2	1..1	PU	Exciter voltage at which exciter saturation is defined (E_{FD2}). Typical Value = 3.15.
seefd2	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, E_{FD2} ($S_E[E_{FD2}]$). Typical Value = 0.068.
exclim	1..1	Boolean	(exclim). IEEE standard is ambiguous about lower limit on exciter output. true = a lower limit of zero is applied to integrator output false = a lower limit of zero is not applied to integrator output. Typical Value = true.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-133 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-133 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.14 ExcIEEEDC4B

The class represents IEEE Std 421.5-2005 type DC4B model. These excitation systems utilize a field-controlled dc commutator exciter with a continuously acting voltage regulator having supplies

obtained from the generator or auxiliary bus. Reference: IEEE Standard 421.5-2005 Section 5.4.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-134 shows all attributes of ExcIEEEDC4B.

Table 9-134 - Attributes of ExcitationSystemDynamics::ExcIEEEDC4B

Attribute name	mult.	Attribute type	Description
ka	1..1	PU	Voltage regulator gain (K_A). Typical Value = 1.
ta	1..1	Seconds	Voltage regulator time constant (T_A). Typical Value = 0.2.
kp	1..1	PU	Regulator proportional gain (K_P). Typical Value = 20.
ki	1..1	PU	Regulator integral gain (K_I). Typical Value = 20.
kd	1..1	PU	Regulator derivative gain (K_D). Typical Value = 20.
td	1..1	Seconds	Regulator derivative filter time constant(T_D). Typical Value = 0.01.
vrmax	1..1	PU	Maximum voltage regulator output (V_{RMAX}). Typical Value = 2.7.
vrmin	1..1	PU	Minimum voltage regulator output (V_{RMIN}). Typical Value = -0.9.
ke	1..1	PU	Exciter constant related to self-excited field (K_E). Typical Value = 1.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (T_E). Typical Value = 0.8.
kf	1..1	PU	Excitation control system stabilizer gain (K_F). Typical Value = 0.
tf	1..1	Seconds	Excitation control system stabilizer time constant (T_F). Typical Value = 1.
efd1	1..1	PU	Exciter voltage at which exciter saturation is defined (E_{FD1}). Typical Value = 1.75.
seefd1	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, E_{FD1} ($S_E[E_{FD1}]$).

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			Typical Value = 0.08.
efd2	1..1	PU	Exciter voltage at which exciter saturation is defined (E_{FD2}). Typical Value = 2.33.
seefd2	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, E_{FD2} ($S_E[E_{FD2}]$). Typical Value = 0.27.
vemin	1..1	PU	Minimum exciter voltage output(V_{EMIN}). Typical Value = 0.
oelin	1..1	Boolean	OEL input (OELin). true = LV gate false = subtract from error signal. Typical Value = true.
uelin	1..1	Boolean	UEL input (UELin). true = HV gate false = add to error signal. Typical Value = true.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-135 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-135 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] ----- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.15 ExcIEEEST1A

The class represents IEEE Std 421.5-2005 type ST1A model. This model represents systems in which excitation power is supplied through a transformer from the generator terminals (or the unit's auxiliary bus) and is regulated by a controlled rectifier. The maximum exciter voltage available from such systems is directly related to the generator terminal voltage. Reference: IEEE Standard 421.5-2005 Section 7.1.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-136 shows all attributes of ExcIEEEST1A.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 9-136 - Attributes of ExcitationSystemDynamics::ExcIEEEST1A

Attribute name	mult.	Attribute type	Description
ilr	1..1	PU	Exciter output current limit reference (I_{LR}). Typical Value = 0.
ka	1..1	PU	Voltage regulator gain (K_A). Typical Value = 190.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (K_C). Typical Value = 0.08.
kf	1..1	PU	Excitation control system stabilizer gains (K_F). Typical Value = 0.
klr	1..1	PU	Exciter output current limiter gain (K_{LR}). Typical Value = 0.
pssin	1..1	Boolean	Selector of the Power System Stabilizer (PSS) input (PSSin). true = PSS input (Vs) added to error signal false = PSS input (Vs) added to voltage regulator output. Typical Value = true.
ta	1..1	Seconds	Voltage regulator time constant (T_A). Typical Value = 0.
tb	1..1	Seconds	Voltage regulator time constant (T_B). Typical Value = 10.
tb1	1..1	Seconds	Voltage regulator time constant (T_{B1}). Typical Value = 0.
tc	1..1	Seconds	Voltage regulator time constant (T_C). Typical Value = 1.
tc1	1..1	Seconds	Voltage regulator time constant (T_{C1}). Typical Value = 0.
tf	1..1	Seconds	Excitation control system stabilizer time constant (T_F). Typical Value = 1.
uelin	1..1	ExcIEEEST1AUELselectorKind	Selector of the connection of the UEL input (UELin). Typical Value = ignoreUELsignal.
vamax	1..1	PU	Maximum voltage regulator output (V_{AMAX}). Typical Value = 14.5.
vamin	1..1	PU	Minimum voltage

			regulator output (V_{AMIN}). Typical Value = -14.5.
vimax	1..1	PU	Maximum voltage regulator input limit (V_{IMAX}). Typical Value = 999.
vimin	1..1	PU	Minimum voltage regulator input limit (V_{IMIN}). Typical Value = -999.
vrmax	1..1	PU	Maximum voltage regulator outputs (V_{RMAX}). Typical Value = 7.8.
vrmin	1..1	PU	Minimum voltage regulator outputs (V_{RMIN}). Typical Value = -6.7.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-137 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-137 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] ----- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.16 ExcIEEEST2A

The class represents IEEE Std 421.5-2005 type ST2A model. Some static systems utilize both current and voltage sources (generator terminal quantities) to comprise the power source. The regulator controls the exciter output through controlled saturation of the power transformer components. These compound-source rectifier excitation systems are designated Type ST2A and are represented by ExcIEEEST2A. Reference: IEEE Standard 421.5-2005 Section 7.2.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-138 shows all attributes of ExcIEEEST2A.

Table 9-138 - Attributes of ExcitationSystemDynamics::ExcIEEEST2A

Attribute name	mult.	Attribute type	Description

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

ka	1..1	PU	Voltage regulator gain (K_A). Typical Value = 120.
ta	1..1	Seconds	Voltage regulator time constant (T_A). Typical Value = 0.15.
vrmax	1..1	PU	Maximum voltage regulator outputs (V_{RMAX}). Typical Value = 1.
vrmin	1..1	PU	Minimum voltage regulator outputs (V_{RMIN}). Typical Value = 0.
ke	1..1	PU	Exciter constant related to self-excited field (K_E). Typical Value = 1.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (T_E). Typical Value = 0.5.
kf	1..1	PU	Excitation control system stabilizer gains (K_F). Typical Value = 0.05.
tf	1..1	Seconds	Excitation control system stabilizer time constant (T_F). Typical Value = 1.
kp	1..1	PU	Potential circuit gain coefficient (K_P). Typical Value = 4.88.
ki	1..1	PU	Potential circuit gain coefficient (K_I). Typical Value = 8.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (K_C). Typical Value = 1.82.
efdmax	1..1	PU	Maximum field voltage (E_{FDMax}). Typical Value = 99.
uelin	1..1	Boolean	UEL input (UELin). true = HV gate false = add to error signal. Typical Value = true.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-139 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-139 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.17 ExcIEEEST3A

The class represents IEEE Std 421.5-2005 type ST3A model. Some static systems utilize a field voltage control loop to linearize the exciter control characteristic. This also makes the output independent of supply source variations until supply limitations are reached. These systems utilize a variety of controlled-rectifier designs: full thyristor complements or hybrid bridges in either series or shunt configurations. The power source may consist of only a potential source, either fed from the machine terminals or from internal windings. Some designs may have compound power sources utilizing both machine potential and current. These power sources are represented as phasor combinations of machine terminal current and voltage and are accommodated by suitable parameters in model Type ST3A which is represented by ExcIEEEST3A.
Reference: IEEE Standard 421.5-2005 Section 7.3.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-140 shows all attributes of ExcIEEEST3A.

Table 9-140 - Attributes of ExcitationSystemDynamics::ExcIEEEST3A

Attribute name	mult.	Attribute type	Description
vimax	1..1	PU	Maximum voltage regulator input limit (V_{IMAX}). Typical Value = 0.2.
vimin	1..1	PU	Minimum voltage regulator input limit (V_{IMIN}). Typical Value = -0.2.
ka	1..1	PU	Voltage regulator gain (K_A). This is parameter K in the IEEE Std. Typical Value = 200.
ta	1..1	Seconds	Voltage regulator time constant (T_A). Typical Value = 0.
tb	1..1	Seconds	Voltage regulator time constant (T_B). Typical Value = 10.
tc	1..1	Seconds	Voltage regulator time constant (T_C). Typical Value = 1.
vrmax	1..1	PU	Maximum voltage regulator output

			(V_{RMAX}). Typical Value = 10.
vrmin	1..1	PU	Minimum voltage regulator output (V_{RMIN}). Typical Value = -10.
km	1..1	PU	Forward gain constant of the inner loop field regulator (K_M). Typical Value = 7.93.
tm	1..1	Seconds	Forward time constant of inner loop field regulator (T_M). Typical Value = 0.4.
vmmax	1..1	PU	Maximum inner loop output (V_{MMax}). Typical Value = 1.
vmmin	1..1	PU	Minimum inner loop output (V_{MMin}). Typical Value = 0.
kg	1..1	PU	Feedback gain constant of the inner loop field regulator (K_G). Typical Value = 1.
kp	1..1	PU	Potential circuit gain coefficient (K_P). Typical Value = 6.15.
thetap	1..1	AngleDegrees	Potential circuit phase angle (thetap). Typical Value = 0.
ki	1..1	PU	Potential circuit gain coefficient (K_I). Typical Value = 0.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (K_C). Typical Value = 0.2.
xl	1..1	PU	Reactance associated with potential source (X_L). Typical Value = 0.081.
vbmax	1..1	PU	Maximum excitation voltage (V_{BMax}). Typical Value = 6.9.
vgmax	1..1	PU	Maximum inner loop feedback voltage (V_{GMax}). Typical Value = 5.8.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-141 shows all association ends of ExcitationSystemDynamics with other classes.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 9-141 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.18 ExcIEEEST4B

The class represents IEEE Std 421.5-2005 type ST4B model. This model is a variation of the Type ST3A model, with a proportional plus integral (PI) regulator block replacing the lag-lead regulator characteristic that is in the ST3A model. Both potential and compound source rectifier excitation systems are modeled. The PI regulator blocks have non-windup limits that are represented. The voltage regulator of this model is typically implemented digitally. Reference: IEEE Standard 421.5-2005 Section 7.4.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-142 shows all attributes of ExcIEEEST4B.

Table 9-142 - Attributes of ExcitationSystemDynamics::ExcIEEEST4B

Attribute name	mult.	Attribute type	Description
kpr	1..1	PU	Voltage regulator proportional gain (K_{PR}). Typical Value = 10.75.
kir	1..1	PU	Voltage regulator integral gain (K_{IR}). Typical Value = 10.75.
ta	1..1	Seconds	Voltage regulator time constant (T_A). Typical Value = 0.02.
vrmax	1..1	PU	Maximum voltage regulator output (V_{RMAX}). Typical Value = 1.
vrmin	1..1	PU	Minimum voltage regulator output (V_{RMIN}). Typical Value = -0.87.
kpm	1..1	PU	Voltage regulator proportional gain output (K_{PM}). Typical Value = 1.
kim	1..1	PU	Voltage regulator integral gain output (K_{IM}). Typical Value = 0.
vmmax	1..1	PU	Maximum inner loop output (V_{MMax}). Typical Value = 99.
vmmin	1..1	PU	Minimum inner loop

			output (V_{MMin}). Typical Value = -99.
kg	1..1	PU	Feedback gain constant of the inner loop field regulator (K_G). Typical Value = 0.
kp	1..1	PU	Potential circuit gain coefficient (K_P). Typical Value = 9.3.
thetap	1..1	AngleDegrees	Potential circuit phase angle (thetap). Typical Value = 0.
ki	1..1	PU	Potential circuit gain coefficient (K_I). Typical Value = 0.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (K_C). Typical Value = 0.113.
xl	1..1	PU	Reactance associated with potential source (X_L). Typical Value = 0.124.
vbmax	1..1	PU	Maximum excitation voltage (V_{BMax}). Typical Value = 11.63.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-143 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-143 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.19 ExcIEEEST5B

The class represents IEEE Std 421.5-2005 type ST5B model. The Type ST5B excitation system is a variation of the Type ST1A model, with alternative overexcitation and underexcitation inputs and additional limits.
Reference: IEEE Standard 421.5-2005 Section 7.5.
Note: the block diagram in the IEEE 421.5 standard has input signal Vc and does not indicate the

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

summation point with Vref. The implementation of the ExcIEEEST5B shall consider summation point with Vref.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-144 shows all attributes of ExcIEEEST5B.

Table 9-144 - Attributes of ExcitationSystemDynamics::ExcIEEEST5B

Attribute name	mult.	Attribute type	Description
kr	1..1	PU	Regulator gain (K_R). Typical Value = 200.
t1	1..1	Seconds	Firing circuit time constant (T1). Typical Value = 0.004.
kc	1..1	PU	Rectifier regulation factor (K_C). Typical Value = 0.004.
vrmax	1..1	PU	Maximum voltage regulator output (V_{RMAX}). Typical Value = 5.
vrmin	1..1	PU	Minimum voltage regulator output (V_{RMIN}). Typical Value = -4.
tc1	1..1	Seconds	Regulator lead time constant (T_{C1}). Typical Value = 0.8.
tb1	1..1	Seconds	Regulator lag time constant (T_{B1}). Typical Value = 6.
tc2	1..1	Seconds	Regulator lead time constant (T_{C2}). Typical Value = 0.08.
tb2	1..1	Seconds	Regulator lag time constant (T_{B2}). Typical Value = 0.01.
toc1	1..1	Seconds	OEL lead time constant (T_{OC1}). Typical Value = 0.1.
tob1	1..1	Seconds	OEL lag time constant (T_{OB1}). Typical Value = 2.
toc2	1..1	Seconds	OEL lead time constant (T_{OC2}). Typical Value = 0.08.
tob2	1..1	Seconds	OEL lag time constant (T_{OB2}). Typical Value = 0.08.
tuc1	1..1	Seconds	UEL lead time constant (T_{UC1}). Typical Value = 2.
tub1	1..1	Seconds	UEL lag time constant (T_{UB1}). Typical Value = 10.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

tuc2	1..1	Seconds	UEL lead time constant (T_{UC2}). Typical Value = 0.1.
tub2	1..1	Seconds	UEL lag time constant (T_{UB2}). Typical Value = 0.05.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-145 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-145 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.20 ExcIEEEST6B

The class represents IEEE Std 421.5-2005 type ST6B model. This model consists of a PI voltage regulator with an inner loop field voltage regulator and pre-control. The field voltage regulator implements a proportional control. The pre-control and the delay in the feedback circuit increase the dynamic response.
Reference: IEEE Standard 421.5-2005 Section 7.6.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-146 shows all attributes of ExcIEEEST6B.

Table 9-146 - Attributes of ExcitationSystemDynamics::ExcIEEEST6B

Attribute name	mult.	Attribute type	Description
ilr	1..1	PU	Exciter output current limit reference (I_{LR}). Typical Value = 4.164.
kci	1..1	PU	Exciter output current limit adjustment (K_{CI}). Typical Value = 1.0577.
kff	1..1	PU	Pre-control gain constant of the inner loop field regulator (K_{FF}). Typical Value = 1.
kg	1..1	PU	Feedback gain constant of the inner loop field regulator (K_G). Typical

			Value = 1.
kia	1..1	PU	Voltage regulator integral gain (K_{IA}). Typical Value = 45.094.
klr	1..1	PU	Exciter output current limiter gain (K_{LR}). Typical Value = 17.33.
km	1..1	PU	Forward gain constant of the inner loop field regulator (K_M). Typical Value = 1.
kpa	1..1	PU	Voltage regulator proportional gain (K_{PA}). Typical Value = 18.038.
oelin	1..1	ExcST6BOELselectorKind	OEL input selector (OELin). Typical Value = noOELinput.
tg	1..1	Seconds	Feedback time constant of inner loop field voltage regulator (T_G). Typical Value = 0.02.
vamax	1..1	PU	Maximum voltage regulator output (V_{AMAX}). Typical Value = 4.81.
vamin	1..1	PU	Minimum voltage regulator output (V_{AMIN}). Typical Value = -3.85.
vrmax	1..1	PU	Maximum voltage regulator output (V_{RMAX}). Typical Value = 4.81.
vrmin	1..1	PU	Minimum voltage regulator output (V_{RMIN}). Typical Value = -3.85.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-147 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-147 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] ----- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.21 ExcIEEEST7B

The class represents IEEE Std 421.5-2005 type ST7B model. This model is representative of static potential-source excitation systems. In this system, the AVR consists of a PI voltage regulator. A phase lead-lag filter in series allows introduction of a derivative function, typically used with brushless excitation systems. In that case, the regulator is of the PID type. In addition, the terminal voltage channel includes a phase lead-lag filter. The AVR includes the appropriate inputs on its reference for overexcitation limiter (OEL1), underexcitation limiter (UEL), stator current limiter (SCL), and current compensator (DROOP). All these limitations, when they work at voltage reference level, keep the PSS (VS signal from Type PSS1A, PSS2A, or PSS2B) in operation. However, the UEL limitation can also be transferred to the high value (HV) gate acting on the output signal. In addition, the output signal passes through a low value (LV) gate for a ceiling overexcitation limiter (OEL2).
Reference: IEEE Standard 421.5-2005 Section 7.7.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-148 shows all attributes of ExcIEEEST7B.

Table 9-148 - Attributes of ExcitationSystemDynamics::ExcIEEEST7B

Attribute name	mult.	Attribute type	Description
kh	1..1	PU	High-value gate feedback gain (K_H). Typical Value 1.
kia	1..1	PU	Voltage regulator integral gain (K_{IA}). Typical Value = 1.
kl	1..1	PU	Low-value gate feedback gain (K_L). Typical Value 1.
kpa	1..1	PU	Voltage regulator proportional gain (K_{PA}). Typical Value = 40.
oelin	1..1	ExcST7BOELselectorKind	OEL input selector (OELin). Typical Value = noOELinput.
tb	1..1	Seconds	Regulator lag time constant (T_B). Typical Value 1.
tc	1..1	Seconds	Regulator lead time constant (T_C). Typical Value 1.
tf	1..1	Seconds	Excitation control system stabilizer time constant (T_F). Typical Value 1.
tg	1..1	Seconds	Feedback time constant of inner loop field voltage regulator (T_G). Typical Value 1.
tia	1..1	Seconds	Feedback time constant (T_{IA}). Typical Value = 3.
uelin	1..1	ExcST7BUELselectorKind	UEL input selector (UELin). Typical Value = noUELinput.

vmax	1..1	PU	Maximum voltage reference signal (V_{MAX}). Typical Value = 1.1.
vmin	1..1	PU	Minimum voltage reference signal (V_{MIN}). Typical Value = 0.9.
vrmax	1..1	PU	Maximum voltage regulator output (V_{RMAX}). Typical Value = 5.
vrmin	1..1	PU	Minimum voltage regulator output (V_{RMIN}). Typical Value = -4.5.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-149 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-149 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] ----- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.22 ExcAC1A

Modified IEEE AC1A alternator-supplied rectifier excitation system with different rate feedback source.

Figure 9-91 class diagram

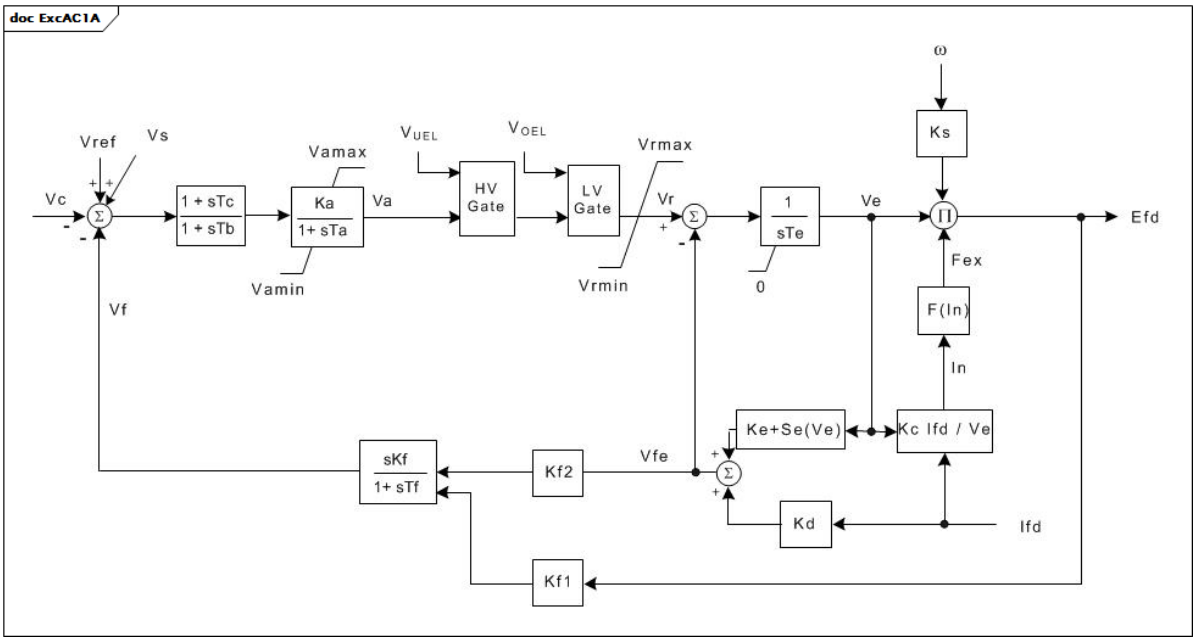


Figure 9-91 - figure ExcAC1A

Parameter Notes:

1. If Ks is set to 1 the output (Efd) is multiplied by generator speed.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-150 shows all attributes of ExcAC1A.

Table 9-150 - Attributes of ExcitationSystemDynamics::ExcAC1A

Attribute name	mult.	Attribute type	Description
tb	1..1	Seconds	Voltage regulator time constant (Tb). Typical Value = 0.
tc	1..1	Seconds	Voltage regulator time constant (T_c). Typical Value = 0.
ka	1..1	PU	Voltage regulator gain (Ka). Typical Value = 400.
ta	1..1	Seconds	Voltage regulator time constant (Ta). Typical Value = 0.02.
vamax	1..1	PU	Maximum voltage regulator output (V_{amax}). Typical Value = 14.5.
vamin	1..1	PU	Minimum voltage regulator output (V_{amin}). Typical Value = -14.5.
te	1..1	Seconds	Exciter time constant,

			integration rate associated with exciter control (Te). Typical Value = 0.8.
kf	1..1	PU	Excitation control system stabilizer gains (Kf). Typical Value = 0.03.
kf1	1..1	PU	Coefficient to allow different usage of the model (Kf1). Typical Value = 0.
kf2	1..1	PU	Coefficient to allow different usage of the model (Kf2). Typical Value = 1.
ks	1..1	PU	Coefficient to allow different usage of the model-speed coefficient (Ks). Typical Value = 0.
tf	1..1	Seconds	Excitation control system stabilizer time constant (Tf). Typical Value = 1.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (Kc). Typical Value = 0.2.
kd	1..1	PU	Demagnetizing factor, a function of exciter alternator reactances (Kd). Typical Value = 0.38.
ke	1..1	PU	Exciter constant related to self-excited field (Ke). Typical Value = 1.
ve1	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (Ve1). Typical Value = 4.18.
seve1	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, Ve1, back of commutating reactance (Se[Ve1]). Typical Value = 0.1.
ve2	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (Ve2). Typical Value = 3.14.
seve2	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, Ve2, back of

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			commutating reactance (Se[Ve2]). Typical Value = 0.03.
vrmax	1..1	PU	Maximum voltage regulator outputs (Vrmax). Typical Value = 6.03.
vrmin	1..1	PU	Minimum voltage regulator outputs (Rrmin). Typical Value = -5.43.
hvlvgates	1..1	Boolean	Indicates if both HV gate and LV gate are active (HVLVgates). true = gates are used false = gates are not used. Typical Value = true.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-151 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-151 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] ----- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.23 ExcAC2A

Modified IEEE AC2A alternator-supplied rectifier excitation system with different field current limit.

Figure 9-92 class diagram

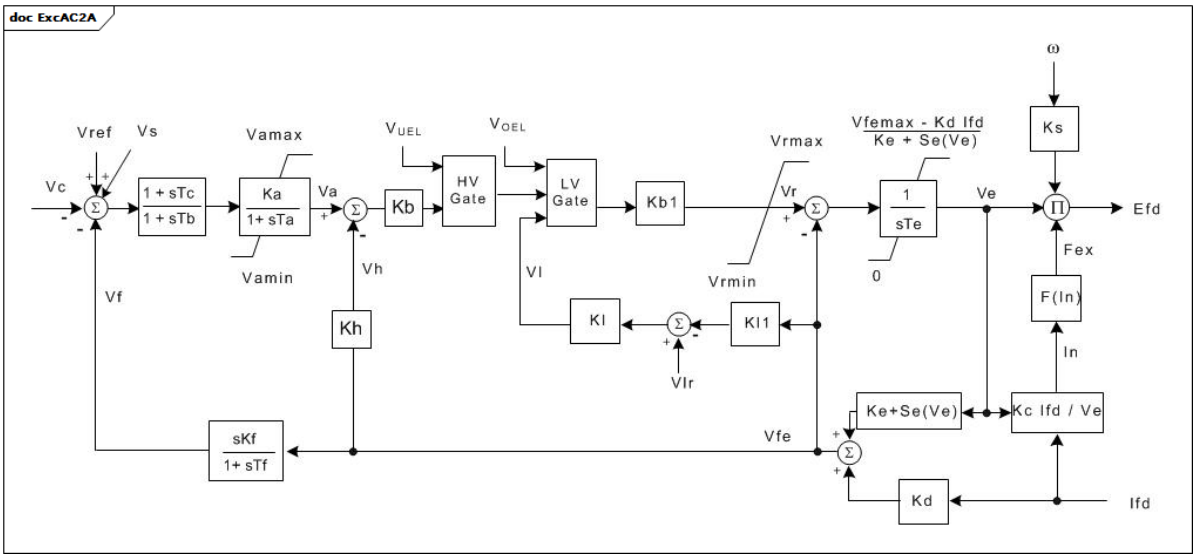


Figure 9-92 - figure ExcAC2A

Parameter Notes:

1. If Ks is set to 1 the output (Efd) is multiplyed by generator speed;
2. The upper limit on Ve represents the effect of the field current limiter. If Vfemax is zero, this limit will not be enforced. The real system, the limiter is implemented by a low value gate just before Kb. The input to this LV gate is Kl * (Vlr – Vfe). If the values of Kl and Vlr are given, Vfemax can be calculated as Vlr*Kl*Kb / (1 + Kl*Kb).

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-152 shows all attributes of ExcAC2A.

Table 9-152 - Attributes of ExcitationSystemDynamics::ExcAC2A

Attribute name	mult.	Attribute type	Description
tb	1..1	Seconds	Voltage regulator time constant (Tb). Typical Value = 0.
tc	1..1	Seconds	Voltage regulator time constant (T_c). Typical Value = 0.
ka	1..1	PU	Voltage regulator gain (Ka). Typical Value = 400.
ta	1..1	Seconds	Voltage regulator time constant (Ta). Typical Value = 0.02.
vamax	1..1	PU	Maximum voltage regulator output (V_{amax}). Typical Value = 8.
vamin	1..1	PU	Minimum voltage regulator output (V_{amin}). Typical Value = -8.

kb	1..1	PU	Second stage regulator gain (Kb) (>0). Exciter field current controller gain. Typical Value = 25.
kb1	1..1	PU	Second stage regulator gain (Kb1). It is exciter field current controller gain used as alternative to Kb to represent a variant of the ExcAC2A model. Typical Value = 25.
vrmax	1..1	PU	Maximum voltage regulator outputs (Vrmax). Typical Value = 105.
vrmin	1..1	PU	Minimum voltage regulator outputs (Vrmin). Typical Value = -95.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (Te). Typical Value = 0.6.
vfemax	1..1	PU	Exciter field current limit reference (Vfemax). Typical Value = 4.4.
kh	1..1	PU	Exciter field current feedback gain (Kh). Typical Value = 1.
kf	1..1	PU	Excitation control system stabilizer gains (Kf). Typical Value = 0.03.
kl	1..1	PU	Exciter field current limiter gain (Kl). Typical Value = 10.
vlr	1..1	PU	Maximum exciter field current (Vlr). Typical Value = 4.4.
kl1	1..1	PU	Coefficient to allow different usage of the model (Kl1). Typical Value = 1.
ks	1..1	PU	Coefficient to allow different usage of the model-speed coefficient (Ks). Typical Value = 0.
tf	1..1	Seconds	Excitation control system stabilizer time constant (Tf). Typical Value = 1.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (Kc). Typical Value =

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			0.28.
kd	1..1	PU	Demagnetizing factor, a function of exciter alternator reactances (Kd). Typical Value = 0.35.
ke	1..1	PU	Exciter constant related to self-excited field (Ke). Typical Value = 1.
ve1	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (Ve_1). Typical Value = 4.4.
seve1	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, Ve_1, back of commutating reactance ($Se[Ve_1]$). Typical Value = 0.037.
ve2	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (Ve_2). Typical Value = 3.3.
seve2	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, Ve_2, back of commutating reactance ($Se[Ve_2]$). Typical Value = 0.012.
hvgate	1..1	Boolean	Indicates if HV gate is active (HVgate). true = gate is used false = gate is not used. Typical Value = true.
lvgate	1..1	Boolean	Indicates if LV gate is active (LVgate). true = gate is used false = gate is not used. Typical Value = true.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-153 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-153 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description

SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.24 ExcAC3A

Modified IEEE AC3A alternator-supplied rectifier excitation system with different field current limit.

Figure 9-93 class diagram

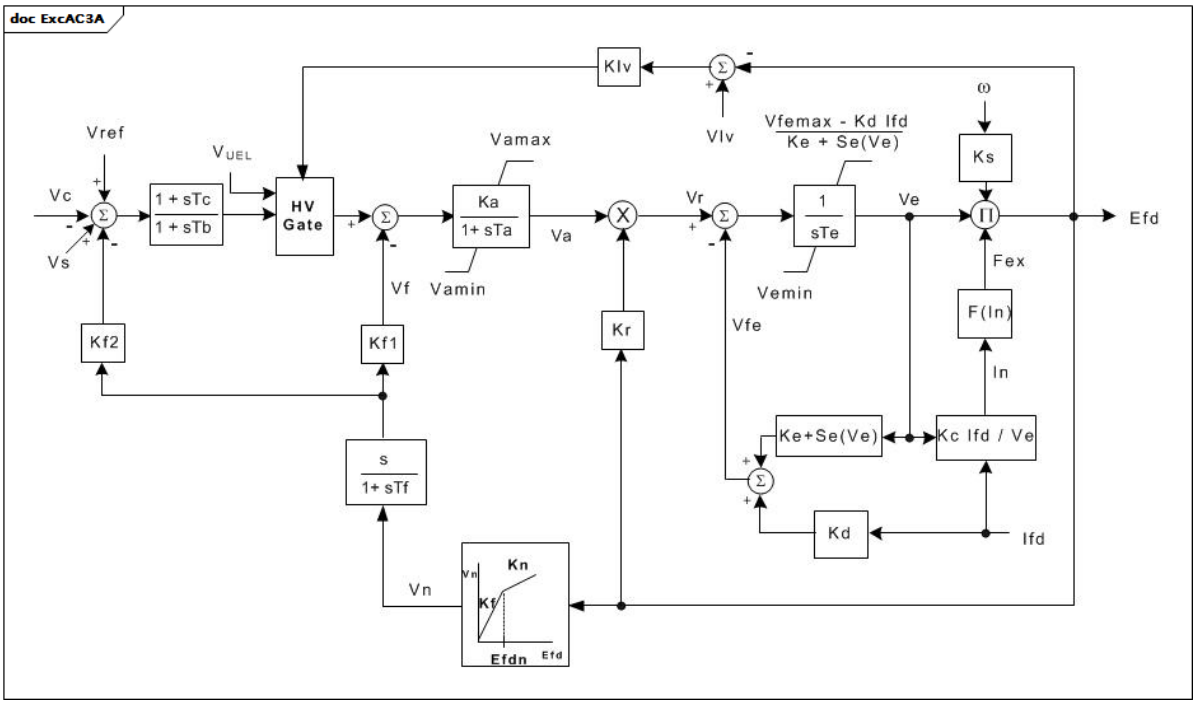


Figure 9-93 - figure ExcAC3A

Parameter Notes:

1. If Ks is set to 1 the output (Efd) is multiplyed by generator speed;
2. The upper limit on Ve is an approximate representation of the effect of the maximum field current limiter. If Vfemax is zero, this limit will not be enforced. The lower limit (Vemin) on Ve is an approximate representation of the minimum field voltage limiter.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-154 shows all attributes of ExcAC3A.

Table 9-154 - Attributes of ExcitationSystemDynamics::ExcAC3A

Attribute name	mult.	Attribute type	Description

tb	1..1	Seconds	Voltage regulator time constant (Tb). Typical Value = 0.
tc	1..1	Seconds	Voltage regulator time constant (T_c). Typical Value = 0.
ka	1..1	Seconds	Voltage regulator gain (Ka). Typical Value = 45.62.
ta	1..1	PU	Voltage regulator time constant (Ta). Typical Value = 0.013.
vamax	1..1	PU	Maximum voltage regulator output (V_{amax}). Typical Value = 1.
vamin	1..1	PU	Minimum voltage regulator output (V_{amin}). Typical Value = -0.95.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (Te). Typical Value = 1.17.
vemin	1..1	PU	Minimum exciter voltage output (Vemin). Typical Value = 0.1.
kr	1..1	PU	Constant associated with regulator and alternator field power supply (Kr). Typical Value =3.77.
kf	1..1	PU	Excitation control system stabilizer gains (Kf). Typical Value = 0.143.
tf	1..1	Seconds	Excitation control system stabilizer time constant (Tf). Typical Value = 1.
kn	1..1	PU	Excitation control system stabilizer gain (Kn). Typical Value =0.05.
efdn	1..1	PU	Value of *EFD* at which feedback gain changes (Efdn). Typical Value = 2.36.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (Kc). Typical Value = 0.104.
kd	1..1	PU	Demagnetizing factor, a function of exciter alternator reactances (Kd). Typical Value = 0.499.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

ke	1..1	PU	Exciter constant related to self-excited field (Ke). Typical Value = 1.
klv	1..1	PU	Gain used in the minimum field voltage limiter loop (Klv). Typical Value = 0.194.
kf1	1..1	PU	Coefficient to allow different usage of the model (Kf1). Typical Value = 1.
kf2	1..1	PU	Coefficient to allow different usage of the model (Kf2). Typical Value = 0.
ks	1..1	PU	Coefficient to allow different usage of the model-speed coefficient (Ks). Typical Value = 0.
vfemax	1..1	PU	Exciter field current limit reference (Vfemax). Typical Value = 16.
ve1	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (Ve1) equals Vemax (Ve1). Typical Value = 6.24.
seve1	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, Ve_1, back of commutating reactance (Se[Ve_1]). Typical Value = 1.143.
ve2	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (Ve_2). Typical Value = 4.68.
seve2	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, Ve_2, back of commutating reactance (Se[Ve_2]). Typical Value = 0.1.
vlv	1..1	PU	Field voltage used in the minimum field voltage limiter loop (Vlv). Typical Value = 0.79.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

name	0..1	String	see IdentifiedObject

Table 9-155 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-155 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] ----- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.25 ExcAC4A

Modified IEEE AC4A alternator-supplied rectifier excitation system with different minimum controller output.

Figure 9-94 class diagram

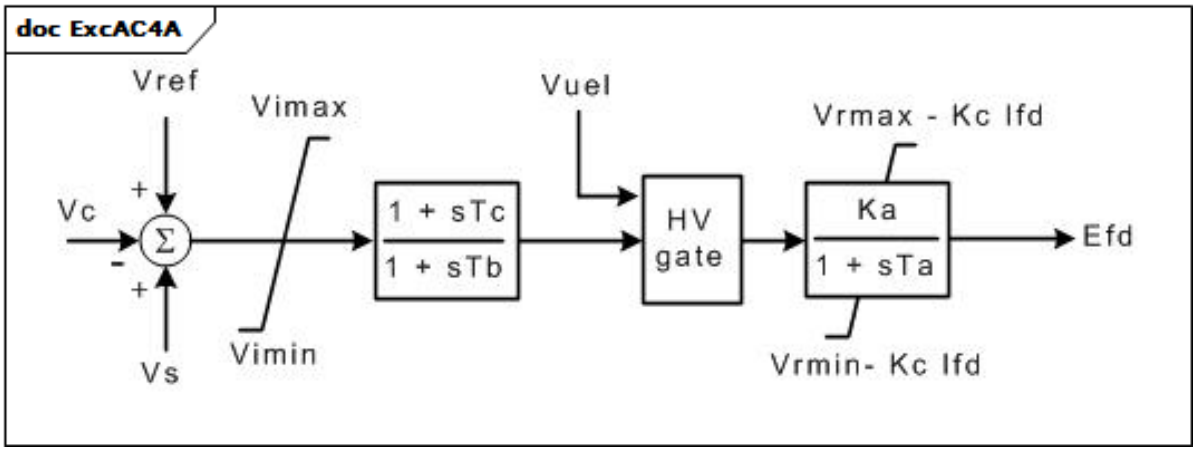


Figure 9-94 - figure ExcAC4A

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-156 shows all attributes of ExcAC4A.

Table 9-156 - Attributes of ExcitationSystemDynamics::ExcAC4A

Attribute name	mult.	Attribute type	Description
vimax	1..1	PU	Maximum voltage regulator input limit (Vimax). Typical Value = 10.
vimin	1..1	PU	Minimum voltage regulator input limit (Vimin). Typical Value = -10.

tc	1..1	Seconds	Voltage regulator time constant (Tc). Typical Value = 1.
tb	1..1	Seconds	Voltage regulator time constant (Tb). Typical Value = 10.
ka	1..1	PU	Voltage regulator gain (Ka). Typical Value = 200.
ta	1..1	Seconds	Voltage regulator time constant (Ta). Typical Value = 0.015.
vrmax	1..1	PU	Maximum voltage regulator output (Vrmax). Typical Value = 5.64.
vrmin	1..1	PU	Minimum voltage regulator output (Vrmin). Typical Value = -4.53.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (Kc). Typical Value = 0.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-157 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-157 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] ----- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.26 ExcAC5A

Modified IEEE AC5A alternator-supplied rectifier excitation system with different minimum controller output.

Figure 9-95 class diagram

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

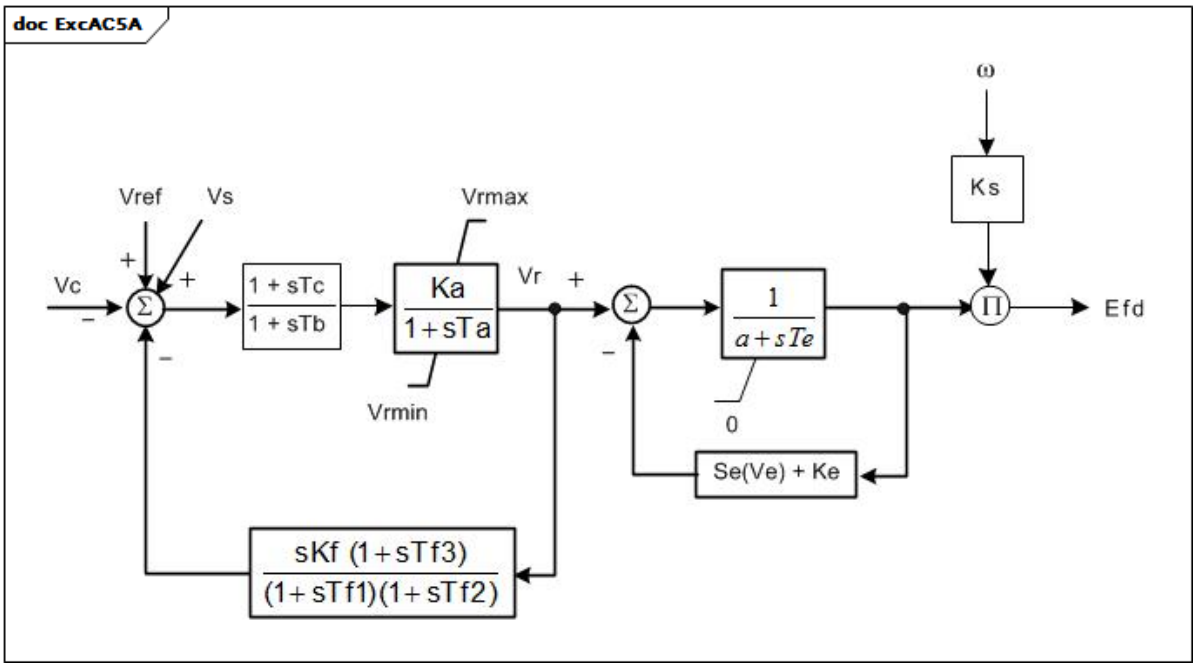


Figure 9-95 - figure ExcAC5A

Parameter Notes:

1. If Ks is set to 1 the output (Efd) is multiplied by generator speed.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-158 shows all attributes of ExcAC5A.

Table 9-158 - Attributes of ExcitationSystemDynamics::ExcAC5A

Attribute name	mult.	Attribute type	Description
ka	1..1	PU	Voltage regulator gain (Ka). Typical Value = 400.
ks	1..1	PU	Coefficient to allow different usage of the model-speed coefficient (Ks). Typical Value = 0.
tb	1..1	Seconds	Voltage regulator time constant (Tb). Typical Value = 0.
tc	1..1	Seconds	Voltage regulator time constant (Tc). Typical Value = 0.
ta	1..1	Seconds	Voltage regulator time constant (Ta). Typical Value = 0.02.
vrmax	1..1	PU	Maximum voltage regulator output (Vrmax). Typical Value

			= 7.3.
vrmin	1..1	PU	Minimum voltage regulator output (Vrmin). Typical Value =-7.3.
ke	1..1	PU	Exciter constant related to self-excited field (Ke). Typical Value = 1.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (Te). Typical Value = 0.8.
kf	1..1	PU	Excitation control system stabilizer gains (Kf). Typical Value = 0.03.
tf1	1..1	Seconds	Excitation control system stabilizer time constant (Tf1). Typical Value = 1.
tf2	1..1	Seconds	Excitation control system stabilizer time constant (Tf2). Typical Value = 0.8.
tf3	1..1	Seconds	Excitation control system stabilizer time constant (Tf3). Typical Value = 0.
efd1	1..1	PU	Exciter voltage at which exciter saturation is defined (Efd1). Typical Value = 5.6.
seefd1	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, Efd1 (S_E[Efd1]). Typical Value = 0.86.
efd2	1..1	PU	Exciter voltage at which exciter saturation is defined (Efd2). Typical Value = 4.2.
seefd2	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, Efd2 (S_E[Efd2]). Typical Value = 0.5.
a	1..1	Simple_Float	Coefficient to allow different usage of the model (a). Typical Value = 1.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

name	0..1	String	see IdentifiedObject

Table 9-159 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-159 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.27 ExcAC6A

Modified IEEE AC6A alternator-supplied rectifier excitation system with speed input.

Figure 9-96 class diagram

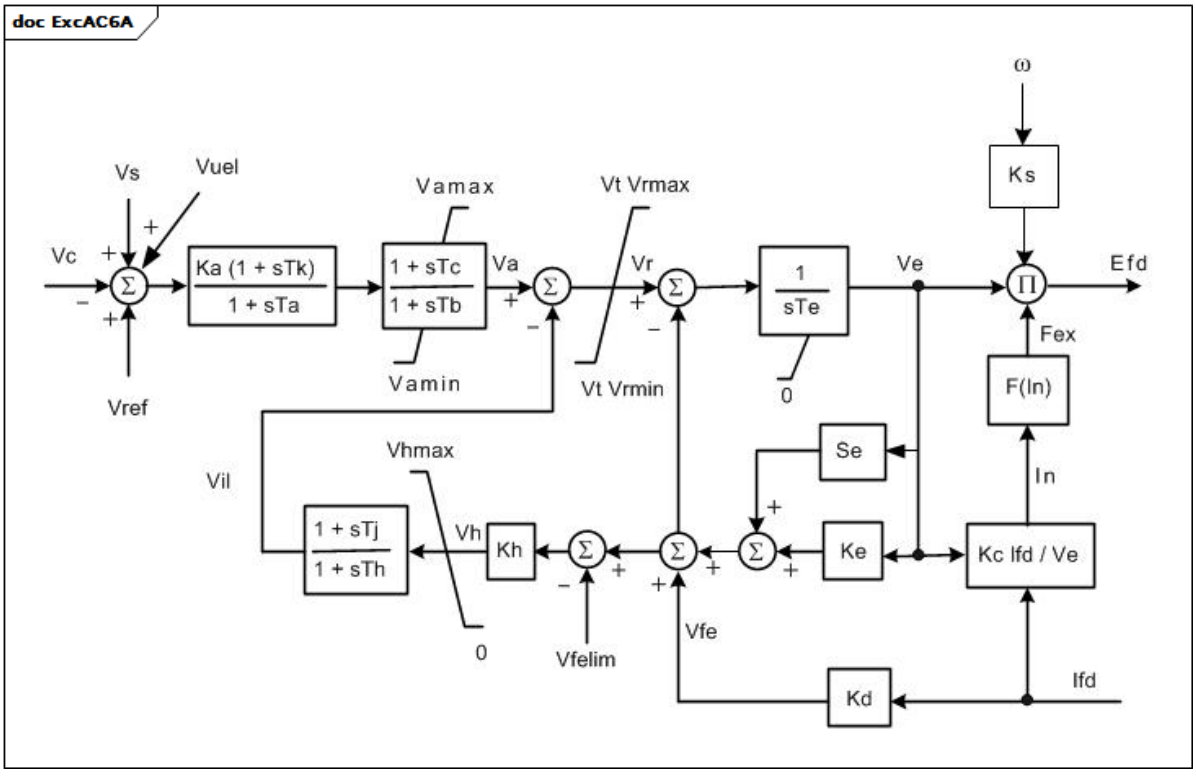


Figure 9-96 - figure ExcAC6A

Parameter Notes:

1. If Ks is set to 1 the output (Efd) is multiplied by generator speed.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-160 shows all attributes of ExcAC6A.

Table 9-160 - Attributes of ExcitationSystemDynamics::ExcAC6A

Attribute name	mult.	Attribute type	Description
ka	1..1	PU	Voltage regulator gain (Ka). Typical Value = 536.
ks	1..1	PU	Coefficient to allow different usage of the model-speed coefficient (Ks). Typical Value = 0.
ta	1..1	Seconds	Voltage regulator time constant (Ta). Typical Value = 0.086.
tk	1..1	Seconds	Voltage regulator time constant (Tk). Typical Value = 0.18.
tb	1..1	Seconds	Voltage regulator time constant (Tb). Typical Value = 9.
tc	1..1	Seconds	Voltage regulator time constant (Tc). Typical Value = 3.
vamax	1..1	PU	Maximum voltage regulator output (Vamax). Typical Value = 75.
vamin	1..1	PU	Minimum voltage regulator output (Vamin). Typical Value = -75.
vrmax	1..1	PU	Maximum voltage regulator output (Vrmax). Typical Value = 44.
vrmin	1..1	PU	Minimum voltage regulator output (Vrmin). Typical Value = -36.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (Te). Typical Value = 1.
kh	1..1	PU	Exciter field current limiter gain (Kh). Typical Value = 92.
tj	1..1	Seconds	Exciter field current limiter time constant (Tj). Typical Value = 0.02.
th	1..1	Seconds	Exciter field current limiter time constant (Th). Typical Value =

			0.08.
vfelim	1..1	PU	Exciter field current limit reference (Vfelim). Typical Value = 19.
vhmax	1..1	PU	Maximum field current limiter signal reference (Vhmax). Typical Value = 75.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (Kc). Typical Value = 0.173.
kd	1..1	PU	Demagnetizing factor, a function of exciter alternator reactances (Kd). Typical Value = 1.91.
ke	1..1	PU	Exciter constant related to self-excited field (Ke). Typical Value = 1.6.
ve1	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (Ve_1). Typical Value = 7.4.
seve1	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, Ve1, back of commutating reactance (Se[Ve1]). Typical Value = 0.214.
ve2	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (Ve2). Typical Value = 5.55.
seve2	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, Ve2, back of commutating reactance (Se[Ve2]). Typical Value = 0.044.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-161 shows all association ends of ExcitationSystemDynamics with other classes.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 9-161 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.28 ExcAC8B

Modified IEEE AC8B alternator-supplied rectifier excitation system with speed input and input limiter.

Figure 9-97 class diagram

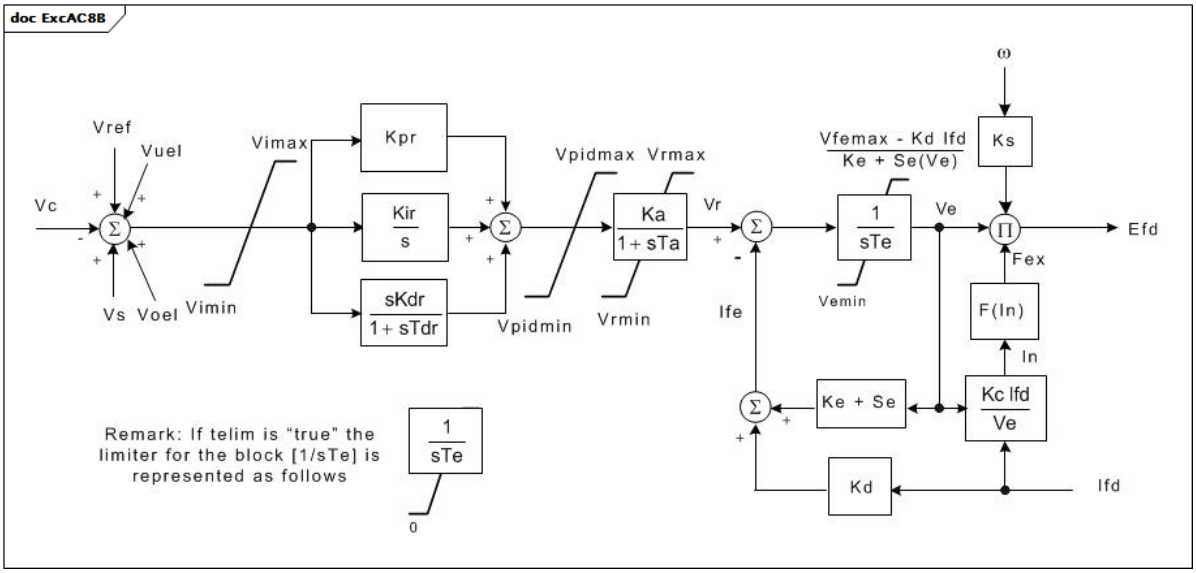


Figure 9-97 - figure ExcAC8B

Parameter Notes:

1. The upper limit on Ve represents the effect of the field current limiter. If Vfemax is zero, this limit will not be enforced.
2. If Ks is set to 1 the output (Efd) is multiplied by generator speed.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-162 shows all attributes of ExcAC8B.

Table 9-162 - Attributes of ExcitationSystemDynamics::ExcAC8B

Attribute name	mult.	Attribute type	Description

inlim	1..1	Boolean	Input limiter indicator. true = input limiter Vimax and Vimin is considered false = input limiter Vimax and Vimin is not considered. Typical Value = true.
ka	1..1	PU	Voltage regulator gain (Ka). Typical Value = 1.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (Kc). Typical Value = 0.55.
kd	1..1	PU	Demagnetizing factor, a function of exciter alternator reactances (Kd). Typical Value = 1.1.
kdr	1..1	PU	Voltage regulator derivative gain (Kdr). Typical Value = 10.
ke	1..1	PU	Exciter constant related to self-excited field (Ke). Typical Value = 1.
kir	1..1	PU	Voltage regulator integral gain (Kir). Typical Value = 5.
kpr	1..1	PU	Voltage regulator proportional gain (Kpr). Typical Value = 80.
ks	1..1	PU	Coefficient to allow different usage of the model-speed coefficient (Ks). Typical Value = 0.
pidlim	1..1	Boolean	PID limiter indicator. true = input limiter Vpidmax and Vpidmin is considered false = input limiter Vpidmax and Vpidmin is not considered. Typical Value = true.
seve1	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, Ve_1, back of commutating reactance (Se[Ve1]). Typical Value = 0.3.
seve2	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, Ve_2, back of commutating reactance (Se[Ve2]). Typical Value = 3.
ta	1..1	Seconds	Voltage regulator time constant (Ta). Typical

			Value = 0.
tdr	1..1	Seconds	Lag time constant (Tdr). Typical Value = 0.1.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (Te). Typical Value = 1.2.
telim	1..1	Boolean	Selector for the limiter on the block [1/sTe]. See diagram for meaning of true and false. Typical Value = false.
ve1	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (Ve_1) equals V_{EMAX} (Ve1). Typical Value = 6.5.
ve2	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (Ve_2). Typical Value = 9.
vemin	1..1	PU	Minimum exciter voltage output (Vemin). Typical Value = 0.
vfemax	1..1	PU	Exciter field current limit reference (Vfemax). Typical Value = 6.
vimax	1..1	PU	Input signal maximum (Vimax). Typical Value = 35.
vimin	1..1	PU	Input signal minimum (Vimin). Typical Value = -10.
vpidmax	1..1	PU	PID maximum controller output (Vpidmax). Typical Value = 35.
vpidmin	1..1	PU	PID minimum controller output (Vpidmin). Typical Value = -10.
vrmax	1..1	PU	Maximum voltage regulator output (Vrmax). Typical Value = 35.
vrmin	1..1	PU	Minimum voltage regulator output (Vrmin). Typical Value = 0.
vtmult	1..1	Boolean	Multiply by generator's terminal voltage indicator. true =the limits Vrmax

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			and Vrmin are multiplied by the generator's terminal voltage to represent a thyristor power stage fed from the generator terminals false = limits are not multiplied by generator's terminal voltage. Typical Value = false.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-163 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-163 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.29 ExcANS

Italian excitation system. It represents static field voltage or excitation current feedback excitation system.

Figure 9-98 class diagram

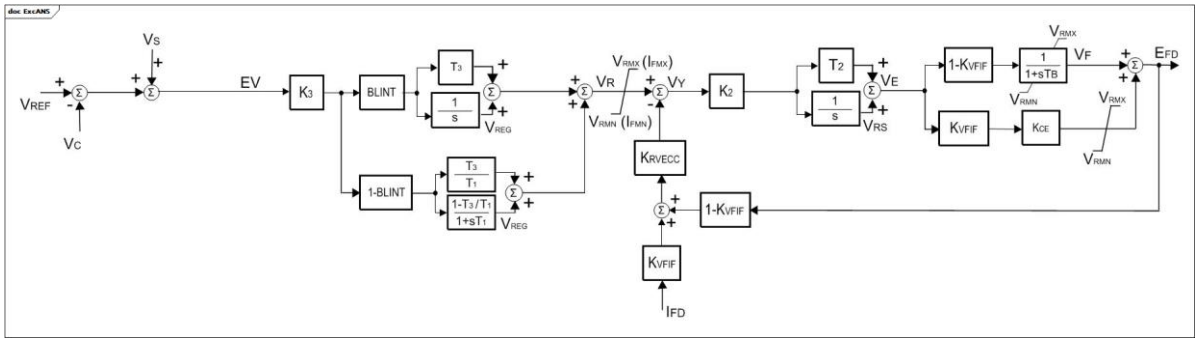


Figure 9-98 - figure ExcANS

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-164 shows all attributes of ExcANS.

Table 9-164 - Attributes of ExcitationSystemDynamics::ExcANS

Attribute name	mult.	Attribute type	Description
k3	1..1	Simple_Float	AVR gain (K_3). Typical Value = 1000.
k2	1..1	Simple_Float	Exciter gain (K_2). Typical Value = 20.
kce	1..1	Simple_Float	Ceiling factor (K_{CE}). Typical Value = 1.
t3	1..1	Seconds	Time constant (T_3). Typical Value = 1.6.
t2	1..1	Seconds	Time constant (T_2). Typical Value = 0.05.
t1	1..1	Seconds	Time constant (T_1). Typical Value = 20.
blint	1..1	Integer	Governor Control Flag (BLINT). 0 = lead-lag regulator 1 = proportional integral regulator. Typical Value = 0.
kvfif	1..1	Integer	Rate feedback signal flag (K_{VFIF}). 0 = output voltage of the exciter 1 = exciter field current. Typical Value = 0.
ifmn	1..1	PU	Minimum exciter current (I_{FMN}). Typical Value = -5.2.
ifmx	1..1	PU	Maximum exciter current (I_{FMX}). Typical Value = 6.5.
vrmn	1..1	PU	Maximum AVR output (V_{RMN}). Typical Value = -5.2.
vrmx	1..1	PU	Minimum AVR output (V_{RMX}). Typical Value = 6.5.
krvecc	1..1	Integer	Feedback enabling (K_{RVECC}). 0 = Open loop control 1 = Closed loop control. Typical Value = 1.
tb	1..1	Seconds	Exciter time constant (T_B). Typical Value = 0.04.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-165 shows all association ends of ExcitationSystemDynamics with other classes.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 9-165 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.30 ExcAVR1

Italian excitation system corresponding to IEEE (1968) Type 1 Model. It represents exciter dynamo and electromechanical regulator.

Figure 9-99 class diagram

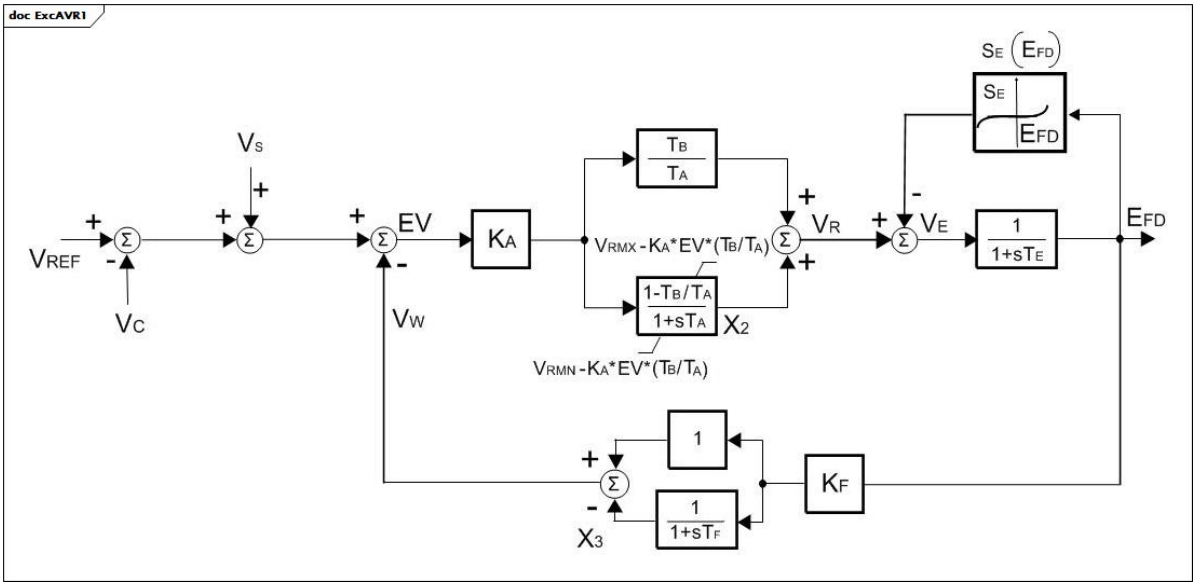


Figure 9-99 - figure ExcAVR1

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-166 shows all attributes of ExcAVR1.

Table 9-166 - Attributes of ExcitationSystemDynamics::ExcAVR1

Attribute name	mult.	Attribute type	Description
ka	1..1	Simple_Float	AVR gain (K_A). Typical Value = 500.
vrmn	1..1	PU	Maximum AVR output (V_{RMN}). Typical Value = -6.
vrmx	1..1	PU	Minimum AVR output (V_{RMX}). Typical Value =

			7.
ta	1..1	Seconds	AVR time constant (T_A). Typical Value = 0.2.
tb	1..1	Seconds	AVR time constant (T_B). Typical Value = 0.
te	1..1	Seconds	Exciter time constant (T_E). Typical Value = 1.
e1	1..1	PU	Field voltage value 1 (E1). Typical Value = 4.18.
se1	1..1	Simple_Float	Saturation factor at E1 (S(E1)). Typical Value = 0.1.
e2	1..1	PU	Field voltage value 2 (E2). Typical Value = 3.14.
se2	1..1	Simple_Float	Saturation factor at E2 (S(E2)). Typical Value = 0.03.
kf	1..1	Simple_Float	Rate feedback gain (K_F). Typical Value = 0.02.
tf	1..1	Seconds	Rate feedback time constant (T_F). Typical Value = 1.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-167 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-167 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] ----- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.31 ExcAVR2

Italian excitation system corresponding to IEEE (1968) Type 2 Model. It represents alternator and rotating diodes and electromechanic voltage regulators.

Figure 9-100 class diagram

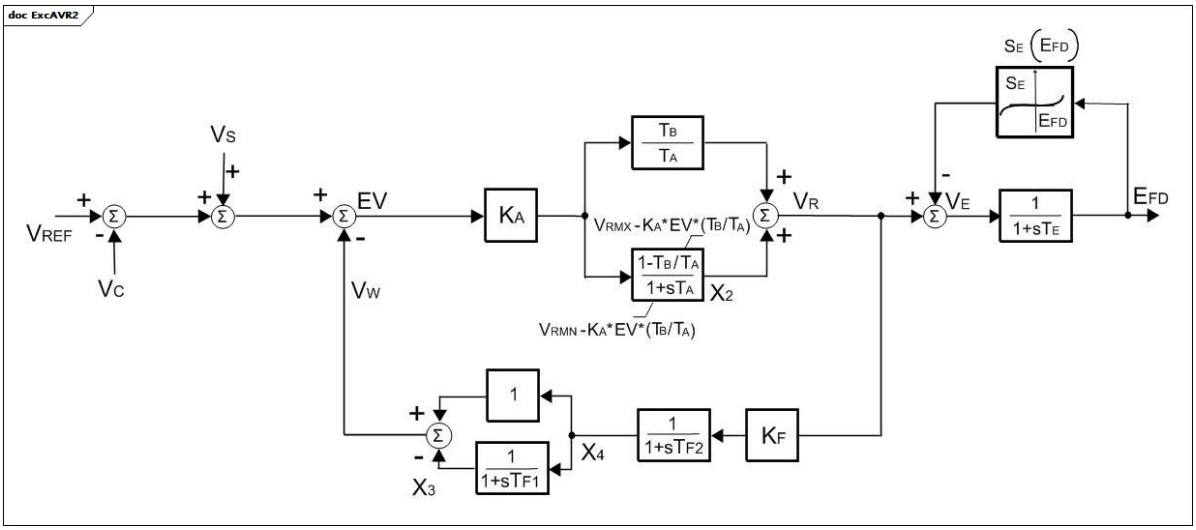


Figure 9-100 - figure ExcAVR2

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-168 shows all attributes of ExcAVR2.

Table 9-168 - Attributes of ExcitationSystemDynamics::ExcAVR2

Attribute name	mult.	Attribute type	Description
ka	1..1	Simple_Float	AVR gain (K_A). Typical Value = 500.
vrmn	1..1	PU	Maximum AVR output (V_{RMN}). Typical Value = -6.
vrmx	1..1	PU	Minimum AVR output (V_{RMX}). Typical Value = 7.
ta	1..1	Seconds	AVR time constant (T_A). Typical Value = 0.02.
tb	1..1	Seconds	AVR time constant (T_B). Typical Value = 0.
te	1..1	Seconds	Exciter time constant (T_E). Typical Value = 1.
e1	1..1	PU	Field voltage value 1 (E1). Typical Value = 4.18.
se1	1..1	Simple_Float	Saturation factor at E1 (S(E1)). Typical Value = 0.1.
e2	1..1	PU	Field voltage value 2 (E2). Typical Value = 3.14.
se2	1..1	Simple_Float	Saturation factor at E2 (S(E2)). Typical Value = 0.03.
kf	1..1	Simple_Float	Rate feedback gain (K_F). Typical Value =

			0.02.
tf1	1..1	Seconds	Rate feedback time constant (T_{F1}). Typical Value = 1.
tf2	1..1	Seconds	Rate feedback time constant (T_{F2}). Typical Value = 1.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-169 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-169 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.32 ExcAVR3

Italian excitation system. It represents exciter dynamo and electric regulator.

Figure 9-101 class diagram

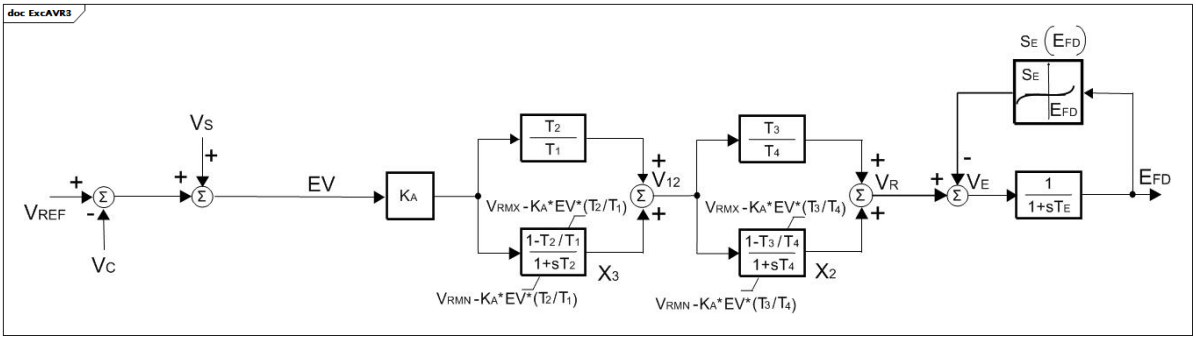


Figure 9-101 - figure ExcAVR3

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-170 shows all attributes of ExcAVR3.

Table 9-170 - Attributes of ExcitationSystemDynamics::ExcAVR3

Attribute name	mult.	Attribute type	Description

ka	1..1	Simple_Float	AVR gain (K_A). Typical Value = 3000.
vrmn	1..1	PU	Maximum AVR output (V_{RMN}). Typical Value = -7.5.
vrmx	1..1	PU	Minimum AVR output (V_{RMX}). Typical Value = 7.5.
t1	1..1	Seconds	AVR time constant (T_1). Typical Value = 220.
t2	1..1	Seconds	AVR time constant (T_2). Typical Value = 1.6.
t3	1..1	Seconds	AVR time constant (T_3). Typical Value = 0.66.
t4	1..1	Seconds	AVR time constant (T_4). Typical Value = 0.07.
te	1..1	Seconds	Exciter time constant (T_E). Typical Value = 1.
e1	1..1	PU	Field voltage value 1 (E1). Typical Value = 4.18.
se1	1..1	Simple_Float	Saturation factor at E1 (S(E1)). Typical Value = 0.1.
e2	1..1	PU	Field voltage value 2 (E2). Typical Value = 3.14.
se2	1..1	Simple_Float	Saturation factor at E2 (S(E2)). Typical Value = 0.03.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-171 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-171 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

9.6.9.33 ExcAVR4

Italian excitation system. It represents static exciter and electric voltage regulator.

Figure 9-102 class diagram

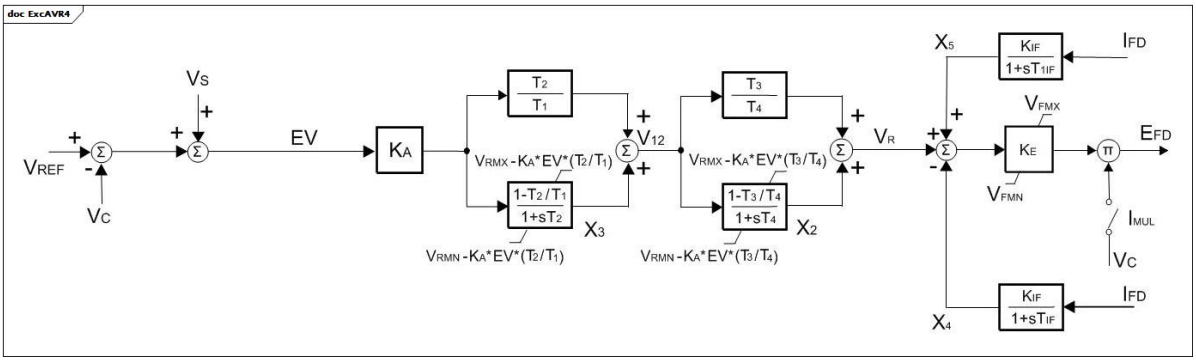


Figure 9-102 - figure ExcAVR4

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-172 shows all attributes of ExcAVR4.

Table 9-172 - Attributes of ExcitationSystemDynamics::ExcAVR4

Attribute name	mult.	Attribute type	Description
ka	1..1	Simple_Float	AVR gain (K_A). Typical Value = 300.
vrmn	1..1	PU	Maximum AVR output (V_{RMN}). Typical Value = 0.
vrmx	1..1	PU	Minimum AVR output (V_{RMX}). Typical Value = 5.
t1	1..1	Seconds	AVR time constant (T_1). Typical Value = 4.8.
t2	1..1	Seconds	AVR time constant (T_2). Typical Value = 1.5.
t3	1..1	Seconds	AVR time constant (T_3). Typical Value = 0.
t4	1..1	Seconds	AVR time constant (T_4). Typical Value = 0.
ke	1..1	Simple_Float	Exciter gain (K_E). Typical Value = 1.
vfmx	1..1	PU	Maximum exciter output (V_{FMX}). Typical Value = 5.
vfmn	1..1	PU	Minimum exciter output (V_{FMN}). Typical Value = 0.
kif	1..1	Simple_Float	Exciter internal reactance (K_{IF}). Typical Value = 0.

tif	1..1	Seconds	Exciter current feedback time constant (T_{IF}). Typical Value = 0.
t1if	1..1	Seconds	Exciter current feedback time constant (T_{1IF}). Typical Value = 60.
imul	1..1	Boolean	AVR output voltage dependency selector (Imul). true = selector is connected false = selector is not connected. Typical Value = true.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-173 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-173 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.34 ExcAVR5

Manual excitation control with field circuit resistance. This model can be used as a very simple representation of manual voltage control.

Figure 9-103 class diagram

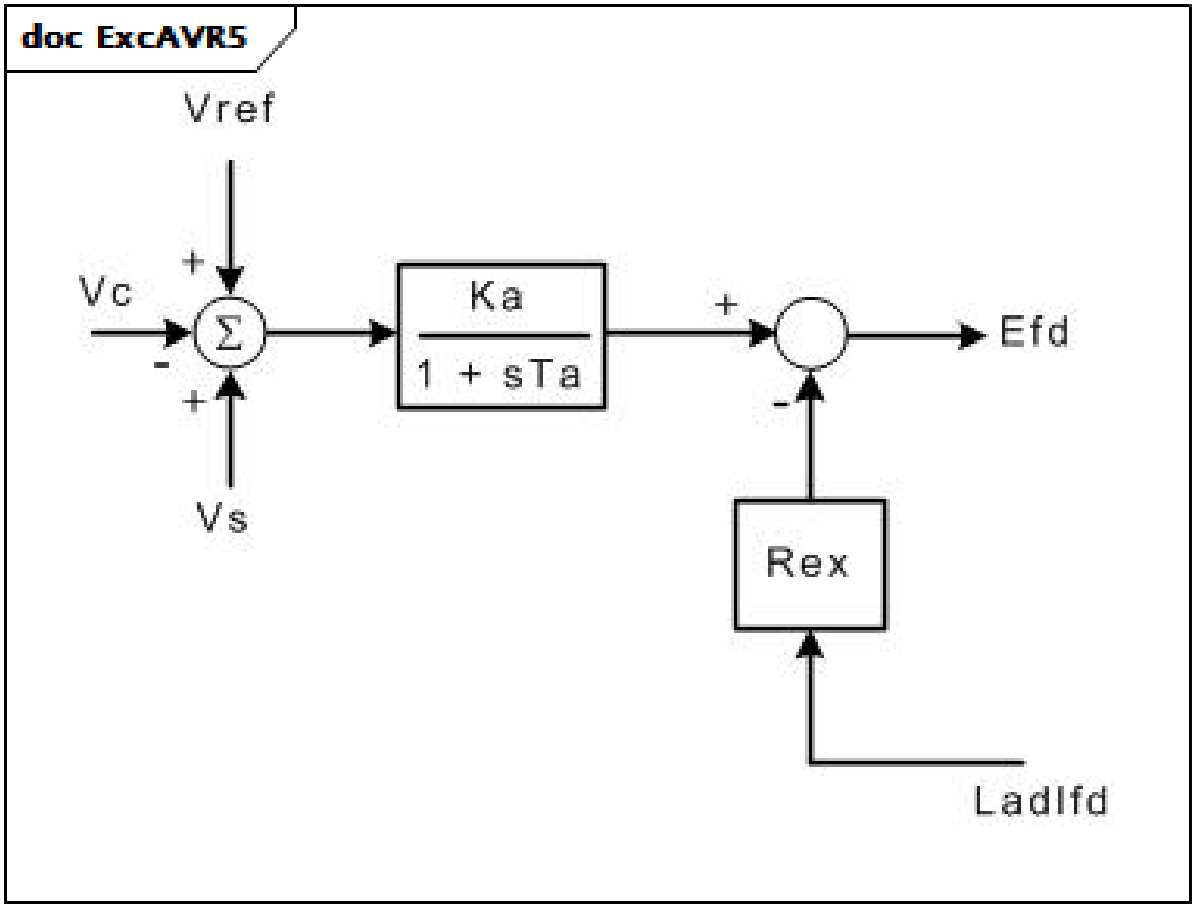


Figure 9-103 - figure ExcAVR5

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-174 shows all attributes of ExcAVR5.

Table 9-174 - Attributes of ExcitationSystemDynamics::ExcAVR5

Attribute name	mult.	Attribute type	Description
ka	1..1	PU	Gain (Ka).
ta	1..1	Seconds	Time constant (Ta).
rex	1..1	PU	Effective Output Resistance (Rex). Rex represents the effective output resistance seen by the excitation system.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-175 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-175 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.35 ExcAVR7

IVO excitation system.

Figure 9-104 class diagram

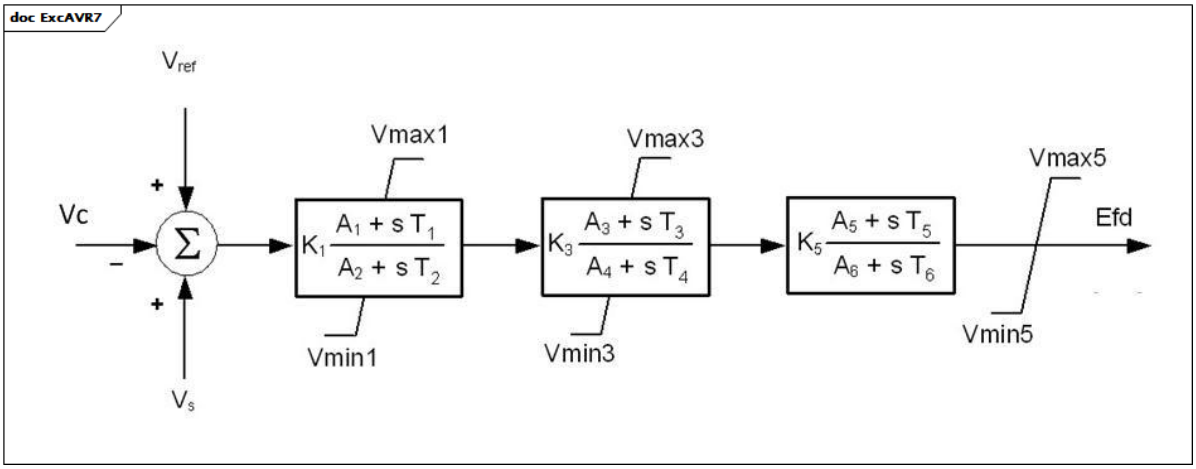


Figure 9-104 - figure ExcAVR7

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-176 shows all attributes of ExcAVR7.

Table 9-176 - Attributes of ExcitationSystemDynamics::ExcAVR7

Attribute name	mult.	Attribute type	Description
k1	1..1	PU	Gain (K1). Typical Value = 1.
a1	1..1	PU	Lead coefficient (A1). Typical Value = 0.5.
a2	1..1	PU	Lag coefficient (A2). Typical Value = 0.5.
t1	1..1	Seconds	Lead time constant (T1). Typical Value = 0.05.
t2	1..1	Seconds	Lag time constant (T2). Typical Value = 0.1.

vmax1	1..1	PU	Lead-lag max. limit (Vmax1). Typical Value = 5.
vmin1	1..1	PU	Lead-lag min. limit (Vmin1). Typical Value = -5.
k3	1..1	PU	Gain (K3). Typical Value = 3.
a3	1..1	PU	Lead coefficient (A3). Typical Value = 0.5.
a4	1..1	PU	Lag coefficient (A4). Typical Value = 0.5.
t3	1..1	Seconds	Lead time constant (T3). Typical Value = 0.1.
t4	1..1	Seconds	Lag time constant (T4). Typical Value = 0.1.
vmax3	1..1	PU	Lead-lag max. limit (Vmax3). Typical Value = 5.
vmin3	1..1	PU	Lead-lag min. limit (Vmin3). Typical Value = -5.
k5	1..1	PU	Gain (K5). Typical Value = 1.
a5	1..1	PU	Lead coefficient (A5). Typical Value = 0.5.
a6	1..1	PU	Lag coefficient (A6). Typical Value = 0.5.
t5	1..1	Seconds	Lead time constant (T5). Typical Value = 0.1.
t6	1..1	Seconds	Lag time constant (T6). Typical Value = 0.1.
vmax5	1..1	PU	Lead-lag max. limit (Vmax5). Typical Value = 5.
vmin5	1..1	PU	Lead-lag min. limit (Vmin5). Typical Value = -2.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-177 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-177 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult	Class type name	Description

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

	.		
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.36 ExcBBC

Transformer fed static excitation system (static with ABB regulator). This model represents a static excitation system in which a gated thyristor bridge fed by a transformer at the main generator terminals feeds the main generator directly.

Figure 9-105 class diagram

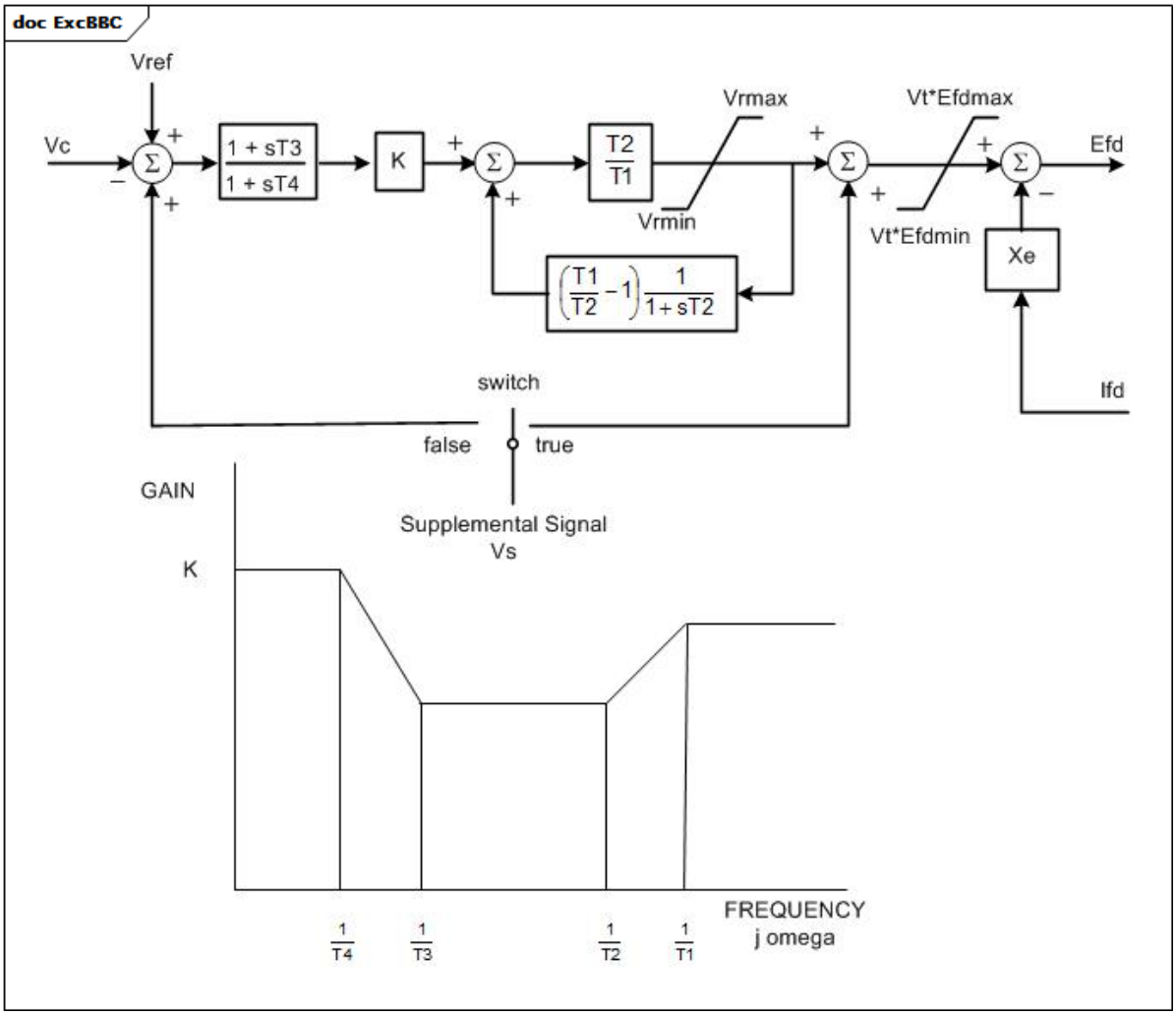


Figure 9-105 - figure ExcBBC

Parameter Notes:

1. K, T1, T2 must be non-zero. If Tf or T4 is zero, the corresponding block is bypassed.
2. Xe models the regulation of the transformer/rectifier unit and must be zero or positive.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-178 shows all attributes of ExcBBC.

Table 9-178 - Attributes of ExcitationSystemDynamics::ExcBBC

Attribute name	mult.	Attribute type	Description
t1	1..1	Seconds	Controller time constant (T1). Typical Value = 6.
t2	1..1	Seconds	Controller time constant (T2). Typical Value = 1.
t3	1..1	Seconds	Lead/lag time constant (T3). Typical Value = 0.05.
t4	1..1	Seconds	Lead/lag time constant (T4). Typical Value = 0.01.
k	1..1	PU	Steady state gain (K). Typical Value = 300.
vrmin	1..1	PU	Minimum control element output (Vrmin). Typical Value = -5.
vrmax	1..1	PU	Maximum control element output (Vrmax). Typical Value = 5.
efdmin	1..1	PU	Minimum open circuit exciter voltage (Efdmin). Typical Value = -5.
efdmax	1..1	PU	Maximum open circuit exciter voltage (Efdmax). Typical Value = 5.
xe	1..1	PU	Effective excitation transformer reactance (Xe). Typical Value = 0.05.
switch	1..1	Boolean	Supplementary signal routing selector (switch). true = Vs connected to 3rd summing point false = Vs connected to 1st summing point (see diagram). Typical Value = true.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-179 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-179 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.37 ExcCZ

Czech Proportion/Integral Exciter.

Figure 9-106 class diagram

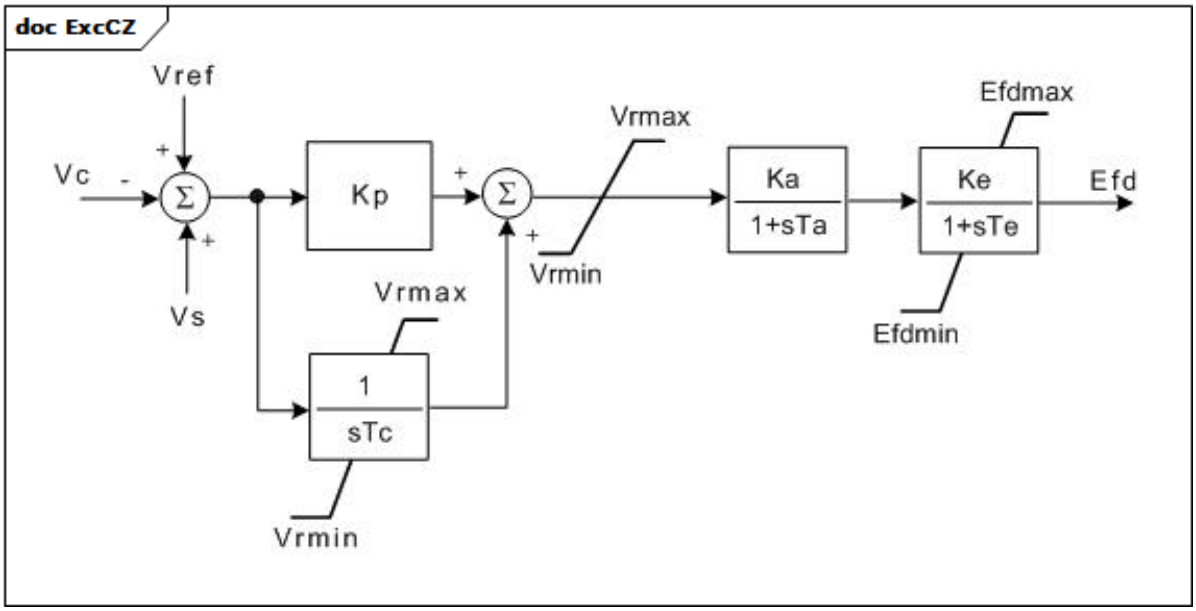


Figure 9-106 - figure ExcCZ

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-180 shows all attributes of ExcCZ.

Table 9-180 - Attributes of ExcitationSystemDynamics::ExcCZ

Attribute name	mult.	Attribute type	Description
kp	1..1	PU	Regulator proportional gain (Kp).
tc	1..1	Seconds	Regulator integral time constant (Tc).
vrmax	1..1	PU	Voltage regulator

			maximum limit (Vrmax).
vrmin	1..1	PU	Voltage regulator minimum limit (Vrmin).
ka	1..1	PU	Regulator gain (Ka).
ta	1..1	Seconds	Regulator time constant (Ta).
ke	1..1	PU	Exciter constant related to self-excited field (Ke).
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (Te).
efdmax	1..1	PU	Exciter output maximum limit (Efdmax).
efdmin	1..1	PU	Exciter output minimum limit (Efdmin).
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-181 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-181 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.38 ExcDC1A

Modified IEEE DC1A direct current commutator exciter with speed input and without underexcitation limiters (UEL) inputs.

Figure 9-107 class diagram

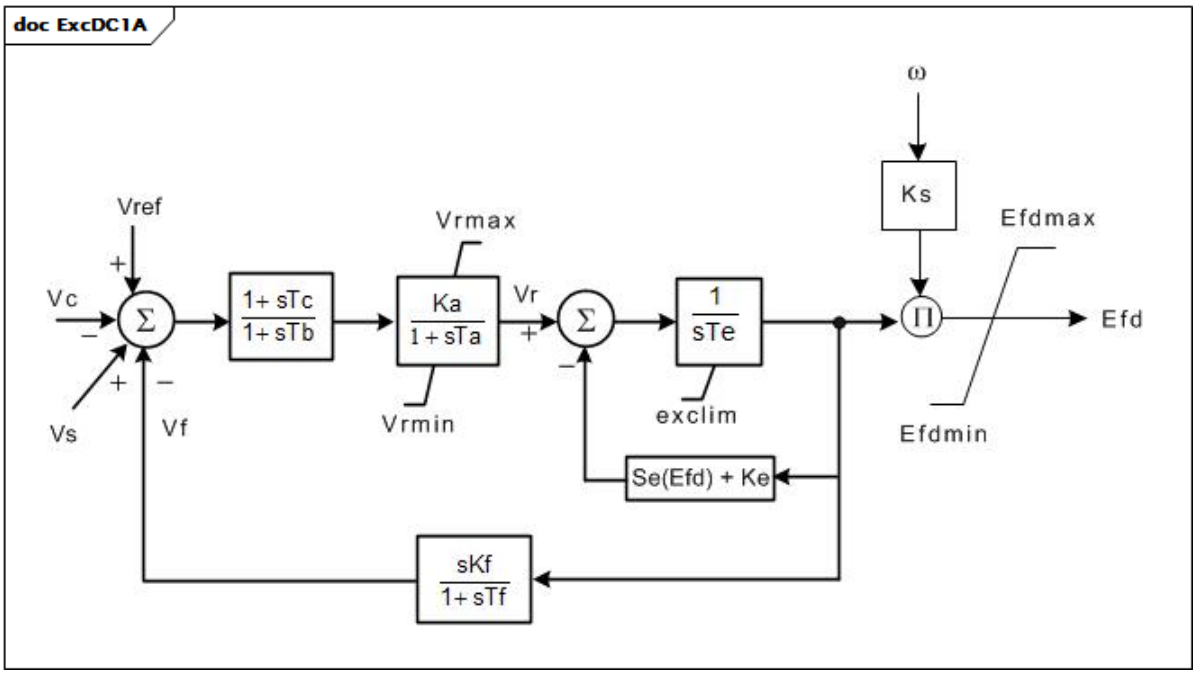


Figure 9-107 - figure ExcDC1A

Parameter **Notes:**

If Ks is set to 1 the output (Efd) is multiplied by generator speed.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-182 shows all attributes of ExcDC1A.

Table 9-182 - Attributes of ExcitationSystemDynamics::ExcDC1A

Attribute name	mult.	Attribute type	Description
ka	1..1	PU	Voltage regulator gain (Ka). Typical Value = 46.
ks	1..1	PU	Coefficient to allow different usage of the model-speed coefficient (Ks). Typical Value = 0.
ta	1..1	Seconds	Voltage regulator time constant (Ta). Typical Value = 0.06.
tb	1..1	Seconds	Voltage regulator time constant (Tb). Typical Value = 0.
tc	1..1	Seconds	Voltage regulator time constant (Tc). Typical Value = 0.
vrmax	1..1	PU	Maximum voltage regulator output (Vrmax). Typical Value = 1.
vrmin	1..1	PU	Minimum voltage regulator output

			(Vrmin). Typical Value = -0.9.
ke	1..1	PU	Exciter constant related to self-excited field (Ke). Typical Value = 0.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (Te). Typical Value = 0.46.
kf	1..1	PU	Excitation control system stabilizer gain (Kf). Typical Value = 0.1.
tf	1..1	Seconds	Excitation control system stabilizer time constant (Tf). Typical Value = 1.
efd1	1..1	PU	Exciter voltage at which exciter saturation is defined (Efd1). Typical Value = 3.1.
seefd1	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, Efd1 (Se[Eefd1]). Typical Value = 0.33.
efd2	1..1	PU	Exciter voltage at which exciter saturation is defined (Efd2). Typical Value = 2.3.
seefd2	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, Efd1 (Se[Eefd1]). Typical Value = 0.33.
exclim	1..1	Boolean	(exclim). IEEE standard is ambiguous about lower limit on exciter output. true = a lower limit of zero is applied to integrator output false = a lower limit of zero is not applied to integrator output. Typical Value = true.
efdmin	1..1	PU	Minimum voltage exciter output limiter (Efdmin). Typical Value = -99.
edfmax	1..1	PU	Maximum voltage exciter output limiter (Efdmax). Typical Value = 99.
enabled	1..1	Boolean	see DynamicsFunctionBlock

description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-183 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-183 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.39 ExcDC2A

Modified IEEE DC2A direct current commutator exciters with speed input, one more leg block in feedback loop and without underexcitation limiters (UEL) inputs. DC type 2 excitation system model with added speed multiplier, added lead-lag, and voltage-dependent limits.

Figure 9-108 class diagram

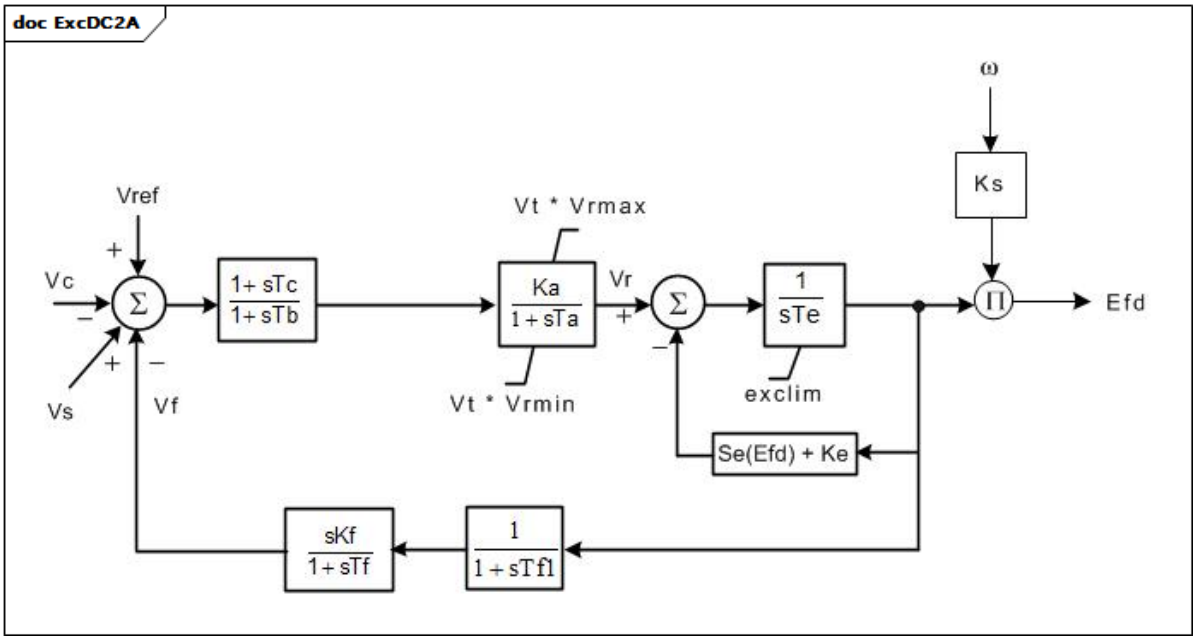


Figure 9-108 - figure ExcDC2A

Parameter Notes:

1. If Ks is set to 1 the output (Efd) is multiplyed by generator speed.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-184 shows all attributes of ExcDC2A.

Table 9-184 - Attributes of ExcitationSystemDynamics::ExcDC2A

Attribute name	mult.	Attribute type	Description
ka	1..1	PU	Voltage regulator gain (Ka). Typical Value = 300.
ks	1..1	PU	Coefficient to allow different usage of the model-speed coefficient (Ks). Typical Value = 0.
ta	1..1	Seconds	Voltage regulator time constant (Ta). Typical Value = 0.01.
tb	1..1	Seconds	Voltage regulator time constant (Tb). Typical Value = 0.
tc	1..1	Seconds	Voltage regulator time constant (Tc). Typical Value = 0.
vrmax	1..1	PU	Maximum voltage regulator output (Vrmax). Typical Value = 4.95.
vrmin	1..1	PU	Minimum voltage regulator output (Vrmin). Typical Value = -4.9.
ke	1..1	PU	Exciter constant related to self-excited field (Ke). If Ke is entered as zero, the model calculates an effective value of Ke such that the initial condition value of Vr is zero. The zero value of Ke is not changed. If Ke is entered as non-zero, its value is used directly, without change. Typical Value = 1.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (Te). Typical Value = 1.33.
kf	1..1	PU	Excitation control system stabilizer gain (Kf). Typical Value = 0.1.
tf	1..1	Seconds	Excitation control system stabilizer time constant (Tf). Typical Value = 0.675.
tf1	1..1	Seconds	Excitation control

			system stabilizer time constant (Tf1). Typical Value = 0.
efd1	1..1	PU	Exciter voltage at which exciter saturation is defined (Efd1). Typical Value = 3.05.
seefd1	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, Efd1 (Se[Eefd1]). Typical Value = 0.279.
efd2	1..1	PU	Exciter voltage at which exciter saturation is defined (Efd2). Typical Value = 2.29.
seefd2	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, Efd2 (Se[Efd2]). Typical Value = 0.117.
exclim	1..1	Boolean	(exclim). IEEE standard is ambiguous about lower limit on exciter output. true = a lower limit of zero is applied to integrator output false = a lower limit of zero is not applied to integrator output. Typical Value = true.
vtlim	1..1	Boolean	(Vtlim). true = limiter at the block [Ka/(1+sTa)] is dependent on Vt false = limiter at the block is not dependent on Vt. Typical Value = true.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-185 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-185 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1]

			----- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.40 ExcDC3A

This is modified IEEE DC3A direct current commutator exciters with speed input, and death band. DC old type 4.

Figure 9-109 class diagram

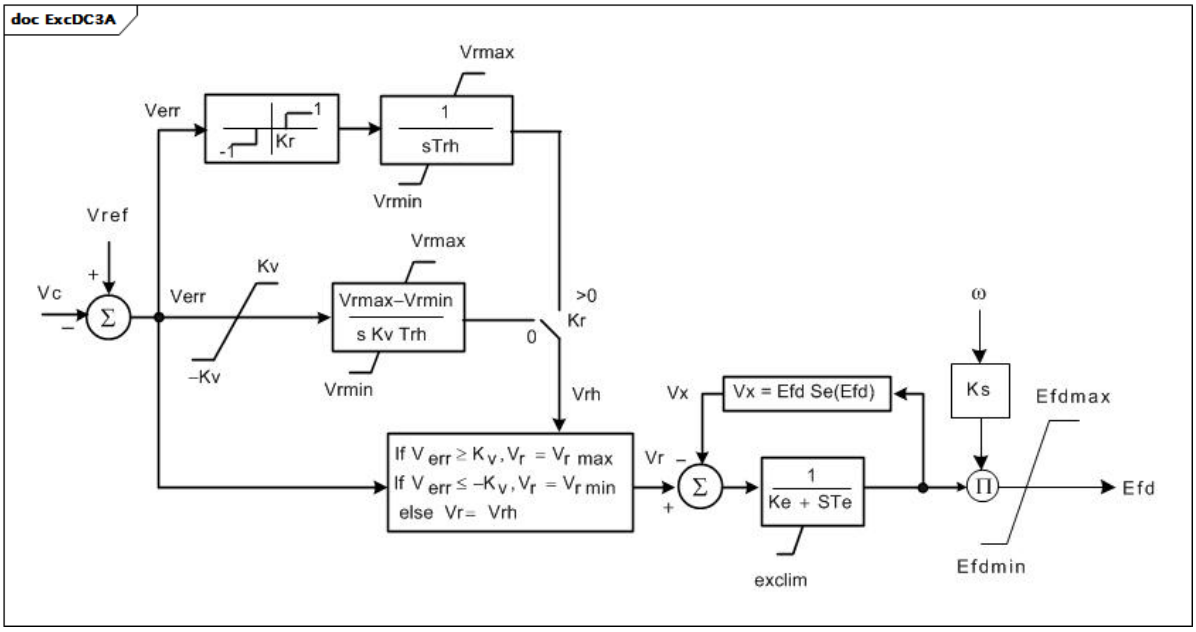


Figure 9-109 - figure ExcDC3A

Parameter Notes:

1. If Ks is set to 1 the output (Efd) is multiplied by generator speed.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-186 shows all attributes of ExcDC3A.

Table 9-186 - Attributes of ExcitationSystemDynamics::ExcDC3A

Attribute name	mult.	Attribute type	Description
trh	1..1	Seconds	Rheostat travel time (Trh). Typical Value = 20.
ks	1..1	PU	Coefficient to allow different usage of the model-speed coefficient (Ks). Typical Value = 0.
kr	1..1	PU	Death band (Kr). If Kr is not zero, the voltage

			regulator input changes at a constant rate if Verr > Kr or Verr < -Kr as per the IEEE (1968) Type 4 model. If Kr is zero, the error signal drives the voltage regulator continuously as per the IEEE (1980) DC3 and IEEE (1992, 2005) DC3A models. Typical Value = 0.
kv	1..1	PU	Fast raise/lower contact setting (Kv). Typical Value = 0.05.
vrmax	1..1	PU	Maximum voltage regulator output (Vrmax). Typical Value = 5.
vrmin	1..1	PU	Minimum voltage regulator output (Vrmin). Typical Value = 0.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (Te). Typical Value = 1.83.
ke	1..1	PU	Exciter constant related to self-excited field (Ke). Typical Value = 1.
efd1	1..1	PU	Exciter voltage at which exciter saturation is defined (Efd1). Typical Value = 2.6.
seefd1	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, Efd1 (Se[Eefd1]). Typical Value = 0.1.
efd2	1..1	PU	Exciter voltage at which exciter saturation is defined (Efd2). Typical Value = 3.45.
seefd2	1..1	Simple_Float	Exciter saturation function value at the corresponding exciter voltage, Efd2 (Se[Efd2]). Typical Value = 0.35.
exclim	1..1	Boolean	(exclim). IEEE standard is ambiguous about lower limit on exciter output. true = a lower limit of zero is applied to integrator output false = a lower limit of zero not applied to integrator output. Typical Value = true.

edfmax	1..1	PU	Maximum voltage exciter output limiter (Efdmax). Typical Value = 99.
efdmin	1..1	PU	Minimum voltage exciter output limiter (Efdmin). Typical Value = -99.
efdlim	1..1	Boolean	(Efdlim). true = exciter output limiter is active false = exciter output limiter not active. Typical Value = true.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-187 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-187 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.41 ExcDC3A1

This is modified old IEEE type 3 excitation system.

Figure 9-110 class diagram

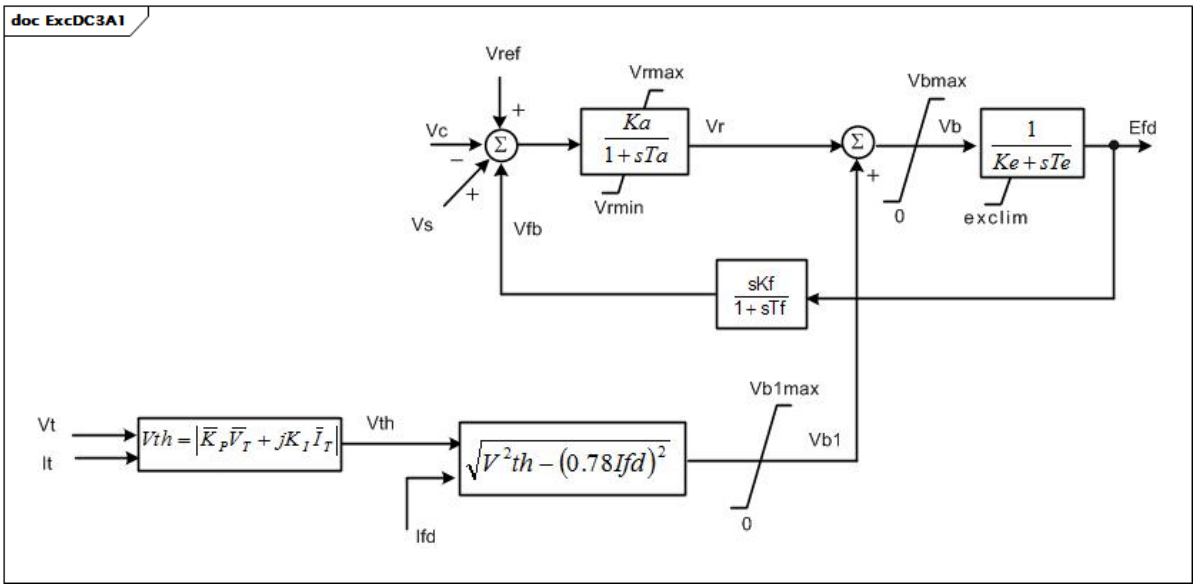


Figure 9-110 - figure ExcDC3A1

Notes:

1. Ifd represents LadIfd (XadIfd) and is a voltage proportional to field current in the per unit system (IEEE 421.5-2005, Annex B). It is equal to Efd in steady-state.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-188 shows all attributes of ExcDC3A1.

Table 9-188 - Attributes of ExcitationSystemDynamics::ExcDC3A1

Attribute name	mult.	Attribute type	Description
ka	1..1	PU	Voltage regulator gain (Ka). Typical Value = 300.
ta	1..1	Seconds	Voltage regulator time constant (Ta). Typical Value = 0.01.
vrmax	1..1	PU	Maximum voltage regulator output (Vrmax). Typical Value = 5.
vrmin	1..1	PU	Minimum voltage regulator output (Vrmin). Typical Value = 0.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (Te). Typical Value = 1.83.
kf	1..1	PU	Excitation control system stabilizer gain

			(Kf). Typical Value = 0.1.
tf	1..1	Seconds	Excitation control system stabilizer time constant (Tf). Typical Value = 0.675.
kp	1..1	PU	Potential circuit gain coefficient (Kp). Typical Value = 4.37.
ki	1..1	PU	Potential circuit gain coefficient (Ki). Typical Value = 4.83.
vbmax	1..1	PU	Available exciter voltage limiter (Vbmax). Typical Value = 11.63.
exclim	1..1	Boolean	(exclim). true = lower limit of zero is applied to integrator output false = lower limit of zero not applied to integrator output. Typical Value = true.
ke	1..1	PU	Exciter constant related to self-excited field (Ke). Typical Value = 1.
vb1max	1..1	PU	Available exciter voltage limiter (Vb1max). Typical Value = 11.63.
vblim	1..1	Boolean	Vb limiter indicator. true = exciter Vbmax limiter is active false = Vb1max is active. Typical Value = true.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-189 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-189 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

9.6.9.42 ExcELIN1

Static PI transformer fed excitation system: ELIN (VATECH) - simplified model. This model represents an all-static excitation system. A PI voltage controller establishes a desired field current set point for a proportional current controller. The integrator of the PI controller has a follow-up input to match its signal to the present field current. A power system stabilizer with power input is included in the model.

Figure 9-111 class diagram

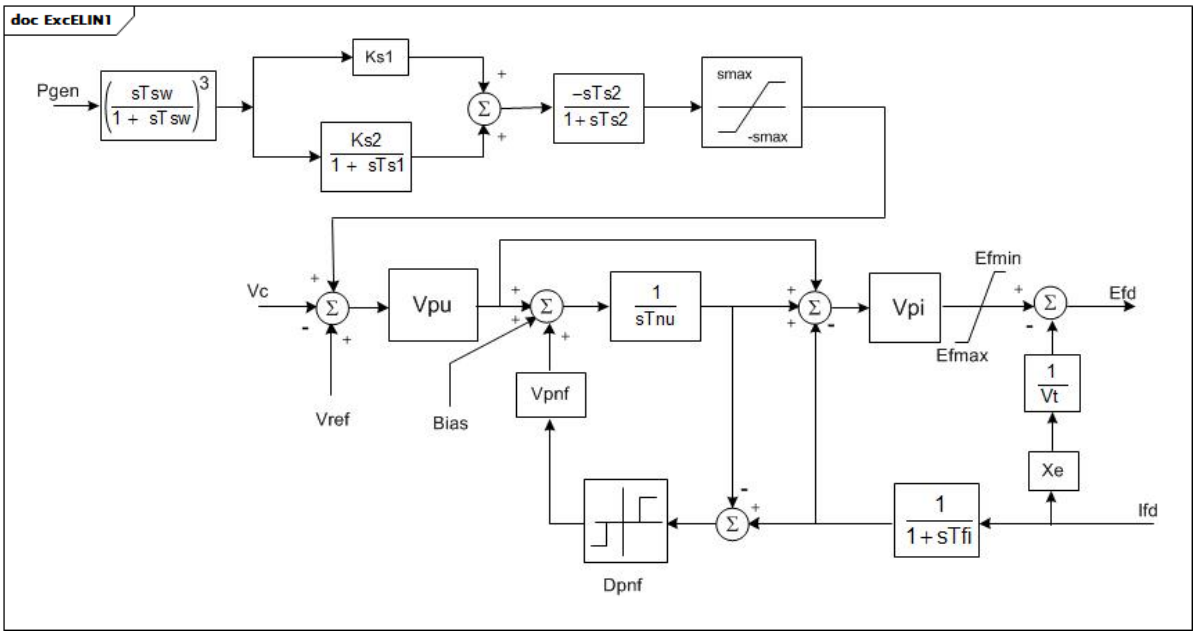


Figure 9-111 - figure ExcELIN1

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-190 shows all attributes of ExcELIN1.

Table 9-190 - Attributes of ExcitationSystemDynamics::ExcELIN1

Attribute name	mult.	Attribute type	Description
tfi	1..1	Seconds	Current transducer time constant (Tfi). Typical Value = 0.
tnu	1..1	Seconds	Controller reset time constant (Tnu). Typical Value = 2.
vpu	1..1	PU	Voltage controller proportional gain (Vpu). Typical Value = 34.5.
vpi	1..1	PU	Current controller gain (Vpi). Typical Value = 12.45.
vpnf	1..1	PU	Controller follow up gain (Vpnf). Typical

			Value = 2.
dpnf	1..1	PU	Controller follow up dead band (Dpnf). Typical Value = 0.
tsw	1..1	Seconds	Stabilizer parameters (Tsw). Typical Value = 3.
efmin	1..1	PU	Minimum open circuit excitation voltage (Efmin). Typical Value = -5.
efmax	1..1	PU	Maximum open circuit excitation voltage (Efmax). Typical Value = 5.
xe	1..1	PU	Excitation transformer effective reactance (Xe) (>=0). Xe represents the regulation of the transformer/rectifier unit. Typical Value = 0.06.
ks1	1..1	PU	Stabilizer Gain 1 (Ks1). Typical Value = 0.
ks2	1..1	PU	Stabilizer Gain 2 (Ks2). Typical Value = 0.
ts1	1..1	Seconds	Stabilizer Phase Lag Time Constant (Ts1). Typical Value = 1.
ts2	1..1	Seconds	Stabilizer Filter Time Constant (Ts2). Typical Value = 1.
smax	1..1	PU	Stabilizer Limit Output (smax). Typical Value = 0.1.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-191 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-191 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDyn

			amics[1]

9.6.9.43 ExcELIN2

Detailed Excitation System Model - ELIN (VATECH). This model represents an all-static excitation system. A PI voltage controller establishes a desired field current set point for a proportional current controller. The integrator of the PI controller has a follow-up input to match its signal to the present field current. Power system stabilizer models used in conjunction with this excitation system model: PssELIN2, PssIEEE2B, Pss2B.

Figure 9-112 class diagram

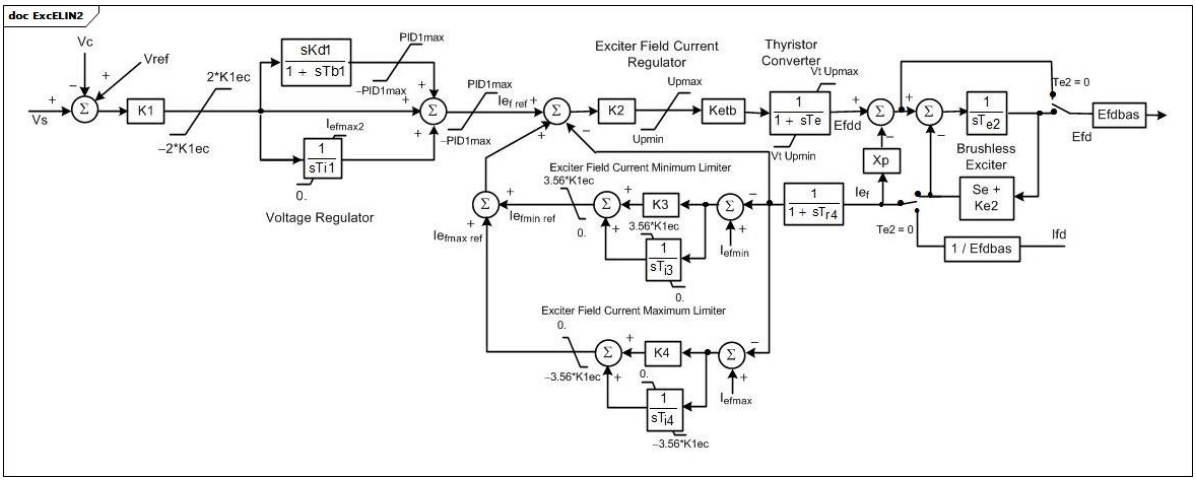


Figure 9-112 - figure ExcELIN2

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-192 shows all attributes of ExcELIN2.

Table 9-192 - Attributes of ExcitationSystemDynamics::ExcELIN2

Attribute name	mult.	Attribute type	Description
k1	1..1	PU	Voltage regulator input gain (K1). Typical Value = 0.
k1ec	1..1	PU	Voltage regulator input limit (K1ec). Typical Value = 2.
kd1	1..1	PU	Voltage controller derivative gain (Kd1). Typical Value = 34.5.
tb1	1..1	Seconds	Voltage controller derivative washout time constant (Tb1). Typical Value = 12.45.
pid1max	1..1	PU	Controller follow up gain (PID1max). Typical Value = 2.
ti1	1..1	PU	Controller follow up dead band (Ti1). Typical Value = 0.

iefmax2	1..1	PU	Minimum open circuit excitation voltage (Iefmax2). Typical Value = -5.
k2	1..1	PU	Gain (K2). Typical Value = 5.
ketb	1..1	PU	Gain (Ketb). Typical Value = 0.06.
upmax	1..1	PU	Limiter (Upmax). Typical Value = 3.
upmin	1..1	PU	Limiter (Upmin). Typical Value = 0.
te	1..1	Seconds	Time constant (Te). Typical Value = 0.
xp	1..1	PU	Excitation transformer effective reactance (Xp). Typical Value = 1.
te2	1..1	Seconds	Time Constant (Te2). Typical Value = 1.
ke2	1..1	PU	Gain (Ke2). Typical Value = 0.1.
ve1	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (Ve1). Typical Value = 3.
seve1	1..1	PU	Exciter saturation function value at the corresponding exciter voltage, Ve1, back of commutating reactance (Se[Ve1]). Typical Value = 0.
ve2	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (Ve2). Typical Value = 0.
seve2	1..1	PU	Exciter saturation function value at the corresponding exciter voltage, Ve2, back of commutating reactance (Se[Ve2]). Typical Value = 1.
tr4	1..1	Seconds	Time constant (Tr4). Typical Value = 1.
k3	1..1	PU	Gain (K3). Typical Value = 0.1.
ti3	1..1	Seconds	Time constant (Ti3). Typical Value = 3.
k4	1..1	PU	Gain (K4). Typical Value = 0.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

ti4	1..1	Seconds	Time constant (Ti4). Typical Value = 0.
iefmax	1..1	PU	Limiter (Iefmax). Typical Value = 1.
iefmin	1..1	PU	Limiter (Iefmin). Typical Value = 1.
efdbas	1..1	PU	Gain (Efdbas). Typical Value = 0.1.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-193 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-193 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.44 ExcHU

Hungarian Excitation System Model, with built-in voltage transducer.

Figure 9-113 class diagram

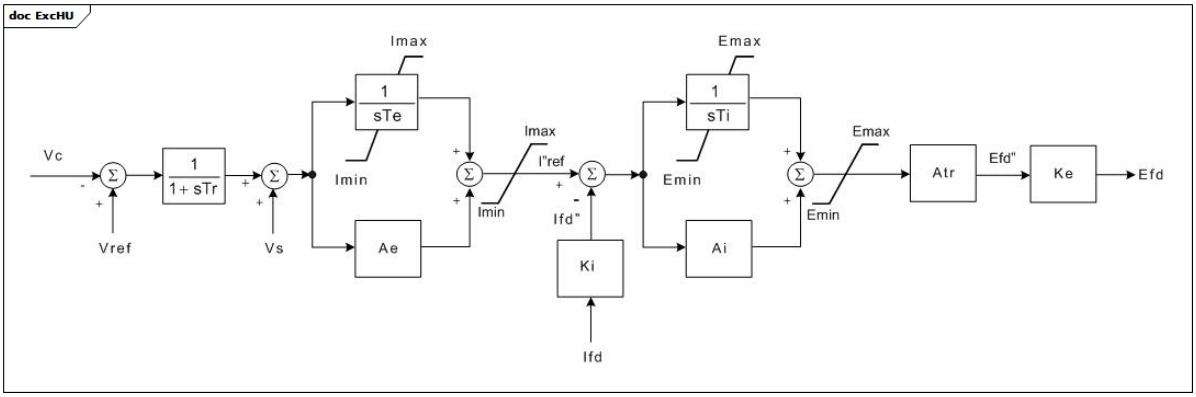


Figure 9-113 - figure ExcHU

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-194 shows all attributes of ExcHU.

Table 9-194 - Attributes of ExcitationSystemDynamics::ExcHU

Attribute name	mult.	Attribute type	Description
tr	1..1	Seconds	Filter time constant (Tr). If a voltage compensator is used in conjunction with this excitation system model, Tr should be set to 0. Typical Value = 0.01.
te	1..1	Seconds	Major loop PI tag integration time constant (Te). Typical Value = 0.154.
imin	1..1	PU	Major loop PI tag output signal lower limit (Imin). Typical Value = 0.1.
imax	1..1	PU	Major loop PI tag output signal upper limit (Imax). Typical Value = 2.19.
ae	1..1	PU	Major loop PI tag gain factor (Ae). Typical Value = 3.
emin	1..1	PU	Field voltage control signal lower limit on AVR base (Emin). Typical Value = -0.866.
emax	1..1	PU	Field voltage control signal upper limit on AVR base (Emax). Typical Value = 0.996.
ki	1..1	Simple_Float	Current base conversion constant (Ki). Typical Value = 0.21428.
ai	1..1	PU	Minor loop PI tag gain factor (Ai). Typical Value = 22.
ti	1..1	Seconds	Minor loop PI control tag integration time constant (Ti). Typical Value = 0.01333.
atr	1..1	PU	AVR constant (Atr). Typical Value = 2.19.
ke	1..1	Simple_Float	Voltage base conversion constant (Ke). Typical Value = 4.666.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 9-195 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-195 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.45 ExcOEX3T

Modified IEEE Type ST1 Excitation System with semi-continuous and acting terminal voltage limiter.

Figure 9-114 class diagram

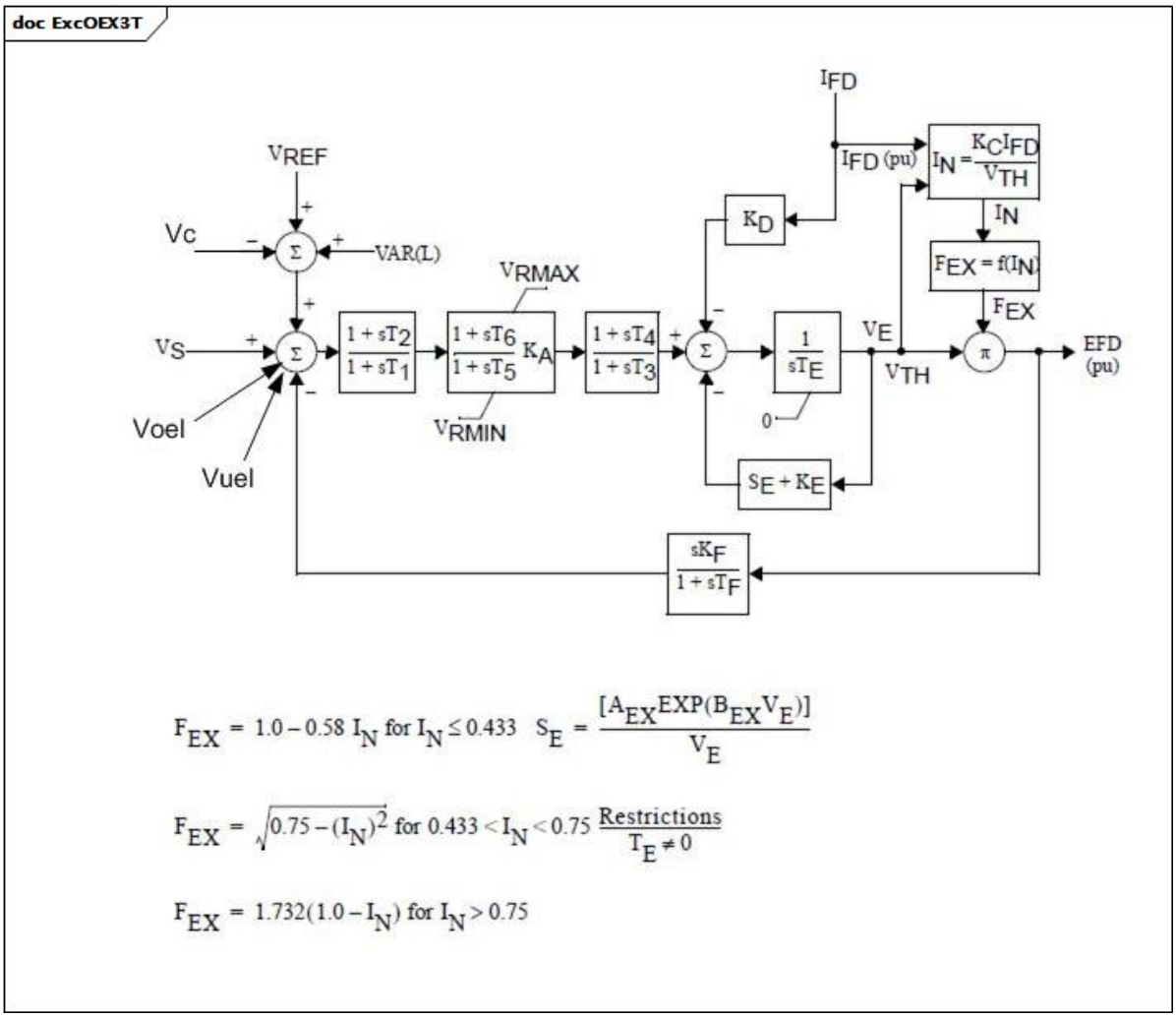


Figure 9-114 - figure ExcOEX3T

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-196 shows all attributes of ExcOEX3T.

Table 9-196 - Attributes of ExcitationSystemDynamics::ExcOEX3T

Attribute name	mult.	Attribute type	Description
t1	1..1	Seconds	Time constant (T_1).
t2	1..1	Seconds	Time constant (T_2).
t3	1..1	Seconds	Time constant (T_3).
t4	1..1	Seconds	Time constant (T_4).
ka	1..1	PU	Gain (K_A).
t5	1..1	Seconds	Time constant (T_5).
t6	1..1	Seconds	Time constant (T_6).
vrmax	1..1	PU	Limiter (V_{RMAX}).
vrmin	1..1	PU	Limiter (V_{RMIN}).
te	1..1	Seconds	Time constant (T_E).
kf	1..1	PU	Gain (K_F).
tf	1..1	Seconds	Time constant (T_F).
kc	1..1	PU	Gain (K_C).
kd	1..1	PU	Gain (K_D).
ke	1..1	PU	Gain (K_E).
e1	1..1	PU	Saturation parameter (E_1).
see1	1..1	PU	Saturation parameter ($S_E(E_1)$).
e2	1..1	PU	Saturation parameter (E_2).
see2	1..1	PU	Saturation parameter ($S_E(E_2)$).
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-197 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-197 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDyna	1..1	SynchronousMachineDyna	

mics		mics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.46 ExcPIC

Proportional/Integral Regulator Excitation System Model. This model can be used to represent excitation systems with a proportional-integral (PI) voltage regulator controller.

Figure 9-115 class diagram

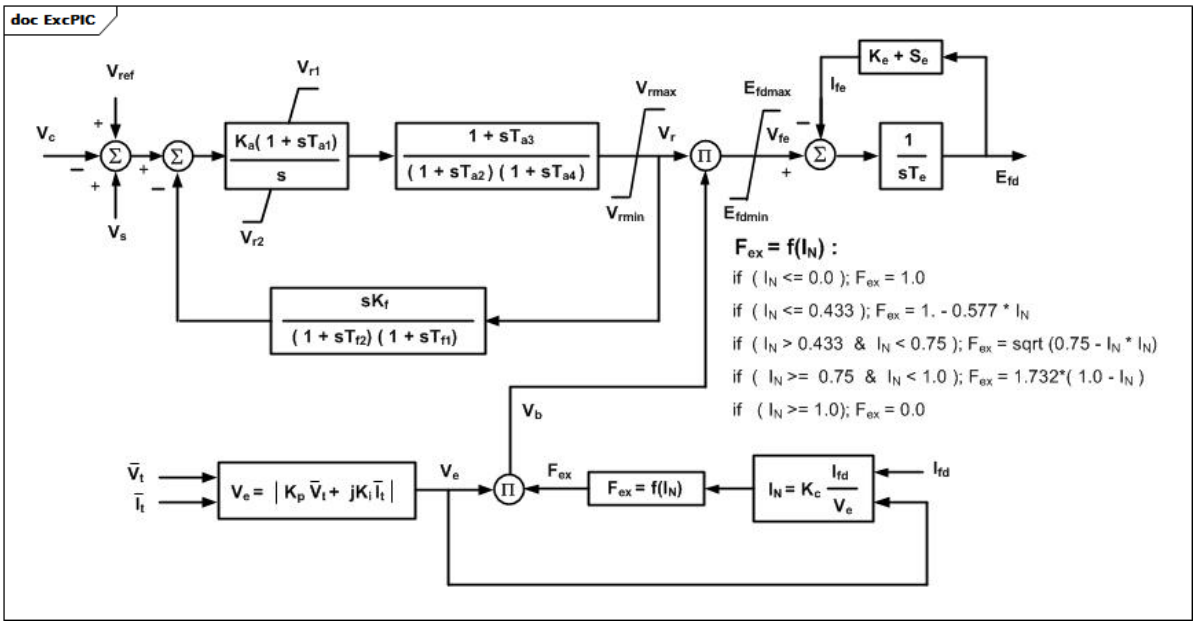


Figure 9-115 - figure ExcPIC

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-198 shows all attributes of ExcPIC.

Table 9-198 - Attributes of ExcitationSystemDynamics::ExcPIC

Attribute name	mult.	Attribute type	Description
ka	1..1	PU	PI controller gain (Ka). Typical Value = 3.15.
ta1	1..1	Seconds	PI controller time constant (Ta1). Typical Value = 1.
vr1	1..1	PU	PI maximum limit (Vr1). Typical Value = 1.
vr2	1..1	PU	PI minimum limit (Vr2). Typical Value = -0.87.
ta2	1..1	Seconds	Voltage regulator time constant (Ta2). Typical Value = 0.01.

ta3	1..1	Seconds	Lead time constant (Ta3). Typical Value = 0.
ta4	1..1	Seconds	Lag time constant (Ta4). Typical Value = 0.
vrmax	1..1	PU	Voltage regulator maximum limit (Vrmax). Typical Value = 1.
vrmin	1..1	PU	Voltage regulator minimum limit (Vrmin). Typical Value = -0.87.
kf	1..1	PU	Rate feedback gain (Kf). Typical Value = 0.
tf1	1..1	Seconds	Rate feedback time constant (Tf1). Typical Value = 0.
tf2	1..1	Seconds	Rate feedback lag time constant (Tf2). Typical Value = 0.
efdmax	1..1	PU	Exciter maximum limit (Efdmax). Typical Value = 8.
efdmin	1..1	PU	Exciter minimum limit (Efdmin). Typical Value = -0.87.
ke	1..1	PU	Exciter constant (Ke). Typical Value = 0.
te	1..1	Seconds	Exciter time constant (Te). Typical Value = 0.
e1	1..1	PU	Field voltage value 1 (E1). Typical Value = 0.
se1	1..1	PU	Saturation factor at E1 (Se1). Typical Value = 0.
e2	1..1	PU	Field voltage value 2 (E2). Typical Value = 0.
se2	1..1	PU	Saturation factor at E2 (Se2). Typical Value = 0.
kp	1..1	PU	Potential source gain (Kp). Typical Value = 6.5.
ki	1..1	PU	Current source gain (Ki). Typical Value = 0.
kc	1..1	PU	Exciter regulation factor (Kc). Typical Value = 0.08.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

name	0..1	String	see IdentifiedObject

Table 9-199 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-199 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.47 ExcREXS

General Purpose Rotating Excitation System Model. This model can be used to represent a wide range of excitation systems whose DC power source is an AC or DC generator. It encompasses IEEE type AC1, AC2, DC1, and DC2 excitation system models.

Figure 9-116 class diagram

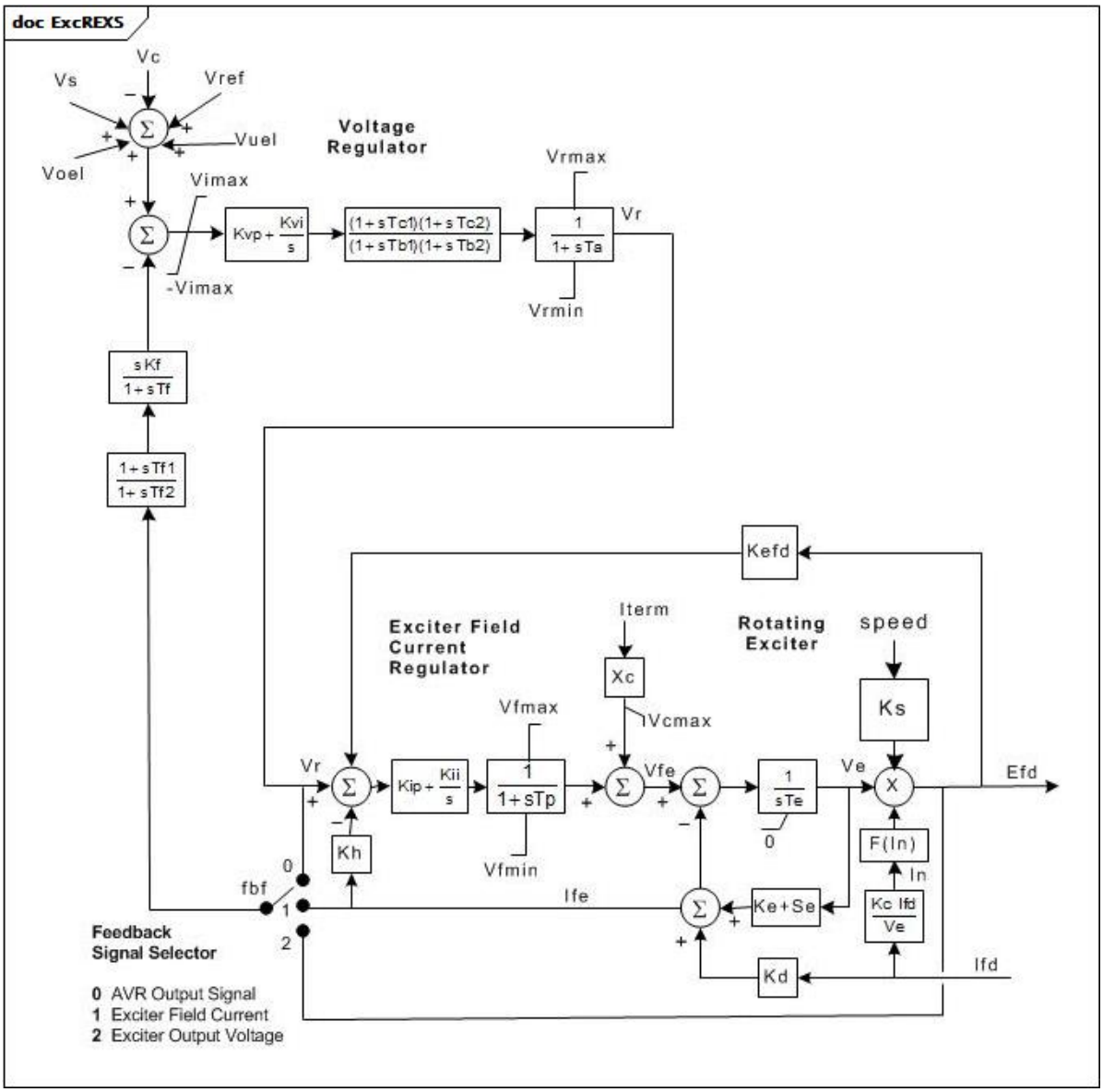


Figure 9-116 - figure ExcREXS

Parameter Notes:

1. The rotating exciter may be of either DC or Brushless AC type. To model a DC exciter, set Kc = Kd = 0
2. An AC exciter can be modeled either by setting Kc and Kd to non- zero values to represent the armature reaction of the exciter and the regulation of the rotating rectifier, respectively, or by specifying the saturation factors, Se1, Se2 to represent the excitation-output-voltage curve of the exciter as loaded by the field resistance of the main generator. It is preferable to use the first approach (Kd = 0) since the later approach does not represent the effect of armature reaction in the exciter on its field current and hence invalidates the model when field current is sensed by the voltage regulator.
3. When this model is used to represent a separately-excited exciter, the value of Ke should normally be close to unity. A shunt-excited DC exciter can be modeled by setting Ke to the

appropriate small negative value needed represent the relationship between the exciter's air gap line and field resistance line. Note though, that this model does not determine the value of Ke automatically when a shunt field DC exciter is being represented

4. The voltage regulator may be of proportional or proportional-plus-integral form. Either Kvp or Kvi, but not both, may be zero.
5. Te must be non-zero. If Tr, Ta, Tb1, Tb2, or Tf2 is zero, the corresponding blocks are bypassed. If Tf is zero, the feedback path is not used.
6. The voltage regulator output may be either the field voltage applied directly to the exciter or the input to an exciter field current regulator by choice of values of Kip, Kii, and Kh. Either or both of Kip and Kii must be non-zero. If Kii is non-zero, Kh should normally be non-zero.
7. The voltage regulator reference is modified by a volts-per-hertz limiter when shaft speed falls below Nvphz. When the shaft speed is below Nvphz the regulator reference is reduced as follows: effective vref = vref (1. – kvphz (speed - nvphz)) The volts-per-hertz limiter can be disabled by setting Nvphz = 0 or Kvphz = 0.
8. Saturation parameters are given by the IEEE saturation factor definition using the open circuit magnetization of the exciter. Either point [E1, S(E1) or E2, S(E2)] may be the higher value and the other the lower.
9. If Ks is set to 1 the output (Efd) is multiplyed by generator speed.
10. The voltage regulator output limits, Vrmax, Vrmin must be stated as multiples of the value of exciter field current needed to maintain the exciter output at its base value. The field current limit, Vlr, must be stated as a multiple of the exciter field current needed to maintain the exciter output at its base value. That is, Vrmax, Vrmin must be stated in terms of the per unit exciter output voltage to which they correspond in the steady state.
11. The regulator output limit flag indicates whether the control power supply for the voltage regulator is a transformer at the generator terminals or an independent source such as a permanent magnet generator. Set Flimf as follows:

- Flimf = 0 the limits on regulator output are Vrmax, Vrmin, Vfmax and Vfmin
- Flimf = 1 the limits on regulator output are (Vrmax*Vterm), (Vrmin*Vterm), (Vfmax*Vterm), and (Vfmin*Vterm).

12. Feedback signal selector uses ExcREXSFeedbackSignalKind

- 0 (AVR Output Signal) = ExcREXSFeedbackSignalKind.fieldVoltage

- 1 (Exciter Field Current) = ExcREXSFeedbackSignalKind.fieldCurent
- 2 (Exciter Output Voltage) = ExcREXSFeedbackSignalKind.outputVoltage.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-200 shows all attributes of ExcREXS.

Table 9-200 - Attributes of ExcitationSystemDynamics::ExcREXS

Attribute name	mult.	Attribute type	Description
e1	1..1	PU	Field voltage value 1 (E1). Typical Value = 3.
e2	1..1	PU	Field voltage value 2 (E2). Typical Value = 4.
fbf	1..1	ExcREXSFeedbackSignalKind	Rate feedback signal flag (Fbf). Typical Value = fieldCurrent.
flimf	1..1	PU	Limit type flag (Flimf).

			Typical Value = 0.
kc	1..1	PU	Rectifier regulation factor (Kc). Typical Value = 0.05.
kd	1..1	PU	Exciter regulation factor (Kd). Typical Value = 2.
ke	1..1	PU	Exciter field proportional constant (Ke). Typical Value = 1.
kefd	1..1	PU	Field voltage feedback gain (Kefd). Typical Value = 0.
kf	1..1	Seconds	Rate feedback gain (Kf). Typical Value = 0.05.
kh	1..1	PU	Field voltage controller feedback gain (Kh). Typical Value = 0.
kii	1..1	PU	Field Current Regulator Integral Gain (Kii). Typical Value = 0.
kip	1..1	PU	Field Current Regulator Proportional Gain (Kip). Typical Value = 1.
ks	1..1	PU	Coefficient to allow different usage of the model-speed coefficient (Ks). Typical Value = 0.
kvi	1..1	PU	Voltage Regulator Integral Gain (Kvi). Typical Value = 0.
kvp	1..1	PU	Voltage Regulator Proportional Gain (Kvp). Typical Value = 2800.
kvphz	1..1	PU	V/Hz limiter gain (Kvphz). Typical Value = 0.
nvphz	1..1	PU	Pickup speed of V/Hz limiter (Nvphz). Typical Value = 0.
se1	1..1	PU	Saturation factor at E1 (Se1). Typical Value = 0.0001.
se2	1..1	PU	Saturation factor at E2 (Se2). Typical Value = 0.001.
ta	1..1	Seconds	Voltage Regulator time constant (Ta). Typical Value = 0.01.
tb1	1..1	Seconds	Lag time constant (Tb1). Typical Value = 0.
tb2	1..1	Seconds	Lag time constant (Tb2). Typical Value =

			0.
tc1	1..1	Seconds	Lead time constant (Tc1). Typical Value = 0.
tc2	1..1	Seconds	Lead time constant (Tc2). Typical Value = 0.
te	1..1	Seconds	Exciter field time constant (Te). Typical Value = 1.2.
tf	1..1	Seconds	Rate feedback time constant (Tf). Typical Value = 1.
tf1	1..1	Seconds	Feedback lead time constant (Tf1). Typical Value = 0.
tf2	1..1	Seconds	Feedback lag time constant (Tf2). Typical Value = 0.
tp	1..1	Seconds	Field current Bridge time constant (Tp). Typical Value = 0.
vcmax	1..1	PU	Maximum compounding voltage (Vcmax). Typical Value = 0.
vfmax	1..1	PU	Maximum Exciter Field Current (Vfmax). Typical Value = 47.
vfmin	1..1	PU	Minimum Exciter Field Current (Vfmin). Typical Value = -20.
vimax	1..1	PU	Voltage Regulator Input Limit (Vimax). Typical Value = 0.1.
vrmax	1..1	PU	Maximum controller output (Vrmax). Typical Value = 47.
vrmin	1..1	PU	Minimum controller output (Vrmin). Typical Value = -20.
xc	1..1	PU	Exciter compounding reactance (Xc). Typical Value = 0.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-201 shows all association ends of ExcitationSystemDynamics with other classes.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 9-201 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.48 ExcSCRX

Simple excitation system model representing generic characteristics of many excitation systems; intended for use where negative field current may be a problem.

Figure 9-117 class diagram

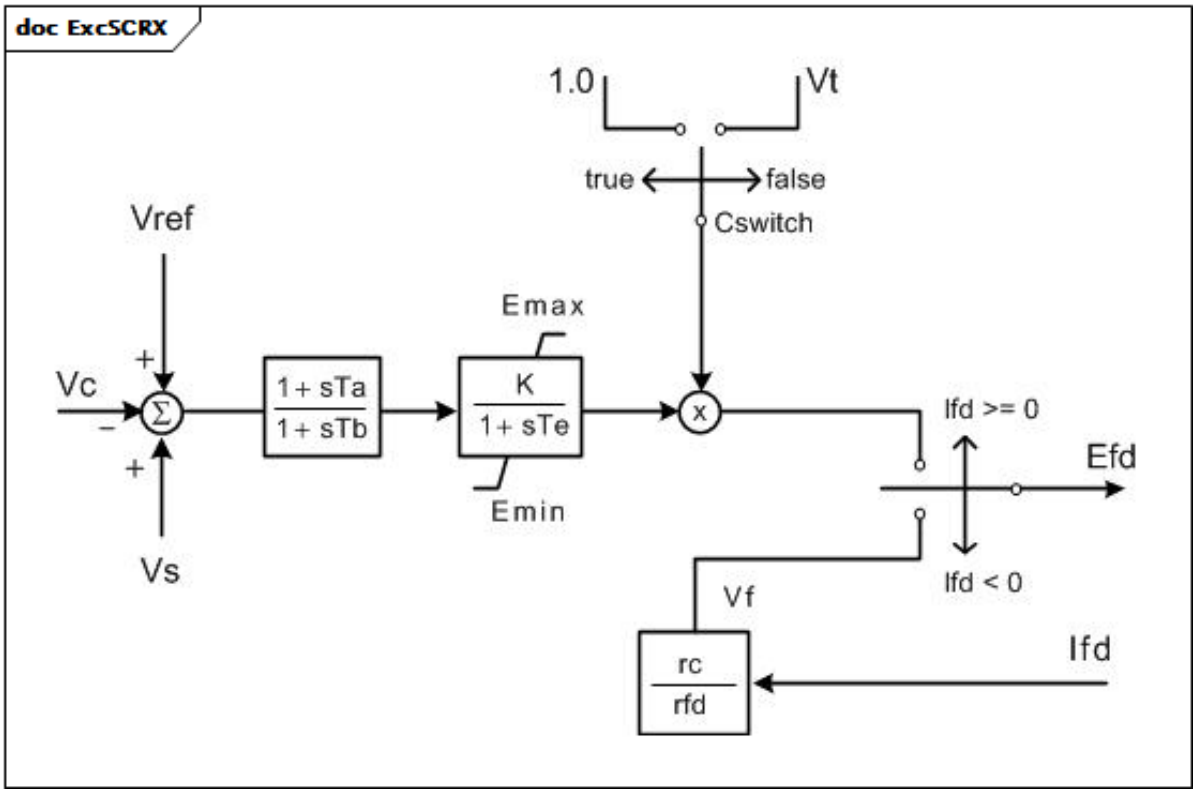


Figure 9-117 - figure ExcSCRX

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-202 shows all attributes of ExcSCRX.

Table 9-202 - Attributes of ExcitationSystemDynamics::ExcSCRX

Attribute name	mult.	Attribute type	Description

tatb	1..1	Simple_Float	Ta/Tb - gain reduction ratio of lag-lead element (TaTb). The parameter Ta is not defined explicitly. Typical Value = 0.1.
tb	1..1	Seconds	Denominator time constant of lag-lead block (Tb). Typical Value = 10.
k	1..1	PU	Gain (K) (>0). Typical Value = 200.
te	1..1	Seconds	Time constant of gain block (Te) (>0). Typical Value = 0.02.
emin	1..1	PU	Minimum field voltage output (Emin). Typical Value = 0.
emax	1..1	PU	Maximum field voltage output (Emax). Typical Value = 5.
cswitch	1..1	Boolean	Power source switch (Cswitch). true = fixed voltage of 1.0 PU false = generator terminal voltage.
rcrfd	1..1	Simple_Float	Rc/Rfd - ratio of field discharge resistance to field winding resistance (RcRfd). Typical Value = 0.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-203 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-203 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.49 ExcSEXS

Simplified Excitation System Model.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Figure 9-118 class diagram

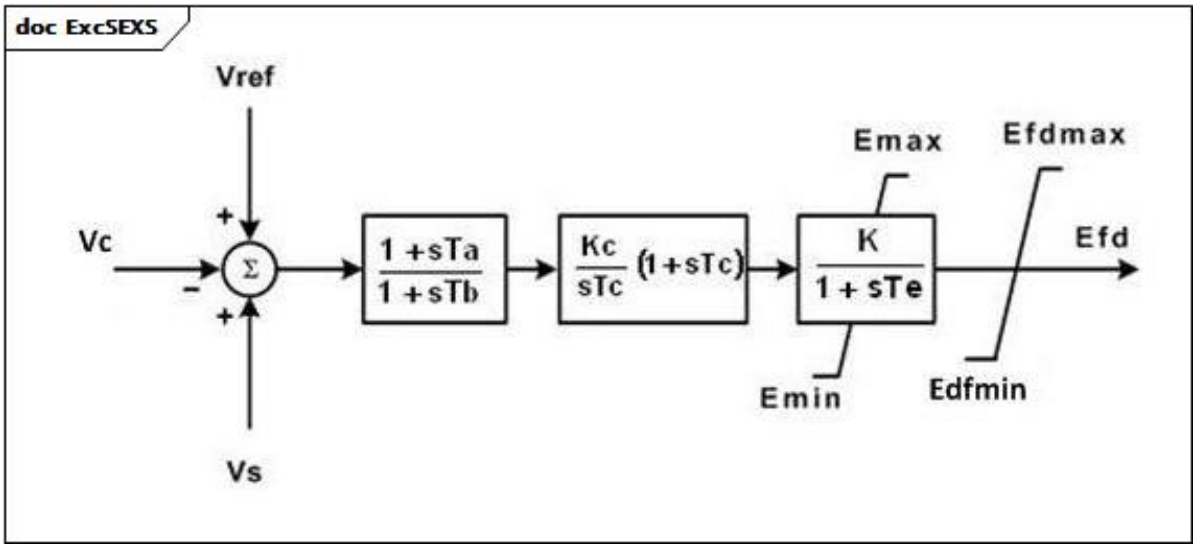


Figure 9-118 - figure ExcSEXS

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-204 shows all attributes of ExcSEXS.

Table 9-204 - Attributes of ExcitationSystemDynamics::ExcSEXS

Attribute name	mult.	Attribute type	Description
tatb	1..1	Simple_Float	Ta/Tb - gain reduction ratio of lag-lead element (TaTb). Typical Value = 0.1.
tb	1..1	Seconds	Denominator time constant of lag-lead block (Tb). Typical Value = 10.
k	1..1	PU	Gain (K) (>0). Typical Value = 100.
te	1..1	Seconds	Time constant of gain block (Te). Typical Value = 0.05.
emin	1..1	PU	Minimum field voltage output (Emin). Typical Value = -5.
emax	1..1	PU	Maximum field voltage output (Emax). Typical Value = 5.
kc	1..1	PU	PI controller gain (Kc). Typical Value = 0.08.
tc	1..1	Seconds	PI controller phase lead time constant (Tc). Typical Value = 0.
efdmin	1..1	PU	Field voltage clipping minimum limit (Efdmin).

			Typical Value = -5.
efdmax	1..1	PU	Field voltage clipping maximum limit (Efdmax). Typical Value = 5.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-205 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-205 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] ----- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.50 ExcSK

Slovakian Excitation System Model. UEL and secondary voltage control are included in this model. When this model is used, there cannot be a separate underexcitation limiter or VAr controller model.

Figure 9-119 class diagram

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

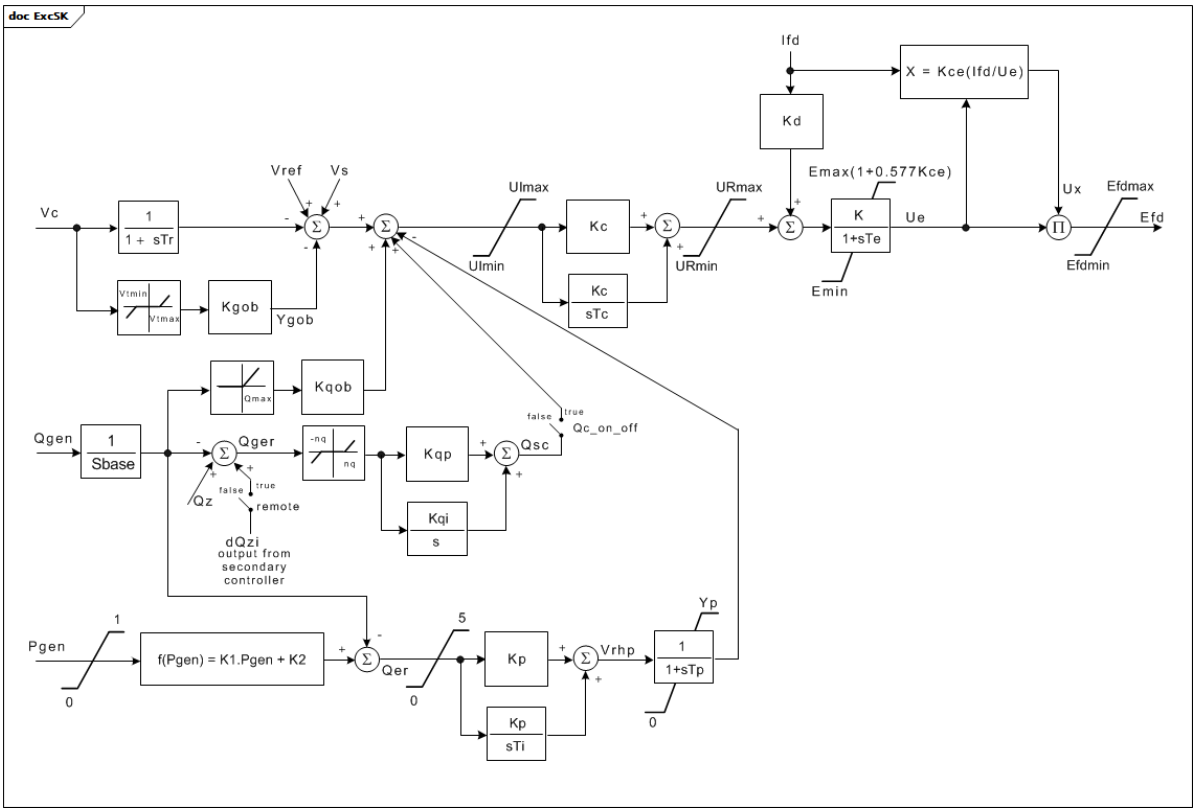


Figure 9-119 - figure ExcSK

Parameter Notes:

1. Qmax is calculated using this equation: Qmax = K1 * Pgen + K2.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-206 shows all attributes of ExcSK.

Table 9-206 - Attributes of ExcitationSystemDynamics::ExcSK

Attribute name	mult.	Attribute type	Description
efdmax	1..1	PU	Field voltage clipping limit (Efdmax).
efdmin	1..1	PU	Field voltage clipping limit (Efdmin).
emax	1..1	PU	Maximum field voltage output (Emax). Typical Value = 20.
emin	1..1	PU	Minimum field voltage output (Emin). Typical Value = -20.
k	1..1	PU	Gain (K). Typical Value = 1.

k1	1..1	PU	Parameter of underexcitation limit (K1). Typical Value = 0.1364.
k2	1..1	PU	Parameter of underexcitation limit (K2). Typical Value = -0.3861.
kc	1..1	PU	PI controller gain (Kc). Typical Value = 70.
kce	1..1	PU	Rectifier regulation factor (Kce). Typical Value = 0.
kd	1..1	PU	Exciter internal reactance (Kd). Typical Value = 0.
kgob	1..1	PU	P controller gain (Kgob). Typical Value = 10.
kp	1..1	PU	PI controller gain (Kp). Typical Value = 1.
kqi	1..1	PU	PI controller gain of integral component (Kqi). Typical Value = 0.
kqob	1..1	PU	Rate of rise of the reactive power (Kqob).
kqp	1..1	PU	PI controller gain (Kqp). Typical Value = 0.
nq	1..1	PU	Dead band of reactive power (nq). Determines the range of sensitivity. Typical Value = 0.001.
qconoff	1..1	Boolean	Secondary voltage control state (Qc_on_off). true = secondary voltage control is ON false = secondary voltage control is OFF. Typical Value = false.
qz	1..1	PU	Desired value (setpoint) of reactive power, manual setting (Qz).
remote	1..1	Boolean	Selector to apply automatic calculation in secondary controller model. true = automatic calculation is activated false = manual set is active; the use of desired value of reactive power (Qz) is required. Typical Value = true.
sbase	1..1	ApparentPower	Apparent power of the unit (Sbase). Unit = MVA. Typical Value =

			259.
tc	1..1	Seconds	PI controller phase lead time constant (Tc). Typical Value = 8.
te	1..1	Seconds	Time constant of gain block (Te). Typical Value = 0.1.
ti	1..1	Seconds	PI controller phase lead time constant (Ti). Typical Value = 2.
tp	1..1	Seconds	Time constant (Tp). Typical Value = 0.1.
tr	1..1	Seconds	Voltage transducer time constant (Tr). Typical Value = 0.01.
uimax	1..1	PU	Maximum error (Uimax). Typical Value = 10.
uimin	1..1	PU	Minimum error (UImin). Typical Value = -10.
urmax	1..1	PU	Maximum controller output (URmax). Typical Value = 10.
urmin	1..1	PU	Minimum controller output (URmin). Typical Value = -10.
vtmax	1..1	PU	Maximum terminal voltage input (Vtmax). Determines the range of voltage dead band. Typical Value = 1.05.
vtmin	1..1	PU	Minimum terminal voltage input (Vtmin). Determines the range of voltage dead band. Typical Value = 0.95.
yp	1..1	PU	Maximum output (Yp). Minimum output = 0. Typical Value = 1.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-207 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-207 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics::

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics::SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.51 ExcST1A

Modification of an old IEEE ST1A static excitation system without overexcitation limiter (OEL) and underexcitation limiter (UEL).

Figure 9-120 class diagram

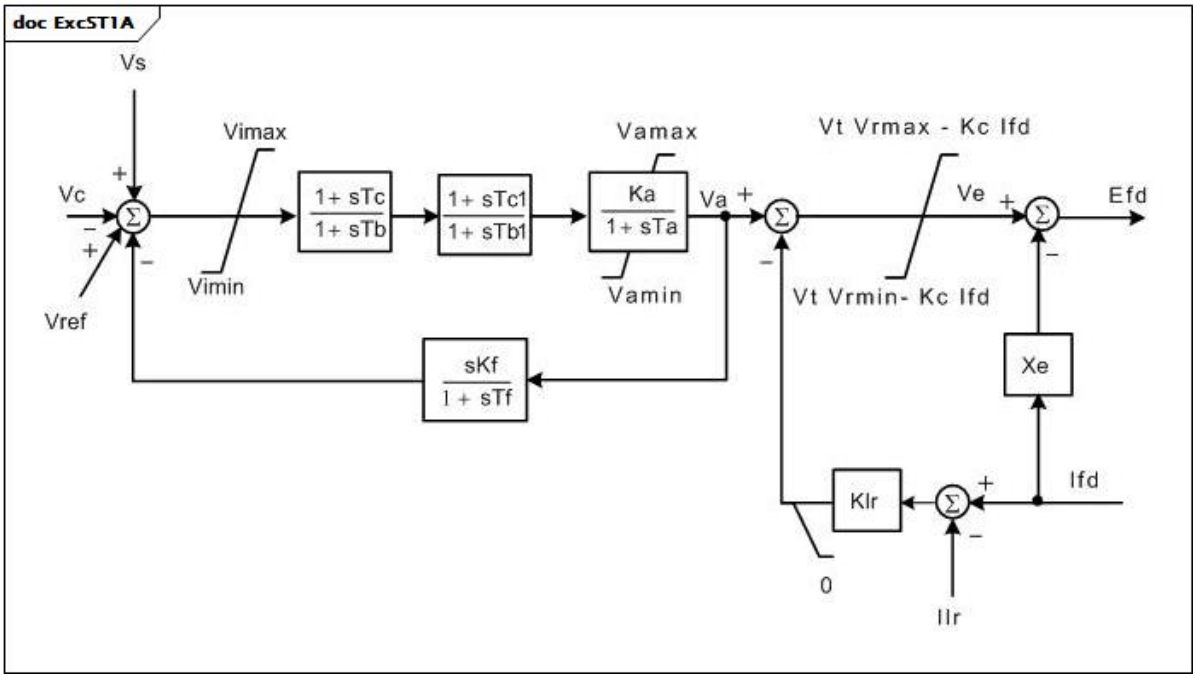


Figure 9-120 - figure ExcST1A

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-208 shows all attributes of ExcST1A.

Table 9-208 - Attributes of ExcitationSystemDynamics::ExcST1A

Attribute name	mult.	Attribute type	Description
vimax	1..1	PU	Maximum voltage regulator input limit (Vimax). Typical Value = 999.
vimin	1..1	PU	Minimum voltage regulator input limit (Vimin). Typical Value = -999.
tc	1..1	Seconds	Voltage regulator time constant (Tc). Typical Value = 1.
tb	1..1	Seconds	Voltage regulator time constant (Tb). Typical

			Value = 10.
ka	1..1	PU	Voltage regulator gain (Ka). Typical Value = 190.
ta	1..1	Seconds	Voltage regulator time constant (Ta). Typical Value = 0.02.
vrmax	1..1	PU	Maximum voltage regulator outputs (Vrmax). Typical Value = 7.8.
vrmin	1..1	PU	Minimum voltage regulator outputs (Vrmin). Typical Value = -6.7.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (Kc). Typical Value = 0.05.
kf	1..1	PU	Excitation control system stabilizer gains (Kf). Typical Value = 0.
tf	1..1	Seconds	Excitation control system stabilizer time constant (Tf). Typical Value = 1.
tc1	1..1	Seconds	Voltage regulator time constant (Tc_1). Typical Value = 0.
tb1	1..1	Seconds	Voltage regulator time constant (Tb_1). Typical Value = 0.
vamax	1..1	PU	Maximum voltage regulator output (Vamax). Typical Value = 999.
vamin	1..1	PU	Minimum voltage regulator output (Vamin). Typical Value = -999.
ilr	1..1	PU	Exciter output current limit reference (Ilr). Typical Value = 0.
klr	1..1	PU	Exciter output current limiter gain (Klr). Typical Value = 0.
xe	1..1	PU	Excitation xfmr effective reactance (Xe). Typical Value = 0.04.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

name	0..1	String	see IdentifiedObject

Table 9-209 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-209 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.52 ExcST2A

Modified IEEE ST2A static excitation system - another lead-lag block added to match the model defined by WECC.

Figure 9-121 class diagram

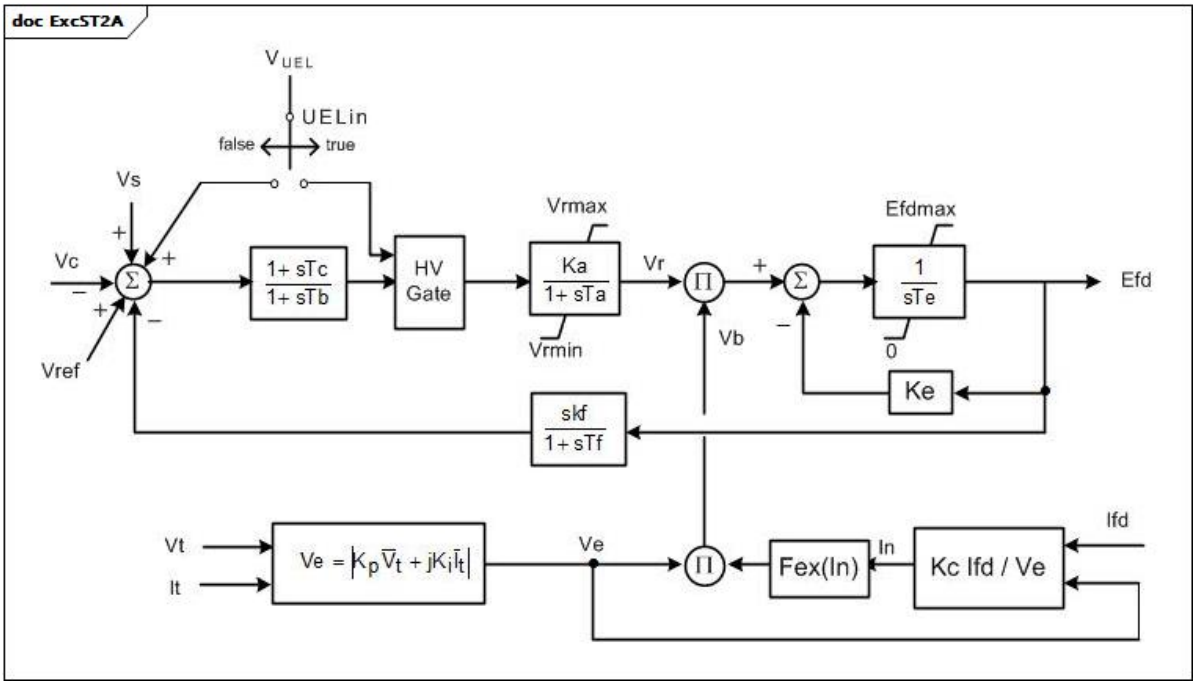


Figure 9-121 - figure ExcST2A

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-210 shows all attributes of ExcST2A.

Table 9-210 - Attributes of ExcitationSystemDynamics::ExcST2A

Attribute name	mult.	Attribute type	Description

ka	1..1	PU	Voltage regulator gain (Ka). Typical Value = 120.
ta	1..1	Seconds	Voltage regulator time constant (Ta). Typical Value = 0.15.
vrmax	1..1	PU	Maximum voltage regulator outputs (Vrmax). Typical Value = 1.
vrmin	1..1	PU	Minimum voltage regulator outputs (Vrmin). Typical Value = -1.
ke	1..1	PU	Exciter constant related to self-excited field (Ke). Typical Value = 1.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (Te). Typical Value = 0.5.
kf	1..1	PU	Excitation control system stabilizer gains (Kf). Typical Value = 0.05.
tf	1..1	Seconds	Excitation control system stabilizer time constant (Tf). Typical Value = 0.7.
kp	1..1	PU	Potential circuit gain coefficient (Kp). Typical Value = 4.88.
ki	1..1	PU	Potential circuit gain coefficient (Ki). Typical Value = 8.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (Kc). Typical Value = 1.82.
efdmax	1..1	PU	Maximum field voltage (Efdmax). Typical Value = 99.
uelin	1..1	Boolean	UEL input (UELin). true = HV gate false = add to error signal. Typical Value = false.
tb	1..1	Seconds	Voltage regulator time constant (Tb). Typical Value = 0.
tc	1..1	Seconds	Voltage regulator time constant (Tc). Typical Value = 0.
enabled	1..1	Boolean	see DynamicsFunctionBlock

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-211 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-211 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.53 ExcST3A

Modified IEEE ST3A static excitation system with added speed multiplier.

Figure 9-122 class diagram

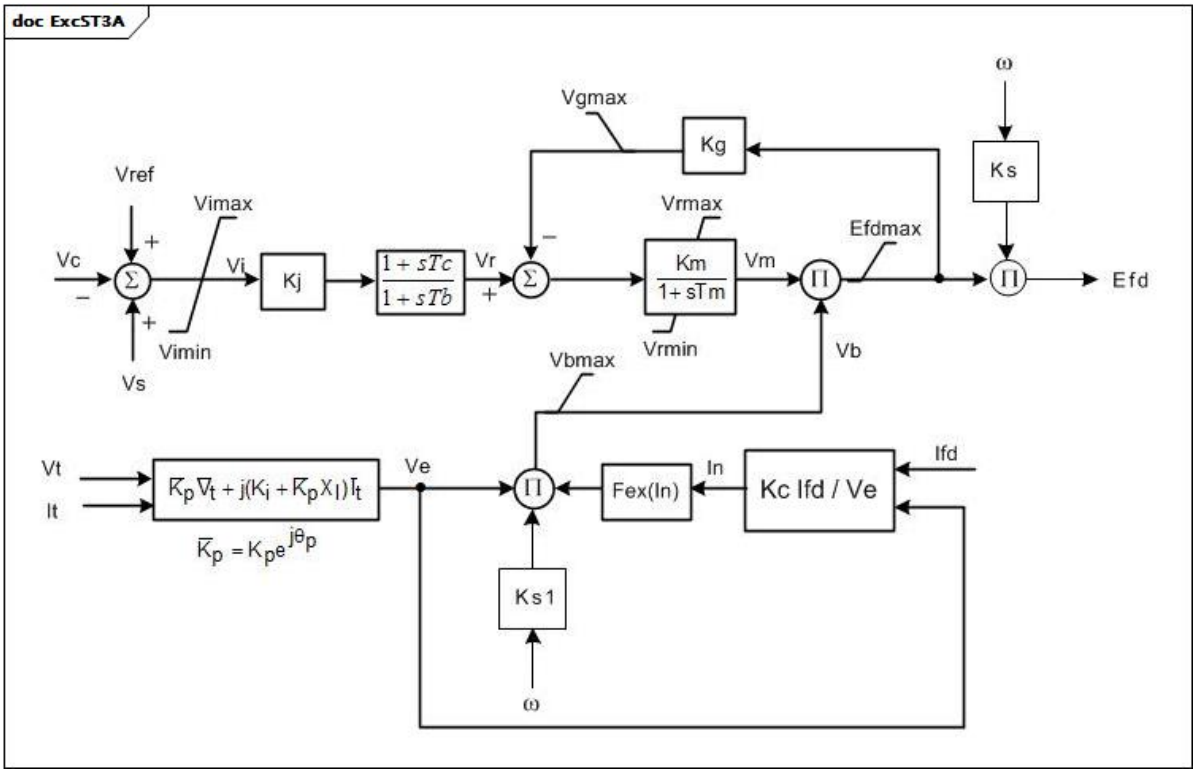


Figure 9-122 - figure ExcST3A

Parameter Notes:

1. If **Ks** or **Ks1** is set to 1 the output (**Efd**) is multiplied by generator speed.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-212 shows all attributes of ExcST3A.

Table 9-212 - Attributes of ExcitationSystemDynamics::ExcST3A

Attribute name	mult.	Attribute type	Description
vimax	1..1	PU	Maximum voltage regulator input limit (Vimax). Typical Value = 0.2.
vimin	1..1	PU	Minimum voltage regulator input limit (Vimin). Typical Value = -0.2.
kj	1..1	PU	AVR gain (Kj). Typical Value = 200.
tb	1..1	Seconds	Voltage regulator time constant (Tb). Typical Value = 6.67.
tc	1..1	Seconds	Voltage regulator time constant (Tc). Typical Value = 1.
efdmax	1..1	PU	Maximum AVR output (Efdmax). Typical Value = 6.9.
km	1..1	PU	Forward gain constant of the inner loop field regulator (Km). Typical Value = 7.04.
tm	1..1	Seconds	Forward time constant of inner loop field regulator (Tm). Typical Value = 1.
vrmax	1..1	PU	Maximum voltage regulator output (Vrmax). Typical Value = 1.
vrmin	1..1	PU	Minimum voltage regulator output (Vrmin). Typical Value = 0.
kg	1..1	PU	Feedback gain constant of the inner loop field regulator (Kg). Typical Value = 1.
kp	1..1	PU	Potential source gain (Kp) (>0). Typical Value = 4.37.
thetap	1..1	AngleDegrees	Potential circuit phase angle (thetap). Typical Value = 20.
ki	1..1	PU	Potential circuit gain coefficient (Ki). Typical

			Value = 4.83.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (Kc). Typical Value = 1.1.
xl	1..1	PU	Reactance associated with potential source (Xl). Typical Value = 0.09.
vbmax	1..1	PU	Maximum excitation voltage (Vbmax). Typical Value = 8.63.
vgmax	1..1	PU	Maximum inner loop feedback voltage (Vgmax). Typical Value = 6.53.
ks	1..1	PU	Coefficient to allow different usage of the model-speed coefficient (Ks). Typical Value = 0.
ks1	1..1	PU	Coefficient to allow different usage of the model-speed coefficient (Ks1). Typical Value = 0.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-213 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-213 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.54 ExcST4B

Modified IEEE ST4B static excitation system with maximum inner loop feedback gain **Vgmax**.

Figure 9-123 class diagram

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

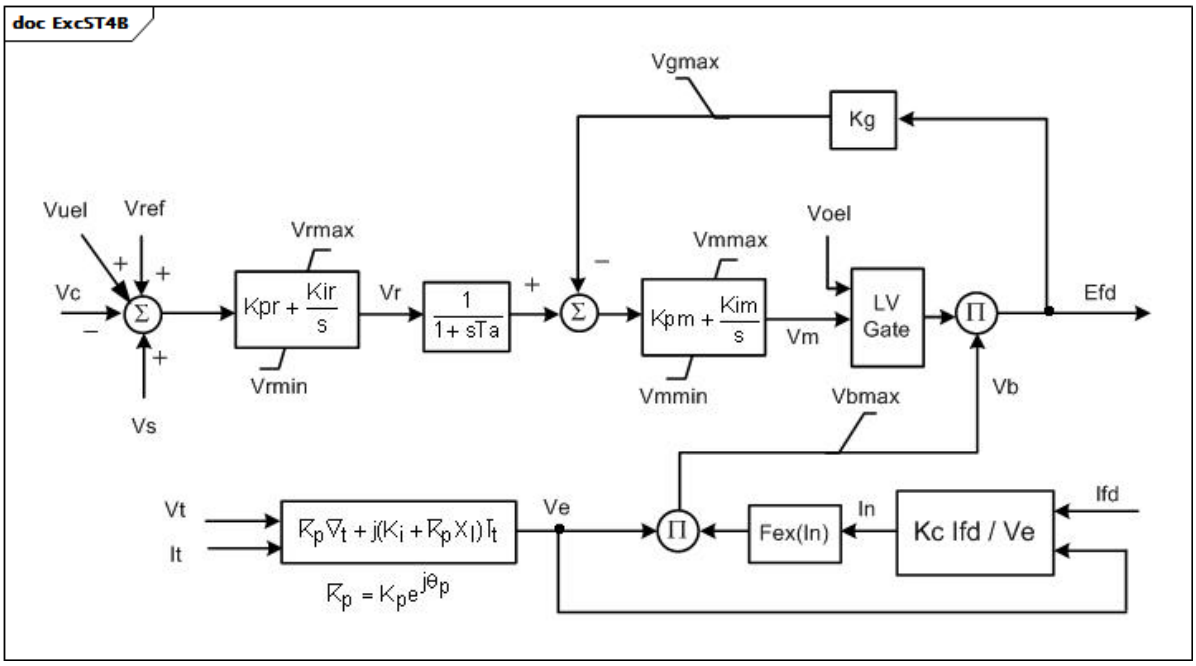


Figure 9-123 - figure ExcST4B

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-214 shows all attributes of ExcST4B.

Table 9-214 - Attributes of ExcitationSystemDynamics::ExcST4B

Attribute name	mult.	Attribute type	Description
kpr	1..1	PU	Voltage regulator proportional gain (Kpr). Typical Value = 10.75.
kir	1..1	PU	Voltage regulator integral gain (Kir). Typical Value = 10.75.
ta	1..1	Seconds	Voltage regulator time constant (Ta). Typical Value = 0.02.
vrmax	1..1	PU	Maximum voltage regulator output (Vrmax). Typical Value = 1.
vrmin	1..1	PU	Minimum voltage regulator output (Vrmin). Typical Value = -0.87.
kpm	1..1	PU	Voltage regulator proportional gain output (Kpm). Typical Value = 1.
kim	1..1	PU	Voltage regulator integral gain output (Kim). Typical Value = 0.

vmmax	1..1	PU	Maximum inner loop output (Vmmax). Typical Value = 99.
vmmin	1..1	PU	Minimum inner loop output (Vmmin). Typical Value = -99.
kg	1..1	PU	Feedback gain constant of the inner loop field regulator (Kg). Typical Value = 0.
kp	1..1	PU	Potential circuit gain coefficient (Kp). Typical Value = 9.3.
thetap	1..1	AngleDegrees	Potential circuit phase angle (thetap). Typical Value = 0.
ki	1..1	PU	Potential circuit gain coefficient (Ki). Typical Value = 0.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (Kc). Typical Value = 0.113.
xl	1..1	PU	Reactance associated with potential source (Xl). Typical Value = 0.124.
vbmax	1..1	PU	Maximum excitation voltage (Vbmax). Typical Value = 11.63.
vgmax	1..1	PU	Maximum inner loop feedback voltage (Vgmax). Typical Value = 5.8.
uel	1..1	Boolean	Selector (Uel). true = UEL is part of block diagram false = UEL is not part of block diagram. Typical Value = false.
lvgate	1..1	Boolean	Selector (LVgate). true = LVgate is part of the block diagram false = LVgate is not part of the block diagram. Typical Value = false.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-215 shows all association ends of ExcitationSystemDynamics with other classes.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 9-215 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.55 ExcST6B

Modified IEEE ST6B static excitation system with PID controller and optional inner feedbacks loop.

Figure 9-124 class diagram

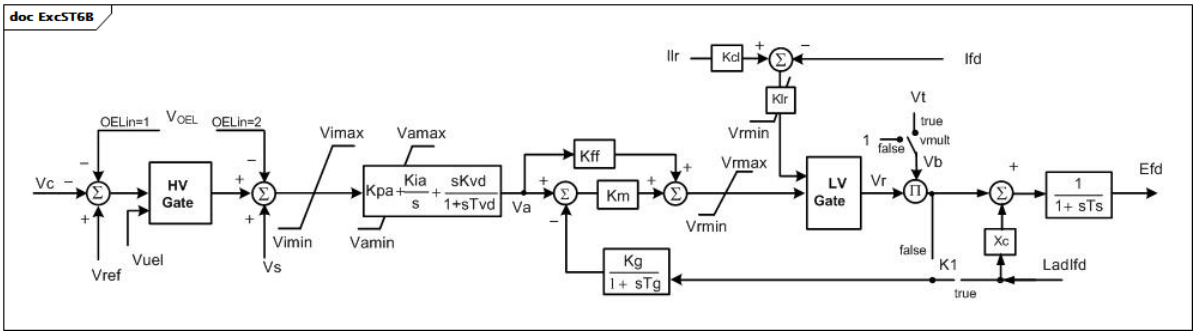


Figure 9-124 - figure ExcST6B

Notes:
1. The connection of the OEL signal is provided by the enumeration ExcST6BOELSelectorKind as follows:

- 0 (no OEL input) = ExcST6BOELSelectorKind.noOELinput
- 1 (before UEL) = ExcST6BOELSelectorKind.beforeUEL
- 2 (after UEL) = ExcST6BOELSelectorKind.afterUEL.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-216 shows all attributes of ExcST6B.

Table 9-216 - Attributes of ExcitationSystemDynamics::ExcST6B

Attribute name	mult.	Attribute type	Description
ilr	1..1	PU	Exciter output current limit reference (Ilr). Typical Value = 4.164.
k1	1..1	Boolean	Selector (K1). true = feedback is from Ifd false = feedback is not from Ifd. Typical Value = true.

kcl	1..1	PU	Exciter output current limit adjustment (Kcl). Typical Value = 1.0577.
kff	1..1	PU	Pre-control gain constant of the inner loop field regulator (Kff). Typical Value = 1.
kg	1..1	PU	Feedback gain constant of the inner loop field regulator (Kg). Typical Value = 1.
kia	1..1	PU	Voltage regulator integral gain (Kia). Typical Value = 45.094.
klr	1..1	PU	Exciter output current limit adjustment (Kcl). Typical Value = 17.33.
km	1..1	PU	Forward gain constant of the inner loop field regulator (Km). Typical Value = 1.
kpa	1..1	PU	Voltage regulator proportional gain (Kpa). Typical Value = 18.038.
kvd	1..1	PU	Voltage regulator derivative gain (Kvd). Typical Value = 0.
oelin	1..1	ExcST6BOELselectorKind	OEL input selector (OELin). Typical Value = noOELinput.
tg	1..1	Seconds	Feedback time constant of inner loop field voltage regulator (Tg). Typical Value = 0.02.
ts	1..1	Seconds	Rectifier firing time constant (Ts). Typical Value = 0.
tvd	1..1	Seconds	Voltage regulator derivative gain (Tvd). Typical Value = 0.
vamax	1..1	PU	Maximum voltage regulator output (Vamax). Typical Value = 4.81.
vamin	1..1	PU	Minimum voltage regulator output (Vamin). Typical Value = -3.85.
vilim	1..1	Boolean	Selector (Vilim). true = Vimin-Vimax limiter is active false = Vimin-Vimax limiter is not active. Typical Value = true.
vimax	1..1	PU	Maximum voltage regulator input limit (Vimax). Typical Value

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			= 10.
vimin	1..1	PU	Minimum voltage regulator input limit (Vimin). Typical Value = -10.
vmult	1..1	Boolean	Selector (Vmult). true = multiply regulator output by terminal voltage false = do not multiply regulator output by terminal voltage. Typical Value = true.
vrmax	1..1	PU	Maximum voltage regulator output (Vrmax). Typical Value = 4.81.
vrmin	1..1	PU	Minimum voltage regulator output (Vrmin). Typical Value = -3.85.
xc	1..1	PU	Excitation source reactance (Xc). Typical Value = 0.05.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-217 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-217 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.56 ExcST7B

Modified IEEE ST7B static excitation system without stator current limiter (SCL) and current compensator (DROOP) inputs.

Figure 9-125 class diagram

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

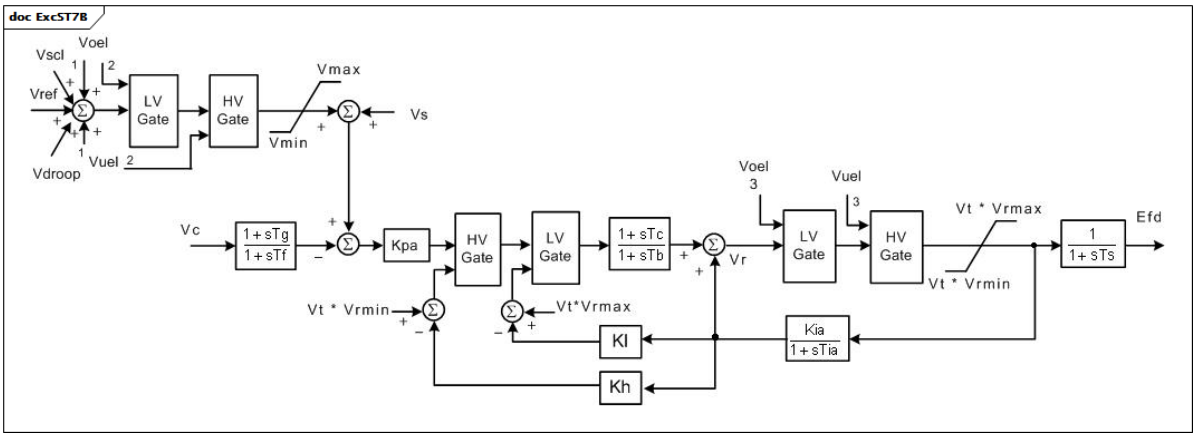


Figure 9-125 - figure ExcST7B

Notes:
1. The connection of the OEL signal is provided by the enumeration ExcST7BOELSelectorKind as follows:

- 0 (no OEL input) = ExcST7BOELSelectorKind.noOELinput
- 1 (add to Vref) = ExcST7BOELSelectorKind.addVref
- 2 (input LV gate) = ExcST7BOELSelectorKind.inputLVgate

- 3 (output LV gate) = ExcST7BOELSelectorKind.outputLVgate.

2. The connection of the UEL signal is provided by the enumeration ExcST7BUELSelectorKind as follows:

- 0 (no OEL input) = ExcST7BUELSelectorKind.noUELinput
- 1 (add to Vref) = ExcST7BUELSelectorKind.addVref
- 2 (input HV gate) = ExcST7BUELSelectorKind.inputHVgate

- 3 (output HV gate) = ExcST7BUELSelectorKind.outputHVgate.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-218 shows all attributes of ExcST7B.

Table 9-218 - Attributes of ExcitationSystemDynamics::ExcST7B

Attribute name	mult.	Attribute type	Description
kh	1..1	PU	High-value gate feedback gain (Kh). Typical Value = 1.
kia	1..1	PU	Voltage regulator integral gain (Kia). Typical Value = 1.
kl	1..1	PU	Low-value gate feedback gain (Kl). Typical Value = 1.
kpa	1..1	PU	Voltage regulator proportional gain (Kpa).

			Typical Value = 40.
oelin	1..1	ExcST7BOELselectorKind	OEL input selector (OELin). Typical Value = noOELinput.
tb	1..1	Seconds	Regulator lag time constant (Tb). Typical Value = 1.
tc	1..1	Seconds	Regulator lead time constant (Tc). Typical Value = 1.
tf	1..1	Seconds	Excitation control system stabilizer time constant (Tf). Typical Value = 1.
tg	1..1	Seconds	Feedback time constant of inner loop field voltage regulator (Tg). Typical Value = 1.
tia	1..1	Seconds	Feedback time constant (Tia). Typical Value = 3.
ts	1..1	Seconds	Rectifier firing time constant (Ts). Typical Value = 0.
uelin	1..1	ExcST7BUELselectorKind	UEL input selector (UELin). Typical Value = noUELinput.
vmax	1..1	PU	Maximum voltage reference signal (Vmax). Typical Value = 1.1.
vmin	1..1	PU	Minimum voltage reference signal (Vmin). Typical Value = 0.9.
vrmax	1..1	PU	Maximum voltage regulator output (Vrmax). Typical Value = 5.
vrmin	1..1	PU	Minimum voltage regulator output (Vrmin). Typical Value = -4.5.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-219 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-219 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] --- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.9.57 Enumerations

9.6.9.57.1 ExcIEEEST1AUELselectorKind

Type of connection for the UEL input used in ExcIEEEST1A.

Literal	Description	Code
ignoreUELsignal	Ignore UEL signal.	
inputHVgateVoltageOutput	UEL input HV gate with voltage regulator output.	
inputHVgateErrorSignal	UEL input HV gate with error signal.	
inputAddedToErrorSignal	UEL input added to error signal.	

9.6.9.57.2 ExcREXSFeedbackSignalKind

Type of rate feedback signals.

Literal	Description	Code
fieldVoltage	The voltage regulator output voltage is used. It is the same as exciter field voltage.	
fieldCurrent	The exciter field current is used.	
outputVoltage	The output voltage of the exciter is used.	

9.6.9.57.3 ExcST6BOELselectorKind

Type of connection for the OEL input used for static excitation systems type 6B.

Literal	Description	Code
noOELinput	No OEL input is used.	
beforeUEL	The connection is before UEL.	
afterUEL	The connection is after UEL.	

9.6.9.57.4 ExcST7BOELselectorKind

Type of connection for the OEL input used for static excitation systems type 7B.

Literal	Description	Code
noOELinput	No OEL input is used.	
addVref	The signal is added to Vref.	
inputLVgate	The signal is connected in the input of the LV gate.	
outputLVgate	The signal is connected in the output of the LV gate.	

9.6.9.57.5 ExcST7BUELselectorKind

Type of connection for the UEL input used for static excitation systems type 7B.

Literal	Description	Code
noUELinput	No UEL input is used.	
addVref	The signal is added to Vref.	
inputHVgate	The signal is connected in the input of the HV gate.	
outputHVgate	The signal is connected in the output of the HV gate.	

9.6.10 OverexcitationLimiterDynamics

9.6.10.1 General

Overexcitation limiters (OELs) are also referred to as *maximum excitation limiters* and *field current limiters.* The possibility of voltage collapse in stressed power systems increases the importance of modelling these limiters in studies of system conditions that cause machines to operate at high levels of excitation for a sustained period, such as voltage collapse or system-islanding. Such events typically occur over a long time frame compared with transient or small-signal stability simulations.

Figure 9-126 shows the diagram.

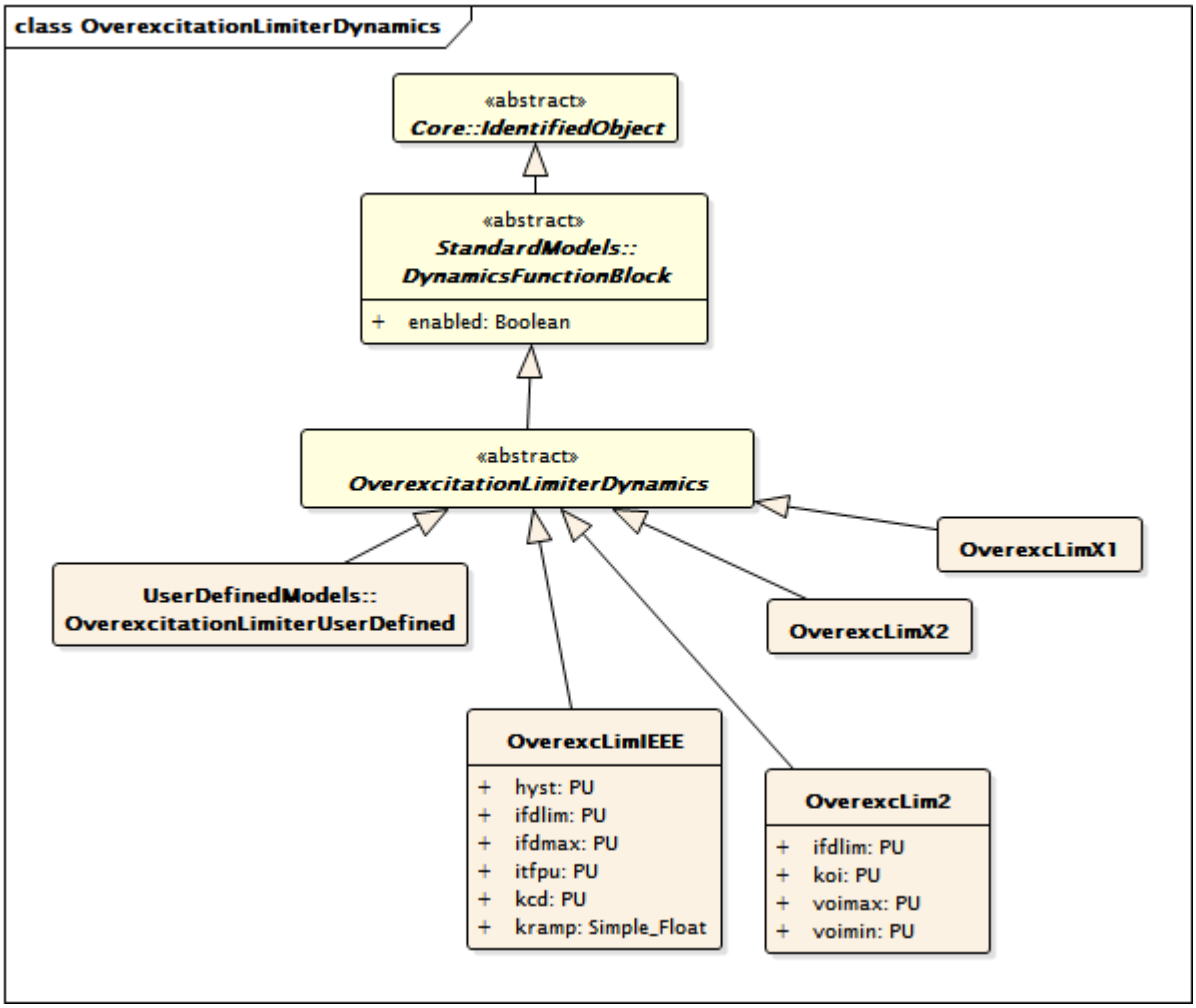


Figure 9-126 - diagram OverexcitationLimiterDynamics

This diagram shows dynamic standard model overexcitation limiter classes and their inheritance from the OverexcitationLimiterDynamics class.

9.6.10.2 OverexcitationLimiterDynamics(Abstract)

OOverexcitation limiter function block whose behaviour is described by reference to a standard model or by definition of a user-defined model.

Inheritance path: ->DynamicsFunctionBlock->IdentifiedObject

Table 9-220 shows all attributes of OverexcitationLimiterDynamics.

Table 9-220 - Attributes of OverexcitationLimiterDynamics::OverexcitationLimiterDynamics

Attribute name	mult.	Attribute type	Description
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-221 shows all association ends of OverexcitationLimiterDynamics with other classes.

Table 9-221 - Association ends of OverexcitationLimiterDynamics::OverexcitationLimiterDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	[1..1]	ExcitationSystemDynamics	Excitation system model with which this overexcitation limiter model is associated. Association Based On : OverexcitationLimiterDynamics:: OverexcitationLimiterDynamics.OverexcitationLimiterDynamics[0..1] ----- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.10.3 OverexcLimIEEE

The over excitation limiter model is intended to represent the significant features of OELs necessary for some large-scale system studies. It is the result of a pragmatic approach to obtain a model that can be widely applied with attainable data from generator owners. An attempt to include all variations in the functionality of OELs and duplicate how they interact with the rest of the excitation systems would likely result in a level of application insufficient for the studies for which they are intended.
Reference: IEEE OEL 421.5-2005 Section 9.

Inheritance path: ->OverexcitationLimiterDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-222 shows all attributes of OverexcLimIEEE.

Table 9-222 - Attributes of OverexcitationLimiterDynamics::OverexcLimIEEE

Attribute name	mult.	Attribute type	Description
itfpu	1..1	PU	OEL timed field current limiter pickup level (I_{TFPU}). Typical Value = 1.05.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

ifdmax	1..1	PU	OEL instantaneous field current limit (I_{FDMAX}). Typical Value = 1.5.
ifdlim	1..1	PU	OEL timed field current limit (I_{FDLIM}). Typical Value = 1.05.
hyst	1..1	PU	OEL pickup/drop-out hysteresis (HYST). Typical Value = 0.03.
kcd	1..1	PU	OEL cooldown gain (K_{CD}). Typical Value = 1.
kramp	1..1	Simple_Float	OEL ramped limit rate (K_{RAMP}). Unit = PU/sec. Typical Value = 10.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-223 shows all association ends of OverexcitationLimiterDynamics with other classes.

Table 9-223 - Association ends of OverexcitationLimiterDynamics::OverexcitationLimiterDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : OverexcitationLimiterDynamics:: OverexcitationLimiterDynamics.OverexcitationLimiterDynamics[0..1] --- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.10.4 OverexcLim2

Different from LimIEEEOEL, LimOEL2 has a fixed pickup threshold and reduces the excitation set-point by mean of non-windup integral regulator. Irated is the rated machine excitation current (calculated from nameplate conditions: V_{nom}, P_{nom}, $CosPhi_{nom}$).

Figure 9-127 class diagram

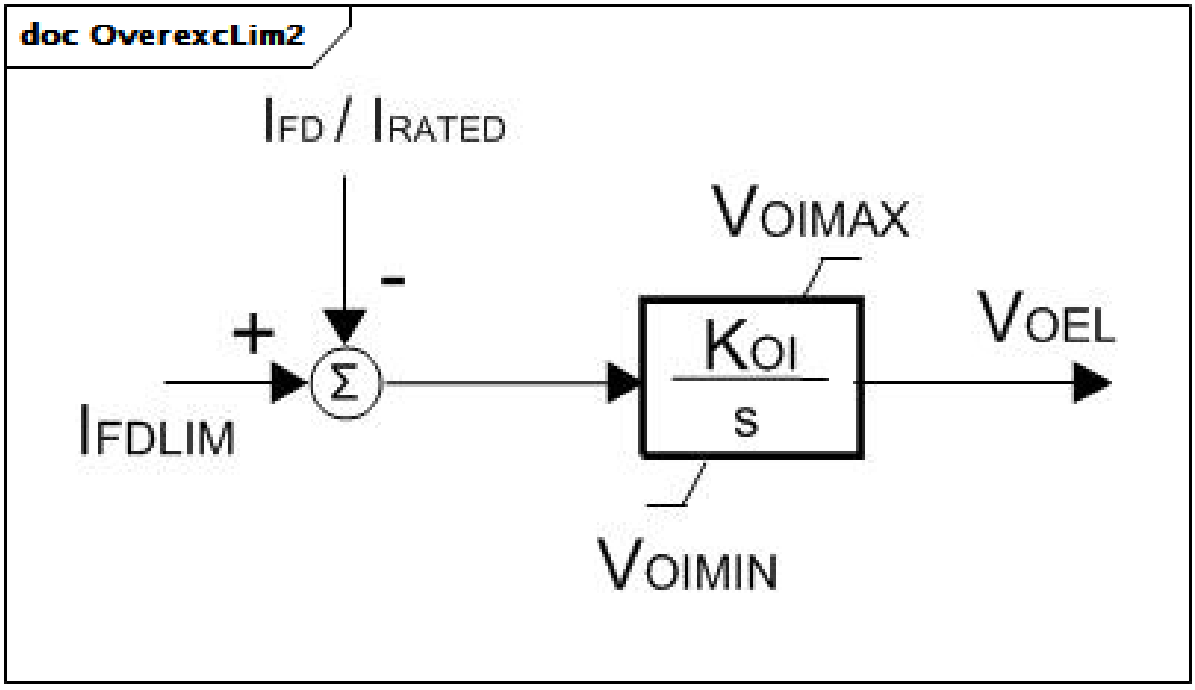


Figure 9-127 - figure OverexcLim2

Inheritance path: ->OverexcitationLimiterDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-224 shows all attributes of OverexcLim2.

Table 9-224 - Attributes of OverexcitationLimiterDynamics::OverexcLim2

Attribute name	mult.	Attribute type	Description
koi	1..1	PU	Gain Over excitation limiter (K_{OI}). Typical Value = 0.1.
voimax	1..1	PU	Maximum error signal (V_{OIMAX}). Typical Value = 0.
voimin	1..1	PU	Minimum error signal (V_{OIMIN}). Typical Value = -9999.
ifdlim	1..1	PU	Limit value of rated field current (I_{FDLIM}). Typical Value = 1.05.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-225 shows all association ends of OverexcitationLimiterDynamics with other classes.

Table 9-225 - Association ends of OverexcitationLimiterDynamics::OverexcitationLimiterDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : OverexcitationLimiterDynamics:: OverexcitationLimiterDynamics.OverexcitationLimiterDynamics[0..1] --- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.10.5 OverexcLimX1

Field voltage over excitation limiter.

Figure 9-128 class diagram

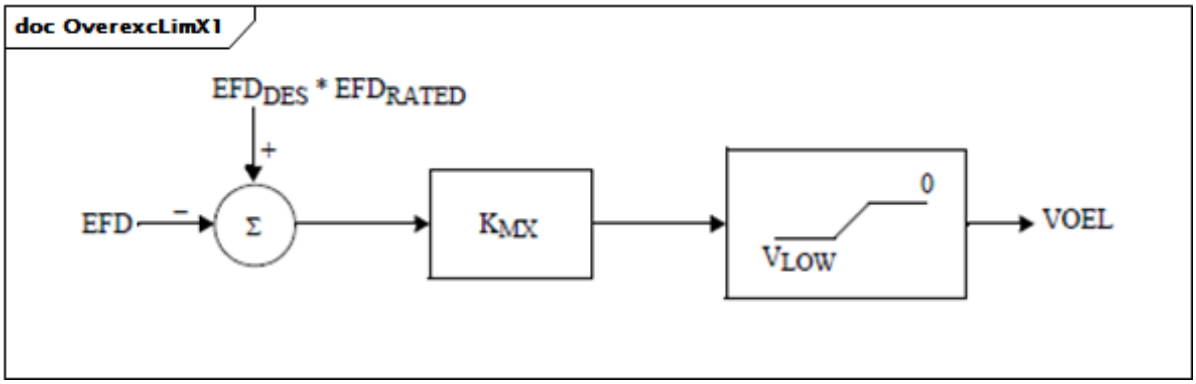


Figure 9-128 - figure OverexcLimX1

Figure 9-129 class diagram

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

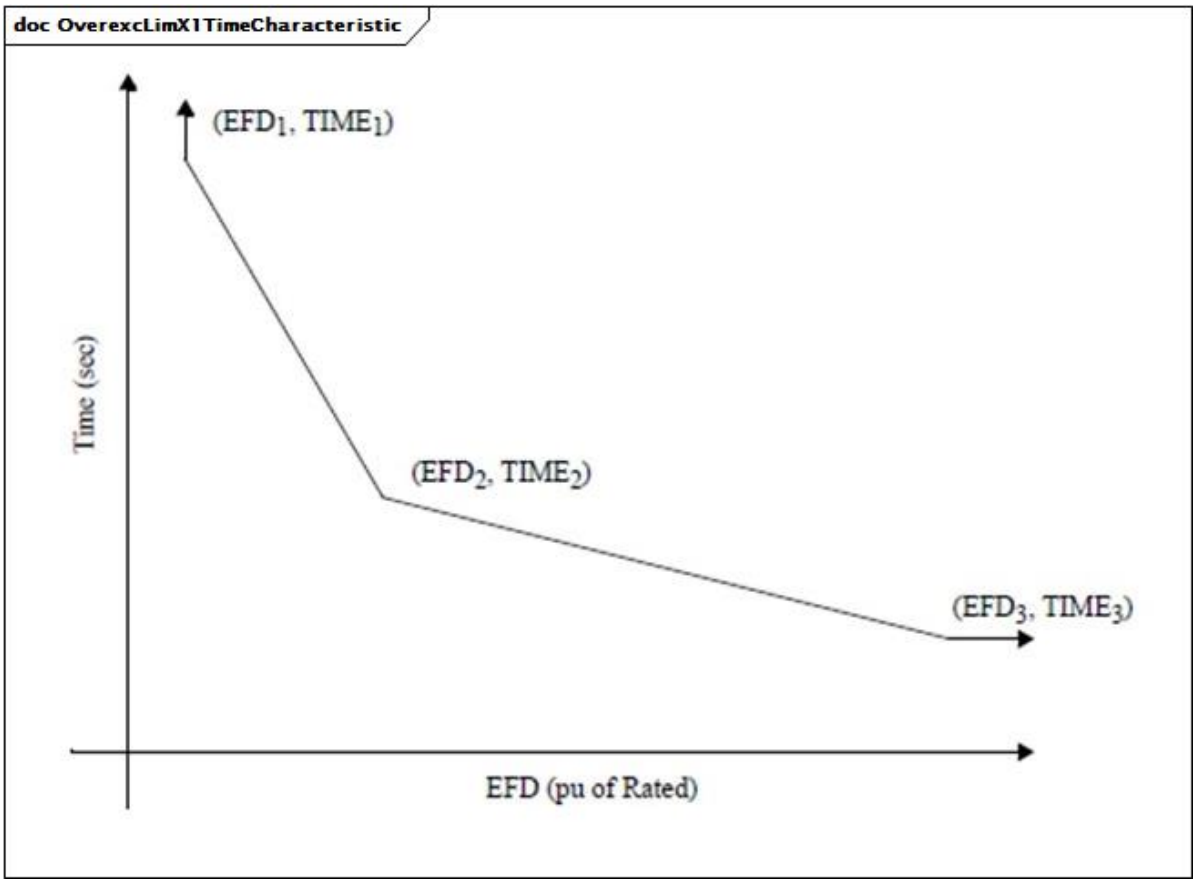


Figure 9-129 - figure OverexcLimX1TimeCharacteristic

This model assumes an inverse time characteristic, where operating time is a function of the magnitude of the overvoltage.

Inheritance path: ->OverexcitationLimiterDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-226 shows all attributes of OverexcLimX1.

Table 9-226 - Attributes of OverexcitationLimiterDynamics::OverexcLimX1

Attribute name	mult.	Attribute type	Description
efdrated	1..1	PU	Rated field voltage (EFD_{RATED}). Typical Value = 1.05.
efd1	1..1	PU	Low voltage point on the inverse time characteristic (EFD_1). Typical Value = 1.1.
t1	1..1	Seconds	Time to trip the exciter at the low voltage point on the inverse time characteristic ($TIME_1$). Typical Value = 120.
efd2	1..1	PU	Mid voltage point on the inverse time characteristic (EFD_2). Typical Value = 1.2.

t2	1..1	Seconds	Time to trip the exciter at the mid voltage point on the inverse time characteristic ($TIME_2$). Typical Value = 40.
efd3	1..1	PU	High voltage point on the inverse time characteristic (EFD_3). Typical Value = 1.5.
t3	1..1	Seconds	Time to trip the exciter at the high voltage point on the inverse time characteristic ($TIME_3$). Typical Value = 15.
efddes	1..1	PU	Desired field voltage (EFD_{DES}). Typical Value = 0.9.
kmx	1..1	PU	Gain (K_{MX}). Typical Value = 0.01.
vlow	1..1	PU	Low voltage limit (V_{LOW}) (>0).
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-227 shows all association ends of OverexcitationLimiterDynamics with other classes.

Table 9-227 - Association ends of OverexcitationLimiterDynamics::OverexcitationLimiterDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : OverexcitationLimiterDynamics:: OverexcitationLimiterDynamics.OverexcitationLimiterDynamics[0..1] --- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.10.6 OverexcLimX2

Field Voltage or Current overexcitation limiter designed to protect the generator field of an AC machine with automatic excitation control from overheating due to prolonged overexcitation.

Figure 9-130 class diagram

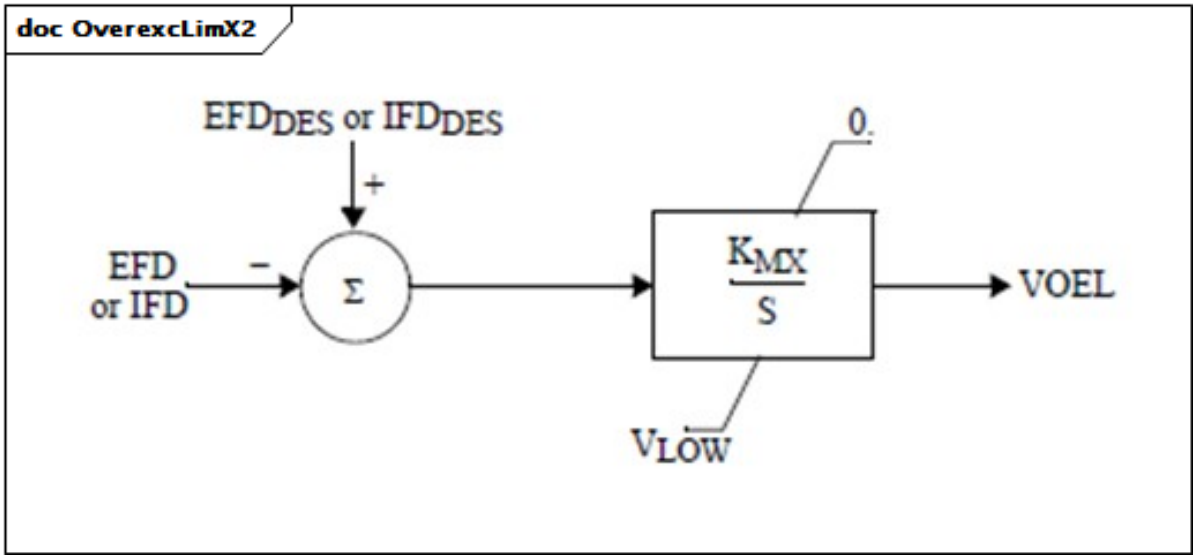


Figure 9-130 - figure OverexcLimX2

Figure 9-131 class diagram

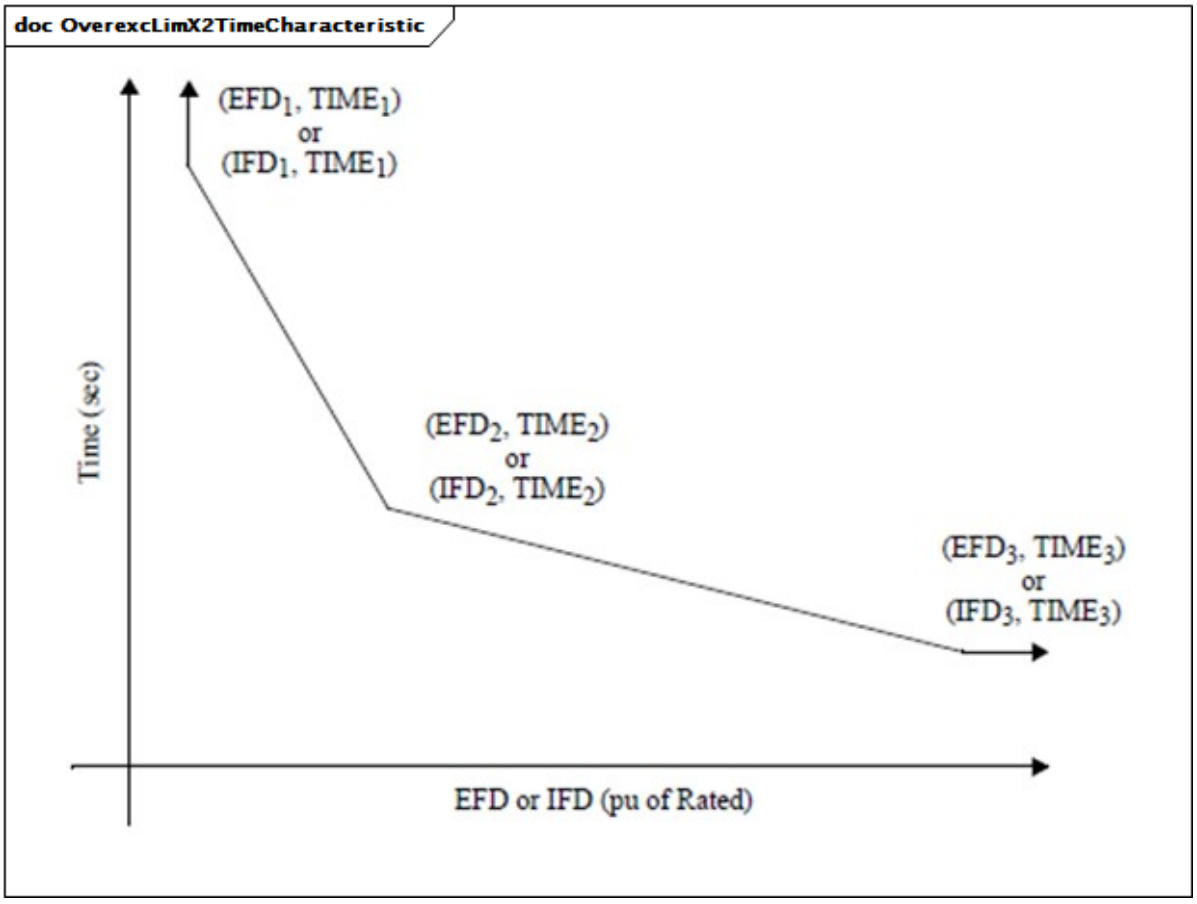


Figure 9-131 - figure OverexcLimX2TimeCharacteristic

This model assumes an inverse time characteristic, where operating time is a function of the magnitude of the overvoltage.

Inheritance path: ->OverexcitationLimiterDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-228 shows all attributes of OverexcLimX2.

Table 9-228 - Attributes of OverexcitationLimiterDynamics::OverexcLimX2

Attribute name	mult.	Attribute type	Description
m	1..1	Boolean	(m). true = IFD limiting false = EFD limiting.
efdrated	1..1	PU	Rated field voltage if m=F or field current if m=T (EFD_{RATED}). Typical Value = 1.05.
efd1	1..1	PU	Low voltage or current point on the inverse time characteristic (EFD_1). Typical Value = 1.1.
t1	1..1	Seconds	Time to trip the exciter at the low voltage or current point on the inverse time characteristic ($TIME_1$). Typical Value = 120.
efd2	1..1	PU	Mid voltage or current point on the inverse time characteristic (EFD_2). Typical Value = 1.2.
t2	1..1	Seconds	Time to trip the exciter at the mid voltage or current point on the inverse time characteristic ($TIME_2$). Typical Value = 40.
efd3	1..1	PU	High voltage or current point on the inverse time characteristic (EFD_3). Typical Value = 1.5.
t3	1..1	Seconds	Time to trip the exciter at the high voltage or current point on the inverse time characteristic ($TIME_3$). Typical Value = 15.
efddes	1..1	PU	Desired field voltage if m=F or field current if m=T (EFD_{DES}). Typical Value = 1.
kmx	1..1	PU	Gain (K_{MX}). Typical Value = 0.002.
vlow	1..1	PU	Low voltage limit (V_{LOW}) (>0).

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-229 shows all association ends of OverexcitationLimiterDynamics with other classes.

Table 9-229 - Association ends of OverexcitationLimiterDynamics::OverexcitationLimiterDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : OverexcitationLimiterDynamics:: OverexcitationLimiterDynamics.OverexcitationLimiterDynamics[0..1] --- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.11 UnderexcitationLimiterDynamics

9.6.11.1 General

Underexcitation limiters (UELs) act to boost excitation. The UEL typically senses either a combination of voltage and current of the synchronous machine or a combination of real and reactive power. Some UELs utilize a temperature or pressure recalibration feature, in which the UEL characteristic is shifted depending upon the generator cooling gas temperature or pressure.

Figure 9-132 shows the diagram.

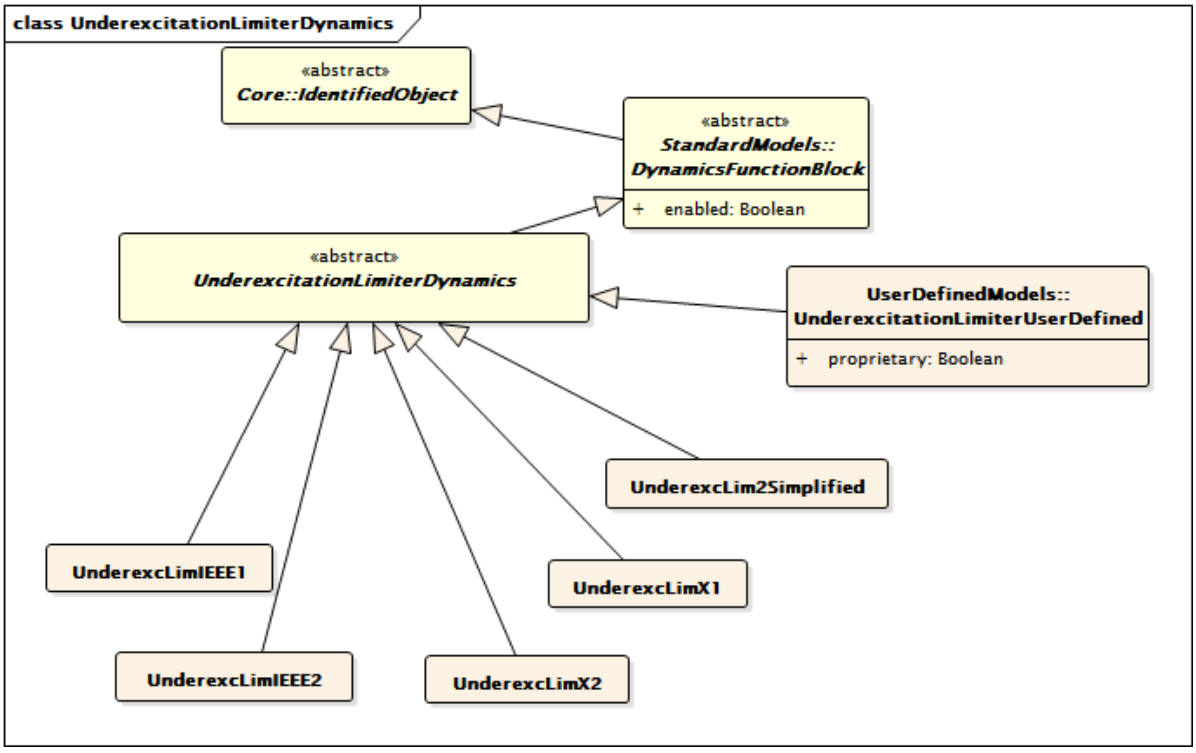


Figure 9-132 - diagram UnderexcitationLimiterDynamics

This diagram shows dynamic standard model underexcitation limiter classes and their inheritance from the UnderexcitationLimiterDynamics class.

9.6.11.2 UnderexcitationLimiterDynamics(Abstract)

Underexcitation limiter function block whose behaviour is described by reference to a standard model or by definition of a user-defined model.

Inheritance path: ->DynamicsFunctionBlock->IdentifiedObject

Table 9-230 shows all attributes of UnderexcitationLimiterDynamics.

Table 9-230 - Attributes of UnderexcitationLimiterDynamics::UnderexcitationLimiterDynamics

Attribute name	mult.	Attribute type	Description
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-231 shows all association ends of UnderexcitationLimiterDynamics with other classes.

Table 9-231 - Association ends of UnderexcitationLimiterDynamics::UnderexcitationLimiterDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	[1..1]	ExcitationSystemDynamics	Excitation system model with which this underexcitation limiter model is associated. Association Based On : UnderexcitationLimiterDynamics:: UnderexcitationLimiterDynamics.UnderexcitationLimiterDynamics [0..1] ----- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.11.3 UnderexcLimIEEE1

The class represents the Type UEL1 model which has a circular limit boundary when plotted in terms of machine reactive power vs. real power output. Reference: IEEE UEL1 421.5-2005 Section 10.1.

Inheritance path: ->UnderexcitationLimiterDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-232 shows all attributes of UnderexcLimIEEE1.

Table 9-232 - Attributes of UnderexcitationLimiterDynamics::UnderexcLimIEEE1

Attribute name	mult.	Attribute type	Description
kur	1..1	PU	UEL radius setting (K_{UR}). Typical Value = 1.95.
kuc	1..1	PU	UEL center setting (K_{UC}). Typical Value = 1.38.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

kuf	1..1	PU	UEL excitation system stabilizer gain (K_{UF}). Typical Value = 3.3.
vurmax	1..1	PU	UEL maximum limit for radius phasor magnitude (V_{URMAX}). Typical Value = 5.8.
vucmax	1..1	PU	UEL maximum limit for operating point phasor magnitude (V_{UCMAX}). Typical Value = 5.8.
kui	1..1	PU	UEL integral gain (K_{UI}). Typical Value = 0.
kul	1..1	PU	UEL proportional gain (K_{UL}). Typical Value = 100.
vuimax	1..1	PU	UEL integrator output maximum limit (V_{UIMAX}).
vuimin	1..1	PU	UEL integrator output minimum limit (V_{UIMIN}).
tu1	1..1	Seconds	UEL lead time constant (T_{U1}). Typical Value = 0.
tu2	1..1	Seconds	UEL lag time constant (T_{U2}). Typical Value = 0.05.
tu3	1..1	Seconds	UEL lead time constant (T_{U3}). Typical Value = 0.
tu4	1..1	Seconds	UEL lag time constant (T_{U4}). Typical Value = 0.
vulmax	1..1	PU	UEL output maximum limit (V_{ULMAX}). Typical Value = 18.
vulmin	1..1	PU	UEL output minimum limit (V_{ULMIN}). Typical Value = -18.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-233 shows all association ends of UnderexcitationLimiterDynamics with other classes.

Table 9-233 - Association ends of UnderexcitationLimiterDynamics::UnderexcitationLimiterDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : UnderexcitationLimiterDynamics:: UnderexcitationLimiterDynamics.UnderexcitationLimiterDynamics[0..1] -----

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.11.4 UnderexcLimIEEE2

The class represents the Type UEL2 which has either a straight-line or multi-segment characteristic when plotted in terms of machine reactive power output vs. real power output. Reference: IEEE UEL2 421.5-2005 Section 10.2. (Limit characteristic lookup table shown in Figure 10.4 (p 32) of the standard).

Inheritance path: ->UnderexcitationLimiterDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-234 shows all attributes of UnderexcLimIEEE2.

Table 9-234 - Attributes of UnderexcitationLimiterDynamics::UnderexcLimIEEE2

Attribute name	mult.	Attribute type	Description
tuv	1..1	Seconds	Voltage filter time constant (T_{UV}). Typical Value = 5.
tup	1..1	Seconds	Real power filter time constant (T_{UP}). Typical Value = 5.
tuq	1..1	Seconds	Reactive power filter time constant (T_{UQ}). Typical Value = 0.
kui	1..1	PU	UEL integral gain (K_{UI}). Typical Value = 0.5.
kul	1..1	PU	UEL proportional gain (K_{UL}). Typical Value = 0.8.
vuimax	1..1	PU	UEL integrator output maximum limit (V_{UIMAX}). Typical Value = 0.25.
vuimin	1..1	PU	UEL integrator output minimum limit (V_{UIMIN}). Typical Value = 0.
kuf	1..1	PU	UEL excitation system stabilizer gain (K_{UF}). Typical Value = 0.
kfb	1..1	PU	Gain associated with optional integrator feedback input signal to UEL (K_{FB}). Typical Value = 0.
tul	1..1	Seconds	Time constant associated with optional integrator feedback input signal to UEL (T_{UL}). Typical Value = 0.
tu1	1..1	Seconds	UEL lead time constant (T_{U1}). Typical Value = 0.
tu2	1..1	Seconds	UEL lag time constant (T_{U2}). Typical Value = 0.
tu3	1..1	Seconds	UEL lead time constant (T_{U3}). Typical Value = 0.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

tu4	1..1	Seconds	UEL lag time constant (T_{U4}). Typical Value = 0.
vulmax	1..1	PU	UEL output maximum limit (V_{ULMAX}). Typical Value = 0.25.
vulmin	1..1	PU	UEL output minimum limit (V_{ULMIN}). Typical Value = 0.
p0	1..1	PU	Real power values for endpoints (P_0). Typical Value = 0.
q0	1..1	PU	Reactive power values for endpoints (Q_0). Typical Value = -0.31.
p1	1..1	PU	Real power values for endpoints (P_1). Typical Value = 0.3.
q1	1..1	PU	Reactive power values for endpoints (Q_1). Typical Value = -0.31.
p2	1..1	PU	Real power values for endpoints (P_2). Typical Value = 0.6.
q2	1..1	PU	Reactive power values for endpoints (Q_2). Typical Value = -0.28.
p3	1..1	PU	Real power values for endpoints (P_3). Typical Value = 0.9.
q3	1..1	PU	Reactive power values for endpoints (Q_3). Typical Value = -0.21.
p4	1..1	PU	Real power values for endpoints (P_4). Typical Value = 1.02.
q4	1..1	PU	Reactive power values for endpoints (Q_4). Typical Value = 0.
p5	1..1	PU	Real power values for endpoints (P_5).
q5	1..1	PU	Reactive power values for endpoints (Q_5).
p6	1..1	PU	Real power values for endpoints (P_6).
q6	1..1	PU	Reactive power values for endpoints (Q_6).
p7	1..1	PU	Real power values for endpoints (P_7).
q7	1..1	PU	Reactive power values for endpoints (Q_7).
p8	1..1	PU	Real power values for endpoints (P_8).

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

q8	1..1	PU	Reactive power values for endpoints (Q_8).
p9	1..1	PU	Real power values for endpoints (P_9).
q9	1..1	PU	Reactive power values for endpoints (Q_9).
p10	1..1	PU	Real power values for endpoints (P_{10}).
q10	1..1	PU	Reactive power values for endpoints (Q_{10}).
k1	1..1	Simple_Float	UEL terminal voltage exponent applied to real power input to UEL limit look-up table (k1). Typical Value = 2.
k2	1..1	Simple_Float	UEL terminal voltage exponent applied to reactive power output from UEL limit look-up table (k2). Typical Value = 2.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-235 shows all association ends of UnderexcitationLimiterDynamics with other classes.

Table 9-235 - Association ends of UnderexcitationLimiterDynamics::UnderexcitationLimiterDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : UnderexcitationLimiterDynamics:: UnderexcitationLimiterDynamics.UnderexcitationLimiterDynamics[0..1] --- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.11.5 UnderexcLim2Simplified

This model can be derived from UnderexcLimIEEE2. The limit characteristic (look –up table) is a single straight-line, the same as UnderexcLimIEEE2 (see Figure 10.4 (p 32), IEEE 421.5-2005 Section 10.2).

Figure 9-133 class diagram

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

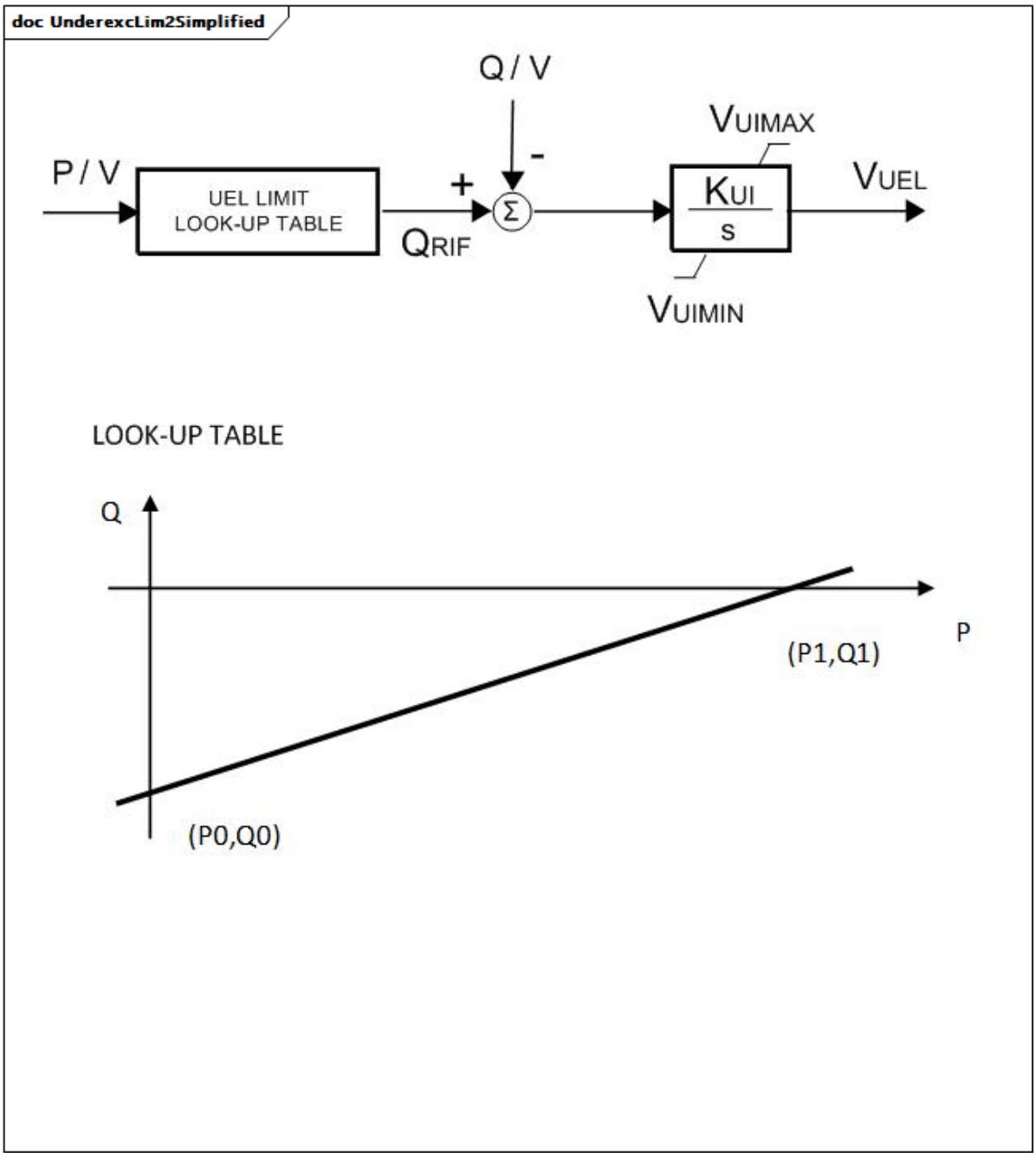


Figure 9-133 - figure UnderexcLim2Simplified

Inheritance path: ->UnderexcitationLimiterDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-236 shows all attributes of UnderexcLim2Simplified.

Table 9-236 - Attributes of UnderexcitationLimiterDynamics::UnderexcLim2Simplified

Attribute name	mult.	Attribute type	Description
q0	1..1	PU	Segment Q initial point (Q0). Typical Value = -0.31.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

q1	1..1	PU	Segment Q end point (Q1). Typical Value = -0.1.
p0	1..1	PU	Segment P initial point (P0). Typical Value = 0.
p1	1..1	PU	Segment P end point (P1). Typical Value = 1.
kui	1..1	PU	Gain Under excitation limiter (Kui). Typical Value = 0.1.
vuimin	1..1	PU	Minimum error signal (V_{UImin}). Typical Value = 0.
vuimax	1..1	PU	Maximum error signal (V_{UImax}). Typical Value = 1.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-237 shows all association ends of UnderexcitationLimiterDynamics with other classes.

Table 9-237 - Association ends of UnderexcitationLimiterDynamics::UnderexcitationLimiterDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : UnderexcitationLimiterDynamics:: UnderexcitationLimiterDynamics.UnderexcitationLimiterDynamics[0..1] --- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.11.6 UnderexcLimX1

Allis-Chalmers minimum excitation limiter.

Figure 9-134 class diagram

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

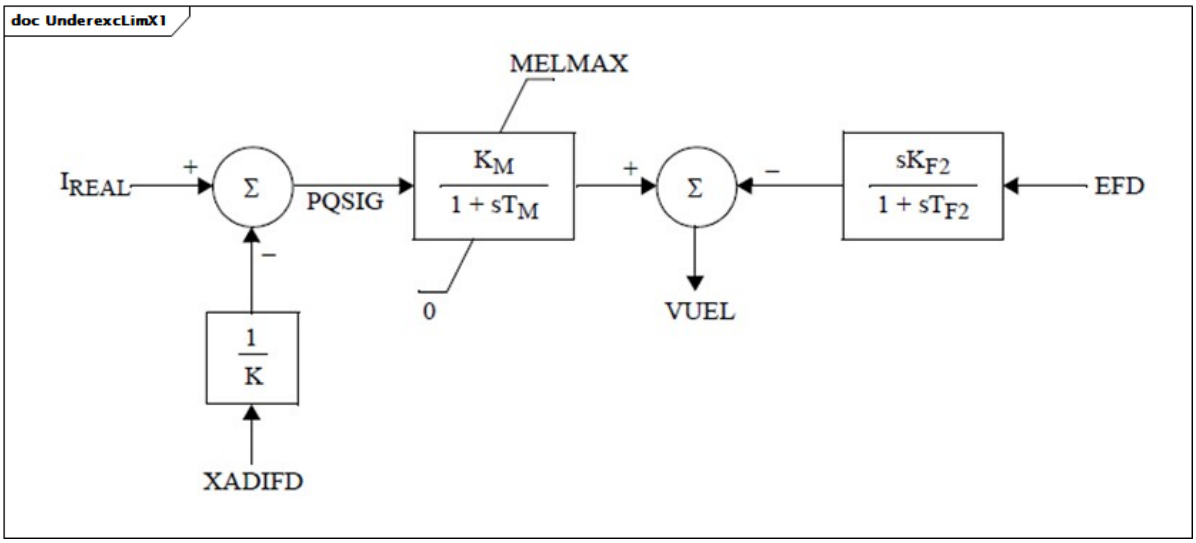


Figure 9-134 - figure UnderexcLimX1

Inheritance path: ->UnderexcitationLimiterDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-238 shows all attributes of UnderexcLimX1.

Table 9-238 - Attributes of UnderexcitationLimiterDynamics::UnderexcLimX1

Attribute name	mult.	Attribute type	Description
kf2	1..1	PU	Differential gain (Kf2).
tf2	1..1	Seconds	Differential time constant (Tf2) (>0).
km	1..1	PU	Minimum excitation limit gain (Km).
tm	1..1	Seconds	Minimum excitation limit time constant (Tm).
melmax	1..1	PU	Minimum excitation limit value (MELMAX).
k	1..1	PU	Minimum excitation limit slope (K) (>0).
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-239 shows all association ends of UnderexcitationLimiterDynamics with other classes.

Table 9-239 - Association ends of UnderexcitationLimiterDynamics::UnderexcitationLimiterDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDyna	1..1	ExcitationSystemDynam	Inherited Association Based On :

mics		ics	UnderexcitationLimiterDynamics:: UnderexcitationLimiterDynamics.UnderexcitationLimiterDynamics[0..1] ----- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.11.7 UnderexcLimX2

Westinghouse minimum excitation limiter.

Figure 9-135 class diagram

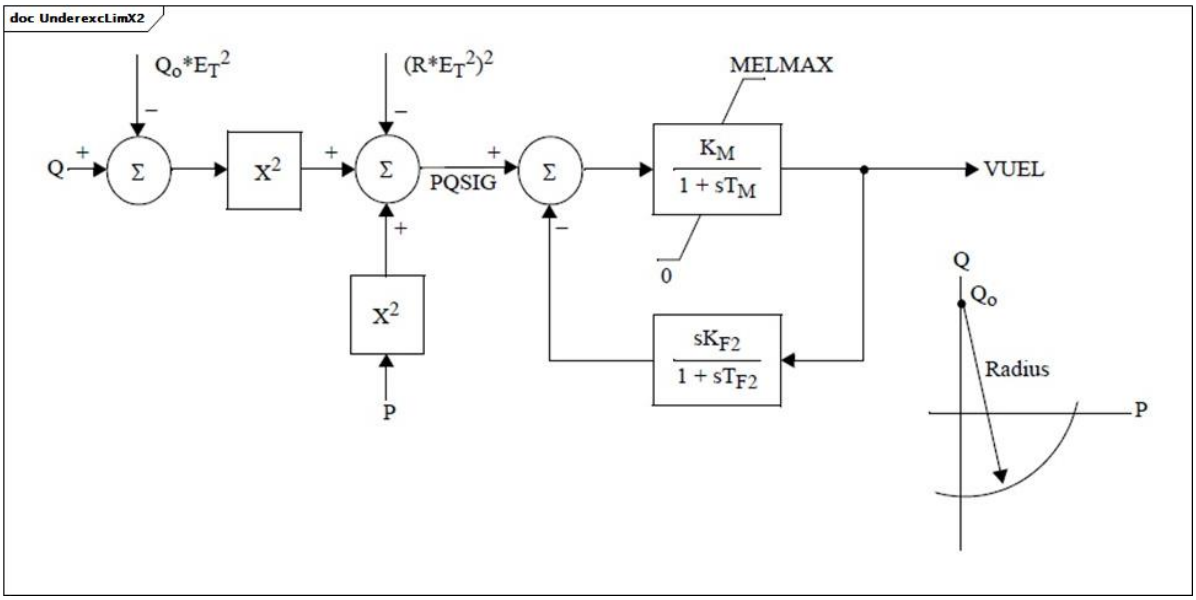


Figure 9-135 - figure UnderexcLimX2

Inheritance path: ->UnderexcitationLimiterDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-240 shows all attributes of UnderexcLimX2.

Table 9-240 - Attributes of UnderexcitationLimiterDynamics::UnderexcLimX2

Attribute name	mult.	Attribute type	Description
kf2	1..1	PU	Differential gain (Kf2).
tf2	1..1	Seconds	Differential time constant (Tf2) (>0).
km	1..1	PU	Minimum excitation limit gain (Km).
tm	1..1	Seconds	Minimum excitation limit time constant (Tm).
melmax	1..1	PU	Minimum excitation limit value (MELMAX).
qo	1..1	PU	Excitation center setting (Qo).
r	1..1	PU	Excitation radius (R).
enabled	1..1	Boolean	see

			DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-241 shows all association ends of UnderexcitationLimiterDynamics with other classes.

Table 9-241 - Association ends of UnderexcitationLimiterDynamics::UnderexcitationLimiterDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : UnderexcitationLimiterDynamics:: UnderexcitationLimiterDynamics.UnderexcitationLimiterDynamics[0..1] --- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.12 PowerSystemStabilizerDynamics

9.6.12.1 General

The power system stabilizer (PSS) model provides an input (Vs) to the excitation system model to improve damping of system oscillations. A variety of input signals may be used depending on the particular design.

Figure 9-136 shows the diagram.

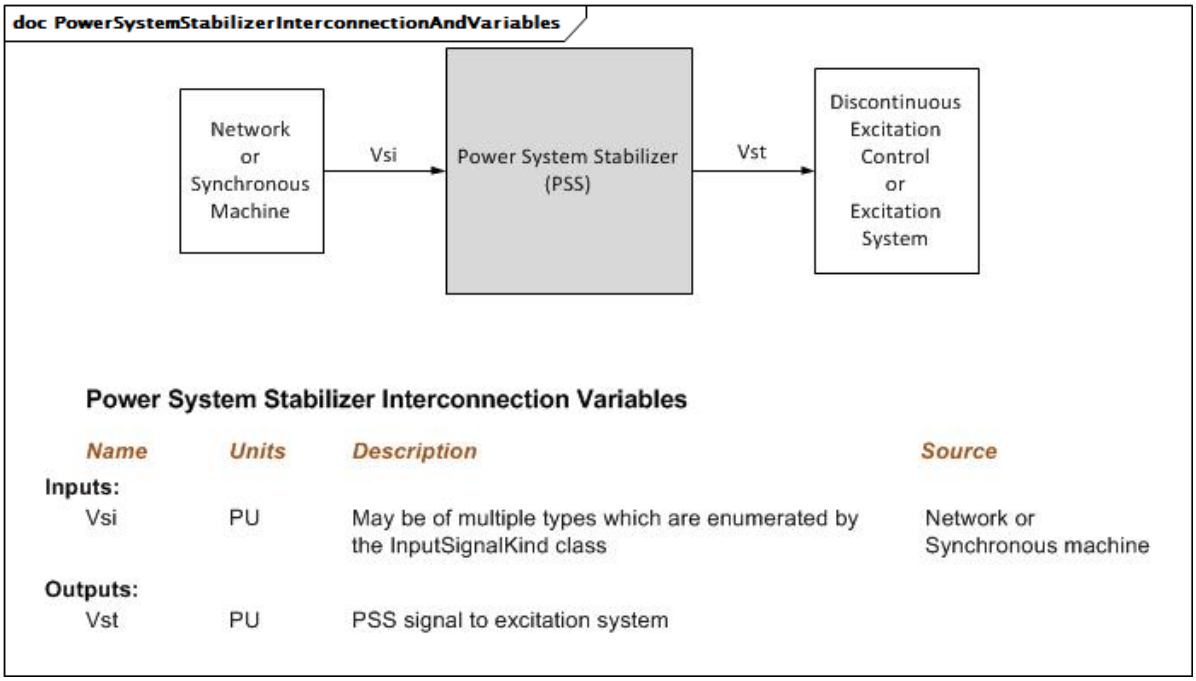


Figure 9-136 - diagram PowerSystemStabilizerInterconnectionAndVariables

This diagram illustrates the inputs and outputs for a power system stabilizer.
Notes:

1. If there is no discontinuous excitation control function block present, then output is **Vs**, not **Vst**
2. **Vs** is always initialized to zero by the excitation system model.
3. If bus voltage frequency is not available from network, the model can calculate it as derivative of the bus voltage angle.

Figure 9-137 shows the diagram.

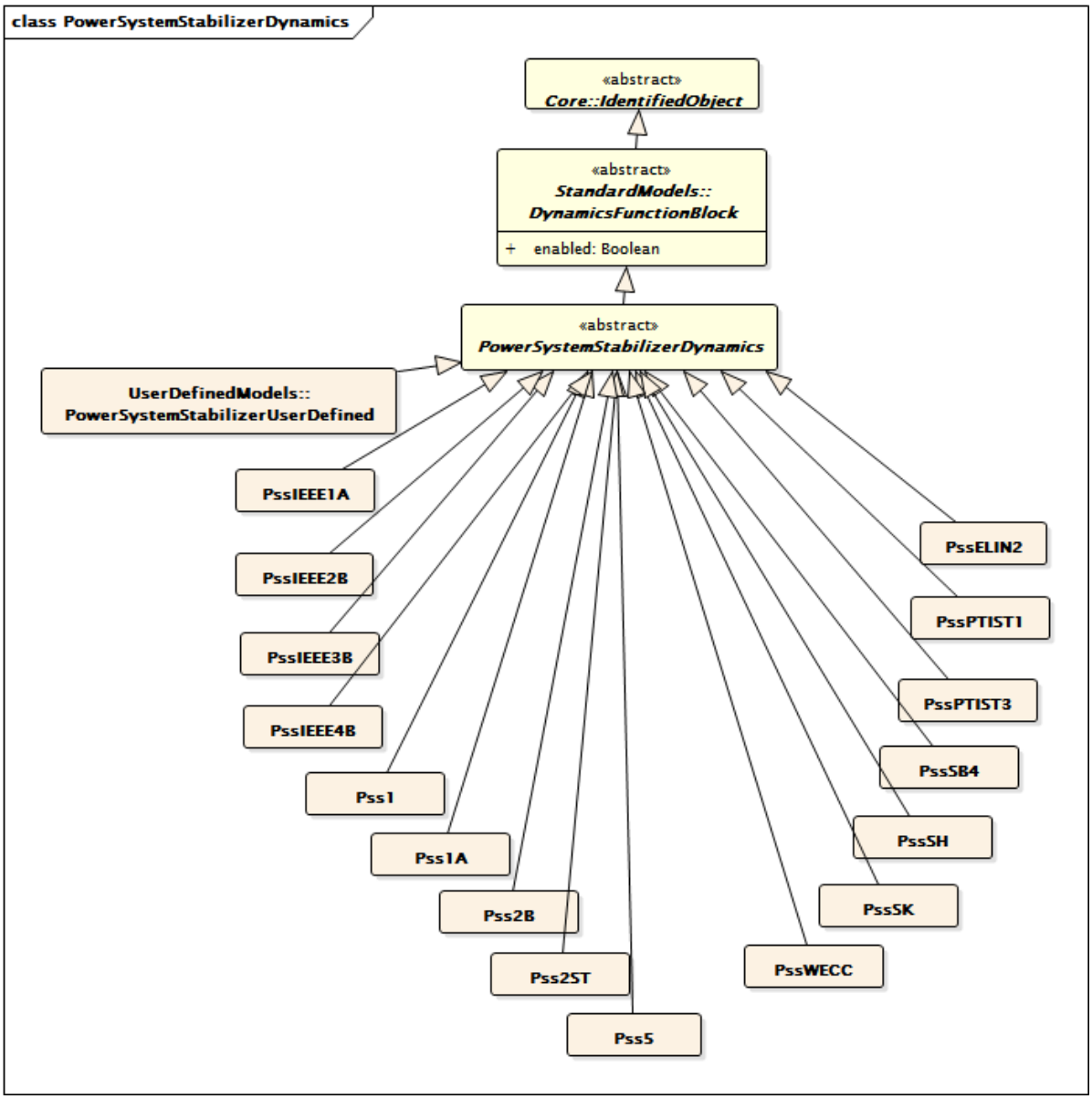


Figure 9-137 - diagram PowerSystemStabilizerDynamics

This diagram shows dynamic standard model power system stabilizer classes and their inheritance from the PowerSystemStabilizerDynamics class.

9.6.12.2 PowerSystemStabilizerDynamics(Abstract)

Power system stabilizer function block whose behaviour is described by reference to a standard model or by definition of a user-defined model.

Inheritance path: ->DynamicsFunctionBlock->IdentifiedObject

Table 9-242 shows all attributes of PowerSystemStabilizerDynamics.

Table 9-242 - Attributes of PowerSystemStabilizerDynamics::PowerSystemStabilizerDynamics

Attribute name	mult.	Attribute type	Description
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-243 shows all association ends of PowerSystemStabilizerDynamics with other classes.

Table 9-243 - Association ends of PowerSystemStabilizerDynamics::PowerSystemStabilizerDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	[1..1]	ExcitationSystemDynamics	Excitation system model with which this power system stabilizer model is associated. Association Based On : PowerSystemStabilizerDynamics:: PowerSystemStabilizerDynamics.PowerSystemStabilizerDynamics[0..1] ----- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.12.3 PssIEEE1A

The class represents IEEE Std 421.5-2005 type PSS1A power system stabilizer model. PSS1A is the generalized form of a PSS with a single input. Some common stabilizer input signals are speed, frequency, and power.
Reference: IEEE 1A 421.5-2005 Section 8.1.

Inheritance path: ->PowerSystemStabilizerDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-244 shows all attributes of PssIEEE1A.

Table 9-244 - Attributes of PowerSystemStabilizerDynamics::PssIEEE1A

Attribute name	mult.	Attribute type	Description
inputSignalType	1..1	InputSignalKind	Type of input signal. Typical Value = rotorAngularFrequencyDeviation.
a1	1..1	PU	PSS signal conditioning frequency filter constant (A1). Typical Value = 0.061.
a2	1..1	PU	PSS signal conditioning frequency filter constant (A2). Typical Value = 0.0017.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

t1	1..1	Seconds	Lead/lag time constant (T1). Typical Value = 0.3.
t2	1..1	Seconds	Lead/lag time constant (T2). Typical Value = 0.03.
t3	1..1	Seconds	Lead/lag time constant (T3). Typical Value = 0.3.
t4	1..1	Seconds	Lead/lag time constant (T4). Typical Value = 0.03.
t5	1..1	Seconds	Washout time constant (T5). Typical Value = 10.
t6	1..1	Seconds	Transducer time constant (T6). Typical Value = 0.01.
ks	1..1	PU	Stabilizer gain (Ks). Typical Value = 5.
vrmax	1..1	PU	Maximum stabilizer output (Vrmax). Typical Value = 0.05.
vrmin	1..1	PU	Minimum stabilizer output (Vrmin). Typical Value = -0.05.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-245 shows all association ends of PowerSystemStabilizerDynamics with other classes.

Table 9-245 - Association ends of PowerSystemStabilizerDynamics::PowerSystemStabilizerDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : PowerSystemStabilizerDynamics:: PowerSystemStabilizerDynamics.PowerSystemStabilizerDynamics[0..1] --- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.12.4 PssIEEE2B

The class represents IEEE Std 421.5-2005 type PSS2B power system stabilizer model. This stabilizer model is designed to represent a variety of dual-input stabilizers, which normally use combinations of power and speed or frequency to derive the stabilizing signal. Reference: IEEE 2B 421.5-2005 Section 8.2.

Inheritance path: ->PowerSystemStabilizerDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-246 shows all attributes of PssIEEE2B.

Table 9-246 - Attributes of PowerSystemStabilizerDynamics::PssIEEE2B

Attribute name	mult.	Attribute type	Description
inputSignal1Type	1..1	InputSignalKind	Type of input signal #1.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			Typical Value = rotorSpeed.
inputSignal2Type	1..1	InputSignalKind	Type of input signal #2. Typical Value = generatorElectricalPower.
vsi1max	1..1	PU	Input signal #1 max limit (Vsi1max). Typical Value = 2.
vsi1min	1..1	PU	Input signal #1 min limit (Vsi1min). Typical Value = -2.
tw1	1..1	Seconds	First washout on signal #1 (Tw1). Typical Value = 2.
tw2	1..1	Seconds	Second washout on signal #1 (Tw2). Typical Value = 2.
vsi2max	1..1	PU	Input signal #2 max limit (Vsi2max). Typical Value = 2.
vsi2min	1..1	PU	Input signal #2 min limit (Vsi2min). Typical Value = -2.
tw3	1..1	Seconds	First washout on signal #2 (Tw3). Typical Value = 2.
tw4	1..1	Seconds	Second washout on signal #2 (Tw4). Typical Value = 0.
t1	1..1	Seconds	Lead/lag time constant (T1). Typical Value = 0.12.
t2	1..1	Seconds	Lead/lag time constant (T2). Typical Value = 0.02.
t3	1..1	Seconds	Lead/lag time constant (T3). Typical Value = 0.3.
t4	1..1	Seconds	Lead/lag time constant (T4). Typical Value = 0.02.
t6	1..1	Seconds	Time constant on signal #1 (T6). Typical Value = 0.
t7	1..1	Seconds	Time constant on signal #2 (T7). Typical Value = 2.
t8	1..1	Seconds	Lead of ramp tracking filter (T8). Typical Value = 0.2.
t9	1..1	Seconds	Lag of ramp tracking filter (T9). Typical Value = 0.1.
t10	1..1	Seconds	Lead/lag time constant (T10). Typical Value = 0.
t11	1..1	Seconds	Lead/lag time constant

			(T11). Typical Value = 0.
ks1	1..1	PU	Stabilizer gain (Ks1). Typical Value = 12.
ks2	1..1	PU	Gain on signal #2 (Ks2). Typical Value = 0.2.
ks3	1..1	PU	Gain on signal #2 input before ramp-tracking filter (Ks3). Typical Value = 1.
n	1..1	Integer	Order of ramp tracking filter (N). Typical Value = 1.
m	1..1	Integer	Denominator order of ramp tracking filter (M). Typical Value = 5.
vstmax	1..1	PU	Stabilizer output max limit (Vstmax). Typical Value = 0.1.
vstmin	1..1	PU	Stabilizer output min limit (Vstmin). Typical Value = -0.1.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-247 shows all association ends of PowerSystemStabilizerDynamics with other classes.

Table 9-247 - Association ends of PowerSystemStabilizerDynamics::PowerSystemStabilizerDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : PowerSystemStabilizerDynamics:: PowerSystemStabilizerDynamics.PowerSystemStabilizerDynamics[0..1] --- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.12.5 PssIEEE3B

The class represents IEEE Std 421.5-2005 type PSS3B power system stabilizer model. The PSS model PSS3B has dual inputs of electrical power and rotor angular frequency deviation. The signals are used to derive an equivalent mechanical power signal. Reference: IEEE 3B 421.5-2005 Section 8.3.

Inheritance path: ->PowerSystemStabilizerDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-248 shows all attributes of PssIEEE3B.

Table 9-248 - Attributes of PowerSystemStabilizerDynamics::PssIEEE3B

Attribute name	mult.	Attribute type	Description
inputSignal1Type	1..1	InputSignalKind	Type of input signal #1. Typical Value = generatorElectricalPower.
inputSignal2Type	1..1	InputSignalKind	Type of input signal #2. Typical Value = rotorSpeed.
t1	1..1	Seconds	Transducer time constant (T1). Typical Value = 0.012.
t2	1..1	Seconds	Transducer time constant (T2). Typical Value = 0.012.
tw1	1..1	Seconds	Washout time constant (Tw1). Typical Value = 0.3.
tw2	1..1	Seconds	Washout time constant (Tw2). Typical Value = 0.3.
tw3	1..1	Seconds	Washout time constant (Tw3). Typical Value = 0.6.
ks1	1..1	PU	Gain on signal # 1 (Ks1). Typical Value = -0.602.
ks2	1..1	PU	Gain on signal # 2 (Ks2). Typical Value = 30.12.
a1	1..1	PU	Notch filter parameter (A1). Typical Value = 0.359.
a2	1..1	PU	Notch filter parameter (A2). Typical Value = 0.586.
a3	1..1	PU	Notch filter parameter (A3). Typical Value = 0.429.
a4	1..1	PU	Notch filter parameter (A4). Typical Value = 0.564.
a5	1..1	PU	Notch filter parameter (A5). Typical Value = 0.001.
a6	1..1	PU	Notch filter parameter (A6). Typical Value = 0.
a7	1..1	PU	Notch filter parameter (A7). Typical Value = 0.031.
a8	1..1	PU	Notch filter parameter (A8). Typical Value = 0.
vstmax	1..1	PU	Stabilizer output max limit (Vstmax). Typical Value = 0.1.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

vstmin	1..1	PU	Stabilizer output min limit (Vstmin). Typical Value = -0.1.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-249 shows all association ends of PowerSystemStabilizerDynamics with other classes.

Table 9-249 - Association ends of PowerSystemStabilizerDynamics::PowerSystemStabilizerDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : PowerSystemStabilizerDynamics:: PowerSystemStabilizerDynamics.PowerSystemStabilizerDynamics[0..1] ----- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.12.6 PssIEEE4B

The class represents IEEE Std 421.5-2005 type PSS2B power system stabilizer model. The PSS4B model represents a structure based on multiple working frequency bands. Three separate bands, respectively dedicated to the low-, intermediate- and high-frequency modes of oscillations, are used in this delta-omega (speed input) PSS. Reference: IEEE 4B 421.5-2005 Section 8.4.

Inheritance path: ->PowerSystemStabilizerDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-250 shows all attributes of PssIEEE4B.

Table 9-250 - Attributes of PowerSystemStabilizerDynamics::PssIEEE4B

Attribute name	mult.	Attribute type	Description
bwh1	1..1	Simple_Float	Notch filter 1 (high-frequency band): Three dB bandwidth (B_{wi}).
bwh2	1..1	Simple_Float	Notch filter 2 (high-frequency band): Three dB bandwidth (B_{wi}).
bwl1	1..1	Simple_Float	Notch filter 1 (low-frequency band): Three dB bandwidth (B_{wi}).
bwl2	1..1	Simple_Float	Notch filter 2 (low-frequency band): Three dB bandwidth (B_{wi}).
kh	1..1	PU	High band gain (K_H). Typical Value = 120.
kh1	1..1	PU	High band differential filter gain (K_{H1}). Typical

			Value = 66.
kh11	1..1	PU	High band first lead-lag blocks coefficient (K_{H11}). Typical Value = 1.
kh17	1..1	PU	High band first lead-lag blocks coefficient (K_{H17}). Typical Value = 1.
kh2	1..1	PU	High band differential filter gain (K_{H2}). Typical Value = 66.
ki	1..1	PU	Intermediate band gain (K_I). Typical Value = 30.
ki1	1..1	PU	Intermediate band differential filter gain (K_{I1}). Typical Value = 66.
ki11	1..1	PU	Intermediate band first lead-lag blocks coefficient (K_{I11}). Typical Value = 1.
ki17	1..1	PU	Intermediate band first lead-lag blocks coefficient (K_{I17}). Typical Value = 1.
ki2	1..1	PU	Intermediate band differential filter gain (K_{I2}). Typical Value = 66.
kl	1..1	PU	Low band gain (K_L). Typical Value = 7.5.
kl1	1..1	PU	Low band differential filter gain (K_{L1}). Typical Value = 66.
kl11	1..1	PU	Low band first lead-lag blocks coefficient (K_{L11}). Typical Value = 1.
kl17	1..1	PU	Low band first lead-lag blocks coefficient (K_{L17}). Typical Value = 1.
kl2	1..1	PU	Low band differential filter gain (K_{L2}). Typical Value = 66.
omeganh1	1..1	Simple_Float	Notch filter 1 (high-frequency band): filter frequency ($omega_{ni}$).
omeganh2	1..1	Simple_Float	Notch filter 2 (high-frequency band): filter frequency ($omega_{ni}$).
omeganl1	1..1	Simple_Float	Notch filter 1 (low-frequency band): filter frequency ($omega_{ni}$).
omeganl2	1..1	Simple_Float	Notch filter 2 (low-frequency band): filter

			frequency (omega$_{ni}$).
th1	1..1	Seconds	High band time constant (T_{H1}). Typical Value = 0.01513.
th10	1..1	Seconds	High band time constant (T_{H10}). Typical Value = 0.
th11	1..1	Seconds	High band time constant (T_{H11}). Typical Value = 0.
th12	1..1	Seconds	High band time constant (T_{H12}). Typical Value = 0.
th2	1..1	Seconds	High band time constant (T_{H2}). Typical Value = 0.01816.
th3	1..1	Seconds	High band time constant (T_{H3}). Typical Value = 0.
th4	1..1	Seconds	High band time constant (T_{H4}). Typical Value = 0.
th5	1..1	Seconds	High band time constant (T_{H5}). Typical Value = 0.
th6	1..1	Seconds	High band time constant (T_{H6}). Typical Value = 0.
th7	1..1	Seconds	High band time constant (T_{H7}). Typical Value = 0.01816.
th8	1..1	Seconds	High band time constant (T_{H8}). Typical Value = 0.02179.
th9	1..1	Seconds	High band time constant (T_{H9}). Typical Value = 0.
ti1	1..1	Seconds	Intermediate band time constant (T_{I1}). Typical Value = 0.173.
ti10	1..1	Seconds	Intermediate band time constant (T_{I11}). Typical Value = 0.
ti11	1..1	Seconds	Intermediate band time constant (T_{I11}). Typical Value = 0.
ti12	1..1	Seconds	Intermediate band time constant (T_{I2}). Typical Value = 0.
ti2	1..1	Seconds	Intermediate band time constant (T_{I2}). Typical Value = 0.2075.
ti3	1..1	Seconds	Intermediate band time constant (T_{I3}). Typical

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			Value = 0.
ti4	1..1	Seconds	Intermediate band time constant (T_{I4}). Typical Value = 0.
ti5	1..1	Seconds	Intermediate band time constant (T_{I5}). Typical Value = 0.
ti6	1..1	Seconds	Intermediate band time constant (T_{I6}). Typical Value = 0.
ti7	1..1	Seconds	Intermediate band time constant (T_{I7}). Typical Value = 0.2075.
ti8	1..1	Seconds	Intermediate band time constant (T_{I8}). Typical Value = 0.2491.
ti9	1..1	Seconds	Intermediate band time constant (T_{I9}). Typical Value = 0.
tl1	1..1	Seconds	Low band time constant (T_{L1}). Typical Value = 1.73.
tl10	1..1	Seconds	Low band time constant (T_{L10}). Typical Value = 0.
tl11	1..1	Seconds	Low band time constant (T_{L11}). Typical Value = 0.
tl12	1..1	Seconds	Low band time constant (T_{L12}). Typical Value = 0.
tl2	1..1	Seconds	Low band time constant (T_{L2}). Typical Value = 2.075.
tl3	1..1	Seconds	Low band time constant (T_{L3}). Typical Value = 0.
tl4	1..1	Seconds	Low band time constant (T_{L4}). Typical Value = 0.
tl5	1..1	Seconds	Low band time constant (T_{L5}). Typical Value = 0.
tl6	1..1	Seconds	Low band time constant (T_{L6}). Typical Value = 0.
tl7	1..1	Seconds	Low band time constant (T_{L7}). Typical Value = 2.075.
tl8	1..1	Seconds	Low band time constant (T_{L8}). Typical Value = 2.491.
tl9	1..1	Seconds	Low band time constant (T_{L9}). Typical Value = 0.
vhmax	1..1	PU	High band output maximum limit (V_{Hmax}). Typical Value = 0.6.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

vhmin	1..1	PU	High band output minimum limit (V_{Hmin}). Typical Value = -0.6.
vimax	1..1	PU	Intermediate band output maximum limit (V_{Imax}). Typical Value = 0.6.
vimin	1..1	PU	Intermediate band output minimum limit (V_{Imin}). Typical Value = -0.6.
vlmax	1..1	PU	Low band output maximum limit (V_{Lmax}). Typical Value = 0.075.
vlmin	1..1	PU	Low band output minimum limit (V_{Lmin}). Typical Value = -0.075.
vstmax	1..1	PU	PSS output maximum limit (V_{STmax}). Typical Value = 0.15.
vstmin	1..1	PU	PSS output minimum limit (V_{STmin}). Typical Value = -0.15.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-251 shows all association ends of PowerSystemStabilizerDynamics with other classes.

Table 9-251 - Association ends of PowerSystemStabilizerDynamics::PowerSystemStabilizerDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : PowerSystemStabilizerDynamics::PowerSystemStabilizerDynamics.PowerSystemStabilizerDynamics[0..1] --- ExcitationSystemDynamics::ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.12.7 Pss1

Italian PSS - three input PSS (speed, frequency, power).

Figure 9-138 class diagram

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

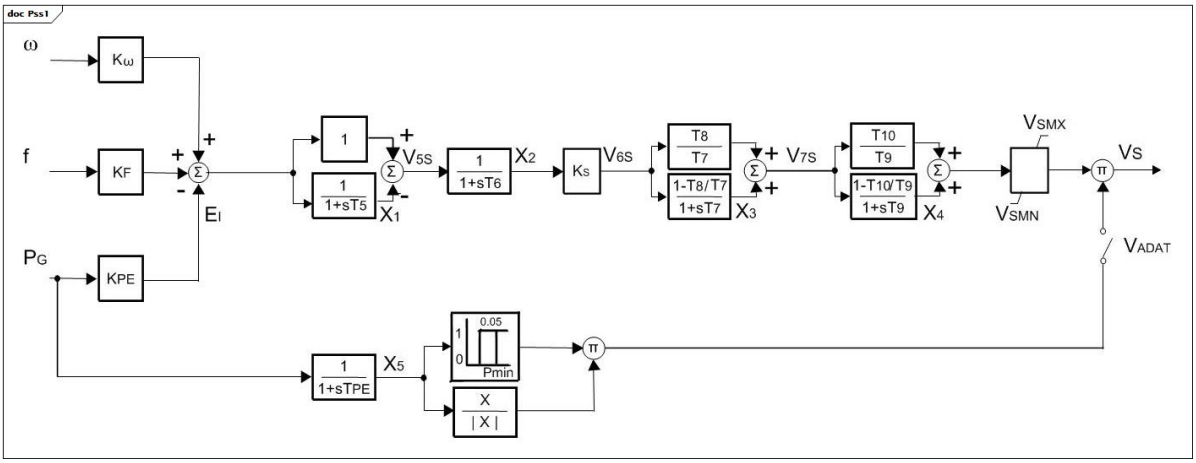


Figure 9-138 - figure Pss1

Parameter Notes:

1. The block [x/|x|] is used when **Pe** is negative, for example for a pumping unit: the output of the block [x/|x|] is the sign of **Pe**.

Inheritance path: ->PowerSystemStabilizerDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-252 shows all attributes of Pss1.

Table 9-252 - Attributes of PowerSystemStabilizerDynamics::Pss1

Attribute name	mult.	Attribute type	Description
kw	1..1	Simple_Float	Shaft speed power input gain (K_W). Typical Value = 0.
kf	1..1	Simple_Float	Frequency power input gain (K_F). Typical Value = 5.
kpe	1..1	Simple_Float	Electric power input gain (K_{PE}). Typical Value = 0.3.
pmin	1..1	PU	Minimum power PSS enabling (P_{MIN}). Typical Value = 0.25.
ks	1..1	Simple_Float	PSS gain (K_S). Typical Value = 1.
vsmn	1..1	PU	Stabilizer output max limit (V_{SMN}). Typical Value = -0.06.
vsmx	1..1	PU	Stabilizer output min limit (V_{SMX}). Typical Value = 0.06.
tpe	1..1	Seconds	Electric power filter time constant (T_{PE}). Typical Value = 0.05.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

t5	1..1	Seconds	Washout (T_5). Typical Value = 3.5.
t6	1..1	Seconds	Filter time constant (T_6). Typical Value = 0.
t7	1..1	Seconds	Lead/lag time constant (T_7). Typical Value = 0.
t8	1..1	Seconds	Lead/lag time constant (T_8). Typical Value = 0.
t9	1..1	Seconds	Lead/lag time constant (T_9). Typical Value = 0.
t10	1..1	Seconds	Lead/lag time constant (T_{10}). Typical Value = 0.
vadat	1..1	Boolean	Signal selector (V_{adAt}). true = closed (Generator Power is greater than Pmin) false = open (Pe is smaller than Pmin). Typical Value = true.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-253 shows all association ends of PowerSystemStabilizerDynamics with other classes.

Table 9-253 - Association ends of PowerSystemStabilizerDynamics::PowerSystemStabilizerDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : PowerSystemStabilizerDynamics:: PowerSystemStabilizerDynamics.PowerSystemStabilizerDynamics[0..1] ----- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.12.8 Pss1A

Single input power system stabilizer. It is a modified version in order to allow representation of various vendors' implementations on PSS type 1A.

Figure 9-139 class diagram

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

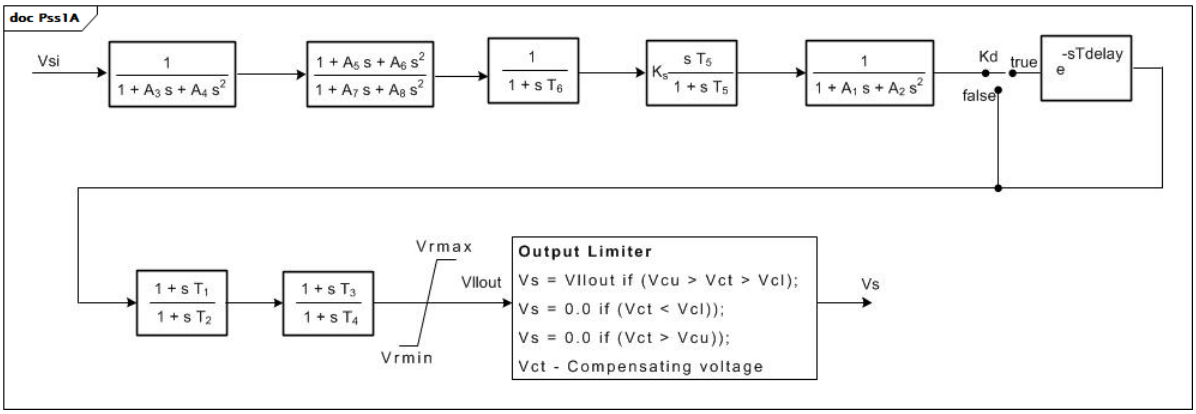


Figure 9-139 - figure Pss1A

Inheritance path: ->PowerSystemStabilizerDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-254 shows all attributes of Pss1A.

Table 9-254 - Attributes of PowerSystemStabilizerDynamics::Pss1A

Attribute name	mult.	Attribute type	Description
inputSignalType	1..1	InputSignalKind	Type of input signal.
a1	1..1	PU	Notch filter parameter (A1).
a2	1..1	PU	Notch filter parameter (A2).
t1	1..1	Seconds	Lead/lag time constant (T1).
t2	1..1	Seconds	Lead/lag time constant (T2).
t3	1..1	Seconds	Lead/lag time constant (T3).
t4	1..1	Seconds	Lead/lag time constant (T4).
t5	1..1	Seconds	Washout time constant (T5).
t6	1..1	Seconds	Transducer time constant (T6).
ks	1..1	PU	Stabilizer gain (Ks).
vrmax	1..1	PU	Maximum stabilizer output (Vrmax).
vrmin	1..1	PU	Minimum stabilizer output (Vrmin).
vcu	1..1	PU	Stabilizer input cutoff threshold (Vcu).
vcl	1..1	PU	Stabilizer input cutoff threshold (Vcl).
a3	1..1	PU	Notch filter parameter (A3).

a4	1..1	PU	Notch filter parameter (A4).
a5	1..1	PU	Notch filter parameter (A5).
a6	1..1	PU	Notch filter parameter (A6).
a7	1..1	PU	Notch filter parameter (A7).
a8	1..1	PU	Notch filter parameter (A8).
kd	1..1	Boolean	Selector (Kd). true = $e^{-sTdelay}$ used false = $e^{-sTdelay}$ not used.
tdelay	1..1	Seconds	Time constant (Tdelay).
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-255 shows all association ends of PowerSystemStabilizerDynamics with other classes.

Table 9-255 - Association ends of PowerSystemStabilizerDynamics::PowerSystemStabilizerDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : PowerSystemStabilizerDynamics:: PowerSystemStabilizerDynamics.PowerSystemStabilizerDynamics[0..1] ----- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.12.9 Pss2B

Modified IEEE PSS2B Model. Extra lead/lag (or rate) block added at end (up to 4 lead/lags total).

Figure 9-140 class diagram

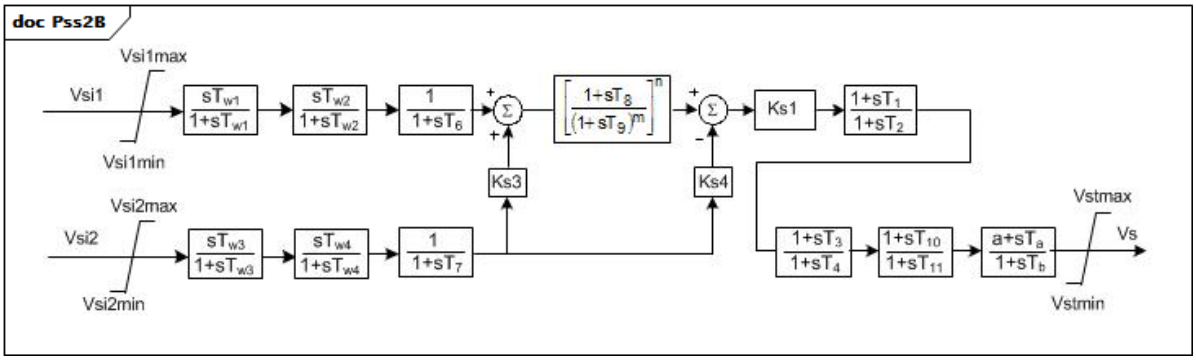


Figure 9-140 - figure Pss2B

Inheritance path: ->PowerSystemStabilizerDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-256 shows all attributes of Pss2B.

Table 9-256 - Attributes of PowerSystemStabilizerDynamics::Pss2B

Attribute name	mult.	Attribute type	Description
inputSignal1Type	1..1	InputSignalKind	Type of input signal #1. Typical Value = rotorSpeed.
inputSignal2Type	1..1	InputSignalKind	Type of input signal #2. Typical Value = generatorElectricalPower.
vsi1max	1..1	PU	Input signal #1 max limit (Vsi1max). Typical Value = 2.
vsi1min	1..1	PU	Input signal #1 min limit (Vsi1min). Typical Value = -2.
tw1	1..1	Seconds	First washout on signal #1 (Tw1). Typical Value = 2.
tw2	1..1	Seconds	Second washout on signal #1 (Tw2). Typical Value = 2.
vsi2max	1..1	PU	Input signal #2 max limit (Vsi2max). Typical Value = 2.
vsi2min	1..1	PU	Input signal #2 min limit (Vsi2min). Typical Value = -2.
tw3	1..1	Seconds	First washout on signal #2 (Tw3). Typical Value = 2.
tw4	1..1	Seconds	Second washout on signal #2 (Tw4). Typical Value = 0.
t1	1..1	Seconds	Lead/lag time constant (T1). Typical Value = 0.12.
t2	1..1	Seconds	Lead/lag time constant (T2). Typical Value = 0.02.
t3	1..1	Seconds	Lead/lag time constant (T3). Typical Value = 0.3.
t4	1..1	Seconds	Lead/lag time constant (T4). Typical Value = 0.02.
t6	1..1	Seconds	Time constant on signal #1 (T6). Typical Value = 0.
t7	1..1	Seconds	Time constant on signal #2 (T7). Typical Value =

			2.
t8	1..1	Seconds	Lead of ramp tracking filter (T8). Typical Value = 0.2.
t9	1..1	Seconds	Lag of ramp tracking filter (T9). Typical Value = 0.1.
t10	1..1	Seconds	Lead/lag time constant (T10). Typical Value = 0.
t11	1..1	Seconds	Lead/lag time constant (T11). Typical Value = 0.
ks1	1..1	PU	Stabilizer gain (Ks1). Typical Value = 12.
ks2	1..1	PU	Gain on signal #2 (Ks2). Typical Value = 0.2.
ks3	1..1	PU	Gain on signal #2 input before ramp-tracking filter (Ks3). Typical Value = 1.
ks4	1..1	PU	Gain on signal #2 input after ramp-tracking filter (Ks4). Typical Value = 1.
n	1..1	Integer	Order of ramp tracking filter (N). Typical Value = 1.
m	1..1	Integer	Denominator order of ramp tracking filter (M). Typical Value = 5.
vstmax	1..1	PU	Stabilizer output max limit (Vstmax). Typical Value = 0.1.
vstmin	1..1	PU	Stabilizer output min limit (Vstmin). Typical Value = -0.1.
a	1..1	Simple_Float	Numerator constant (a). Typical Value = 1.
ta	1..1	Seconds	Lead constant (Ta). Typical Value = 0.
tb	1..1	Seconds	Lag time constant (Tb). Typical Value = 0.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-257 shows all association ends of PowerSystemStabilizerDynamics with other classes.

Table 9-257 - Association ends of PowerSystemStabilizerDynamics::PowerSystemStabilizerDynamics with other classes

mult.	mult.	Class type name	Description

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : PowerSystemStabilizerDynamics:: PowerSystemStabilizerDynamics.PowerSystemStabilizerDynamics[0..1] --- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.12.10 Pss2ST

PTI Microprocessor-Based Stabilizer type 1.

Figure 9-141 class diagram

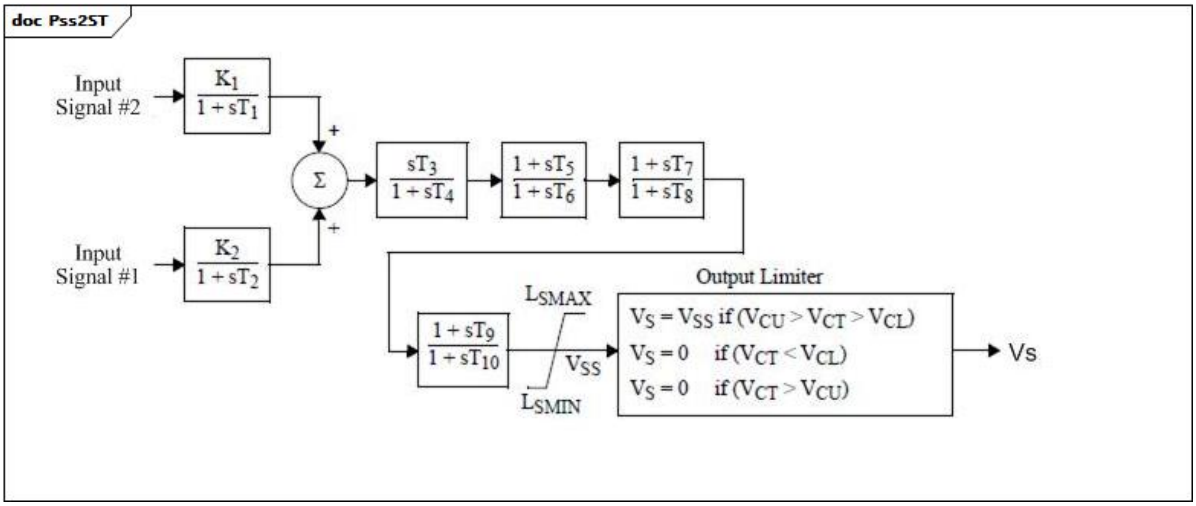


Figure 9-141 - figure Pss2ST

Inheritance path: ->PowerSystemStabilizerDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-258 shows all attributes of Pss2ST.

Table 9-258 - Attributes of PowerSystemStabilizerDynamics::Pss2ST

Attribute name	mult.	Attribute type	Description
inputSignal1Type	1..1	InputSignalKind	Type of input signal #1. Typical Value = rotorAngularFrequencyDeviation.
inputSignal2Type	1..1	InputSignalKind	Type of input signal #2. Typical Value = generatorElectricalPower.
k1	1..1	PU	Gain (K1).
k2	1..1	PU	Gain (K2).
t1	1..1	Seconds	Time constant (T1).
t2	1..1	Seconds	Time constant (T2).
t3	1..1	Seconds	Time constant (T3).
t4	1..1	Seconds	Time constant (T4).
t5	1..1	Seconds	Time constant (T5).

t6	1..1	Seconds	Time constant (T6).
t7	1..1	Seconds	Time constant (T7).
t8	1..1	Seconds	Time constant (T8).
t9	1..1	Seconds	Time constant (T9).
t10	1..1	Seconds	Time constant (T10).
lsmax	1..1	PU	Limiter (Lsmax).
lsmin	1..1	PU	Limiter (Lsmin).
vcu	1..1	PU	Cutoff limiter (Vcu).
vcl	1..1	PU	Cutoff limiter (Vcl).
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-259 shows all association ends of PowerSystemStabilizerDynamics with other classes.

Table 9-259 - Association ends of PowerSystemStabilizerDynamics::PowerSystemStabilizerDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : PowerSystemStabilizerDynamics:: PowerSystemStabilizerDynamics.PowerSystemStabilizerDynamics[0..1] --- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.12.11 Pss5

Italian PSS - Detailed PSS.

Figure 9-142 class diagram

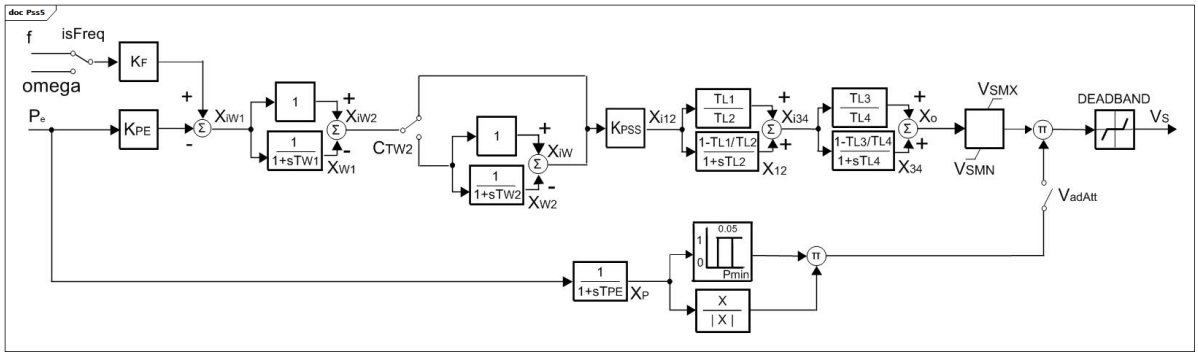


Figure 9-142 - figure Pss5

Parameter Notes:

1. The block [x/|x|] is used when **Pe** is negative, for example for a pumping unit: the output of the block [x/|x|] is the sign of **Pe**.

Inheritance path: ->PowerSystemStabilizerDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-260 shows all attributes of Pss5.

Table 9-260 - Attributes of PowerSystemStabilizerDynamics::Pss5

Attribute name	mult.	Attribute type	Description
kpe	1..1	Simple_Float	Electric power input gain (K_{PE}). Typical Value = 0.3.
kf	1..1	Simple_Float	Frequency/shaft speed input gain (K_F). Typical Value = 5.
isfreq	1..1	Boolean	Selector for Frequency/shaft speed input (IsFreq). true = speed false = frequency. Typical Value = true.
kpss	1..1	Simple_Float	PSS gain (K_{PSS}). Typical Value = 1.
ctw2	1..1	Boolean	Selector for Second washout enabling (C_{TW2}). true = second washout filter is bypassed false = second washout filter in use. Typical Value = true.
tw1	1..1	Seconds	First WashOut (T_{w1}). Typical Value = 3.5.
tw2	1..1	Seconds	Second WashOut (T_{w2}). Typical Value = 0.
tl1	1..1	Seconds	Lead/lag time constant (T_{L1}). Typical Value = 0.
tl2	1..1	Seconds	Lead/lag time constant (T_{L2}). Typical Value = 0.
tl3	1..1	Seconds	Lead/lag time constant (T_{L3}). Typical Value = 0.
tl4	1..1	Seconds	Lead/lag time constant (T_{L4}). Typical Value = 0.
vsmn	1..1	PU	Stabilizer output max limit (V_{SMN}). Typical Value = -0.1.
vsmx	1..1	PU	Stabilizer output min limit (V_{SMX}). Typical Value = 0.1.
tpe	1..1	Seconds	Electric power filter time constant (T_{PE}). Typical Value = 0.05.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

pmm	1..1	PU	Minimum power PSS enabling (P_{mn}). Typical Value = 0.25.
deadband	1..1	PU	Stabilizer output dead band (DeadBand). Typical Value = 0.
vadat	1..1	Boolean	Signal selector (V_{adAtt}). true = closed (Generator Power is greater than Pmin) false = open (Pe is smaller than Pmin). Typical Value = true.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-261 shows all association ends of PowerSystemStabilizerDynamics with other classes.

Table 9-261 - Association ends of PowerSystemStabilizerDynamics::PowerSystemStabilizerDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : PowerSystemStabilizerDynamics:: PowerSystemStabilizerDynamics.PowerSystemStabilizerDynamics[0..1] ----- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.12.12 PssELIN2

Power system stabilizer typically associated with ExcELIN2 (though PssIEEE2B or Pss2B can also be used).

Figure 9-143 class diagram

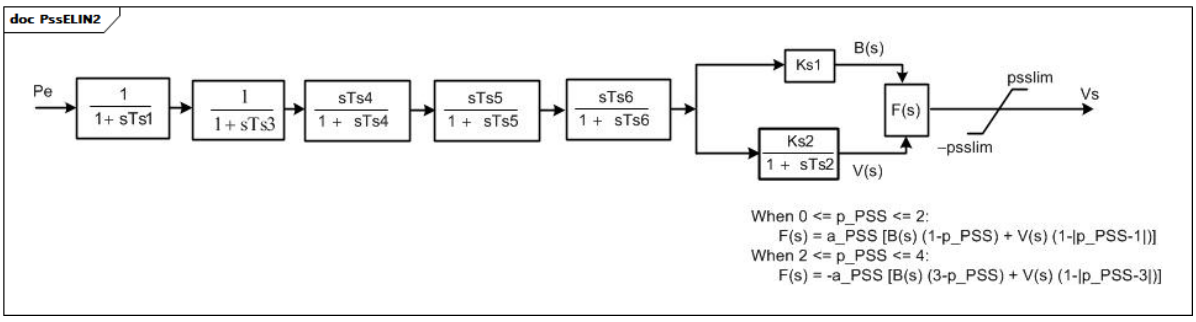


Figure 9-143 - figure PssELIN2

Inheritance path: ->PowerSystemStabilizerDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-262 shows all attributes of PssELIN2.

Table 9-262 - Attributes of PowerSystemStabilizerDynamics::PssELIN2

Attribute name	mult.	Attribute type	Description
ts1	1..1	Seconds	Time constant (Ts1). Typical Value = 0.
ts2	1..1	Seconds	Time constant (Ts2). Typical Value = 1.
ts3	1..1	Seconds	Time constant (Ts3). Typical Value = 1.
ts4	1..1	Seconds	Time constant (Ts4). Typical Value = 0.1.
ts5	1..1	Seconds	Time constant (Ts5). Typical Value = 0.
ts6	1..1	Seconds	Time constant (Ts6). Typical Value = 1.
ks1	1..1	PU	Gain (Ks1). Typical Value = 1.
ks2	1..1	PU	Gain (Ks2). Typical Value = 0.1.
ppss	1..1	PU	Coefficient (p_PSS) (>=0 and <=4). Typical Value = 0.1.
apss	1..1	PU	Coefficient (a_PSS). Typical Value = 0.1.
psslim	1..1	PU	PSS limiter (psslim). Typical Value = 0.1.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-263 shows all association ends of PowerSystemStabilizerDynamics with other classes.

Table 9-263 - Association ends of PowerSystemStabilizerDynamics::PowerSystemStabilizerDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : PowerSystemStabilizerDynamics:: PowerSystemStabilizerDynamics.PowerSystemStabilizerDynamics[0..1] ----- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.12.13 PssPTIST1

PTI Microprocessor-Based Stabilizer type 1.

Figure 9-144 class diagram

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

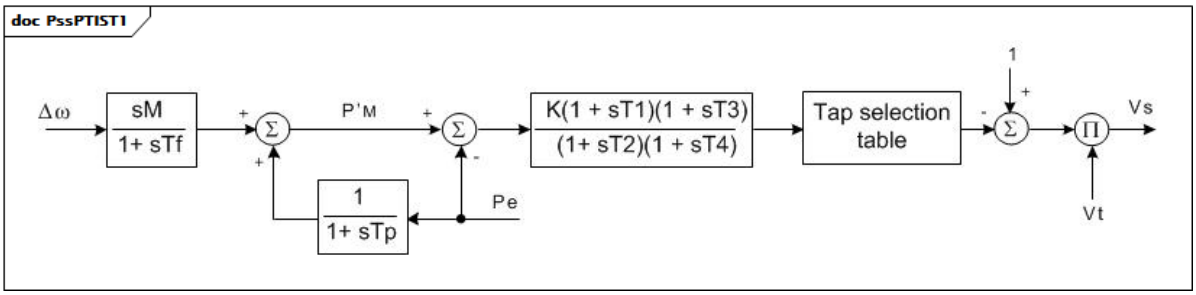


Figure 9-144 - figure PssPTIST1

Parameter Notes:

1. Tap selection table – the output control signal is used to select an autotransformer tap position, which, in turn, modulates the feedback terminal voltage (digital output option).

Inheritance path: ->PowerSystemStabilizerDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-264 shows all attributes of PssPTIST1.

Table 9-264 - Attributes of PowerSystemStabilizerDynamics::PssPTIST1

Attribute name	mult.	Attribute type	Description
m	1..1	PU	(M). M=2*H. Typical Value = 5.
tf	1..1	Seconds	Time constant (Tf). Typical Value = 0.2.
tp	1..1	Seconds	Time constant (Tp). Typical Value = 0.2.
t1	1..1	Seconds	Time constant (T1). Typical Value = 0.3.
t2	1..1	Seconds	Time constant (T2). Typical Value = 1.
t3	1..1	Seconds	Time constant (T3). Typical Value = 0.2.
t4	1..1	Seconds	Time constant (T4). Typical Value = 0.05.
k	1..1	PU	Gain (K). Typical Value = 9.
dtf	1..1	Seconds	Time step frequency calculation (Dtf). Typical Value = 0.025.
dtc	1..1	Seconds	Time step related to activation of controls (Dtc). Typical Value = 0.025.
dtp	1..1	Seconds	Time step active power calculation (Dtp). Typical Value = 0.0125.
enabled	1..1	Boolean	see

			DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-265 shows all association ends of PowerSystemStabilizerDynamics with other classes.

Table 9-265 - Association ends of PowerSystemStabilizerDynamics::PowerSystemStabilizerDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : PowerSystemStabilizerDynamics:: PowerSystemStabilizerDynamics.PowerSystemStabilizerDynamics[0..1] --- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.12.14 PssPTIST3

PTI Microprocessor-Based Stabilizer type 3.

Figure 9-145 class diagram

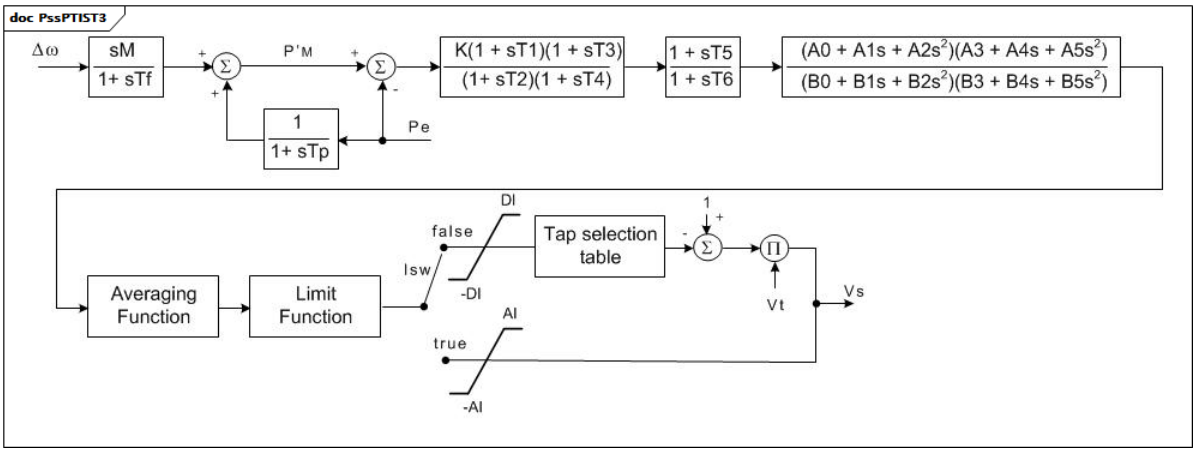


Figure 9-145 - figure PssPTIST3

Parameter Notes:

1. Tap selection table – the output control signal is used to select an autotransformer tap position, which, in turn, modulates the feedback terminal voltage (digital output option).
2. Averaging function – The tap averaging function keeps a running average of the last, NAV, control outputs (clicks) up to a maximum of 16. This feature can be disabled by setting NAV equal to 1 (i.e., average 1 output). The last NAV outputs are stored in the output averaging table, VAR(1) through VAR(16). The sum of the last outputs is stored in the output accumulator. The averaging feature was added to reduce tap chattering under steady-state conditions. If the control output at any given click exceeds the threshold specified in, Athres, the averaging function will be automatically bypassed, and that output

signal will be passed directly to the limit function. This feature has been added such that under transient conditions the stabilizer reverts to normal operation. When the absolute value of output signal is beyond the threshold, the output averaging table entries and accumulator are zeroed out. When the signal falls below the threshold, the table is rebuilt starting with the first signal within the threshold. If the output averaging is enabled, then the threshold value would probably be specified in the range from 0.002 to 0.02.

3. Limit function – Counts the number of control outputs that the output signal continuously exceeds a user-defined threshold as set in Lthres and After the output signal has continuously exceeded the threshold for a number of user defined counts as specified by NCL, the function ramps the tap position to nominal or the analog output to zero over a user-specified number of counts as defined in NCR.

Inheritance path: ->PowerSystemStabilizerDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-266 shows all attributes of PssPTIST3.

Table 9-266 - Attributes of PowerSystemStabilizerDynamics::PssPTIST3

Attribute name	mult.	Attribute type	Description
m	1..1	PU	(M). M=2*H. Typical Value = 5.
tf	1..1	Seconds	Time constant (Tf). Typical Value = 0.2.
tp	1..1	Seconds	Time constant (Tp). Typical Value = 0.2.
t1	1..1	Seconds	Time constant (T1). Typical Value = 0.3.
t2	1..1	Seconds	Time constant (T2). Typical Value = 1.
t3	1..1	Seconds	Time constant (T3). Typical Value = 0.2.
t4	1..1	Seconds	Time constant (T4). Typical Value = 0.05.
k	1..1	PU	Gain (K). Typical Value = 9.
dtf	1..1	Seconds	Time step frequency calculation (0.03 for 50 Hz) (Dtf). Typical Value = 0.025.
dtc	1..1	Seconds	Time step related to activation of controls (0.03 for 50 Hz) (Dtc). Typical Value = 0.025.
dtp	1..1	Seconds	Time step active power calculation (0.015 for 50 Hz) (Dtp). Typical Value = 0.0125.
t5	1..1	Seconds	Time constant (T5).
t6	1..1	Seconds	Time constant (T6).
a0	1..1	PU	Filter coefficient (A0).
a1	1..1	PU	Limiter (Al).
a2	1..1	PU	Filter coefficient (A2).

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

b0	1..1	PU	Filter coefficient (B0).
b1	1..1	PU	Filter coefficient (B1).
b2	1..1	PU	Filter coefficient (B2).
a3	1..1	PU	Filter coefficient (A3).
a4	1..1	PU	Filter coefficient (A4).
a5	1..1	PU	Filter coefficient (A5).
b3	1..1	PU	Filter coefficient (B3).
b4	1..1	PU	Filter coefficient (B4).
b5	1..1	PU	Filter coefficient (B5).
athres	1..1	PU	Threshold value above which output averaging will be bypassed (Athres). Typical Value = 0.005.
dl	1..1	PU	Limiter (Dl).
al	1..1	PU	Limiter (Al).
lthres	1..1	PU	Threshold value (Lthres).
pmin	1..1	PU	(Pmin).
isw	1..1	Boolean	Digital/analog output switch (Isw). true = produce analog output false = convert to digital output, using tap selection table.
nav	1..1	Simple_Float	Number of control outputs to average (Nav) (1 <= Nav <= 16). Typical Value = 4.
ncl	1..1	Simple_Float	Number of counts at limit to active limit function (Ncl) (>0).
ncr	1..1	Simple_Float	Number of counts until reset after limit function is triggered (Ncr).
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-267 shows all association ends of PowerSystemStabilizerDynamics with other classes.

Table 9-267 - Association ends of PowerSystemStabilizerDynamics::PowerSystemStabilizerDynamics with other classes

mult.	mult.	Class type name	Description

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : PowerSystemStabilizerDynamics:: PowerSystemStabilizerDynamics.PowerSystemStabilizerDynamics[0..1] --- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.12.15 PssSB4

Power sensitive stabilizer model.

Figure 9-146 class diagram

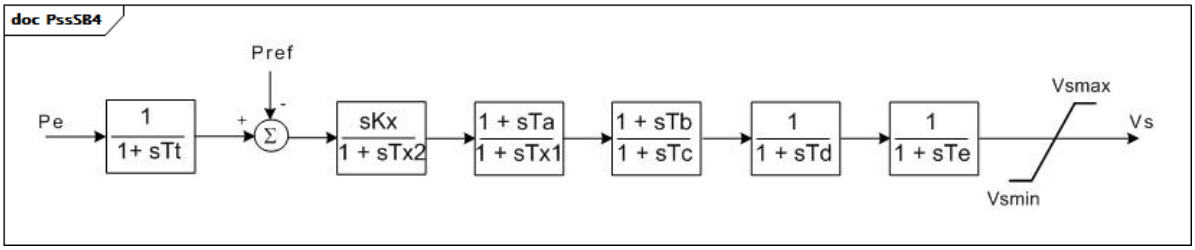


Figure 9-146 - figure PssSB4

Inheritance path: ->PowerSystemStabilizerDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-268 shows all attributes of PssSB4.

Table 9-268 - Attributes of PowerSystemStabilizerDynamics::PssSB4

Attribute name	mult.	Attribute type	Description
tt	1..1	Seconds	Time constant (Tt).
kx	1..1	PU	Gain (Kx).
tx2	1..1	Seconds	Time constant (Tx2).
ta	1..1	Seconds	Time constant (Ta).
tx1	1..1	Seconds	Reset time constant (Tx1).
tb	1..1	Seconds	Time constant (Tb).
tc	1..1	Seconds	Time constant (Tc).
td	1..1	Seconds	Time constant (Td).
te	1..1	Seconds	Time constant (Te).
vsmax	1..1	PU	Limiter (Vsmax).
vsmin	1..1	PU	Limiter (Vsmin).
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-269 shows all association ends of PowerSystemStabilizerDynamics with other classes.

Table 9-269 - Association ends of PowerSystemStabilizerDynamics::PowerSystemStabilizerDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : PowerSystemStabilizerDynamics:: PowerSystemStabilizerDynamics.PowerSystemStabilizerDynamics[0..1] --- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.12.16 PssSH

Model for Siemens “H infinity” power system stabilizer with generator electrical power input.

Figure 9-147 class diagram

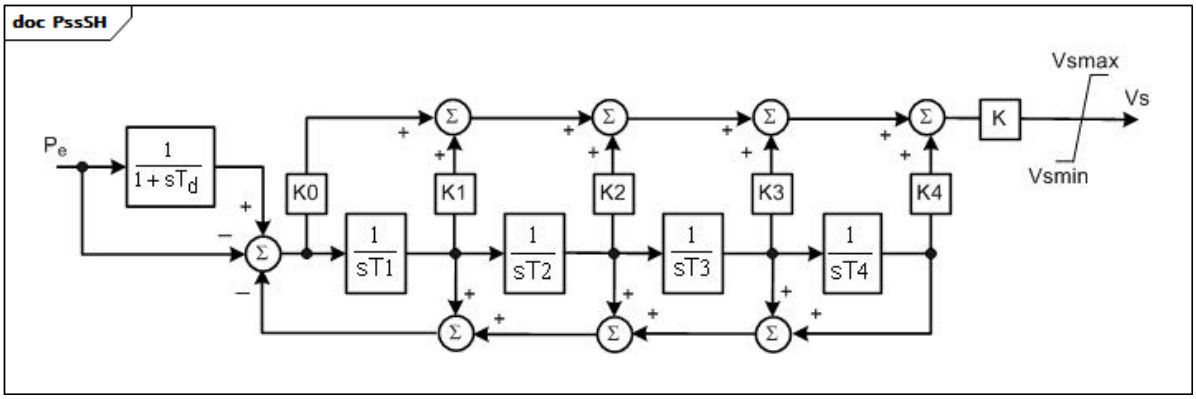


Figure 9-147 - figure PssSH

Inheritance path: ->PowerSystemStabilizerDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-270 shows all attributes of PssSH.

Table 9-270 - Attributes of PowerSystemStabilizerDynamics::PssSH

Attribute name	mult.	Attribute type	Description
k	1..1	PU	Main gain (K). Typical Value = 1.
k0	1..1	PU	Gain 0 (K0). Typical Value = 0.012.
k1	1..1	PU	Gain 1 (K1). Typical Value = 0.488.
k2	1..1	PU	Gain 2 (K2). Typical Value = 0.064.
k3	1..1	PU	Gain 3 (K3). Typical Value = 0.224.
k4	1..1	PU	Gain 4 (K4). Typical Value = 0.1.
td	1..1	Seconds	Input time constant (Td). Typical Value =

			10.
t1	1..1	Seconds	Time constant 1 (T1). Typical Value = 0.076.
t2	1..1	Seconds	Time constant 2 (T2). Typical Value = 0.086.
t3	1..1	Seconds	Time constant 3 (T3). Typical Value = 1.068.
t4	1..1	Seconds	Time constant 4 (T4). Typical Value = 1.913.
vsmax	1..1	PU	Output maximum limit (Vsmax). Typical Value = 0.1.
vsmin	1..1	PU	Output minimum limit (Vsmin). Typical Value = -0.1.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-271 shows all association ends of PowerSystemStabilizerDynamics with other classes.

Table 9-271 - Association ends of PowerSystemStabilizerDynamics::PowerSystemStabilizerDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : PowerSystemStabilizerDynamics:: PowerSystemStabilizerDynamics.PowerSystemStabilizerDynamics[0..1] ----- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.12.17 PssSK

PSS Slovakian type – three inputs.

Figure 9-148 class diagram

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

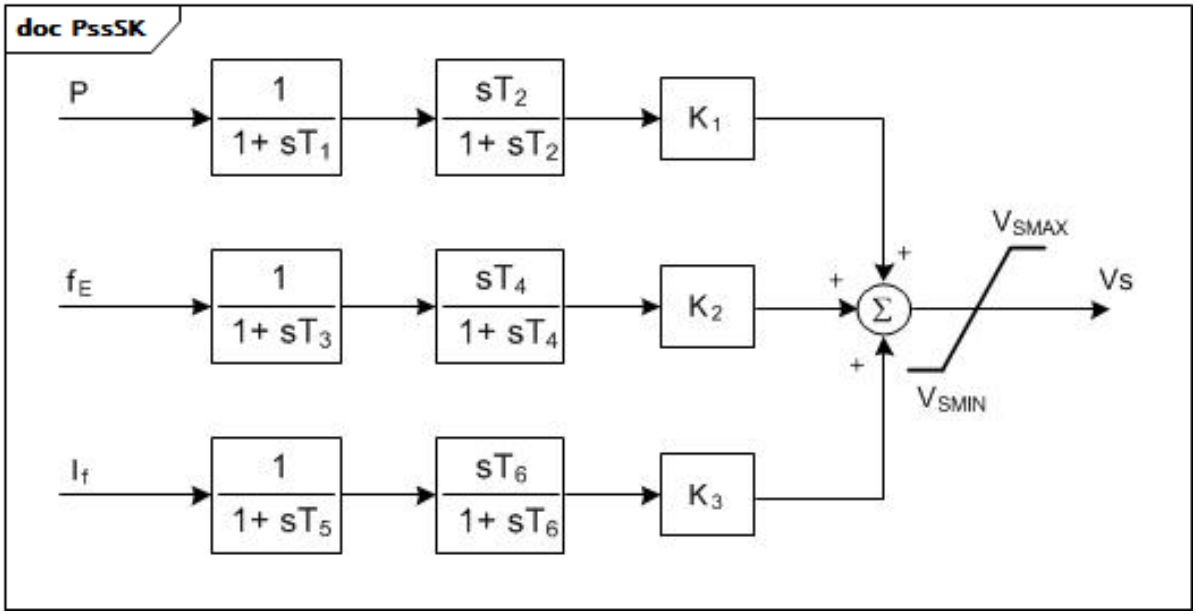


Figure 9-148 - figure PssSK

Parameter Notes:

1. **T1**, **T2**, **T3**, **T4**, **T5** and **T6** must be greater than 0.005 sec.

Inheritance path: ->PowerSystemStabilizerDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-272 shows all attributes of PssSK.

Table 9-272 - Attributes of PowerSystemStabilizerDynamics::PssSK

Attribute name	mult.	Attribute type	Description
k1	1..1	PU	Gain P (K1). Typical Value = -0.3.
k2	1..1	PU	Gain fe (K2). Typical Value = -0.15.
k3	1..1	PU	Gain If (K3). Typical Value = 10.
t1	1..1	Seconds	Denominator time constant (T1). Typical Value = 0.3.
t2	1..1	Seconds	Filter time constant (T2). Typical Value = 0.35.
t3	1..1	Seconds	Denominator time constant (T3). Typical Value = 0.22.
t4	1..1	Seconds	Filter time constant (T4). Typical Value = 0.02.
t5	1..1	Seconds	Denominator time

			constant (T5). Typical Value = 0.02.
t6	1..1	Seconds	Filter time constant (T6). Typical Value = 0.02.
vsmax	1..1	PU	Stabilizer output max limit (Vsmax). Typical Value = 0.4.
vsmin	1..1	PU	Stabilizer output min limit (Vsmin). Typical Value = -0.4.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-273 shows all association ends of PowerSystemStabilizerDynamics with other classes.

Table 9-273 - Association ends of PowerSystemStabilizerDynamics::PowerSystemStabilizerDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : PowerSystemStabilizerDynamics:: PowerSystemStabilizerDynamics.PowerSystemStabilizerDynamics[0..1] ----- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.12.18 PssWECC

Dual input Power System Stabilizer, based on IEEE type 2, with modified output limiter defined by WECC (Western Electricity Coordinating Council, USA).

Figure 9-149 class diagram

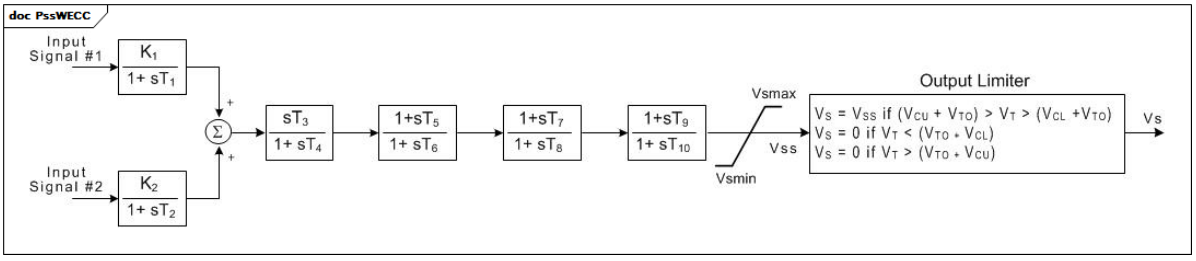


Figure 9-149 - figure PssWECC

Notes:

1. V_T is the terminal voltage.
2. V_{TO} is the initial terminal voltage.

Inheritance path: ->PowerSystemStabilizerDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-274 shows all attributes of PssWECC.

Table 9-274 - Attributes of PowerSystemStabilizerDynamics::PssWECC

Attribute name	mult.	Attribute type	Description
inputSignal1Type	1..1	InputSignalKind	Type of input signal #1.
inputSignal2Type	1..1	InputSignalKind	Type of input signal #2.
k1	1..1	PU	Input signal 1 gain (K_1).
t1	1..1	Seconds	Input signal 1 transducer time constant (T_1).
k2	1..1	PU	Input signal 2 gain (K_2).
t2	1..1	Seconds	Input signal 2 transducer time constant (T_2).
t3	1..1	Seconds	Stabilizer washout time constant (T_3).
t4	1..1	Seconds	Stabilizer washout time lag constant (T_4) (>0).
t5	1..1	Seconds	Lead time constant (T_5).
t6	1..1	Seconds	Lag time constant (T_6).
t7	1..1	Seconds	Lead time constant (T_7).
t8	1..1	Seconds	Lag time constant (T_8).
t10	1..1	Seconds	Lag time constant (T_{10}).
t9	1..1	Seconds	Lead time constant (T_9).
vsmax	1..1	PU	Maximum output signal (Vsmax).
vsmin	1..1	PU	Minimum output signal (Vsmin).
vcu	1..1	PU	Maximum value for voltage compensator output (V_{CU}).
vcl	1..1	PU	Minimum value for voltage compensator output (V_{CL}).
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-275 shows all association ends of PowerSystemStabilizerDynamics with other classes.

Table 9-275 - Association ends of PowerSystemStabilizerDynamics::PowerSystemStabilizerDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : PowerSystemStabilizerDynamics:: PowerSystemStabilizerDynamics.PowerSystemStabilizerDynamics[0..1] --- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.12.19 Enumerations

9.6.12.19.1 InputSignalKind

Input signal type. In Dynamics modelling, commonly represented by j parameter.

Literal	Description	Code
rotorSpeed	Input signal is rotor or shaft speed (angular frequency).	
rotorAngularFrequencyDeviation	Input signal is rotor or shaft angular frequency deviation.	
busFrequency	Input signal is bus voltage frequency. This could be a terminal frequency or remote frequency.	
busFrequencyDeviation	Input signal is deviation of bus voltage frequency. This could be a terminal frequency deviation or remote frequency deviation.	
generatorElectricalPower	Input signal is generator electrical power on rated S.	
generatorAcceleratingPower	Input signal is generating accelerating power.	
busVoltage	Input signal is bus voltage. This could be a terminal voltage or remote voltage.	
busVoltageDerivative	Input signal is derivative of bus voltage. This could be a terminal voltage derivative or remote voltage derivative.	
branchCurrent	Input signal is amplitude of remote branch current.	
fieldCurrent	Input signal is generator field current.	

9.6.13 DiscontinuousExcitationControlDynamics

9.6.13.1 General

In some particular system configurations, continuous excitation control with terminal voltage and power system stabilizing regulator input signals does not ensure that the potential of the excitation system for improving system stability is fully exploited. For these situations, discontinuous excitation control signals may be employed to enhance stability following large transient disturbances.
For additional information please refer to IEEE Standard 421.5-2005, Section 12.

Figure 9-150 shows the diagram.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

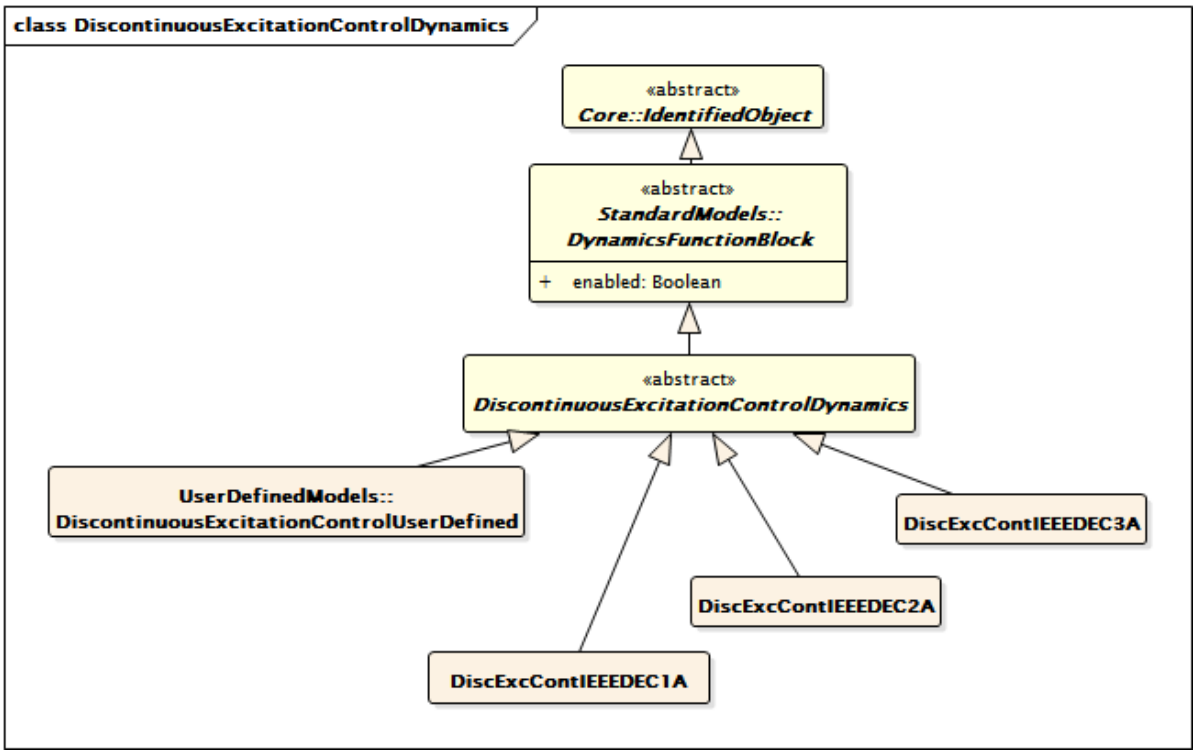


Figure 9-150 - diagram DiscontinuousExcitationControlDynamics

This diagram illustrates the inputs and outputs for a discontinuous excitation control.
Notes:

1. **Vs** is always initialized to zero by the excitation system model.
2. If bus voltage frequency is not available from network, the model can calculate it as derivative of the bus voltage angle.

Figure 9-151 shows the diagram.

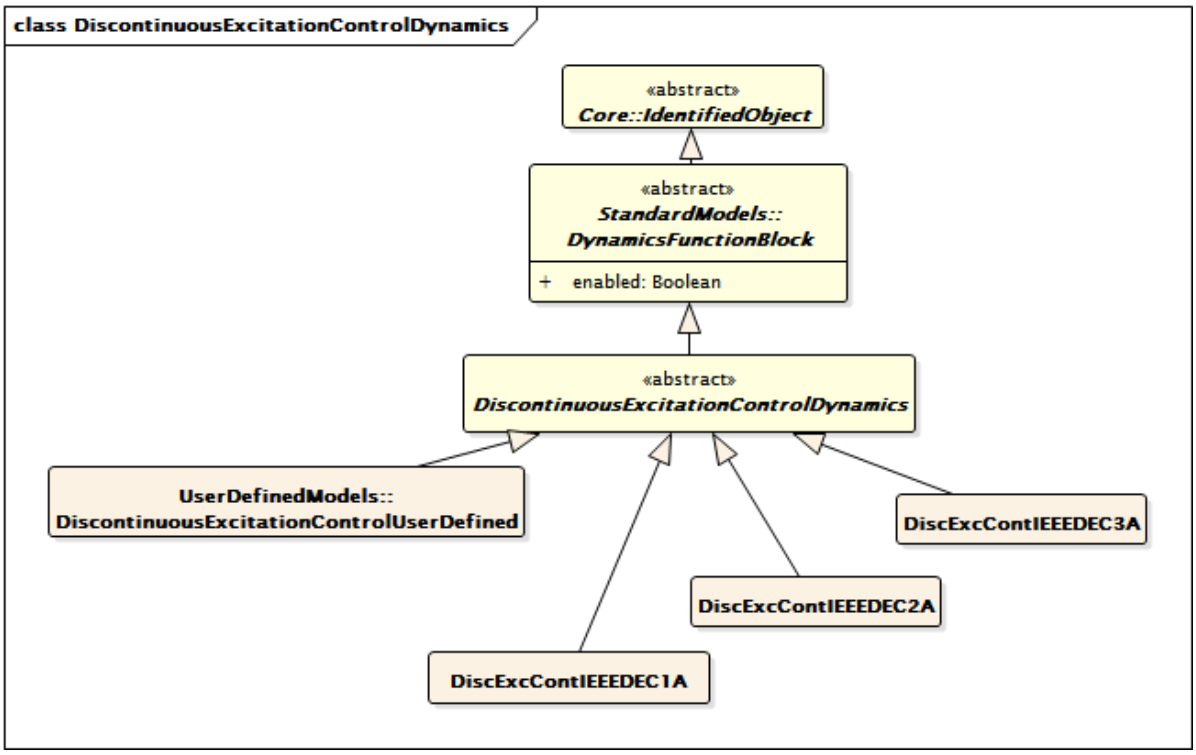


Figure 9-151 - diagram DiscontinuousExcitationControlDynamics

This diagram shows dynamic standard model discontinuous excitation control classes and their inheritance from the DiscontinuousExcitationControlDynamics class.

9.6.13.2 DiscontinuousExcitationControlDynamics(Abstract)

Discontinuous excitation control function block whose behaviour is described by reference to a standard model or by definition of a user-defined model.

Inheritance path: ->DynamicsFunctionBlock->IdentifiedObject

Table 9-276 shows all attributes of DiscontinuousExcitationControlDynamics.

Table 9-276 - Attributes of DiscontinuousExcitationControlDynamics::DiscontinuousExcitationControlDynamics

Attribute name	mult.	Attribute type	Description
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-277 shows all association ends of DiscontinuousExcitationControlDynamics with other classes.

Table 9-277 - Association ends of DiscontinuousExcitationControlDynamics::DiscontinuousExcitationControlDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	[1..1]	ExcitationSystemDynamics	Excitation system model with which this discontinuous excitation control model is associated. Association Based On : DiscontinuousExcitationControlDynamics:: DiscontinuousExcitationControlDynamics.DiscontinuousExcitationControlDynamics[0..1] ----- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.13.3 DiscExcContIEEEDEC1A

The class represents IEEE Type DEC1A discontinuous excitation control model that boosts generator excitation to a level higher than that demanded by the voltage regulator and stabilizer immediately following a system fault. Reference: IEEE Standard 421.5-2005 Section 12.2.

Inheritance path: ->DiscontinuousExcitationControlDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-278 shows all attributes of DiscExcContIEEEDEC1A.

Table 9-278 - Attributes of DiscontinuousExcitationControlDynamics::DiscExcContIEEEDEC1A

Attribute name	mult.	Attribute type	Description
vtlmt	1..1	PU	Voltage reference (V_{TLMT}). Typical Value = 1.1.
vomax	1..1	PU	Limiter (V_{OMAX}). Typical Value = 0.3.
vomin	1..1	PU	Limiter (V_{OMIN}). Typical Value = 0.1.
ketl	1..1	PU	Terminal voltage limiter gain (K_{ETL}). Typical Value = 47.
vtc	1..1	PU	Terminal voltage level reference (V_{TC}). Typical Value = 0.95.
val	1..1	PU	Regulator voltage reference (V_{AL}). Typical Value = 5.5.
esc	1..1	PU	Speed change reference (E_{SC}). Typical Value = 0.0015.
kan	1..1	PU	Discontinuous controller gain (K_{AN}). Typical Value = 400.
tan	1..1	Seconds	Discontinuous controller time constant (T_{AN}). Typical Value = 0.08.
tw5	1..1	Seconds	DEC washout time

			constant (T_{W5}). Typical Value = 5.
vsmax	1..1	PU	Limiter (V_{SMAX}). Typical Value = 0.2.
vsmin	1..1	PU	Limiter (V_{SMIN}). Typical Value = -0.066.
td	1..1	Seconds	Time constant (T_D). Typical Value = 0.03.
tl1	1..1	Seconds	Time constant (T_{L1}). Typical Value = 0.025.
tl2	1..1	Seconds	Time constant (T_{L2}). Typical Value = 1.25.
vtm	1..1	PU	Voltage limits (V_{TM}). Typical Value = 1.13.
vtn	1..1	PU	Voltage limits (V_{TN}). Typical Value = 1.12.
vanmax	1..1	PU	Limiter for Van (V_{ANMAX}).
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-279 shows all association ends of DiscontinuousExcitationControlDynamics with other classes.

Table 9-279 - Association ends of DiscontinuousExcitationControlDynamics::DiscontinuousExcitationControlDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : DiscontinuousExcitationControlDynamics::DiscontinuousExcitationControlDynamics.DiscontinuousExcitationControlDynamics[0..1] --- ExcitationSystemDynamics::ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.13.4 DiscExcContIEEEDEC2A

The class represents IEEE Type DEC2A model for the discontinuous excitation control. This system provides transient excitation boosting via an open-loop control as initiated by a trigger signal generated remotely.
Reference: IEEE Standard 421.5-2005 Section 12.3.

Inheritance path: ->DiscontinuousExcitationControlDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-280 shows all attributes of DiscExcContIEEEDEC2A.

Table 9-280 - Attributes of DiscontinuousExcitationControlDynamics::DiscExcContIEEEDEC2A

Attribute name	mult.	Attribute type	Description
vk	1..1	PU	Discontinuous controller input reference (V_K).
td1	1..1	Seconds	Discontinuous controller time constant (T_{D1}).
td2	1..1	Seconds	Discontinuous controller washout time constant (T_{D2}).
vdmin	1..1	PU	Limiter (V_{DMIN}).
vdmax	1..1	PU	Limiter (V_{DMAX}).
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-281 shows all association ends of DiscontinuousExcitationControlDynamics with other classes.

Table 9-281 - Association ends of DiscontinuousExcitationControlDynamics::DiscontinuousExcitationControlDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : DiscontinuousExcitationControlDynamics:: DiscontinuousExcitationControlDynamics.DiscontinuousExcitationControlDynamics[0..1] --- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.13.5 DiscExcContIEEEDEC3A

The class represents IEEE Type DEC3A model. In some systems, the stabilizer output is disconnected from the regulator immediately following a severe fault to prevent the stabilizer from competing with action of voltage regulator during the first swing. Reference: IEEE Standard 421.5-2005 Section 12.4.

Inheritance path: ->DiscontinuousExcitationControlDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-282 shows all attributes of DiscExcContIEEEDEC3A.

Table 9-282 - Attributes of DiscontinuousExcitationControlDynamics::DiscExcContIEEEDEC3A

Attribute name	mult.	Attribute type	Description
vtmin	1..1	PU	Terminal undervoltage comparison level

			(V_{TMIN}).
tdr	1..1	Seconds	Reset time delay (T_{DR}).
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-283 shows all association ends of DiscontinuousExcitationControlDynamics with other classes.

Table 9-283 - Association ends of DiscontinuousExcitationControlDynamics::DiscontinuousExcitationControlDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : DiscontinuousExcitationControlDynamics::DiscontinuousExcitationControlDynamics.DiscontinuousExcitationControlDynamics[0..1] --- ExcitationSystemDynamics::ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.14 PFVArControllerType1Dynamics

9.6.14.1 General

Excitation systems for synchronous machines are sometimes supplied with an optional means of automatically adjusting generator output reactive power (VAr) or power factor (PF) to a user-specified value This can be accomplished with either a reactive power or power factor controller or regulator. A reactive power or power factor controller is defined as a PF/VAr controller in IEEE Std 421.1 as “A control function that acts through the reference adjuster to modify the voltage regulator set point to maintain the synchronous machine steady-state power factor or reactive power at a predetermined value.”
For additional information please refer to IEEE Standard 421.5-2005, Section 11.

Figure 9-152 shows the diagram.

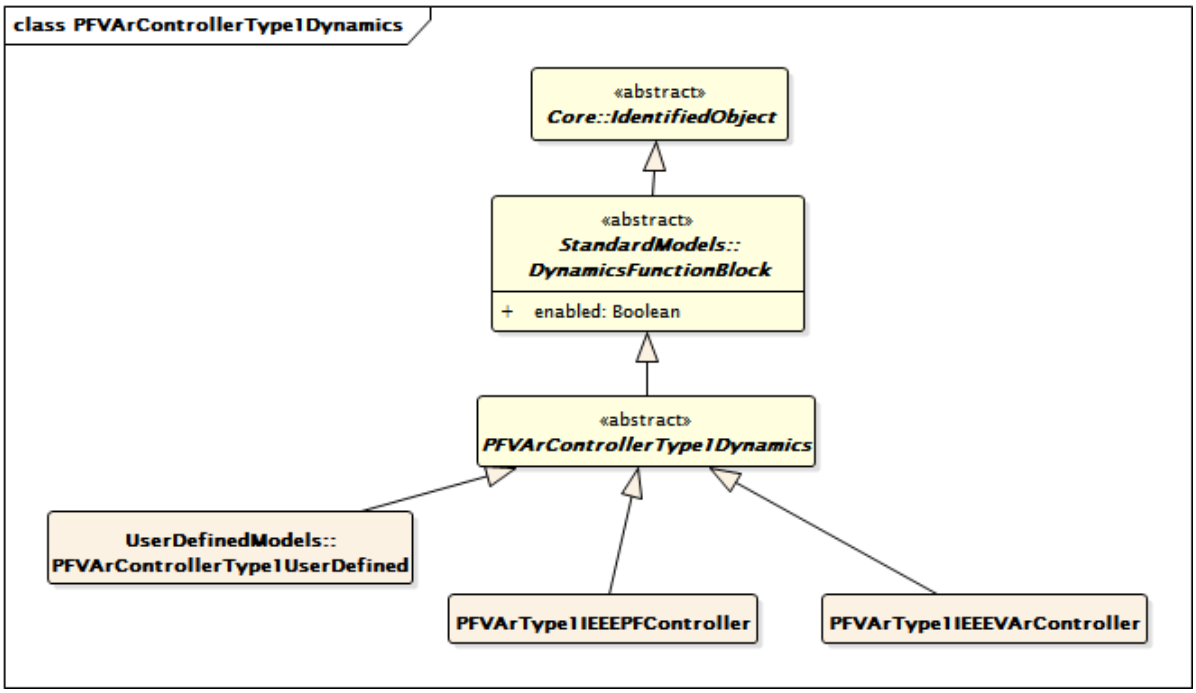


Figure 9-152 - diagram PFVArControllerType1Dynamics

This diagram shows dynamic standard model Power Factor or VAr controller Type I classes and their inheritance from the PRVArControllerTypeIDynamics class.

9.6.14.2 PFVArControllerType1Dynamics(Abstract)

Power Factor or VAr controller Type I function block whose behaviour is described by reference to a standard model or by definition of a user-defined model.

Inheritance path: ->DynamicsFunctionBlock->IdentifiedObject

Table 9-284 shows all attributes of PFVArControllerType1Dynamics.

Table 9-284 - Attributes of PFVArControllerType1Dynamics::PFVArControllerType1Dynamics

Attribute name	mult.	Attribute type	Description
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-285 shows all association ends of PFVArControllerType1Dynamics with other classes.

Table 9-285 - Association ends of PFVArControllerType1Dynamics::PFVArControllerType1Dynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	[1..1]	ExcitationSystemDynamics	Excitation system model with which this Power Factor or VAr controller Type I model is associated. Association Based On : PFVArControllerType1Dynamics::

			PFVArControllerType1Dynamics.PFVArControllerType1Dynamics [0..1] ----- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.14.3 PFVArType1IEEEPFController

The class represents IEEE PF Controller Type 1 which operates by moving the voltage reference directly.
Reference: IEEE Standard 421.5-2005 Section 11.2.

Inheritance path: ->PFVArControllerType1Dynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-286 shows all attributes of PFVArType1IEEEPFController.

Table 9-286 - Attributes of PFVArControllerType1Dynamics::PFVArType1IEEEPFController

Attribute name	mult.	Attribute type	Description
ovex	1..1	Boolean	Overexcitation Flag (*OVEX*) true = overexcited false = underexcited.
tpfc	1..1	Seconds	PF controller time delay (T_{PFC}). Typical Value = 5.
vitmin	1..1	PU	Minimum machine terminal current needed to enable pf/var controller (V_{ITMIN}).
vpf	1..1	PU	Synchronous machine power factor (V_{PF}).
vpfcbw	1..1	Simple_Float	PF controller dead band (V_{PFC_BW}). Typical Value = 0.05.
vpfref	1..1	PU	PF controller reference (V_{PFREF}).
vvtmax	1..1	PU	Maximum machine terminal voltage needed for pf/var controller to be enabled (V_{VTMAX}).
vvtmin	1..1	PU	Minimum machine terminal voltage needed to enable pf/var controller (V_{VTMIN}).
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-287 shows all association ends of PFVArControllerType1Dynamics with other classes.

Table 9-287 - Association ends of PFVArControllerType1Dynamics::PFVArControllerType1Dynamics with other classes

mult.	mult	Class type name	Description

	.		
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : PFVArControllerType1Dynamics:: PFVArControllerType1Dynamics.PFVArControllerType1Dynamics[0..1] ----- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.14.4 PFVArType1IEEEVArController

The class represents IEEE VAR Controller Type 1 which operates by moving the voltage reference directly.
Reference: IEEE Standard 421.5-2005 Section 11.3.

Inheritance path: ->PFVArControllerType1Dynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-288 shows all attributes of PFVArType1IEEEVArController.

Table 9-288 - Attributes of PFVArControllerType1Dynamics::PFVArType1IEEEVArController

Attribute name	mult.	Attribute type	Description
tvarc	1..1	Seconds	Var controller time delay (T_{VARC}). Typical Value = 5.
vvar	1..1	PU	Synchronous machine power factor (V_{VAR}).
vvarcbw	1..1	Simple_Float	Var controller dead band (V_{VARC_BW}). Typical Value = 0.02.
vvarref	1..1	PU	Var controller reference (V_{VARREF}).
vvtmax	1..1	PU	Maximum machine terminal voltage needed for pf/var controller to be enabled (V_{VTMAX}).
vvtmin	1..1	PU	Minimum machine terminal voltage needed to enable pf/var controller (V_{VTMIN}).
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-289 shows all association ends of PFVArControllerType1Dynamics with other classes.

Table 9-289 - Association ends of PFVArControllerType1Dynamics::PFVArControllerType1Dynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : PFVArControllerType1Dynamics::

			PFVArControllerType1Dynamics.PFVArControllerType1Dynamics[0..1] --- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.15 VoltageAdjusterDynamics

9.6.15.1 General

A voltage adjuster is a reference adjuster that uses inputs from a reactive power or power factor controller to modify the voltage regulator set point to maintain the synchronous machine steady-state power factor or reactive power at a predetermined value. For additional information please refer to IEEE Standard 421.5-2005, Section 11.

Figure 9-153 shows the diagram.

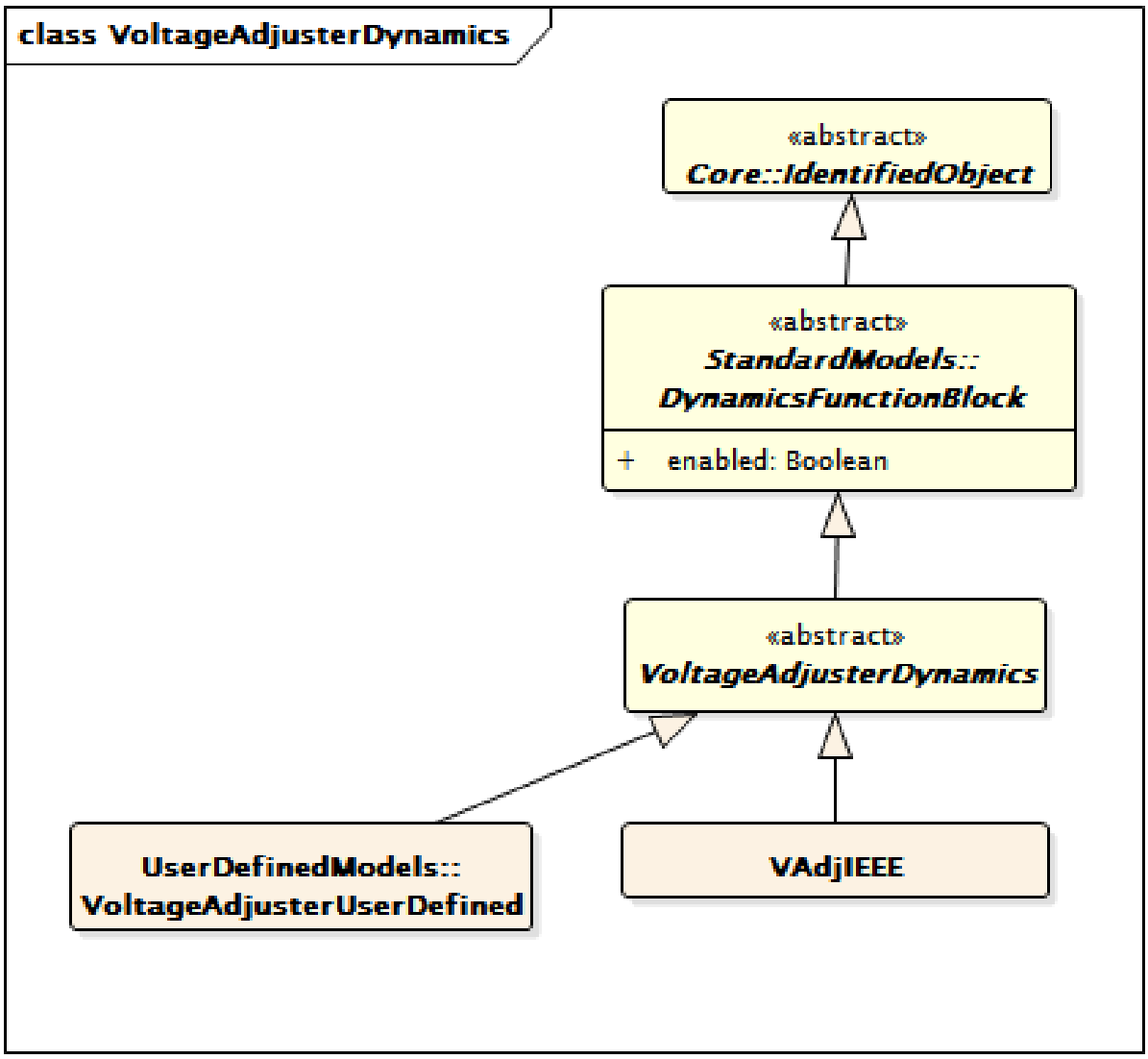


Figure 9-153 - diagram VoltageAdjusterDynamics

This diagram shows dynamic standard model voltage adjuster classes and their inheritance from the VoltageAdjusterDynamics class.

9.6.15.2 VoltageAdjusterDynamics(Abstract)

Voltage adjuster function block whose behaviour is described by reference to a standard model or by definition of a user-defined model.

Inheritance path: ->DynamicsFunctionBlock->IdentifiedObject

Table 9-290 shows all attributes of VoltageAdjusterDynamics.

Table 9-290 - Attributes of VoltageAdjusterDynamics::VoltageAdjusterDynamics

Attribute name	mult.	Attribute type	Description
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-291 shows all association ends of VoltageAdjusterDynamics with other classes.

Table 9-291 - Association ends of VoltageAdjusterDynamics::VoltageAdjusterDynamics with other classes

mult.	mult.	Class type name	Description
PFVArControllerType1Dynamics	[1..1]	PFVArControllerType1Dynamics	Power Factor or VAr controller Type I model with which this voltage adjuster is associated. Association Based On : VoltageAdjusterDynamics:: VoltageAdjusterDynamics.VoltageAdjusterDynamics[0..1] ----- PFVArControllerType1Dynamics:: PFVArControllerType1Dynamics.PFVArControllerType1Dynamics[1]

9.6.15.3 VAdjIEEE

The class represents IEEE Voltage Adjuster which is used to represent the voltage adjuster in either a power factor or var control system. Reference: IEEE Standard 421.5-2005 Section 11.1.

Inheritance path: ->VoltageAdjusterDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-292 shows all attributes of VAdjIEEE.

Table 9-292 - Attributes of VoltageAdjusterDynamics::VAdjIEEE

Attribute name	mult.	Attribute type	Description
vadjf	1..1	Simple_Float	Set high to provide a continuous raise or lower (V_{ADJF}).
adjslew	1..1	Simple_Float	Rate at which output of adjuster changes (*ADJ_SLEW*). Unit = sec./PU. Typical Value = 300.
vadjmax	1..1	PU	Maximum output of the adjuster (V_{ADJMAX}). Typical Value = 1.1.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

vadjmin	1..1	PU	Minimum output of the adjuster (V_{ADJMIN}). Typical Value = 0.9.
taon	1..1	Seconds	Time that adjuster pulses are on (T_{AON}). Typical Value = 0.1.
taoff	1..1	Seconds	Time that adjuster pulses are off (T_{AOFF}). Typical Value = 0.5.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-293 shows all association ends of VoltageAdjusterDynamics with other classes.

Table 9-293 - Association ends of VoltageAdjusterDynamics::VoltageAdjusterDynamics with other classes

mult.	mult.	Class type name	Description
PFVArControllerType1Dynamics	1..1	PFVArControllerType1Dynamics	Inherited Association Based On : VoltageAdjusterDynamics:: VoltageAdjusterDynamics.VoltageAdjusterDynamics[0..1] ----- PFVArControllerType1Dynamics:: PFVArControllerType1Dynamics.PFVArControllerType1Dynamics[1]

9.6.16 PFVArControllerType2Dynamics

9.6.16.1 General

A var/pf regulator is defined as “A synchronous machine regulator that functions to maintain the power factor or reactive component of power at a predetermined value.” For additional information please refer to IEEE Standard 421.5-2005, Section 11.

Figure 9-154 shows the diagram.

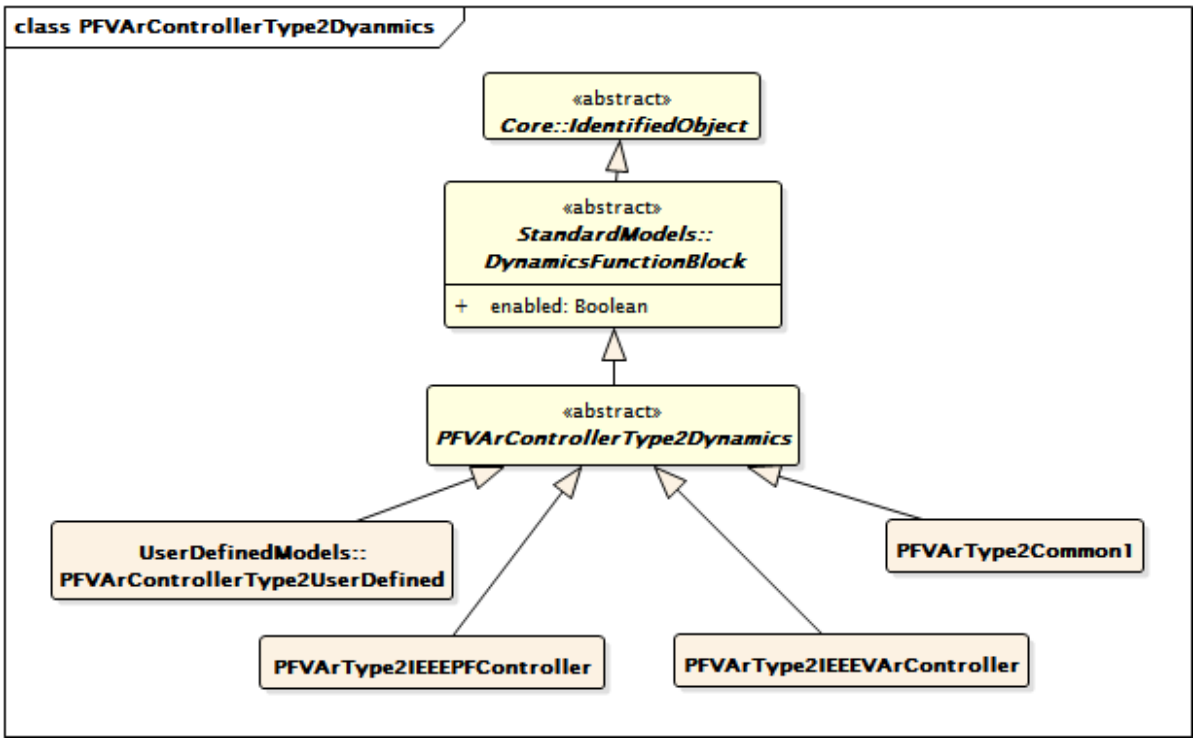


Figure 9-154 - diagram PFVArControllerType2Dyanmics

This diagram shows dynamic standard model Power Factor or VAr controller Type II classes and their inheritance from the PRVArControllerTypeIIDynamics class.

9.6.16.2 PFVArControllerType2Dynamics(Abstract)

Power Factor or VAr controller Type II function block whose behaviour is described by reference to a standard model or by definition of a user-defined model.

Inheritance path: ->DynamicsFunctionBlock->IdentifiedObject

Table 9-294 shows all attributes of PFVArControllerType2Dynamics.

Table 9-294 - Attributes of PFVArControllerType2Dynamics::PFVArControllerType2Dynamics

Attribute name	mult.	Attribute type	Description
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-295 shows all association ends of PFVArControllerType2Dynamics with other classes.

Table 9-295 - Association ends of PFVArControllerType2Dynamics::PFVArControllerType2Dynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	[1..1]	ExcitationSystemDynamics	Excitation system model with which this Power Factor or VAr controller Type II is associated.

			Association Based On : PFVArControllerType2Dynamics:: PFVArControllerType2Dynamics.PFVArControllerType2Dynamics [0..1] ----- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.16.3 PFVArType2IEEEPFController

The class represents IEEE PF Controller Type 2 which is a summing point type controller and makes up the outside loop of a two-loop system. This controller is implemented as a slow PI type controller. The voltage regulator forms the inner loop and is implemented as a fast controller. Reference: IEEE Standard 421.5-2005 Section 11.4.

Inheritance path: ->PFVArControllerType2Dynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-296 shows all attributes of PFVArType2IEEEPFController.

Table 9-296 - Attributes of PFVArControllerType2Dynamics::PFVArType2IEEEPFController

Attribute name	mult.	Attribute type	Description
pfref	1..1	PU	Power factor reference (P_{FREF}).
vref	1..1	PU	Voltage regulator reference (V_{REF}).
vclmt	1..1	PU	Maximum output of the pf controller (V_{CLMT}). Typical Value = 0.1.
kp	1..1	PU	Proportional gain of the pf controller (K_P). Typical Value = 1.
ki	1..1	PU	Integral gain of the pf controller (K_I). Typical Value = 1.
vs	1..1	Simple_Float	Generator sensing voltage (V_S).
exlon	1..1	Boolean	Overexcitation or under excitation flag (*EXLON*) true = 1 (not in the overexcitation or underexcitation state, integral action is active) false = 0 (in the overexcitation or underexcitation state, so integral action is disabled to allow the limiter to play its role).
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-297 shows all association ends of PFVArControllerType2Dynamics with other classes.

Table 9-297 - Association ends of PFVArControllerType2Dynamics::PFVArControllerType2Dynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : PFVArControllerType2Dynamics:: PFVArControllerType2Dynamics.PFVArControllerType2Dynamics[0..1] --- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.16.4 PFVArType2IEEEVArController

The class represents IEEE VAR Controller Type 2 which is a summing point type controller. It makes up the outside loop of a two-loop system. This controller is implemented as a slow PI type controller, and the voltage regulator forms the inner loop and is implemented as a fast controller. Reference: IEEE Standard 421.5-2005 Section 11.5.

Inheritance path: ->PFVArControllerType2Dynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-298 shows all attributes of PFVArType2IEEEVArController.

Table 9-298 - Attributes of PFVArControllerType2Dynamics::PFVArType2IEEEVArController

Attribute name	mult.	Attribute type	Description
qref	1..1	PU	Reactive power reference (Q_{REF}).
vref	1..1	PU	Voltage regulator reference (V_{REF}).
vclmt	1..1	PU	Maximum output of the pf controller (V_{CLMT}).
kp	1..1	PU	Proportional gain of the pf controller (K_P).
ki	1..1	PU	Integral gain of the pf controller (K_I).
vs	1..1	Simple_Float	Generator sensing voltage (V_S).
exlon	1..1	Boolean	Overexcitation or under excitation flag (*EXLON*) true = 1 (not in the overexcitation or underexcitation state, integral action is active) false = 0 (in the overexcitation or underexcitation state, so integral action is disabled to allow the limiter to play its role).
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-299 shows all association ends of PFVArControllerType2Dynamics with other classes.

Table 9-299 - Association ends of PFVArControllerType2Dynamics::PFVArControllerType2Dynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : PFVArControllerType2Dynamics:: PFVArControllerType2Dynamics.PFVArControllerType2Dynamics[0..1] --- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.16.5 PFVArType2Common1

Power factor / Reactive power regulator. This model represents the power factor or reactive power controller such as the Basler SCP-250. The controller measures power factor or reactive power (PU on generator rated power) and compares it with the operator's set point.

Figure 9-155 class diagram

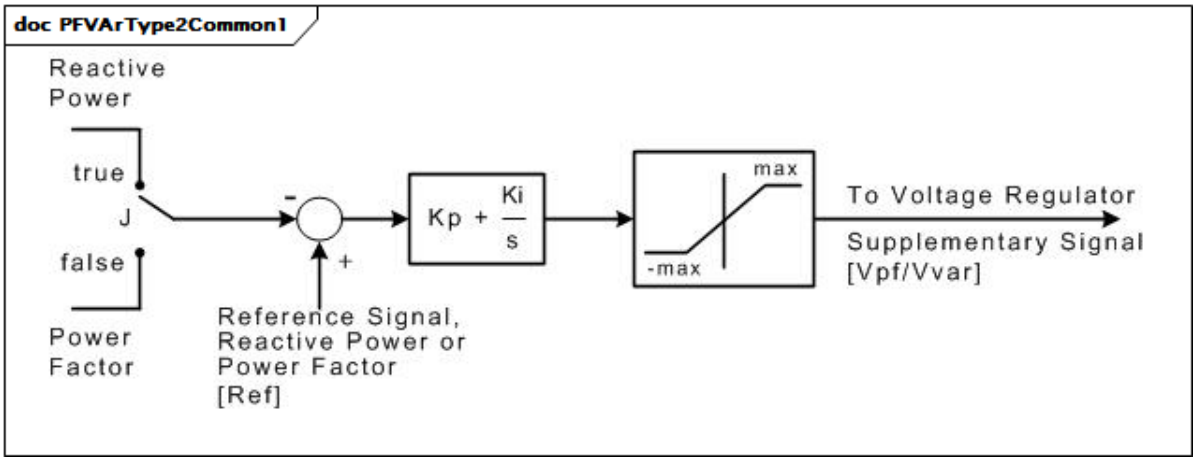


Figure 9-155 - figure PFVArType2Common1

Inheritance path: ->PFVArControllerType2Dynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-300 shows all attributes of PFVArType2Common1.

Table 9-300 - Attributes of PFVArControllerType2Dynamics::PFVArType2Common1

Attribute name	mult.	Attribute type	Description
j	1..1	Boolean	Selector (J). true = control mode for reactive power false = control mode for power factor.
kp	1..1	PU	Proportional gain (Kp).
ki	1..1	PU	Reset gain (Ki).
max	1..1	PU	Output limit (max).

ref	1..1	PU	Reference value of reactive power or power factor (Ref). The reference value is initialised by this model. This initialisation may override the value exchanged by this attribute to represent a plant operator's change of the reference setting.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-301 shows all association ends of PFVArControllerType2Dynamics with other classes.

Table 9-301 - Association ends of PFVArControllerType2Dynamics::PFVArControllerType2Dynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : PFVArControllerType2Dynamics:: PFVArControllerType2Dynamics.PFVArControllerType2Dynamics[0..1] --- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

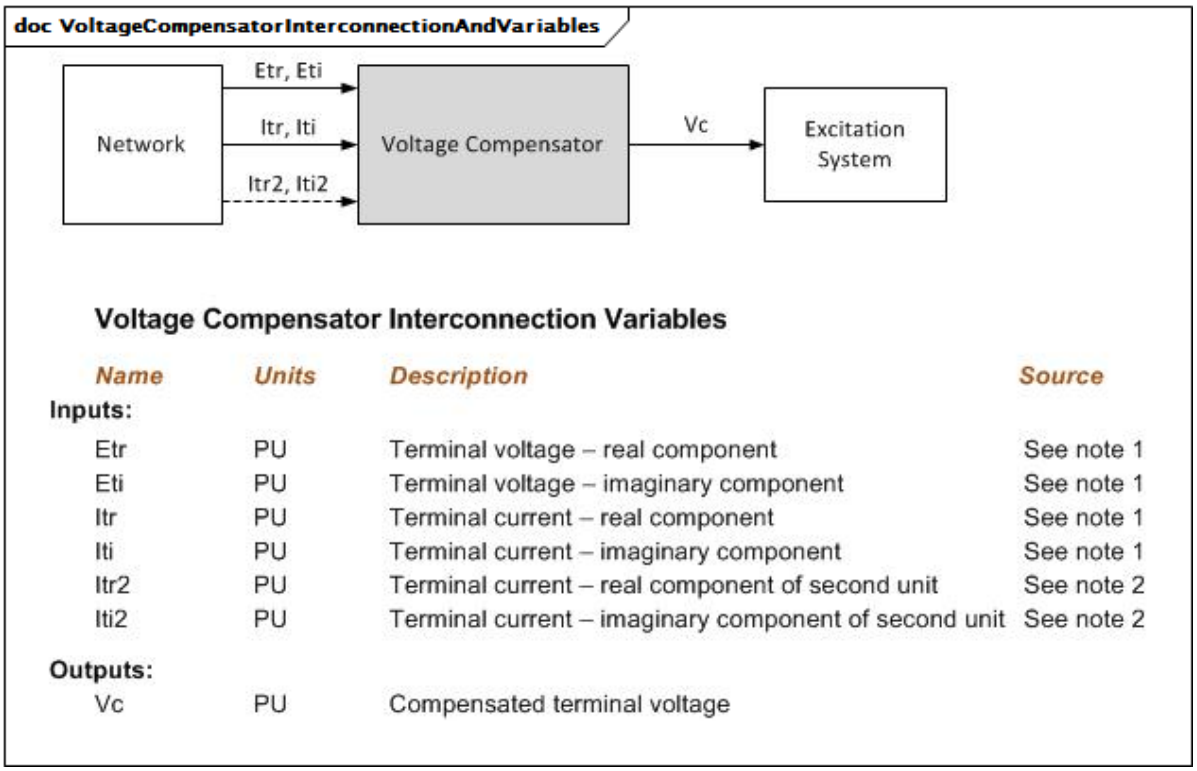
9.6.17 VoltageCompensatorDynamics

9.6.17.1 General

Synchronous machine terminal voltage transducer and current compensator models adjust the terminal voltage feedback to the excitation system by adding a quantity that is proportional to the terminal current of the generator. It is linked to a specific generator (synchronous machine). Several types of compensation are available on most excitation systems. Synchronous machine active and reactive current compensation are the most common. Either reactive droop compensation and/or line-drop compensation may be used, simulating an impedance drop and effectively regulating at some point other than the terminals of the machine. The impedance or range of adjustment and type of compensation should be specified for different types. Care must be taken to ensure that a consistent pu system is utilized for the compensator parameters and the synchronous machine current base. For further information see IEEE Standard 421.5-2005, Section 4.

Figure 9-156 shows the diagram.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu



Voltage Compensator Interconnection Variables

Name	Units	Description	Source
Inputs:			
Etr	PU	Terminal voltage – real component	See note 1
Eti	PU	Terminal voltage – imaginary component	See note 1
Itr	PU	Terminal current – real component	See note 1
Iti	PU	Terminal current – imaginary component	See note 1
Itr2	PU	Terminal current – real component of second unit	See note 2
Iti2	PU	Terminal current – imaginary component of second unit	See note 2
Outputs:			
Vc	PU	Compensated terminal voltage	

Figure 9-156 - diagram VoltageCompensatorInterconnectionAndVariables

This diagram illustrates the inputs and outputs for a voltage compensator.
Notes:

1. Source of the generator complex voltage components is application dependent. It may come from the generator or from the network equations.
2. Itr2 and Iti2 relate to a second generator connected to the same terminal bus, usually the other unit of a cross-compound pair. These inputs are not used by all voltage compensation models.

Figure 9-157 shows the diagram.

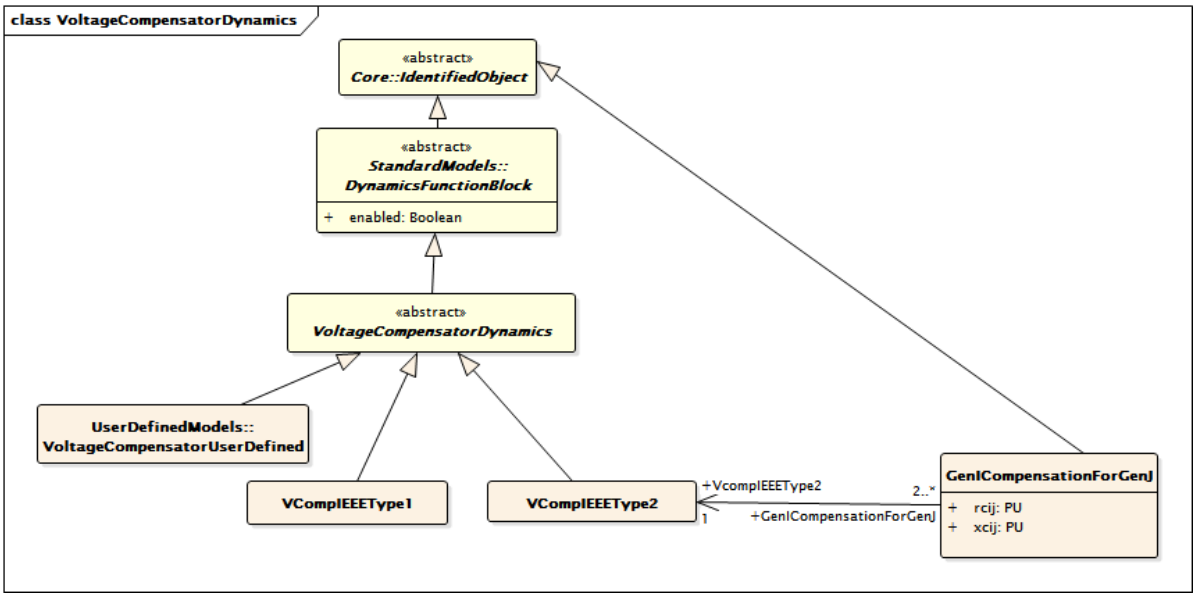


Figure 9-157 - diagram VoltageCompensatorDynamics

This diagram shows dynamic standard model voltage compensator classes and their inheritance from the VoltageCompensatorDynamics class.

9.6.17.2 VoltageCompensatorDynamics(Abstract)

Voltage compensator function block whose behaviour is described by reference to a standard model or by definition of a user-defined model.

Inheritance path: ->DynamicsFunctionBlock->IdentifiedObject

Table 9-302 shows all attributes of VoltageCompensatorDynamics.

Table 9-302 - Attributes of VoltageCompensatorDynamics::VoltageCompensatorDynamics

Attribute name	mult.	Attribute type	Description
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-303 shows all association ends of VoltageCompensatorDynamics with other classes.

Table 9-303 - Association ends of VoltageCompensatorDynamics::VoltageCompensatorDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	[1..1]	ExcitationSystemDynamics	Excitation system model with which this voltage compensator is associated. Association Based On : VoltageCompensatorDynamics:: VoltageCompensatorDynamics.VoltageCompensatorDynamics[0..1] ----- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.17.3 VCompIEEEType1

The class represents the terminal voltage transducer and the load compensator as defined in the IEEE Std 421.5-2005, Section 4. This model is common to all excitation system models described in the IEEE Standard.
Reference: IEEE Standard 421.5-2005 Section 4.

Inheritance path: ->VoltageCompensatorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-304 shows all attributes of VCompIEEEType1.

Table 9-304 - Attributes of VoltageCompensatorDynamics::VCompIEEEType1

Attribute name	mult.	Attribute type	Description
rc	1..1	PU	Resistive component of compensation of a generator (Rc).
xc	1..1	PU	Reactive component of compensation of a generator (Xc).
tr	1..1	Seconds	Time constant which is used for the combined voltage sensing and compensation signal (Tr).
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-305 shows all association ends of VoltageCompensatorDynamics with other classes.

Table 9-305 - Association ends of VoltageCompensatorDynamics::VoltageCompensatorDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : VoltageCompensatorDynamics:: VoltageCompensatorDynamics.VoltageCompensatorDynamics[0..1] --- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.17.4 VCompIEEEType2

The class represents the terminal voltage transducer and the load compensator as defined in the IEEE Std 421.5-2005, Section 4. This model is designed to cover the following types of compensation:

- reactive droop
- transformer-drop or line-drop compensation
- reactive differential compensation known also as cross-current compensation.

Reference: IEEE Standard 421.5-2005, Section 4.

Inheritance path: ->VoltageCompensatorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-306 shows all attributes of VCompIEEEType2.

Table 9-306 - Attributes of VoltageCompensatorDynamics::VCompIEEEType2

Attribute name	mult.	Attribute type	Description
tr	1..1	Seconds	Time constant which is used for the combined voltage sensing and compensation signal (Tr).
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-307 shows all association ends of VoltageCompensatorDynamics with other classes.

Table 9-307 - Association ends of VoltageCompensatorDynamics::VoltageCompensatorDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : VoltageCompensatorDynamics:: VoltageCompensatorDynamics.VoltageCompensatorDynamics[0..1] --- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.6.17.5 GenICompensationForGenJ

This class provides the resistive and reactive components of compensation for the generator associated with the IEEE Type 2 voltage compensator for current flow out of one of the other generators in the interconnection.

Inheritance path: ->IdentifiedObject

Table 9-308 shows all attributes of GenICompensationForGenJ.

Table 9-308 - Attributes of VoltageCompensatorDynamics::GenICompensationForGenJ

Attribute name	mult.	Attribute type	Description
rcij	1..1	PU	Resistive component of compensation of generator associated with this IEEE Type 2 voltage compensator for current flow out of another generator (Rcij).
xcij	1..1	PU	Reactive component of compensation of

			generator associated with this IEEE Type 2 voltage compensator for current flow out of another generator (Xcij).
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-309 shows all association ends of GenICompensationForGenJ with other classes.

Table 9-309 - Association ends of VoltageCompensatorDynamics::GenICompensationForGenJ with other classes

mult.	mult.	Class type name	Description
VcompIEEEType2	[1..1]	VCompIEEEType2	The standard IEEE Type 2 voltage compensator of this compensation. Association Based On : VoltageCompensatorDynamics::GenICompensationForGenJ.GenICompensationForGenJ[2..*] ----- VoltageCompensatorDynamics::VCompIEEEType2.VcompIEEEType2[1]
SynchronousMachineDynamics	[1..1]	SynchronousMachineDynamics	Standard synchronous machine out of which current flow is being compensated for. Association Based On : VoltageCompensatorDynamics::GenICompensationForGenJ.GenICompensationForGenJ[0..*] ----- SynchronousMachineDynamics::SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.6.18 WindDynamics

9.6.18.1 General

Wind turbines are generally divided into 4 types, which are currently significant in power systems. The 4 types have the following characteristics:

- Type 1: Wind turbine with directly grid connected asynchronous generator with fixed rotor resistance (typically squirrel cage)
- Type 2: Wind turbine with directly grid connected asynchronous generator with variable rotor resistance
- Type 3: Wind turbines with doubly-fed asynchronous generators (directly connected stator and rotor connected through power converter)
- Type 4: Wind turbines connected fully through a power converter.

Models included in this package are according to IEC 61400-27-1.

Figure 9-158 shows the diagram.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

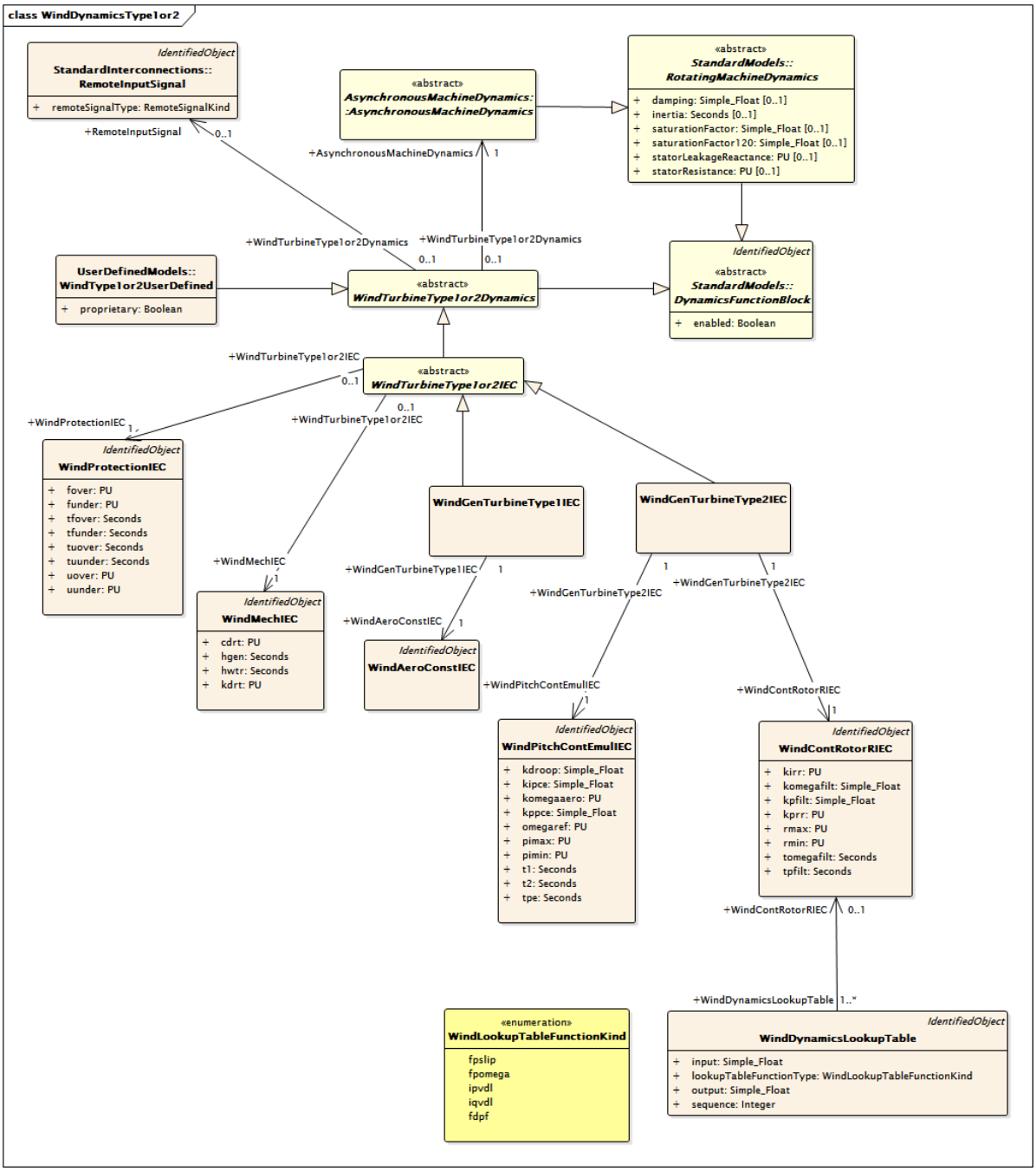


Figure 9-158 - diagram WindDynamicsType1or2

This diagram shows dynamic standard model wind classes (for wind turbines Type 1 and Type 2) and their relationships. Note that classes in the Wind package are not grouped by function block classes. This is because specific standard model classes not abstract function block classes are specified in the wind interconnection models.

Figure 9-159 shows the diagram.

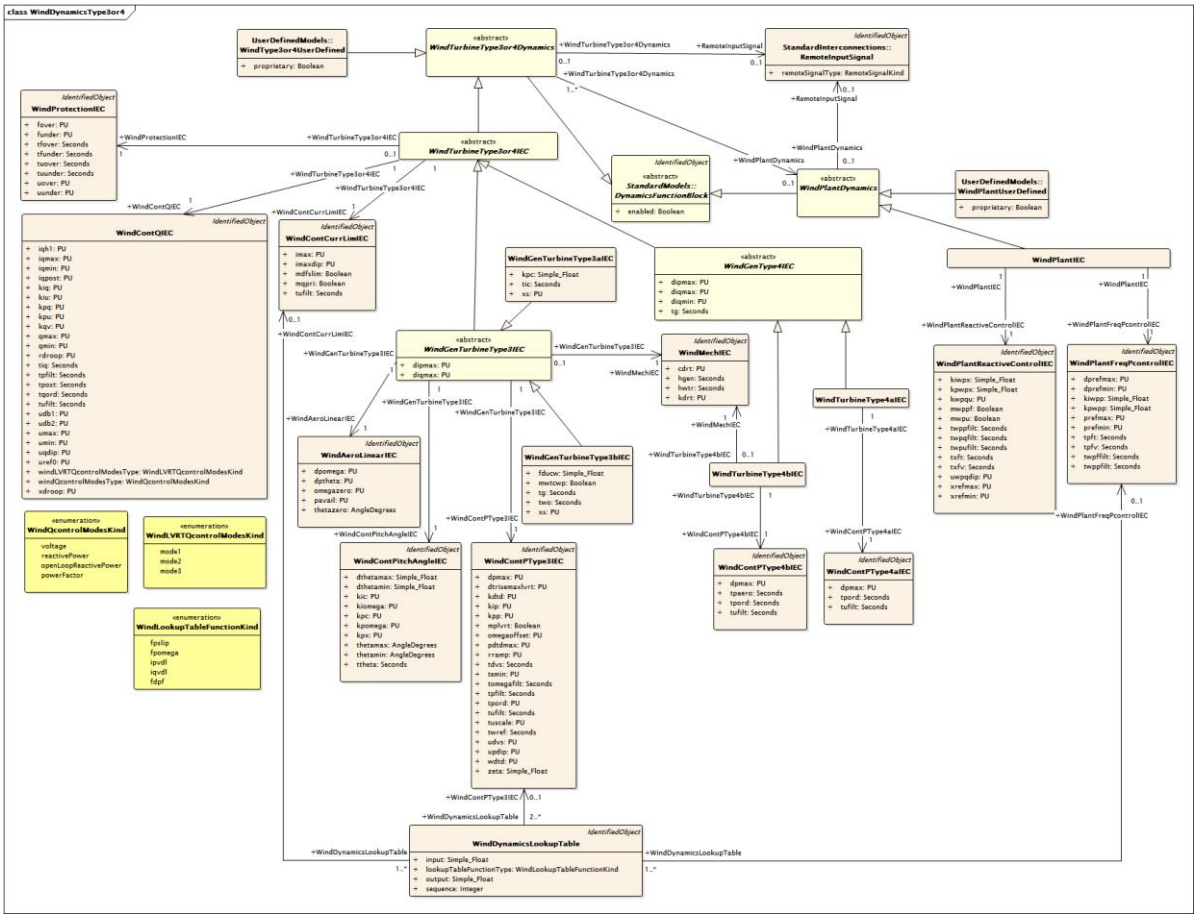


Figure 9-159 - diagram WindDynamicsType3or4

This diagram shows dynamic standard model wind classes (for wind turbines Type 3 and Type 4) and their relationships. Note that classes in the Wind package are not grouped by function block classes. This is because specific standard model classes not abstract function block classes are specified in the wind interconnection models.

9.6.18.2 WindAeroConstIEC

The constant aerodynamic torque model assumes that the aerodynamic torque is constant. Reference: IEC Standard 61400-27-1 Section 6.6.1.1.

Inheritance path: ->IdentifiedObject

Table 9-310 shows all attributes of WindAeroConstIEC.

Table 9-310 - Attributes of WindDynamics::WindAeroConstIEC

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

9.6.18.3 WindAeroLinearIEC

The linearised aerodynamic model. Reference: IEC Standard 614000-27-1 Section 6.6.1.2.

Inheritance path: ->IdentifiedObject

Table 9-311 shows all attributes of WindAeroLinearIEC.

Table 9-311 - Attributes of WindDynamics::WindAeroLinearIEC

Attribute name	mult.	Attribute type	Description
dpomega	1..1	PU	Partial derivative of aerodynamic power with respect to changes in WTR speed (dp_{omega}). It is case dependent parameter.
dptheta	1..1	PU	Partial derivative of aerodynamic power with respect to changes in pitch angle (dp_{theta}). It is case dependent parameter.
omegazero	1..1	PU	Rotor speed if the wind turbine is not derated ($omega_0$). It is case dependent parameter.
pavail	1..1	PU	Available aerodynamic power (p_{avail}). It is case dependent parameter.
thetazero	1..1	AngleDegrees	Pitch angle if the wind turbine is not derated ($theta_0$). It is case dependent parameter.
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

9.6.18.4 WindContCurrLimIEC

Current limitation model. The current limitation model combines the physical limits. Reference: IEC Standard 61400-27-1 Section 6.6.5.7.

Inheritance path: ->IdentifiedObject

Table 9-312 shows all attributes of WindContCurrLimIEC.

Table 9-312 - Attributes of WindDynamics::WindContCurrLimIEC

Attribute name	mult.	Attribute type	Description
imax	1..1	PU	Maximum continuous current at the wind turbine terminals (i_{max}). It is type dependent parameter.
imaxdip	1..1	PU	Maximum current during voltage dip at the wind

			turbine terminals ($i_{max,dip}$). It is project dependent parameter.
mdfslim	1..1	Boolean	Limitation of type 3 stator current (M_{DFSLim}): - false=0: total current limitation, - true=1: stator current limitation). It is type dependent parameter.
mqpri	1..1	Boolean	Prioritisation of q control during LVRT (M_{qpri}): - true = 1: reactive power priority, - false = 0: active power priority. It is project dependent parameter.
tufilt	1..1	Seconds	Voltage measurement filter time constant (T_{ufilt}). It is type dependent parameter.
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

9.6.18.5 WindContPitchAngleIEC

Pitch angle control model. Reference: IEC Standard 61400-27-1 Section 6.6.5.8.

Inheritance path: ->IdentifiedObject

Table 9-313 shows all attributes of WindContPitchAngleIEC.

Table 9-313 - Attributes of WindDynamics::WindContPitchAngleIEC

Attribute name	mult.	Attribute type	Description
dthetamax	1..1	Simple_Float	Maximum pitch positive ramp rate (d$theta_{max}$). It is type dependent parameter. Unit = degrees/sec.
dthetamin	1..1	Simple_Float	Maximum pitch negative ramp rate (d$theta_{min}$). It is type dependent parameter. Unit = degrees/sec.
kic	1..1	PU	Power PI controller integration gain (K_{Ic}). It is type dependent parameter.
kiomega	1..1	PU	Speed PI controller integration gain (K_{Iomega}). It is type dependent parameter.
kpc	1..1	PU	Power PI controller

			proportional gain (K_{Pc}). It is type dependent parameter.
kpomega	1..1	PU	Speed PI controller proportional gain (K_{Pomega}). It is type dependent parameter.
kpx	1..1	PU	Pitch cross coupling gain (K_{PX}). It is type dependent parameter.
thetamax	1..1	AngleDegrees	Maximum pitch angle ($theta_{max}$). It is type dependent parameter.
thetamin	1..1	AngleDegrees	Minimum pitch angle ($theta_{min}$). It is type dependent parameter.
ttheta	1..1	Seconds	Pitch time constant ($ttheta$). It is type dependent parameter.
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

9.6.18.6 WindContPType3IEC

P control model Type 3.
Reference: IEC Standard 61400-27-1 Section 6.6.5.3.

Inheritance path: ->IdentifiedObject

Table 9-314 shows all attributes of WindContPType3IEC.

Table 9-314 - Attributes of WindDynamics::WindContPType3IEC

Attribute name	mult.	Attribute type	Description
dpmax	1..1	PU	Maximum wind turbine power ramp rate (dp_{max}). It is project dependent parameter.
dtrisemaxlvrt	1..1	PU	Limitation of torque rise rate during LVRT for S_1 ($dT_{risemaxLVRT}$). It is project dependent parameter.
kdtd	1..1	PU	Gain for active drive train damping (K_{DTD}). It is type dependent parameter.
kip	1..1	PU	PI controller integration parameter (K_{Ip}). It is type dependent parameter.
kpp	1..1	PU	PI controller proportional gain (K_{Pp}). It is type dependent parameter.

mplvrt	1..1	Boolean	Enable LVRT power control mode ($M_{pLVRT)}$. true = 1: voltage control false = 0: reactive power control. It is project dependent parameter.
omegaoffset	1..1	PU	Offset to reference value that limits controller action during rotor speed changes ($omega_{offset}$). It is case dependent parameter.
pdtdmax	1..1	PU	Maximum active drive train damping power (p_{DTDmax}). It is type dependent parameter.
rramp	1..1	PU	Ramp limitation of torque, required in some grid codes (R_{ramp}). It is project dependent parameter.
tdvs	1..1	Seconds	Time delay after deep voltage sags (T_{DVS}). It is project dependent parameter.
temin	1..1	PU	Minimum electrical generator torque (T_{emin}). It is type dependent parameter.
tomegafilt	1..1	Seconds	Filter time constant for generator speed measurement ($T_{omegafilt}$). It is type dependent parameter.
tpfilt	1..1	Seconds	Filter time constant for power measurement (T_{pfilt}). It is type dependent parameter.
tpord	1..1	PU	Time constant in power order lag (T_{pord}). It is type dependent parameter.
tufilt	1..1	Seconds	Filter time constant for voltage measurement (T_{ufilt}). It is type dependent parameter.
tuscale	1..1	PU	Voltage scaling factor of reset-torque (T_{uscale}). It is project dependent parameter.
twref	1..1	Seconds	Time constant in speed reference filter ($T_{omega,ref}$). It is type dependent parameter.
udvs	1..1	PU	Voltage limit for hold LVRT status after deep voltage sags (u_{DVS}). It is project dependent parameter.

updip	1..1	PU	Voltage dip threshold for P-control (u_{Pdip}). Part of turbine control, often different (e.g 0.8) from converter thresholds. It is project dependent parameter.
wdtd	1..1	PU	Active drive train damping frequency ($omega_{DTD}$). It can be calculated from two mass model parameters. It is type dependent parameter.
zeta	1..1	Simple_Float	Coefficient for active drive train damping (zeta). It is type dependent parameter.
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

9.6.18.7 WindContPType4aIEC

P control model Type 4A.
Reference: IEC Standard 61400-27-1 Section 6.6.5.4.

Inheritance path: ->IdentifiedObject

Table 9-315 shows all attributes of WindContPType4aIEC.

Table 9-315 - Attributes of WindDynamics::WindContPType4aIEC

Attribute name	mult.	Attribute type	Description
dpmax	1..1	PU	Maximum wind turbine power ramp rate (dp_{max}). It is project dependent parameter.
tpord	1..1	Seconds	Time constant in power order lag (T_{pord}). It is type dependent parameter.
tufilt	1..1	Seconds	Voltage measurement filter time constant (T_{ufilt}). It is type dependent parameter.
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

9.6.18.8 WindContPType4bIEC

P control model Type 4B.
Reference: IEC Standard 61400-27-1 Section 6.6.5.5.

Inheritance path: ->IdentifiedObject

Table 9-316 shows all attributes of WindContPType4bIEC.

Table 9-316 - Attributes of WindDynamics::WindContPType4bIEC

Attribute name	mult.	Attribute type	Description
dpmax	1..1	PU	Maximum wind turbine power ramp rate (dp_{max}). It is project dependent parameter.
tpaero	1..1	Seconds	Time constant in aerodynamic power response (T_{paero}). It is type dependent parameter.
tpord	1..1	Seconds	Time constant in power order lag (T_{pord}). It is type dependent parameter.
tufilt	1..1	Seconds	Voltage measurement filter time constant (T_{ufilt}). It is type dependent parameter.
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

9.6.18.9 WindContQIEC

Q control model.
Reference: IEC Standard 61400-27-1 Section 6.6.5.6.

Inheritance path: ->IdentifiedObject

Table 9-317 shows all attributes of WindContQIEC.

Table 9-317 - Attributes of WindDynamics::WindContQIEC

Attribute name	mult.	Attribute type	Description
iqh1	1..1	PU	Maximum reactive current injection during dip (i_{qh1}). It is type dependent parameter.
iqmax	1..1	PU	Maximum reactive current injection (i_{qmax}). It is type dependent parameter.
iqmin	1..1	PU	Minimum reactive current injection (i_{qmin}). It is type dependent parameter.
iqpost	1..1	PU	Post fault reactive current injection (i_{qpost}). It is project dependent parameter.
kiq	1..1	PU	Reactive power PI controller integration gain ($K_{I,q}$). It is type dependent parameter.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

kiu	1..1	PU	Voltage PI controller integration gain ($K_{I,u}$). It is type dependent parameter.
kpq	1..1	PU	Reactive power PI controller proportional gain ($K_{P,q}$). It is type dependent parameter.
kpu	1..1	PU	Voltage PI controller proportional gain ($K_{P,u}$). It is type dependent parameter.
kqv	1..1	PU	Voltage scaling factor for LVRT current (K_{qv}). It is project dependent parameter.
qmax	1..1	PU	Maximum reactive power (q_{max}). It is type dependent parameter.
qmin	1..1	PU	Minimum reactive power (q_{min}). It is type dependent parameter.
rdroop	1..1	PU	Resistive component of voltage drop impedance (r_{droop}). It is project dependent parameter.
tiq	1..1	Seconds	Time constant in reactive current lag (T_{iq}). It is type dependent parameter.
tpfilt	1..1	Seconds	Power measurement filter time constant (T_{pfilt}). It is type dependent parameter.
tpost	1..1	Seconds	Length of time period where post fault reactive power is injected (T_{post}). It is project dependent parameter.
tqord	1..1	Seconds	Time constant in reactive power order lag (T_{qord}). It is type dependent parameter.
tufilt	1..1	Seconds	Voltage measurement filter time constant (T_{ufilt}). It is type dependent parameter.
udb1	1..1	PU	Voltage dead band lower limit (u_{db1}). It is type dependent parameter.
udb2	1..1	PU	Voltage dead band upper limit (u_{db2}). It is type dependent parameter.
umax	1..1	PU	Maximum voltage in voltage PI controller

			integral term (u_{max}). It is type dependent parameter.
umin	1..1	PU	Minimum voltage in voltage PI controller integral term (u_{min}). It is type dependent parameter.
uqdip	1..1	PU	Voltage threshold for LVRT detection in q control (u_{qdip}). It is type dependent parameter.
uref0	1..1	PU	User defined bias in voltage reference (u_{ref0}), used when $M_{qG} = M_{G,u}$. It is case dependent parameter.
windLVRTQcontrolModesType	1..1	WindLVRTQcontrolModesKind	Types of LVRT Q control modes (M_{qLVRT}). It is project dependent parameter.
windQcontrolModesType	1..1	WindQcontrolModesKind	Types of general wind turbine Q control modes (M_{qG}). It is project dependent parameter.
xdroop	1..1	PU	Inductive component of voltage drop impedance (x_{droop}). It is project dependent parameter.
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

9.6.18.10 WindContRotorRIEC

Rotor resistance control model.
Reference: IEC Standard 61400-27-1 Section 6.6.5.2.

Inheritance path: ->IdentifiedObject

Table 9-318 shows all attributes of WindContRotorRIEC.

Table 9-318 - Attributes of WindDynamics::WindContRotorRIEC

Attribute name	mult.	Attribute type	Description
kirr	1..1	PU	Integral gain in rotor resistance PI controller (K_{Irr}). It is type dependent parameter.
komegafilt	1..1	Simple_Float	Filter gain for generator speed measurement ($K_{omegafilt}$). It is type dependent parameter.
kpfilt	1..1	Simple_Float	Filter gain for power measurement (K_{pfilt}). It is type dependent parameter.

kprr	1..1	PU	Proportional gain in rotor resistance PI controller (K_{Prr}). It is type dependent parameter.
rmax	1..1	PU	Maximum rotor resistance (r_{max}). It is type dependent parameter.
rmin	1..1	PU	Minimum rotor resistance (r_{min}). It is type dependent parameter.
tomegafilt	1..1	Seconds	Filter time constant for generator speed measurement ($T_{omegafilt}$). It is type dependent parameter.
tpfilt	1..1	Seconds	Filter time constant for power measurement (T_{pfilt}). It is type dependent parameter.
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

9.6.18.11 WindDynamicsLookupTable

The class models a look up table for the purpose of wind standard models.

Inheritance path: ->IdentifiedObject

Table 9-319 shows all attributes of WindDynamicsLookupTable.

Table 9-319 - Attributes of WindDynamics::WindDynamicsLookupTable

Attribute name	mult.	Attribute type	Description
input	1..1	Simple_Float	Input value (x) for the lookup table function.
lookupTableFunctionType	1..1	WindLookupTableFunctionKind	Type of the lookup table function.
output	1..1	Simple_Float	Output value (y) for the lookup table function.
sequence	1..1	Integer	Sequence numbers of the pairs of the input (x) and the output (y) of the lookup table function.
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-320 shows all association ends of WindDynamicsLookupTable with other classes.

Table 9-320 - Association ends of WindDynamics::WindDynamicsLookupTable with other classes

mult.	mult.	Class type name	Description
WindPlantFreqPcontrolIEC	[0..1]	WindPlantFreqPcontrolIEC	The wind dynamics lookup table associated with this frequency and active power wind plant model. Association Based On : WindDynamics:: WindDynamicsLookupTable.WindDynamicsLookupTable[1..*] ----- WindDynamics:: WindPlantFreqPcontrolIEC.WindPlantFreqPcontrolIEC[0..1]
WindContRotorRIEC	[0..1]	WindContRotorRIEC	The rotor resistance control model with which this wind dynamics lookup table is associated. Association Based On : WindDynamics:: WindDynamicsLookupTable.WindDynamicsLookupTable[1..*] ----- WindDynamics:: WindContRotorRIEC.WindContRotorRIEC[0..1]
WindContPType3IEC	[0..1]	WindContPType3IEC	The wind dynamics lookup table associated with this P control type 3 model. Association Based On : WindDynamics:: WindDynamicsLookupTable.WindDynamicsLookupTable[2..*] ----- WindDynamics:: WindContPType3IEC.WindContPType3IEC[0..1]
WindContCurrLimIEC	[0..1]	WindContCurrLimIEC	The wind dynamics lookup table associated with this current control limitation model. Association Based On : WindDynamics:: WindDynamicsLookupTable.WindDynamicsLookupTable[1..*] ----- WindDynamics:: WindContCurrLimIEC.WindContCurrLimIEC[0..1]

9.6.18.12 WindGenTurbineType1IEC

Wind turbine IEC Type 1.
Reference: IEC Standard 61400-27-1, section 6.5.2.

Inheritance path: ->WindTurbineType1or2IEC->WindTurbineType1or2Dynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-321 shows all attributes of WindGenTurbineType1IEC.

Table 9-321 - Attributes of WindDynamics::WindGenTurbineType1IEC

Attribute name	mult.	Attribute type	Description
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-322 shows all association ends of WindGenTurbineType1IEC with other classes.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 9-322 - Association ends of WindDynamics::WindGenTurbineType1IEC with other classes

mult.	mult.	Class type name	Description
WindAeroConstIEC	[1..1]	WindAeroConstIEC	Wind aerodynamic model associated with this wind turbine type 1 model. Association Based On : WindDynamics:: WindAeroConstIEC.WindAeroConstIEC[1] ----- WindDynamics:: WindGenTurbineType1IEC.WindGenTurbineType1IEC[1]
WindProtectionIEC	1..1	WindProtectionIEC	Inherited Association Based On : WindDynamics:: WindProtectionIEC.WindProtectionIEC[1] ----- WindDynamics:: WindTurbineType1or2IEC.WindTurbineType1or2IEC[0..1]
WindMechIEC	1..1	WindMechIEC	Inherited Association Based On : WindDynamics:: WindMechIEC.WindMechIEC[1] ----- WindDynamics:: WindTurbineType1or2IEC.WindTurbineType1or2IEC[0..1]
AsynchronousMachineDynamics	1..1	AsynchronousMachineDynamics	Inherited Association Based On : WindDynamics:: WindTurbineType1or2Dynamics.WindTurbineType1or2Dynamics[0..1] ----- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[1]
RemoteInputSignal	0..1	RemoteInputSignal	Inherited Association Based On : WindDynamics:: WindTurbineType1or2Dynamics.WindTurbineType1or2Dynamics[0..1] ----- StandardInterconnections:: RemoteInputSignal.RemoteInputSignal[0..1]

9.6.18.13 WindGenTurbineType2IEC

Wind turbine IEC Type 2.
Reference: IEC Standard 61400-27-1, section 6.5.3.

Inheritance path: ->WindTurbineType1or2IEC->WindTurbineType1or2Dynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-323 shows all attributes of WindGenTurbineType2IEC.

Table 9-323 - Attributes of WindDynamics::WindGenTurbineType2IEC

Attribute name	mult.	Attribute type	Description
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-324 shows all association ends of WindGenTurbineType2IEC with other classes.

Table 9-324 - Association ends of WindDynamics::WindGenTurbineType2IEC with other classes

mult.	mult.	Class type name	Description
WindPitchContEmulIEC	[1..1]	WindPitchContEmulIEC	Pitch control emulator model associated with this wind turbine type 2 model. Association Based On : WindDynamics:: WindPitchContEmulIEC.WindPitchContEmulIEC[1] ----- WindDynamics:: WindGenTurbineType2IEC.WindGenTurbineType2IEC[1]
WindContRotorRIEC	[1..1]	WindContRotorRIEC	Wind control rotor resistance model associated with wind turbine type 2 model. Association Based On : WindDynamics:: WindContRotorRIEC.WindContRotorRIEC[1] ----- WindDynamics:: WindGenTurbineType2IEC.WindGenTurbineType2IEC[1]
WindProtectionIEC	1..1	WindProtectionIEC	Inherited Association Based On : WindDynamics:: WindProtectionIEC.WindProtectionIEC[1] ----- WindDynamics:: WindTurbineType1or2IEC.WindTurbineType1or2IEC[0..1]
WindMechIEC	1..1	WindMechIEC	Inherited Association Based On : WindDynamics:: WindMechIEC.WindMechIEC[1] ----- WindDynamics:: WindTurbineType1or2IEC.WindTurbineType1or2IEC[0..1]
AsynchronousMachineDynamics	1..1	AsynchronousMachineDynamics	Inherited Association Based On : WindDynamics:: WindTurbineType1or2Dynamics.WindTurbineType1or2Dynamics[0..1] ----- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[1]
RemoteInputSignal	0..1	RemoteInputSignal	Inherited Association Based On : WindDynamics:: WindTurbineType1or2Dynamics.WindTurbineType1or2Dynamics[0..1] ----- StandardInterconnections:: RemoteInputSignal.RemoteInputSignal[0..1]

9.6.18.14 WindGenTurbineType3aIEC

IEC Type 3A generator set model. Reference: IEC Standard 61400-27-1 Section 6.6.3.2.

Inheritance path: ->WindGenTurbineType3IEC->WindTurbineType3or4IEC->WindTurbineType3or4Dynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-325 shows all attributes of WindGenTurbineType3aIEC.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 9-325 - Attributes of WindDynamics::WindGenTurbineType3aIEC

Attribute name	mult.	Attribute type	Description
kpc	1..1	Simple_Float	Current PI controller proportional gain (K_{Pc}). It is type dependent parameter.
xs	1..1	PU	Electromagnetic transient reactance (x_S). It is type dependent parameter.
tic	1..1	Seconds	Current PI controller integration time constant (T_{Ic}). It is type dependent parameter.
dipmax	1..1	PU	see WindGenTurbineType3IEC
diqmax	1..1	PU	see WindGenTurbineType3IEC
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-326 shows all association ends of WindGenTurbineType3IEC with other classes.

Table 9-326 - Association ends of WindDynamics::WindGenTurbineType3IEC with other classes

mult.	mult.	Class type name	Description
WindMechIEC	1..1	WindMechIEC	Inherited Association Based On : WindDynamics:: WindMechIEC.WindMechIEC[1] ----- WindDynamics:: WindGenTurbineType3IEC.WindGenTurbineType3IEC[0..1]
WindContPitchAngleIEC	1..1	WindContPitchAngleIEC	Inherited Association Based On : WindDynamics:: WindContPitchAngleIEC.WindContPitchAngleIEC[1] ----- WindDynamics:: WindGenTurbineType3IEC.WindGenTurbineType3IEC[1]
WindContPType3IEC	1..1	WindContPType3IEC	Inherited Association Based On : WindDynamics:: WindContPType3IEC.WindContPType3IEC[1] ----- WindDynamics:: WindGenTurbineType3IEC.WindGenTurbineType3IEC[1]
WindAeroLinearIEC	1..1	WindAeroLinearIEC	Inherited Association Based On : WindDynamics:: WindAeroLinearIEC.WindAeroLinearIEC[1] ----- WindDynamics:: WindGenTurbineType3IEC.WindGenTurbineType3IEC[1]

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

WindProtectionIEC	1..1	WindProtectionIEC	Inherited Association Based On : WindDynamics:: WindProtectionIEC.WindProtectionIEC[1] ----- WindDynamics:: WindTurbineType3or4IEC.WindTurbineType3or4IEC[0..1]
WIndContQIEC	1..1	WindContQIEC	Inherited Association Based On : WindDynamics:: WindContQIEC.WIndContQIEC[1] ----- WindDynamics:: WindTurbineType3or4IEC.WindTurbineType3or4IEC[1]
WindContCurrLimIEC	1..1	WindContCurrLimIEC	Inherited Association Based On : WindDynamics:: WindContCurrLimIEC.WindContCurrLimIEC[1] ----- WindDynamics:: WindTurbineType3or4IEC.WindTurbineType3or4IEC[1]
WindPlantDynamics	0..1	WindPlantDynamics	Inherited Association Based On : WindDynamics:: WindPlantDynamics.WindPlantDynamics[0..1] ----- WindDynamics:: WindTurbineType3or4Dynamics.WindTurbineType3or4Dynamics[1..*]
RemoteInputSignal	0..1	RemoteInputSignal	Inherited Association Based On : WindDynamics:: WindTurbineType3or4Dynamics.WindTurbineType3or4Dynamics[0..1] ----- StandardInterconnections:: RemoteInputSignal.RemoteInputSignal[0..1]
EnergySource	1..1	EnergySource	Inherited Association Based On : WindDynamics:: WindTurbineType3or4Dynamics.WindTurbineType3or4Dynamics[0..1] ----- Wires:: EnergySource.EnergySource[1]

9.6.18.15 WindGenTurbineType3bIEC

IEC Type 3B generator set model. Reference: IEC Standard 61400-27-1 Section 6.6.3.3.

Inheritance path: ->WindGenTurbineType3IEC->WindTurbineType3or4IEC->WindTurbineType3or4Dynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-327 shows all attributes of WindGenTurbineType3bIEC.

Table 9-327 - Attributes of WindDynamics::WindGenTurbineType3bIEC

Attribute name	mult.	Attribute type	Description
fducw	1..1	Simple_Float	Crowbar duration versus voltage variation look-up table (f_{duCW}()). It is case dependent parameter.
tg	1..1	Seconds	Current generation Time constant (T_g). It is type dependent parameter.

two	1..1	Seconds	Time constant for crowbar washout filter (T_{wo}). It is case dependent parameter.
mwtcwp	1..1	Boolean	Crowbar control mode (M_{WTcwp}). true = 1 in the model false = 0 in the model. The parameter is case dependent parameter.
xs	1..1	PU	Electromagnetic transient reactance (x_S). It is type dependent parameter.
dipmax	1..1	PU	see WindGenTurbineType3IEC
diqmax	1..1	PU	see WindGenTurbineType3IEC
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-328 shows all association ends of WindGenTurbineType3IEC with other classes.

Table 9-328 - Association ends of WindDynamics::WindGenTurbineType3IEC with other classes

mult.	mult.	Class type name	Description
WindMechIEC	1..1	WindMechIEC	Inherited Association Based On : WindDynamics:: WindMechIEC.WindMechIEC[1] ----- WindDynamics:: WindGenTurbineType3IEC.WindGenTurbineType3IEC[0..1]
WindContPitchAngleIEC	1..1	WindContPitchAngleIEC	Inherited Association Based On : WindDynamics:: WindContPitchAngleIEC.WindContPitchAngleIEC[1] ----- WindDynamics:: WindGenTurbineType3IEC.WindGenTurbineType3IEC[1]
WindContPType3IEC	1..1	WindContPType3IEC	Inherited Association Based On : WindDynamics:: WindContPType3IEC.WindContPType3IEC[1] ----- WindDynamics:: WindGenTurbineType3IEC.WindGenTurbineType3IEC[1]
WindAeroLinearIEC	1..1	WindAeroLinearIEC	Inherited Association Based On : WindDynamics:: WindAeroLinearIEC.WindAeroLinearIEC[1] ----- WindDynamics:: WindGenTurbineType3IEC.WindGenTurbineType3IEC[1]
WindProtectionIEC	1..1	WindProtectionIEC	Inherited Association Based On :

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			WindDynamics:: WindProtectionIEC.WindProtectionIEC[1] ----- WindDynamics:: WindTurbineType3or4IEC.WindTurbineType3or4IEC[0..1]
WIndContQIEC	1..1	WindContQIEC	Inherited Association Based On : WindDynamics:: WindContQIEC.WIndContQIEC[1] ----- WindDynamics:: WindTurbineType3or4IEC.WindTurbineType3or4IEC[1]
WindContCurrLimIEC	1..1	WindContCurrLimIEC	Inherited Association Based On : WindDynamics:: WindContCurrLimIEC.WindContCurrLimIEC[1] ----- WindDynamics:: WindTurbineType3or4IEC.WindTurbineType3or4IEC[1]
WindPlantDynamics	0..1	WindPlantDynamics	Inherited Association Based On : WindDynamics:: WindPlantDynamics.WindPlantDynamics[0..1] ----- WindDynamics:: WindTurbineType3or4Dynamics.WindTurbineType3or4Dynamics[1..*]
RemoteInputSignal	0..1	RemoteInputSignal	Inherited Association Based On : WindDynamics:: WindTurbineType3or4Dynamics.WindTurbineType3or4Dynamics[0..1] ----- StandardInterconnections:: RemoteInputSignal.RemoteInputSignal[0..1]
EnergySource	1..1	EnergySource	Inherited Association Based On : WindDynamics:: WindTurbineType3or4Dynamics.WindTurbineType3or4Dynamics[0..1] ----- Wires:: EnergySource.EnergySource[1]

9.6.18.16 WindGenTurbineType3IEC(Abstract)

Generator model for wind turbines of IEC type 3A and 3B.

Inheritance path: ->WindTurbineType3or4IEC->WindTurbineType3or4Dynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-329 shows all attributes of WindGenTurbineType3IEC.

Table 9-329 - Attributes of WindDynamics::WindGenTurbineType3IEC

Attribute name	mult.	Attribute type	Description
dipmax	1..1	PU	Maximum active current ramp rate (di_{pmax}). It is project dependent parameter.
diqmax	1..1	PU	Maximum reactive current ramp rate (di_{qmax}). It is project dependent parameter.
enabled	1..1	Boolean	see DynamicsFunctionBlock

description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-330 shows all association ends of WindGenTurbineType3IEC with other classes.

Table 9-330 - Association ends of WindDynamics::WindGenTurbineType3IEC with other classes

mult.	mult.	Class type name	Description
WindMechIEC	[1..1]	WindMechIEC	Wind mechanical model associated with this wind turbine Type 3 model. Association Based On : WindDynamics:: WindMechIEC.WindMechIEC[1] ----- WindDynamics:: WindGenTurbineType3IEC.WindGenTurbineType3IEC[0..1]
WindContPitchAngleIEC	[1..1]	WindContPitchAngleIEC	Wind control pitch angle model associated with this wind turbine type 3. Association Based On : WindDynamics:: WindContPitchAngleIEC.WindContPitchAngleIEC[1] ----- WindDynamics:: WindGenTurbineType3IEC.WindGenTurbineType3IEC[1]
WindContPType3IEC	[1..1]	WindContPType3IEC	Wind control P type 3 model associated with this wind turbine type 3 model. Association Based On : WindDynamics:: WindContPType3IEC.WindContPType3IEC[1] ----- WindDynamics:: WindGenTurbineType3IEC.WindGenTurbineType3IEC[1]
WindAeroLinearIEC	[1..1]	WindAeroLinearIEC	Wind aerodynamic model associated with this wind generator type 3 model. Association Based On : WindDynamics:: WindAeroLinearIEC.WindAeroLinearIEC[1] ----- WindDynamics:: WindGenTurbineType3IEC.WindGenTurbineType3IEC[1]
WindProtectionIEC	1..1	WindProtectionIEC	Inherited Association Based On : WindDynamics:: WindProtectionIEC.WindProtectionIEC[1] ----- WindDynamics:: WindTurbineType3or4IEC.WindTurbineType3or4IEC[0..1]
WIndContQIEC	1..1	WindContQIEC	Inherited Association Based On : WindDynamics:: WindContQIEC.WIndContQIEC[1] ----- WindDynamics:: WindTurbineType3or4IEC.WindTurbineType3or4IEC[1]
WindContCurrLimIEC	1..1	WindContCurrLimIEC	Inherited Association Based On : WindDynamics:: WindContCurrLimIEC.WindContCurrLimIEC[1] ----- WindDynamics:: WindTurbineType3or4IEC.WindTurbineType3or4IEC[1]
WindPlantDynamics	0..1	WindPlantDynamics	Inherited Association Based On : WindDynamics:: WindPlantDynamics.WindPlantDynamics[0..1] -----

			WindDynamics:: WindTurbineType3or4Dynamics.WindTurbineType3or4Dynamics[1..*]
RemoteInputSignal	0..1	RemoteInputSignal	Inherited Association Based On : WindDynamics:: WindTurbineType3or4Dynamics.WindTurbineType3or4Dynamics[0..1] ----- StandardInterconnections:: RemoteInputSignal.RemoteInputSignal[0..1]
EnergySource	1..1	EnergySource	Inherited Association Based On : WindDynamics:: WindTurbineType3or4Dynamics.WindTurbineType3or4Dynamics[0..1] ----- Wires:: EnergySource.EnergySource[1]

9.6.18.17 WindGenType4IEC(Abstract)

IEC Type 4 generator set model. Reference: IEC Standard 61400-27-1 Section 6.6.3.4.

Inheritance path: ->WindTurbineType3or4IEC->WindTurbineType3or4Dynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-331 shows all attributes of WindGenType4IEC.

Table 9-331 - Attributes of WindDynamics::WindGenType4IEC

Attribute name	mult.	Attribute type	Description
dipmax	1..1	PU	Maximum active current ramp rate (di_{pmax}). It is project dependent parameter.
diqmin	1..1	PU	Minimum reactive current ramp rate (di_{qmin}). It is case dependent parameter.
diqmax	1..1	PU	Maximum reactive current ramp rate (di_{qmax}). It is project dependent parameter.
tg	1..1	Seconds	Time constant (T_g). It is type dependent parameter.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-332 shows all association ends of WindTurbineType3or4IEC with other classes.

Table 9-332 - Association ends of WindDynamics::WindTurbineType3or4IEC with other classes

mult.	mult.	Class type name	Description
WindProtectionIEC	1..1	WindProtectionIEC	Inherited Association Based On : WindDynamics:: WindProtectionIEC.WindProtectionIEC[1] ----- WindDynamics:: WindTurbineType3or4IEC.WindTurbineType3or4IEC[0..1]
WIndContQIEC	1..1	WindContQIEC	Inherited Association Based On : WindDynamics:: WindContQIEC.WIndContQIEC[1] ----- WindDynamics:: WindTurbineType3or4IEC.WindTurbineType3or4IEC[1]
WindContCurrLimIEC	1..1	WindContCurrLimIEC	Inherited Association Based On : WindDynamics:: WindContCurrLimIEC.WindContCurrLimIEC[1] ----- WindDynamics:: WindTurbineType3or4IEC.WindTurbineType3or4IEC[1]
WindPlantDynamics	0..1	WindPlantDynamics	Inherited Association Based On : WindDynamics:: WindPlantDynamics.WindPlantDynamics[0..1] ----- WindDynamics:: WindTurbineType3or4Dynamics.WindTurbineType3or4Dynamics[1..*]
RemoteInputSignal	0..1	RemoteInputSignal	Inherited Association Based On : WindDynamics:: WindTurbineType3or4Dynamics.WindTurbineType3or4Dynamics[0..1] ----- StandardInterconnections:: RemoteInputSignal.RemoteInputSignal[0..1]
EnergySource	1..1	EnergySource	Inherited Association Based On : WindDynamics:: WindTurbineType3or4Dynamics.WindTurbineType3or4Dynamics[0..1] ----- Wires:: EnergySource.EnergySource[1]

9.6.18.18 WindMechIEC

Two mass model.
Reference: IEC Standard 61400-27-1 Section 6.6.2.1.

Inheritance path: ->IdentifiedObject

Table 9-333 shows all attributes of WindMechIEC.

Table 9-333 - Attributes of WindDynamics::WindMechIEC

Attribute name	mult.	Attribute type	Description
cdrt	1..1	PU	Drive train damping (c_{drt}). It is type dependent parameter.
hgen	1..1	Seconds	Inertia constant of generator (H_{gen}). It is type dependent

			parameter.
hwtr	1..1	Seconds	Inertia constant of wind turbine rotor (H_{WTR}). It is type dependent parameter.
kdrt	1..1	PU	Drive train stiffness (k_{drt}). It is type dependent parameter.
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

9.6.18.19 WindPitchContEmulIEC

Pitch control emulator model. Reference: IEC Standard 61400-27-1 Section 6.6.5.1.

Inheritance path: ->IdentifiedObject

Table 9-334 shows all attributes of WindPitchContEmulIEC.

Table 9-334 - Attributes of WindDynamics::WindPitchContEmulIEC

Attribute name	mult.	Attribute type	Description
kdroop	1..1	Simple_Float	Power error gain (K_{droop}). It is case dependent parameter.
kipce	1..1	Simple_Float	Pitch control emulator integral constant ($K_{I,pce}$). It is type dependent parameter.
komegaaero	1..1	PU	Aerodynamic power change vs. $omega_{WTR}$ change ($K_{omegaaero}$). It is case dependent parameter.
kppce	1..1	Simple_Float	Pitch control emulator proportional constant ($K_{P,pce}$). It is type dependent parameter.
omegaref	1..1	PU	Rotor speed in initial steady state ($omega_{ref}$). It is case dependent parameter.
pimax	1..1	PU	Maximum steady state power (p_{imax}). It is case dependent parameter.
pimin	1..1	PU	Minimum steady state power (p_{imin}). It is case dependent parameter.
t1	1..1	Seconds	First time constant in pitch control lag (T_1). It is type dependent parameter.
t2	1..1	Seconds	Second time constant in pitch control lag (T_2). It

			is type dependent parameter.
tpe	1..1	Seconds	Time constant in generator air gap power lag (T_{pe}). It is type dependent parameter.
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

9.6.18.20 WindPlantDynamics(Abstract)

Parent class supporting relationships to wind turbines Type 3 and 4 and wind plant IEC and user defined wind plants including their control models.

Inheritance path: ->DynamicsFunctionBlock->IdentifiedObject

Table 9-335 shows all attributes of WindPlantDynamics.

Table 9-335 - Attributes of WindDynamics::WindPlantDynamics

Attribute name	mult.	Attribute type	Description
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-336 shows all association ends of WindPlantDynamics with other classes.

Table 9-336 - Association ends of WindDynamics::WindPlantDynamics with other classes

mult.	mult.	Class type name	Description
RemoteInputSignal	[0..1]	RemoteInputSignal	The wind plant using the remote signal. Association Based On : WindDynamics:: WindPlantDynamics.WindPlantDynamics[0..1] ----- StandardInterconnections:: RemoteInputSignal.RemoteInputSignal[0..1]

9.6.18.21 WindPlantFreqPcontrolIEC

Frequency and active power controller model. Reference: IEC Standard 61400-27-1 Annex E.

Inheritance path: ->IdentifiedObject

Table 9-337 shows all attributes of WindPlantFreqPcontrolIEC.

Table 9-337 - Attributes of WindDynamics::WindPlantFreqPcontrolIEC

Attribute name	mult.	Attribute type	Description
dprefmax	1..1	PU	Maximum ramp rate of p_{WTref} request from the plant controller to the wind turbines (dp_{refmax}). It is project dependent parameter.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

dprefmin	1..1	PU	Minimum (negative) ramp rate of p_{WTref} request from the plant controller to the wind turbines (dp_{refmin}). It is project dependent parameter.
kiwpp	1..1	Simple_Float	Plant P controller integral gain (K_{IWPp}). It is type dependent parameter.
kpwpp	1..1	Simple_Float	Plant P controller proportional gain (K_{PWPp}). It is type dependent parameter.
prefmax	1..1	PU	Maximum p_{WTref} request from the plant controller to the wind turbines (p_{refmax}). It is type dependent parameter.
prefmin	1..1	PU	Minimum p_{WTref} request from the plant controller to the wind turbines (p_{refmin}). It is type dependent parameter.
tpft	1..1	Seconds	Lead time constant in reference value transfer function (T_{pft}). It is type dependent parameter.
tpfv	1..1	Seconds	Lag time constant in reference value transfer function (T_{pfv}). It is type dependent parameter.
twpffilt	1..1	Seconds	Filter time constant for frequency measurement ($T_{WPffilt}$). It is type dependent parameter.
twppfilt	1..1	Seconds	Filter time constant for active power measurement ($T_{WPpfilt}$). It is type dependent parameter.
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

9.6.18.22 WindPlantIEC

Simplified IEC type plant level model. Reference: IEC 61400-27-1, AnnexE.

Inheritance path: ->WindPlantDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-338 shows all attributes of WindPlantIEC.

Table 9-338 - Attributes of WindDynamics::WindPlantIEC

Attribute name	mult.	Attribute type	Description

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-339 shows all association ends of WindPlantIEC with other classes.

Table 9-339 - Association ends of WindDynamics::WindPlantIEC with other classes

mult.	mult.	Class type name	Description
WindPlantReactiveControlIEC	[1..1]	WindPlantReactiveControlIEC	Wind plant reactive control model associated with this wind plant. Association Based On : WindDynamics:: WindPlantIEC.WindPlantIEC[1] ----- WindDynamics:: WindPlantReactiveControlIEC.WindPlantReactiveControlIEC[1]
WindPlantFreqPcontrolIEC	[1..1]	WindPlantFreqPcontrolIEC	Wind plant frequency and active power control model associated with this wind plant. Association Based On : WindDynamics:: WindPlantFreqPcontrolIEC.WindPlantFreqPcontrolIEC[1] ----- WindDynamics:: WindPlantIEC.WindPlantIEC[1]
RemoteInputSignal	0..1	RemoteInputSignal	Inherited Association Based On : WindDynamics:: WindPlantDynamics.WindPlantDynamics[0..1] ----- StandardInterconnections:: RemoteInputSignal.RemoteInputSignal[0..1]

9.6.18.23 WindPlantReactiveControlIEC

Simplified plant voltage and reactive power control model for use with type 3 and type 4 wind turbine models.
Reference: IEC Standard 61400-27-1 Annex E.

Inheritance path: ->IdentifiedObject

Table 9-340 shows all attributes of WindPlantReactiveControlIEC.

Table 9-340 - Attributes of WindDynamics::WindPlantReactiveControlIEC

Attribute name	mult.	Attribute type	Description
kiwpx	1..1	Simple_Float	Plant Q controller integral gain (K_{IWPx}). It is type dependent parameter.
kpwpx	1..1	Simple_Float	Plant Q controller proportional gain (K_{PWPx}). It is type dependent parameter.
kwpqu	1..1	PU	Plant voltage control droop (K_{WPqu}). It is project dependent parameter.

mwppf	1..1	Boolean	Power factor control modes selector (M_{WPpf}). Used only if mwpu is set to false. true = 1: power factor control false = 0: reactive power control. It is project dependent parameter.
mwpu	1..1	Boolean	Reactive power control modes selector (M_{WPu}). true = 1: voltage control false = 0: reactive power control. It is project dependent parameter.
twppfilt	1..1	Seconds	Filter time constant for active power measurement ($T_{WPpfilt}$). It is type dependent parameter.
twpqfilt	1..1	Seconds	Filter time constant for reactive power measurement ($T_{WPqfilt}$). It is type dependent parameter.
twpufilt	1..1	Seconds	Filter time constant for voltage measurement ($T_{WPufilt}$). It is type dependent parameter.
txft	1..1	Seconds	Lead time constant in reference value transfer function (T_{xft}). It is type dependent parameter.
txfv	1..1	Seconds	Lag time constant in reference value transfer function (T_{xfv}). It is type dependent parameter.
uwpqdip	1..1	PU	Voltage threshold for LVRT detection in q control (u_{WPqdip}). It is type dependent parameter.
xrefmax	1..1	PU	Maximum x_{WTref} (q_{WTref} or delta u_{WTref}) request from the plant controller (x_{refmax}). It is project dependent parameter.
xrefmin	1..1	PU	Minimum x_{WTref} (q_{WTref} or deltau_{WTref}) request from the plant controller (x_{refmin}). It is project dependent parameter.
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

9.6.18.24 WindProtectionIEC

The grid protection model includes protection against over and under voltage, and against over and under frequency.
Reference: IEC Standard 614000-27-1 Section 6.6.6.

Inheritance path: ->IdentifiedObject

Table 9-341 shows all attributes of WindProtectionIEC.

Table 9-341 - Attributes of WindDynamics::WindProtectionIEC

Attribute name	mult.	Attribute type	Description
fover	1..1	PU	Set of wind turbine over frequency protection levels (f_{over}). It is project dependent parameter.
funder	1..1	PU	Set of wind turbine under frequency protection levels (f_{under}). It is project dependent parameter.
tfover	1..1	Seconds	Set of corresponding wind turbine over frequency protection disconnection times (T_{fover}). It is project dependent parameter.
tfunder	1..1	Seconds	Set of corresponding wind turbine under frequency protection disconnection times (T_{funder}). It is project dependent parameter.
tuover	1..1	Seconds	Set of corresponding wind turbine over voltage protection disconnection times (T_{uover}). It is project dependent parameter.
tuunder	1..1	Seconds	Set of corresponding wind turbine under voltage protection disconnection times (T_{uunder}). It is project dependent parameter.
uover	1..1	PU	Set of wind turbine over voltage protection levels (u_{over}). It is project dependent parameter.
uunder	1..1	PU	Set of wind turbine under voltage protection levels (u_{under}). It is project dependent parameter.
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

9.6.18.25 WindTurbineType1or2Dynamics(Abstract)

Parent class supporting relationships to wind turbines Type 1 and 2 and their control models.

Inheritance path: ->DynamicsFunctionBlock->IdentifiedObject

Table 9-342 shows all attributes of WindTurbineType1or2Dynamics.

Table 9-342 - Attributes of WindDynamics::WindTurbineType1or2Dynamics

Attribute name	mult.	Attribute type	Description
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-343 shows all association ends of WindTurbineType1or2Dynamics with other classes.

Table 9-343 - Association ends of WindDynamics::WindTurbineType1or2Dynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	[1..1]	AsynchronousMachineDynamics	Asynchronous machine model with which this wind generator type 1 or 2 model is associated. Association Based On : WindDynamics:: WindTurbineType1or2Dynamics.WindTurbineType1or2Dynamics[0..1] ----- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[1]
RemoteInputSignal	[0..1]	RemoteInputSignal	Remote input signal used by this wind generator Type 1 or Type 2 model. Association Based On : WindDynamics:: WindTurbineType1or2Dynamics.WindTurbineType1or2Dynamics[0..1] ----- StandardInterconnections:: RemoteInputSignal.RemoteInputSignal[0..1]

9.6.18.26 WindTurbineType1or2IEC(Abstract)

Generator model for wind turbine of IEC Type 1 or Type 2 is a standard asynchronous generator model.
Reference: IEC Standard 614000-27-1 Section 6.6.3.1.

Inheritance path: ->WindTurbineType1or2Dynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-344 shows all attributes of WindTurbineType1or2IEC.

Table 9-344 - Attributes of WindDynamics::WindTurbineType1or2IEC

Attribute name	mult.	Attribute type	Description
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

name	0..1	String	see IdentifiedObject

Table 9-345 shows all association ends of WindTurbineType1or2IEC with other classes.

Table 9-345 - Association ends of WindDynamics::WindTurbineType1or2IEC with other classes

mult.	mult.	Class type name	Description
WindProtectionIEC	[1..1]	WindProtectionIEC	Wind turbune protection model associated with this wind generator type 1 or 2 model. Association Based On : WindDynamics:: WindProtectionIEC.WindProtectionIEC[1] ----- WindDynamics:: WindTurbineType1or2IEC.WindTurbineType1or2IEC[0..1]
WindMechIEC	[1..1]	WindMechIEC	Wind mechanical model associated with this wind generator type 1 or 2 model. Association Based On : WindDynamics:: WindMechIEC.WindMechIEC[1] ----- WindDynamics:: WindTurbineType1or2IEC.WindTurbineType1or2IEC[0..1]
AsynchronousMachineDynamics	1..1	AsynchronousMachineDynamics	Inherited Association Based On : WindDynamics:: WindTurbineType1or2Dynamics.WindTurbineType1or2Dynamics[0..1] ----- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[1]
RemoteInputSignal	0..1	RemoteInputSignal	Inherited Association Based On : WindDynamics:: WindTurbineType1or2Dynamics.WindTurbineType1or2Dynamics[0..1] ----- StandardInterconnections:: RemoteInputSignal.RemoteInputSignal[0..1]

9.6.18.27 WindTurbineType3or4Dynamics(Abstract)

Parent class supporting relationships to wind turbines Type 3 and 4 and wind plant including their control models.

The class WindGeneratorTupe3or4Dynamics is associated with EnergySource as these types use current source in the load flow representation.

Inheritance path: ->DynamicsFunctionBlock->IdentifiedObject

Table 9-346 shows all attributes of WindTurbineType3or4Dynamics.

Table 9-346 - Attributes of WindDynamics::WindTurbineType3or4Dynamics

Attribute name	mult.	Attribute type	Description
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject

name	0..1	String	see IdentifiedObject

Table 9-347 shows all association ends of WindTurbineType3or4Dynamics with other classes.

Table 9-347 - Association ends of WindDynamics::WindTurbineType3or4Dynamics with other classes

mult.	mult.	Class type name	Description
WindPlantDynamics	[0..1]	WindPlantDynamics	The wind plant with which the wind turbines type 3 or 4 are associated. Association Based On : WindDynamics:: WindPlantDynamics.WindPlantDynamics[0..1] ----- WindDynamics:: WindTurbineType3or4Dynamics.WindTurbineType3or4Dynamics[1..*]
RemoteInputSignal	[0..1]	RemoteInputSignal	Wind turbine Type 3 or 4 models using this remote input signal. Association Based On : WindDynamics:: WindTurbineType3or4Dynamics.WindTurbineType3or4Dynamics[0..1] ----- StandardInterconnections:: RemoteInputSignal.RemoteInputSignal[0..1]
EnergySource	[1..1]	EnergySource	Energy Source (current source) with which this wind Type 3 or 4 dynamics model is asoociated. Association Based On : WindDynamics:: WindTurbineType3or4Dynamics.WindTurbineType3or4Dynamics[0..1] ----- Wires:: EnergySource.EnergySource[1]

9.6.18.28 WindTurbineType3or4IEC(Abstract)

Parent class supporting relationships to IEC wind turbines Type 3 and 4 and wind plant including their control models.

Inheritance path: ->WindTurbineType3or4Dynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-348 shows all attributes of WindTurbineType3or4IEC.

Table 9-348 - Attributes of WindDynamics::WindTurbineType3or4IEC

Attribute name	mult.	Attribute type	Description
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-349 shows all association ends of WindTurbineType3or4IEC with other classes.

Table 9-349 - Association ends of WindDynamics::WindTurbineType3or4IEC with other classes

mult.	mult.	Class type name	Description
WindProtectionIEC	[1..1]	WindProtectionIEC	Wind turbune protection model associated with this wind generator type 3 or 4 model. Association Based On : WindDynamics:: WindProtectionIEC.WindProtectionIEC[1] ----- WindDynamics::

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			WindTurbineType3or4IEC.WindTurbineType3or4IEC[0..1]
WIndContQIEC	[1..1]	WindContQIEC	Wind control Q model associated with this wind turbine type 3 or 4 model. Association Based On : WindDynamics:: WindContQIEC.WIndContQIEC[1] ----- WindDynamics:: WindTurbineType3or4IEC.WindTurbineType3or4IEC[1]
WindContCurrLimIEC	[1..1]	WindContCurrLimIEC	Wind control current limitation model associated with this wind turbine type 3 or 4 model. Association Based On : WindDynamics:: WindContCurrLimIEC.WindContCurrLimIEC[1] ----- WindDynamics:: WindTurbineType3or4IEC.WindTurbineType3or4IEC[1]
WindPlantDynamics	0..1	WindPlantDynamics	Inherited Association Based On : WindDynamics:: WindPlantDynamics.WindPlantDynamics[0..1] ----- WindDynamics:: WindTurbineType3or4Dynamics.WindTurbineType3or4Dynamics[1..*]
RemoteInputSignal	0..1	RemoteInputSignal	Inherited Association Based On : WindDynamics:: WindTurbineType3or4Dynamics.WindTurbineType3or4Dynamics[0..1] ----- StandardInterconnections:: RemoteInputSignal.RemoteInputSignal[0..1]
EnergySource	1..1	EnergySource	Inherited Association Based On : WindDynamics:: WindTurbineType3or4Dynamics.WindTurbineType3or4Dynamics[0..1] ----- Wires:: EnergySource.EnergySource[1]

9.6.18.29 WindTurbineType4aIEC

Wind turbine IEC Type 4A.
Reference: IEC Standard 61400-27-1, section 6.5.5.2.

Inheritance path: ->WindGenType4IEC->WindTurbineType3or4IEC->WindTurbineType3or4Dynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-350 shows all attributes of WindTurbineType4aIEC.

Table 9-350 - Attributes of WindDynamics::WindTurbineType4aIEC

Attribute name	mult.	Attribute type	Description
dipmax	1..1	PU	see WindGenType4IEC
diqmin	1..1	PU	see WindGenType4IEC
diqmax	1..1	PU	see WindGenType4IEC
tg	1..1	Seconds	see WindGenType4IEC
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 9-351 shows all association ends of WindTurbineType4aIEC with other classes.

Table 9-351 - Association ends of WindDynamics::WindTurbineType4aIEC with other classes

mult.	mult.	Class type name	Description
WindContPType4aIEC	[1..1]	WindContPType4aIEC	Wind control P type 4A model associated with this wind turbine type 4A model. Association Based On : WindDynamics:: WindContPType4aIEC.WindContPType4aIEC[1] ----- WindDynamics:: WindTurbineType4aIEC.WindTurbineType4aIEC[1]
WindProtectionIEC	1..1	WindProtectionIEC	Inherited Association Based On : WindDynamics:: WindProtectionIEC.WindProtectionIEC[1] ----- WindDynamics:: WindTurbineType3or4IEC.WindTurbineType3or4IEC[0..1]
WIndContQIEC	1..1	WindContQIEC	Inherited Association Based On : WindDynamics:: WindContQIEC.WIndContQIEC[1] ----- WindDynamics:: WindTurbineType3or4IEC.WindTurbineType3or4IEC[1]
WindContCurrLimIEC	1..1	WindContCurrLimIEC	Inherited Association Based On : WindDynamics:: WindContCurrLimIEC.WindContCurrLimIEC[1] ----- WindDynamics:: WindTurbineType3or4IEC.WindTurbineType3or4IEC[1]
WindPlantDynamics	0..1	WindPlantDynamics	Inherited Association Based On : WindDynamics:: WindPlantDynamics.WindPlantDynamics[0..1] ----- WindDynamics:: WindTurbineType3or4Dynamics.WindTurbineType3or4Dynamics[1..*]
RemoteInputSignal	0..1	RemoteInputSignal	Inherited Association Based On : WindDynamics:: WindTurbineType3or4Dynamics.WindTurbineType3or4Dynamics[0..1] ----- StandardInterconnections:: RemoteInputSignal.RemoteInputSignal[0..1]
EnergySource	1..1	EnergySource	Inherited Association Based On : WindDynamics:: WindTurbineType3or4Dynamics.WindTurbineType3or4Dynamics[0..1] ----- Wires:: EnergySource.EnergySource[1]

9.6.18.30 WindTurbineType4bIEC

Wind turbine IEC Type 4A.
Reference: IEC Standard 61400-27-1, section 6.5.5.3.

Inheritance path: ->WindGenType4IEC->WindTurbineType3or4IEC->WindTurbineType3or4Dynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-352 shows all attributes of WindTurbineType4bIEC.

Table 9-352 - Attributes of WindDynamics::WindTurbineType4bIEC

Attribute name	mult.	Attribute type	Description
dipmax	1..1	PU	see WindGenType4IEC
diqmin	1..1	PU	see WindGenType4IEC
diqmax	1..1	PU	see WindGenType4IEC
tg	1..1	Seconds	see WindGenType4IEC
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-353 shows all association ends of WindTurbineType4bIEC with other classes.

Table 9-353 - Association ends of WindDynamics::WindTurbineType4bIEC with other classes

mult.	mult.	Class type name	Description
WindMechIEC	[1..1]	WindMechIEC	Wind mechanical model associated with this wind turbine Type 4B model. Association Based On : WindDynamics:: WindMechIEC.WindMechIEC[1] ----- WindDynamics:: WindTurbineType4bIEC.WindTurbineType4bIEC[0..1]
WindContPType4bIEC	[1..1]	WindContPType4bIEC	Wind control P type 4B model associated with this wind turbine type 4B model. Association Based On : WindDynamics:: WindContPType4bIEC.WindContPType4bIEC[1] ----- WindDynamics:: WindTurbineType4bIEC.WindTurbineType4bIEC[1]
WindProtectionIEC	1..1	WindProtectionIEC	Inherited Association Based On : WindDynamics:: WindProtectionIEC.WindProtectionIEC[1] ----- WindDynamics:: WindTurbineType3or4IEC.WindTurbineType3or4IEC[0..1]
WIndContQIEC	1..1	WindContQIEC	Inherited Association Based On : WindDynamics:: WindContQIEC.WIndContQIEC[1] ----- WindDynamics:: WindTurbineType3or4IEC.WindTurbineType3or4IEC[1]
WindContCurrLimIEC	1..1	WindContCurrLimIEC	Inherited Association Based On : WindDynamics:: WindContCurrLimIEC.WindContCurrLimIEC[1] ----- WindDynamics:: WindTurbineType3or4IEC.WindTurbineType3or4IEC[1]
WindPlantDynamics	0..1	WindPlantDynamics	Inherited Association Based On : WindDynamics:: WindPlantDynamics.WindPlantDynamics[0..1] ----- WindDynamics:: WindTurbineType3or4Dynamics.WindTurbineType3or4Dynamics[1..*]

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

RemoteInputSignal	0..1	RemoteInputSignal	Inherited Association Based On : WindDynamics:: WindTurbineType3or4Dynamics.WindTurbineType3or4Dynamics[0..1] ----- StandardInterconnections:: RemoteInputSignal.RemoteInputSignal[0..1]
EnergySource	1..1	EnergySource	Inherited Association Based On : WindDynamics:: WindTurbineType3or4Dynamics.WindTurbineType3or4Dynamics[0..1] ----- Wires:: EnergySource.EnergySource[1]

9.6.18.31 Enumerations

9.6.18.31.1 WindLookupTableFunctionKind

Function of the lookup table.

Literal	Description	Code
fpslip	Power versus slip lookup table ($f_{pslip}()$). It is used for rotor resistance control model, IEC 61400-27-1, section 6.6.5.2.	
fpomega	Power vs. speed lookup table ($f_{pomega}()$). It is used for P control model type 3, IEC 61400-27-1, section 6.6.5.3.	
ipvdl	Lookup table for voltage dependency of active current limits ($i_{pVDL}()$). It is used for current limitation model, IEC 61400-27-1, section 6.6.5.7.	
iqvdl	Lookup table for voltage dependency of reactive current limits ($i_{qVDL}()$). It is used for current limitation model, IEC 61400-27-1, section 6.6.5.7.	
fdpf	Power vs. frequency lookup table ($f_{dpf}()$). It is used for wind power plant frequency and active power control model, IEC 61400-27-1, Annex E.	

9.6.18.31.2 WindLVRTQcontrolModesKind

LVRT Q control modes M_{qLVRT}.

Literal	Description	Code
mode1	Voltage dependent reactive current injection ($M_{LVRT,1}$).	
mode2	Reactive current injection controlled as the pre-fault value plus an additional voltage dependent reactive current injection ($M_{LVRT,2}$).	
mode3	Reactive current injection controlled as the pre-fault value plus an additional voltage dependent reactive current injection during fault, and as the pre-fault value plus an additional constant reactive current injection post fault ($M_{LVRT,3}$).	

9.6.18.31.3 WindQcontrolModesKind

General wind turbine Q control modes M_{qG}.

Literal	Description	Code
voltage	Voltage control ($M_{G,u}$).	
reactivePower	Reactive power control ($M_{G,q}$).	

openLoopReactivePower	Open loop reactive power control (only used with closed loop at plant level) ($M_{G,qol}$).	
powerFactor	Power factor control ($M_{G,pf}$).	

9.6.19 LoadDynamics

9.6.19.1 General

Dynamic load models are used to represent the dynamic real and reactive load behaviour of a load from the static power flow model.
Dynamic load models can be defined as applying either to a single load (energy consumer) or to a group of energy consumers.
Large industrial motors or groups of similar motors may be represented by individual motor models (synchronous or asynchronous) which are usually represented as generators with negative Pgen in the static (power flow) data.
In the CIM, such individual modelling is handled by child classes of either the SynchronousMachineDynamics or AsynchronousMachineDynamics classes.

Figure 9-160 shows the diagram.

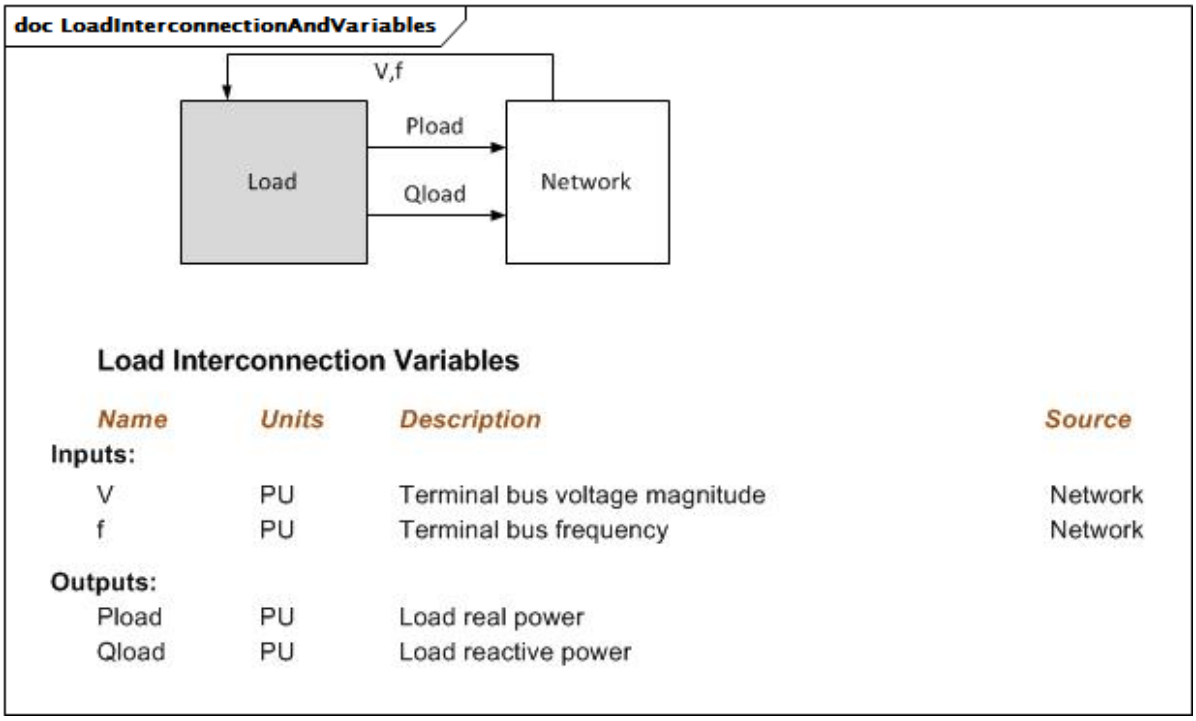


Figure 9-160 - diagram LoadInterconnectionAndVariables

This diagram illustrates the inputs and outputs for a load.
Notes:

1. The application program converts the P and Q of the load into a current injection at the bus.

Figure 9-161 shows the diagram.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

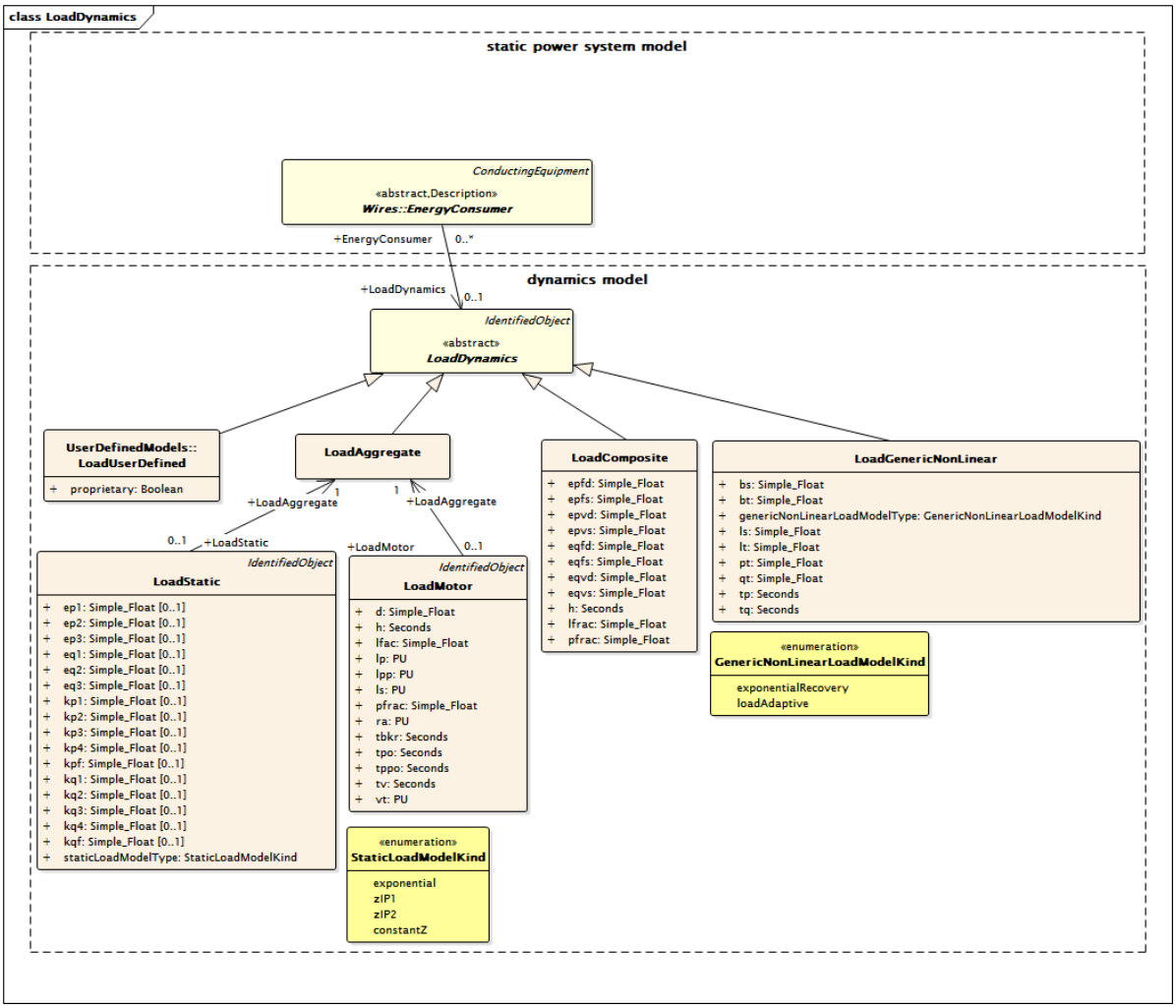


Figure 9-161 - diagram LoadDynamics

This diagram shows static power system model classes and dynamics model classes related to modeling of load for dynamics. An association between the EnergyConsumerDynamics class and the EnergyConsumer class provides a means for relating the dynamics model to the static model.

9.6.19.2 LoadComposite

This models combines static load and induction motor load effects. The dynamics of the motor are simplified by linearizing the induction machine equations.

Figure 9-162 class diagram

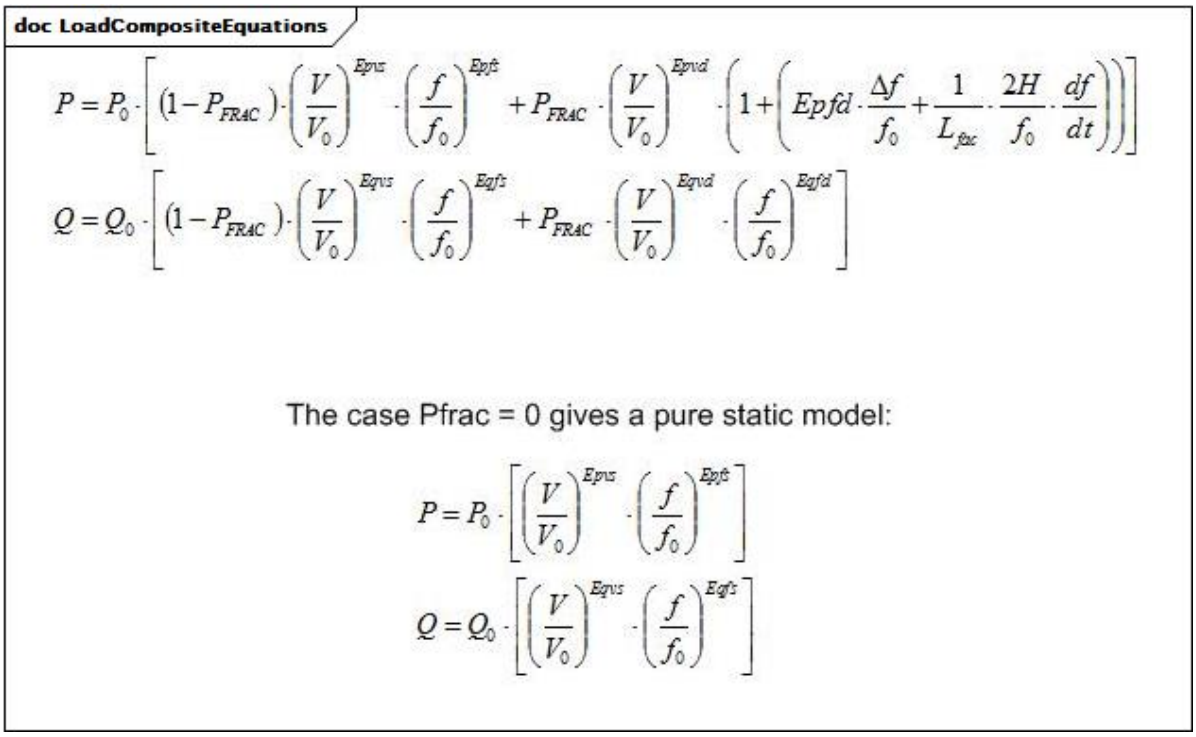


Figure 9-162 - figure LoadCompositeEquations

Inheritance path: ->LoadDynamics->IdentifiedObject

Table 9-354 shows all attributes of LoadComposite.

Table 9-354 - Attributes of LoadDynamics::LoadComposite

Attribute name	mult.	Attribute type	Description
epvs	1..1	Simple_Float	Active load-voltage dependence index (static) (Epvs). Typical Value = 0.7.
epfs	1..1	Simple_Float	Active load-frequency dependence index (static) (Epfs). Typical Value = 1.5.
eqvs	1..1	Simple_Float	Reactive load-voltage dependence index (static) (Eqvs). Typical Value = 2.
eqfs	1..1	Simple_Float	Reactive load-frequency dependence index (static) (Eqfs). Typical Value = 0.
epvd	1..1	Simple_Float	Active load-voltage dependence index (dynamic) (Epvd). Typical Value = 0.7.
epfd	1..1	Simple_Float	Active load-frequency dependence index (dynamic) (Epfd).

			Typical Value = 1.5.
eqvd	1..1	Simple_Float	Reactive load-voltage dependence index (dynamic) (Eqvd). Typical Value = 2.
eqfd	1..1	Simple_Float	Reactive load-frequency dependence index (dynamic) (Eqfd). Typical Value = 0.
lfrac	1..1	Simple_Float	Loading factor – ratio of initial P to motor MVA base (Lfrac). Typical Value = 0.8.
h	1..1	Seconds	Inertia constant (H). Typical Value = 2.5.
pfrac	1..1	Simple_Float	Fraction of constant-power load to be represented by this motor model (Pfrac) (>=0.0 and <=1.0). Typical Value = 0.5.
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

9.6.19.3 LoadGenericNonLinear

These load models (known also as generic non-linear dynamic (GNLD) load models) can be used in mid-term and long-term voltage stability simulations (i.e., to study voltage collapse), as they can replace a more detailed representation of aggregate load, including induction motors, thermostatically controlled and static loads.

Figure 9-163 class diagram

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

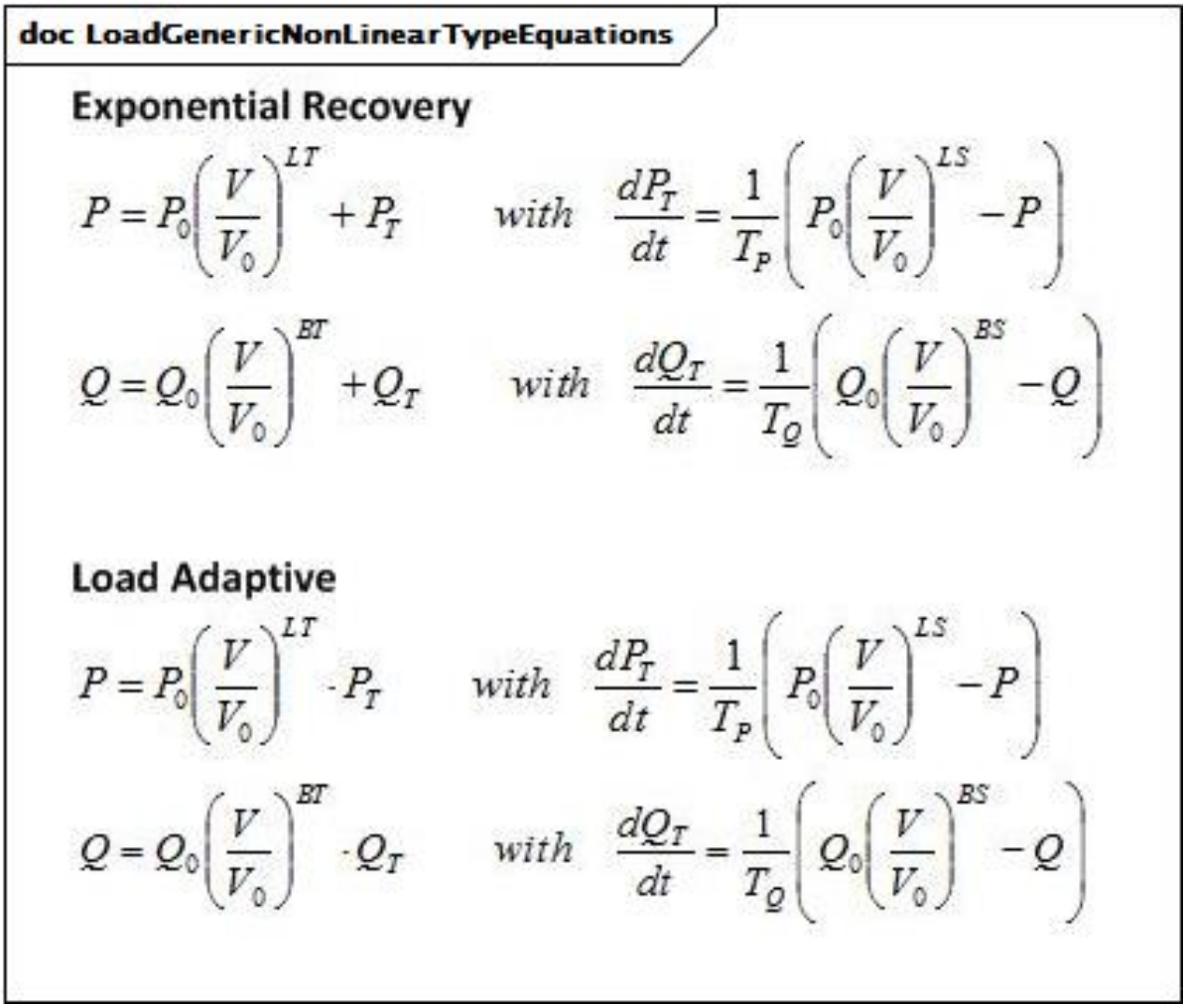


Figure 9-163 - figure LoadGenericNonLinearTypeEquations

Notes:
P_0 and **Q_0** are initial active and reactive power from the steady state solution (SvPowerFlow.p and SvPowerFlow.q).
V_0 is the initial voltage from the steady state solution (SvPowerFlow.v). **V** is the implied voltage signal from the bus to which load is attached.

Inheritance path: ->LoadDynamics->IdentifiedObject

Table 9-355 shows all attributes of LoadGenericNonLinear.

Table 9-355 - Attributes of LoadDynamics::LoadGenericNonLinear

Attribute name	mult.	Attribute type	Description
genericNonLinearLoadModelType	1..1	GenericNonLinearLoadModelKind	Type of generic non-linear load model.
pt	1..1	Simple_Float	Dynamic portion of active load (P_T).
qt	1..1	Simple_Float	Dynamic portion of reactive load (Q_T).

tp	1..1	Seconds	Time constant of lag function of active power (T_P).
tq	1..1	Seconds	Time constant of lag function of reactive power (T_Q).
ls	1..1	Simple_Float	Steady state voltage index for active power (LS).
lt	1..1	Simple_Float	Transient voltage index for active power (LT).
bs	1..1	Simple_Float	Steady state voltage index for reactive power (BS).
bt	1..1	Simple_Float	Transient voltage index for reactive power (BT).
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

9.6.19.4 LoadDynamics(Abstract)

Load whose behaviour is described by reference to a standard model or by definition of a user-defined model.
A standard feature of dynamic load behaviour modelling is the ability to associate the same behaviour to multiple energy consumers by means of a single aggregate load definition. Aggregate loads are used to represent all or part of the real and reactive load from one or more loads in the static (power flow) data. This load is usually the aggregation of many individual load devices and the load model is approximate representation of the aggregate response of the load devices to system disturbances. The load model is always applied to individual bus loads (energy consumers) but a single set of load model parameters can used for all loads in the grouping.

Inheritance path: ->IdentifiedObject

Table 9-356 shows all attributes of LoadDynamics.

Table 9-356 - Attributes of LoadDynamics::LoadDynamics

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

9.6.19.5 LoadAggregate

Standard aggregate load model comprised of static and/or dynamic components. A static load model represents the sensitivity of the real and reactive power consumed by the load to the amplitude and frequency of the bus voltage. A dynamic load model can used to represent the aggregate response of the motor components of the load.

Inheritance path: ->LoadDynamics->IdentifiedObject

Table 9-357 shows all attributes of LoadAggregate.

Table 9-357 - Attributes of LoadDynamics::LoadAggregate

Attribute name	mult.	Attribute type	Description
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

9.6.19.6 LoadStatic

General static load model representing the sensitivity of the real and reactive power consumed by the load to the amplitude and frequency of the bus voltage.

Figure 9-164 class diagram

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

doc LoadStaticTypeEquations

Exponential

$$P = P_0\left[K_{p1}\left(\frac{V}{V_0}\right)^{Ep1} + K_{p2}\left(\frac{V}{V_0}\right)^{Ep2} + K_{p3}\left(\frac{V}{V_0}\right)^{Ep3}\right]\left(1. + K_{pf}\Delta f\right)$$

$$Q = Q_0\left[K_{q1}\left(\frac{V}{V_0}\right)^{Eq1} + K_{q2}\left(\frac{V}{V_0}\right)^{Eq2} + K_{q3}\left(\frac{V}{V_0}\right)^{Eq3}\right]\left(1. + K_{qf}\Delta f\right)$$

ZIP1

$$P = P_0\left[K_{p1}\left(\frac{V}{V_0}\right)^{2} + K_{p2}\left(\frac{V}{V_0}\right)^{1} + K_{p3}\right]\left(1. + K_{pf}\Delta f\right)$$

$$Q = Q_0\left[K_{q1}\left(\frac{V}{V_0}\right)^{2} + K_{q2}\left(\frac{V}{V_0}\right)^{1} + K_{q3}\right]\left(1. + K_{qf}\Delta f\right)$$

ZIP2

$$P = P_0\left[K_{p1}\left(\frac{V}{V_0}\right)^{2} + K_{p2}\left(\frac{V}{V_0}\right)^{1} + K_{p3}\right] + K_{p4}\left(1. + K_{pf}\Delta f\right)$$

$$Q = Q_0\left[K_{q1}\left(\frac{V}{V_0}\right)^{2} + K_{q2}\left(\frac{V}{V_0}\right)^{1} + K_{q3}\right] + K_{q4}\left(1. + K_{qf}\Delta f\right)$$

ConstantZ

$$P = P_0\left(\frac{V}{V_0}\right)^{2}$$

$$Q = Q_0\left(\frac{V}{V_0}\right)^{2}$$

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Figure 9-164 - figure LoadStaticTypeEquations

Several variations of the static load model are used in various programs. The staticLoadModelType attribute is used to specify these variations.

Inheritance path: ->IdentifiedObject

Table 9-358 shows all attributes of LoadStatic.

Table 9-358 - Attributes of LoadDynamics::LoadStatic

Attribute name	mult.	Attribute type	Description
staticLoadModelType	1..1	StaticLoadModelKind	Type of static load model. Typical Value = constantZ.
kp1	0..1	Simple_Float	First term voltage coefficient for active power (Kp1). Not used when .staticLoadModelType = constantZ.
kp2	0..1	Simple_Float	Second term voltage coefficient for active power (Kp2). Not used when .staticLoadModelType = constantZ.
kp3	0..1	Simple_Float	Third term voltage coefficient for active power (Kp3). Not used when .staticLoadModelType = constantZ.
kp4	0..1	Simple_Float	Frequency coefficient for active power (Kp4). Must be non-zero when .staticLoadModelType = ZIP2. Not used for all other values of .staticLoadModelType.
ep1	0..1	Simple_Float	First term voltage exponent for active power (Ep1). Used only when .staticLoadModelType = exponential.
ep2	0..1	Simple_Float	Second term voltage exponent for active power (Ep2). Used only when .staticLoadModelType = exponential.
ep3	0..1	Simple_Float	Third term voltage exponent for active power (Ep3). Used only when .staticLoadModelType = exponential.
kpf	0..1	Simple_Float	Frequency deviation coefficient for active power (Kpf). Not used when .staticLoadModelType = constantZ.
kq1	0..1	Simple_Float	First term voltage coefficient for reactive power (Kq1). Not used when .staticLoadModelType

			= constantZ.
kq2	0..1	Simple_Float	Second term voltage coefficient for reactive power (Kq2). Not used when .staticLoadModelType = constantZ.
kq3	0..1	Simple_Float	Third term voltage coefficient for reactive power (Kq3). Not used when .staticLoadModelType = constantZ.
kq4	0..1	Simple_Float	Frequency coefficient for reactive power (Kq4). Must be non-zero when .staticLoadModelType = ZIP2. Not used for all other values of .staticLoadModelType.
eq1	0..1	Simple_Float	First term voltage exponent for reactive power (Eq1). Used only when .staticLoadModelType = exponential.
eq2	0..1	Simple_Float	Second term voltage exponent for reactive power (Eq2). Used only when .staticLoadModelType = exponential.
eq3	0..1	Simple_Float	Third term voltage exponent for reactive power (Eq3). Used only when .staticLoadModelType = exponential.
kqf	0..1	Simple_Float	Frequency deviation coefficient for reactive power (Kqf). Not used when .staticLoadModelType = constantZ.
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-359 shows all association ends of LoadStatic with other classes.

Table 9-359 - Association ends of LoadDynamics::LoadStatic with other classes

mult.	mult.	Class type name	Description
LoadAggregate	[1..1]	LoadAggregate	Aggregate load to which this aggregate static load belongs. Association Based On : LoadDynamics:: LoadStatic.LoadStatic[0..1] ----- LoadDynamics:: LoadAggregate.LoadAggregate[1]

9.6.19.7 LoadMotor

Aggregate induction motor load. This model is used to represent a fraction of an ordinary load as "induction motor load". It allows load that is treated as ordinary constant power in power flow

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

analysis to be represented by an induction motor in dynamic simulation. If **Lpp** = 0. or **Lpp** = **Lp**, or **Tppo** = 0., only one cage is represented. Magnetic saturation is not modelled. Either a "one-cage" or "two-cage" model of the induction machine can be modelled. Magnetic saturation is not modelled.
This model is intended for representation of aggregations of many motors dispersed through a load represented at a high voltage bus but where there is no information on the characteristics of individual motors.
This model treats a fraction of the constant power part of a load as a motor. During initialisation, the initial power drawn by the motor is set equal to **Pfrac** times the constant **P** part of the static load. The remainder of the load is left as static load.
The reactive power demand of the motor is calculated during initialisation as a function of voltage at the load bus. This reactive power demand may be less than or greater than the constant **Q** component of the load. If the motor's reactive demand is greater than the constant **Q** component of the load, the model inserts a shunt capacitor at the terminal of the motor to bring its reactive demand down to equal the constant **Q** reactive load.
If a motor model and a static load model are both present for a load, the motor **Pfrac** is assumed to be subtracted from the power flow constant **P** load before the static load model is applied. The remainder of the load, if any, is then represented by the static load model.

Figure 9-165 class diagram

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

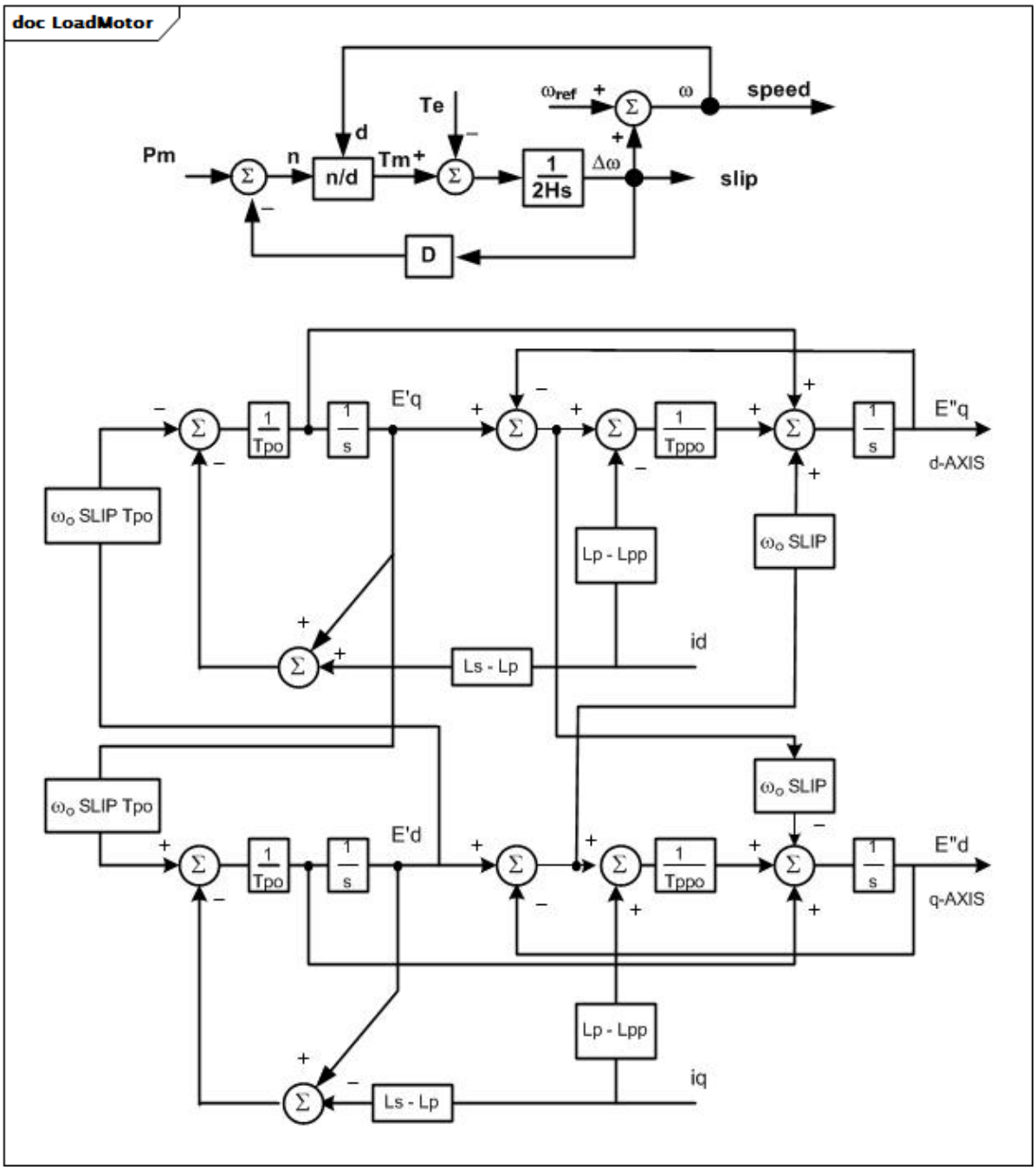


Figure 9-165 - figure LoadMotor

Parameter notes:

1. **Ls**, **Lp**, **Lpp** must all be specified.
2. The initial value of **Pm** or the value coming from an separate mechanical load model, if used, is multiplied by speed raised to the power **D** to determine the effective mechanical load seen by the motor.

3. The parameters **Vt** and **Tv** define under-voltage tripping logic. A timer is started if the terminal voltage falls below **Vt** per unit and continues to run until voltage rises above **Vt**. If voltage has not risen above **Vt** when the timer reaches **Tv** the motor is tripped immediately.
4. Per unit parameters are on the motor MVA base. The MVA base is calculated during initialization as the initial motor **P** in MW divided by the loading factor (**Lfac**) parameter.

Inheritance path: ->IdentifiedObject

Table 9-360 shows all attributes of LoadMotor.

Table 9-360 - Attributes of LoadDynamics::LoadMotor

Attribute name	mult.	Attribute type	Description
pfrac	1..1	Simple_Float	Fraction of constant-power load to be represented by this motor model (Pfrac) (>=0.0 and <=1.0). Typical Value = 0.3.
lfac	1..1	Simple_Float	Loading factor – ratio of initial P to motor MVA base (Lfac). Typical Value = 0.8.
ls	1..1	PU	Synchronous reactance (Ls). Typical Value = 3.2.
lp	1..1	PU	Transient reactance (Lp). Typical Value = 0.15.
lpp	1..1	PU	Subtransient reactance (Lpp). Typical Value = 0.15.
ra	1..1	PU	Stator resistance (Ra). Typical Value = 0.
tpo	1..1	Seconds	Transient rotor time constant (Tpo) (not=0). Typical Value = 1.
tppo	1..1	Seconds	Subtransient rotor time constant (Tppo). Typical Value = 0.02.
h	1..1	Seconds	Inertia constant (H) (not=0). Typical Value = 0.4.
d	1..1	Simple_Float	Damping factor (D). Unit = delta P/delta speed. Typical Value = 2.
vt	1..1	PU	Voltage threshold for tripping (Vt). Typical Value = 0.7.
tv	1..1	Seconds	Voltage trip pickup time (Tv). Typical Value = 0.1.
tbkr	1..1	Seconds	Circuit breaker operating time (Tbkr). Typical Value = 0.08.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-361 shows all association ends of LoadMotor with other classes.

Table 9-361 - Association ends of LoadDynamics::LoadMotor with other classes

mult.	mult.	Class type name	Description
LoadAggregate	[1..1]	LoadAggregate	Aggregate load to which this aggregate motor (dynamic) load belongs. Association Based On : LoadDynamics:: LoadMotor.LoadMotor[0..1] ----- LoadDynamics:: LoadAggregate.LoadAggregate[1]

9.6.19.8 Enumerations

9.6.19.8.1 GenericNonLinearLoadModelKind

Type of generic non-linear load model.

Literal	Description	Code
exponentialRecovery	Exponential recovery model.	
loadAdaptive	Load adaptive model.	

9.6.19.8.2 StaticLoadModelKind

Type of static load model.

Literal	Description	Code
exponential	Exponential P and Q equations are used and the following attributes are required: kp1, kp2, kp3, kpf, ep1, ep2, ep3 kq1, kq2, kq3, kqf, eq1, eq2, eq3.	
zIP1	ZIP1 P and Q equations are used and the following attributes are required: kp1, kp2, kp3, kpf kq1, kq2, kq3, kqf.	
zIP2	This model separates the frequency-dependent load (primarily motors) from other load. ZIP2 P and Q equations are used and the following attributes are required: kp1, kp2, kp3, kq4, kpf kq1, kq2, kq3, kq4, kqf.	
constantZ	The load is represented as a constant impedance. ConstantZ P and Q equations are used and no attributes are required.	

9.7 UserDefinedModels

9.7.1 General

This package contains user-defined dynamic model classes to support the exchange of both proprietary and explicitly defined models. Proprietary models have behavior which, while not defined by a standard model class, is mutually understood by the sending and receiving applications based on the name passed in the .name attribute of the xxxUserDefined class. Parameters are passed as general attributes using as many instances of the ProprietaryParameterDynamics class as there are parameters. Explicitly defined models describe dynamic behavior in detail in terms of control blocks and their

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

input and output signals. NOTE: The classes to support explicitly defined modeling are not currently defined - it is future work. They are described here to document the current thinking on where they would be added.

Figure 9-166 shows the diagram.

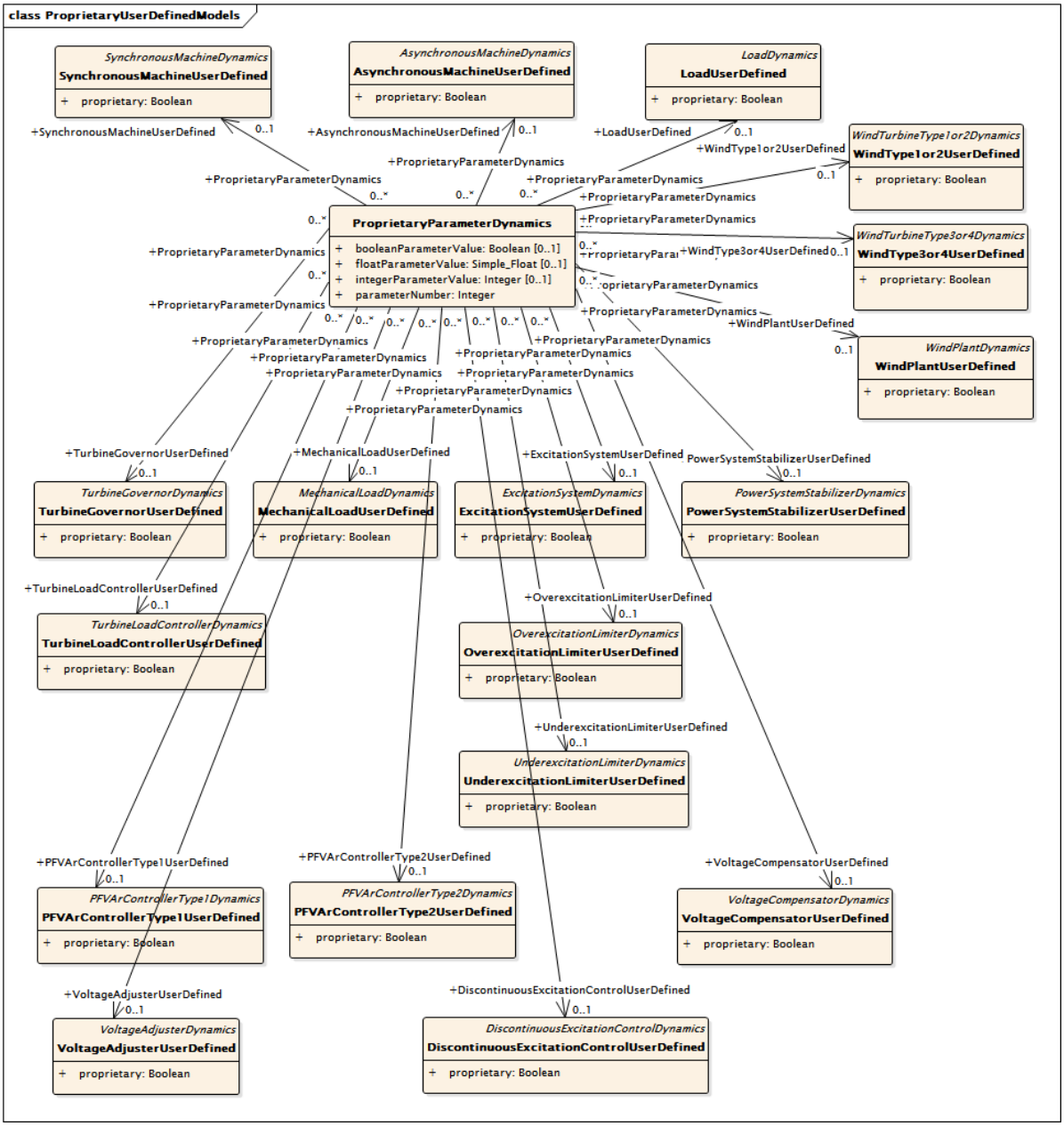


Figure 9-166 - diagram ProprietaryUserDefinedModels

This diagram shows the user-defined model classes and their relationship to the ProprietaryParameterDynamics class that supports the definition of multiple, appropriately datatyped attributes for each user-defined model.

9.7.2 WindPlantUserDefined

Wind plant function block whose dynamic behaviour is described by a user-defined model.

Inheritance path: ->WindPlantDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-362 shows all attributes of WindPlantUserDefined.

Table 9-362 - Attributes of UserDefinedModels::WindPlantUserDefined

Attribute name	mult.	Attribute type	Description
proprietary	1..1	Boolean	Behaviour is based on proprietary model as opposed to detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-363 shows all association ends of WindPlantDynamics with other classes.

Table 9-363 - Association ends of WindDynamics::WindPlantDynamics with other classes

mult.	mult.	Class type name	Description
RemoteInputSignal	0..1	RemoteInputSignal	Inherited Association Based On : WindDynamics:: WindPlantDynamics.WindPlantDynamics[0..1] --- StandardInterconnections:: RemoteInputSignal.RemoteInputSignal[0..1]

9.7.3 WindType1or2UserDefined

Wind Type 1 or Type 2 function block whose dynamic behaviour is described by a user-defined model.

Inheritance path: ->WindTurbineType1or2Dynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-364 shows all attributes of WindType1or2UserDefined.

Table 9-364 - Attributes of UserDefinedModels::WindType1or2UserDefined

Attribute name	mult.	Attribute type	Description
proprietary	1..1	Boolean	Behaviour is based on proprietary model as opposed to detailed model.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-365 shows all association ends of WindTurbineType1or2Dynamics with other classes.

Table 9-365 - Association ends of WindDynamics::WindTurbineType1or2Dynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	1..1	AsynchronousMachineDynamics	Inherited Association Based On : WindDynamics:: WindTurbineType1or2Dynamics.WindTurbineType1or2Dynamics[0..1] --- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[1]
RemoteInputSignal	0..1	RemoteInputSignal	Inherited Association Based On : WindDynamics:: WindTurbineType1or2Dynamics.WindTurbineType1or2Dynamics[0..1] --- StandardInterconnections:: RemoteInputSignal.RemoteInputSignal[0..1]

9.7.4 WindType3or4UserDefined

Wind Type 3 or Type 4 function block whose dynamic behaviour is described by a user-defined model.

Inheritance path: ->WindTurbineType3or4Dynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-366 shows all attributes of WindType3or4UserDefined.

Table 9-366 - Attributes of UserDefinedModels::WindType3or4UserDefined

Attribute name	mult.	Attribute type	Description
proprietary	1..1	Boolean	Behaviour is based on proprietary model as opposed to detailed model.

			true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-367 shows all association ends of WindTurbineType3or4Dynamics with other classes.

Table 9-367 - Association ends of WindDynamics::WindTurbineType3or4Dynamics with other classes

mult.	mult.	Class type name	Description
WindPlantDynamics	0..1	WindPlantDynamics	Inherited Association Based On : WindDynamics:: WindPlantDynamics.WindPlantDynamics[0..1] ----- WindDynamics:: WindTurbineType3or4Dynamics.WindTurbineType3or4Dynamics[1..*]
RemoteInputSignal	0..1	RemoteInputSignal	Inherited Association Based On : WindDynamics:: WindTurbineType3or4Dynamics.WindTurbineType3or4Dynamics[0..1] ----- StandardInterconnections:: RemoteInputSignal.RemoteInputSignal[0..1]
EnergySource	1..1	EnergySource	Inherited Association Based On : WindDynamics:: WindTurbineType3or4Dynamics.WindTurbineType3or4Dynamics[0..1] ----- Wires:: EnergySource.EnergySource[1]

9.7.5 SynchronousMachineUserDefined

Synchronous machine whose dynamic behaviour is described by a user-defined model.

Inheritance path: ->SynchronousMachineDynamics->RotatingMachineDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-368 shows all attributes of SynchronousMachineUserDefined.

Table 9-368 - Attributes of UserDefinedModels::SynchronousMachineUserDefined

Attribute name	mult.	Attribute type	Description
proprietary	1..1	Boolean	Behaviour is based on

			proprietary model as opposed to detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
damping	0..1	Simple_Float	see RotatingMachineDynamics
inertia	0..1	Seconds	see RotatingMachineDynamics
saturationFactor	0..1	Simple_Float	see RotatingMachineDynamics
saturationFactor120	0..1	Simple_Float	see RotatingMachineDynamics
statorLeakageReactance	0..1	PU	see RotatingMachineDynamics
statorResistance	0..1	PU	see RotatingMachineDynamics
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-369 shows all association ends of SynchronousMachineDynamics with other classes.

Table 9-369 - Association ends of SynchronousMachineDynamics::SynchronousMachineDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachine	1..1	SynchronousMachine	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..1] ----- Wires:: SynchronousMachine.SynchronousMachine[1]

9.7.6 AsynchronousMachineUserDefined

Asynchronous machine whose dynamic behaviour is described by a user-defined model.

Inheritance path: ->AsynchronousMachineDynamics->RotatingMachineDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-370 shows all attributes of AsynchronousMachineUserDefined.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 9-370 - Attributes of UserDefinedModels::AsynchronousMachineUserDefined

Attribute name	mult.	Attribute type	Description
proprietary	1..1	Boolean	Behaviour is based on proprietary model as opposed to detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
damping	0..1	Simple_Float	see RotatingMachineDynamics
inertia	0..1	Seconds	see RotatingMachineDynamics
saturationFactor	0..1	Simple_Float	see RotatingMachineDynamics
saturationFactor120	0..1	Simple_Float	see RotatingMachineDynamics
statorLeakageReactance	0..1	PU	see RotatingMachineDynamics
statorResistance	0..1	PU	see RotatingMachineDynamics
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-371 shows all association ends of AsynchronousMachineDynamics with other classes.

Table 9-371 - Association ends of AsynchronousMachineDynamics::AsynchronousMachineDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachine	1..1	AsynchronousMachine	Inherited Association Based On : AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1] --- Wires:: AsynchronousMachine.AsynchronousMachine[1]

9.7.7 TurbineGovernorUserDefined

Turbine-governor function block whose dynamic behaviour is described by a user-defined model.

Inheritance path: ->TurbineGovernorDynamics->DynamicsFunctionBlock->IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 9-372 shows all attributes of TurbineGovernorUserDefined.

Table 9-372 - Attributes of UserDefinedModels::TurbineGovernorUserDefined

Attribute name	mult.	Attribute type	Description
proprietary	1..1	Boolean	Behaviour is based on proprietary model as opposed to detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-373 shows all association ends of TurbineGovernorDynamics with other classes.

Table 9-373 - Association ends of TurbineGovernorDynamics::TurbineGovernorDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..1] --- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..*	SynchronousMachineDynamics	Inherited Association Based On : SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..*] --- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[0..*]

9.7.8 TurbineLoadControllerUserDefined

Turbine load controller function block whose dynamic behaviour is described by a user-defined model.

Inheritance path: ->TurbineLoadControllerDynamics->DynamicsFunctionBlock->IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 9-374 shows all attributes of TurbineLoadControllerUserDefined.

Table 9-374 - Attributes of UserDefinedModels::TurbineLoadControllerUserDefined

Attribute name	mult.	Attribute type	Description
proprietary	1..1	Boolean	Behaviour is based on proprietary model as opposed to detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-375 shows all association ends of TurbineLoadControllerDynamics with other classes.

Table 9-375 - Association ends of TurbineLoadControllerDynamics::TurbineLoadControllerDynamics with other classes

mult.	mult.	Class type name	Description
TurbineGovernorDynamics	1..1	TurbineGovernorDynamics	Inherited Association Based On : TurbineLoadControllerDynamics:: TurbineLoadControllerDynamics.TurbineLoadControllerDynamics[0..1] --- TurbineGovernorDynamics:: TurbineGovernorDynamics.TurbineGovernorDynamics[1]

9.7.9 MechanicalLoadUserDefined

Mechanical load function block whose dynamic behaviour is described by a user-defined model.

Inheritance path: ->MechanicalLoadDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-376 shows all attributes of MechanicalLoadUserDefined.

Table 9-376 - Attributes of UserDefinedModels::MechanicalLoadUserDefined

Attribute name	mult.	Attribute type	Description
proprietary	1..1	Boolean	Behaviour is based on proprietary model as opposed to detailed model. true = user-defined

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-377 shows all association ends of MechanicalLoadDynamics with other classes.

Table 9-377 - Association ends of MechanicalLoadDynamics::MechanicalLoadDynamics with other classes

mult.	mult.	Class type name	Description
AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Inherited Association Based On : MechanicalLoadDynamics:: MechanicalLoadDynamics.MechanicalLoadDynamics[0..1] ----- AsynchronousMachineDynamics:: AsynchronousMachineDynamics.AsynchronousMachineDynamics[0..1]
SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	Inherited Association Based On : MechanicalLoadDynamics:: MechanicalLoadDynamics.MechanicalLoadDynamics[0..1] ----- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[0..1]

9.7.10 ExcitationSystemUserDefined

Excitation system function block whose dynamic behaviour is described by a user-defined model.

Inheritance path: ->ExcitationSystemDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-378 shows all attributes of ExcitationSystemUserDefined.

Table 9-378 - Attributes of UserDefinedModels::ExcitationSystemUserDefined

Attribute name	mult.	Attribute type	Description
proprietary	1..1	Boolean	Behaviour is based on proprietary model as opposed to detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending

			and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-379 shows all association ends of ExcitationSystemDynamics with other classes.

Table 9-379 - Association ends of ExcitationSystemDynamics::ExcitationSystemDynamics with other classes

mult.	mult.	Class type name	Description
SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Inherited Association Based On : ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[0..1] ----- SynchronousMachineDynamics:: SynchronousMachineDynamics.SynchronousMachineDynamics[1]

9.7.11 OverexcitationLimiterUserDefined

Overexcitation limiter system function block whose dynamic behaviour is described by a user-defined model.

Inheritance path: ->OverexcitationLimiterDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-380 shows all attributes of OverexcitationLimiterUserDefined.

Table 9-380 - Attributes of UserDefinedModels::OverexcitationLimiterUserDefined

Attribute name	mult.	Attribute type	Description
proprietary	1..1	Boolean	Behaviour is based on proprietary model as opposed to detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output

			signals.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-381 shows all association ends of OverexcitationLimiterDynamics with other classes.

Table 9-381 - Association ends of OverexcitationLimiterDynamics::OverexcitationLimiterDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : OverexcitationLimiterDynamics:: OverexcitationLimiterDynamics.OverexcitationLimiterDynamics[0..1] --- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.7.12 UnderexcitationLimiterUserDefined

Underexcitation limiter function block whose dynamic behaviour is described by a user-defined model.

Inheritance path: ->UnderexcitationLimiterDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-382 shows all attributes of UnderexcitationLimiterUserDefined.

Table 9-382 - Attributes of UserDefinedModels::UnderexcitationLimiterUserDefined

Attribute name	mult.	Attribute type	Description
proprietary	1..1	Boolean	Behaviour is based on proprietary model as opposed to detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-383 shows all association ends of UnderexcitationLimiterDynamics with other classes.

Table 9-383 - Association ends of UnderexcitationLimiterDynamics::UnderexcitationLimiterDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : UnderexcitationLimiterDynamics:: UnderexcitationLimiterDynamics.UnderexcitationLimiterDynamics[0..1] --- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.7.13 PowerSystemStabilizerUserDefined

Power system stabilizer function block whose dynamic behaviour is described by a user-defined model.

Inheritance path: ->PowerSystemStabilizerDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-384 shows all attributes of PowerSystemStabilizerUserDefined.

Table 9-384 - Attributes of UserDefinedModels::PowerSystemStabilizerUserDefined

Attribute name	mult.	Attribute type	Description
proprietary	1..1	Boolean	Behaviour is based on proprietary model as opposed to detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-385 shows all association ends of PowerSystemStabilizerDynamics with other classes.

Table 9-385 - Association ends of PowerSystemStabilizerDynamics::PowerSystemStabilizerDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDyna	1..1	ExcitationSystemDynam	

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

mics		ics	Inherited Association Based On : PowerSystemStabilizerDynamics:: PowerSystemStabilizerDynamics.PowerSystemStabilizerDynamics[0..1] ----- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.7.14 DiscontinuousExcitationControlUserDefined

Discontinuous excitation control function block whose dynamic behaviour is described by a user-defined model.

Inheritance path: ->DiscontinuousExcitationControlDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-386 shows all attributes of DiscontinuousExcitationControlUserDefined.

Table 9-386 - Attributes of UserDefinedModels::DiscontinuousExcitationControlUserDefined

Attribute name	mult.	Attribute type	Description
proprietary	1..1	Boolean	Behaviour is based on proprietary model as opposed to detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-387 shows all association ends of DiscontinuousExcitationControlDynamics with other classes.

Table 9-387 - Association ends of DiscontinuousExcitationControlDynamics::DiscontinuousExcitationControlDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : DiscontinuousExcitationControlDynamics:: DiscontinuousExcitationControlDynamics.DiscontinuousExcitationControlDynamics[0..1] ----- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[

			1]

9.7.15 PFVArControllerType1UserDefined

Power Factor or VAr controller Type I function block whose dynamic behaviour is described by a user-defined model.

Inheritance path: ->PFVArControllerType1Dynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-388 shows all attributes of PFVArControllerType1UserDefined.

Table 9-388 - Attributes of UserDefinedModels::PFVArControllerType1UserDefined

Attribute name	mult.	Attribute type	Description
proprietary	1..1	Boolean	Behaviour is based on proprietary model as opposed to detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-389 shows all association ends of PFVArControllerType1Dynamics with other classes.

Table 9-389 - Association ends of PFVArControllerType1Dynamics::PFVArControllerType1Dynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : PFVArControllerType1Dynamics:: PFVArControllerType1Dynamics.PFVArControllerType1Dynamics[0..1] --- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.7.16 VoltageAdjusterUserDefined

Voltage adjuster function block whose dynamic behaviour is described by a user-defined model.

Inheritance path: ->VoltageAdjusterDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-390 shows all attributes of VoltageAdjusterUserDefined.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 9-390 - Attributes of UserDefinedModels::VoltageAdjusterUserDefined

Attribute name	mult.	Attribute type	Description
proprietary	1..1	Boolean	Behaviour is based on proprietary model as opposed to detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-391 shows all association ends of VoltageAdjusterDynamics with other classes.

Table 9-391 - Association ends of VoltageAdjusterDynamics::VoltageAdjusterDynamics with other classes

mult.	mult.	Class type name	Description
PFVArControllerType1Dynamics	1..1	PFVArControllerType1Dynamics	Inherited Association Based On : VoltageAdjusterDynamics:: VoltageAdjusterDynamics.VoltageAdjusterDynamics[0..1] --- PFVArControllerType1Dynamics:: PFVArControllerType1Dynamics.PFVArControllerType1Dynamics[1]

9.7.17 PFVArControllerType2UserDefined

Power Factor or VAr controller Type II function block whose dynamic behaviour is described by a user-defined model.

Inheritance path: ->PFVArControllerType2Dynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-392 shows all attributes of PFVArControllerType2UserDefined.

Table 9-392 - Attributes of UserDefinedModels::PFVArControllerType2UserDefined

Attribute name	mult.	Attribute type	Description
proprietary	1..1	Boolean	Behaviour is based on proprietary model as opposed to detailed model. true = user-defined model is proprietary

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-393 shows all association ends of PFVArControllerType2Dynamics with other classes.

Table 9-393 - Association ends of PFVArControllerType2Dynamics::PFVArControllerType2Dynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : PFVArControllerType2Dynamics:: PFVArControllerType2Dynamics.PFVArControllerType2Dynamics[0..1] --- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.7.18 VoltageCompensatorUserDefined

Voltage compensator function block whose dynamic behaviour is described by a user-defined model.

Inheritance path: ->VoltageCompensatorDynamics->DynamicsFunctionBlock->IdentifiedObject

Table 9-394 shows all attributes of VoltageCompensatorUserDefined.

Table 9-394 - Attributes of UserDefinedModels::VoltageCompensatorUserDefined

Attribute name	mult.	Attribute type	Description
proprietary	1..1	Boolean	Behaviour is based on proprietary model as opposed to detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their

			input and output signals.
enabled	1..1	Boolean	see DynamicsFunctionBlock
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

Table 9-395 shows all association ends of VoltageCompensatorDynamics with other classes.

Table 9-395 - Association ends of VoltageCompensatorDynamics::VoltageCompensatorDynamics with other classes

mult.	mult.	Class type name	Description
ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Inherited Association Based On : VoltageCompensatorDynamics:: VoltageCompensatorDynamics.VoltageCompensatorDynamics[0..1] --- ExcitationSystemDynamics:: ExcitationSystemDynamics.ExcitationSystemDynamics[1]

9.7.19 LoadUserDefined

Load whose dynamic behaviour is described by a user-defined model.

Inheritance path: ->LoadDynamics->IdentifiedObject

Table 9-396 shows all attributes of LoadUserDefined.

Table 9-396 - Attributes of UserDefinedModels::LoadUserDefined

Attribute name	mult.	Attribute type	Description
proprietary	1..1	Boolean	Behaviour is based on proprietary model as opposed to detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
description	0..1	String	see IdentifiedObject
mRID	0..1	String	see IdentifiedObject
name	0..1	String	see IdentifiedObject

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

9.7.20 ProprietaryParameterDynamics

Supports definition of one or more parameters of several different datatypes for use by proprietary user-defined models. NOTE: This class does not inherit from IdentifiedObject since it is not intended that a single instance of it be referenced by more than one proprietary user-defined model instance.

Table 9-397 shows all attributes of ProprietaryParameterDynamics.

Table 9-397 - Attributes of UserDefinedModels::ProprietaryParameterDynamics

Attribute name	mult.	Attribute type	Description
parameterNumber	1..1	Integer	Sequence number of the parameter among the set of parameters associated with the related proprietary user-defined model.
booleanParameterValue	0..1	Boolean	Used for boolean parameter value. If this attribute is populated, integerParameterValue and floatParameterValue will not be.
integerParameterValue	0..1	Integer	Used for integer parameter value. If this attribute is populated, booleanParameterValue and floatParameterValue will not be.
floatParameterValue	0..1	Simple_Float	Used for floating point parameter value. If this attribute is populated, booleanParameterValue and integerParameterValue will not be.

Table 9-398 shows all association ends of ProprietaryParameterDynamics with other classes.

Table 9-398 - Association ends of UserDefinedModels::ProprietaryParameterDynamics with other classes

mult.	mult.	Class type name	Description
LoadUserDefined	[0..1]	LoadUserDefined	Proprietary user-defined model with which this parameter is associated. Association Based On : UserDefinedModels:: ProprietaryParameterDynamics.ProprietaryParameterDynamics[0..*] ----- UserDefinedModels:: LoadUserDefined.LoadUserDefined[0..1]
VoltageCompensatorUserDefined	[0..1]	VoltageCompensatorUserDefined	Proprietary user-defined model with which this parameter is associated. Association Based On : UserDefinedModels:: ProprietaryParameterDynamics.ProprietaryParameterDynamics[0..*] ----- UserDefinedModels:: VoltageCompensatorUserDefined.VoltageCompensator

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

			UserDefined[0..1]
PFVArControllerType2UserDefined	[0..1]	PFVArControllerType2UserDefined	Proprietary user-defined model with which this parameter is associated. Association Based On : UserDefinedModels::ProprietaryParameterDynamics.ProprietaryParameterDynamics[0..*] ----- UserDefinedModels::PFVArControllerType2UserDefined.PFVArControllerType2UserDefined[0..1]
VoltageAdjusterUserDefined	[0..1]	VoltageAdjusterUserDefined	Proprietary user-defined model with which this parameter is associated. Association Based On : UserDefinedModels::ProprietaryParameterDynamics.ProprietaryParameterDynamics[0..*] ----- UserDefinedModels::VoltageAdjusterUserDefined.VoltageAdjusterUserDefined[0..1]
PFVArControllerType1UserDefined	[0..1]	PFVArControllerType1UserDefined	Proprietary user-defined model with which this parameter is associated. Association Based On : UserDefinedModels::ProprietaryParameterDynamics.ProprietaryParameterDynamics[0..*] ----- UserDefinedModels::PFVArControllerType1UserDefined.PFVArControllerType1UserDefined[0..1]
DiscontinuousExcitationControlUserDefined	[0..1]	DiscontinuousExcitationControlUserDefined	Proprietary user-defined model with which this parameter is associated. Association Based On : UserDefinedModels::ProprietaryParameterDynamics.ProprietaryParameterDynamics[0..*] ----- UserDefinedModels::DiscontinuousExcitationControlUserDefined.DiscontinuousExcitationControlUserDefined[0..1]
PowerSystemStabilizerUserDefined	[0..1]	PowerSystemStabilizerUserDefined	Proprietary user-defined model with which this parameter is associated. Association Based On : UserDefinedModels::ProprietaryParameterDynamics.ProprietaryParameterDynamics[0..*] ----- UserDefinedModels::PowerSystemStabilizerUserDefined.PowerSystemStabilizerUserDefined[0..1]
UnderexcitationLimiterUserDefined	[0..1]	UnderexcitationLimiterUserDefined	Proprietary user-defined model with which this parameter is associated. Association Based On : UserDefinedModels::ProprietaryParameterDynamics.ProprietaryParameterDynamics[0..*] ----- UserDefinedModels::UnderexcitationLimiterUserDefined.UnderexcitationLimiterUserDefined[0..1]
OverexcitationLimiterUserDefined	[0..1]	OverexcitationLimiterUserDefined	Proprietary user-defined model with which this parameter is associated. Association Based On : UserDefinedModels::ProprietaryParameterDynamics.ProprietaryParameterDynamics[0..*] ----- UserDefinedModels::OverexcitationLimiterUserDefined.OverexcitationLimiterUserDefined[0..1]

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

ExcitationSystemUserDefined	[0..1]	ExcitationSystemUserDefined	Proprietary user-defined model with which this parameter is associated. Association Based On : UserDefinedModels::ProprietaryParameterDynamics.ProprietaryParameterDynamics[0..*] ----- UserDefinedModels::ExcitationSystemUserDefined.ExcitationSystemUserDefined[0..1]
MechanicalLoadUserDefined	[0..1]	MechanicalLoadUserDefined	Proprietary user-defined model with which this parameter is associated. Association Based On : UserDefinedModels::ProprietaryParameterDynamics.ProprietaryParameterDynamics[0..*] ----- UserDefinedModels::MechanicalLoadUserDefined.MechanicalLoadUserDefined[0..1]
TurbineLoadControllerUserDefined	[0..1]	TurbineLoadControllerUserDefined	Proprietary user-defined model with which this parameter is associated. Association Based On : UserDefinedModels::ProprietaryParameterDynamics.ProprietaryParameterDynamics[0..*] ----- UserDefinedModels::TurbineLoadControllerUserDefined.TurbineLoadControllerUserDefined[0..1]
TurbineGovernorUserDefined	[0..1]	TurbineGovernorUserDefined	Proprietary user-defined model with which this parameter is associated. Association Based On : UserDefinedModels::ProprietaryParameterDynamics.ProprietaryParameterDynamics[0..*] ----- UserDefinedModels::TurbineGovernorUserDefined.TurbineGovernorUserDefined[0..1]
AsynchronousMachineUserDefined	[0..1]	AsynchronousMachineUserDefined	Proprietary user-defined model with which this parameter is associated. Association Based On : UserDefinedModels::ProprietaryParameterDynamics.ProprietaryParameterDynamics[0..*] ----- UserDefinedModels::AsynchronousMachineUserDefined.AsynchronousMachineUserDefined[0..1]
SynchronousMachineUserDefined	[0..1]	SynchronousMachineUserDefined	Proprietary user-defined model with which this parameter is associated. Association Based On : UserDefinedModels::ProprietaryParameterDynamics.ProprietaryParameterDynamics[0..*] ----- UserDefinedModels::SynchronousMachineUserDefined.SynchronousMachineUserDefined[0..1]
WindType3or4UserDefined	[0..1]	WindType3or4UserDefined	Proprietary user-defined model with which this parameter is associated. Association Based On : UserDefinedModels::WindType3or4UserDefined.WindType3or4UserDefined[0..1] ----- UserDefinedModels::ProprietaryParameterDynamics.ProprietaryParameterDynamics[0..*]
WindType1or2UserDe	[0..1]	WindType1or2UserDe	Proprietary user-defined model with which this

fined		fined	parameter is associated. Association Based On : UserDefinedModels:: WindType1or2UserDefined.WindType1or2UserDefined[0..1] ----- UserDefinedModels:: ProprietaryParameterDynamics.ProprietaryParameterDynamics[0..*]
WindPlantUserDefined	[0..1]	WindPlantUserDefined	Proprietary user-defined model with which this parameter is associated. Association Based On : UserDefinedModels:: WindPlantUserDefined.WindPlantUserDefined[0..1] ----- UserDefinedModels:: ProprietaryParameterDynamics.ProprietaryParameterDynamics[0..*]

9.8 Global Used Datatypes

9.8.1 Datatypes

9.8.1.1 Simple_Float

A floating point number. The range is unspecified and not limited.

In ENTSO-E profile, Simple_Float range is the IEEE754 simple precision floating point one. It correspond to xs:float datatype

Table 9-399 shows all attributes of Simple_Float.

Table 9-399 - Attributes of DomainProfile::Simple_Float.

Attribute name	mult.	Attribute type	Description
value	1..1	Float	

9.8.1.2 Seconds

Time, in seconds.

Value type is IEEE 754 simple precision floating point

Table 9-400 shows all attributes of Seconds.

Table 9-400 - Attributes of DomainProfile::Seconds.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	Time, in seconds
unit	0..1	UnitSymbol	ConstantValue= s
multiplier	0..1	UnitMultiplier	ConstantValue= none

9.8.1.3 PU

Per Unit - a positive or negative value referred to a defined base. Values typically range from -10 to +10.

Value type is IEEE 754 simple precision floating point

Table 9-401 shows all attributes of PU.

Table 9-401 - Attributes of DomainProfile::PU.

Attribute name	mult.	Attribute type	Description

value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= none
multiplier	0..1	UnitMultiplier	ConstantValue= none

9.8.1.4 CurrentFlow

Electrical current with sign convention: positive flow is out of the conducting equipment into the connectivity node. Can be both AC and DC.

Value type is IEEE 754 simple precision floating point

Table 9-402 shows all attributes of CurrentFlow.

Table 9-402 - Attributes of DomainProfile::CurrentFlow.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= A
multiplier	0..1	UnitMultiplier	ConstantValue= none

9.8.1.5 ActivePower

Product of RMS value of the voltage and the RMS value of the in-phase component of the current.

Value type is IEEE 754 simple precision floating point

Table 9-403 shows all attributes of ActivePower.

Table 9-403 - Attributes of DomainProfile::ActivePower.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= W
multiplier	0..1	UnitMultiplier	ConstantValue= M

9.8.1.6 Frequency

Cycles per second.

Value type is IEEE 754 simple precision floating point

Table 9-404 shows all attributes of Frequency.

Table 9-404 - Attributes of DomainProfile::Frequency.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= Hz
multiplier	0..1	UnitMultiplier	DefaultValue= none

9.8.1.7 Temperature

Value of temperature in degrees Celsius.

Value type is IEEE 754 simple precision floating point

Table 9-405 shows all attributes of Temperature.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Table 9-405 - Attributes of DomainProfile::Temperature.

Attribute name	mult.	Attribute type	Description
multiplier	0..1	UnitMultiplier	ConstantValue= none
unit	0..1	UnitSymbol	ConstantValue= degC
value	0..1	Float	

9.8.1.8 Area

Area.

Table 9-406 shows all attributes of Area.

Table 9-406 - Attributes of DomainProfile::Area.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= m2
multiplier	0..1	UnitMultiplier	

9.8.1.9 Length

Unit of length. Never negative.

Value type is IEEE 754 simple precision floating point

Table 9-407 shows all attributes of Length.

Table 9-407 - Attributes of DomainProfile::Length.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= m
multiplier	0..1	UnitMultiplier	ConstantValue= k

9.8.1.10 VolumeFlowRate

Volume per time.

Table 9-408 shows all attributes of VolumeFlowRate.

Table 9-408 - Attributes of DomainProfile::VolumeFlowRate.

Attribute name	mult.	Attribute type	Description
denominatorMultiplier	0..1	UnitMultiplier	ConstantValue= none
denominatorUnit	0..1	UnitSymbol	ConstantValue= s
multiplier	0..1	UnitMultiplier	ConstantValue= none
unit	0..1	UnitSymbol	ConstantValue= m3
value	0..1	Float	

9.8.1.11 AngleDegrees

Measurement of angle in degrees.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Value type is IEEE 754 simple precision floating point

Table 9-409 shows all attributes of AngleDegrees.

Table 9-409 - Attributes of DomainProfile::AngleDegrees.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= deg
multiplier	0..1	UnitMultiplier	ConstantValue= none

9.8.1.12 ApparentPower

Product of the RMS value of the voltage and the RMS value of the current.

Value type is IEEE 754 simple precision floating point

Table 9-410 shows all attributes of ApparentPower.

Table 9-410 - Attributes of DomainProfile::ApparentPower.

Attribute name	mult.	Attribute type	Description
value	0..1	Float	
unit	0..1	UnitSymbol	ConstantValue= VA
multiplier	0..1	UnitMultiplier	ConstantValue= M

9.8.2 Enumerations

9.8.2.1 RemoteSignalKind

Type of input signal coming from remote bus.

Literal	Description	Code
remoteBusVoltageFrequency	Input is voltage frequency from remote terminal bus.	
remoteBusVoltageFrequencyDeviation	Input is voltage frequency deviation from remote terminal bus.	
remoteBusFrequency	Input is frequency from remote terminal bus.	
remoteBusFrequencyDeviation	Input is frequency deviation from remote terminal bus.	
remoteBusVoltageAmplitude	Input is voltage amplitude from remote terminal bus.	
remoteBusVoltage	Input is voltage from remote terminal bus.	
remoteBranchCurrentAmplitude	Input is branch current amplitude from remote terminal bus.	
remoteBusVoltageAmplitudeDerivative	Input is branch current amplitude derivative from remote terminal bus.	
remotePuBusVoltageDerivative	Input is PU voltage derivative from remote terminal bus.	

9.8.2.2 UnitSymbol

The units defined for usage in the CIM.

Literal	Description	Code
VA	Apparent power in volt ampere.	
W	Active power in watt.	

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

VAr	Reactive power in volt ampere reactive.	
VAh	Apparent energy in volt ampere hours.	
Wh	Real energy in what hours.	
VArh	Reactive energy in volt ampere reactive hours.	
V	Voltage in volt.	
ohm	Resistance in ohm.	
A	Current in ampere.	
F	Capacitance in farad.	
H	Inductance in henry.	
degC	Relative temperature in degrees Celsius. In the SI unit system the symbol is ºC. Electric charge is measured in coulomb that has the unit symbol C. To distinguish degree Celsius form coulomb the symbol used in the UML is degC. Reason for not using ºC is the special character º is difficult to manage in software.	
s	Time in seconds.	
min	Time in minutes.	
h	Time in hours.	
deg	Plane angle in degrees.	
rad	Plane angle in radians.	
J	Energy in joule.	
N	Force in newton.	
S	Conductance in siemens.	
none	Dimension less quantity, e.g. count, per unit, etc.	
Hz	Frequency in hertz.	
g	Mass in gram.	
Pa	Pressure in pascal (n/m2).	
m	Length in meter.	
m2	Area in square meters.	
m3	Volume in cubic meters.	

9.8.2.3 UnitMultiplier

The unit multipliers defined for the CIM.

Literal	Description	Code
p	Pico 10**-12.	
n	Nano 10**-9.	
micro	Micro 10**-6.	
m	Milli 10**-3.	
c	Centi 10**-2.	
d	Deci 10**-1.	
k	Kilo 10**3.	

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

M	Mega 10**6.	
G	Giga 10**9.	
T	Tera 10**12.	
none	No multiplier or equivalently multiply by 1.	

9.8.2.4 IfdBaseKind

Excitation base system mode.

Literal	Description	Code
ifag	Air gap line mode. ifdBaseValue is computed, not defined by the user, in this mode.	
ifnl	No load system with saturation mode. ifdBaseValue is computed, not defined by the user, in this mode.	
iffl	Full load system mode. ifdBaseValue is computed, not defined by the user, in this mode.	
other	Free mode. ifdBaseValue is defined by the user in this mode.	

9.8.2.5 RotorKind

Type of rotor on physical machine.

Literal	Description	Code
roundRotor	Round rotor type of synchronous machine.	
salientPole	Salient pole type of synchronous machine.	

9.8.2.6 SynchronousMachineModelKind

Type of synchronous machine model used in Dynamic simulation applications.

Literal	Description	Code
subtransient	Subtransient synchronous machine model.	
subtransientTypeF	WECC Type F variant of subtransient synchronous machine model.	
subtransientTypeJ	WECC Type J variant of subtransient synchronous machine model.	
subtransientSimplified	Simplified version of subtransient synchronous machine model where magnetic coupling between the direct and quadrature axes is ignored.	
subtransientSimplifiedDirectAxis	Simplified version of a subtransient synchronous machine model with no damper circuit on d-axis.	

9.8.2.7 DroopSignalFeedbackKind

Governor droop signal feedback source.

Literal	Description	Code
electricalPower	Electrical power feedback (connection indicated as 1 in the block diagrams of models, e.g. GovCT1, GovCT2).	
none	No droop signal feedback, is isochronous governor.	
fuelValveStroke	Fuel valve stroke feedback (true stroke) (connection indicated as 2 in the block diagrams of model, e.g. GovCT1, GovCT2).	
governorOutput	Governor output feedback (requested stroke) (connection indicated as 3 in the block diagrams of models, e.g. GovCT1, GovCT2).	

9.8.2.8 FrancisGovernorControlKind

Governor control flag for Francis hydro model.

Literal	Description	Code
mechanicHydrolicTachoAccelerator	Mechanic-hydraulic regulator with tacho-accelerometer (Cflag = 1).	
mechanicHydraulicTransientFeedback	Mechanic-hydraulic regulator with transient feedback (Cflag=2).	
electromechanicalElectrohydraulic	Electromechanical and electrohydraulic regulator (Cflag=3).	

9.8.2.9 ExcIEEEST1AUELselectorKind

Type of connection for the UEL input used in ExcIEEEST1A.

Literal	Description	Code
ignoreUELsignal	Ignore UEL signal.	
inputHVgateVoltageOutput	UEL input HV gate with voltage regulator output.	
inputHVgateErrorSignal	UEL input HV gate with error signal.	
inputAddedToErrorSignal	UEL input added to error signal.	

9.8.2.10 ExcST6BOELselectorKind

Type of connection for the OEL input used for static excitation systems type 6B.

Literal	Description	Code
noOELinput	No OEL input is used.	
beforeUEL	The connection is before UEL.	
afterUEL	The connection is after UEL.	

9.8.2.11 ExcST7BOELselectorKind

Type of connection for the OEL input used for static excitation systems type 7B.

Literal	Description	Code
noOELinput	No OEL input is used.	
addVref	The signal is added to Vref.	
inputLVgate	The signal is connected in the input of the LV gate.	
outputLVgate	The signal is connected in the output of the LV gate.	

9.8.2.12 ExcST7BUELselectorKind

Type of connection for the UEL input used for static excitation systems type 7B.

Literal	Description	Code
noUELinput	No UEL input is used.	
addVref	The signal is added to Vref.	
inputHVgate	The signal is connected in the input of the HV gate.	
outputHVgate	The signal is connected in the output of the HV gate.	

9.8.2.13 ExcREXSFeedbackSignalKind

Type of rate feedback signals.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

Literal	Description	Code
fieldVoltage	The voltage regulator output voltage is used. It is the same as exciter field voltage.	
fieldCurrent	The exciter field current is used.	
outputVoltage	The output voltage of the exciter is used.	

9.8.2.14 InputSignalKind

Input signal type. In Dynamics modelling, commonly represented by j parameter.

Literal	Description	Code
rotorSpeed	Input signal is rotor or shaft speed (angular frequency).	
rotorAngularFrequencyDeviation	Input signal is rotor or shaft angular frequency deviation.	
busFrequency	Input signal is bus voltage frequency. This could be a terminal frequency or remote frequency.	
busFrequencyDeviation	Input signal is deviation of bus voltage frequency. This could be a terminal frequency deviation or remote frequency deviation.	
generatorElectricalPower	Input signal is generator electrical power on rated S.	
generatorAcceleratingPower	Input signal is generating accelerating power.	
busVoltage	Input signal is bus voltage. This could be a terminal voltage or remote voltage.	
busVoltageDerivative	Input signal is derivative of bus voltage. This could be a terminal voltage derivative or remote voltage derivative.	
branchCurrent	Input signal is amplitude of remote branch current.	
fieldCurrent	Input signal is generator field current.	

9.8.2.15 WindLVRTQcontrolModesKind

LVRT Q control modes M_{qLVRT}.

Literal	Description	Code
mode1	Voltage dependent reactive current injection ($M_{LVRT,1}$).	
mode2	Reactive current injection controlled as the pre-fault value plus an additional voltage dependent reactive current injection ($M_{LVRT,2}$).	
mode3	Reactive current injection controlled as the pre-fault value plus an additional voltage dependent reactive current injection during fault, and as the pre-fault value plus an additional constant reactive current injection post fault ($M_{LVRT,3}$).	

9.8.2.16 WindQcontrolModesKind

General wind turbine Q control modes M_{qG}.

Literal	Description	Code
voltage	Voltage control ($M_{G,u}$).	
reactivePower	Reactive power control ($M_{G,q}$).	
openLoopReactivePower	Open loop reactive power control (only used with closed loop at plant level) ($M_{G,qol}$).	
powerFactor	Power factor control ($M_{G,pf}$).	

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

9.8.2.17 WindLookupTableFunctionKind

Function of the lookup table.

Literal	Description	Code
fpslip	Power versus slip lookup table (f_{pslip}()). It is used for rotor resistance control model, IEC 61400-27-1, section 6.6.5.2.	
fpomega	Power vs. speed lookup table (f_{pomega}()). It is used for P control model type 3, IEC 61400-27-1, section 6.6.5.3.	
ipvdl	Lookup table for voltage dependency of active current limits (i_{pVDL}()). It is used for current limitation model, IEC 61400-27-1, section 6.6.5.7.	
iqvdl	Lookup table for voltage dependency of reactive current limits (i_{qVDL}()). It is used for current limitation model, IEC 61400-27-1, section 6.6.5.7.	
fdpf	Power vs. frequency lookup table (f_{dpf}()). It is used for wind power plant frequency and active power control model, IEC 61400-27-1, Annex E.	

9.8.2.18 GenericNonLinearLoadModelKind

Type of generic non-linear load model.

Literal	Description	Code
exponentialRecovery	Exponential recovery model.	
loadAdaptive	Load adaptive model.	

9.8.2.19 StaticLoadModelKind

Type of static load model.

Literal	Description	Code
exponential	Exponential P and Q equations are used and the following attributes are required: kp1, kp2, kp3, kpf, ep1, ep2, ep3 kq1, kq2, kq3, kqf, eq1, eq2, eq3.	
zIP1	ZIP1 P and Q equations are used and the following attributes are required: kp1, kp2, kp3, kpf kq1, kq2, kq3, kqf.	
zIP2	This model separates the frequency-dependent load (primarily motors) from other load. ZIP2 P and Q equations are used and the following attributes are required: kp1, kp2, kp3, kq4, kpf kq1, kq2, kq3, kq4, kqf.	
constantZ	The load is represented as a constant impedance. ConstantZ P and Q equations are used and no attributes are required.	

9.8.3 Primitives

9.8.3.1 String

A string consisting of a sequence of characters. The character encoding is UTF-8. The string length is unspecified and unlimited.

9.8.3.2 Date

Date as "yyyy-mm-dd", which conforms with ISO 8601. UTC time zone is specified as "yyyy-mm-ddZ". A local timezone relative UTC is specified as "yyyy-mm-dd(+/-)hh:mm".

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu

9.8.3.3 Boolean

A type with the value space "true" and "false".

9.8.3.4 Float

A floating point number. The range is unspecified and not limited.

9.8.3.5 Integer

An integer number. The range is unspecified and not limited.

ENTSO-E AISBL • Avenue de Cortenbergh 100 • 1000 Brussels • Belgium • Tel + 32 2 741 09 50 • Fax + 32 2 741 09 51 • info@entsoe.eu • www. entsoe.eu